# SUPPLEMENTARY MATERIAL FOR "SHORT-LIFETIME $H_0$ PERSISTENCE FOR THE THREE-DIMENSIONAL SIDE-24 PERIODIZED BARGMANN–FOCK FIELD"

DYLAN ROY

Abstract. This supplement contains the proofs that the main text cites. Part I proves the leading short-lifetime law, its rate and the side-24 correction: the setting and pair coordinates (§S1), the contact measure (§S2), the finite-side coefficient (§S3), the side-24 correction (§S4), elder selection (§S5), the limit (§S6) and the bounded remainder (§S7). Part II states the three-term law and the results behind it, with their proofs (§§S8–S11). Part III states and proves the results of the companion record that Parts I and II use: its three theorems in every dimension $d \geq 2$ and for every side $L > 0$ (§S12), and the deterministic separating-cylinder theorem on which its elder selection rests (§S13).

## Status of this draft (for the author; remove before submission)

- *What this is.* Checkpoint draft 15 of 21 October 2026, an author-side assembly of the proofs that the V3 main text cites, written as numbered statements with proofs. Draft 1 (13 October) was the first assembly. Draft 2 applies the repairs listed below. Draft 3 adds a proof of the far elder rate (Proposition S8.13), which Part II previously cited to a repository record. Draft 4 makes the repairs to §S5 listed below and adds the jet bounds (3.29) of the main text to Lemma S4.29. Draft 5 adds Part III (§§S12–S13), described below, drafts 6 and 7 write out steps of §§S12–S13 that the record states without derivation, draft 8 records the reading, draft 9 repairs the finite-scale witness densities of §S5, draft 10 makes the repairs of a same-provider audit and proves the identification of $c_2$ that drafts 1–9 stated as a hypothesis, draft 11 proves the cap-region bound that drafts 1–10 imported, draft 12 proves the equal-height statements that drafts 1–11 imported, draft 13 adds a proof of Proposition S12.24, makes two item-level dependencies explicit, and ships a proof-dependency check, and draft 14 removes the remaining references to earlier drafts and to repository paths from Sections S1–S13 (below).
  - Every statement and display is mapped to its source in the separate file `CONCORDANCE.md`.
  - That file also lists each correction applied, each step added and each issue found during assembly. Its sections "Checkpoint draft 2" to "Checkpoint draft 13" give the disposition of each issue.
  - The draft has not been read as a whole, by the author or by anyone else. Every section has had a bounded nonauthor read, in slices of drafts 4 to 7, by a provider-distinct AI agent posting from the same account; each requested change was applied and read back. A second provider-distinct read of §S5.4, by an OpenAI agent on draft 8, found that the densities at a fixed positive scale near the pair axis were not proved; draft 9 repairs this, and the repair has a nonauthor read by a Grok Bot support agent (PASS) and a provider-distinct read by OpenAI (PASS_SCOPED, bound to the TEX source); the other slices of §S5 have provider-distinct reads by xAI (PASS_SCOPED). The pull request of the main text records the verdicts. No organizational independence is claimed.

*Date*: Draft, October 21, 2026.

- *Part I* (§§S1–S7) follows the main text's proof of Theorem 2.1 and of the correction (2.10). *Part II* (§§S8–S11) states the results behind Theorem 2.3. Its proofs are given in full or condensed, as `CONCORDANCE.md` marks for each.
- *Author correction.* The elder mark is defined through the maximin connection level (Definition S5.2). The two-sector formula of the 1 August supplement required $\mathfrak{m}_f(u_+, u_-) < h$, but the segment through $S$ gives $\mathfrak{m}_f(u_+, u_-) \geq h$, so that mark was identically zero as written.
- *Repairs in draft 2, §S5.* The selection bound $1 - p_r \leq Cr^3$ on compact marks is Theorem A of [7] (Theorem S12.3) for every $d \geq 2$ and $L > 0$, and the main text cites it. §S5 now gives a second proof in $d = 3$ (Theorem S5.36), whose finite-scale witness densities draft 9 completes (below):
  - the three witness regions are measured from the pair midpoint;
  - self-attachment is same-maximum self-attachment, and the remaining cases are witnesses by Lemma S5.6;
  - the fixed-distance count (main text (3.17)) is stated and proved on compact marks, with the conditional pin density;
  - the singular-near envelope (main text (3.16)) is proved by Taylor's formula with the value window, for every angle (Lemma S5.22; the symbolic check is `checks/s5_envelope_check.py`). The elimination (B.1), whose denominators vanish at $c = 0$, is no longer used there;
  - the collar count uses (B.1) only where $|c|$ is bounded below, and Hessian collision zeros near $c = 0$ (superseded in draft 9, where the collar count no longer uses (B.1));
  - the capture–escape lemmas contain the five local repairs (§S5.5);
  - the Morse event under the Palm law is addressed in Remark S5.38.
- *Repairs in draft 4, §S5.*
  - The singular-near residual of $V_0$ has a random term $-\frac{\eta^2}{16}(Yf_{112} + Zf_{113})$, which drafts 1–3 omitted. The planar covariances now depend on $\eta$, and the axis suppression used is $\exp(-c_*/\rho^2)$ (Lemma S5.20). The conclusions are unchanged.
  - The positivity of $A_{\mathrm{col}}$ and $A_{\mathrm{sing}}$ on the side-24 field is proved by exact factorizations and the derivative nondegeneracy of §S1 (Lemmas S5.18 and S5.21). Drafts 1–3 cited a derivative bound $10^{40}$ to §S4, which does not contain it.
  - The level terms of Lemma S5.28 are displayed. The symbolic check is `checks/s5_factor_check.py`.
- *Repairs in draft 9, §S5*, after the provider-distinct read just mentioned (its findings F1–F3).
  - F1: drafts 1–8 stated in (S5.56) a bound for $\int_{S^2} \rho^{-m}$ times the singular-near density, which is false at a fixed positive scale because the density is positive on the pair axis; the collar count passed from contact limits to finite scale in the same way. The new §S5.8 proves the finite-scale densities near the axis by a radial blow-up of the parameters (Lemmas S5.39–S5.41), and Lemmas S5.19, S5.20, S5.22 and S5.23 now use them. The conditional moments are bounded through the Mahalanobis norm of the conditioning values, and the collar envelopes come from the Hessian collision zero alone. The check is `checks/s5_finite_scale_check.py`.
  - F2: $V_0$ subtracts $U^{(0)} = \frac{1}{2}(f(M) + f(S))$, which the pins fix, in place of $f(x)$, which they do not. The contact limits are unchanged.
  - F3: Lemma S5.15 now says that the product of the three determinants has degree at most six in $\kappa$, and exactly six off the collision strata.
  - No statement or display number of draft 8 changes; the new statements are Lemmas S5.39–S5.41, with displays (S5.107)–(S5.112), at the end of §S5.
- *Repairs in draft 10*, from a same-provider audit of §§S2–S4 and S6–S13 against the main text, which found no mathematical error (`CONCORDANCE.md`, "Checkpoint draft 10").

  - Citations: Proposition S6.4 takes $p_r \to 1$ from Theorem S12.3; the marked Kac–Rice formula with the global elder mark is cited to Proposition S12.22; the limit (S3.27) cites the contact lemmas of §S2.
  - Statements made exact: the canonical density version does not depend on the separation $\varrho$ (after (S6.5)); (14.1) of the record holds at every separation (after Lemma S12.29); Theorem S3.10 holds for every $L$ without hypothesis; Lemma S12.21 allows every real $b$; Theorem S12.5 names its density versions; Corollary S12.25 says which unrestricted statements hold; Definition S11.1 covers $k = 0$.
  - Wording: the all-typed measures of §S2 count maxima with index-$(d-1)$ critical points; the moments after Lemma S2.9 are polynomial in the marks; Proposition S3.19 is derived; Remark S3.21 marks its remainder as orientation; Remark S7.22 and Lemma S7.1(b) are made accurate; the proof of Theorem S11.5 calls $q_0$ the pin quadratic form; three misprints $A_2$ for $\mathcal{A}_2$ in §S10 are corrected.
  - Hypothesis S11.11 of drafts 1–9 is now Corollary S11.11, proved from §S10, so Corollary S11.12 is unconditional.
  - Repairs in §§S8–S9: Theorem S9.13 states the unrounded enclosures of $c_1$ from which its ratios are formed, and its method is described exactly; Proposition S8.2 takes $r_R \le \frac{1}{2}$, and Proposition S8.3 states the embedding condition $r < L/(4\sqrt{2})$; the proof of Theorem S8.6 cites Lemma S7.14. Draft 11 proves the bound of Theorem K of record [R7] of the main text that Proposition S8.3 imported (Lemma S8.16, appended as §S8.5); no earlier statement or display number changes.
  - No statement number or display number of draft 9 changes. Three check scripts are shipped: `checks/s1s2_verify_check.py`, `checks/s6_limit_check.py` and `checks/s10_s11_fold_check.py`.

- *Draft 11* proves the second bound of Proposition S8.3, on which the term $c_1\ell^{1/4}$ rests, in a new last subsection §S8.5 (Lemma S8.16); drafts 1–10 imported it from Theorem K of record [R7] of the main text. The proof is that of Lemma S7.18 with the axial bound (S7.14), which keeps the factor $k^2$, and with a shifted Gaussian bound for the density of the transverse Hessian. The check script is `checks/s8_capK_check.py`.
- *Draft 12* proves items (i) and (ii) of Proposition S8.4 (the equal-height mass $0 < B_{d,L} < \infty$, the limit $\rho_{\mathrm{rej}} \to B_{d,L}$, and the fixed-separation coupling), which drafts 1–11 imported from Theorem Z of record [R7] of the main text, in a new last subsection §S8.6, with no new numbered statement. The coupling and the bounds on $B_{d,L}$ are direct, from Lemma S12.8, Lemma S7.14 and (S8.33); the limit follows from (S8.16) and (S8.20), whose proofs do not use it. With Lemma S8.16, Part II now imports no analytic statement from a repository record.
- *Draft 15* makes radius and parameter matching explicit at six places that a whole-document consistency read identified (normalizer radius and the radius of Theorem S5.36; Theorem S12.3 in the variables of §S6; the radii of §S7 against §S12; $r_R$ in §§S8 and S10; Proposition S8.2(iv) at $k = 0$; the density citation for the unrestricted densities). No statement is strengthened except Proposition S8.2(iv), now stated for $k \ge 0$, and no number changes.
- *Draft 14* is documentary. Outside this section, ten clauses that named earlier drafts and one sentence that named a repository path are removed or restated without the history; the mathematics they carried is kept. The concordance records the old wording. No statement, number or label changes.
- *Draft 13* is documentary. Proposition S12.24, which drafts 1–12 stated without a proof, gets its two-line proof from Proposition S12.23(b) and the change of variables $\ell = kr^3$. The proofs of Propositions S12.22 and S12.23 and of Lemma S5.20 now say which parts of each other they use, so that the dependency is visibly acyclic item by item. The check

script `checks/dep_graph_check.py` builds the proof-dependency graph of the 194 numbered statements from the sources (396 references from proofs and statements to other statements), finds seven statement-level strongly connected components, each explained in the script and in the concordance as a pointer, a definition used from another statement or an item-level acyclic dependency, and exits with an error if a later draft adds one that is not explained. It judges no mathematics.

- *Part III, added in draft 5.* §S12 transcribes the companion record [7], read as reconciled in [6], in every dimension $d \geq 2$ and for every side $L > 0$; §S13 proves the deterministic separating-cylinder theorem on which its elder selection rests. Every citation of [7] in §§S5–S10 that uses a statement of the record now names its counterpart there, so Part I and Part II no longer depend on the record for any statement they use. The non-use sentence in Remark S5.37 is unchanged.
  - Lemma S12.19(b), the genericity of the unpinned field, is added: the record's proof of Theorem C uses it without proof.
  - Two statements are worded precisely: in Lemma S12.27(c) the births range over $\mathbb{R}$, and Lemma S12.8 allows several functionals at each site.
  - The finite algebra of §§S12–S13 is checked by `checks/s12_parent_check.py`. Where the record only sketches a step, the step is given as sketched there; `CONCORDANCE.md` lists these places.
  - Drafts 6 and 7 write out the proofs of Lemmas S12.10 and S12.14, the averaging formulas in the proof of Lemma S12.11(a), the measurability in Lemma S12.20(a) and the Hermite identity of Lemma S13.3. The identities are checked by `checks/s12_derivations_check.py`.
- *Other repairs in draft 2.* Convention S1.6 on pointwise representatives, and the $\beta$-normalization of $D_n$ in §S3, which matters only when $\beta \neq 1$.
- *Open items*, with details in `CONCORDANCE.md`.
  - The remaining issues of §§S7–S11 are notational or concern the sources; draft 3 resolves the far elder rate, the pointer of Corollary S11.11(a) (a hypothesis through draft 9) and the DLMF citation.
  - `CONCORDANCE.md` (section "CONC_s12", issues I-7 to I-13, with the dispositions of "Checkpoint draft 6") lists the remaining steps of §S12 that the record states without derivation. None is above low severity.
- *Review status* is not stated in this supplement. It is given result by result in §7 of the main text.
- *References from the main text* use this supplement's numbering, as of draft 4. Draft 5 changes no statement or display number of §§S1–S11, drafts 9 and 10 change none at all, draft 11 adds §S8.5, Lemma S8.16 and its displays after every existing number of §S8, draft 12 adds only the unnumbered §S8.6, draft 13 changes no number (a proof carries none), and neither does draft 14.

## Part I. The leading law, its rate, and the side-24 correction

### S1. Setting, pair coordinates and multiplicity

This section collects the facts about the field and about ordered pairs of critical points on which the contact analysis of Section S2 rests: the spectral representation of the covariance (§S1.1), the positivity of every finite Gram matrix of derivative evaluations (§S1.2), the density convention (§S1.3), the coordinates of ordered typed pairs and their Jacobian (§S1.4), and the unit multiplicity of the elder pairing (§S1.5).

S1.1. **The field and its spectral representation.** Write $\mathbb{T}^3_{24} = (\mathbb{R}/24\mathbb{Z})^3$ and, for $m \in \mathbb{Z}^3$, $e_m(z) = \exp(2\pi i\, m\cdot z/24)$. As in (1.1) of the main text, the normalized periodized Bargmann–Fock covariance is

$$K_{24}(z) = Z_{24}^{-1} \sum_{n\in\mathbb{Z}^3} \exp\Big(-\frac{|z+24n|^2}{2}\Big), \qquad Z_{24} = \sum_{n\in\mathbb{Z}^3} e^{-288|n|^2}, \tag{S1.1}$$

and $f = f_{24}$ is a centered stationary Gaussian field on $\mathbb{T}^3_{24}$ with $\mathbb{E}[f(x)f(y)] = K_{24}(x-y)$. Since $|24n|^2/2 = 288|n|^2$, the normalization gives $K_{24}(0) = 1$.

**Lemma S1.1** (spectral representation)**.** *For every $z \in \mathbb{R}^3$,*

$$K_{24}(z) = Z_{24}^{-1} \sum_{m\in\mathbb{Z}^3} a_m\, e_m(z), \qquad a_m = \frac{(2\pi)^{3/2}}{24^3} \exp\Big(-\frac{2\pi^2|m|^2}{24^2}\Big) > 0. \tag{S1.2}$$

*Thus the Fourier coefficients of $K_{24}$ in the basis $(e_m)$ are $\widehat{K}_{24}(m) = Z_{24}^{-1} a_m$, as in (1.2) of the main text. All of them are strictly positive, and $\sum_m \widehat{K}_{24}(m)(1+|m|)^N < \infty$ for every $N \ge 0$.*

*Proof.* Let $g(z) = e^{-|z|^2/2}$ on $\mathbb{R}^3$, with Fourier transform $\hat g(k) = \int_{\mathbb{R}^3} g(z) e^{-ik\cdot z}\, dz = (2\pi)^{3/2} e^{-|k|^2/2}$. The Poisson summation formula for the lattice $24\mathbb{Z}^3$ gives

$$\sum_{n\in\mathbb{Z}^3} g(z+24n) = 24^{-3} \sum_{m\in\mathbb{Z}^3} \hat g\Big(\frac{2\pi m}{24}\Big) e_m(z) = \frac{(2\pi)^{3/2}}{24^3} \sum_{m\in\mathbb{Z}^3} e^{-2\pi^2|m|^2/24^2}\, e_m(z),$$

where both series converge absolutely and uniformly because $g$ and $\hat g$ are Schwartz functions. Dividing by $Z_{24}$ gives (S1.2). The summability statement follows from the Gaussian decay of $a_m$. □

S1.2. **Full-spectrum Gram positivity.** For a multi-index $\alpha$ and $x \in \mathbb{T}^3_{24}$ write $L_{\alpha,x}(f) = \partial^\alpha f(x)$. A *finite derivative evaluation* is a finite real linear combination

$$T(f) = \sum_{j,\alpha} c_{j,\alpha}\, L_{\alpha,x_j}(f) = \sum_{j,\alpha} c_{j,\alpha}\, \partial^\alpha f(x_j). \tag{S1.3}$$

Its *associated distribution* is the finite distribution

$$\mathcal{T} = \sum_{j,\alpha} c_{j,\alpha}\, (-1)^{|\alpha|}\, \partial^\alpha \delta_{x_j}, \qquad \text{so that} \qquad \langle \mathcal{T}, \varphi\rangle = T(\varphi) \quad (\varphi \in C^\infty(\mathbb{T}^3_{24})). \tag{S1.4}$$

We fix the Fourier convention $\widehat{T}(m) = T(e_m) = \langle \mathcal{T}, e_m\rangle$; up to the factor $24^{-3}$ and the reflection $m \mapsto -m$, this is the $m$-th Fourier coefficient of $\mathcal{T}$.

**Lemma S1.2** (spectral Gram identity)**.** *For every finite derivative evaluation $T$,*

$$\mathrm{Var}[T(f)] = Z_{24}^{-1} \sum_{m\in\mathbb{Z}^3} a_m\, |\widehat{T}(m)|^2 = \sum_{m\in\mathbb{Z}^3} \widehat{K}_{24}(m)\, |\widehat{T}(m)|^2. \tag{S1.5}$$

*Because every $a_m$ is strictly positive,*

$$\mathrm{Var}[T(f)] = 0 \iff \widehat{T}(m) = 0 \ \text{ for every } m \in \mathbb{Z}^3 \iff \mathcal{T} = 0 \ \text{ as a distribution.} \tag{S1.6}$$

*Proof.* By Lemma S1.1 the series (S1.2) may be differentiated term by term any number of times, so the mean-square derivatives of $f$ satisfy

$$\mathrm{Cov}(\partial^\alpha f(x), \partial^\beta f(y)) = \partial_x^\alpha \partial_y^\beta\, K_{24}(x-y) = \sum_{m\in\mathbb{Z}^3} \widehat{K}_{24}(m)\, \partial^\alpha e_m(x)\, \overline{\partial^\beta e_m(y)},$$

using $e_m(x-y) = e_m(x)\overline{e_m(y)}$. Multiplying by the real coefficients $c_{j,\alpha}c_{j',\beta}$ and summing gives $\mathrm{Var}[T(f)] = \sum_m \widehat{K}_{24}(m)\, T(e_m)\, \overline{T(e_m)}$, which is (S1.5). Since every $a_m$ is strictly positive, the variance vanishes if and only if $\widehat{T}(m) = 0$ for all $m$. The last equivalence in (S1.6) is the uniqueness

of Fourier series for distributions on the compact torus: trigonometric polynomials are dense in $C^\infty(\mathbb{T}^3_{24})$, so a distribution that annihilates every $e_m$ is zero. □

**Theorem S1.3** (full-spectrum Gram positivity)**.** *Let $\Lambda_1, \dots, \Lambda_N$ be finite-order constant-coefficient derivative-evaluation functionals on $\mathbb{T}^3_{24}$, $\Lambda_i(f) = (P_i(\partial)f)(x_i)$ with polynomial symbols $P_i$, and let $\mathcal{T}_i = P_i(-\partial)\delta_{x_i}$ be the associated distributions, $\langle \mathcal{T}_i, \varphi\rangle = \Lambda_i(\varphi)$. The covariance matrix $\big(\mathrm{Cov}(\Lambda_i f, \Lambda_j f)\big)_{i,j\le N}$ is positive definite if and only if $\mathcal{T}_1, \dots, \mathcal{T}_N$ are linearly independent.*

*In particular, if at each of the distinct support points the polynomial symbols of the differential operators evaluated there are linearly independent, then the covariance matrix is positive definite.*

*Proof.* For $c \in \mathbb{R}^N$, $c^\top(\mathrm{Cov}(\Lambda_i f, \Lambda_j f))c$ is the variance (S1.5) of the finite derivative evaluation $\sum_i c_i\Lambda_i$, whose associated distribution is $\sum_i c_i\mathcal{T}_i$. By (S1.6) it vanishes exactly when $\sum_i c_i\mathcal{T}_i = 0$, which proves the equivalence.

For the second statement, suppose $\sum_i c_i\mathcal{T}_i = 0$. Distributions supported at distinct points have disjoint supports, so the partial sum $\sum_{i:\, x_i=x} c_i\mathcal{T}_i$ vanishes at each support point $x$. At one point, the derivatives $\partial^\alpha\delta_x$ indexed by distinct multi-indices are linearly independent; hence $\sum_{i:\, x_i=x} c_i P_i(-\partial)\delta_x = 0$ holds exactly when $\sum_{i:\, x_i=x} c_i P_i = 0$ as a polynomial. In other words, operator functionals at one point are independent exactly when their polynomial symbols are. By hypothesis all $c_i$ vanish. □

**Corollary S1.4** (compact uniform eigenfloors)**.** *Let $\mathcal{P}$ be a compact metric space of parameters and, for $\pi \in \mathcal{P}$, let $\Lambda_1^\pi, \dots, \Lambda_N^\pi$ be derivative-evaluation functionals of order at most $k$,*

$$\Lambda_i^\pi(f) = \sum_{|\alpha|\le k} c_{i,\alpha}(\pi)\, \partial^\alpha f(x_i(\pi)),$$

*whose coefficients $c_{i,\alpha}$ and support points $x_i$ depend continuously on $\pi$; for instance, a fixed finite derivative basis in directions from a compact set, evaluated at support points whose distinct members keep a mutual distance at least $\delta > 0$. If the associated distributions are linearly independent for every $\pi \in \mathcal{P}$, then the smallest eigenvalue of $\mathrm{Cov}(\Lambda_1^\pi f, \dots, \Lambda_N^\pi f)$ is a continuous and strictly positive function of $\pi$, and hence has a positive lower bound on $\mathcal{P}$.*

*Proof.* The entries $\mathrm{Cov}(\Lambda_i^\pi f, \Lambda_j^\pi f) = \sum_{\alpha,\beta} c_{i,\alpha}(\pi)c_{j,\beta}(\pi)(-1)^{|\beta|}(\partial^{\alpha+\beta}K_{24})(x_i(\pi) - x_j(\pi))$ are continuous in $\pi$ because $K_{24}$ is smooth. The smallest eigenvalue of a symmetric matrix is 1-Lipschitz in the operator norm (Weyl's inequality), hence continuous in $\pi$; it is strictly positive by Theorem S1.3; and a positive continuous function on a compact set has a positive minimum. □

**Proposition S1.5** (independence of the corrected pair frame)**.** *Let $x \in \mathbb{T}^3_{24}$, $t \in S^2$, and let $(t, v, w)$ be an orthonormal frame.*

*(i) The eight derivative evaluations at $x$*

$$f, \quad \partial_t f, \quad \partial_t^2 f, \quad \partial_t^3 f, \quad \partial_v f, \quad \partial_t\partial_v f, \quad \partial_w f, \quad \partial_t\partial_w f \tag{S1.7}$$

*have a positive definite covariance matrix. For the Bargmann–Fock kernel $e^{-|z|^2/2}$ on $\mathbb{R}^3$ in place of $K_{24}$, the determinant of this covariance matrix equals 12.*

*(ii) Let $\delta > 0$ and let $y \in \mathbb{T}^3_{24}$ with $\mathrm{dist}(y, x) \ge \delta$. Append to (S1.7) any standard coordinate basis of*

$$f(y), \qquad \nabla f(y), \qquad \mathrm{Hess}\, f(y). \tag{S1.8}$$

*The covariance matrix of the enlarged family is positive definite, and its smallest eigenvalue has a positive lower bound depending only on $\delta$, uniformly in $x$, $y$ and the frame.*

*Proof.* (i) In the orthonormal coordinates $(\xi_t, \xi_v, \xi_w) = (t\cdot\xi, v\cdot\xi, w\cdot\xi)$ the polynomial symbols of (S1.7) are the distinct monomials $1, \xi_t, \xi_t^2, \xi_t^3, \xi_v, \xi_t\xi_v, \xi_w, \xi_t\xi_w$, which are linearly independent. All eight evaluations are supported at $x$, so Theorem S1.3 gives positive definiteness. For the kernel

$e^{-|z|^2/2}$, which is rotation invariant and factorizes over the coordinates $(t, v, w)$, $\mathrm{Cov}(\partial^\alpha f, \partial^\beta f) = (-1)^{|\beta|}\partial^{\alpha+\beta}e^{-|z|^2/2}|_{z=0}$ vanishes unless $\alpha+\beta$ is even in each coordinate. The covariance matrix therefore splits into the blocks $\{f, \partial_t^2 f\}$ with covariance $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 3 \end{smallmatrix}\right)$, $\{\partial_t f, \partial_t^3 f\}$ with covariance $\left(\begin{smallmatrix} 1 & -3 \\ -3 & 15 \end{smallmatrix}\right)$, and the four singletons $\partial_v f$, $\partial_t\partial_v f$, $\partial_w f$, $\partial_t\partial_w f$ of variance 1, and its determinant is $2 \cdot 6 \cdot 1 = 12$.

(ii) The distributions of (S1.7) are supported at $x$ and those of (S1.8) at $y \neq x$. The symbols at $y$ are the distinct monomials 1, $\xi_i$ and $\xi_i\xi_j$ $(i \le j)$, so Theorem S1.3 gives positive definiteness. The parameters $(x, y, (t, v, w))$ with $\mathrm{dist}(y, x) \ge \delta$ range over a compact set on which the support points stay at distance at least $\delta$ and the coefficients depend continuously on the frame; Corollary S1.4 gives the uniform lower bound. □

The list (S1.7) is the contact limit of the corrected pair pins. By Lemma S2.3, the corrected divided-difference pin vector $Y_r$ of Definition S2.1 is obtained from the raw pins by an invertible linear transformation for every $r > 0$, and its contact limit consists, up to the order of the entries and the factor $-1/12$ on $\partial_t^3 f$, of the derivative coordinates (S1.7) at the pair midpoint. This coordinate transformation does not depend on the covariance kernel. Proposition S1.5(i) therefore gives strict contact-limit nondegeneracy of the corrected pair frame, and Proposition S1.5(ii) gives it jointly with a remote third point; this is the nondegeneracy used for the fixed-distance witnesses of Section S5.

S1.3. **The density convention.**

**Convention S1.6** (density)**.** $\nu_{3,24}$ is the *first-moment intensity density per unit ordinary torus volume*: $\nu_{3,24}(\ell)\, d\ell$ is the expected number of finite nonessential superlevel $H_0$ bars per unit volume with lifetime in $[\ell, \ell + d\ell]$. It is not the probability density of the lifetime of a randomly sampled bar; such a law requires normalizing by the expected bar count per unit volume and specifying the sampling rule, neither of which is used in Theorem 2.1 of the main text. The asymptotics of Theorem 2.1 of the main text hold for the pointwise Kac–Rice representative of the Radon–Nikodym derivative of Proposition 1.2 of the main text. A pointwise asymptotic is a property of that representative: a change of the density on a null set need not preserve it.

S1.4. **Ordered typed pairs and their coordinates.** In this subsection and the next, $d \ge 2$ and $\mathbb{T}_L^d = (\mathbb{R}/L\mathbb{Z})^d$ is the flat torus of side $L > 0$ with its quotient distance dist. The index of a critical point is the number of negative eigenvalues of its Hessian. The side-24 field is the case $d = 3$, $L = 24$.

**Definition S1.7** (ordered typed pair space)**.** Fix $0 < \rho < L/2$. For a Morse function $f$ on $\mathbb{T}_L^d$ with distinct critical values, the *typed ordered local pair space* is

$$\mathcal{P}_\rho(f) = \Big\{(M, S) : \begin{array}{l} M \text{ is a local maximum, } S \text{ is an index-}(d-1) \text{ critical point,} \\ 0 < \mathrm{dist}(M, S) < \rho, \ f(M) > f(S) \end{array}\Big\}. \tag{S1.9}$$

**Lemma S1.8** (midpoint, radius and direction)**.** *For points $M, S \in \mathbb{T}_L^d$ with $0 < \mathrm{dist}(M, S) < \rho$, the short torus displacement $u = \mathrm{Log}_M(S) \in \mathbb{R}^d$ is unique. Put*

$$x = M + \frac{u}{2} \bmod L, \qquad r = |u|, \qquad t = \frac{u}{r} \in S^{d-1}. \tag{S1.10}$$

*Conversely,*

$$M = x - \frac{r}{2}\, t, \qquad S = x + \frac{r}{2}\, t. \tag{S1.11}$$

*These formulas define a bijection between ordered pairs of points at distance in $(0, \rho)$ and triples $(x, r, t) \in \mathbb{T}_L^d \times (0, \rho) \times S^{d-1}$; in particular $r = \mathrm{dist}(M, S)$, and $\mathcal{P}_\rho(f)$ is identified with a subset of $\mathbb{T}_L^d \times (0, \rho) \times S^{d-1}$.*

*Proof.* The lifts of $S - M$ to $\mathbb{R}^d$ form a coset $\tilde{u} + L\mathbb{Z}^d$, and $\operatorname{dist}(M, S) = \min_n |\tilde{u} + Ln| < \rho < L/2$. Two lifts of length below $L/2$ differ by an element of $L\mathbb{Z}^d$ of length below $L$, hence coincide; so the lift $u = \operatorname{Log}_M(S)$ of length $\operatorname{dist}(M, S)$ is unique, (S1.10) is well defined, and (S1.11) recovers $(M, S)$. Conversely, for $(x, r, t)$ with $0 < r < \rho$, the points (S1.11) satisfy $S - M \equiv rt$ modulo $L\mathbb{Z}^d$, and $rt$ is the unique lift of length below $L/2$; hence $\operatorname{dist}(M, S) = r$, $\operatorname{Log}_M(S) = rt$, and (S1.10) returns $(x, r, t)$. □

**Proposition S1.9** (exact Jacobian and absence of a half factor)**.** *In the coordinates of Lemma S1.8,*

$$dM\, dS = dx\, r^{d-1}\, dr\, d\sigma_{d-1}(t), \tag{S1.12}$$

*with no numerical prefactor, where $\sigma_{d-1}$ is the surface measure of $S^{d-1}$. The antipodal coordinate $(x, r, -t)$ is mapped by (S1.11) to $(S, M)$, which is not another element of $\mathcal{P}_\rho(f)$. Hence integration over the full directed sphere counts each typed ordered pair once.*

*Proof.* Work in the unique local Euclidean lift and set $u = S - M$, $x = (M + S)/2$. The inverse linear map is

$$(x, u) \longmapsto (M, S) = \Big(x - \frac{u}{2},\ x + \frac{u}{2}\Big). \tag{S1.13}$$

Its $2d \times 2d$ block matrix is

$$\Theta = \begin{pmatrix} I & -I/2 \\ I & I/2 \end{pmatrix}. \tag{S1.14}$$

Subtracting the first block row from the second gives the block-triangular matrix $\left(\begin{smallmatrix} I & -I/2 \\ 0 & I \end{smallmatrix}\right)$, so

$$|\det \Theta| = 1, \qquad dM\, dS = dx\, du. \tag{S1.15}$$

Standard polar coordinates in the directed displacement give

$$du = r^{d-1}\, dr\, d\sigma_{d-1}(t), \tag{S1.16}$$

and (S1.12) follows. By (S1.11), $(x, r, -t)$ is mapped to $(x + \frac{r}{2}t,\ x - \frac{r}{2}t) = (S, M)$. This is not another atom of $\mathcal{P}_\rho(f)$, because $S$ is not a maximum and $M$ is not an index-$(d-1)$ critical point. A factor $1/2$ belongs only to a different quotient convention using unoriented axes $t \sim -t$; the coefficient formula (2.9) of the main text (Section S3) integrates over the full directed sphere and does not use that quotient. □

**Corollary S1.10** (the side-24 pair coordinates)**.** *Let $d = 3$, $L = 24$ and $0 < \rho < 12$. For a maximum $M$ and an index-two saddle $S$ of $f_{24}$ at distance $r < \rho$, the coordinates of Lemma S1.8 are $x = (M + S)/2$ and $S - M = rt$, $t \in S^2$, with inverse $(x, u) \mapsto (M, S) = (x - u/2,\ x + u/2)$ of absolute determinant one, so that*

$$dM\, dS = dx\, du = dx\, r^2\, dr\, d\sigma_2(t). \tag{S1.17}$$

*The antipodal direction $-t$ exchanges $M$ and $S$ and does not represent the same typed ordered pair; the full directed sphere $S^2$ counts each maximum–saddle pair once, with no factor $1/2$.*

*Proof.* This is Lemma S1.8 and Proposition S1.9 with $d = 3$ and $L = 24$. □

S1.5. **Unit multiplicity.**

**Proposition S1.11** ($H_0$ unit multiplicity)**.** *Let $f$ be a Morse function with pairwise distinct critical values on a compact connected $d$-manifold, $d \geq 2$, and consider the superlevel filtration $X_a = \{f \geq a\}$. Then the elder death map $M \mapsto D_f(M)$ induces bijections*

$$\{\textit{non-global maxima}\} \longleftrightarrow \{\textit{finite superlevel } H_0 \textit{ bars}\} \longleftrightarrow \{\textit{index-}(d-1)\ H_0\textit{-merge saddles}\}, \tag{S1.18}$$

*and every finite nonessential $H_0$ bar has exactly one ordered pair $(M, D_f(M))$.*

*Proof.* As $a$ decreases, $X_a$ changes topology only at critical values, and, the critical values being distinct, at one critical point at a time; crossing a critical point of index $k$ attaches a $(d-k)$-handle to the superlevel set. Hence:

- every local maximum ($k = d$) creates one $H_0$ component;
- crossing an index-$(d-1)$ critical point attaches a one-handle to the superlevel set; that handle either merges two distinct components or creates a loop inside one component, and a merge lowers $\beta_0$ by exactly one while a loop leaves $\beta_0$ unchanged;
- a critical point of index $k \le d-2$ attaches a handle of dimension $d-k \ge 2$ along a connected attaching region, and does not change $\beta_0$.

At a merge, the two components immediately above the saddle have unique elder maxima, because critical values are distinct. The elder rule retains the higher maximum and kills the lower one. Thus each merge kills exactly one $H_0$ class. Every non-global maximum eventually dies: below the global minimum the connected manifold is one superlevel component, represented by the global maximum. No maximum can die twice, and no generic binary merge can kill two classes. Therefore the three sets in (S1.18) are in bijection through the elder death map, and every finite nonessential $H_0$ bar has one ordered pair $(M, D_f(M))$. □

The proof uses only connectedness, Morse nondegeneracy, distinct critical values, and the one-handle description of an index-$(d-1)$ superlevel saddle. By Proposition 1.1 of the main text, $f_{24}$ is almost surely Morse with pairwise distinct critical values, so Proposition S1.11 applies on $\mathbb{T}^3_{24}$ with probability one: each finite nonessential bar has one creating maximum and one index-two merging death saddle.

*Remark* S1.12 (branch count). An index-$(d-1)$ superlevel merge saddle has a one-dimensional unstable manifold under gradient ascent, and hence two half-branches. The pair process counts the critical-point pair $(M, S)$, not branch incidences. Direct adjacency is an existential Boolean mark:

$$A(M,S) = 1 \iff \text{at least one unstable half-branch of } S \text{ converges to } M. \tag{S1.19}$$

If both half-branches converge to $M$, the same ordered pair still contributes one atom. No branch factor 2 enters the contact density $J_r$ of Theorem S2.12 or the coefficient $c_{3,24}$.

## S2. The contact measure

This section derives the Kac–Rice measure of ordered pairs (maximum, index-$(d-1)$ saddle) at small separation $r$: the corrected pins and their Jacobian (§S2.1), the soft endpoint factors and the Palm normalizer (§S2.2), the cubic-gap Jacobian (§S2.3), the cancellation of the ambient dimension (§S2.4) and the lifetime pushforward (§S2.5). The general statements concern a centered $C^4$ Gaussian field $f$ in dimension $d \ge 1$, on $\mathbb{R}^d$, on a flat $d$-torus, or in a Euclidean chart of a smooth $d$-manifold; the side-24 field is the case $d = 3$, $f = f_{24}$. The pair coordinates $(x, r, t)$ are those of §S1.4. For a stationary field, in particular $f_{24}$, none of the laws below depends on the midpoint $x$; in a chart they may, and we suppress $x$ from the notation.

### S2.1. **Corrected pins.**

**Definition S2.1** (contact pair, pins and corrected frame)**.** Fix a directed unit vector $t \in S^{d-1}$ and an orthonormal transverse frame $n_1, \dots, n_{d-1}$; for $d = 3$ we write $(t, v, w) = (t, n_1, n_2)$. For $r > 0$ put

$$M = x - \frac{r}{2}\,t, \qquad S = x + \frac{r}{2}\,t, \tag{S2.1}$$

and impose the exact critical-pair pins

$$f(M) = b, \qquad f(S) = b - \frac{\kappa r^3}{6}, \qquad \nabla f(M) = \nabla f(S) = 0, \qquad \kappa > 0. \tag{S2.2}$$

The intended types are a maximum at $M$ and a critical point of index $d-1$ at $S$: the transverse Hessian is negative definite and the unique axial eigenvalue changes from negative to positive. Such a point merges two superlevel components or creates a loop (Proposition S1.11); the weights below test only the index, and only the elder mark of §S5 selects the merging pairs. The raw pin vector and its target are

$$\Pi_r = \Pi_{r,x,t} = (f(M), f(S), \nabla f(M), \nabla f(S)) \in \mathbb{R}^{2d+2}, \qquad z_r(b,\kappa) = \Big(b,\, b - \frac{\kappa r^3}{6},\, 0,\, 0\Big). \tag{S2.3}$$

The corrected divided-difference pin vector is $Y_r = (V^+, V^-_{\mathrm{corr}}, G^+_t, G^-_t, G^+_{n_1}, G^-_{n_1}, \dots, G^+_{n_{d-1}}, G^-_{n_{d-1}})$, where

$$V^+ = \frac{f(S)+f(M)}{2}, \qquad V^-_{\mathrm{corr}} = \frac{f(S)-f(M)-\frac{r}{2}\big(\partial_t f(S) + \partial_t f(M)\big)}{r^3}, \tag{S2.4}$$

and, for every frame component $e \in \{t, n_1, \dots, n_{d-1}\}$,

$$G^+_e = \frac{\partial_e f(S) + \partial_e f(M)}{2}, \qquad G^-_e = \frac{\partial_e f(S) - \partial_e f(M)}{r}. \tag{S2.5}$$

For $d = 3$, (S2.1)–(S2.5) are (3.4)–(3.5) and (3.7)–(3.8) of the main text. The construction is invariant under orthogonal changes of the transverse frame, which act orthogonally on the transverse coordinates $G^{\pm}_{n_i}$ and leave the pins, the targets and the endpoint types unchanged. A finite frame atlas on $S^{d-1}$ is sufficient for global angular integration; on each chart of it we take the frame to depend continuously on $t$.

**Lemma S2.2** (exact pin Jacobian)**.** *Let $\Delta_r$ be the linear map $\Pi_r \mapsto Y_r$. Then*

$$|\det \Delta_r| = r^{-(d+3)}. \tag{S2.6}$$

*On the target (S2.2),*

$$Y_r = y_r(b,\kappa) := \Delta_r z_r(b,\kappa) = \Big(b - \frac{\kappa r^3}{12},\, -\frac{\kappa}{6},\, 0, \dots, 0\Big). \tag{S2.7}$$

*Whenever $\mathrm{Cov}(\Pi_r)$ is nondegenerate, let $p_\Pi(r,t,b,\kappa) = p_\Pi(z_r(b,\kappa))$ be the Gaussian density of $\Pi_r$ at $z_r(b,\kappa)$ and $p^{\mathrm{corr}}_r(t,b,\kappa)$ that of $Y_r$ at $y_r(b,\kappa)$. Then exactly*

$$p_\Pi(r,t,b,\kappa) = r^{-(d+3)}\, p^{\mathrm{corr}}_r(t,b,\kappa). \tag{S2.8}$$

*Assume that $\Sigma_r(t) = \mathrm{Cov}(Y_r)$ converges, as $r \downarrow 0$ and uniformly in $t$, to a positive-definite limit $\Sigma_0(t)$, continuous in $t$. Then $p^{\mathrm{corr}}_r \to p^{\mathrm{corr}}_0$ continuously and locally uniformly, where $p^{\mathrm{corr}}_0(t,b,\kappa)$ is the Gaussian density with covariance $\Sigma_0(t)$ at $y_0(b,\kappa) = (b, -\kappa/6, 0, \dots, 0)$.*

*Proof.* Write the gradients in the orthonormal frame $(t, n_1, \dots, n_{d-1})$; this orthogonal change of coordinates does not change $|\det \Delta_r|$. Order the raw coordinates as the value pair $(f(M), f(S))$ followed by the $d$ gradient pairs $(\partial_e f(M), \partial_e f(S))$. The value pair contributes the block $\left(\begin{smallmatrix} 1/2 & 1/2 \\ -r^{-3} & r^{-3} \end{smallmatrix}\right)$, of determinant magnitude $r^{-3}$, and each of the $d$ gradient pairs contributes the block $\left(\begin{smallmatrix} 1/2 & 1/2 \\ -r^{-1} & r^{-1} \end{smallmatrix}\right)$, of determinant magnitude $r^{-1}$. The tangential-gradient terms in $V^-_{\mathrm{corr}}$ are entries of the value rows in the columns of $\partial_t f(M), \partial_t f(S)$; they are off-diagonal block entries of a block-triangular matrix and do not change the determinant. This proves (S2.6). On (S2.2), $V^+ = b - \kappa r^3/12$, $V^-_{\mathrm{corr}} = (f(S) - f(M))/r^3 = -\kappa/6$, and every $G^{\pm}_e$ vanishes, which is (S2.7). Since $Y_r = \Delta_r \Pi_r$, the change-of-variables formula for densities gives $p_\Pi(z) = |\det \Delta_r|\, p^{\mathrm{corr}}_r(\Delta_r z)$, which is (S2.8). Finally,

$$p^{\mathrm{corr}}_r(t,b,\kappa) = (2\pi)^{-(d+1)} \det\big(\Sigma_r(t)\big)^{-1/2} \exp\Big(-\tfrac{1}{2}\, y_r(b,\kappa)^\top \Sigma_r(t)^{-1} y_r(b,\kappa)\Big);$$

under the stated convergence $\Sigma_r(t) \to \Sigma_0(t) \succ 0$ uniformly in $t$, and $y_r(b,\kappa) \to y_0(b,\kappa)$ uniformly on compact sets of $(b,\kappa)$, which gives the last assertion. □

**Lemma S2.3** (the corrected pins of the side-24 field)**.** *Let $d = 3$ and $f = f_{24}$, and put $\phi(s) = f(x + st)$ and $\psi_e(s) = \partial_e f(x + st)$ for $e \in \{t, v, w\}$ and $|s| \le r/2$.*

*(i) For every $0 < r < 12$ and $t \in S^2$, $\mathrm{Cov}(\Pi_r)$ and $\Sigma_r(t) = \mathrm{Cov}(Y_r)$ are positive definite.*

*(ii) The corrected coordinates are averages over the segment from $M$ to $S$: $V^+$ and $G_e^+$ are the endpoint averages of $\phi$ and $\psi_e$, and*

$$V_{\mathrm{corr}}^- = -\frac{1}{2r^3}\int_{-r/2}^{r/2}\Big(\frac{r^2}{4} - s^2\Big)\phi'''(s)\,ds, \qquad G_e^- = \frac{1}{r}\int_{-r/2}^{r/2}\psi_e'(s)\,ds. \tag{S2.9}$$

*Consequently, as $r \downarrow 0$, $Y_r$ converges in $L^2$, with error $O(r)$ uniformly in $(x, t)$, to the contact limit*

$$Y_0 = \Big(f, -\frac{\partial_t^3 f}{12}, \partial_t f, \partial_t^2 f, \partial_v f, \partial_t\partial_v f, \partial_w f, \partial_t\partial_w f\Big)(x), \tag{S2.10}$$

*which is the list (S1.7) up to the order of its entries and the factor $-1/12$.*

*(iii) $\Sigma_0(t) = \mathrm{Cov}(Y_0)$ is positive definite, $\Sigma_r(t) \to \Sigma_0(t)$ uniformly in $t$, and for every $r_1 \in (0, 12)$ there is $\lambda_1 > 0$ with $\Sigma_r(t) \succeq \lambda_1 I$ for all $0 < r \le r_1$ and $t \in S^2$. In particular the assumption of the last part of Lemma S2.2 holds, $p_r^{\mathrm{corr}} \to p_0^{\mathrm{corr}}$ locally uniformly, and the limiting corrected target $y_0(b, \kappa) = (b, -\kappa/6, 0, \dots, 0)$ pins $\partial_t^3 f(x) = 2\kappa$.*

*(iv) For $\delta > 0$ and $0 < r_0 < \delta$ there is $\lambda_\delta > 0$ such that the covariance matrix of $Y_r$ joined with the remote family (S1.8) at a point $y$ with $\mathrm{dist}(y, x) \ge \delta$ is bounded below by $\lambda_\delta I$, for all $0 < r \le r_0$, $t \in S^2$ and such $y$.*

*Proof.* (i) By Lemma S1.8, $\mathrm{dist}(M, S) = r > 0$. The raw pins are the evaluations with symbols $1, \xi_1, \xi_2, \xi_3$ at the two distinct points $M$ and $S$, so $\mathrm{Cov}(\Pi_r)$ is positive definite by Theorem S1.3, and so is $\Sigma_r(t) = \Delta_r \mathrm{Cov}(\Pi_r)\Delta_r^\top$.

(ii) For $h = r/2$, two integrations by parts give $\int_{-h}^{h}(h^2 - s^2)\phi'''(s)\,ds = 2h(\phi'(h) + \phi'(-h)) - 2(\phi(h) - \phi(-h))$, which is the first identity in (S2.9); the second is the fundamental theorem of calculus. Since $\int_{-h}^{h}(h^2 - s^2)\,ds = r^3/6$, every coordinate of $Y_r$ is the average, with respect to a probability measure on $[-r/2, r/2]$, of $Df(x + st)$ for a derivative $D$ of order at most three: $D = 1$ or $\partial_e$ (endpoint averages), $D = \partial_t\partial_e$ (uniform average) or $D = -\frac{1}{12}\partial_t^3$ (average with density $6(r^2/4 - s^2)/r^3$). For each such $D$, $\mathbb{E}|Df(x + st) - Df(x)|^2 \le Cs^2$, with $C$ bounded by the second derivatives of the covariance function of $Df$, uniformly in $t$, because $K_{24}$ is smooth. Minkowski's integral inequality gives $\|Y_r - Y_0\|_{L^2} \le Cr$.

(iii) The coordinates of $Y_0$ are the evaluations (S1.7), rescaled, so $\Sigma_0(t)$ is positive definite by Proposition S1.5(i). By (ii), $\Sigma_r(t) = \Sigma_0(t) + O(r)$ uniformly in $t$. The eigenvalues of $\Sigma_r(t)$ do not depend on $x$ (stationarity) or on the choice of transverse frame. For $r > 0$ the entries of $\Sigma_r(t)$ are finite combinations of derivatives of $K_{24}$ at 0 and $\pm rt$, divided by powers of $r$; together with the $O(r)$ bound, the smallest eigenvalue of $\Sigma_r(t)$, with $\Sigma_0(t)$ at $r = 0$, is a continuous function on the compact set $[0, r_1] \times S^2$. It is positive by (i) and Proposition S1.5(i), hence bounded below by some $\lambda_1 > 0$. The remaining assertions follow from Lemma S2.2; the second coordinate of $Y_0$ is $-\partial_t^3 f(x)/12$, so the pin $-\kappa/6$ means $\partial_t^3 f(x) = 2\kappa$.

(iv) For $0 < r \le r_0 < \delta$ and $\mathrm{dist}(y, x) \ge \delta$, both $\mathrm{dist}(y, M)$ and $\mathrm{dist}(y, S)$ are at least $\delta - r_0/2 > 0$, so $M$, $S$ and $y$ are distinct and the raw family consisting of $\Pi_r$ and (S1.8) has positive definite covariance by Theorem S1.3; applying the invertible map $\Delta_r \oplus I$ gives the corrected family. At $r = 0$, positive definiteness is Proposition S1.5(ii). By (ii) and stationarity, the covariance is a continuous function of $(r, t, y - x)$ on the compact set $[0, r_0] \times S^2 \times \{z : \mathrm{dist}(z, 0) \ge \delta\}$, and we conclude as in (iii). □

**Definition S2.4** (pinned law, endpoint weight and normalizer)**.** Whenever $\mathrm{Cov}(\Pi_r)$ is nondegenerate (for $f_{24}$, for every $0 < r < 12$, by Lemma S2.3(i)), let

$$\mathbb{E}^0 = \mathbb{E}^0_{r,t,b,\kappa} = \mathbb{E}\big[\,\cdot \mid \Pi_r = z_r(b,\kappa)\big]$$

be the regular conditional Gaussian law given the pins; conditioning on $Y_r = y_r(b,\kappa)$ is equivalent. With $H_p$ the Hessian of $f$ at $p$, put

$$W_r = |\det H_M|\,|\det H_S|\,\mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S)=d-1\}}, \qquad Z_r(t,b,\kappa) = \mathbb{E}^0[W_r], \qquad K_r^{\mathrm{all}} = p_\Pi\, Z_r. \tag{S2.11}$$

For $d = 3$, $W_r$ and $Z_r$ are those of (3.3a) of the main text. The determinant-weighted Palm law $\mathbb{E}_r^{MS}[F] = \mathbb{E}^0[W_r F]/Z_r$ built on $\mathbb{E}^0$, the elder mark, and the selected intensity $K_r^{\mathrm{sel}} = p_r K_r^{\mathrm{all}}$ are defined in Section S5.

## S2.2. **Soft endpoint factors and the Palm normalizer.**

**Lemma S2.5** (exact endpoint factor $r$)**.** *In the frame $(t, n_1, \dots, n_{d-1})$ write the endpoint Hessians as*

$$H_p = \begin{pmatrix} a_p & b_p^\top \\ b_p & Q_p \end{pmatrix}, \qquad p \in \{M, S\}, \tag{S2.12}$$

*with $a_p = \partial_t^2 f(p)$, $b_p = (\partial_t \partial_{n_i} f(p))_i$ and $Q_p = (\partial_{n_i}\partial_{n_j} f(p))_{i,j}$. Suppose $\nabla f(M) = \nabla f(S) = 0$. For each frame direction $e$, the function $g_e(s) = \partial_e f(x + st)$ satisfies $g_e(-r/2) = g_e(r/2) = 0$, and Taylor's integral formula gives the exact identities*

$$\begin{aligned} \partial_t \partial_e f(M) &= -\frac{1}{r}\int_{-r/2}^{r/2} \Big(\frac{r}{2} - s\Big)\, \partial_t^2 \partial_e f(x + st)\, ds, \\ \partial_t \partial_e f(S) &= \frac{1}{r}\int_{-r/2}^{r/2} \Big(s + \frac{r}{2}\Big)\, \partial_t^2 \partial_e f(x + st)\, ds. \end{aligned} \tag{S2.13}$$

*Thus the first row and column of each endpoint Hessian carry one exact factor $r$:*

$$a_M = r\,\alpha_M, \qquad b_M = r\,\beta_M, \qquad a_S = r\,\alpha_S, \qquad b_S = r\,\beta_S, \tag{S2.14}$$

*where $\alpha_p$ $(e = t)$ and $\beta_p$ $(e = n_i)$ are the right sides of (S2.13) divided by $r$; they are linear functionals of the third-derivative process on the shrinking segment. Expanding along the first row,*

$$\det H_M = r\, D_M, \qquad \det H_S = r\, D_S, \qquad D_p = \alpha_p \det Q_p - r\,\beta_p^\top \mathrm{adj}(Q_p)\,\beta_p, \tag{S2.15}$$

*so $D_M$ and $D_S$ are polynomials of degree $d$ (degree three for $d = 3$) in the scaled endpoint variables and the transverse Hessian entries, with coefficients polynomial in $r$. Type indicators can only decrease absolute values:*

$$W_r = r^2\,|D_M D_S|\,\mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S)=d-1\}} \le r^2\,|D_M D_S|. \tag{S2.16}$$

*We call $U_r = (\alpha_M, \alpha_S, \beta_M, \beta_S, Q_M, Q_S)$ the endpoint variables.*

*Proof.* For $g \in C^2$ and real $a, c$, Taylor's formula reads $g(c) = g(a) + (c-a)g'(a) + \int_a^c (c-s)g''(s)\,ds$. With $g = g_e$, $(a,c) = (-r/2, r/2)$ and $g_e(\pm r/2) = 0$ this gives $0 = r\,g_e'(-r/2) + \int_{-r/2}^{r/2}(r/2 - s)g_e''(s)\,ds$, the first identity in (S2.13), because $g_e'(-r/2) = \partial_t\partial_e f(M)$ and $g_e''(s) = \partial_t^2\partial_e f(x+st)$. With $(a,c) = (r/2, -r/2)$ it gives $0 = -r\,g_e'(r/2) + \int_{-r/2}^{r/2}(s + r/2)g_e''(s)\,ds$, the second. Taking $e = t$ and $e = n_i$ gives (S2.14). For a block matrix with scalar corner, $\det \left(\begin{smallmatrix} a & c^\top \\ b & Q \end{smallmatrix}\right) = a \det Q - c^\top \mathrm{adj}(Q)\, b$; the first row of $H_p$ is $r(\alpha_p, \beta_p^\top)$, and pulling out the factor $r$ gives (S2.15). Then (S2.16) follows from (S2.11). □

**Lemma S2.6** (Gaussian polynomial moments)**.** *Let $G$ be a Gaussian vector in $\mathbb{R}^n$, possibly degenerate, with mean $\mu$ and covariance $\Gamma$, and suppose that $\|\mu\| \le C(1+\|a\|)$ and $\|\Gamma\|_{\mathrm{op}} \le C$ for a parameter vector $a$. For every polynomial $P$ of degree at most $k$ there is $C_P$, depending only on $P$, $C$, $n$ and $k$, such that*

$$\mathbb{E}\,|P(G)| \le C_P(1+\|a\|^k). \tag{S2.17}$$

*In particular,* $\det H_M \det H_S$ *is a polynomial of degree six in the Hessian coordinates of the two endpoints, and $D_M D_S$ is a polynomial of degree six in the endpoint variables; a product of three $3\times 3$ Hessian determinants is a polynomial of degree nine in the concatenated Hessian coordinates. The bound* (S2.17) *applies with these degrees.*

*Proof.* Write $G = \mu + \Gamma^{1/2}N$ with $N$ standard Gaussian in $\mathbb{R}^n$; $\Gamma^{1/2}$ may be singular. Expanding $P(\mu+\Gamma^{1/2}N)$ gives a finite sum of monomials $c\,\mu^\eta N^\gamma$ with $|\eta| \le k$, whose coefficients $c$ are bounded in terms of $P$ and $\|\Gamma^{1/2}\| \le C^{1/2}$. Since $\mathbb{E}|N^\gamma| < \infty$ and $\|\mu\|^{|\eta|} \le C^{|\eta|}(1+\|a\|)^{|\eta|} \le (2C+2)^k(1+\|a\|^k)$, the bound follows. The degree statements are immediate, since the determinant of a $3\times 3$ matrix is homogeneous of degree three in its entries. □

The general contact statements below use the following hypotheses on the field.

**Hypothesis S2.7** (corrected-pin nondegeneracy and endpoint moment control)**.** Let $\mathcal{K} \subset \mathbb{R}\times(0,\infty)$ be a compact set of marks $(b,\kappa)$.

(a) (Corrected-pin nondegeneracy.) As $r \downarrow 0$, $\Sigma_r(t) = \mathrm{Cov}(Y_r)$ converges, uniformly in $t \in S^{d-1}$, to a positive-definite limit $\Sigma_0(t)$.

(b) (Uniform Gaussian moment control.) As $r \downarrow 0$, the joint covariance of $(U_r, Y_r)$ converges, uniformly in $t$, to the joint covariance of $(U_0, Y_0)$, and this limit is continuous in $t$. Here $Y_0 = (f, -\frac{1}{12}\partial_t^3 f, (\partial_e f, \partial_t\partial_e f)_e)(x)$ and

$$U_0 = \Big(-\tfrac{1}{2}\partial_t^3 f,\ \tfrac{1}{2}\partial_t^3 f,\ -\tfrac{1}{2}\partial_t^2\partial_\perp f,\ \tfrac{1}{2}\partial_t^2\partial_\perp f,\ Q,\ Q\Big)(x),$$

with $\partial_t^2\partial_\perp f = (\partial_t^2\partial_{n_i} f)_i$ and $Q = (\partial_{n_i}\partial_{n_j} f)_{i,j}$ the transverse Hessian at the midpoint.

(c) For every $t \in S^{d-1}$ and $(b,\kappa) \in \mathcal{K}$, the boundary $\{\det Q = 0\}$ has probability zero under the conditional law of $Q$ given $Y_0 = y_0(b,\kappa)$.

By Lemma S2.2, (a) and (b) imply $p_r^{\mathrm{corr}} \to p_0^{\mathrm{corr}}$ locally uniformly, with $\Sigma_0(t) = \mathrm{Cov}(Y_0)$.

**Proposition S2.8** (contact limits of the endpoint factors)**.** *Assume Hypothesis S2.7 for a compact $\mathcal{K}$. There are $r_0 > 0$ and $C < \infty$ such that, for $0 < r \le r_0$, $t \in S^{d-1}$ and $(b,\kappa) \in \mathcal{K}$, the conditional law of $U_r$ under $\mathbb{E}^0_{r,t,b,\kappa}$ is Gaussian, with mean bounded by $C(1+|b|+\kappa)$ and covariance bounded by $C$. As $r \downarrow 0$, in conditional law: the transverse blocks $Q_M, Q_S$ converge jointly to one symmetric $(d-1)\times(d-1)$ matrix $Q$, distributed as the transverse Hessian at the midpoint given $Y_0 = y_0(b,\kappa)$; the corrected pin limit forces*

$$\alpha_M \to -\kappa, \qquad \alpha_S \to +\kappa; \tag{S2.18}$$

*away from the probability-zero boundary $\{\det Q = 0\}$, the block determinant formula gives*

$$r^{-1}\det H_M = D_M \to -\kappa \det Q, \qquad r^{-1}\det H_S = D_S \to +\kappa \det Q; \tag{S2.19}$$

*and the maximum/saddle type indicator $\mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S)=d-1\}}$ converges to $\mathbf{1}_{\{Q\prec 0\}}$. Moreover*

$$\begin{aligned} Z_r = r^2\zeta_r, \qquad \zeta_r &:= \mathbb{E}^0\big[|D_M D_S|\,\mathbf{1}_{\{H_M\prec 0,\ \mathrm{ind}(H_S)=d-1\}}\big], \\ \zeta_r \ \longrightarrow\ \zeta_0 &:= \kappa^2\,\mathbb{E}\big[(\det Q)^2\,\mathbf{1}_{\{Q\prec 0\}} \mid Y_0 = y_0(b,\kappa)\big], \end{aligned} \tag{S2.20}$$

*uniformly on $S^{d-1}\times\mathcal{K}$, and $\zeta_0$ is continuous on $S^{d-1}\times\mathcal{K}$.*

*Proof.* Let $\Xi_r(t)$ be the joint covariance of $(U_r, Y_r)$, with blocks $\Xi_r^{UU}$, $\Xi_r^{UY}$ and $\Xi_r^{YY} = \Sigma_r(t)$, and let $\Xi_0(t)$ be that of $(U_0, Y_0)$. By Hypothesis S2.7(a)–(b) and compactness of $S^{d-1}$, there are $r_0 > 0$ and $\lambda > 0$ with $\Sigma_r(t) \succeq \lambda I$ and $\|\Xi_r(t)\| \le C$ for $0 \le r \le r_0$; moreover $\Xi_{r_n}(t_n) \to \Xi_0(t)$ whenever $r_n \to 0$ and $t_n \to t$. Under $\mathbb{E}^0_{r,t,b,\kappa}$ the vector $U_r$ is Gaussian with

$$m_r = \Xi_r^{UY}\,\Sigma_r^{-1}\,y_r(b,\kappa), \qquad \Gamma_r = \Xi_r^{UU} - \Xi_r^{UY}\,\Sigma_r^{-1}\,\Xi_r^{YU},$$

so $|m_r| \le C(1+|b|+\kappa)$ by (S2.7), and $\|\Gamma_r\| \le C$. At $r = 0$ let $(m_0, \Gamma_0)$ be the mean and covariance of the conditional law of $U_0$ given $Y_0 = y_0(b,\kappa)$. Along every sequence $(r_n, t_n, b_n, \kappa_n) \to (0, t, b, \kappa)$ with $0 \le r_n \le r_0$ we have $(m_{r_n}, \Gamma_{r_n}) \to (m_0, \Gamma_0)$ at $(t, b, \kappa)$.

Under the limiting conditional law, $\alpha_{M,0} = -\frac{1}{2}\partial_t^3 f(x)$ is six times the second coordinate $-\frac{1}{12}\partial_t^3 f(x)$ of $Y_0$, whose pinned value is $-\kappa/6$; hence $\alpha_{M,0} = -\kappa$ almost surely, and likewise $\alpha_{S,0} = +\kappa$, while $Q_{M,0} = Q_{S,0} = Q$. This gives the convergence in law of $(Q_M, Q_S)$ to $(Q, Q)$ and (S2.18).

For the remaining assertions, fix a sequence as above and realize the conditional laws on one probability space as $U^{(n)} = m_{r_n} + \Gamma_{r_n}^{1/2} N$ with a single standard Gaussian vector $N$. Since the positive square root is continuous on positive semidefinite matrices, $U^{(n)} \to U^{(\infty)} = m_0 + \Gamma_0^{1/2} N$ almost surely, and $U^{(\infty)}$ has the limiting conditional law at $(t, b, \kappa)$; let $Q^{(\infty)}$ be its transverse block. On the event $\{\det Q^{(\infty)} \neq 0\}$, of probability one by Hypothesis S2.7(c), the formula for $D_p$ in (S2.15) gives $D_M^{(n)} \to -\kappa \det Q^{(\infty)}$ and $D_S^{(n)} \to +\kappa \det Q^{(\infty)}$, which is (S2.19). For the type indicator, let $r_n > 0$. For $n$ large, $Q_M^{(n)}$ and $Q_S^{(n)}$ are nonsingular with the inertia of $Q^{(\infty)}$. By Sylvester's law of inertia, $H_p$ is congruent to $\mathrm{diag}(s_p, Q_p)$ with Schur complement $s_p = r_n(\alpha_p - r_n \beta_p^\top Q_p^{-1} \beta_p)$, whose sign is eventually that of $\alpha_p^{(\infty)}$: negative at $M$ and positive at $S$. Hence, for $n$ large, $H_M \prec 0$ if and only if $Q^{(\infty)} \prec 0$, and $\mathrm{ind}(H_S) = \mathrm{ind}(Q^{(\infty)})$, which equals $d-1$ if and only if $Q^{(\infty)} \prec 0$. So the indicator converges to $\mathbf{1}_{\{Q^{(\infty)} \prec 0\}}$, and

$$F_n = |D_M^{(n)} D_S^{(n)}|\,\mathbf{1}_{\{H_M^{(n)} \prec 0,\ \mathrm{ind}(H_S^{(n)}) = d-1\}} \longrightarrow \kappa^2 (\det Q^{(\infty)})^2\,\mathbf{1}_{\{Q^{(\infty)} \prec 0\}} \qquad \text{almost surely.}$$

For $r_n = 0$, read $F_n = \kappa_n^2 (\det Q^{(n)})^2 \mathbf{1}_{\{Q^{(n)} \prec 0\}}$, which converges in the same way. By Lemma S2.6 with degree $4d$, $\sup_n \mathbb{E}[F_n^2] < \infty$, so $(F_n)$ is uniformly integrable and $\mathbb{E}F_n \to \zeta_0(t, b, \kappa)$. Sequences with $r_n = 0$ show that $\zeta_0$ is continuous; sequences with $r_n > 0$ show that $\zeta_{r_n}(t_n, b_n, \kappa_n) \to \zeta_0(t, b, \kappa)$. Since $\zeta_0$ is continuous on the compact set $S^{d-1} \times \mathcal{K}$, the convergence is uniform there. Finally $Z_r = r^2 \zeta_r$ by (S2.16). □

The power $r^2$ in (S2.20) is independent of $d$ because a generic fold has exactly one soft eigenvalue at each endpoint.

**Lemma S2.9** (the endpoint normalizer of the side-24 field)**.** *Let $d = 3$ and $f = f_{24}$.*

*(i) For every $0 < r < 12$, $t \in S^2$, $b \in \mathbb{R}$ and $\kappa > 0$, $Z_r(t, b, \kappa) > 0$.*

*(ii) $f_{24}$ satisfies Hypothesis S2.7 for every compact $\mathcal{K} \subset \mathbb{R} \times (0, \infty)$, and $\zeta_0 > 0$ on $S^2 \times \mathbb{R} \times (0, \infty)$.*

*(iii) For every compact $\mathcal{K} \subset \mathbb{R} \times (0, \infty)$ there are $r_0 \in (0, 1]$ and constants $0 < c_Z \le C_Z < \infty$ and $C_W < \infty$ such that, for $0 < r \le r_0$, $t \in S^2$ and $(b, \kappa) \in \mathcal{K}$,*

$$c_Z\, r^2 \le Z_r(t, b, \kappa) \le C_Z\, r^2, \qquad \mathbb{E}^0[W_r^2] \le C_W\, r^4. \tag{S2.21}$$

*Proof.* (i) The family consisting of $\Pi_r$ and the Hessian coordinates of $f$ at $M$ and at $S$ has, at each of the two distinct points, the distinct symbols $1$, $\xi_i$, $\xi_i \xi_j$ $(i \le j)$. By Theorem S1.3 its covariance is positive definite, and so is the conditional covariance of $(H_M, H_S)$ given $\Pi_r$, a Schur complement of it. Under $\mathbb{E}^0$, $(H_M, H_S)$ therefore has a nondegenerate Gaussian law on pairs of symmetric

$3 \times 3$ matrices, which charges the nonempty open set where $H_M \prec 0$ and $H_S$ is nonsingular with exactly two negative eigenvalues. On that set $W_r > 0$, so $Z_r = \mathbb{E}^0[W_r] > 0$.

(ii) Hypothesis S2.7(a) is Lemma S2.3(iii). For (b): by (S2.13), $\alpha_M$ and $\beta_M$ are $-\frac{1}{2}$ times averages of $\partial_t^3 f$ and $\partial_t^2 \partial_{n_i} f$ over the segment with respect to the probability density $2(r/2 - s)/r^2$, and $\alpha_S$, $\beta_S$ are $+\frac{1}{2}$ times averages with respect to $2(s + r/2)/r^2$; $Q_M$ and $Q_S$ are second derivatives evaluated at $x \mp \frac{r}{2}t$. As in the proof of Lemma S2.3(ii), $(U_r, Y_r) \to (U_0, Y_0)$ in $L^2$ with error $O(r)$ uniformly in $(x, t)$, so the joint covariances converge uniformly in $t$; the limit is continuous in $t$ because $K_{24}$ is smooth. For (c): the family consisting of $Y_0$ and the three entries $\partial_v^2 f, \partial_v \partial_w f, \partial_w^2 f$ of $Q$ at $x$ has, up to scaling, distinct monomial symbols at one point, so its covariance is positive definite (Theorem S1.3); this nondegeneracy of the limiting pins includes the positivity of the odd Schur variance, the conditional variance of $-\frac{1}{12}\partial_t^3 f$ given $(\partial_t f, \partial_v f, \partial_w f)$. Hence the conditional law of $Q$ given $Y_0 = y_0(b, \kappa)$ is a nondegenerate Gaussian law on symmetric $2 \times 2$ matrices. It gives probability zero to $\{\det Q = 0\}$, and the negative-definite transverse cone $\{Q \prec 0\}$, which is open and nonempty, has positive Gaussian measure; since $(\det Q)^2 > 0$ on it and $\kappa > 0$, $\zeta_0 > 0$.

(iii) By (S2.16), under the corrected pins $r^{-2}W_r = |D_M D_S|\, \mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S)=2\}}$. By Proposition S2.8, which applies by (ii), it converges in every compact positive mark family to the nonnegative contact-limit weight $\kappa^2(\det Q)^2 \mathbf{1}_{\{Q \prec 0\}}$, and $\zeta_r = r^{-2} Z_r \to \zeta_0$ uniformly on $S^2 \times \mathcal{K}$. Its expectation $\zeta_0$ is strictly positive by (ii), and continuous on the compact set $S^2 \times \mathcal{K}$; with $2c_Z = \min_{S^2 \times \mathcal{K}} \zeta_0 > 0$ we get $\zeta_r \geq c_Z$, that is $Z_r \geq c_Z r^2$, for $0 < r \leq r_0$ after decreasing $r_0$. Its second moment is finite by Gaussian polynomial moments: under $\mathbb{E}^0$ the endpoint variables have mean bounded by $C(1 + |b| + \kappa)$ and covariance bounded by $C$ (Proposition S2.8), and for $r \leq 1$, $|D_M D_S| \leq C|U_r|^6$, a polynomial of degree six. Lemma S2.6 with degrees six and twelve gives, on $\mathcal{K}$,

$$\zeta_r \leq \mathbb{E}^0|D_M D_S| \leq C\big(1 + (|b| + \kappa)^6\big) \leq C_Z, \qquad \mathbb{E}^0\big[(D_M D_S)^2\big] \leq C\big(1 + (|b| + \kappa)^{12}\big) \leq C_W.$$

With (S2.16), $Z_r = r^2 \zeta_r \leq C_Z r^2$ and $\mathbb{E}^0[W_r^2] \leq r^4\, \mathbb{E}^0[(D_M D_S)^2] \leq C_W r^4$. □

The factor $r^2$ in (S2.21) is the same endpoint soft-mode factor as in (S2.16). For $d = 3$, (S2.15) and Lemma S2.9 are (3.9) of the main text and the statement following it: conditionally on the pins, $\det H_M = r D_M$ and $\det H_S = r D_S$, with conditional Gaussian moments bounded by a polynomial in $|b| + \kappa$, uniformly in $t$ and in $r$ below any fixed $\varrho < 12$ (the proof of Proposition S6.4), and hence uniformly bounded on compact mark sets, so that the two endpoint determinant weights contribute $r^2$.

### S2.3. **The cubic-gap Jacobian.**

**Lemma S2.10** (cubic-gap change of variables)**.** *At fixed $x, t, r, b$, write the saddle value as*

$$h = f(S) = b - \frac{\kappa r^3}{6}, \qquad \kappa > 0. \tag{S2.22}$$

*The map $\kappa \mapsto h$ is affine and strictly decreasing, with*

$$\frac{dh}{d\kappa} = -\frac{r^3}{6}, \qquad |dh| = \frac{r^3}{6}\, d\kappa. \tag{S2.23}$$

*Therefore any Kac–Rice density originally written with respect to the saddle-value differential $dh$ acquires exactly the factor $r^3/6$ when it is rewritten with respect to $d\kappa$.*

*Proof.* Differentiate (S2.22) in $\kappa$ at fixed $(x, t, r, b)$. □

*Remark* S2.11 (the factor 1/6). The factor 1/6 in (S2.23) is not an orientation factor, a branch multiplicity, or a Morse-index factor. It is solely the absolute Jacobian of (S2.22). All ordered-pair, directed-axis and type conventions of Section S1 remain unchanged. Combining (S2.23) with the

corrected-pin density factor $r^{-(d+3)}$ of (S2.8) and the two endpoint soft factors $r^2$ of (S2.20) gives the contact-density factor

$$\frac{r^3}{6}\, r^{-(d+3)}\, r^2 = \frac{1}{6}\, r^{2-d}. \tag{S2.24}$$

After multiplication by the polar displacement measure $r^{d-1}dr$, the universal radial measure is

$$\frac{1}{6}\, r\, dr. \tag{S2.25}$$

S2.4. **Dimension cancellation.**

**Theorem S2.12** (generic-fold dimension cancellation)**.** *Let $d \geq 1$. Assume Hypothesis S2.7 for every compact $\mathcal{K} \subset \mathbb{R} \times (0, \infty)$, and that the two-point Kac–Rice formula holds for the ordered typed pair process. Then:*

*(i) The Kac–Rice density of the all-typed pair measure of maxima and index-$(d-1)$ critical points with respect to $dx\, r^{d-1}dr\, d\sigma_{d-1}(t)\, db\, d\kappa$ is*

$$\frac{r^3}{6}\, K_r^{\mathrm{all}} = \frac{r^3}{6}\, p_\Pi\, Z_r, \tag{S2.26}$$

*where $r^3/6$ is the exact Jacobian from the saddle value to $\kappa$.*

*(ii) Put*

$$J_r := r^{d-2}\, \frac{r^3}{6}\, K_r^{\mathrm{all}}. \tag{S2.27}$$

*Then, exactly,*

$$J_r = \frac{1}{6}\, p_r^{\mathrm{corr}}\, \zeta_r. \tag{S2.28}$$

*(iii) $J_r \to J_0$ locally uniformly on $S^{d-1} \times \mathbb{R} \times (0, \infty)$, where*

$$J_0(t, b, \kappa) = \frac{1}{6}\, p_0^{\mathrm{corr}}(t, b, \kappa)\, \kappa^2\, \mathbb{E}\big[(\det Q)^2\, \mathbf{1}_{\{Q \prec 0\}} \mid Y_0 = y_0(b, \kappa)\big]. \tag{S2.29}$$

*(iv) The full radial measure satisfies exactly*

$$\frac{r^3}{6}\, K_r^{\mathrm{all}}\, r^{d-1} dr = J_r\, r\, dr. \tag{S2.30}$$

*Consequently, the all-typed contact measure of maxima and index-$(d-1)$ critical points has the universal effective radial form $J_0(t, b, \kappa)\, r\, dr\, d\sigma_{d-1}(t)\, db\, d\kappa$ at leading order, for every ambient dimension $d \geq 1$. The ambient dimension changes the coefficient law through $p_0^{\mathrm{corr}}$, $Q$ and the sphere $S^{d-1}$, but not the surviving power of $r$.*

*Proof.* (i) By the two-point Kac–Rice formula, the expected number of typed ordered pairs $(M, S)$ with $(M, S, f(M), f(S))$ in a Borel set $B$ is

$$\int_B p_\Pi(b, h, 0, 0)\, \mathbb{E}\big[|\det H_M|\, |\det H_S|\, \mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S) = d-1\}} \mid \Pi_r = (b, h, 0, 0)\big]\, dM\, dS\, db\, dh,$$

with $p_\Pi(b, h, 0, 0)$ the density of $\Pi_r$ at $(b, h, 0, 0)$. Change variables by $dM\, dS = dx\, r^{d-1}dr\, d\sigma_{d-1}(t)$ (Proposition S1.9; for $d = 1$, (S1.15)–(S1.12) hold verbatim with $\sigma_0$ the counting measure on $S^0 = \{\pm 1\}$) and by $|dh| = (r^3/6)\, d\kappa$ (Lemma S2.10). At $h = b - \kappa r^3/6$ the integrand is $p_\Pi(r, t, b, \kappa)\, Z_r(t, b, \kappa) = K_r^{\mathrm{all}}$, which gives (S2.26).

(ii) By (S2.8) and (S2.20), $J_r = r^{d-2} \cdot \frac{r^3}{6} \cdot r^{-(d+3)} p_r^{\mathrm{corr}} \cdot r^2 \zeta_r = \frac{1}{6}\, p_r^{\mathrm{corr}}\, \zeta_r$.

(iii) By Lemma S2.2 and Hypothesis S2.7(a)–(b), $p_r^{\mathrm{corr}} \to p_0^{\mathrm{corr}}$ locally uniformly; by Proposition S2.8, $\zeta_r \to \zeta_0$ uniformly on $S^{d-1} \times \mathcal{K}$ for every compact $\mathcal{K}$. Both limits are continuous, hence bounded on compact sets, and $\frac{1}{6} p_0^{\mathrm{corr}} \zeta_0$ is (S2.29).

(iv) $\frac{r^3}{6} K_r^{\mathrm{all}}\, r^{d-1} = r^{d-2} \frac{r^3}{6} K_r^{\mathrm{all}} \cdot r = J_r\, r$. □

For a stationary field, $K_r^{\mathrm{all}}$ does not depend on $x$; per unit volume (for $f_{24}$, $24^{-3}\int_{\mathbb{T}^3_{24}} dx = 1$) the $x$-integration drops out. For $f_{24}$ the hypotheses of Theorem S2.12 hold: Hypothesis S2.7 by Lemma S2.9(ii), and the two-point Kac–Rice formula on each set $\{\delta \le r < \rho\}$, $\rho < 12$, by the uniform nondegeneracy of Lemma S2.3(i),(iii) and the analyticity of the sample paths, as in the proof of Proposition 1.2 of the main text, hence on $\{0 < r < \rho\}$ by monotone convergence as $\delta \downarrow 0$. With a bounded Borel mark of the whole field, such as the elder mark, the formula is Proposition S12.22.

**Corollary S2.13** (the three-dimensional power count)**.** *For $d = 3$ the complete local power count is: gap Jacobian $r^3/6$ (Lemma S2.10), corrected pins $r^{-6}$ (Lemma S2.2), endpoint weights $r^2$ (Lemma S2.9) and pair polar measure $r^2\,dr\,d\sigma_2$ (Corollary S1.10), whose radial product is*

$$\frac{r^3}{6}\; r^{-6}\; r^2\; r^2\, dr\, d\sigma_2(t) = \frac{1}{6}\, r\, dr\, d\sigma_2(t). \tag{S2.31}$$

*With $G_r^{\mathrm{all}} = r^4 K_r^{\mathrm{all}}$, the all-typed analogue of $G_r^{\mathrm{sel}} = r^4 K_r^{\mathrm{sel}}$ in Proposition 3.3 of the main text, one has $J_r = G_r^{\mathrm{all}}/6 = \frac{1}{6} p_r^{\mathrm{corr}} \zeta_r$. No additional ordering, antipodal, branch, merge-index or factor-$1/2$ multiplier remains.*

*Proof.* For $d = 3$, (S2.24) is $\frac{r^3}{6} r^{-6} r^2 = \frac{1}{6} r^{-1}$, and multiplication by $r^2\,dr\,d\sigma_2(t)$ gives (S2.31); this is (3.10) of the main text. By (S2.27) with $d = 3$, $J_r = r \cdot \frac{r^3}{6} K_r^{\mathrm{all}} = \frac{1}{6} G_r^{\mathrm{all}}$, and (S2.28) gives the second expression. The measure $dx\, r^2 dr\, d\sigma_2(t)$ is the exact ordered-pair coordinate measure over the full directed sphere (Corollary S1.10); each finite bar has one ordered pair (Proposition S1.11); adjacency is a Boolean mark without branch factor (Remark S1.12); and the factor $1/6$ is the gap Jacobian alone (Remark S2.11). □

S2.5. **Lifetime pushforward.**

**Lemma S2.14** (lifetime pushforward)**.** *The persistence lifetime on the cubic fold channel is*

$$\ell = b - h = \frac{\kappa r^3}{6}. \tag{S2.32}$$

*For fixed $\kappa > 0$,*

$$r = r_\ell(\kappa) := \Big(\frac{6\ell}{\kappa}\Big)^{1/3}, \tag{S2.33}$$

$$dr = \Big(\frac{6}{\kappa}\Big)^{1/3} \frac{1}{3}\, \ell^{-2/3}\, d\ell, \tag{S2.34}$$

*and*

$$r\, dr = \frac{6^{2/3}}{3}\, \kappa^{-2/3}\, \ell^{-1/3}\, d\ell. \tag{S2.35}$$

*Thus the lifetime factor is $6^{2/3}/3$, and the exponent is $-1/3$.*

*Proof.* Equation (S2.32) is (S2.22) with $\ell = b - h$; (S2.33) solves it for $r$, and (S2.34) is its derivative. Multiplying (S2.33) and (S2.34) gives (S2.35). An independent inverse-derivative check gives

$$\frac{d\ell}{dr} = \frac{\kappa r^2}{2}, \qquad r\, dr = \frac{2}{\kappa r}\, d\ell, \tag{S2.36}$$

which becomes exactly (S2.35) after substituting (S2.33), since

$$\frac{2}{\kappa}\Big(\frac{6\ell}{\kappa}\Big)^{-1/3} = 2 \cdot 6^{-1/3}\, \kappa^{-2/3}\, \ell^{-1/3} = \frac{6^{2/3}}{3}\, \kappa^{-2/3}\, \ell^{-1/3}.$$ □

For $d = 3$, (S2.32) and (S2.35) are (3.11) and (3.12) of the main text.

**Proposition S2.15** (lifetime density of the contact measure)**.** *Let $d \geq 1$ and let $s_r(t,b,\kappa) \geq 0$ be a measurable mark multiplier: $s_r \equiv 1$ for the all-typed intensity, and $s_r = p_r$ of Section S5 for the selected one. Fix $\varrho > 0$, with $\varrho < L/2$ on a torus of side $L$. Let $\nu_{s,\varrho}$ be the density of the image under $(r,t,b,\kappa) \mapsto \kappa r^3/6$ of the measure $J_r s_r \mathbf{1}_{\{r<\varrho\}}\, r\, dr\, d\sigma_{d-1}(t)\, db\, d\kappa$, which by (S2.30) is the contact measure per unit volume with separation below $\varrho$ and mark multiplier $s_r$. Then, for $\ell > 0$,*

$$\nu_{s,\varrho}(\ell) = \frac{6^{2/3}}{3}\, \ell^{-1/3} \int_{S^{d-1}} \int_{\mathbb{R}} \int_{6\ell/\varrho^3}^{\infty} J_{r_\ell(\kappa)}\, s_{r_\ell(\kappa)}\, \kappa^{-2/3}\, d\kappa\, db\, d\sigma_{d-1}(t). \tag{S2.37}$$

*Assume that $J_r s_r \to J_0 s_0$ pointwise as $r \downarrow 0$, and that $\kappa^{-2/3} J_r s_r$ has an integrable majorant on $S^{d-1} \times \mathbb{R} \times (0,\infty)$ uniform in $0 < r < \varrho$. Then dominated convergence gives $\ell^{1/3}\nu_{s,\varrho}(\ell) \to c$ as $\ell \downarrow 0$, where*

$$c = \frac{6^{2/3}}{3} \int_{S^{d-1}} \int_{\mathbb{R}} \int_0^{\infty} J_0(t,b,\kappa)\, s_0(t,b,\kappa)\, \kappa^{-2/3}\, d\kappa\, db\, d\sigma_{d-1}(t). \tag{S2.38}$$

*Proof.* For a nonnegative Borel function $\varphi$ on $(0,\infty)$,

$$\int_0^{\infty} \varphi(\ell)\, \nu_{s,\varrho}(\ell)\, d\ell = \int \varphi\Big(\frac{\kappa r^3}{6}\Big)\, J_r s_r \mathbf{1}_{\{r<\varrho\}}\, r\, dr\, d\sigma_{d-1}(t)\, db\, d\kappa.$$

At fixed $(t,b,\kappa)$ substitute $r = r_\ell(\kappa)$ by (S2.33) and (S2.35); the condition $r < \varrho$ becomes $\kappa > 6\ell/\varrho^3$. Tonelli's theorem gives (S2.37). As $\ell \downarrow 0$, $r_\ell(\kappa) \to 0$ for every $\kappa > 0$ and $\mathbf{1}_{\{\kappa > 6\ell/\varrho^3\}} \to 1$, so the integrand of (S2.37) converges pointwise to $J_0 s_0 \kappa^{-2/3}$ and is dominated by the majorant; dominated convergence gives $\ell^{1/3}\nu_{s,\varrho}(\ell) \to c$. □

**Theorem S2.16** (dimension-independent generic-fold exponent)**.** *In the setting of Proposition S2.15, whenever the limiting selected mass $c$ in (S2.38) is finite and strictly positive,*

$$\nu_{s,\varrho}(\ell) = c\, \ell^{-1/3}(1 + o(1)), \qquad \ell \downarrow 0, \tag{S2.39}$$

*so the short-lifetime density exponent is $-1/3$ in every ambient dimension. Selection affects the coefficient through $s_0$; it does not affect the exponent unless the limiting selected mass vanishes or introduces an additional singular scale.*

*Proof.* By Proposition S2.15, $\ell^{1/3}\nu_{s,\varrho}(\ell) \to c \in (0,\infty)$, which is (S2.39). The exponent comes from (S2.35) alone; the dimension enters only through $J_0$ and $S^{d-1}$. □

The identification of the elder density $\nu_{3,24}$ with the selected contact density, and the verification of the hypotheses of Proposition S2.15 for $f_{24}$ with $s_r = p_r$, are carried out in Section S6.

*Remark* S2.17 (no double counting of 1/6). The limiting contact density (S2.29) already contains the Kac–Rice factor 1/6:

$$J_0 = \frac{1}{6}\, p_0^{\mathrm{corr}}\, \kappa^2\, \mathbb{E}\big[(\det Q)^2 \mathbf{1}_{\{Q \prec 0\}} \,\big|\, \text{limiting pins}\big].$$

The lifetime pushforward multiplies $J_0$ by the lifetime factor $6^{2/3}/3$ of (S2.35); it does not introduce another factor 1/6. The combined universal scalar factor carried from the gap variable and the radial pushforward is therefore

$$\frac{1}{6} \cdot \frac{6^{2/3}}{3} = \frac{6^{2/3}}{18}. \tag{S2.40}$$

For $d = 3$, with $J_r = G_r^{\mathrm{all}}/6$ (Corollary S2.13), (S2.38) reads $c = \frac{6^{2/3}}{18} \int \kappa^{-2/3}\, G_0^{\mathrm{all}}\, s_0\, d\kappa\, db\, d\sigma_2(t)$, which carries the scalar of (3.12b) of the main text.

## S3. The finite-side coefficient

This section evaluates the coefficient of the short-lifetime law in closed finite-dimensional form. By Theorem 3.9 of the main text, $c_{3,24}$ is the integral of the limiting all-typed contact density against $\kappa^{-2/3}$. Theorem S3.10 reduces this integral to the Gaussian-cone formula (2.9) of the main text. We then give the cone–sphere and cubic-symmetry reductions of the formula (§S3.5), compute the covariance objects at the Bargmann–Fock jet and their limits as the side tends to infinity (§S3.6), prove the closed forms of the cone constant $D_2$ and of $c_{3,\infty}$ (§S3.7), and record the isotropic form of the coefficient (§S3.8).

**Convention S3.1** (kernels, fields and jets)**.** For $0 < L < \infty$ let

$$K_L(z) = Z_L^{-1} \sum_{n\in\mathbb{Z}^3} \exp\Big(-\frac{|z+Ln|^2}{2}\Big), \qquad Z_L = \sum_{n\in\mathbb{Z}^3} e^{-L^2|n|^2/2}, \tag{S3.1}$$

be the normalized side-$L$ periodized Bargmann–Fock covariance on $\mathbb{T}^3_L = (\mathbb{R}/L\mathbb{Z})^3$, and let $K_\infty(z) = e^{-|z|^2/2}$ on $\mathbb{R}^3$. Thus $K_{24}$ is the covariance (1.1) of the main text. For $L \in (0,\infty]$, $f$ denotes a centered stationary Gaussian field with covariance $K_L$; for $L = 24$, $f = f_{24}$. All derivatives of $f$ below are taken at one point $x$; by stationarity their joint law does not depend on $x$, and

$$\mathrm{Cov}\big(\partial^\alpha f(x), \partial^\beta f(x)\big) = (-1)^{|\beta|}\, \partial^{\alpha+\beta} K_L(0). \tag{S3.2}$$

We write $J_L$ for the even covariance jet of $K_L$ at the origin through order six (its derivatives of orders two, four and six) and, following (3.25) of the main text, $J_0$ for the jet of $K_\infty$. On $S^2$, $d\sigma_2$ denotes surface measure; it is written $dt$ in (2.9) of the main text.

### S3.1. The directional contact frame.

**Definition S3.2** (directional contact frame)**.** Fix $t \in S^2$ and an orthonormal transverse frame $n_1, n_2$, so that $(t, n_1, n_2)$ is an orthonormal basis of $\mathbb{R}^3$. The *odd pin variables* are

$$u_t = -\frac{\partial_t^3 f}{12}, \qquad g_t = (\partial_t f, \partial_{n_1} f, \partial_{n_2} f), \tag{S3.3}$$

the *even Hessian-row pins* are

$$h_t = (\partial_{tt} f, \partial_{tn_1} f, \partial_{tn_2} f), \tag{S3.4}$$

and the transverse Hessian vector and matrix are

$$q_t = (\partial_{n_1n_1} f, \partial_{n_1n_2} f, \partial_{n_2n_2} f), \qquad Q_t(q) = \begin{pmatrix} q_1 & q_2 \\ q_2 & q_3 \end{pmatrix}, \quad q \in \mathbb{R}^3. \tag{S3.5}$$

The complete limiting corrected pin vector is

$$Y_t = (f, u_t, g_t, h_t) \in \mathbb{R}^8, \tag{S3.6}$$

with target

$$(b,\ -\kappa/6,\ 0,\ 0), \qquad b \in \mathbb{R},\ \kappa > 0. \tag{S3.7}$$

For $f = f_{24}$, (S3.3)–(S3.5) are (2.2)–(2.4) of the main text. The quantities entering the coefficient formula below do not depend on the choice of transverse frame (Lemma S3.8).

**Lemma S3.3** (even/odd independence and the limiting pins)**.** *Let $L \in (0,\infty]$.*

*(i) All even-order derivatives of $f$ at $x$ are jointly independent of all odd-order derivatives of $f$ at $x$. In particular $(f, h_t, q_t)$ is independent of $(u_t, g_t)$.*

*(ii) Let $f$ be $C^5$ near $x$, put $M = x - \frac{r}{2}t$, $S = x + \frac{r}{2}t$, and let $V^+, V^-_{\mathrm{corr}}, G^\pm_e$ ($e \in \{t, n_1, n_2\}$) be the corrected divided-difference pin frame (3.7)–(3.8) of the main text. As $r \downarrow 0$,*

$$V^+ \to f(x), \qquad V^-_{\mathrm{corr}} \to u_t, \qquad (G^+_t, G^+_{n_1}, G^+_{n_2}) \to g_t, \qquad (G^-_t, G^-_{n_1}, G^-_{n_2}) \to h_t.$$

*If $f$ satisfies the pins (3.5) of the main text, then $V^+ = b - \kappa r^3/12$, $V^-_{\mathrm{corr}} = -\kappa/6$ and $G^\pm_e = 0$. Thus $Y_t$ is the contact limit of the corrected pin vector, and (S3.7) is the limit of its target.*

*Proof.* (i) Since $K_L$ is even, every derivative of $K_L$ of odd order is an odd function and vanishes at the origin. By (S3.2), $\partial^\alpha f(x)$ and $\partial^\beta f(x)$ are uncorrelated whenever $|\alpha| + |\beta|$ is odd. The derivatives are jointly Gaussian, so the even-order family is independent of the odd-order family. The components of $(f, h_t, q_t)$ have orders 0 and 2, those of $(u_t, g_t)$ orders 3 and 1.

(ii) Put $\varphi(s) = f(x + st)$ and $\psi_e(s) = \partial_e f(x + st)$. Taylor expansion at $s = 0$ gives

$$f(S) - f(M) = r\varphi'(0) + \frac{r^3}{24}\varphi'''(0) + O(r^5), \qquad \frac{r}{2}\big(\partial_t f(S) + \partial_t f(M)\big) = r\varphi'(0) + \frac{r^3}{8}\varphi'''(0) + O(r^5),$$

so $V^-_{\mathrm{corr}} = -\varphi'''(0)/12 + O(r^2) \to -\partial_t^3 f(x)/12 = u_t$. Similarly $V^+ = \varphi(0) + O(r^2)$, $G^+_e = \psi_e(0) + O(r^2)$ and $G^-_e = \psi'_e(0) + O(r^2)$ with $\psi'_e(0) = \partial_t \partial_e f(x)$. On the pins, $V^+ = \frac{1}{2}(b + b - \kappa r^3/6) = b - \kappa r^3/12$, $V^-_{\mathrm{corr}} = (-\kappa r^3/6)/r^3 = -\kappa/6$, and all gradient terms vanish. □

S3.2. **Covariance objects.**

**Definition S3.4** (covariance objects)**.** Put

$$\Sigma_g(t) = \mathrm{Cov}(g_t), \tag{S3.8}$$

$$s_u(t) = \mathrm{Var}(u_t), \qquad c_{ug}(t) = \mathrm{Cov}(u_t, g_t), \qquad c_{gu}(t) = c_{ug}(t)^\top, \tag{S3.9}$$

and define the conditional odd variance

$$\sigma_t^2 = s_u(t) - c_{ug}(t)\,\Sigma_g(t)^{-1} c_{gu}(t). \tag{S3.10}$$

For the even block put

$$\Sigma_h(t) = \mathrm{Cov}(h_t), \tag{S3.11}$$

$$\Sigma_q(t) = \mathrm{Cov}(q_t), \qquad C_{qh}(t) = \mathrm{Cov}(q_t, h_t), \qquad C_{hq}(t) = C_{qh}(t)^\top, \tag{S3.12}$$

and

$$\Gamma_t = \Sigma_q(t) - C_{qh}(t)\,\Sigma_h(t)^{-1} C_{hq}(t). \tag{S3.13}$$

Finally let

$$F(q) = \big(\det Q_t(q)\big)^2\, \mathbf{1}_{\{Q_t(q) \prec 0\}}, \qquad q \in \mathbb{R}^3, \tag{S3.14}$$

where $\prec 0$ means negative definite, and

$$D_t = \mathbb{E}\Big[\big(\det Q_t(q_t)\big)^2\, \mathbf{1}_{\{Q_t(q_t) \prec 0\}} \;\Big|\; h_t = 0\Big]. \tag{S3.15}$$

For $f = f_{24}$ these are (2.5)–(2.8) of the main text. Since $|\det Q_t(q)| = |q_1 q_3 - q_2^2| \le |q|^2$, we have $0 \le F(q) \le |q|^4$.

**Lemma S3.5** (nondegeneracy and conditional laws)**.** *Let $L \in (0, \infty]$ and $t \in S^2$. Every entry of $\Sigma_g(t)$, $s_u(t)$, $c_{ug}(t)$, $\Sigma_h(t)$, $\Sigma_q(t)$ and $C_{qh}(t)$ is, by (S3.2), a constant multiple of a directional derivative of $K_L$ at the origin of order two, four or six; for $L < \infty$ it is an exact lattice sum obtained by differentiating (S3.1) term by term. The quantities $\sigma_t^2$ and $\Gamma_t$ are rational functions of these entries. The matrices $\Sigma_g(t)$, $\Sigma_h(t)$ and $\Gamma_t$ are positive definite, $\sigma_t^2 > 0$, and*

$$q_t \mid \{h_t = 0\} \;\sim\; N(0, \Gamma_t). \tag{S3.16}$$

*In particular $D_t = \mathbb{E}F(q)$ for $q \sim N(0, \Gamma_t)$, and all objects (S3.8)–(S3.15) are determined by $J_L$, $t$ and the transverse frame.*

*Proof.* The orders are read off from (S3.3)–(S3.5): $\Sigma_g$ involves second derivatives of $K_L$, $c_{ug}$, $\Sigma_h$, $\Sigma_q$ and $C_{qh}$ fourth derivatives, and $s_u$ sixth derivatives.

For $L<\infty$, Poisson summation shows that $K_L$ has strictly positive Fourier coefficients, exactly as in (1.2) of the main text with 24 replaced by $L$, and the variance identity (1.3) of the main text holds with the same proof. As shown there, the covariance of a finite family of derivative evaluations at one point is positive definite as soon as the corresponding derivative distributions are linearly independent, that is, as soon as the polynomial symbols of the differential operators are linearly independent. For $L=\infty$ the same conclusion holds because $K_\infty$ has the everywhere positive spectral density $(2\pi)^{-3/2}e^{-|\xi|^2/2}$: the variance of $\sum_\alpha c_\alpha\partial^\alpha f(x)$ is $\int|\sum_\alpha c_\alpha(i\xi)^\alpha|^2(2\pi)^{-3/2}e^{-|\xi|^2/2}\,d\xi$, which vanishes only if the polynomial vanishes identically.

In the orthonormal coordinates $(t,n_1,n_2)$ the symbols of $\partial_t,\partial_{n_1},\partial_{n_2},\partial_t^3$ are distinct monomials, so $\mathrm{Cov}(g_t,u_t)$ is positive definite; hence $\Sigma_g(t)$ is positive definite and its Schur complement $\sigma_t^2$ is positive. Likewise the six second derivatives $\partial_e\partial_{e'}$, $e,e'\in\{t,n_1,n_2\}$, have distinct monomial symbols, so $\mathrm{Cov}(h_t,q_t)$ is positive definite; hence $\Sigma_h(t)$ and the Schur complement $\Gamma_t$ are positive definite. Finally (S3.16) is Gaussian regression: the conditional law of a centered Gaussian vector given that another component vanishes is centered Gaussian with the Schur-complement covariance. □

S3.3. **Birth integration and the normalized gap.** Conditional expectations given Gaussian coordinates are taken in their Gaussian-regression versions, which are continuous in the conditioning values. The *even contact factor before birth integration* is

$$p_{(f,h_t)}(b,0)\,\mathbb{E}\big[F(q_t)\mid f=b,\ h_t=0\big],\qquad b\in\mathbb{R},\tag{S3.17}$$

where $p_{(f,h_t)}$ is the Gaussian density of $(f,h_t)$.

**Lemma S3.6** (birth-integration collapse)**.** *For every $L\in(0,\infty]$ and $t\in S^2$,*

$$\int_{\mathbb{R}} p_{(f,h_t)}(b,0)\,\mathbb{E}\big[F(q_t)\mid f=b,\ h_t=0\big]\,db=p_{h_t}(0)\,\mathbb{E}\big[F(q_t)\mid h_t=0\big]=p_{h_t}(0)\,D_t,\tag{S3.18}$$

*and*

$$p_{h_t}(0)\,D_t=(2\pi)^{-3/2}\big[\det\Sigma_h(t)\big]^{-1/2}D_t.\tag{S3.19}$$

*Thus the field value $f$ and every covariance involving $f$ disappear from the coefficient.*

*Proof.* The vector $(f,h_t)$ is nondegenerate (its symbols 1 and $\xi_t\xi_e$ are linearly independent; see the proof of Lemma S3.5), so $p_{(f,h_t)}(b,0)=p_{h_t}(0)\,p_{f|h_t=0}(b)$, where $p_{f|h_t=0}$ is the centered normal density of $f$ given $h_t=0$. The conditional law of $q_t$ given $h_t=0$ is the mixture, over $b$ distributed according to $p_{f|h_t=0}$, of the conditional laws of $q_t$ given $(f,h_t)=(b,0)$. Since $0\le F(q)\le|q|^4$ is integrable under these Gaussian laws, the law of total expectation gives

$$\int_{\mathbb{R}} p_{f|h_t=0}(b)\,\mathbb{E}\big[F(q_t)\mid f=b,\ h_t=0\big]\,db=\mathbb{E}\big[F(q_t)\mid h_t=0\big]=D_t.$$

Multiplying by $p_{h_t}(0)=(2\pi)^{-3/2}[\det\Sigma_h(t)]^{-1/2}$ gives (S3.18) and (S3.19). □

Equation (S3.18) is the anisotropic finite-side form of the birth-integration collapse; it is (3.20) of the main text.

**Lemma S3.7** (normalized-gap integral)**.** *For every $L\in(0,\infty]$ and $t\in S^2$: conditionally on $g_t=0$,*

$$u_t\sim N(0,\sigma_t^2).\tag{S3.20}$$

*Therefore the odd pin density at the target $u_t=-\kappa/6$ is*

$$p_{(u_t,g_t)}(-\kappa/6,0)=p_{g_t}(0)\,\big[2\pi\sigma_t^2\big]^{-1/2}\exp\Big[-\frac{\kappa^2}{72\sigma_t^2}\Big],\tag{S3.21}$$

*where*

$$p_{g_t}(0) = (2\pi)^{-3/2}\big[\det \Sigma_g(t)\big]^{-1/2}, \tag{S3.22}$$

*and*

$$\int_0^\infty p_{(u_t,g_t)}(-\kappa/6, 0)\,\kappa^{4/3}\,d\kappa = p_{g_t}(0)\,\frac{\Gamma(7/6)\,72^{7/6}\,(\sigma_t^2)^{2/3}}{2\sqrt{2\pi}}. \tag{S3.23}$$

*Proof.* Equation (S3.20) is Gaussian regression with the Schur complement (S3.10), and (S3.21)–(S3.22) follow from $p_{(u_t,g_t)}(u,0) = p_{g_t}(0)\,p_{u_t|g_t=0}(u)$ and $(\kappa/6)^2/(2\sigma_t^2) = \kappa^2/(72\sigma_t^2)$. For (S3.23), substitute $\kappa = (72\sigma_t^2 y)^{1/2}$:

$$\int_0^\infty \kappa^{4/3} e^{-\kappa^2/(72\sigma_t^2)}\,d\kappa = \frac{1}{2}\,(72\sigma_t^2)^{7/6}\int_0^\infty y^{1/6}e^{-y}\,dy = \frac{1}{2}\,(72\sigma_t^2)^{7/6}\,\Gamma(7/6),$$

and multiply by $p_{g_t}(0)(2\pi\sigma_t^2)^{-1/2}$. □

Equation (S3.23) is (3.21) of the main text.

## S3.4. **The coefficient formula.**

**Lemma S3.8** (frame invariance)**.** *Let $L \in (0,\infty]$ and $t \in S^2$, and change the transverse frame by $R \in O(2)$, that is, $(n_1', n_2') = (n_1, n_2)R$. Then $g_t$ and $h_t$ transform by $U = \operatorname{diag}(1, R^\top)$, $q_t$ transforms by the induced representation $\rho(R)$ on $\mathbb{R}^3$ defined by $Q_t(\rho(R)q) = R^\top Q_t(q) R$, $\Sigma_g(t)$ and $\Sigma_h(t)$ change by congruence with $U$, $\Gamma_t$ changes by congruence with $\rho(R)$, and $\det Q_t$ together with the negative-definite cone is invariant. Consequently $\det\Sigma_g(t)$, $\det\Sigma_h(t)$, $\sigma_t^2$ and $D_t$ do not depend on the transverse frame, and*

$$I(t) = \frac{(\sigma_t^2)^{2/3}\,D_t}{\sqrt{\det\Sigma_g(t)\,\det\Sigma_h(t)}} \tag{S3.24}$$

*is a well-defined function on $S^2$. It is continuous and strictly positive.*

*Proof.* Since $\partial_{n_j'} = \sum_i R_{ij}\partial_{n_i}$, the new vectors are $g_t' = Ug_t$, $h_t' = Uh_t$ and $Q_t(q_t') = R^\top Q_t(q_t) R$, while $u_t$ depends only on $t$. Hence $\Sigma_g' = U\Sigma_g U^\top$ and $\Sigma_h' = U\Sigma_h U^\top$ with $U$ orthogonal, so their determinants are unchanged; $c_{ug}' = c_{ug}U^\top$ and $s_u' = s_u$, so $\sigma_t^2$ is unchanged. As $h_t' = Uh_t$, conditioning on $h_t' = 0$ is conditioning on $h_t = 0$, and Gaussian regression gives $\Gamma_t' = \rho(R)\Gamma_t\rho(R)^\top$ with $q_t' \mid \{h_t' = 0\}$ the image of $q_t \mid \{h_t = 0\}$ under $\rho(R)$. Since $\det(R^\top Q R) = \det Q$ and $R^\top Q R \prec 0$ if and only if $Q \prec 0$, we have $F \circ \rho(R) = F$ and $D_t' = D_t$.

Near any point of $S^2$ there is a continuous orthonormal transverse frame. In such a frame the entries of (S3.8)–(S3.12) are polynomials in the coordinates of $(t, n_1, n_2)$ with coefficients in $J_L$, so $\Sigma_g$, $\Sigma_h$, $\sigma_t^2$ and $\Gamma_t$ are continuous in $t$. The map $\Gamma \mapsto \int F(q)\,\phi_\Gamma(q)\,dq$, with $\phi_\Gamma$ the $N(0,\Gamma)$ density, is continuous on positive definite matrices by dominated convergence, so $D_t$ is continuous; by frame invariance $I$ is well defined and continuous on $S^2$. It is strictly positive: $\sigma_t^2 > 0$ by Lemma S3.5, and $D_t > 0$ because $N(0,\Gamma_t)$ has an everywhere positive density and $F > 0$ on the nonempty open cone $\{q : Q_t(q) \prec 0\}$. □

**Definition S3.9** (the coefficient functional)**.** For $L \in (0,\infty]$ put

$$c_{3,L} = B_3 \int_{S^2} \frac{(\sigma_t^2)^{2/3}\,D_t}{\sqrt{\det\Sigma_g(t)\,\det\Sigma_h(t)}}\,d\sigma_2(t), \tag{S3.25}$$

where

$$B_3 = \frac{3\,\Gamma(7/6)}{2^{4/3}\pi^{7/2}}, \tag{S3.26}$$

and the covariance objects are those of $K_L$. By Lemma S3.8, $0 < c_{3,L} < \infty$. In particular $c_{3,\infty}$ is the value of (2.9) of the main text for the Bargmann–Fock field on $\mathbb{R}^3$, as in (2.11) there. In terms of the coefficient functional $\Phi$ of (3.25) of the main text, $c_{3,L} = \Phi(J_L)$ and $c_{3,\infty} = \Phi(J_0)$.

*The limiting contact density.* By the two-point Kac–Rice formula, the all-typed contact intensity $K_r^{\mathrm{all}}$ of (3.3b) of the main text is the raw pin density at the pins (3.5) there, times the normalizer $Z_r$ of (3.3a) there. The raw pin density is $r^{-6}$ times the corrected-pin density (the raw-to-corrected Jacobian of (3.7)–(3.8) of the main text), and by Lemma S3.3(ii) the corrected-pin density converges to the density $p_{Y_t}$ of $Y_t$ at the target (S3.7), since the corrected-pin covariance converges and has a uniform eigenvalue floor (Lemma S2.3(iii)). On the pins, the endpoint determinants of (3.9) of the main text satisfy $r^{-1}\det H_M \to -\kappa \det Q_t(q_t)$ and $r^{-1}\det H_S \to \kappa \det Q_t(q_t)$, where $Q_t(q_t)$ is the common limit of the transverse blocks of $H_M$ and $H_S$, and the type indicators converge to $\mathbf{1}_{\{Q_t(q_t)\prec 0\}}$; thus $r^{-2}Z_r \to \kappa^2\,\mathbb{E}[F(q_t) \mid Y_t = (b, -\kappa/6, 0, 0)]$, by Proposition S2.8, which applies by Lemma S2.9(ii) (see also (S6.19) in the proof of Proposition S6.4). Accordingly the rescaled all-typed contact density $G_r^{\mathrm{all}} = r^4 K_r^{\mathrm{all}}$ has the pointwise limit

$$G_0^{\mathrm{all}}(t, b, \kappa) = p_{Y_t}(b, -\kappa/6, 0, 0)\,\kappa^2\,\mathbb{E}\big[F(q_t) \mid Y_t = (b, -\kappa/6, 0, 0)\big]. \tag{S3.27}$$

**Theorem S3.10** (exact finite-side coefficient reduction)**.** *Let $L \in (0, \infty]$, and let $G_0^{\mathrm{all}}$ be given by (S3.27) for the covariance $K_L$. Then*

$$\frac{6^{2/3}}{18}\int_{S^2}\int_{\mathbb{R}}\int_0^\infty \kappa^{-2/3}\,G_0^{\mathrm{all}}(t, b, \kappa)\,d\kappa\,db\,d\sigma_2(t) \tag{S3.28}$$

*is exactly $c_{3,L}$ of (S3.25)–(S3.26). When the contact limit and the compact elder selection hold for the side-$L$ field, (S3.28) is its complete selected $H_0$ short-lifetime coefficient, as in Theorem 3.9 of the main text; for $L = 24$ this is Proposition S6.4 and Corollary S6.5. In particular the coefficient $c_{3,24}$ of Theorem 2.1 of the main text is given by (2.9) there. Every input of (S3.25) is an explicit finite-dimensional covariance object derived from $K_L$ at the origin, and the formula is independent of the chosen transverse frame (Lemma S3.8).*

*Proof.* By Lemma S3.3(i), $(f, h_t, q_t)$ is independent of $(u_t, g_t)$. Hence the density of $Y_t$ factors as $p_{Y_t}(b, -\kappa/6, 0, 0) = p_{(f,h_t)}(b, 0)\,p_{(u_t,g_t)}(-\kappa/6, 0)$, and conditioning $q_t$ on $Y_t = (b, -\kappa/6, 0, 0)$ is conditioning on $(f, h_t) = (b, 0)$. By (S3.27),

$$G_0^{\mathrm{all}}(t, b, \kappa) = \Big[p_{(f,h_t)}(b, 0)\,\mathbb{E}\big[F(q_t) \mid f = b,\ h_t = 0\big]\Big] \cdot \Big[p_{(u_t,g_t)}(-\kappa/6, 0)\,\kappa^2\Big],$$

the product of the even contact factor (S3.17) and an odd factor. The lifetime pushforward supplies $\kappa^{-2/3}$ and the two soft endpoint determinants supply $\kappa^2$, so the $\kappa$-integrand in (S3.28) is $p_{(u_t,g_t)}(-\kappa/6, 0)\,\kappa^{4/3}$ times the even factor. All factors are nonnegative; by Tonelli's theorem, Lemma S3.6 and Lemma S3.7,

$$\int_{\mathbb{R}}\int_0^\infty \kappa^{-2/3} G_0^{\mathrm{all}}\,d\kappa\,db = (2\pi)^{-3/2}\big[\det \Sigma_h(t)\big]^{-1/2} D_t \cdot (2\pi)^{-3/2}\big[\det \Sigma_g(t)\big]^{-1/2}\,\frac{\Gamma(7/6)\,72^{7/6}\,(\sigma_t^2)^{2/3}}{2\sqrt{2\pi}}.$$

The Kac–Rice contact factor is $1/6$ and the lifetime Jacobian is $6^{2/3}/3$, whose product is the prefactor $6^{2/3}/18$ of (S3.28). Since $6^{2/3}\,72^{7/6} = 2^{25/6}3^3$ and $18 \cdot 2 \cdot (2\pi)^{7/2} = 2^{11/2}3^2\pi^{7/2}$,

$$\frac{6^{2/3}}{18}\,(2\pi)^{-3}\,\frac{\Gamma(7/6)\,72^{7/6}}{2\sqrt{2\pi}} = \frac{2^{25/6}\,3^3\,\Gamma(7/6)}{2^{11/2}\,3^2\,\pi^{7/2}} = \frac{3\,\Gamma(7/6)}{2^{4/3}\pi^{7/2}} = B_3.$$

Integrating over $t \in S^2$ gives (S3.25). The inputs are the covariance objects of Lemma S3.5, and frame independence is Lemma S3.8. □

S3.5. **Cone–sphere and cubic-symmetry reductions.**

**Proposition S3.11** (deterministic cone–sphere reduction)**.** *Let $L \in (0,\infty]$ and $t \in S^2$. Choose any $3\times 3$ matrix $A_t$ with*

$$A_t A_t^\top = \Gamma_t, \tag{S3.29}$$

*and for $v \in S^2$ put*

$$q = A_t v, \qquad \mathcal{Q}_t(v) = Q_t(q). \tag{S3.30}$$

*Then*

$$D_t = \frac{15}{4\pi}\int_{S^2} \big[\det \mathcal{Q}_t(v)\big]^2\, \mathbf{1}_{\{\operatorname{tr}\mathcal{Q}_t(v)<0,\ \det\mathcal{Q}_t(v)>0\}}\, d\sigma_2(v), \tag{S3.31}$$

*because*

$$(2\pi)^{-3/2}\int_0^\infty \varrho^6 e^{-\varrho^2/2}\, d\varrho = \frac{15}{4\pi}. \tag{S3.32}$$

*Consequently*

$$c_{3,L} = \frac{45\,\Gamma(7/6)}{2^{10/3}\pi^{9/2}}\int_{S^2} \frac{(\sigma_t^2)^{2/3}}{\sqrt{\det\Sigma_g(t)\det\Sigma_h(t)}}\int_{S^2}\big[\det \mathcal{Q}_t(v)\big]^2\, \mathbf{1}_{\{\operatorname{tr}\mathcal{Q}_t(v)<0,\ \det\mathcal{Q}_t(v)>0\}}\, d\sigma_2(v)\, d\sigma_2(t). \tag{S3.33}$$

*Proof.* By (S3.16), $q_t$ given $h_t = 0$ has the law of $A_t\xi$ with $\xi$ standard Gaussian in $\mathbb{R}^3$. Write $\xi = \varrho v$ with $\varrho = |\xi|$ and $v \in S^2$. The map $q \mapsto \det Q_t(q)$ is homogeneous of degree two and the negative-definite set is a cone, so $F(A_t\varrho v) = \varrho^4 F(A_t v)$. Polar integration gives

$$D_t = (2\pi)^{-3/2}\int_0^\infty \varrho^6 e^{-\varrho^2/2}\, d\varrho \int_{S^2} F(A_t v)\, d\sigma_2(v),$$

and $\int_0^\infty \varrho^6 e^{-\varrho^2/2}\, d\varrho = 15\sqrt{\pi/2}$ gives (S3.32). A symmetric $2\times 2$ matrix is negative definite if and only if its trace is negative and its determinant positive, which gives (S3.31). Finally $B_3 \cdot 15/(4\pi) = 45\,\Gamma(7/6)/(2^{10/3}\pi^{9/2})$, and substitution into (S3.25) gives (S3.33). □

Equation (S3.31) is (3.22) of the main text. Thus (S3.33) is a compact deterministic quadrature problem on $S^2 \times S^2$, and $c_{3,L}$ is a deterministic nested angular integral involving only the order-six covariance jet $J_L$ at the origin.

**Proposition S3.12** (cubic-symmetry reduction)**.** *Let $L \in (0,\infty)$ and let $\mathcal{H}_3 \subset O(3)$ be the group of signed permutation matrices (the hyperoctahedral group, of order 48). The outer integrand $I$ of (S3.24) satisfies $I(Pt) = I(t)$ for $P \in \mathcal{H}_3$ and $t \in S^2$. Therefore*

$$\int_{S^2} I(t)\, d\sigma_2(t) = 48\int_{\mathcal{F}} I(t)\, d\sigma_2(t), \tag{S3.34}$$

*where a fundamental spherical chamber is*

$$\mathcal{F} = \{t \in S^2 :\ t_1 \ge t_2 \ge t_3 \ge 0\}. \tag{S3.35}$$

*Proof.* Each $P \in \mathcal{H}_3$ is orthogonal and maps $\mathbb{Z}^3$ onto itself, so $|Pz + Ln| = |z + LP^\top n|$ and reindexing (S3.1) gives $K_L(Pz) = K_L(z)$. Differentiating, the multilinear derivatives satisfy $D^k K_L(0)[Pv_1,\dots,Pv_k] = D^k K_L(0)[v_1,\dots,v_k]$, so by (S3.2) the joint law of the directional derivatives in the orthonormal frame $(Pt, Pn_1, Pn_2)$ equals that in $(t, n_1, n_2)$. All covariance objects at $Pt$, computed in the transverse frame $(Pn_1, Pn_2)$, therefore coincide with those at $t$, and Lemma S3.8 gives $I(Pt) = I(t)$.

For $t$ outside the union $N$ of the great circles $\{t_i = 0\}$ and $\{t_i = \pm t_j\}$ $(i \ne j)$, ordering $|t_1|, |t_2|, |t_3|$ decreasingly and making the entries nonnegative determines the unique $P \in \mathcal{H}_3$ with $P^{-1}t \in \mathcal{F}$. Thus the 48 sets $P\mathcal{F}$ cover $S^2$ and overlap only in $N$; the boundaries have zero area. Hence $\int_{S^2} I\, d\sigma_2 = \sum_{P\in\mathcal{H}_3}\int_{P\mathcal{F}} I\, d\sigma_2 = 48\int_{\mathcal{F}} I\, d\sigma_2$. □

*Remark* S3.13 (quadrature). By Proposition S3.12 only one forty-eighth of the outer sphere needs evaluation. The inner cone integral in (S3.33) may be evaluated by deterministic spherical quadrature with exact cone tests.

S3.6. **The Bargmann–Fock jet and the planar limit.** A *standard* $\mathrm{GOE}_2$ *matrix* is $G_2 = \begin{pmatrix} g_1 & g_2 \\ g_2 & g_3 \end{pmatrix}$ with $g_1, g_3 \sim N(0,2)$, $g_2 \sim N(0,1)$, the three entries independent.

**Lemma S3.14** (covariance objects at the Bargmann–Fock jet)**.** *Let $L = \infty$. For every $t \in S^2$ and every transverse frame,*

$$\Sigma_g(t) = I_3, \qquad \Sigma_h(t) = \mathrm{diag}(3,1,1), \qquad \sigma_t^2 = \frac{1}{24}, \tag{S3.36}$$

$$\Gamma_t = \Gamma^{(\infty)} := \begin{pmatrix} 8/3 & 0 & 2/3 \\ 0 & 1 & 0 \\ 2/3 & 0 & 8/3 \end{pmatrix}, \tag{S3.37}$$

*which has eigenvalues 1, 2 and 10/3. Moreover $q_t \mid \{h_t = 0\}$ has the law of $(g_1 + \sqrt{2/3}\, Z,\ g_2,\ g_3 + \sqrt{2/3}\, Z)$, where $Z \sim N(0,1)$ is independent of the entries of $G_2$; equivalently,*

$$Q_t(q_t) \mid \{h_t = 0\} \ \overset{d}{=}\ G_2 + \sqrt{\tfrac{2}{3}}\, Z\, I_2. \tag{S3.38}$$

*In particular, for every $t$, $D_t = D_2$, where*

$$D_2 = \mathbb{E}\big[(\det Q)^2\, \mathbf{1}_{\{Q \prec 0\}}\big], \qquad Q = G_2 + \sqrt{\tfrac{2}{3}}\, Z\, I_2. \tag{S3.39}$$

*Proof.* $K_\infty(Rz) = K_\infty(z)$ for $R \in O(3)$, so the joint law of the derivatives of $f$ in the frame $(t, n_1, n_2)$ is that in the standard frame; take $t = e_1$, $n_1 = e_2$, $n_2 = e_3$ and write $f_i = \partial_i f$, $f_{ij} = \partial_i \partial_j f$, $f_{111} = \partial_1^3 f$. Since $K_\infty(z) = \prod_i e^{-z_i^2/2}$ and $\frac{d^{2k}}{ds^{2k}} e^{-s^2/2}\big|_{s=0} = (-1)^k (2k-1)!!$, while odd derivatives vanish at 0, (S3.2) gives, for $\gamma = \alpha + \beta$,

$$\mathrm{Cov}(\partial^\alpha f, \partial^\beta f) = (-1)^{|\beta|} \prod_{i=1}^{3} (-1)^{\gamma_i/2} (\gamma_i - 1)!! \quad \text{if all } \gamma_i \text{ are even},$$

and 0 otherwise. Hence $\mathrm{Var}(f_i) = 1$ and the first derivatives are uncorrelated, so $\Sigma_g = I_3$. Next $\mathrm{Var}(f_{11}) = 3$, $\mathrm{Var}(f_{12}) = \mathrm{Var}(f_{13}) = 1$ and the mixed covariances of $h_t = (f_{11}, f_{12}, f_{13})$ vanish, so $\Sigma_h = \mathrm{diag}(3,1,1)$. For $q_t = (f_{22}, f_{23}, f_{33})$,

$$\Sigma_q = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}, \qquad C_{qh} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

so $\Gamma_t = \Sigma_q - \frac{1}{3} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \Gamma^{(\infty)}$. Its eigenvectors are $e_2$ (eigenvalue 1) and $(1, 0, \pm 1)$ (eigenvalues $8/3 \pm 2/3$, that is 10/3 and 2). For the odd block, $\mathrm{Var}(f_{111}) = 15$ and $\mathrm{Cov}(f_{111}, f_1) = -3$, while $f_{111}$ is uncorrelated with $f_2, f_3$; thus $s_u = 15/144 = 5/48$, $c_{ug} = (1/4, 0, 0)$ and $\sigma_t^2 = 5/48 - 1/16 = 1/24$.

By (S3.16), $q_t \mid \{h_t = 0\} \sim N(0, \Gamma^{(\infty)})$. The vector $(g_1 + sZ, g_2, g_3 + sZ)$ with $s = \sqrt{2/3}$ is centered Gaussian with covariance $\begin{pmatrix} 2+s^2 & 0 & s^2 \\ 0 & 1 & 0 \\ s^2 & 0 & 2+s^2 \end{pmatrix} = \Gamma^{(\infty)}$, so the two laws coincide, and $Q_t$ of this vector is $G_2 + sZI_2$. The values do not depend on the frame, and $D_t = D_2$ follows from (S3.15). □

Equations (S3.36) and (S3.38) are (3.23) and the first part of (3.24) of the main text.

**Proposition S3.15** (planar limit)**.** *Write* $\Sigma_g^{(L)}(t)$, $\Sigma_h^{(L)}(t)$, $\sigma_t^{2,(L)}$, $\Gamma_t^{(L)}$, $D_t^{(L)}$ *for the objects of* $K_L$*. As* $L \to \infty$*, uniformly in* $t \in S^2$ *and in the transverse frame,*

$$\Sigma_g^{(L)}(t) \to I_3, \qquad \det \Sigma_g^{(L)}(t) \to 1, \tag{S3.40}$$

$$\Sigma_h^{(L)}(t) \to \operatorname{diag}(3,1,1), \qquad \det \Sigma_h^{(L)}(t) \to 3, \tag{S3.41}$$

$$\sigma_t^{2,(L)} \to \frac{1}{24}, \tag{S3.42}$$

*and* $\Gamma_t^{(L)} \to \Gamma^{(\infty)}$*, so that the birth-integrated transverse matrix law becomes*

$$Q_t(q_t) \mid \{h_t = 0\} \;\longrightarrow\; G_2 + \sqrt{\tfrac{2}{3}}\, Z\, I_2 \quad \textit{in law.} \tag{S3.43}$$

*Hence*

$$D_t^{(L)} \to D_2 = \frac{29}{6} - \sqrt{6}, \tag{S3.44}$$

*the value of* $D_2$ *being Theorem S3.16, and* $c_{3,L} \to c_{3,\infty}$*.*

*Proof.* Let $\gamma(z) = e^{-|z|^2/2}$. Each $\partial^\alpha \gamma$ is $\gamma$ times a polynomial of degree $|\alpha|$, so $|\partial^\alpha \gamma(y)| \le C_\alpha (1+|y|)^{|\alpha|} e^{-|y|^2/2}$. By (S3.1), $\partial^\alpha K_L(0) = Z_L^{-1}\big[\partial^\alpha \gamma(0) + \sum_{n \neq 0} \partial^\alpha \gamma(Ln)\big]$, and for $L \ge 1$ and $|\alpha| \le 6$ the sum over $n \neq 0$ is $O(L^6 e^{-L^2/2})$, while $Z_L = 1 + O(e^{-L^2/2})$. Hence $J_L \to J_0$. In any orthonormal frame the entries of (S3.8)–(S3.12) are multilinear in the frame vectors with coefficients in the jet, and inverses and Schur complements are continuous on positive definite matrices; since orthonormal frames form the compact set $O(3)$, the objects of $K_L$ converge to those of $K_\infty$ uniformly in $t$ and the frame. By Lemma S3.14 the limits are (S3.36) and $\Gamma^{(\infty)}$ in every frame, which gives (S3.40)–(S3.43). As in the proof of Lemma S3.8, $\Gamma \mapsto \mathbb{E}F(q)$, $q \sim N(0,\Gamma)$, is continuous on positive definite matrices, so $D_t^{(L)} \to D_2$, and the value of $D_2$ is Theorem S3.16. The integrands of (S3.25) therefore converge uniformly on $S^2$, and $c_{3,L} \to c_{3,\infty}$. □

S3.7. **The closed forms of $D_2$ and $c_{3,\infty}$.** Throughout this subsection $\Phi$ and $\phi$ denote the standard normal distribution function and density, and

$$\mu_k(m) = \mathbb{E}\big[N^k \mathbf{1}_{\{N<m\}}\big], \qquad N \sim N(0,1), \tag{S3.45}$$

are the truncated normal moments; integration by parts gives

$$\mu_0 = \Phi(m), \qquad \mu_2 = \mu_0 - m\phi(m), \qquad \mu_4 = 3\mu_0 - (m^3 + 3m)\phi(m). \tag{S3.46}$$

**Theorem S3.16** (the cone constant)**.** *Let* $G_2$ *be a standard* $\mathrm{GOE}_2$ *matrix,* $Z \sim N(0,1)$ *independent of* $G_2$*, and* $Q = G_2 + \sqrt{2/3}\, Z\, I_2$*. Then*

$$D_2 = \mathbb{E}\big[(\det Q)^2\, \mathbf{1}_{\{Q \prec 0\}}\big] = \frac{29}{6} - \sqrt{6}. \tag{S3.47}$$

*Proof.* Put $s = \sqrt{2/3}$ and $\theta = \sqrt{3/20}$.

*Step 1 (eigenvalues).* Let $\lambda_1, \lambda_2$ be the eigenvalues of $G_2$. By orthogonal invariance the scalar shift $sZ$ translates both eigenvalues, so $Q$ has eigenvalues $\lambda_i + sZ$, $\det Q = \prod_i (\lambda_i + sZ)$, and $Q \prec 0$ if and only if $Z < -\max_i \lambda_i / s$:

$$D_2 = \mathbb{E}\Big[\prod_{i=1}^{2} (\lambda_i + sZ)^2\, \mathbf{1}_{\{Z < -\max_i \lambda_i / s\}}\Big]. \tag{S3.48}$$

*Step 2 (trace and difference).* The density of $G_2$ is proportional to $e^{-\operatorname{tr}(G_2^2)/4}$, and the eigenvalues have joint density $C|\lambda_1 - \lambda_2| e^{-(\lambda_1^2+\lambda_2^2)/4}$. The integrand of (S3.48) is symmetric in $(\lambda_1, \lambda_2)$; ordering $\lambda_1 \ge \lambda_2$ and passing to $T = \lambda_1 + \lambda_2$ and $D = \lambda_1 - \lambda_2 \ge 0$ (Jacobian $1/2$), and using $\lambda_1^2 + \lambda_2^2 = (T^2 + D^2)/2$, the density becomes proportional to $D\, e^{-(T^2+D^2)/8}$ on $\mathbb{R} \times [0,\infty)$. Thus

$T \sim N(0,4)$ is independent of $D$, and $D$ has density $\frac{D}{4}e^{-D^2/8}$ on $[0,\infty)$. (Equivalently, in terms of the entries, $T = g_1+g_3$ and $D = ((g_1-g_3)^2+4g_2^2)^{1/2}$, where $g_1+g_3$, $g_1-g_3$ and $2g_2$ are independent $N(0,4)$ variables.) Put $W = T + 2sZ$. Then $\lambda_1 + sZ = (W+D)/2$ and $\lambda_2 + sZ = (W-D)/2$, so

$$\det Q = \frac{W^2 - D^2}{4}, \qquad Q \prec 0 \iff W < -D, \tag{S3.49}$$

and $W \sim N(0, 4+4s^2) = N(0, 20/3)$ is independent of $D$.

*Step 3 (the $W$-integration).* For fixed $D \geq 0$ write $W = (20/3)^{1/2}N$ with $N \sim N(0,1)$; then $W < -D$ if and only if $N < m := -\theta D$. Since $(20/3)^2 = 400/9$ and $2 \cdot 20/3 = 40/3$,

$$\mathbb{E}\big[(W^2-D^2)^2 \mathbf{1}_{\{W<-D\}}\big] = \tfrac{400}{9}\mu_4(m) - \tfrac{40}{3}D^2\mu_2(m) + D^4\mu_0(m),$$

and therefore

$$D_2 = \frac{1}{16}\int_0^\infty \frac{D\,e^{-D^2/8}}{4}\Big[\tfrac{400}{9}\mu_4 - \tfrac{40}{3}D^2\mu_2 + D^4\mu_0\Big]dD, \qquad m = -\theta D. \tag{S3.50}$$

*Step 4 (separation of the $\Phi$- and $\phi$-terms).* Insert (S3.46) with $m = -\theta D$, $\phi(m) = \phi(\theta D)$ and $\frac{400}{9}\theta^2 = \frac{20}{3}$. The $\phi$-coefficient is $-\frac{400}{9}(m^3+3m) + \frac{40}{3}D^2 m = \theta D(\frac{400}{3} - \frac{20}{3}D^2)$, so the bracket in (S3.50) equals

$$\Phi(-\theta D)\Big(D^4 - \tfrac{40}{3}D^2 + \tfrac{400}{3}\Big) + \theta D\,\phi(\theta D)\Big(\tfrac{400}{3} - \tfrac{20}{3}D^2\Big).$$

With

$$I_k = \int_0^\infty D^k\,\Phi(-\theta D)\,e^{-D^2/8}\,dD \qquad (k \text{ odd}), \tag{S3.51}$$

and since the combined exponent of the $\phi$-terms is $\frac{1}{8} + \frac{\theta^2}{2} = \frac{1}{5}$,

$$D_2 = \frac{1}{64}\Big[I_5 - \tfrac{40}{3}I_3 + \tfrac{400}{3}I_1\Big] + \frac{\theta}{64\sqrt{2\pi}}\int_0^\infty D^2 e^{-D^2/5}\Big(\tfrac{400}{3} - \tfrac{20}{3}D^2\Big)dD. \tag{S3.52}$$

*Step 5 (the $\phi$-terms).* They are pure Gaussian moments: $\int_0^\infty D^2 e^{-D^2/5}dD = \frac{\sqrt{\pi}}{4}5^{3/2}$ and $\int_0^\infty D^4 e^{-D^2/5}dD = \frac{3\sqrt{\pi}}{8}5^{5/2}$. With $\theta/\sqrt{2\pi} = \sqrt{30}/(20\sqrt{\pi})$ the second term of (S3.52) equals

$$\frac{\sqrt{30}}{1280}\,5^{3/2}\Big(\frac{100}{3} - \frac{25}{2}\Big) = \frac{625\sqrt{6}}{1536}.$$

*Step 6 (the $\Phi$-terms, by parts).* For $j \geq 0$ let

$$R_j(D) = \int_D^\infty x^{2j+1}e^{-x^2/8}\,dx = \frac{1}{2}\,e^{-D^2/8}\sum_{i=0}^{j}\frac{j!}{i!}\,8^{\,j-i+1}D^{2i}.$$

Since $R_j'(D) = -D^{2j+1}e^{-D^2/8}$ and $\frac{d}{dD}\Phi(-\theta D) = -\theta\phi(\theta D)$, integration by parts gives

$$I_{2j+1} = \Phi(0)R_j(0) - \theta\int_0^\infty R_j(D)\,\phi(\theta D)\,dD = \frac{j!\,8^{\,j+1}}{4} - \frac{\theta}{2\sqrt{2\pi}}\sum_{i=0}^{j}\frac{j!}{i!}\,8^{\,j-i+1}\int_0^\infty D^{2i}e^{-D^2/5}\,dD.$$

Here $\frac{\theta}{2\sqrt{2\pi}} = \frac{\sqrt{30}}{40\sqrt{\pi}}$ and $\int_0^\infty D^{2i}e^{-D^2/5}dD = (2i-1)!!\,\sqrt{\pi}\,5^{i+1/2}/2^{i+1}$ (with $(-1)!! = 1$), so the $i$-th subtracted term equals $\frac{\sqrt{6}}{8}\cdot\frac{j!}{i!}\,8^{\,j-i+1}(2i-1)!!\,5^i/2^{i+1}$. The leading terms are $\Phi(0)R_j(0) = 2, 16, 256$ for $j = 0,1,2$, and the coefficients of $\frac{\sqrt{6}}{8}$ in the subtracted sums are 4 for $j=0$, $32+10 = 42$ for $j=1$ and $512+160+75 = 747$ for $j=2$. This yields

$$I_1 = 2 - \frac{\sqrt{6}}{2}, \qquad I_3 = 16 - \frac{21\sqrt{6}}{4}, \qquad I_5 = 256 - \frac{747\sqrt{6}}{8}. \tag{S3.53}$$

*Step 7 (recombination).* By (S3.53),

$$I_5 - \tfrac{40}{3} I_3 + \tfrac{400}{3} I_1 = \Big(256 - \tfrac{640}{3} + \tfrac{800}{3}\Big) + \Big(-\tfrac{747}{8} + 70 - \tfrac{200}{3}\Big)\sqrt{6} = \frac{928}{3} - \frac{2161}{24}\sqrt{6},$$

so the first term of (S3.52) is $\frac{29}{6} - \frac{2161\sqrt{6}}{1536}$. Adding Step 5,

$$D_2 = \frac{29}{6} - \frac{2161\sqrt{6}}{1536} + \frac{625\sqrt{6}}{1536} = \frac{29}{6} - \sqrt{6}. \qquad \square$$

*Remark* S3.17. Step 2 is the trace and difference reduction stated with (3.24) of the main text: $T \sim N(0,4)$ is independent of $D \geq 0$, $\det Q = (W^2 - D^2)/4$, and the cone condition is $W < -D$ for the single Gaussian $W = T + 2\sqrt{2/3}\, Z \sim N(0, 20/3)$. Numerically $D_2 = 2.38384359055015523513\ldots$.

**Corollary S3.18** (the coefficient on $\mathbb{R}^3$)**.**

$$c_{3,\infty} = \frac{2^{1/6}\, 3^{1/3}\, (29\sqrt{6} - 36)\, \Gamma(1/6)}{432\, \pi^{5/2}} = 2^{-23/6}\, 3^{-8/3}\, (29\sqrt{6} - 36)\, \Gamma(1/6)\, \pi^{-5/2}, \tag{S3.54}$$

*and*

$$c_{3,\infty} = 0.04177593184059834334293666542857555646668151966 1\ldots. \tag{S3.55}$$

*Proof.* By Lemma S3.14 and Theorem S3.16, the integrand of (S3.25) at $L = \infty$ is the constant $(1/24)^{2/3}(29/6 - \sqrt{6})/\sqrt{1 \cdot 3}$. Substitution into (S3.25), using $|S^2| = 4\pi$, gives

$$c_{3,\infty} = B_3 \cdot 4\pi \cdot 24^{-2/3}\, 3^{-1/2} \Big(\frac{29}{6} - \sqrt{6}\Big).$$

By $\Gamma(7/6) = \Gamma(1/6)/6$, $B_3 = \Gamma(1/6)/(2^{7/3}\pi^{7/2})$; with $24^{-2/3} = 2^{-2}3^{-2/3}$ this gives $c_{3,\infty} = 2^{-7/3}3^{-7/6}(\frac{29}{6} - \sqrt{6})\Gamma(1/6)\pi^{-5/2}$. Since $\frac{29}{6} - \sqrt{6} = (29\sqrt{6} - 36)/(6\sqrt{6}) = 2^{-3/2}3^{-3/2}(29\sqrt{6} - 36)$, we obtain the second form of (S3.54); the first follows from $432 = 2^4 3^3$. Evaluating gives (S3.55). $\square$

Equation (S3.54) is (2.11) of the main text.

S3.8. **Isotropic form of the coefficient.** The following formula expresses the coefficient of an isotropic kernel in dimension $d$ through scalar jet parameters. Let $K$ be an isotropic covariance on $\mathbb{R}^d$ and let $K^{(1)}(s) = K(se)$, for a unit vector $e$, be its one-dimensional sectional covariance. Put $n = d - 1$ and

$$a = -(K^{(1)})''(0) > 0, \qquad m_4 = (K^{(1)})^{(4)}(0) > 0, \qquad \chi = -(K^{(1)})^{(6)}(0) > 0, \qquad \Omega = a\chi - m_4^2. \tag{S3.56}$$

Thus $a = \mathrm{Var}(\partial_t f)$, $m_4 = \mathrm{Var}(\partial_{tt} f)$, $\chi = \mathrm{Var}(\partial_{ttt} f)$, and $\Omega/a = \mathrm{Var}(\partial_{ttt} f \mid \partial_t f)$ is the conditional third-derivative variance. Further, $\beta = m_4/3$ is the transverse second-derivative variance parameter, and

$$D_n = \mathbb{E}\big[(\det M_n)^2\, \mathbf{1}_{\{M_n \prec 0\}}\big] \tag{S3.57}$$

is the centered negative-definite determinant moment of the $\beta$-normalized conditional transverse Hessian $M_n = \beta^{-1/2} Q_n$ in dimension $n$, where $Q_n$ is the conditional transverse Hessian of (S3.15). Without the normalization the moment would be $\beta^n D_n$, and the formulas below would be off by that factor when $\beta \neq 1$. For the Bargmann–Fock kernel $\beta = 1$, so nothing at $d = 3$ changes. For the Bargmann–Fock kernel $K_\infty$, $a = 1$, $m_4 = 3$, $\chi = 15$ and $\Omega = 6$, consistently with $\sigma_t^2 = 1/24$ through $\Omega/(2a) = 3 = 72\sigma_t^2$.

**Proposition S3.19** (isotropic recombination)**.** *For an isotropic kernel in dimension $d$ with the parameters* (S3.56)–(S3.57)*, the birth-integrated even block is*

$$\mathcal{B}_d = D_n\, (2\pi m_4)^{-1/2}\, (2\pi)^{-n}\, a^{-n/2}\, \beta^{n/2}, \tag{S3.58}$$

*and the odd-block density and $\kappa$ integral are*

$$\mathcal{K}_d = \frac{3\,\Gamma(7/6)}{\pi\sqrt{\Omega}}\Big(\frac{\Omega}{2a}\Big)^{7/6}. \tag{S3.59}$$

*Multiplying the Kac–Rice factor $1/6$, the lifetime factor $6^{2/3}/3$, the angular area $|S^{d-1}|$, $\mathcal{B}_d$ and $\mathcal{K}_d$, and using*

$$\Gamma(7/6) = \Gamma(1/6)/6, \tag{S3.60}$$

*gives*

$$c_d^{\mathrm{iso}} = A_d^{\mathrm{jet}}\,\frac{\Omega^{2/3}\,m_4^{(d-2)/2}}{a^{(3d+4)/6}}, \tag{S3.61}$$

*where*

$$A_d^{\mathrm{jet}} = \frac{|S^{d-1}|\,D_{d-1}\,\Gamma(1/6)}{2^{d+2}\,3^{(3d+5)/6}\,\pi^{d+1/2}}. \tag{S3.62}$$

*Proof.* Here $c_d^{\mathrm{iso}}$ denotes the $d$-dimensional analogue of (S3.28), that is, $c_d[K]$ of Definition S4.2. By Lemma S4.20(b), $\mathrm{Cov}(g_t) = aI_d$, $\mathrm{Cov}(h_t) = \mathrm{diag}(m_4, \beta, \dots, \beta)$ and $\mathrm{Var}(\partial_t^3 f \mid g_t) = \Omega/a$, so that $p_{g_t}(0) = (2\pi a)^{-d/2}$, $p_{h_t}(0) = (2\pi)^{-d/2} m_4^{-1/2}\beta^{-n/2}$, $\sigma_t^2 = \Omega/(144a)$ and $D_t = \beta^n D_n$. The block $\mathcal{B}_d = (2\pi a)^{-n/2}\,p_{h_t}(0)\,D_t$ is the even factor of the proof of Theorem S3.10 together with the density of the $n$ transverse gradient components at zero, which is (S3.58); and $\mathcal{K}_d$ is $(2\pi a)^{-1/2}$, the density of the axial gradient component at zero, times $\Gamma(7/6)\,72^{7/6}(\sigma_t^2)^{2/3}/(2\sqrt{2\pi})$, the gap integral of Lemma S3.7 divided by $p_{g_t}(0)$, which is (S3.59). By (S3.60), $3\Gamma(7/6) = \Gamma(1/6)/2$, and with $\beta = m_4/3$,

$$\mathcal{K}_d = \frac{\Gamma(1/6)\,\Omega^{2/3}}{2^{13/6}\,\pi\,a^{7/6}}, \qquad \mathcal{B}_d = \frac{D_n\,m_4^{(n-1)/2}}{2^{n+1/2}\,\pi^{n+1/2}\,3^{n/2}\,a^{n/2}}, \qquad \frac{1}{6}\cdot\frac{6^{2/3}}{3} = 2^{-1/3}3^{-4/3}.$$

Hence the product of the five factors is

$$|S^{d-1}|\,D_n\,\Gamma(1/6)\,\Omega^{2/3}\,m_4^{(n-1)/2}\,a^{-n/2-7/6}\cdot 2^{-1/3-13/6-1/2-n}\,3^{-4/3-n/2}\,\pi^{-n-3/2}.$$

With $n = d-1$ the exponents are $-1/3-13/6-1/2-n = -(d+2)$, $-4/3-n/2 = -(3d+5)/6$, $-n-3/2 = -(d+1/2)$, $(n-1)/2 = (d-2)/2$ and $-n/2-7/6 = -(3d+4)/6$, which is (S3.61)–(S3.62). For $d=3$ at the Bargmann–Fock jet, (S3.58) and (S3.59) are compared with (S3.19) and (S3.23) in Corollary S3.20. □

**Corollary S3.20** (Bargmann–Fock specialization)**.** *For $d=3$ and the Bargmann–Fock kernel,*

$$D_2 = \frac{29}{6} - \sqrt{6}, \qquad a = 1, \qquad m_4 = 3, \qquad \Omega = 6, \tag{S3.63}$$

*and substitution into (S3.61)–(S3.62) yields*

$$c_3^{\mathrm{BF}} = \frac{2^{1/6}\,3^{1/3}\,(29\sqrt{6}-36)\,\Gamma(1/6)}{432\,\pi^{5/2}}, \tag{S3.64}$$

*which is $c_{3,\infty}$ of (S3.54). At this jet,*

$$\mathcal{B}_3 = (2\pi)^{-1}\,p_{h_t}(0)\,D_t, \qquad \mathcal{K}_3 = (2\pi)^{-1/2}\,p_{g_t}(0)^{-1}\int_0^\infty p_{(u_t,g_t)}(-\kappa/6, 0)\,\kappa^{4/3}\,d\kappa;$$

*thus $\mathcal{B}_3\mathcal{K}_3$ is the product of the factors (S3.19) and (S3.23) of the proof of Theorem S3.10, regrouped so that the density $(2\pi)^{-1}$ of the two transverse gradient components at zero is carried by $\mathcal{B}_3$ and the density $(2\pi)^{-1/2}$ of the axial component by $\mathcal{K}_3$. For the side-24 field,*

$$c_{3,24} = c_3^{\mathrm{BF}}\big[1+\delta_{24}\big], \tag{S3.65}$$

*where $\delta_{24} = c_{3,24}/c_{3,\infty} - 1$ is the relative correction treated in Section S4. The normalization factors $1/6$ and $6^{2/3}/3$ multiply both the finite-side and the planar coefficient and therefore cancel identically from $\delta_{24}$.*

*Proof.* At $d = 3$, $|S^2| = 4\pi$, and (S3.62) and (S3.63) give

$$A_3^{\mathrm{jet}} = \frac{4\pi\, D_2\, \Gamma(1/6)}{2^5\, 3^{7/3}\, \pi^{7/2}}, \qquad \frac{\Omega^{2/3} m_4^{1/2}}{a^{13/6}} = 6^{2/3}\, 3^{1/2},$$

whose product is $2^{-7/3} 3^{-7/6} D_2\, \Gamma(1/6)\, \pi^{-5/2}$, the expression obtained in the proof of Corollary S3.18; this proves (S3.64). At the Bargmann–Fock jet, Lemma S3.14 gives $\det \Sigma_h = 3$, $D_t = D_2$, $\det \Sigma_g = 1$ and $\sigma_t^2 = 1/24$. Hence, by (S3.19),

$$(2\pi)^{-1} p_{h_t}(0) D_t = 3^{-1/2} (2\pi)^{-5/2} D_2 = \mathcal{B}_3,$$

and, by (S3.23) with $p_{g_t}(0) = (2\pi)^{-3/2}$,

$$(2\pi)^{-1/2}\, \frac{\Gamma(7/6)\, 72^{7/6}\, 24^{-2/3}}{2\sqrt{2\pi}} = \frac{2^{-1/2}\, 3^{5/3}\, \Gamma(7/6)}{\pi} = \mathcal{K}_3.$$

The normalization factors enter (S3.25) only through the constant $B_3$, which is the same for every $L$; hence they cancel from $\delta_{24}$. □

*Remark* S3.21 (first-image expansion). Expanding the covariance objects in (S3.25) around their planar values (Lemma S3.14) and applying rotational projection gives

$$\frac{c_{3,L} - c_{3,\infty}}{c_{3,\infty}} = -\frac{L^2(10L^4 - 147L^2 + 315)}{105}\, e^{-L^2/2} + O(L^{12} e^{-L^2}), \qquad L \to \infty. \tag{S3.66}$$

Thus (S3.25) is the exact finite-$L$ parent formula and (S3.66) is its first-image asymptotic expansion. The order of the remainder is stated for orientation and is not proved here for general $L$. The polynomial factor is $P_3(L)$ of (3.31) of the main text, and $e^{-L^2/2} = e^{-288}$ at $L = 24$. The side-24 statement, (2.10) of the main text, together with its remainder bound, is proved in Section S4.

## S4. The side-24 correction

This section proves (2.10) of the main text: the relative difference between the coefficient $c_{3,24}$ of (2.9) of the main text and its value $c_{3,\infty}$ for the Bargmann–Fock covariance equals $P_3(24)e^{-288}$ up to an error of absolute value below $10^{-216}$. The argument applies to the coefficient expression in dimensions $d = 2$ and $d = 3$ simultaneously, and has three parts. First, the angular integrand of the coefficient is a Gaussian integral of a fixed homogeneous weight over a fixed linear subspace; the exact radial moments of a $\chi^2$ law then give explicit relative bounds on its first and second covariance derivatives. Second, the periodized covariance jet is compared with the Euclidean one, with the $2d$ nearest images and their share of the normalizer separated from all remaining images and from the normalization products. Third, the first variation along the nearest-image jet is computed exactly by Haar projection onto isotropic jets; this yields the polynomial $P_d(L)$ for every $d \geq 2$.

**Convention S4.1** (notation in this section)**.** Throughout this section $d \geq 2$ is an integer and $m = d - 1$. We write $\varphi(x) = e^{-|x|^2/2}$ for $x \in \mathbb{R}^d$, and, for $L > 0$,

$$q_L = e^{-L^2/2}, \qquad Z_L = \sum_{n \in \mathbb{Z}^d} q_L^{|n|^2}, \qquad K_L(x) = Z_L^{-1} \sum_{n \in \mathbb{Z}^d} \varphi(x + Ln), \qquad q = q_{24} = e^{-288}.$$

For $d = 3$, $K_{24}$ is the covariance (1.1) of the main text. The surface measure on $S^{d-1}$ is $\sigma$, and $|S^{d-1}|$ is its total mass. A *frame at* $t \in S^{d-1}$ is an orthonormal basis $\mathfrak{f} = (e_0, e_1, \ldots, e_m)$ of $\mathbb{R}^d$ with $e_0 = t$; we also write $n_i = e_i$. For a symmetric matrix $U$, $\operatorname{svec} U$ lists the diagonal entries of $U$ and

$\sqrt{2}$ times its entries above the diagonal; svec is an isometry for the Frobenius inner product. The operator norm of a matrix is $\|\cdot\|$, the Loewner order is $\preccurlyeq$, and $U \prec 0$ means that $U$ is negative definite. For a positive definite $N \times N$ matrix $C$, $\varphi_C$ is the centered Gaussian density on $\mathbb{R}^N$ with covariance $C$. Finally $M_{4/3} = \mathbb{E}|Z|^{4/3} = 2^{2/3}\Gamma(7/6)/\sqrt{\pi}$ for a standard normal variable $Z$.

S4.1. **The coefficient expression and the result.**

**Definition S4.2** (the coefficient expression)**.** Let $K$ be an even covariance function on $\mathbb{R}^d$ of class $C^6$, and let $f$ be a centered stationary Gaussian field on $\mathbb{R}^d$ with covariance $K$. For a frame $(t, n_1, \dots, n_m)$ at $t \in S^{d-1}$ put

$$\begin{aligned} g_t &= (\partial_t f, \partial_{n_1} f, \dots, \partial_{n_m} f)(0), \qquad \zeta_t = \partial_t^3 f(0), \\ h_t &= (\partial_t^2 f, \partial_t\partial_{n_1} f, \dots, \partial_t\partial_{n_m} f)(0), \qquad A_t = \big(\partial_{n_i}\partial_{n_j} f(0)\big)_{i,j=1}^{m}, \end{aligned} \tag{S4.1}$$

let $H_t$ be the Hessian of $f$ at 0 in the frame (so that $h_t$ is its first row and $A_t$ its lower right $m \times m$ block), and let

$$X_t = (g_t, \zeta_t, \operatorname{svec} H_t) \in \mathbb{R}^N, \qquad N = d + 1 + \tfrac{d(d+1)}{2}.$$

Call $K$ *admissible* if $\operatorname{Cov}(X_t)$ is positive definite for every frame. For admissible $K$ put

$$\tau_t^2 = \operatorname{Var}(\zeta_t \mid g_t = 0), \qquad D_t = \mathbb{E}\big[(\det A_t)^2\, \mathbf{1}_{\{A_t \prec 0\}} \mid h_t = 0\big], \tag{S4.2}$$

and define

$$c_d[K] = \frac{\Gamma(7/6)}{24^{1/3}\sqrt{\pi}} \int_{S^{d-1}} p_{g_t}(0)\, p_{h_t}(0)\, \tau_t^{4/3}\, D_t\, d\sigma(t), \tag{S4.3}$$

where $p_{g_t}$ and $p_{h_t}$ are the densities of $g_t$ and $h_t$. By Lemma S4.8 below the integrand depends only on $t$, and it is continuous. We put

$$c_{d,\mathrm{ref}} = c_d[\varphi], \qquad c_{d,24} = c_d[K_{24}];$$

$\varphi$ is admissible for every $d$ (Lemma S4.20), and $K_{24}$ is admissible for $d \in \{2, 3\}$ (Lemma S4.17).

The number $c_{d,\mathrm{ref}}$ is the value of the coefficient expression at the Euclidean covariance; its definition involves no persistence statement for a field on $\mathbb{R}^d$. No isotropy of $K$ is assumed in Definition S4.2: the periodized covariance $K_{24}$ need not be rotation invariant, and its coefficient is evaluated with the full covariance of $X_t$ in every frame; in particular it is not claimed to depend only on three isotropic moment scalars.

**Lemma S4.3** (the case $d = 3$)**.** *Let $d = 3$ and let $K$ be admissible. With $u_t = -\zeta_t/12$, $\Sigma_g(t) = \operatorname{Cov}(g_t)$, $\Sigma_h(t) = \operatorname{Cov}(h_t)$, $\sigma_t^2$ as in (2.6) and $D_t$ as in (2.8) of the main text,*

$$c_3[K] = B_3 \int_{S^2} \frac{(\sigma_t^2)^{2/3} D_t}{\sqrt{\det \Sigma_g(t)\, \det \Sigma_h(t)}}\, d\sigma(t), \qquad B_3 = \frac{3\,\Gamma(7/6)}{2^{4/3}\pi^{7/2}}. \tag{S4.4}$$

*Thus $c_3[K]$ is the expression (2.9) of the main text for the field with covariance $K$. In particular $c_{3,24}$ is the coefficient of Theorem 2.1 of the main text, and $c_{3,\mathrm{ref}} = c_{3,\infty}$ is its value for the Bargmann–Fock covariance, given in closed form by (2.11) of the main text.*

*Proof.* For $d = 3$ the vectors $g_t$ and $h_t$ of (S4.1) are those of (2.2) and (2.3) of the main text, and $A_t = Q_t(q_t)$ in the notation of (2.4); hence the quantity $D_t$ of (S4.2) is the quantity $D_t$ of (2.8). The variable $\sigma_t^2$ of (2.6) is the Schur complement of $\Sigma_g(t)$ in the covariance of $(u_t, g_t)$, that is, $\sigma_t^2 = \operatorname{Var}(u_t \mid g_t = 0) = \tau_t^2/144$. Moreover $p_{g_t}(0) = (2\pi)^{-3/2}(\det \Sigma_g(t))^{-1/2}$ and $p_{h_t}(0) = (2\pi)^{-3/2}(\det \Sigma_h(t))^{-1/2}$. Substituting in (S4.3),

$$c_3[K] = \frac{\Gamma(7/6)\, 144^{2/3}}{24^{1/3}\sqrt{\pi}\,(2\pi)^3} \int_{S^2} \frac{(\sigma_t^2)^{2/3} D_t}{\sqrt{\det \Sigma_g(t)\, \det \Sigma_h(t)}}\, d\sigma(t),$$

and $144^{2/3}/24^{1/3} = (144^2/24)^{1/3} = 864^{1/3} = 6 \cdot 2^{2/3}$, so the prefactor is $6 \cdot 2^{2/3}\Gamma(7/6)/(8\pi^{7/2}) = 3\Gamma(7/6)/(2^{4/3}\pi^{7/2}) = B_3$. □

**Theorem S4.4** (the side-24 correction)**.** *Let $d \in \{2, 3\}$ and $q = e^{-288}$. Then*

$$\left|\frac{c_{d,24}}{c_{d,\mathrm{ref}}} - 1 - P_d(24)\, q\right| < 10^{-216}, \tag{S4.5}$$

*where*

$$\begin{aligned} P_d(L) &= -\frac{10L^6}{3(d+2)(d+4)} + \frac{(d+4)L^4}{d+2} - dL^2, \\ P_2(24) &= -26045568, \qquad P_3(24) = -\frac{620813376}{35}. \end{aligned} \tag{S4.6}$$

*In particular $c_{d,24} < c_{d,\mathrm{ref}}$ for $d = 2$ and for $d = 3$.*

The number $P_d(L)q_L$ is the normalized first variation of the coefficient along the nearest-image jet (Proposition S4.24). The proof of Theorem S4.4 is completed at the end of this section; all constants in it are explicit integers and rationals.

**Corollary S4.5** ((2.10) of the main text)**.** *With $P_3(L) = -L^2(10L^4 - 147L^2 + 315)/105$,*

$$\frac{c_{3,24}}{c_{3,\infty}} - 1 = P_3(24)\, e^{-288} + \varepsilon_{24} = -\frac{620813376}{35}\, e^{-288} + \varepsilon_{24}, \qquad |\varepsilon_{24}| < 10^{-216}, \tag{S4.7}$$

*and $c_{3,24} < c_{3,\infty}$.*

*Proof.* For $d = 3$, $P_d(L) = -\frac{2}{21}L^6 + \frac{7}{5}L^4 - 3L^2 = -L^2(10L^4 - 147L^2 + 315)/105$. The claim is Theorem S4.4 for $d = 3$, with $c_{3,\mathrm{ref}} = c_{3,\infty}$ by Lemma S4.3. □

## S4.2. **A Gaussian integral over a fixed subspace.**

**Definition S4.6** (pinned subspace and weight)**.** Write the coordinates of $\mathbb{R}^N$ as $\xi = (\xi_g, \xi_\zeta, \xi_H)$, matching $X_t = (g_t, \zeta_t, \operatorname{svec} H_t)$. Let $\xi_h$ be the $d$ coordinates of $\xi_H$ that correspond to the first row of $H_t$ (the entry $H_{00}$ and the $\sqrt{2}$-scaled entries $H_{0i}$), and $\xi_A$ the $m(m+1)/2$ coordinates that correspond to the block $A_t$. The *pinned subspace* is

$$\mathcal{P} = \{\xi \in \mathbb{R}^N : \ \xi_g = 0, \ \xi_h = 0\};$$

it has dimension $w = 1 + m(m+1)/2$ and coordinates $(\xi_\zeta, \xi_A)$, and $d\xi_{\mathcal{P}}$ is Lebesgue measure in these coordinates. For $\xi \in \mathcal{P}$ let $A(\xi)$ be the symmetric $m \times m$ matrix with $\operatorname{svec} A(\xi) = \xi_A$, and put

$$\psi(\xi) = |\xi_\zeta|^{4/3}\, (\det A(\xi))^2\, \mathbf{1}_{\{A(\xi) \prec 0\}}, \qquad p = \tfrac{4}{3} + 2m, \qquad \nu = w + p. \tag{S4.8}$$

For a positive definite $N \times N$ matrix $C$ define

$$\Psi(C) = \int_{\mathcal{P}} \psi(\xi)\, \varphi_C(\xi)\, d\xi_{\mathcal{P}}. \tag{S4.9}$$

The weight $\psi$ is nonnegative and measurable, bounded on the unit sphere of $\mathcal{P}$, and positively homogeneous of degree $p$: $\psi(\lambda\xi) = \lambda^p \psi(\xi)$ for $\lambda > 0$. It is never differentiated; the covariance enters (S4.9) only through the Gaussian density, so the cone $\{A \prec 0\}$ is a fixed part of the integrand and there is no moving boundary. The dimensions are

$$(N, w, p, \nu) = (6, 2, \tfrac{10}{3}, \tfrac{16}{3}) \ \ (d = 2), \qquad (N, w, p, \nu) = (10, 4, \tfrac{16}{3}, \tfrac{28}{3}) \ \ (d = 3), \tag{S4.10}$$

and $\nu - N = -\frac{2}{3}$ for every $d \geq 2$.

**Lemma S4.7** (fixed-subspace representation)**.** *Let $C$ be a positive definite $N \times N$ matrix whose entries between the coordinates $(\xi_g, \xi_\zeta)$ and the coordinates $\xi_H$ vanish, and let $X_t = (g_t, \zeta_t, \operatorname{svec} H_t)$ be a centered Gaussian vector with covariance $C$, with $h_t$ and $A_t$ the first row and the lower right block of $H_t$ and with $\tau_t^2$ and $D_t$ defined by (S4.2). Then*

$$\Psi(C) = 2^{-m/2}\, M_{4/3}\, p_{g_t}(0)\, p_{h_t}(0)\, \tau_t^{4/3}\, D_t. \tag{S4.11}$$

*If $K$ is admissible and $C_t = \operatorname{Cov}(X_t)$ is the covariance of Definition S4.2 in a frame at $t$, then $C_t$ has this block structure, and*

$$c_d[K] = 2^{m/2}\, 96^{-1/3} \int_{S^{d-1}} \Psi(C_t)\, d\sigma(t). \tag{S4.12}$$

*Proof.* For an even covariance $K$, the odd derivatives of $K$ vanish at 0, so

$$\operatorname{Cov}(\partial^\alpha f(0), \partial^\beta f(0)) = (-1)^{|\beta|} \partial^{\alpha+\beta} K(0) = 0 \qquad \text{whenever } |\alpha| + |\beta| \text{ is odd;}$$

since $g_t$ and $\zeta_t$ have odd order and $H_t$ has order two, $C_t$ has the stated block structure. Under this structure the Gaussian vectors $(g_t, \zeta_t)$ and $H_t$ are uncorrelated, hence independent. For $\xi \in \mathcal{P}$ factor the joint density as

$$\varphi_C(\xi) = p_{(g_t, \xi_h(X_t))}(0,0) \cdot p(\xi_\zeta, \xi_A),$$

where $p(\cdot)$ is the conditional density of $(\zeta_t, \operatorname{svec} A_t)$ given $g_t = 0$ and $\xi_h(X_t) = 0$. The coordinates $\xi_h(X_t) = (H_{00}, \sqrt{2}H_{01}, \dots, \sqrt{2}H_{0m})$ are the image of $h_t$ under a linear map of determinant $2^{m/2}$, so by independence $p_{(g_t, \xi_h(X_t))}(0,0) = 2^{-m/2} p_{g_t}(0) p_{h_t}(0)$, and conditioning on $\xi_h(X_t) = 0$ is conditioning on $h_t = 0$. Given $g_t = 0$ and $h_t = 0$, the variable $\zeta_t$ is centered Gaussian with variance $\tau_t^2$ and is independent of $A_t$, whose law is the conditional law given $h_t = 0$. Therefore

$$\Psi(C) = 2^{-m/2} p_{g_t}(0) p_{h_t}(0)\, \mathbb{E}\big[|\zeta_t|^{4/3} \mid g_t = 0\big]\, \mathbb{E}\big[(\det A_t)^2 \mathbf{1}_{\{A_t \prec 0\}} \mid h_t = 0\big],$$

and $\mathbb{E}[|\zeta_t|^{4/3} \mid g_t = 0] = M_{4/3} \tau_t^{4/3}$. This is (S4.11). Since $M_{4/3} = 2^{2/3}\Gamma(7/6)/\sqrt{\pi}$ and $24^{1/3} 2^{2/3} = 96^{1/3}$, the prefactor in (S4.3) is $\Gamma(7/6)/(24^{1/3}\sqrt{\pi}) = 96^{-1/3} M_{4/3}$, and (S4.12) follows. □

**Lemma S4.8** (transverse-frame invariance)**.** *Let $(t, n_1, \dots, n_m)$ and $(t, n_1', \dots, n_m')$ be frames at $t$, with $n_j' = \sum_i O_{ij} n_i$ for some $O \in O(m)$, and let $X_t, X_t'$ be the corresponding coordinate vectors. Then $X_t' = S_O X_t$ for an orthogonal $N \times N$ matrix $S_O$ that maps $\mathcal{P}$ onto itself and satisfies $\psi(S_O \xi) = \psi(\xi)$ for $\xi \in \mathcal{P}$. Consequently $\Psi(S_O C S_O^\top) = \Psi(C)$ for every positive definite $C$, the value $\Psi(C_t)$ and the integrand of (S4.3) do not depend on the transverse frame, and $t \mapsto \Psi(C_t)$ is continuous on $S^{d-1}$.*

*Proof.* In the new frame $g_t' = \operatorname{diag}(1, O^\top) g_t$, $\zeta_t' = \zeta_t$, $h_t' = \operatorname{diag}(1, O^\top) h_t$ and $A_t' = O^\top A_t O$. The map $U \mapsto O^\top U O$ preserves the Frobenius norm, and svec is an isometry, so $X_t' = S_O X_t$ with $S_O$ orthogonal. The matrix $S_O$ maps $\{\xi_g = 0,\ \xi_h = 0\}$ onto itself and acts on $\mathcal{P}$ by $(\xi_\zeta, A) \mapsto (\xi_\zeta, O^\top A O)$; since $(\det O^\top A O)^2 = (\det A)^2$ and $O^\top A O \prec 0$ if and only if $A \prec 0$, the weight is preserved. As $S_O$ is orthogonal, $\varphi_{S_O C S_O^\top}(\xi) = \varphi_C(S_O^\top \xi)$, and the substitution $\xi = S_O \eta$ on $\mathcal{P}$, whose restriction to $\mathcal{P}$ is orthogonal, gives $\Psi(S_O C S_O^\top) = \Psi(C)$. Applied to $C = C_t$, and combined with (S4.11), this shows that $\Psi(C_t)$ and $p_{g_t}(0) p_{h_t}(0) \tau_t^{4/3} D_t$ depend only on $t$. Finally, $\Psi$ is continuous on positive definite matrices by dominated convergence (a locally uniform Gaussian bound), frames can be chosen continuously near each point of $S^{d-1}$, and the entries of $C_t$ are derivatives of $K$ at 0 contracted with frame vectors; hence $t \mapsto \Psi(C_t)$ is continuous. □

S4.3. **Relative bounds for the covariance derivatives.**

**Lemma S4.9** (Gaussian score identities)**.** *Let $C_0$ be positive definite, $\Delta$ symmetric, and $C_s = C_0 + s\Delta$ positive definite for $s$ in an interval. For $\xi \in \mathbb{R}^N$ put $B_s = C_s^{-1/2}\Delta C_s^{-1/2}$ and $y_s = C_s^{-1/2}\xi$. Then*

$$\partial_s \log \varphi_{C_s}(\xi) = -\tfrac{1}{2}\operatorname{tr} B_s + \tfrac{1}{2}\, y_s^\top B_s y_s, \qquad \partial_s^2 \log \varphi_{C_s}(\xi) = \tfrac{1}{2}\operatorname{tr}(B_s^2) - y_s^\top B_s^2 y_s. \tag{S4.13}$$

*Proof.* We have $\log \varphi_{C_s}(\xi) = -\frac{N}{2}\log(2\pi) - \frac{1}{2}\log\det C_s - \frac{1}{2}\xi^\top C_s^{-1}\xi$, and

$$\partial_s \log\det C_s = \operatorname{tr}(C_s^{-1}\Delta) = \operatorname{tr} B_s, \qquad \partial_s^2 \log\det C_s = -\operatorname{tr}(C_s^{-1}\Delta C_s^{-1}\Delta) = -\operatorname{tr}(B_s^2),$$
$$\partial_s C_s^{-1} = -C_s^{-1}\Delta C_s^{-1}, \qquad \partial_s^2 C_s^{-1} = 2C_s^{-1}\Delta C_s^{-1}\Delta C_s^{-1}.$$

Hence $\partial_s(\xi^\top C_s^{-1}\xi) = -y_s^\top B_s y_s$ and $\partial_s^2(\xi^\top C_s^{-1}\xi) = 2y_s^\top B_s^2 y_s$. The sign of the last term in (S4.13) is essential. □

**Lemma S4.10** (radial law)**.** *Let $C$ be positive definite and let $\xi$ be a random point of $\mathcal{P}$ with density proportional to $\psi(\xi)\varphi_C(\xi)$ with respect to $d\xi_{\mathcal{P}}$. Then $R = (\xi^\top C^{-1}\xi)^{1/2}$ has density proportional to $R^{\nu-1}e^{-R^2/2}$ on $(0,\infty)$, that is, $R^2$ has the $\chi^2$ law with parameter $\nu$ (not necessarily an integer). In particular*

$$\mathbb{E}R^2 = \nu, \qquad \mathbb{E}R^4 = \nu(\nu+2). \tag{S4.14}$$

*Proof.* The linear map $C^{-1/2}$ maps $\mathcal{P}$ onto the $w$-dimensional subspace $\mathcal{P}' = C^{-1/2}\mathcal{P}$ with a constant Jacobian. For $\eta \in \mathcal{P}'$, $\varphi_C(C^{1/2}\eta) = (2\pi)^{-N/2}(\det C)^{-1/2}e^{-|\eta|^2/2}$ and, by homogeneity, $\psi(C^{1/2}\eta) = |\eta|^p\,\psi(C^{1/2}\eta/|\eta|)$. In polar coordinates $\eta = R\omega$ on $\mathcal{P}'$, where $d\eta = R^{w-1}dR\,d\omega$, the law of $\eta = C^{-1/2}\xi$ is therefore proportional to $R^{w+p-1}e^{-R^2/2}dR \times \psi(C^{1/2}\omega)\,d\omega$. The angular factor has finite positive mass, because $\psi$ is bounded on bounded sets and positive on a nonempty open subset of $\mathcal{P}$; the constant Jacobian and the angular factor cancel on normalizing. Hence $R = |\eta|$ has density proportional to $R^{\nu-1}e^{-R^2/2}$. Integration by parts gives $\int_0^\infty R^{\nu+1}e^{-R^2/2}dR = \nu\int_0^\infty R^{\nu-1}e^{-R^2/2}dR$, so $\mathbb{E}R^2 = \nu$, and in the same way $\mathbb{E}R^4 = (\nu+2)\mathbb{E}R^2$. □

**Proposition S4.11** (second-order bound)**.** *Let $C_0$ be positive definite, $\Delta$ symmetric,*

$$\theta = \|C_0^{-1/2}\Delta C_0^{-1/2}\| < 1,$$

*and $C_s = C_0 + s\Delta$ for $0 \le s \le 1$. Then $s \mapsto \Psi(C_s)$ is twice continuously differentiable on $[0,1]$, its derivatives may be computed under the integral sign, and for $0 \le s \le 1$:*

*(a) $(1-\theta)C_0 \preccurlyeq C_s \preccurlyeq (1+\theta)C_0$ and $\|B_s\| \le \theta/(1-\theta)$;*
*(b) with $R_s^2 = \xi^\top C_s^{-1}\xi$,*

$$|\partial_s^2 \varphi_{C_s}(\xi)| \le \Big(\frac{\theta}{1-\theta}\Big)^2 \varphi_{C_s}(\xi)\Big[\frac{(N+R_s^2)^2}{4} + \frac{N}{2} + R_s^2\Big]; \tag{S4.15}$$

*(c) with $\mathcal{K}(N,\nu) = \big[N^2 + 2N\nu + \nu(\nu+2)\big]/4 + N/2 + \nu$,*

$$|\partial_s^2 \Psi(C_s)| \le \mathcal{K}(N,\nu)\Big(\frac{\theta}{1-\theta}\Big)^2 \Psi(C_s); \tag{S4.16}$$

*(d)*

$$\Psi(C_s) \le (1-\theta)^{-N/2}(1+\theta)^{\nu/2}\,\Psi(C_0). \tag{S4.17}$$

*If moreover $d \in \{2,3\}$ and $\theta \le 1/4$, then, with $D\Psi(C_0)[\Delta] = \partial_s\Psi(C_s)|_{s=0}$,*

$$\big|\Psi(C_0+\Delta) - \Psi(C_0) - D\Psi(C_0)[\Delta]\big| \le 2000\,\theta^2\,\Psi(C_0). \tag{S4.18}$$

*Proof.* (a) We have $C_s = C_0^{1/2}(I + sC_0^{-1/2}\Delta C_0^{-1/2})C_0^{1/2}$ and $\|sC_0^{-1/2}\Delta C_0^{-1/2}\| \le \theta$, which gives the two-sided bound. For every vector $\eta$, $|\eta^\top \Delta \eta| \le \theta\, \eta^\top C_0 \eta \le \theta(1-\theta)^{-1}\eta^\top C_s \eta$, so $\|B_s\| \le \theta/(1-\theta)$.

(b) By Lemma S4.9, $\partial_s^2 \varphi_{C_s}/\varphi_{C_s} = (\partial_s \log \varphi_{C_s})^2 + \partial_s^2 \log \varphi_{C_s}$. Since $|\operatorname{tr} B| \le N\|B\|$, $|y^\top B y| \le \|B\|\,|y|^2$, $|\operatorname{tr} B^2| \le N\|B\|^2$ and $y^\top B^2 y \le \|B\|^2|y|^2$, and $|y_s|^2 = R_s^2$, we get $|\partial_s \log \varphi_{C_s}| \le \|B_s\|(N + R_s^2)/2$ and $|\partial_s^2 \log \varphi_{C_s}| \le \|B_s\|^2(N/2 + R_s^2)$. Now use (a).

Differentiation under the integral sign. By (a), for $0 \le s \le 1$ and $\xi \in \mathcal{P}$,

$$\varphi_{C_s}(\xi) \le (1-\theta)^{-N/2}(2\pi)^{-N/2}(\det C_0)^{-1/2}\exp\Big(-\frac{\xi^\top C_0^{-1}\xi}{2(1+\theta)}\Big), \qquad R_s^2 \le \frac{\xi^\top C_0^{-1}\xi}{1-\theta},$$

so, by (b) and the analogous bound for the first derivative, $|\partial_s^j \varphi_{C_s}(\xi)|$ $(j = 1, 2)$ is bounded uniformly in $s$ by a constant multiple of $(1 + \xi^\top C_0^{-1}\xi)^2 \exp(-\xi^\top C_0^{-1}\xi/(2(1+\theta)))$. Its product with $\psi$ is integrable on $\mathcal{P}$, because $\psi(\xi) \le |\xi|^p$. Dominated convergence shows that $s \mapsto \Psi(C_s)$ is twice continuously differentiable and that the derivatives are the integrals of $\psi\, \partial_s^j \varphi_{C_s}$.

(c) Integrate (S4.15) against $\psi$. By Lemma S4.10 applied to $C_s$, $\int_{\mathcal{P}} \psi \varphi_{C_s}[(N + R_s^2)^2/4 + N/2 + R_s^2]\, d\xi_{\mathcal{P}} = \Psi(C_s)([N^2 + 2N\nu + \nu(\nu+2)]/4 + N/2 + \nu)$.

(d) By (a), $\det C_s \ge (1-\theta)^N \det C_0$ and $\xi^\top C_s^{-1}\xi \ge (1+\theta)^{-1}\xi^\top C_0^{-1}\xi$, so $\varphi_{C_s}(\xi) \le (1-\theta)^{-N/2}\varphi_{C_0}((1+\theta)^{-1/2}\xi)$. The substitution $\xi = (1+\theta)^{1/2}\eta$ on $\mathcal{P}$, with $d\xi_{\mathcal{P}} = (1+\theta)^{w/2} d\eta_{\mathcal{P}}$ and $\psi(\xi) = (1+\theta)^{p/2}\psi(\eta)$, gives $\int_{\mathcal{P}} \psi(\xi)\varphi_{C_0}((1+\theta)^{-1/2}\xi)\, d\xi_{\mathcal{P}} = (1+\theta)^{\nu/2}\Psi(C_0)$.

Finally, by Taylor's formula with integral remainder, $\Psi(C_1) - \Psi(C_0) - \partial_s \Psi(C_s)|_{s=0} = \int_0^1 (1-s)\, \partial_s^2 \Psi(C_s)\, ds$, so by (c) and (d) the left side of (S4.18) is at most

$$\tfrac{1}{2}\,\mathcal{K}(N,\nu)\,(1-\theta)^{-2}\theta^2\,(1-\theta)^{-N/2}(1+\theta)^{\nu/2}\,\Psi(C_0).$$

For $d = 3$, $\mathcal{K}(10, \frac{28}{3}) = \frac{1012}{9}$; for $\theta \le \frac{1}{4}$, $(1-\theta)^{-2} \le \frac{16}{9}$ and $(1-\theta)^{-5}(1+\theta)^{14/3} \le (\frac{4}{3})^5(\frac{5}{4})^{14/3} \le (\frac{5}{3})^5 = \frac{3125}{243}$. For $d = 2$, $\mathcal{K}(6, \frac{16}{3}) = \frac{388}{9}$ and $(1-\theta)^{-3}(1+\theta)^{8/3} \le (1-\theta)^{-5}(1+\theta)^{14/3}$, so each factor is bounded by its value for $d = 3$. Hence the bound is at most

$$\frac{1}{2}\cdot\frac{1012}{9}\cdot\frac{16}{9}\cdot\frac{3125}{243}\,\theta^2\Psi(C_0) = \frac{25300000}{19683}\,\theta^2\Psi(C_0) \le 2000\,\theta^2\Psi(C_0). \qquad \square$$

**Lemma S4.12** (first-derivative bound)**.** *Let $C_0$ be positive definite and $R_0$ symmetric, and put $B = C_0^{-1/2}R_0C_0^{-1/2}$. Then*

$$D\Psi(C_0)[R_0] = \int_{\mathcal{P}} \psi(\xi)\varphi_{C_0}(\xi)\Big[-\tfrac{1}{2}\operatorname{tr} B + \tfrac{1}{2}\,\xi^\top C_0^{-1}R_0C_0^{-1}\xi\Big]d\xi_{\mathcal{P}},$$

*which is linear in $R_0$, and*

$$\big|D\Psi(C_0)[R_0]\big| \le \frac{N+\nu}{2}\,\|C_0^{-1/2}R_0C_0^{-1/2}\|\,\Psi(C_0). \tag{S4.19}$$

*For $d = 3$, $(N+\nu)/2 = \frac{29}{3}$; for $d = 2$, $(N+\nu)/2 = \frac{17}{3}$.*

*Proof.* Apply Proposition S4.11 to $\Delta = \epsilon R_0$ with $\epsilon > 0$ so small that $\epsilon\|B\| < 1$, and Lemma S4.9 at $s = 0$; this gives the formula. The bracket is at most $\|B\|(N + R^2)/2$ in absolute value, with $R^2 = \xi^\top C_0^{-1}\xi$, and $\mathbb{E}R^2 = \nu$ by Lemma S4.10. $\square$

*Remark* S4.13 (a homogeneity check). For $a > 0$, $\varphi_{aC}(\xi) = a^{-N/2}\varphi_C(a^{-1/2}\xi)$, and the substitution $\xi = a^{1/2}\eta$ on $\mathcal{P}$ gives

$$\Psi(aC) = a^{(w+p-N)/2}\,\Psi(C) = a^{-1/3}\,\Psi(C). \tag{S4.20}$$

Along $\Delta = C_0$ the score identities (S4.13) have $B_0 = I$, and Lemma S4.10 gives the first derivative $\mathbb{E}[(R^2 - N)/2] = (\nu - N)/2 = -\frac{1}{3}$ and the second derivative $\mathbb{E}[(R^2 - N)^2/4 + N/2 - R^2] = [\nu(\nu+2) - 2N\nu + N^2]/4 + N/2 - \nu = \frac{4}{9}$ for $d = 2, 3$, in agreement with $\frac{d}{da}a^{-1/3} = -\frac{1}{3}$ and $\frac{d^2}{da^2}a^{-1/3} = \frac{4}{9}$ at $a = 1$. Replacing the ambient dimension $N$ by the subspace dimension $w$ in these

computations (for instance in $\operatorname{tr} B_0 = N$) would violate this check. By (S4.12), (S4.20) says that replacing $f$ by $Af$ ($A > 0$) multiplies the coefficient by $A^{-2/3}$. This is consistent with lifetimes: the lifetime density of $Af$ at $\ell$ is $A^{-1}$ times that of $f$ at $\ell/A$, which turns $c\,\ell^{-1/3}$ into $A^{-2/3}c\,\ell^{-1/3}$.

S4.4. **The side-24 covariance jet.**

**Lemma S4.14** (derivatives of the Gaussian)**.** *For $k \geq 0$, unit vectors $v_1, \dots, v_k$ and $x \in \mathbb{R}^d$,*

$$D^k\varphi(x)[v_1, \dots, v_k] = \varphi(x) \sum_{\mathsf{P}} \prod_{\{a,b\}\in\mathsf{P}} (-v_a \cdot v_b) \prod_{c\notin\cup\mathsf{P}} (-x \cdot v_c), \tag{S4.21}$$

*the sum running over all partial matchings $\mathsf{P}$ of $\{1, \dots, k\}$. Consequently*

$$\left|D^k\varphi(x)[v_1, \dots, v_k]\right| \leq e^{-|x|^2/2} \sum_{i=0}^{\lfloor k/2\rfloor} \frac{k!\,|x|^{k-2i}}{2^i\, i!\,(k-2i)!},$$

*and for $k \leq 6$*

$$\left|D^k\varphi(x)[v_1, \dots, v_k]\right| \leq 76\,|x|^6 e^{-|x|^2/2} \quad (|x| \geq 1), \qquad \left|D^k\varphi(0)[v_1, \dots, v_k]\right| \leq 15. \tag{S4.22}$$

*Moreover, for a unit vector $v$, $D^k\varphi(x)[v, \dots, v] = (-1)^k \operatorname{He}_k(x \cdot v)\,\varphi(x)$, where $\operatorname{He}_k$ is the probabilists' Hermite polynomial.*

*Proof.* Formula (S4.21) follows by induction on $k$: differentiating $\varphi$ in the direction $v_{k+1}$ produces the factor $-x \cdot v_{k+1}$ (the index $k+1$ unmatched), and differentiating a factor $-x \cdot v_c$ produces $-v_c \cdot v_{k+1}$ (the index $k+1$ matched with $c$). There are $k!/(2^i i!(k-2i)!)$ partial matchings with $i$ pairs, and $|v_a \cdot v_b| \leq 1$, $|x \cdot v_c| \leq |x|$. For $k = 0, 1, \dots, 6$ the coefficient sums are $1, 1, 2, 4, 10, 26, 76$; the largest is $1 + 15 + 45 + 15 = 76$ at $k = 6$, and for $|x| \geq 1$ every power $|x|^{k-2i}$ with $k \leq 6$ is at most $|x|^6$. At $x = 0$ only perfect matchings contribute, and there are $(k-1)!! \leq 15$ of them for even $k \leq 6$. For the last statement, $\varphi(x + sv) = \varphi(x)\exp(-s\,x \cdot v - s^2/2)$, and the generating function $\exp(us - s^2/2) = \sum_k \operatorname{He}_k(u)s^k/k!$ with $u = -x \cdot v$, together with $\operatorname{He}_k(-u) = (-1)^k \operatorname{He}_k(u)$, gives the claim. □

**Lemma S4.15** (lattice sums)**.** *Let $d \in \{2, 3\}$. Then $10^{-126} < q < 10^{-125}$, and*

$$\sum_{n\in\mathbb{Z}^d\setminus\{0\}} |n|^6 q^{|n|^2} \leq 1458\,q, \qquad Z_{24} - 1 = \sum_{n\in\mathbb{Z}^d\setminus\{0\}} q^{|n|^2} \leq 1458\,q. \tag{S4.23}$$

*Proof.* The Taylor polynomial of $e^x$ of degree 20 at $x = 288/125$, a sum of positive rational terms, exceeds 10; hence $e^{288} = (e^{288/125})^{125} > 10^{125}$. The Taylor polynomial of degree 40 at $x = 16/7$ plus the geometric bound $(x^{41}/41!)/(1 - x/42)$ for the remaining terms is below 10; hence $e^{288} = (e^{16/7})^{126} < 10^{126}$. (Both inequalities are verified in exact rational arithmetic.) Group the points $n \neq 0$ by $j = |n|_\infty \geq 1$. For $d \leq 3$ there are at most $(2j+1)^3 - (2j-1)^3 = 24j^2 + 2 \leq 27j^3$ of them, and each satisfies $j^2 \leq |n|^2 \leq 3j^2$, so $|n|^6 \leq 27j^6$ and $q^{|n|^2} \leq q^{j^2}$. Hence the first sum is at most $729 \sum_{j\geq 1} j^9 q^{j^2}$. The ratio of consecutive terms is $((j+1)/j)^9 q^{2j+1} \leq 512q^3 < \frac{1}{2}$, so $\sum_{j\geq 1} j^9 q^{j^2} \leq 2q$ and the first sum is at most $1458q$. The second sum is bounded by the first. □

**Lemma S4.16** (jet comparison)**.** *Let $d \in \{2, 3\}$ and*

$$B_0 = 76 \cdot 24^6 + 15 = 14523826191, \qquad E = 1458\,B_0 \cdot 10^{-125} = 21175738586478 \cdot 10^{-125}. \tag{S4.24}$$

*For every $k \leq 6$ and all unit vectors $v_1, \dots, v_k$,*

$$\left|D^k K_{24}(0)[v_1, \dots, v_k] - D^k\varphi(0)[v_1, \dots, v_k]\right| \leq 1458\,B_0\,q < E. \tag{S4.25}$$

*Proof.* The series $\sum_n \varphi(x+24n)$ converges, with all its derivatives, uniformly on compact sets, so

$$D^k K_{24}(0) - D^k\varphi(0) = Z_{24}^{-1}\Big[\sum_{n\neq 0} D^k\varphi(24n) - (Z_{24}-1)D^k\varphi(0)\Big],$$

evaluated at $(v_1,\dots,v_k)$. Since $|24n| \geq 24 \geq 1$ and $\varphi(24n) = q^{|n|^2}$, Lemma S4.14 gives $|D^k\varphi(24n)[\dots]| \leq 76\cdot 24^6|n|^6 q^{|n|^2}$ and $|D^k\varphi(0)[\dots]| \leq 15$. By Lemma S4.15 and $Z_{24}\geq 1$, the difference is at most $(76\cdot 24^6+15)\cdot 1458q = 1458B_0q$, and $1458B_0q < E$ because $q < 10^{-125}$. This bound concerns the infinite image sum and holds for all directions, including mixed ones, through total order six, with the exact variance-one normalization. □

**Lemma S4.17** (reference floor and covariance comparison)**.** *Let $d\in\{2,3\}$, let $\mathfrak{f}$ be a frame, and let $C_0$ and $C_{24}$ be the covariance matrices of $X_t$ for $K=\varphi$ and $K=K_{24}$.*

*(a) $C_0$ does not depend on the frame and $C_0 \succcurlyeq \frac{1}{3}I$. Precisely, $C_0$ is block diagonal: the block of $(\partial_t f(0),\zeta_t)$ is $\left(\begin{smallmatrix}1 & -3\\ -3 & 15\end{smallmatrix}\right)$, the transverse gradient has covariance $I_m$, and on the coordinates $\operatorname{svec} H_t$ the covariance is the operator $\operatorname{svec} U \mapsto \operatorname{svec}((\operatorname{tr}U)I+2U)$, with eigenvalue $d+2$ on $\operatorname{svec} I$ and 2 on traceless matrices.*

*(b) Every entry of $C_{24}-C_0$ has absolute value less than $2E$, and $\|C_{24}-C_0\|\leq 20E$.*

*(c)*

$$\theta = \|C_0^{-1/2}(C_{24}-C_0)C_0^{-1/2}\| \leq 60E =: \theta_0 = 87480\,B_0\cdot 10^{-125} < \tfrac{1}{4}. \tag{S4.26}$$

*In particular $(1-\theta_0)C_0 \preccurlyeq C_{24} \preccurlyeq (1+\theta_0)C_0$ in every frame; $\varphi$ and $K_{24}$ are admissible.*

*Proof.* The entry of $\operatorname{Cov}(X_t)$ between coordinates built from frame-direction derivatives $\partial^\alpha$ and $\partial^\beta$ is $c_\alpha c_\beta(-1)^{|\beta|}D^{|\alpha|+|\beta|}K(0)[e_\alpha,e_\beta]$, with $c\in\{1,\sqrt{2}\}$ the svec scale, and it vanishes for odd $|\alpha|+|\beta|$.

(a) By (S4.21) at $x=0$, $D^2\varphi(0)[u,v] = -u\cdot v$, $D^4\varphi(0)$ is the sum of the three products of inner products over the perfect matchings, and $D^6\varphi(0)$ is minus the sum over the fifteen perfect matchings. Hence $\operatorname{Var}(\partial_t f)=1$, $\operatorname{Cov}(\partial_t f,\zeta_t) = -D^4\varphi(0)[t,t,t,t] = -3$, $\operatorname{Var}(\zeta_t) = -D^6\varphi(0)[t,\dots,t] = 15$, $\operatorname{Cov}(\partial_{n_i}f,\partial_{n_j}f)=\delta_{ij}$, and the remaining odd-block entries vanish because every perfect matching then pairs two orthogonal frame vectors. In the even block, $\operatorname{Cov}(H_{ab},H_{cd}) = \delta_{ab}\delta_{cd}+\delta_{ac}\delta_{bd}+\delta_{ad}\delta_{bc}$, i.e. $\operatorname{Cov}(\langle H_t,U\rangle,\langle H_t,V\rangle) = \operatorname{tr}U\operatorname{tr}V + 2\operatorname{tr}(UV)$ for symmetric $U,V$; since svec is an isometry, this is the stated operator, with eigenvalues $d+2$ and 2. Only inner products of frame vectors occur, so $C_0$ is the same in every frame. The matrix $\left(\begin{smallmatrix}1 & -3\\ -3 & 15\end{smallmatrix}\right) - \frac{1}{3}I$ has first principal minor $\frac{2}{3}$ and determinant $\frac{2}{3}\cdot\frac{44}{3} - 9 = \frac{7}{9}$, both positive; the transverse gradient variances are 1 and the even-block eigenvalues are at least 2. Hence $C_0\succcurlyeq\frac{1}{3}I$.

(b) Each entry of $C_{24}-C_0$ is $c_\alpha c_\beta$ times a difference as in (S4.25), so it is at most $2\cdot 1458B_0q < 2E$ in absolute value. The dimension is $N\leq 10$, so $\|C_{24}-C_0\|\leq N\max|\text{entry}|\leq 20E$.

(c) $\|C_0^{-1/2}MC_0^{-1/2}\| \leq \|C_0^{-1}\|\,\|M\| \leq 3\|M\|$, so $\theta\leq 60E = 60\cdot 1458B_0\cdot 10^{-125} = 87480B_0\cdot 10^{-125}$, which is about $1.27\cdot 10^{-110} < \frac{1}{4}$. The final assertions follow from Proposition S4.11(a). □

## S4.5. **The first normalized nearest-image variation.**

**Definition S4.18** (jets, frame matrices and the coefficient functional)**.** Let $\mathcal{J}_d$ be the real vector space of triples $J=(J_2,J_4,J_6)$ of symmetric multilinear forms on $\mathbb{R}^d$ of ranks $2,4,6$. The jet of an even $C^6$ function $K$ is $J(K) = (D^2K(0),D^4K(0),D^6K(0))$; put $J_0 = J(\varphi)$ and $J_{24} = J(K_{24})$. The orthogonal group acts by $(g\cdot J)_k(v_1,\dots,v_k) = J_k(g^{-1}v_1,\dots,g^{-1}v_k)$, so that $J(K\circ g^{-1}) = g\cdot J(K)$. For a frame $\mathfrak{f}=(e_0,\dots,e_m)$, index the coordinates of $\mathbb{R}^N$ by derivative words $\alpha$ in the frame: $\alpha=(a)$ for the $a$-th entry of $g_t$, $\alpha=(0,0,0)$ for $\zeta_t$, and $\alpha=(a,b)$, $a\leq b$, for the svec entry of $H_{ab}$, with scale $c_\alpha=\sqrt{2}$ if $a<b$ and $c_\alpha=1$ otherwise. For $J\in\mathcal{J}_d$ let $C(J;\mathfrak{f})$ be the symmetric

$N \times N$ matrix

$$C(J;\mathfrak{f})_{\alpha\beta} = \begin{cases} c_\alpha c_\beta (-1)^{|\beta|} J_{|\alpha|+|\beta|}(e_\alpha, e_\beta), & |\alpha|+|\beta| \text{ even}, \\ 0, & |\alpha|+|\beta| \text{ odd}, \end{cases} \tag{S4.27}$$

where $e_\alpha = (e_{\alpha_1}, e_{\alpha_2}, \dots)$; for admissible $K$, $C(J(K);\mathfrak{f}) = \mathrm{Cov}(X_t)$ at $t = e_0$. Let $\mathcal{U}_d$ be the open set of $J$ with $C(J;\mathfrak{f})$ positive definite for every frame, and for $J \in \mathcal{U}_d$ define

$$\Phi_d(J) = 2^{m/2}\, 96^{-1/3} \int_{S^{d-1}} \Psi\big(C(J;\mathfrak{f}_t)\big)\, d\sigma(t), \tag{S4.28}$$

where $\mathfrak{f}_t$ is any frame at $t$ (Lemma S4.8, whose proof applies verbatim to $C(J;\cdot)$). Every $C(J;\mathfrak{f})$ has the block structure of Lemma S4.7, so (S4.11) applies to it, with $g_t, \zeta_t, H_t$ the coordinates of a centered Gaussian vector with covariance $C(J;\mathfrak{f})$; likewise Lemma S4.3 and its proof apply to $\Phi_3(J)$. By Lemma S4.7, $c_d[K] = \Phi_d(J(K))$ for admissible $K$. For $d = 3$, $\Phi_3$ is the coefficient functional $\Phi$ of (3.25) of the main text, and $K(0)$ does not enter. We write $C_0 = C(J_0;\mathfrak{f})$; by Lemma S4.20 below, $C_0$ is positive definite and the same in every frame.

**Lemma S4.19** (rotations)**.** *The matrix $C(J;\mathfrak{f})$ is linear in $J$; $C(g\cdot J; g\mathfrak{f}) = C(J;\mathfrak{f})$ for $g \in O(d)$, where $g\mathfrak{f} = (ge_0, \dots, ge_m)$; and $\Phi_d(g\cdot J) = \Phi_d(J)$ on $\mathcal{U}_d$.*

*Proof.* Linearity is clear from (S4.27), and $(g\cdot J)_k(ge_\alpha, ge_\beta) = J_k(e_\alpha, e_\beta)$. Hence $\Psi(C(g\cdot J;\mathfrak{f}_t)) = \Psi(C(J; g^{-1}\mathfrak{f}_t))$, and $g^{-1}\mathfrak{f}_t$ is a frame at $g^{-1}t$. By transverse-frame invariance and the rotation invariance of $\sigma$, the integral (S4.28) is unchanged. □

**Lemma S4.20** (isotropic jets)**.** *For $j = 1, 2, 3$ let $S_{2j}(v_1, \dots, v_{2j}) = \sum \prod (v_a \cdot v_b)$, the sum over the $(2j-1)!!$ perfect matchings of $\{1, \dots, 2j\}$.*

*(a) $J \in \mathcal{J}_d$ is $O(d)$-invariant if and only if*

$$J = J^{\mathrm{iso}}(a, m_4, \chi) := \big(-aS_2,\ \tfrac{m_4}{3}S_4,\ -\tfrac{\chi}{15}S_6\big) \tag{S4.29}$$

*for real $a, m_4, \chi$; then $a = -J_2(v,v)$, $m_4 = J_4(v,v,v,v)$ and $\chi = -J_6(v,\dots,v)$ for every unit vector $v$. Moreover $J_0 = J^{\mathrm{iso}}(1,3,15)$, and the matrix $C(J^{\mathrm{iso}};\mathfrak{f})$ does not depend on the frame.*

*(b) If $a > 0$, $m_4 > 0$ and $\Omega := a\chi - m_4^2 > 0$, then $J^{\mathrm{iso}}(a, m_4, \chi) \in \mathcal{U}_d$ and, in every frame,*

$$\mathrm{Cov}(g_t) = aI_d, \qquad \mathrm{Cov}(h_t) = \mathrm{diag}(m_4, \tfrac{m_4}{3}, \dots, \tfrac{m_4}{3}), \qquad \tau_t^2 = \chi - \frac{m_4^2}{a} = \frac{\Omega}{a},$$

$$\mathrm{Cov}\big((A_t)_{ij}, (A_t)_{kl} \mid h_t = 0\big) = \frac{m_4}{3}\Big[\frac{2}{3}\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\Big], \tag{S4.30}$$

*so that $p_{g_t}(0) \propto a^{-d/2}$, $p_{h_t}(0) \propto m_4^{-d/2}$ and $D_t \propto m_4^{d-1}$. Consequently*

$$\Psi\big(C(J^{\mathrm{iso}}(a, m_4, \chi);\mathfrak{f})\big) = \kappa_d\, a^{-d/2-2/3}\, m_4^{d/2-1}\, \Omega^{2/3} \tag{S4.31}$$

*with a constant $\kappa_d > 0$ depending only on $d$.*

*(c) At $(a, m_4, \chi) = (1, 3, 15)$, where $\Omega = 6$,*

$$d\log\Psi = \Big(1 - \frac{d}{2}\Big)da + \Big(\frac{d}{6} - 1\Big)dm_4 + \frac{d\chi}{9}. \tag{S4.32}$$

*Proof.* (a) If $J_{2j}$ is invariant, the polynomial $P(x) = J_{2j}(x, \dots, x)$ satisfies $P(gx) = P(x)$, so $P(x) = P(e)|x|^{2j}$ for any unit $e$; since $S_{2j}(x, \dots, x) = (2j-1)!!\,|x|^{2j}$ and a symmetric form is determined by its values on the diagonal, $J_{2j} = P(e)S_{2j}/(2j-1)!!$. Conversely each $S_{2j}$ is invariant. With $(2j-1)!! = 1, 3, 15$ this gives (S4.29) and the formulas for $a, m_4, \chi$. By (S4.21), $D^{2j}\varphi(0) = (-1)^j S_{2j}$, so $J_0 = J^{\mathrm{iso}}(1,3,15)$. The entries of $C(J^{\mathrm{iso}};\mathfrak{f})$ involve only inner products of frame vectors.

(b) By (S4.27): $\mathrm{Var}(\partial_{e_a} f) = a$; $\mathrm{Cov}(\partial_t f, \zeta_t) = -J_4(t,t,t,t) = -m_4$; $\mathrm{Var}(\zeta_t) = -J_6(t,\dots,t) = \chi$; the transverse gradient components are uncorrelated with $\partial_t f$ and with $\zeta_t$; and $\mathrm{Cov}(H_{ab}, H_{cd}) =$

$\frac{m_4}{3}(\delta_{ab}\delta_{cd}+\delta_{ac}\delta_{bd}+\delta_{ad}\delta_{bc})$. The odd block is positive definite if and only if $a>0$ and $a\chi-m_4^2>0$, and the even block has eigenvalues $\frac{m_4}{3}(d+2)$ and $\frac{2m_4}{3}$; this proves $J^{\mathrm{iso}}\in\mathcal{U}_d$ and the formulas for $\mathrm{Cov}(g_t)$, $\mathrm{Cov}(h_t)$ and $\tau_t^2$. Among the entries of $h_t$ only $H_{00}$ is correlated with $A_t$, with $\mathrm{Cov}((A_t)_{ij},H_{00})=\frac{m_4}{3}\delta_{ij}$ and $\mathrm{Var}(H_{00})=m_4$, so conditioning on $h_t=0$ subtracts $(\frac{m_4}{3})^2\delta_{ij}\delta_{kl}/m_4=\frac{m_4}{9}\delta_{ij}\delta_{kl}$ from the unconditional covariance; this is (S4.30). Thus $p_{g_t}(0)=(2\pi a)^{-d/2}$ and $p_{h_t}(0)=(2\pi)^{-d/2}3^{m/2}m_4^{-d/2}$, and the conditional law of $A_t$ is the law for $m_4=3$ scaled by $(m_4/3)^{1/2}$. Since $(\det A)^2$ is homogeneous of degree $2m$ and the cone $\{A\prec 0\}$ is scale invariant, $D_t=(m_4/3)^m D^{(3)}$, where $D^{(3)}>0$ is the value at $m_4=3$. Inserting these factors in (S4.11) gives (S4.31) with $\kappa_d=2^{-m/2}M_{4/3}(2\pi)^{-d}3^{-m/2}D^{(3)}$.

(c) By (S4.31), $d\log\Psi=(-\frac{d}{2}-\frac{2}{3})\frac{da}{a}+(\frac{d}{2}-1)\frac{dm_4}{m_4}+\frac{2}{3}\frac{d\Omega}{\Omega}$ with $d\Omega=\chi\,da+a\,d\chi-2m_4\,dm_4$. At $(1,3,15)$ this is $(-\frac{d}{2}-\frac{2}{3}+\frac{5}{3})da+(\frac{d}{6}-\frac{1}{3}-\frac{2}{3})dm_4+\frac{1}{9}d\chi$. □

**Lemma S4.21** (Haar projection of the first variation)**.** *For $H\in\mathcal{J}_d$ put*

$$\ell(H)=\int_{S^{d-1}}D\Psi(C_0)\big[C(H;\mathfrak{f}_t)\big]\,d\sigma(t),\qquad \Pi H=\int_{O(d)}g\cdot H\,dg,$$

*with $dg$ the Haar probability measure. Then:*

*(a) $D\Psi(C_0)[C(H;\mathfrak{f}_t)]$ depends only on $t$, $\ell(g\cdot H)=\ell(H)$ for $g\in O(d)$, and $\ell(H)=\ell(\Pi H)$;*
*(b) $\Pi H=J^{\mathrm{iso}}(\delta a,\delta m_4,\delta\chi)$, where, for $v$ uniformly distributed on $S^{d-1}$,*

$$\delta a=-\mathbb{E}H_2(v,v),\qquad \delta m_4=\mathbb{E}H_4(v,v,v,v),\qquad \delta\chi=-\mathbb{E}H_6(v,\dots,v);$$

*(c) $\ell(H)=|S^{d-1}|\,\Psi(C_0)\big[(1-\frac{d}{2})\delta a+(\frac{d}{6}-1)\delta m_4+\frac{1}{9}\delta\chi\big]$;*
*(d) $\Phi_d$ is differentiable at $J_0$ in the direction $H$, with $D\Phi_d(J_0)[H]=2^{m/2}96^{-1/3}\ell(H)$; hence*

$$\frac{D\Phi_d(J_0)[H]}{\Phi_d(J_0)}=\Big(1-\frac{d}{2}\Big)\delta a+\Big(\frac{d}{6}-1\Big)\delta m_4+\frac{\delta\chi}{9}.\tag{S4.33}$$

*Proof.* (a) The proof of Lemma S4.8 gives $C(J;\mathfrak{f}')=S_OC(J;\mathfrak{f})S_O^\top$ for two frames at $t$, and $S_OC_0S_O^\top=C_0$ because $C_0$ is the same in every frame (Lemma S4.20(a)). Since $\Psi(S_OCS_O^\top)=\Psi(C)$, differentiation at $C_0$ gives $D\Psi(C_0)[S_O\Delta S_O^\top]=D\Psi(C_0)[\Delta]$, which is the first claim. By Lemma S4.19, $C(g\cdot H;\mathfrak{f}_t)=C(H;g^{-1}\mathfrak{f}_t)$, a matrix of $H$ in a frame at $g^{-1}t$; the rotation invariance of $\sigma$ gives $\ell(g\cdot H)=\ell(H)$. The functional $\ell$ is linear on the finite-dimensional space $\mathcal{J}_d$, so $\ell(\Pi H)=\int\ell(g\cdot H)\,dg=\ell(H)$.

(b) $\Pi H$ is invariant by the invariance of Haar measure, so Lemma S4.20(a) applies. For a unit vector $e$, $(\Pi H)_2(e,e)=\int H_2(g^{-1}e,g^{-1}e)\,dg=\mathbb{E}H_2(v,v)$, because $g^{-1}e$ is uniformly distributed on $S^{d-1}$; similarly for ranks four and six.

(c) Since $\Pi H$ is isotropic, $C(\Pi H;\mathfrak{f}_t)$ is the same matrix for all $t$, so, by linearity of $C$ in $J$,

$$\ell(\Pi H)=|S^{d-1}|\,D\Psi(C_0)[C(\Pi H;\mathfrak{f})]=|S^{d-1}|\,\partial_\epsilon\Psi\big(C(J_0+\epsilon\Pi H;\mathfrak{f})\big)\big|_{\epsilon=0}.$$

Now $J_0+\epsilon\Pi H=J^{\mathrm{iso}}(1+\epsilon\delta a,3+\epsilon\delta m_4,15+\epsilon\delta\chi)$, and (S4.32) gives the claim; combine with (a).

(d) Put $\Delta_t=C(H;\mathfrak{f}_t)$ and $\theta_*=\sup_t\|C_0^{-1/2}\Delta_tC_0^{-1/2}\|$, which is finite because the entries of $\Delta_t$ are contractions of $H$ with unit vectors. For $0<|\epsilon|\theta_*\le\frac12$, $J_0+\epsilon H\in\mathcal{U}_d$ by Proposition S4.11(a), and Taylor's formula with integral remainder together with Proposition S4.11(c),(d), applied to $\epsilon\Delta_t$ (so that $\theta\le|\epsilon|\theta_*\le\frac12$), give, uniformly in $t$,

$$\big|\Psi(C_0+\epsilon\Delta_t)-\Psi(C_0)-\epsilon D\Psi(C_0)[\Delta_t]\big|\le\tfrac12\mathcal{K}(N,\nu)\,(2\epsilon\theta_*)^2\,2^{N/2}(3/2)^{\nu/2}\,\Psi(C_0).$$

Dividing by $\epsilon$ and integrating over $t$ gives the derivative of $\Phi_d$, and (c) together with $\Phi_d(J_0)=2^{m/2}96^{-1/3}|S^{d-1}|\Psi(C_0)$ gives (S4.33). □

This is a first-derivative statement only: $\Phi_d$ at an anisotropic jet is not replaced by its value at the isotropic projection. In the proof of Theorem S4.4 the second-order term is controlled with the full anisotropic matrices.

**Definition S4.22** (the nearest-image jet)**.** For $L > 0$ put

$$J_L^{(1)} = \sum_{i=1}^{d} \sum_{\varsigma=\pm 1} J(\varphi(\cdot + \varsigma L e_i)) - 2d\, q_L\, J_0 \ \in \mathcal{J}_d, \tag{S4.34}$$

where $J(\varphi(\cdot + y)) = (D^2\varphi(y), D^4\varphi(y), D^6\varphi(y))$ and $e_1, \dots, e_d$ is the standard basis. Since $\varphi(x + \varsigma L e_i) = q_L e^{-|x|^2/2} e^{-\varsigma L x_i}$, $J_L^{(1)}$ is the jet of the function

$$x \longmapsto 2q_L\, e^{-|x|^2/2} \sum_{i=1}^{d} (\cosh(Lx_i) - 1).$$

The jet $J_L^{(1)}$ contains the $2d$ nearest images $\pm L e_i$ together with their share $2dq_L$ of the normalizer $Z_L$; it is not the raw nearest-image sum. For $d = 3$ and $L = 24$ it is the term $J_{\text{near}}^{(1)}$ of (3.28) of the main text, and $R_{\text{jet}} = J_{24} - J_0 - J_{24}^{(1)}$.

**Lemma S4.23** (projected image variations)**.** *For every unit vector $v$ and $j = 1, 2, 3$,*

$$(J_L^{(1)})_{2j}(v, \dots, v) = 2q_L \sum_{i=1}^{d} [\mathrm{He}_{2j}(Lv_i) - \mathrm{He}_{2j}(0)]. \tag{S4.35}$$

*For $v$ uniform on $S^{d-1}$, $\mathbb{E}v_i^2 = \frac{1}{d}$, $\mathbb{E}v_i^4 = \frac{3}{d(d+2)}$ and $\mathbb{E}v_i^6 = \frac{15}{d(d+2)(d+4)}$. Consequently $\Pi J_L^{(1)} = q_L\, J^{\text{iso}}(\delta a, \delta m_4, \delta\chi)$ with*

$$\delta a = -2L^2, \qquad \delta m_4 = \frac{6L^4}{d+2} - 12L^2, \qquad \delta\chi = -\frac{30L^6}{(d+2)(d+4)} + \frac{90L^4}{d+2} - 90L^2. \tag{S4.36}$$

*Proof.* By Lemma S4.14 with $x = \varsigma L e_i$, $x \cdot v = \varsigma L v_i$, $\varphi(x) = q_L$ and the evenness of $\mathrm{He}_{2j}$, $D^{2j}\varphi(\varsigma L e_i)[v, \dots, v] = q_L\, \mathrm{He}_{2j}(Lv_i)$, while $D^{2j}\varphi(0)[v, \dots, v] = \mathrm{He}_{2j}(0)$; summing over $i$ and $\varsigma$ gives (S4.35). For the moments, write a standard Gaussian vector $G$ in $\mathbb{R}^d$ as $G = |G|v$ with $|G|$ and $v$ independent; then $\mathbb{E}G_1^{2j} = \mathbb{E}|G|^{2j}\, \mathbb{E}v_1^{2j}$, with $\mathbb{E}G_1^2, \mathbb{E}G_1^4, \mathbb{E}G_1^6 = 1, 3, 15$ and $\mathbb{E}|G|^2, \mathbb{E}|G|^4, \mathbb{E}|G|^6 = d,\ d(d+2),\ d(d+2)(d+4)$. With $\mathrm{He}_2(u) = u^2 - 1$, $\mathrm{He}_4(u) = u^4 - 6u^2 + 3$ and $\mathrm{He}_6(u) = u^6 - 15u^4 + 45u^2 - 15$,

$$\mathbb{E}\sum_i [\mathrm{He}_2(Lv_i) - \mathrm{He}_2(0)] = L^2,$$

$$\mathbb{E}\sum_i [\mathrm{He}_4(Lv_i) - \mathrm{He}_4(0)] = \frac{3L^4}{d+2} - 6L^2,$$

$$\mathbb{E}\sum_i [\mathrm{He}_6(Lv_i) - \mathrm{He}_6(0)] = \frac{15L^6}{(d+2)(d+4)} - \frac{45L^4}{d+2} + 45L^2.$$

Multiplying by $2q_L$ and applying Lemma S4.21(b) (with the signs $-, +, -$ for ranks $2, 4, 6$) gives (S4.36). □

For $v = e_i$, (S4.35) reduces to $2q_L(\mathrm{He}_{2j}(L) - \mathrm{He}_{2j}(0))$: since $K_L$ factorizes over the coordinate axes, this is the first-order variation of the normalized one-dimensional even moments of an axis factor, and (S4.36) is its rotational projection.

**Proposition S4.24** (the first normalized nearest-image variation)**.** *For every integer $d \geq 2$ and every $L > 0$, $\Phi_d$ is differentiable at $J_0$ in the direction $J_L^{(1)}$, and*

$$\frac{D\Phi_d(J_0)[J_L^{(1)}]}{\Phi_d(J_0)} = \frac{1}{|S^{d-1}|\,\Psi(C_0)} \int_{S^{d-1}} D\Psi(C_0)\big[C(J_L^{(1)}; \mathfrak{f}_t)\big]\, d\sigma(t) = P_d(L)\, q_L, \tag{S4.37}$$

*with $P_d$ as in* (S4.6).

*Proof.* By Lemma S4.21(c),(d) and Lemma S4.23, the left side equals $q_L$ times

$$\Big(1 - \frac{d}{2}\Big)(-2L^2) + \Big(\frac{d}{6} - 1\Big)\Big(\frac{6L^4}{d+2} - 12L^2\Big) + \frac{1}{9}\Big(-\frac{30L^6}{(d+2)(d+4)} + \frac{90L^4}{d+2} - 90L^2\Big).$$

The coefficient of $L^2$ is $-2+d-2d+12-10 = -d$, that of $L^4$ is $(d-6+10)/(d+2) = (d+4)/(d+2)$, and that of $L^6$ is $-10/(3(d+2)(d+4))$. This is $P_d(L)$. □

**Corollary S4.25** (dimension three)**.** *Let $d = 3$ and $F = \Phi_3/\Phi_3(J_0)$ as in (3.25) of the main text. Then*

$$P_3(L) = -\frac{L^2(10L^4 - 147L^2 + 315)}{105} = -\frac{2}{21}L^6 + \frac{7}{5}L^4 - 3L^2, \qquad P_3(24) = -\frac{620813376}{35}, \tag{S4.38}$$

*and $DF(J_0)[J_{24}^{(1)}] = P_3(24)\, e^{-288}$. In the coordinates $(a, m_4, \chi)$ and $\Omega = a\chi - m_4^2$, the variations per unit $q_L$ are*

$$\delta a = -2L^2, \qquad \delta m_4 = \frac{6}{5}L^4 - 12L^2, \qquad \delta\chi = -\frac{6}{7}L^6 + 18L^4 - 90L^2, \tag{S4.39}$$

*the pure partial derivatives of $\log c$, where $c(a, m_4, \chi) = \Phi_3(J^{\mathrm{iso}}(a, m_4, \chi))$ is the coefficient along isotropic jets, are at $(1, 3, 15)$*

$$\partial_a \log c = -\tfrac{1}{2}, \qquad \partial_{m_4} \log c = -\tfrac{1}{2}, \qquad \partial_\chi \log c = \tfrac{1}{9}, \tag{S4.40}$$

*and, with $\delta\Omega = \chi\,\delta a + a\,\delta\chi - 2m_4\,\delta m_4$,*

$$\delta \log c = -\tfrac{1}{2}\,\delta a - \tfrac{1}{2}\,\delta m_4 + \tfrac{1}{9}\,\delta\chi = \frac{\delta\Omega}{9} + \frac{\delta m_4}{6} - \frac{13}{6}\,\delta a = P_3(L). \tag{S4.41}$$

*Also $P_2(24) = -26045568$.*

*Proof.* The formulas (S4.38), (S4.39) and $DF(J_0)[J_{24}^{(1)}] = P_3(24)q$ are Proposition S4.24 and Lemma S4.23 at $d = 3$; the values $P_3(24)$ and $P_2(24)$ are exact rational arithmetic. For the partial derivatives, use the form (S4.4) of the coefficient along isotropic jets: $\det\Sigma_g = a^3$, $\det\Sigma_h = m_4^3/9$ and $\sigma_t^2 = \tau_t^2/144 = \Omega/(144a)$ by Lemma S4.20(b), and $D_t = D(\Gamma_t)$ with $\Gamma_t$ proportional to $m_4$ and the cone moment $D$ homogeneous of degree 2 in the covariance. Hence

$$\partial_a \log c = -\frac{3}{2a} + \frac{2}{3}\Big(\frac{\chi}{\Omega} - \frac{1}{a}\Big), \qquad \partial_{m_4} \log c = -\frac{3}{2m_4} + \frac{2}{3}\cdot\frac{-2m_4}{\Omega} + \frac{2}{m_4}, \qquad \partial_\chi \log c = \frac{2}{3}\cdot\frac{a}{\Omega},$$

where the term $2/m_4$ comes from the homogeneity of $D$. At $(1, 3, 15)$, $\Omega = 6$, these are $-\frac{1}{2}$, $-\frac{1}{2}$ and $\frac{1}{9}$, in agreement with (S4.32) at $d = 3$. The two expressions in (S4.41) agree identically: with $\delta\Omega = 15\delta a + \delta\chi - 6\delta m_4$ at $(1, 3, 15)$, $\frac{\delta\Omega}{9} + \frac{\delta m_4}{6} - \frac{13}{6}\delta a = (\frac{15}{9} - \frac{13}{6})\delta a + (\frac{1}{6} - \frac{6}{9})\delta m_4 + \frac{1}{9}\delta\chi = -\frac{1}{2}\delta a - \frac{1}{2}\delta m_4 + \frac{1}{9}\delta\chi$. Substituting (S4.39) gives $P_3(L)$. □

*Remark* S4.26 (a scale check). Along the scale family $K_\lambda(x) = e^{-\lambda|x|^2/2}$, $\lambda > 0$, which is the covariance of $f(\sqrt{\lambda}\,\cdot)$, one has $J(K_\lambda) = J^{\mathrm{iso}}(\lambda, 3\lambda^2, 15\lambda^3)$, and (S4.31) with $\Omega = 6\lambda^4$ gives $c_3[K_\lambda] = \lambda^{-13/6+1+8/3} c_3[\varphi] = \lambda^{3/2} c_3[\varphi]$. In terms of (S4.40), at $\lambda = 1$,

$$\frac{d\log c}{d\log\lambda} = -\tfrac{1}{2}\cdot 1 - \tfrac{1}{2}\cdot 6 + \tfrac{1}{9}\cdot 45 = \tfrac{3}{2},$$

in agreement with the scaling of densities per unit volume under $x \mapsto \sqrt{\lambda}\,x$ in dimension three.

S4.6. **Deep images and the normalization product.**

**Lemma S4.27** (normalization identity)**.** *Let $d \in \{2,3\}$, let $k \in \{2,4,6\}$, fix unit vectors $v_1,\dots,v_k$, and evaluate every $k$-form below at $(v_1,\dots,v_k)$. Put*

$$\alpha_0 = D^k\varphi(0), \quad \alpha_1 = \sum_{|n|=1} D^k\varphi(24n), \quad \alpha_2 = \sum_{|n|^2\ge 2} D^k\varphi(24n), \quad \beta_1 = 2dq, \quad \beta_2 = \sum_{|n|^2\ge 2} q^{|n|^2},$$

*and $\gamma_1 = \alpha_1 - \beta_1\alpha_0$. Then $D^kK_{24}(0) = (\alpha_0+\alpha_1+\alpha_2)/(1+\beta_1+\beta_2)$, $\gamma_1 = (J_{24}^{(1)})_k(v_1,\dots,v_k)$, and*

$$D^kK_{24}(0) - \alpha_0 - \gamma_1 = \frac{\alpha_2 - \beta_2\alpha_0 - \gamma_1(\beta_1+\beta_2)}{1+\beta_1+\beta_2}. \tag{S4.42}$$

*Proof.* There are $2d$ points $n \in \mathbb{Z}^d$ with $|n| = 1$, so $Z_{24} = 1+\beta_1+\beta_2$; the formula for $\gamma_1$ is (S4.34). Then $D^kK_{24}(0) - \alpha_0 - \gamma_1$ equals $[\alpha_1+\alpha_2 - (\beta_1+\beta_2)\alpha_0 - \gamma_1(1+\beta_1+\beta_2)]/(1+\beta_1+\beta_2)$, and $\alpha_1 - \beta_1\alpha_0 - \gamma_1 = 0$. □

*Remark* S4.28 (the normalization product). The product $\gamma_1(\beta_1+\beta_2)$ in (S4.42) is of order $q^2$ and must be retained. Already in one dimension, keeping only the images $\pm L$, the normalized second derivative is

$$\frac{-1+2(L^2-1)q_L}{1+2q_L} = -1 + 2L^2q_L - \frac{4L^2q_L^2}{1+2q_L},$$

which has a term of order $q_L^2$, although the next images $\pm 2L$ carry the weight $q_L^4$.

**Lemma S4.29** (deep images)**.** *Let $d \in \{2,3\}$ and $T_6 = \sum_{|n|^2\ge 2} |n|^6 q^{|n|^2}$. Then $T_6 < 541q^2$, and for every $k \in \{2,4,6\}$ and all unit vectors $v_1,\dots,v_k$,*

$$\left|(J_{24} - J_0 - J_{24}^{(1)})_k(v_1,\dots,v_k)\right| < (541 + 6\cdot 1458)\, B_0 q^2 = 9289\, B_0\, q^2. \tag{S4.43}$$

*Consequently, for every frame $\mathfrak{f}$ the matrix $R_{\mathfrak{f}} = C(R_{\mathrm{jet}};\mathfrak{f})$ of $R_{\mathrm{jet}} = J_{24} - J_0 - J_{24}^{(1)}$ satisfies*

$$\|C_0^{-1/2} R_{\mathfrak{f}} C_0^{-1/2}\| < 60\cdot 9289\, B_0 \cdot 10^{-250} =: \rho_0. \tag{S4.44}$$

*Moreover $|(J_{24}^{(1)})_k(v_1,\dots,v_k)| \le 6B_0q$, so that*

$$\left|(J_{24}-J_0)_k(v_1,\dots,v_k)\right| < 6B_0q + 9289B_0q^2 < 10^{-113}, \qquad \left|(R_{\mathrm{jet}})_k(v_1,\dots,v_k)\right| < 10^{-230}.$$

*For $d = 3$, with the max norm on coordinate entries, these are the bounds (3.29) of the main text.*

*Proof.* Let $d = 3$. The shell $|n|_\infty = 1$ consists of 26 points: the 6 with $|n| = 1$, and 20 with $|n|^2 \in \{2,3\}$, each with $|n|^6 \le 27$; their contribution to $T_6$ is at most $540q^2$. The shells $j = |n|_\infty \ge 2$ contribute at most $729\sum_{j\ge 2} j^9 q^{j^2}$, as in the proof of Lemma S4.15; for $j \ge 2$ the ratio of consecutive terms is at most $(\frac{3}{2})^9 q^5 < \frac{1}{2}$, so this is at most $729\cdot 2\cdot 2^9 q^4 = 746496q^4 < q^2$. Hence $T_6 < 541q^2$. For $d = 2$, the terms of $T_6$ form a subset of those for $d = 3$ under $n \mapsto (n,0)$, so the same bound holds.

In Lemma S4.27, $|\alpha_2| \le 76\cdot 24^6\, T_6$ by Lemma S4.14, and $|\beta_2\alpha_0| \le 15\beta_2 \le 15T_6$, so $|\alpha_2 - \beta_2\alpha_0| \le B_0T_6 < 541B_0q^2$. Also $|\gamma_1| \le |\alpha_1| + \beta_1|\alpha_0| \le 2d(76\cdot 24^6+15)q \le 6B_0q$, and $\beta_1+\beta_2 = Z_{24}-1 \le 1458q$ by Lemma S4.15. The denominator in (S4.42) is at least 1. Hence the left side of (S4.43) is less than $541B_0q^2 + 6B_0q\cdot 1458q = 9289B_0q^2$. By Lemma S4.27, $(J_{24}^{(1)})_k = \gamma_1$ and $(J_{24}-J_0)_k = \gamma_1 + (R_{\mathrm{jet}})_k$, which gives the two final bounds, since $6B_0 = 87142957146 < 8.72\cdot 10^{10}$, $9289B_0 < 1.35\cdot 10^{14}$ and $q < 10^{-125}$.

Every entry of $R_{\mathfrak{f}}$ is $c_\alpha c_\beta$ times $\pm$ a contraction as in (S4.43), hence less than $2\cdot 9289B_0q^2$ in absolute value; with $N \le 10$ and $C_0 \succcurlyeq \frac{1}{3}I$ (Lemma S4.17), $\|C_0^{-1/2}R_{\mathfrak{f}}C_0^{-1/2}\| < 3\cdot 10\cdot 2\cdot 9289B_0q^2$, and $q^2 < 10^{-250}$. □

S4.7. **Proof of Theorem S4.4.**

*Proof of Theorem S4.4.* Let $d \in \{2,3\}$. By Lemma S4.7 and Definition S4.18,

$$\frac{c_{d,24}}{c_{d,\mathrm{ref}}} = \frac{\Phi_d(J_{24})}{\Phi_d(J_0)} = \frac{1}{|S^{d-1}|\,\Psi(C_0)} \int_{S^{d-1}} \Psi(C_0+\Delta_t)\,d\sigma(t), \qquad \Delta_t = C(J_{24}-J_0;\mathfrak{f}_t),$$

where $C_0+\Delta_t$ is the covariance of $X_t$ for $K_{24}$ and $C_0$ is the reference matrix, the same in every frame (Lemma S4.17(a)).

*Second-order term.* By Lemma S4.17(c), the full exact perturbation $\Delta_t$ satisfies

$$\theta_t = \|C_0^{-1/2}\Delta_t C_0^{-1/2}\| \le \theta_0 < \tfrac{1}{4},$$

so Proposition S4.11 gives

$$\Psi(C_0+\Delta_t) = \Psi(C_0) + D\Psi(C_0)[\Delta_t] + r_t, \qquad |r_t| \le 2000\,\theta_0^2\,\Psi(C_0). \tag{S4.45}$$

*Linear term.* By linearity of $C$ in $J$, $\Delta_t = C(J_{24}^{(1)};\mathfrak{f}_t) + C(R_{\mathrm{jet}};\mathfrak{f}_t)$. By Lemma S4.12 (with $(N+\nu)/2 \le \frac{29}{3}$) and Lemma S4.29,

$$|D\Psi(C_0)[C(R_{\mathrm{jet}};\mathfrak{f}_t)]| < \tfrac{29}{3}\,\rho_0\,\Psi(C_0), \tag{S4.46}$$

and by Proposition S4.24 with $L=24$,

$$\int_{S^{d-1}} D\Psi(C_0)\big[C(J_{24}^{(1)};\mathfrak{f}_t)\big]\,d\sigma(t) = P_d(24)\,q\,|S^{d-1}|\,\Psi(C_0). \tag{S4.47}$$

*Assembly.* The bounds (S4.45) and (S4.46) are uniform in $t$. Integrating over $S^{d-1}$ and dividing by $|S^{d-1}|\Psi(C_0)$ gives

$$\Big|\frac{c_{d,24}}{c_{d,\mathrm{ref}}} - 1 - P_d(24)\,q\Big| < 2000\,\theta_0^2 + \tfrac{29}{3}\,\rho_0 = R_{\mathrm{rat}}, \tag{S4.48}$$

where, by (S4.26) and (S4.44),

$$R_{\mathrm{rat}} = \big[2000\,(87480\,B_0)^2 + 580\cdot 9289\,B_0\big]\cdot 10^{-250}. \tag{S4.49}$$

With $B_0 = 14523826191$, exact integer arithmetic gives

$$R_{\mathrm{rat}} = 32285657137165437337368567479554 20\cdot 10^{-250} < 10^{-216}.$$

This proves (S4.5).

*Sign.* By Lemma S4.15, $q > 10^{-126}$. Since $|P_2(24)| = 26045568$ and $|P_3(24)| = 620813376/35$, both $|P_2(24)|\cdot 10^{-126}$ and $|P_3(24)|\cdot 10^{-126}$ exceed $R_{\mathrm{rat}}$. As $P_d(24) < 0$, (S4.48) gives $c_{d,24}/c_{d,\mathrm{ref}} - 1 < P_d(24)q + R_{\mathrm{rat}} < 0$, that is, $c_{d,24} < c_{d,\mathrm{ref}}$. □

*Remark* S4.30 (numerical values). The number $q = e^{-288}$ can be enclosed in rational arithmetic with outward rounding (on a grid of mesh $10^{-260}$): enclose $e^{9/32}$ between its degree-100 Taylor sum and that sum plus an exact geometric bound for the tail, take reciprocals, and square ten times ($1024\cdot\frac{9}{32} = 288$). Multiplying by the negative number $P_d(24)$ reverses the endpoints, and adding $\pm R_{\mathrm{rat}}$ encloses the full relative correction. This gives

$$\begin{aligned} q &= 8.37894253381936897136\ldots\cdot 10^{-126},\\ P_3(24)\,q &= -1.48621702906582760722\ldots\cdot 10^{-118},\\ P_2(24)\,q &= -2.18234317532684674260\ldots\cdot 10^{-118}. \end{aligned}$$

Hence $c_{3,24}/c_{3,\infty} - 1 = -1.48621702906582760722\ldots\cdot 10^{-118} + \varepsilon_{24}$ with $|\varepsilon_{24}| < 10^{-216} < 10^{-98}\,|P_3(24)q|$, so the leading term $P_3(24)e^{-288}$ gives the correction to a relative precision better than $10^{-98}$.

## S5. Elder selection

This section concerns the selection bound (3.13) of the main text: under the determinant-weighted Palm law of a typed maximum–index-two-saddle contact pair at separation $r$, the probability that the pair is not the elder persistence pair is at most $C_{\mathrm{sel}} r^3$, uniformly on compact positive marks. The main text takes (3.13) from Theorem A of [7] (Theorem S12.3 of this supplement), which proves it for every $d \geq 2$ and $L > 0$ (Remark S5.37). This section proves the statements that the main text quotes for the Palm law and the elder mark (Definitions S5.1, S5.2 and S5.4, Lemma S5.3) and the trichotomy (Lemma S5.7). It then gives a second, three-dimensional proof of (3.13) through explicit witnesses (Theorem S5.36): the regional witness counts behind (3.14)–(3.17) of the main text (§S5.4), with the finite-scale densities near the pair axis of §S5.8, and the capture–escape theorem (Theorem S5.31).

Throughout, $f = f_{24}$ is the normalized side-24 periodized Bargmann–Fock field on $\mathbb{T}^3_{24}$, $0 < r < 12$, $t \in S^2$, $b \in \mathbb{R}$ and $\kappa > 0$; the pair is $M = x - \frac{r}{2}t$, $S = x + \frac{r}{2}t$, with midpoint $x$, and $h = f(S)$. We use an orthonormal frame $(e_1, e_2, e_3) = (t, v, w)$ whose first vector is $t$, and write $f_{ij\cdots}$ for partial derivatives in this frame. Constants $C, c$ depend only on the compact mark set and on the fixed charts, and may change from line to line.

### S5.1. The Palm law and the elder mark.

**Definition S5.1** (pins, weight and Palm law)**.** For $0 < r < 12$, $t \in S^2$, $b \in \mathbb{R}$ and $\kappa > 0$ let

$$\Pi_{r,x,t} = \big(f(M), f(S), \nabla f(M), \nabla f(S)\big), \qquad z_r(b,\kappa) = \big(b,\, b - \kappa r^3/6,\, 0,\, 0\big).$$

The conditional law $\mathbb{E}^0_{r,t,b,\kappa}[\,\cdot\,] = \mathbb{E}[\,\cdot \mid \Pi_{r,x,t} = z_r(b,\kappa)]$ is the regular conditional Gaussian distribution. It exists because the pin vector is Gaussian, and since the corrected-pin covariance is nondegenerate (Section S1), the conditional distributions have a version that is continuous in $(r, t, b, \kappa)$. Put

$$W_r = |\det H_M|\,|\det H_S|\,\mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S)=2\}}, \qquad Z_r(t,b,\kappa) = \mathbb{E}^0_{r,t,b,\kappa}[W_r].$$

By Lemma S2.9, $Z_r > 0$ for $0 < r < 12$, and $c_Z r^2 \leq Z_r \leq C_Z r^2$ on a compact positive mark set for $0 < r \leq r_0$ of Lemma S2.9(iii). The *all-typed determinant-weighted Palm law* is

$$\mathbb{E}^{MS}_{r,t,b,\kappa}[F] = \frac{\mathbb{E}^0_{r,t,b,\kappa}[W_r F]}{Z_r(t,b,\kappa)}. \tag{S5.1}$$

We write $\mathbb{P}^{MS}_r$ and $\mathbb{P}^0$ for the corresponding probabilities, and drop the subscripts $t, b, \kappa$ when they are fixed.

**Definition S5.2** (maximin level and elder mark)**.** Let $\Omega_{\mathrm{Mo}}$ be the event that $f$ is Morse with pairwise distinct critical values; $\mathbb{P}(\Omega_{\mathrm{Mo}}) = 1$ by Proposition 1.1 of the main text. For continuous $g : \mathbb{T}^3_{24} \to \mathbb{R}$ and $x \in \mathbb{T}^3_{24}$, the *maximin connection level* of $x$ is

$$d_g(x) = \sup\Big\{ \min_{s\in[0,1]} g(\gamma(s)) \ :\ \gamma : [0,1] \to \mathbb{T}^3_{24} \text{ continuous},\ \gamma(0) = x,\ g(\gamma(1)) > g(x)\Big\}, \tag{S5.2}$$

with $\sup \emptyset = -\infty$. Since $\mathbb{T}^3_{24}$ is path-connected, $d_g(x) = -\infty$ exactly when $x$ is a global maximum of $g$. For a typed pair $(M, S)$ with $h = f(S) < f(M)$, the *elder mark* is

$$e(f; M, S) = \mathbf{1}\{d_f(M) = f(S)\}. \tag{S5.3}$$

**Lemma S5.3** (the mark is correct and Borel)**.** *On $\Omega_{\mathrm{Mo}}$, let $M$ be a local maximum and $S$ an index-two critical point with $h = f(S) < f(M)$. Then $e(f; M, S) = 1$ if and only if*

*(i) the two upper local sectors of $S$ lie in distinct components of $\{f > h\}$;*
*(ii) $M$ is the birth maximum of one of them, that is, the maximum of $f$ over that component; and*

*(iii) the birth value at $M$ is the smaller of the two.*

*Conditions (i)–(iii) are exactly the elder-pair condition $S = D_f(M)$ of Section 1 of the main text. Moreover, $(g, x) \mapsto d_g(x)$ is lower semicontinuous on $C(\mathbb{T}^3_{24}) \times \mathbb{T}^3_{24}$, so $(f, M, S) \mapsto e(f; M, S)$ is Borel.*

*Proof.* Let $\mathcal{U}_1$ be the component of the open set $\{f > h\}$ that contains $M$. A path from $M$ that stays in $\{f > h\}$ stays in $\mathcal{U}_1$, and a compact path in $\{f > h\}$ has minimum $> h$.

*(i)–(iii) imply $e = 1$.* Let $\mathcal{U}_2 \neq \mathcal{U}_1$ be the component containing the other upper sector of $S$, with birth value $b_2 > f(M)$. Going from $M$ inside $\mathcal{U}_1$ to the first sector of $S$, through $S$, and inside $\mathcal{U}_2$ to its birth maximum gives a path to a point higher than $f(M)$ whose minimum is $f(S) = h$; hence $d_f(M) \geq h$. If some path from $M$ to a point higher than $f(M)$ had minimum $> h$, it would lie in $\mathcal{U}_1$, contradicting (ii). Hence $d_f(M) = h = f(S)$.

*$e = 1$ implies (i)–(iii).* Suppose $d_f(M) = h$. A point of $\mathcal{U}_1$ higher than $f(M)$ would give a path in $\mathcal{U}_1$ with minimum $> h$, so $f \leq f(M)$ on $\mathcal{U}_1$ and $M$ is the birth maximum of $\mathcal{U}_1$. For $t < h$ close to $h$, the definition of $d_f(M)$ gives a path from $M$ to a point higher than $f(M)$ with minimum $> t$, so the component of $M$ in $\{f > t\}$ contains a point higher than $f(M)$, while for $t \in (h, f(M))$ it does not. Since $h$ is the only critical value in a neighbourhood of $h$ and $S$ is the only critical point at level $h$, the components of $\{f > t\}$ change between $t > h$ and $t < h$ only by attaching the one-handle at $S$, which joins the components of its two upper sectors. Hence the two sectors lie in distinct components, one of which is $\mathcal{U}_1$ and the other, $\mathcal{U}_2$, contains a point higher than $f(M)$; this is (i)–(iii).

*Measurability.* For a fixed continuous path $\gamma$ from 0, the functional $g \mapsto \min_s g(\gamma(s))$ on the open set $\{g(\gamma(1)) > g(0)\}$, extended by $-\infty$ elsewhere, is lower semicontinuous in the uniform norm, and $d_g(0)$ is the supremum of these functionals over $\gamma$; a supremum of lower semicontinuous functions is lower semicontinuous. Since $d_g(x) = d_{g(\cdot+x)}(0)$ and $(g, x) \mapsto g(\cdot + x)$ is continuous, $(g, x) \mapsto d_g(x)$ is lower semicontinuous, hence Borel, and so is $\{d_f(M) = f(S)\}$. □

**Definition S5.4** (selection probability)**.** Let $p_r(t, b, \kappa) = \mathbb{E}^{MS}_{r,t,b,\kappa}[e(f; M, S)]$. With $p_\Pi$ the Gaussian density of $\Pi_{r,x,t}$ and $K^{\text{all}}_r = p_\Pi(z_r(b, \kappa))\, \mathbb{E}^0[W_r]$, the selected contact intensity is

$$K^{\text{sel}}_r(t, b, \kappa) = p_r(t, b, \kappa)\, K^{\text{all}}_r(t, b, \kappa). \tag{S5.4}$$

Neither adjacency nor selection enters the base law (S5.1).

## S5.2. **The trichotomy.**

**Definition S5.5** (selection events)**.** On $\Omega_{\text{Mo}}$ fix an ordered typed pair $(M, S)$ at distance $r$, and let $h = f(S)$ and $b = f(M)$.

(a) $A_r$ (*adjacency*): at least one unstable half-branch of $S$ under gradient ascent converges to $M$. This is a Boolean mark: if both half-branches converge to $M$, the pair still counts once (Section S1).

(b) $E_r = \{e(f; M, S) = 1\}$ (*actual elder selection*, (S5.3)).

(c) $L_{\text{same},r}$ (*same-maximum self-attachment*): both unstable half-branches of $S$ under gradient ascent converge to $M$.

(d) $B_r$ (*third-critical witness*): there is a critical point $y \notin \{M, S\}$ of $f$ with $f(y) \in [h, b]$ (value in the pair window). Its distance from the pair midpoint $x$ places it in exactly one of the regions

  (i) *collar*: $\operatorname{dist}(y, x) \leq C_1 r$;

  (ii) *singular-near*: $C_1 r < \operatorname{dist}(y, x) \leq \delta$ (so that $\eta = r/\operatorname{dist}(y, x) < 1/C_1 < 1$, a compact $\eta$-chart);

  (iii) *fixed-distance*: $\operatorname{dist}(y, x) > \delta$, and then $\operatorname{dist}(y, \{M, S\}) > \delta - r/2 \geq \delta/2$ for $r \leq \delta$.

  Here $C_1 > 1$ and $\delta > 0$ are fixed constants of the chart decomposition, and $r \leq \delta/C_1$.

**Lemma S5.6** (merge level)**.** *On $\Omega_{\mathrm{Mo}}$, let $M \neq M'$ be local maxima of $f$ that lie in one component of $\{f > h\}$, and let*

$$\tau = \sup\Big\{ \min_{s\in[0,1]} f(\gamma(s)) \ : \ \gamma : [0,1] \to \mathbb{T}^3_{24} \text{ continuous, } \gamma(0) = M, \ \gamma(1) = M' \Big\}.$$

*Then $h < \tau < \min(f(M), f(M'))$, and $\tau = f(y)$ for a critical point $y$ of $f$. In particular $y \notin \{M, M', S\}$, and $f(y) \in (h, f(M))$.*

*Proof.* A path from $M$ to $M'$ inside their common component of $\{f > h\}$ has compact image, hence minimum larger than $h$; so $\tau > h$. Since $M$ is a strict local maximum, there is a small sphere around $M$ on which $f \leq f(M) - \varepsilon$ for some $\varepsilon > 0$, and every path from $M$ to $M'$ crosses it; so $\tau \leq f(M) - \varepsilon$, and likewise for $M'$. Suppose that $\tau$ is not a critical value. The critical values are finite in number, so there is $\varepsilon > 0$ with no critical value in $[\tau - \varepsilon, \tau + \varepsilon]$ and with $\tau + \varepsilon < \min(f(M), f(M'))$. The flow of $\nabla f/|\nabla f|^2$, stopped on reaching level $\tau + \varepsilon$, deforms $\{f \geq \tau - \varepsilon\}$ into $\{f \geq \tau + \varepsilon\}$ and fixes $\{f \geq \tau + \varepsilon\}$, which contains $M$ and $M'$. It carries a path from $M$ to $M'$ with minimum larger than $\tau - \varepsilon$ to one with minimum at least $\tau + \varepsilon > \tau$, contradicting the definition of $\tau$. Hence $\tau = f(y)$ for a critical point $y$. Its value lies strictly between $f(S) = h$ and $\min(f(M), f(M'))$, so $y \notin \{M, M', S\}$. □

**Lemma S5.7** (exhaustive trichotomy)**.** *On $\Omega_{\mathrm{Mo}}$,*

$$A_r \cap E_r^c \subset B_r \cup L_{\mathrm{same},r}. \tag{S5.5}$$

*Proof.* Work on $A_r$. A half-branch of $S$ ascends to $M$; it leaves $S$ through one upper local sector of $S$ and stays in $\{f > h\}$, so $M$ lies in the component $\mathcal{U}_1$ of $\{f > h\}$ that contains this sector. Let $\mathcal{U}_2$ be the component containing the other upper sector. It suffices to find a critical point $y \notin \{M, S\}$ with $f(y) \in (h, b)$; the regions (i)–(iii) of Definition S5.5 then place it in $B_r$.

*Case 0: $\mathcal{U}_1 = \mathcal{U}_2$.* The other half-branch also stays in $\mathcal{U}_1$ and converges to a critical point $z$ of $f$ with $f(z) > h$. If $z = M$, both half-branches converge to $M$, which is $L_{\mathrm{same},r}$. Otherwise $z \neq M$, and $z \neq S$ because $f(z) > h$. If $f(z) < f(M) = b$, then $y = z$ is the required point. If $f(z) > f(M)$, then $\mathcal{U}_1$ contains a point higher than $M$. On the boundary of $\mathcal{U}_1$, $f = h < f(M)$, so the maximum of $f$ over the closure of $\mathcal{U}_1$ is attained inside $\mathcal{U}_1$, at a local maximum $M'$ with $f(M') > f(M)$. Lemma S5.6, applied to $M$ and $M'$, gives the required point $y$.

For Cases 1 and 2, assume $\mathcal{U}_1 \neq \mathcal{U}_2$, with birth values $b_1, b_2$ and birth maxima $M_1, M_2$; distinct critical values make these unique.

*Case 1: $M \neq M_1$.* Then $M_1$ and $M$ are distinct maxima in the component $\mathcal{U}_1$ of $\{f > h\}$, and Lemma S5.6 gives a critical point $y \notin \{M, S\}$ with $f(y) \in (h, f(M)) = (h, b)$, a window witness.

*Case 2: $M = M_1$ but $e = 0$.* Since $\mathcal{U}_1 \neq \mathcal{U}_2$ and condition (ii) of Lemma S5.3 holds, failure of (S5.3) forces $f(M) > \min(b_1, b_2)$, that is, $b_2 < f(M)$: $\mathcal{U}_2$ is younger and is the component killed at $S$. The required witness is the birth maximum of $\mathcal{U}_2$. Indeed, $y = M_2$ is a critical point (a local maximum), and its value satisfies $h < b_2 = f(M_2) < f(M) = b$; the lower bound holds because $\mathcal{U}_2$ is a component of the *open* superlevel set $\{f > h\}$ containing an upper sector of $S$, so its birth value exceeds $h$. Moreover $M_2 \notin \{M, S\}$: $M_2 \neq M$ because they lie in distinct components, and $M_2 \neq S$ because $f(S) = h < b_2$. So $y = M_2$ is a window witness. (The merge saddle of the surviving class of $M$ would not serve here: its value can lie below $h$, outside the window.)

Outside $L_{\mathrm{same},r}$, each case gives a critical point $y \notin \{M, S\}$ with value in the pair window, and the regions (i)–(iii) of Definition S5.5 place it in $B_r$. □

**Corollary S5.8** (reduction of the selection loss)**.** *Let $N_{\mathrm{col}}$ be the number of critical points $y \notin \{M, S\}$ of $f$ in the collar region (i), and let $N_{\mathrm{sing}}$ and $N_{\mathrm{far}}$ be the numbers of critical points $y$ with $f(y) \in [h, b]$ in the singular-near region (ii) and in the fixed-distance region (iii). Under the Palm law (S5.1),*

$$1 - p_r \leq \mathbb{P}_r^{MS}(A_r^c) + \mathbb{P}_r^{MS}(B_r) + \mathbb{P}_r^{MS}(L_{\mathrm{same},r}), \tag{S5.6}$$

*and*

$$\mathbb{P}_r^{MS}(B_r) \le \mathbb{E}_r^{MS} N_{\mathrm{col}} + \mathbb{E}_r^{MS} N_{\mathrm{sing}} + \mathbb{E}_r^{MS} N_{\mathrm{far}}. \tag{S5.7}$$

*Proof.* Since $e \in \{0,1\}$, $1 - p_r = \mathbb{P}_r^{MS}(E_r^c) \le \mathbb{P}_r^{MS}(A_r^c) + \mathbb{P}_r^{MS}(E_r^c \cap A_r)$, and Lemma S5.7 gives (S5.6). On $B_r$ at least one of the three counts is at least one, so Markov's inequality gives (S5.7). □

The three terms on the right of (S5.6) are $O(r^3)$ uniformly on compact positive mark sets: the three regional expectations in (S5.7) by Lemmas S5.19, S5.23 and S5.25, and the adjacency and self-attachment failures by Proposition S5.35, which rests on the capture–escape Theorem S5.31. The assembly is Theorem S5.36.

S5.3. **The endpoint factor and the Palm normalization.** Recall from Lemma S2.9(iii) that, on a compact positive mark set and for $0 < r \le r_0$ of that lemma,

$$c_Z r^2 \le Z_r = \mathbb{E}^0[W_r] \le C_Z r^2, \qquad \mathbb{E}^0[W_r^2] \le C_W r^4. \tag{S5.8}$$

Each Palm expectation below is an unnormalized expectation under $\mathbb{E}^0$ divided explicitly by $Z_r$; the lower bound in (S5.8) is the Palm normalization used in every witness count.

**Lemma S5.9** (exact endpoint factor $r^2$)**.** *In the frame $(t, v, w)$ write*

$$H_p = \begin{pmatrix} a_p & m_p^\top \\ m_p & Q_p \end{pmatrix}, \qquad p \in \{M, S\}. \tag{S5.9}$$

*Under the corrected pair pins $\nabla f(M) = \nabla f(S) = 0$, $f(M) - f(S) = \kappa r^3/6$,*

$$a_M = r\alpha_M, \qquad m_M = r\beta_M, \qquad a_S = r\alpha_S, \qquad m_S = r\beta_S, \tag{S5.10}$$

*where the scaled variables are linear functionals of the third-derivative process on the segment $\{x + \vartheta t : |\vartheta| \le r/2\}$, with conditional Gaussian moments uniformly bounded on compact marks. Moreover*

$$\det H_M = rD_M, \qquad \det H_S = rD_S, \tag{S5.11}$$

*where $D_M, D_S$ are polynomials of degree three in the scaled endpoint variables and the transverse Hessian entries. In particular $W_r \le r^2 |D_M|\,|D_S|$.*

*Proof.* For each coordinate direction $e_i$ put $g_i(\vartheta) = \partial_i f(x + \vartheta t)$. Since $g_i(-r/2) = g_i(r/2) = 0$, Taylor's integral formula gives the exact identities

$$\begin{aligned} \partial_{ti} f(M) &= -\frac{1}{r} \int_{-r/2}^{r/2} \Big(\frac{r}{2} - \vartheta\Big)\, \partial_{tti} f(x + \vartheta t)\, d\vartheta, \\ \partial_{ti} f(S) &= \frac{1}{r} \int_{-r/2}^{r/2} \Big(\vartheta + \frac{r}{2}\Big)\, \partial_{tti} f(x + \vartheta t)\, d\vartheta. \end{aligned} \tag{S5.12}$$

Thus the first row and column of each endpoint Hessian carry one exact factor $r$, which is (S5.10). The conditional Gaussian moments of the scaled variables are uniformly bounded on compact marks because the corrected pin covariance has a uniform eigenfloor and the side-24 field has finite derivative moments. Expanding (S5.9) along the first row gives (S5.11). Type indicators can only decrease absolute values, so $W_r \le |\det H_M|\,|\det H_S| = r^2 |D_M|\,|D_S|$. □

This is the same endpoint soft-mode factor that yields the two-sided bound in (S5.8).

**Lemma S5.10** (conditional polynomial moments)**.** *Let $H$ be a Gaussian vector whose conditional law given $\mathcal{A} = a$ has mean $\mu(a)$ with $\|\mu(a)\| \le C(1 + \|a\|)$ and covariance of operator norm at most $C$. Then for every polynomial $\Psi$ of degree at most nine,*

$$\mathbb{E}\big[|\Psi(H)| \;\big|\; \mathcal{A} = a\big] \le C_\Psi (1 + \|a\|^9). \tag{S5.13}$$

*The product* $\det H_M \det H_S \det H_y$ *of three* $3 \times 3$ *Hessian determinants is a polynomial of total degree nine in the concatenated Hessian coordinates, and the product* $\det H_M \det H_S$ *of two such determinants is a polynomial of degree six.*

*Proof.* Write $H = \mu(a) + G$, where $G$ is a centred Gaussian vector, possibly degenerate, with covariance of operator norm at most $C$. Expand $\Psi$ and use uniform Gaussian moments. □

S5.4. **Regional witness counts.** The collar and singular-near charts are first computed at the planar Bargmann–Fock jet and then transferred to the side-24 field through exact factorizations of the conditional determinants. These factorizations hold for every kernel satisfying the standing hypotheses below, and their remaining coefficients are positive by the derivative nondegeneracy of Section S1. They are contact limits. The passage to densities at a fixed positive scale is not uniform near the pair axis; it is made in §S5.8, for every such kernel. The fixed-distance count is proved directly for the side-24 field.

S5.4.1. *Standing hypotheses and the contact-limit pair law.* Let $K_0(z) = e^{-|z|^2/2}$ be the planar Bargmann–Fock covariance and $J_0$ its even covariance jet through order six. For a stationary even kernel $\mathcal{K}$ let $J_6(\mathcal{K})$ be its even jet through order six, with the max norm on coordinate entries.

**Hypothesis S5.11** (jet ball and standing hypotheses)**.** (1) (*Jet ball.*)

$$B = \{J : \ \|J - J_0\|_{\max} \le 10^{-110}\}. \tag{S5.14}$$

(2) (*Standing hypotheses.*) $\mathcal{K}$ is a stationary, even, real-analytic covariance kernel with $J_6(\mathcal{K}) \in B$; the associated centred Gaussian field has the finite-family derivative nondegeneracy of Section S1 (full positive Fourier support); and all compact charts, mark sets and constants are those of the planar charts below.

*Remark* S5.12. The normalized side-24 kernel $K_{24}$ satisfies Hypothesis S5.11. It has full positive Fourier support (Section S1), so Theorem S1.3 gives the derivative nondegeneracy, and $\|J_6(K_{24}) - J_0\|_{\max} < 10^{-113}$ by Lemma S4.29. The planar kernel $K_0$ has an everywhere positive spectral density, so a family of derivative evaluations at one point is nondegenerate for $K_0$ as soon as its polynomial symbols are linearly independent. The structural coefficients $A_{\mathrm{col}}$ and $A_{\mathrm{sing}}$ of Lemmas S5.18 and S5.21 are bounded below by exact factorizations and this nondegeneracy; no derivative bound for them on the jet ball is used. The jet ball enters through Lemma S5.14.

**Lemma S5.13** (contact-limit pair pin law at the planar jet)**.** *In the contact limit the raw pin vector of the pair may be written*

$$P = (f, f_{111}, f_1, f_{11}, f_2, f_{12}, f_3, f_{13}), \tag{S5.15}$$

*all derivatives taken at* $x$*, with target*

$$(b, 2\kappa, 0, 0, 0, 0, 0, 0), \tag{S5.16}$$

*up to the invertible rescaling* $f_{111} \mapsto -f_{111}/12$*. For the planar Bargmann–Fock covariance*

$$\det \mathrm{Cov}(P) = 12, \tag{S5.17}$$

*so the limiting pair pin law is nondegenerate. The corrected finite-*$r$ *pair frame converges analytically to* (S5.15)*, so compact covariance eigenfloors and Gaussian density envelopes follow for sufficiently small* $r$*. Conditioned on* (S5.16)*, the transverse Hessian*

$$Q = \begin{pmatrix} f_{22} & f_{23} \\ f_{23} & f_{33} \end{pmatrix} \qquad \textit{has law} \qquad Q = -bI_2 + G_2, \tag{S5.18}$$

*where* $G_2$ *is standard* $GOE_2$*: its diagonal entries are* $N(0, 2)$*, its off-diagonal entry is* $N(0, 1)$*, and these three variables are independent.*

*Proof.* All covariances follow from $\mathrm{Cov}(\partial^\alpha f, \partial^\beta f) = (-1)^{|\beta|}\partial^{\alpha+\beta}K_0(0)$, and (S5.17) and (S5.18) are exact Gaussian computations from these covariances. □

**Lemma S5.14** (pair-pin stability on the jet ball)**.** *Let $P_r$ be the corrected pair pin vector. At contact its planar covariance has determinant* 12*, and every entry is a contraction of $J_6$ with $\ell^1$ coefficient at most $3^6 = 729$. A crude row-sum bound of* 120 *gives*

$$\lambda_{\min}\,\mathrm{Cov}_0(P) \geq \frac{12}{120^7} > 3.34 \times 10^{-14}. \tag{S5.19}$$

*On the jet ball* (S5.14) *the operator perturbation of the $8 \times 8$ pin covariance is below*

$$8 \cdot 729 \cdot 10^{-110} < 5.84 \times 10^{-107}. \tag{S5.20}$$

*Hence the pair pin covariance, its inverse, all conditional means and all conditional covariance blocks are analytic and uniformly bounded throughout the neighbourhood. This also gives uniform Gaussian polynomial moments and density envelopes.*

*Proof.* For a positive definite $8 \times 8$ matrix, $\lambda_{\min} \geq \det / \lambda_{\max}^7$, and the row-sum bound gives $\lambda_{\max} \leq 120$; with (S5.17) this is (S5.19). Each entry moves by at most $729 \cdot 10^{-110}$ on the jet ball, which gives (S5.20). Since (S5.20) is far below (S5.19), Weyl's inequality keeps the covariance uniformly positive definite, and the remaining assertions follow from the analytic dependence of Gaussian regression on the covariance. □

S5.4.2. *The two-dimensional triple-contact elimination.*

**Lemma S5.15** (exact triple-contact elimination)**.** *Let $c, \rho \neq 0$ with $c^2+\rho^2 = 1$, and let $\eta > 0$, $\kappa > 0$ and $q_M, q_S \in \mathbb{R}$. There is a unique real cubic polynomial $p(X, Y)$ on $\mathbb{R}^2$ such that $M = (-\frac{1}{2}, 0)$, $S = (\frac{1}{2}, 0)$ and $y = (c/\eta, \rho/\eta)$ are critical points, $p(M) = 0$, $p(S) = -\kappa/6$, $p_{YY}(M) = q_M$ and $p_{YY}(S) = q_S$. Its Hessian determinants at the three critical points are*

$$-\det H_M = \frac{\kappa^2(4c^2-\eta^2)^2 + 4\kappa\rho^2\big[(12c^2+\eta^2)q_M + (4c^2-\eta^2)q_S\big] + 4\rho^4(q_M-q_S)^2}{64\,c^2\rho^2}, \tag{S5.21}$$

$$-\det H_S = \frac{\kappa^2(4c^2-\eta^2)^2 - 4\kappa\rho^2\big[(4c^2-\eta^2)q_M + (12c^2+\eta^2)q_S\big] + 4\rho^4(q_M-q_S)^2}{64\,c^2\rho^2}, \tag{S5.22}$$

$$\det H_y = \frac{N_y}{64\,c^2\eta\rho^2}, \tag{S5.23}$$

*where*

$$N_y = \eta\kappa^2(4c^2-\eta^2)^2 + 4\kappa\rho^2(4c^2+\eta^2)\big[\eta(q_M-q_S) - 2c(q_M+q_S)\big] \\ - 16c\rho^4(q_M^2-q_S^2) + 4\eta\rho^4(q_M-q_S)^2. \tag{S5.24}$$

*In particular, the endpoint determinants are exactly quadratic in $(\kappa, q_M, q_S)$, with the common $\kappa^2$ coefficient $(4c^2-\eta^2)^2/(64c^2\rho^2)$ and the common $\kappa$-free term $4\rho^4(q_M-q_S)^2/(64c^2\rho^2)$; the product $\det H_M \det H_S \det H_y$ has degree at most six in $\kappa$, with $\kappa^6$ coefficient $(4c^2-\eta^2)^6/(64c^2\rho^2)^3$, so its degree is exactly six off the collision strata $2c = \pm\eta$, on which it can vanish identically (for $c = \frac{1}{2}$, $\rho = \frac{\sqrt{3}}{2}$, $\eta = 1$ and $q_M = q_S = 0$ the cubic is $\kappa(X-1)(2X+1)^2/12$, and all three determinants vanish); the denominators vanish only on the strata $c = 0$ and $\rho = 0$ (and $\eta = 0$ for $\det H_y$); and on the collision strata $2c = \pm\eta$ the $\kappa^2$ coefficient $(4c^2-\eta^2)^2$ vanishes.*

*Proof.* The ten conditions (six station equations, two value equations and two curvature prescriptions) are linear in the ten coefficients of $p$. The determinant of this linear system is $-24c\rho^3/\eta^4 \neq 0$, so the solution is unique. Substituting it into the Hessians at $M$, $S$ and $y$ gives (S5.21)–(S5.24) by direct exact computation, and the stated structure is read off from these formulas. □

In the witness charts below, the plane is spanned by the pair axis and the direction of the third point, the chart is centred at the pair midpoint, lengths are measured in units of $r$ and values in units of $r^3$ (so that the pair sits at $(\mp\frac{1}{2},0)$ and the value gap is $\kappa/6$), $\eta = r/\operatorname{dist}(y,x)$, and $q_M, q_S$ are the in-plane transverse curvatures at the endpoints; in the application they are the $O(1)$ contact-law blocks. Physical two-dimensional Hessian determinants are therefore $r^2$ times the chart determinants (S5.21)–(S5.23). In the chart the endpoint determinants are generically of order $\kappa^2$ as $\eta \to 0$. Neither the collar count (Lemma S5.19) nor the singular-near envelope (Lemma S5.22) uses these expressions; both are proved from Taylor's formula.

S5.4.3. *The collar count.*

**Lemma S5.16** (exact collar scaled-gradient chart at the planar jet)**.** *Let a third point be*

$$y = x + r(Xe_1 + Ye_2 + Ze_3), \qquad \rho^2 = Y^2 + Z^2, \tag{S5.25}$$

*and define the physical scaled third gradient*

$$G_{\mathrm{col}} = (r^{-2}f_1(y),\, r^{-1}f_2(y),\, r^{-1}f_3(y)). \tag{S5.26}$$

*The pair pins make* (S5.26) *converge to* $(L_1, L_\perp)$*, where*

$$L_1 = \tfrac{1}{2}(X^2 - \tfrac{1}{4})f_{111} + XYf_{112} + XZf_{113} + \tfrac{Y^2}{2}f_{122} + YZf_{123} + \tfrac{Z^2}{2}f_{133}, \tag{S5.27}$$

$$L_\perp = Q(Y,Z)^\top. \tag{S5.28}$$

*The axial and transverse blocks are conditionally independent. Their exact conditional means are*

$$\mathbb{E}[L_1 \mid P] = \kappa(X^2 - \tfrac{1}{4}), \qquad \mathbb{E}[L_\perp \mid P] = -b(Y,Z)^\top, \tag{S5.29}$$

*the exact conditional axial variance is*

$$\operatorname{Var}(L_1 \mid P) = \frac{\rho^2(4X^2+\rho^2)}{2}, \tag{S5.30}$$

*and the transverse covariance is*

$$C_\perp = \begin{pmatrix} 2Y^2+Z^2 & YZ \\ YZ & Y^2+2Z^2 \end{pmatrix}, \qquad \det C_\perp = 2\rho^4, \qquad (Y,Z)\,C_\perp^{-1}(Y,Z)^\top = \tfrac{1}{2}. \tag{S5.31}$$

*Therefore*

$$\det \operatorname{Cov}\big(\lim G_{\mathrm{col}} \,\big|\, P\big) = \rho^6(4X^2+\rho^2). \tag{S5.32}$$

*For $\rho > 0$ the exact limiting density at zero is*

$$\hat{p}_{\mathrm{col}}(X,Y,Z;b,\kappa) = (2\pi)^{-3/2}\big[\rho^6(4X^2+\rho^2)\big]^{-1/2}\exp\Big[-\frac{b^2}{4} - \frac{\kappa^2(X^2-\frac{1}{4})^2}{\rho^2(4X^2+\rho^2)}\Big], \tag{S5.33}$$

*and consequently*

$$p_{\nabla f(y)|\textit{pair pins}}(0) = r^{-4}\big[\hat{p}_{\mathrm{col}} + o(1)\big] \tag{S5.34}$$

*uniformly on compact subcharts away from the removable collision and axis strata. Near those strata the limit is not uniform, and the collar count uses the finite-scale bounds of Lemma S5.41. For the contact-limit density $\hat{p}_{\mathrm{col}}$: if $X$ stays away from the pair endpoints, the mean mismatch supplies exponential suppression at least $\exp(-c/\rho^2)$, and at $X = 0$ it supplies $\exp(-c/\rho^4)$; near $X = \pm\frac{1}{2}$, lack of mean suppression occurs only inside a collision cone $|X \mp \frac{1}{2}| = O(\rho)$.*

*Proof.* The conditional moments (S5.29)–(S5.31) are exact Gaussian regressions computed from $\operatorname{Cov}(\partial^\alpha f, \partial^\beta f) = (-1)^{|\beta|}\partial^{\alpha+\beta}K_0(0)$, and (S5.32) is the product of the axial variance (S5.30) and $\det C_\perp$. The factor $r^{-4}$ in (S5.34) is the Jacobian of the scaling (S5.26). The suppression statements are read off from the exponent in (S5.33). □

**Lemma S5.17** (Hessian collision zero)**.** *If $\nabla f(p) = \nabla f(q) = 0$ and $f$ has bounded third derivatives between $p$ and $q$, then*

$$\|H_p(q-p)\| \le C|q-p|^2, \qquad \|H_q(q-p)\| \le C|q-p|^2. \tag{S5.35}$$

*Hence each Hessian has a singular value $O(|q-p|)$, and its determinant has an additional factor $|q-p|$ times a polynomial bound in the remaining Hessian entries. The statement is dimension independent.*

*Proof.* Subtract $\nabla f(q) - \nabla f(p) = 0$ and use the integral formula $0 = \int_0^1 H_{p+s(q-p)}(q-p)\,ds$. Subtract $H_p(q-p)$ and use the Lipschitz continuity of $H$. The statement at $q$ is identical. □

**Lemma S5.18** (kernel-independent collar powers)**.** *Write a third point in the pair-scale collar as*

$$y = x + r(Xt + \rho v), \tag{S5.36}$$

*where $v$ is a unit transverse direction, $|X|$ is bounded on the collar chart and $\rho \ge 0$, and let $C_{\mathrm{col}}(\mathcal{K}; t, X, \rho, v)$ be the contact-limit conditional covariance of the scaled third gradient*

$$(r^{-2}\partial_t f(y),\ r^{-1}\partial_v f(y),\ r^{-1}\partial_w f(y)), \tag{S5.37}$$

*with $w$ the second transverse direction. For every $\mathcal{K}$ satisfying Hypothesis S5.11,*

$$\det C_{\mathrm{col}} = \rho^6(X^2+\rho^2)\, A_{\mathrm{col}}(\mathcal{K}; t, X, \rho, v), \tag{S5.38}$$

*where $A_{\mathrm{col}}$ extends continuously to $\rho = 0$ and to $(X,\rho) = (0,0)$ after the ordinary radial blow-up, and the extension is analytic in $J_6(\mathcal{K})$. At the planar jet,*

$$A_{\mathrm{col}}(K_0) = \frac{4X^2+\rho^2}{X^2+\rho^2}, \tag{S5.39}$$

*whose radial-blow-up minimum is one. For every $\mathcal{K}$ satisfying Hypothesis S5.11 there is $a_{\mathrm{col}}(\mathcal{K}) > 0$ such that*

$$A_{\mathrm{col}}(\mathcal{K}; t, X, \rho, v) \ge a_{\mathrm{col}}(\mathcal{K}) \qquad \text{for all } t,\ v \text{ and } (X,\rho) \ne (0,0). \tag{S5.40}$$

*Proof.* Take derivatives in the orthonormal frame $(t, v, w)$, so that $y = x + r(Xt + \rho v)$; in this frame the pins are the eight evaluations of (S1.7), up to an invertible linear map that does not depend on the kernel. Put

$$U_1 = f_{ttv}, \qquad U_2 = \tfrac{1}{2} f_{tvv}, \qquad W = f_{vvv},$$

all at $x$. By (S5.27)–(S5.28), read in this frame, the contact limit of the scaled gradient (S5.37) is

$$\Big(\tfrac{1}{2}(X^2 - \tfrac{1}{4}) f_{ttt} + \rho\,(XU_1 + \rho U_2),\ \ \rho f_{vv},\ \ \rho f_{vw}\Big).$$

This is an identity between Taylor coefficients, so it holds for every kernel. The term $f_{ttt}$ is a pin. Taking out the factor $\rho$ of each of the three entries gives

$$\det C_{\mathrm{col}} = \rho^6 D, \qquad D = \det \mathrm{Cov}(XU_1 + \rho U_2,\ f_{vv},\ f_{vw} \mid P).$$

Since $XU_1 + \rho U_2$ is linear in $(X,\rho)$, $D$ is a quadratic form in $(X,\rho)$, and its coefficients are continuous in $(t,v)$ and analytic in $J_6(\mathcal{K})$. Put $A_{\mathrm{col}} = D/(X^2+\rho^2)$. It is homogeneous of degree zero in $(X,\rho)$, so it is a continuous function of $(t,v)$ and of the direction of $(X,\rho)$; this is the radial blow-up, and (S5.38) holds.

For $(X,\rho) \ne (0,0)$ the polynomial symbols of the eleven variables $P$, $XU_1 + \rho U_2$, $f_{vv}$, $f_{vw}$ are linearly independent, up to constant factors. The pins have the eight distinct monomials of (S1.7). The variable $XU_1 + \rho U_2$ has the nonzero symbol $X\xi_t^2\xi_v + \frac{\rho}{2}\xi_t\xi_v^2$. The last two have the monomials $\xi_v^2$ and $\xi_v\xi_w$. None of these monomials occurs twice. By Hypothesis S5.11(2) (for $K_{24}$, Theorem S1.3; for $K_0$, Remark S5.12), the joint covariance of these variables is positive definite. Hence so is its Schur complement given $P$, and $D > 0$. Thus $A_{\mathrm{col}}$ is continuous and positive on a compact set: $t \in S^2$, $v$ a unit vector orthogonal to $t$, and the direction of $(X,\rho)$ with $\rho \ge 0$. Its

minimum $a_{\mathrm{col}}(\mathcal{K})$ is therefore positive, which is (S5.40). At the planar jet, (S5.32) gives (S5.39): there $\mathrm{Var}(U_1 \mid P) = 2$, $\mathrm{Var}(U_2 \mid P) = \frac{1}{2}$, $\mathrm{Cov}(U_1, U_2 \mid P) = 0$, and $f_{vv}$, $f_{vw}$ are conditionally independent of $U_1, U_2$ with $\det \mathrm{Cov}(f_{vv}, f_{vw} \mid P) = 2$, so $D = 4X^2 + \rho^2$. The symbolic check `checks/s5_factor_check.py` verifies these identities. □

Equation (S5.38) is (3.14) of the main text.

**Lemma S5.19** (collar count)**.** *For the side-24 field, uniformly for $t \in S^2$ and on compact positive mark sets,*

$$\mathbb{E}_r^{MS} N_{\mathrm{col}} \le C_{\mathrm{col}}\, r^3. \tag{S5.41}$$

*Proof.* By the Kac–Rice formula, $\mathbb{E}^0[W_r N_{\mathrm{col}}] = \int J(y)\, dy$ over the collar with $M$ and $S$ removed, where

$$J(y) = \mathbb{E}^0\left[W_r |\det H_y| \mid \nabla f(y) = 0\right] p_{\nabla f(y)|\mathrm{pins}}(0).$$

Let $\Psi \ge 1$ be a fixed polynomial in $\kappa$ and in the suprema over $B_{3\delta}(x)$ of the norms of the derivatives of $f$ of orders two and three.

*Deterministic envelopes.* Since $M$, $S$ and $y$ are critical points, Lemma S5.17 gives $\|H_p u\| \le C|p' - p|\Psi$ for each vertex $p$ of the triangle $MSy$, each other vertex $p'$ and $u = (p' - p)/|p' - p|$. For a $3 \times 3$ matrix $H$ and unit vectors $u_1, u_2$,

$$|\det H| \le \|Hu_1\|\, \|H\|^2, \qquad |\det H| \le \|Hu_1\|\, \|Hu_2\|\, \|H\| / \sin \angle(u_1, u_2).$$

With $W_r \le |\det H_M|\, |\det H_S|$ and Lemma S5.39(b), $J(y) \le C\, E(y)\, (1 + \mathcal{M})^q\, p_{\nabla f(y)|\mathrm{pins}}(0)$. Here $E(y)$ is the deterministic bound for $|\det H_M \det H_S \det H_y| / \Psi^9$ given below, $\mathcal{M}$ is the Mahalanobis norm of $\nabla f(y)$ at 0 given the pins, $q$ is fixed, and the pin term $\mathcal{M}_\Pi$ is bounded on compact marks (§S5.8). Fix $\rho_1 \in (0, \frac{1}{8}]$ and $\gamma_1 \in (0, 1)$, and then $r_0$, with $\rho_1^2 + r_0^2 \le \varepsilon_1^2$ and $\gamma_1^2 + r_0^2 \le \varepsilon_1^2$, where $\varepsilon_1$ is that of Lemmas S5.40 and S5.41, and with $r_0$ at most the $r_0$ of Lemma S5.41(a) for $\rho_1$ and of Lemma S5.41(c) for $\gamma_1$. In the frame adapted to $y - x$ write $y = x + r(Xt + \rho v)$, and split the collar into three parts.

*(i) $\rho \ge \rho_1$, $|y - M| \ge r/4$ and $|y - S| \ge r/4$.* The sines of the three angles of the triangle are at least $\rho/(C_1 + 1)^2$, so $|\det H_p| \le Cr^2\Psi^3/\rho_1$ at each vertex, and $E \le Cr^6$. By Lemma S5.41(a) and (S5.107), $(1 + \mathcal{M})^q p_{\nabla f(y)|\mathrm{pins}}(0) = r^{-4}(1 + \mathcal{M})^q p_{G|\mathrm{pins}}(0) \le Cr^{-4}$. Hence $J \le Cr^2$, and this part, of volume at most $Cr^3$, contributes at most $Cr^5$.

*(ii) $\rho < \rho_1$, $|y - M| \ge r/4$ and $|y - S| \ge r/4$.* The same angle bound, and one row of each Hessian, give

$$|\det H_M \det H_S \det H_y| \le C \min\{r^3,\ r^6\rho^{-3}\}\, \Psi^9. \tag{S5.42}$$

By Lemma S5.41(b), $r^4(1 + \mathcal{M})^q p_{\nabla f(y)|\mathrm{pins}}(0) \le C\varepsilon^{-2} e^{-c_*/\varepsilon^2}$, with $\varepsilon^2 = \rho^2 + r^2$. Where $\rho \ge r$, $\varepsilon^2 \le 2\rho^2$ and $J \le Cr^2\rho^{-5} e^{-c_*/(2\rho^2)}$; with $dy = r^3\, dX\, \rho\, d\rho\, d\phi$ over a bounded range of $X$, this contributes at most $Cr^5 \int_0^{\rho_1} \rho^{-4} e^{-c_*/(2\rho^2)} d\rho \le Cr^5$. Where $\rho < r$, $\varepsilon^2 \le 2r^2$ and $J \le Cr^{-3} e^{-c_*/(2r^2)}$ on a set of volume at most $Cr^5$, which contributes at most $Cr^2 e^{-c_*/(2r^2)} \le Cr^5$.

*(iii) $|y - p| < r/4$ for $p = M$ or $p = S$.* Take $p = M$; the case $p = S$ is the same. Write $y = M + r\tau\hat{y}$ with $\hat{y} = \cos\alpha\, t + \sin\alpha\, v$ and $0 < \tau < \frac{1}{4}$, so that $dy = r^3\tau^2\, d\tau\, \sin\alpha\, d\alpha\, d\phi$ and $p_{\nabla f(y)|\mathrm{pins}}(0) = r^{-4}\tau^{-3} p_{\tilde{G}|\mathrm{pins}}(0)$. Two soft directions at each vertex give the refined bound: at $M$, $t$ and $\hat{y}$, with $\|H_M t\| \le Cr\Psi$ and $\|H_M \hat{y}\| \le Cr\tau\Psi$; at $S$, $t$ and $\hat{y}$, with $\|H_S \hat{y}\| \le \|H_M \hat{y}\| + Cr\Psi \le Cr\Psi$; at $y$, $\hat{y}$ and $u = (S - y)/|S - y|$, with $\|H_y \hat{y}\| \le Cr\tau\Psi$, $\|H_y u\| \le Cr\Psi$ and $\sin\angle(\hat{y}, u) \ge \sin\alpha/(1 + \tau)$. One soft row at each vertex gives the crude bound. Hence $E \le C \min\{r^3\tau^2,\ r^6\tau^2 \sin^{-3}\alpha\}$. By Lemma S5.41(c), $(1 + \mathcal{M})^q p_{\tilde{G}|\mathrm{pins}}(0)$ is at most $C$ where $\sin\alpha \ge \gamma_1$, and at most $C\varepsilon'^{-2} e^{-c_*/\varepsilon'^2}$ where $\sin\alpha < \gamma_1$, with $\varepsilon'^2 = \sin^2\alpha + r^2$. Where $\sin\alpha \ge \gamma_1$, $J \le Cr^2\tau^{-1}$, which contributes at most $Cr^5 \int_0^{1/4} \tau\, d\tau$. Where $r \le \sin\alpha < \gamma_1$, $J \le Cr^2\tau^{-1} \sin^{-5}\alpha\, e^{-c_*/(2\sin^2\alpha)}$, which contributes

at most $Cr^5 \int_0^\pi \sin^{-4}\alpha\, e^{-c_*/(2\sin^2\alpha)} d\alpha \le Cr^5$. Where $\sin\alpha < r$, $J \le Cr^{-3}\tau^{-1}e^{-c_*/(2r^2)}$, which contributes at most $Cr^2 e^{-c_*/(2r^2)}$.

Summing, $\mathbb{E}^0[W_r N_{\mathrm{col}}] \le Cr^5$, and dividing by $Z_r \ge c_Z r^2$ from (S5.8) gives (S5.41). Every step uses only Taylor's formula and Lemmas S5.18, S5.39 and S5.41, so the bound holds for every kernel satisfying Hypothesis S5.11, with constants depending on the kernel; in particular for $K_{24}$ (Remark S5.12). The exact elimination of Lemma S5.15 is not used. □

S5.4.4. *The singular-near count.*

**Lemma S5.20** (singular-near corrected value–gradient frame at the planar jet)**.** *Let*

$$y = x + s\,\omega, \qquad \omega = (c, Y, Z), \qquad c^2 + \rho^2 = 1, \qquad \eta = r/s, \tag{S5.43}$$

*with $\rho^2 = Y^2 + Z^2$ and $\eta$ in a fixed compact interval $[0, \eta_0]$, $\eta_0 < 1$. Let the third value lie in the pair window,*

$$f(y) = b - \alpha\kappa r^3/6, \qquad \alpha \in [0, 1], \tag{S5.44}$$

*and use the corrected variables*

$$V_0 = s^{-3}\Big[f(y) - U^{(0)} - \frac{s}{2}\big(Y f_2(y) + Z f_3(y)\big) - \frac{sc}{3} f_1(y)\Big], \tag{S5.45}$$

$$V_1 = s^{-2} f_1(y), \qquad V_2 = s^{-1} f_2(y), \qquad V_3 = s^{-1} f_3(y). \tag{S5.46}$$

*where $U^{(0)} = \frac{1}{2}\big(f(M) + f(S)\big)$ is a combination of the pins, equal to $b - \kappa r^3/12$ under them. Given the pins, the transformation from $(f(y), \nabla f(y))$ to $(V_0, V_1, V_2, V_3)$ is affine, with determinant of magnitude $s^{-7}$. At contact, with $\eta$ fixed, the random residual limits are*

$$L_0 = -\tfrac{1}{12}\big[Y\, f_{2ij}\omega_i\omega_j + Z\, f_{3ij}\omega_i\omega_j\big] - \tfrac{\eta^2}{16}\big[Y f_{112} + Z f_{113}\big], \tag{S5.47}$$

$$L_1 = \tfrac{1}{2} f_{1ij}\omega_i\omega_j, \tag{S5.48}$$

$$(L_2, L_3) = Q(Y, Z)^\top, \tag{S5.49}$$

*with summation over repeated indices, and with the pinned term $\frac{1}{2}c^2 f_{111}$ removed from $L_1$. The other $\eta$-dependent terms are deterministic: $-\kappa\eta^2/4$ in $V_1$ and $-c\kappa\eta^2/6$ in $V_0$. The last term of $L_0$ is random. It arises because the pins force $f_k(x) = -\frac{r^2}{8} f_{k11}(x) + O(r^4)$ for $k = 2, 3$, and $V_0$ multiplies $f_2(x)$ and $f_3(x)$ by $Y/(2s^2)$ and $Z/(2s^2)$. The exact conditional covariance of $(L_0, L_1)$ is*

$$C_{00} = \frac{\rho^2\big[(c^2 + \frac{3}{4}\eta^2)^2 + 4c^2\rho^2 + 3\rho^4\big]}{72}, \qquad C_{01} = -\frac{c\rho^2(c^2 + \rho^2 + \frac{3}{4}\eta^2)}{6}, \qquad C_{11} = \frac{\rho^2(4c^2 + \rho^2)}{2}, \tag{S5.50}$$

*so that*

$$\det C_{(L_0, L_1)} = \frac{\rho^6 Q_\eta}{144}, \qquad Q_\eta = \big(3c^2 - \tfrac{3}{4}\eta^2\big)^2 + 12c^2\rho^2 + 3\rho^4. \tag{S5.51}$$

*Combining this with $\det C_\perp = 2\rho^4$ gives*

$$\det \mathrm{Cov}(L_0, L_1, L_2, L_3 \mid P) = \frac{\rho^{10} Q_\eta}{72}. \tag{S5.52}$$

*At $\eta = 0$, $Q_0 = 3(c^2 + \rho^2)(3c^2 + \rho^2)$, so on the unit sphere*

$$\det \mathrm{Cov}(L_0, L_1, L_2, L_3 \mid P)\big|_{\eta=0} = \frac{\rho^{10}(3c^2 + \rho^2)}{24}. \tag{S5.53}$$

*The conditional mean mismatch in $V_1$ is*

$$\kappa(c^2 - \eta^2/4). \tag{S5.54}$$

*Consequently, in the contact limit, the density $p_{\mathrm{corr}}$ of $V$ at $(v_0,0,0,0)$, with $v_0$ the value that the window (S5.44) gives to $V_0$, satisfies near the axis*

$$p_{\mathrm{corr}} \le C\rho^{-5}\exp(-c_*/\rho^2) \tag{S5.55}$$

*for a constant $c_*>0$. This limit is not uniform near the axis at a fixed positive scale. At a fixed positive scale, for every kernel satisfying Hypothesis S5.11, with $\varepsilon^2=\rho^2+s^2$, the Mahalanobis norm $\mathcal{M}(u)$ of Lemma S5.40, every $q\ge 0$, and $\rho_0,s_0\in(0,\frac12]$ with $\rho_0^2+s_0^2\le\varepsilon_1^2$ (Lemma S5.40; proved in §S5.8),*

$$(1+\mathcal{M}(u))^q\, p_{(f(y),\nabla f(y))|\textit{pair pins}}(u,0) \le C_q\, s^{-7}\times\begin{cases}\varepsilon^{-5}\exp(-c_*/\varepsilon^2), & \rho<\rho_0 \text{ and } s<s_0,\\ 1, & \text{otherwise.}\end{cases} \tag{S5.56}$$

*Proof.* Given the pins, $U^{(0)}$ is fixed, so the transformation (S5.45)–(S5.46) is affine and triangular, with diagonal scalings $s^{-3},s^{-2},s^{-1},s^{-1}$, whence the determinant $s^{-7}$. The residual limits follow from Taylor's formula at $x$ and the pins, with $s=r/\eta$; `checks/s5_factor_check.py` verifies them, including the last term of $L_0$, for a polynomial field. That script subtracts $f(x)$, but the pins do not fix $f(x)$. Under the pins $f(x)-U^{(0)}=\frac{r^4}{384}f_{tttt}(x)+O(r^6)$ (`checks/s5_finite_scale_check.py`), which changes $V_0$ by $O(\eta^3 r)$, so the limits are the same. The covariances (S5.50)–(S5.52) are exact Gaussian regressions computed from $\mathrm{Cov}(\partial^\alpha f,\partial^\beta f)=(-1)^{|\beta|}\partial^{\alpha+\beta}K_0(0)$, and (S5.53) uses $c^2+\rho^2=1$. Near the pair axis, $|c|$ tends to one and $\eta\le\eta_0<1$, so (S5.54) is bounded away from zero. For any positive definite covariance, the Mahalanobis penalty is at least the squared mismatch of one coordinate divided by its variance (Lemma S5.39(a)). With the $V_1$ coordinate and $C_{11}\le 2\rho^2$ on the unit sphere, the penalty is at least $c_0\kappa^2/\rho^2$. Also $Q_\eta\ge(3c^2-\frac34\eta^2)^2\ge(\frac94-3\rho^2)^2$ for $\rho^2<\frac34$, which gives (S5.55). Away from the axis, $Q_\eta\ge 3\rho^4$ gives a positive compact eigenfloor, uniformly in $\eta\in[0,\eta_0]$. These are statements about the contact limit. The finite-scale bound (S5.56) is Lemma S5.40(d),(e); the proof of (d) and (e) uses from the present one only the window (S5.44), the chart (S5.45)–(S5.46) with its Jacobian, the residual limits (S5.47)–(S5.49) and, through the determinant bound of Lemma S5.21, the covariances (S5.50)–(S5.52). The proof of (c) cites, as agreement, the deterministic terms $-\kappa\eta^2/4$ in $V_1$ and $-c\kappa\eta^2/6$ in $V_0$. The means paragraph of Lemma S5.21 also uses the contact-limit Mahalanobis argument above, including (S5.54); it is not an input of (d) or (e). All of these are stated or proved above, so there is no circularity. An integral of $\rho^{-m}$ times the density over $S^2$ would not serve in its place: at a fixed positive scale that integral diverges for $m\ge 2$, because the density is positive on the axis. □

The bound (S5.56) is the density half of the singular-near witness bound; the determinant half is Lemma S5.22.

**Lemma S5.21** (kernel-independent extra $\rho^2$ cancellation)**.** *Write $y=x+s\omega$ with $\omega=(c,\rho v)$, $c^2+\rho^2=1$ and $\eta=r/s\in[0,\eta_0]$, $\eta_0<1$. Use the corrected variables $V_0,\dots,V_3$ of Lemma S5.20, and let $C_{\mathrm{sing}}$ denote their contact-limit conditional covariance. For every $\mathcal{K}$ satisfying Hypothesis S5.11,*

$$\det C_{\mathrm{sing}} = \rho^{10}(3c^2+\rho^2)\,A_{\mathrm{sing}}(\mathcal{K};t,c,\rho,v,\eta), \tag{S5.57}$$

*where $A_{\mathrm{sing}}$ extends continuously to $\rho=0$ and is analytic in $J_6(\mathcal{K})$. At the planar jet,*

$$A_{\mathrm{sing}}(K_0)=\frac{Q_\eta}{72(3c^2+\rho^2)}, \qquad \textit{which is } \frac{1}{24} \textit{ at } \eta=0, \tag{S5.58}$$

*with $Q_\eta$ as in (S5.51). For every $\mathcal{K}$ satisfying Hypothesis S5.11 there is $a_{\mathrm{sing}}(\mathcal{K})>0$ such that*

$$A_{\mathrm{sing}}(\mathcal{K};t,c,\rho,v,\eta)\ge a_{\mathrm{sing}}(\mathcal{K}) \qquad \textit{for all } t,\ v,\ c^2+\rho^2=1 \textit{ and } \eta\in[0,\eta_0]. \tag{S5.59}$$

*Thus, in the contact limit, the $\rho^{-5}$ density factor and the $\exp(-c_*/\rho^2)$ axis suppression persist for small $\rho$, with changed constants; at a fixed positive scale near the axis, Lemma S5.40 replaces*

*them. The conditional mean mismatch in $V_1$ is $\kappa(c^2-\eta^2/4)+O(\rho\kappa)$, which is separated from zero in the axis chart because $\eta_0<1$.*

*Proof.* Take derivatives in the frame $(t,v,w)$, so that $\omega=ct+\rho v$, and let $U_1$, $U_2$ and $W$ be as in the proof of Lemma S5.18. By (S5.47)–(S5.49), read in this frame, the random residuals are, for every kernel,

$$L_0=-\tfrac{\rho}{12}\big((c^2+\tfrac34\eta^2)U_1+4c\rho\,U_2+\rho^2W\big),\qquad L_1=\rho\,(cU_1+\rho U_2),\qquad (L_2,L_3)=\rho\,(f_{vv},f_{vw}). \tag{S5.60}$$

Thus $(L_0,L_1)=(-\frac{\rho}{12}\tilde L_0,\ \rho\tilde L_1)$ with $(\tilde L_0,\tilde L_1)^\top=A\,(U_1,U_2,W)^\top$ and

$$A=\begin{pmatrix} c^2+\frac34\eta^2 & 4c\rho & \rho^2\\ c & \rho & 0\end{pmatrix}.$$

Every $2\times2$ minor of $A$ is divisible by $\rho$: the minors are $\rho\,\tilde m$ with $\tilde m=(\frac34\eta^2-3c^2,\ -c\rho,\ -\rho^2)$. Let $\Sigma=\mathrm{Cov}(U_1,U_2,W\mid P)$. By the Cauchy–Binet formula,

$$\det\mathrm{Cov}(\tilde L_0,\tilde L_1\mid P)=\det(A\Sigma A^\top)=\rho^2Q_\eta(\mathcal K),\qquad Q_\eta(\mathcal K)=\sum_{S,T}\tilde m_S\,\tilde m_T\det\Sigma_{S,T},$$

the sum running over pairs of two-element column sets. For an even kernel, odd and even derivatives are uncorrelated, so $(L_0,L_1)$ and $(L_2,L_3)$ are conditionally independent given $P$. Hence

$$\det C_{\mathrm{sing}}=\frac{\rho^{10}}{144}\,Q_\eta(\mathcal K)\,\det\mathrm{Cov}(f_{vv},f_{vw}\mid P).$$

The nominal power $\rho^8$ is thus $\rho^{10}$; this is the cancellation, and it holds for every even kernel. Put $A_{\mathrm{sing}}=Q_\eta(\mathcal K)\det\mathrm{Cov}(f_{vv},f_{vw}\mid P)/(144(3c^2+\rho^2))$. It is continuous in $(t,v,c,\rho,\eta)$ and analytic in $J_6(\mathcal K)$, and (S5.57) holds.

The symbols of $P$, $U_1$, $U_2$, $W$, $f_{vv}$ and $f_{vw}$ are distinct monomials. As in the proof of Lemma S5.18, $\Sigma$ and $\mathrm{Cov}(f_{vv},f_{vw}\mid P)$ are therefore positive definite. Since $A\Sigma A^\top\succcurlyeq\lambda_{\min}(\Sigma)AA^\top$ and $\det(AA^\top)=\rho^2|\tilde m|^2$, we get $Q_\eta(\mathcal K)\ge\lambda_{\min}(\Sigma)^2|\tilde m|^2$. Moreover $\tilde m\neq0$: if $\rho=0$ then $c^2=1$, and $\frac34\eta^2-3\neq0$ because $\eta<1$. So $A_{\mathrm{sing}}$ is continuous and positive on the compact set given by $t\in S^2$, $v$ a unit vector orthogonal to $t$, $c^2+\rho^2=1$, $\rho\ge0$ and $\eta\in[0,\eta_0]$. Its minimum $a_{\mathrm{sing}}(\mathcal K)$ is positive, which is (S5.59). At the planar jet $\Sigma=\mathrm{diag}(2,\frac12,6)$ and $\det\mathrm{Cov}(f_{vv},f_{vw}\mid P)=2$, so $Q_\eta(K_0)$ is the $Q_\eta$ of (S5.51), and (S5.52) gives (S5.58). The check `checks/s5_factor_check.py` verifies these identities.

The means. The random parts in (S5.60) are odd, and the only odd pin with a nonzero target is $f_{ttt}$. So the conditional mean of $V_1$ differs from its planar value $\kappa(c^2-\eta^2/4)$ by the difference of two regressions of $\rho(cU_1+\rho U_2)$ on the pins, which is $O(\rho\kappa)$ on compact positive mark sets. For $\rho\le\rho_0(\mathcal K)$ the mismatch is therefore separated from zero. As in the proof of Lemma S5.20, the Mahalanobis penalty is at least the squared $V_1$ mismatch over $C_{11}=\mathrm{Var}(L_1\mid P)\le C\rho^2$, hence at least $c_*\kappa^2/\rho^2$. □

Equation (S5.57) is (3.15) of the main text.

**Lemma S5.22** (anisotropic triple-determinant envelope)**.** *Let $y=x+s\omega$ be a critical point of $f$ in the singular-near region, $C_1r<s\le\delta$, with $\omega=ct+\rho n$, where $n\perp t$ is a unit vector, $c^2+\rho^2=1$ and $\rho>0$, and let $f(y)$ lie in the pair window (S5.44). Let $\Psi\ge1$ be a fixed polynomial in $\kappa$ and in the suprema over the ball $B_{3\delta}(x)$ of the norms of the derivatives of $f$ of orders two to five. Under the pair pins,*

$$|\det H_M|+|\det H_S|\le C\rho^{-2}\,r(r+s^2)\,\Psi, \tag{S5.61}$$

$$|\det H_y|\le C\rho^{-2}\,(r^2+s^3)\,\Psi, \tag{S5.62}$$

*where $C$ depends only on $C_1$. Without the factor $\rho^{-2}$, $|\det H_M| + |\det H_S| \le Cr\Psi$ and $|\det H_y| \le Cs\Psi$, and these crude bounds hold also for $\rho = 0$. Consequently, with $\mathcal{M}(u)$ the Mahalanobis norm of Lemma S5.40, a fixed $q$, and the second entry of the minimum read as $+\infty$ when $\rho = 0$,*

$$\mathbb{E}[W_r |\det H_y| \mid \text{pins},\ f(y) = u,\ \nabla f(y) = 0] \le C(1 + \mathcal{M}(u))^q \min\{r^2 s,\ \rho^{-6}[r(r+s^2)]^2(r^2+s^3)\}, \tag{S5.63}$$

*uniformly on compact marks and over the window. The same bounds hold for every kernel satisfying Hypothesis S5.11, in particular for $K_{24}$, with constants depending on the kernel.*

*Proof.* All estimates are deterministic consequences of Taylor's formula on $B_{3\delta}(x)$, which contains every segment used below. We write $O(X)$ for a quantity bounded by $C\Psi X$. Since $C_1 > 1$ and $r \le \delta$, the distances from $y$ to $M$ and to $S$ lie in $[s/2, 2s]$, and $r < s$; these inequalities absorb the lower-order terms below. The elimination of Lemma S5.15 is not used.

*Expansions.* For a critical point $p$ of $f$ put $A = H_p$, $T = D^3 f(p)$, $\Theta = D^4 f(p)$ and $F_p(v) = f(p+v) - f(p) - \frac{1}{2}\nabla f(p+v)\cdot v$. For $|v| \le 2s$,

$$\nabla f(p+v) = Av + \tfrac{1}{2}T[v,v] + \tfrac{1}{6}\Theta[v,v,v] + R_p(v), \tag{S5.64}$$

$$F_p(v) = -\tfrac{1}{12}T[v,v,v] - \tfrac{1}{24}\Theta[v,v,v,v] + R'_p(v), \tag{S5.65}$$

with $|R_p(v)| + |v|\,|DR_p(v)| \le C\Psi|v|^4$ and $|R'_p(v)| + |v|\,|DR'_p(v)| \le C\Psi|v|^5$; the coefficients of $F_p$ are $\frac{1}{k!}(1 - \frac{k}{2})$, which vanish for $k = 2$. If $p + v$ is also a critical point, the left side of (S5.64) vanishes and $F_p(v) = f(p+v) - f(p)$.

*The endpoint $M$.* Take $p = M$. Then $S = M + rt$ and $y = M + w$ with $w = \frac{r}{2}t + s\omega$ and $|w| \in [s/2, 2s]$; put $\hat{w} = w/|w|$. The gradient equation at $S$ gives $At = -\frac{r}{2}T[t,t] + O(r^2)$, so $A[t,e] = O(r)$ for every unit vector $e$. The gradient equation at $y$ gives $Aw = -\frac{1}{2}T[w,w] + O(s^3)$, so $A[\hat{w}, e] = O(s)$. Dotting it with $w$ and eliminating $T[w,w,w]$ with (S5.65) gives

$$A[w,w] = 6F_M(w) + \tfrac{1}{12}\Theta[w,w,w,w] + O(s^5),$$

and the window gives $F_M(w) = f(y) - f(M) \in [-\kappa r^3/6, 0]$; hence $A[\hat{w},\hat{w}] = O(r^3 s^{-2} + s^2)$. Let $e_h$ be a unit vector orthogonal to $t$ and $\omega$. In the basis $(t, \hat{w}, e_h)$ the entries of $A$ are $O(r)$ in the first row and column, $A[\hat{w},\hat{w}] = O(r^3 s^{-2} + s^2)$, $A[\hat{w}, e_h] = O(s)$ and $A[e_h, e_h] = O(1)$. Expanding along the first row, every term of $\det(A[u_i, u_j])$ is $O(r(r+s^2))$. That determinant equals $\det H_M$ times the Gram determinant $\sin^2\angle(t,\hat{w})$, and $\sin\angle(t,\hat{w}) = s\rho/|w| \ge \rho/2$. This gives (S5.61) at $M$. The endpoint $S$ is the same with $t$ replaced by $-t$, using the window in the form $f(y) - f(S) \in [0, \kappa r^3/6]$.

*The third point.* Take $p = y$, $a = M - y$ and $b = S - y$, so that $b - a = rt$ and $\frac{1}{2}(a+b) = x - y = -s\omega$. Subtracting and adding the gradient equations at $a$ and $b$, with $|R_y(b) - R_y(a)| \le C\Psi r s^3$, gives

$$At = s\,T[\omega,t] - \tfrac{s^2}{2}\Theta[\omega,\omega,t] + O(r^2 + s^3), \tag{S5.66}$$

$$A\omega = \tfrac{s}{2}T[\omega,\omega] + \tfrac{r^2}{8s}T[t,t] - \tfrac{s^2}{6}\Theta[\omega,\omega,\omega] + O(r^2 + s^3). \tag{S5.67}$$

The pins give $F_y(a) - F_y(b) = f(M) - f(S) = \kappa r^3/6$, and the window gives $0 \le F_y(a) \le \kappa r^3/6$. Expanding (S5.65) at $a$ and $b$, with $|R'_y(a) - R'_y(b)| \le C\Psi r s^4$, gives

$$T[\omega,\omega,t] = \tfrac{2}{3}s\,\Theta[\omega,\omega,\omega,t] + O(r^2 s^{-2} + s^2), \tag{S5.68}$$

$$T[\omega,\omega,\omega] = \tfrac{1}{2}s\,\Theta[\omega,\omega,\omega,\omega] + O(r^2 s^{-2} + s^2); \tag{S5.69}$$

in (S5.69), (S5.68) cancels the term $\frac{1}{12}s^3 r\,\Theta[\omega,\omega,\omega,t]$. Inserting these into (S5.66)–(S5.67),

$$A[t,t] = O(s), \qquad A[\omega,t] = \tfrac{1}{6}s^2\Theta[\omega,\omega,\omega,t] + O(r^2 s^{-1} + s^3),$$
$$A[\omega,\omega] = \tfrac{1}{12}s^2\Theta[\omega,\omega,\omega,\omega] + O(r^2 s^{-1} + s^3),$$

so $A[t,t]A[\omega,\omega]-A[\omega,t]^2=O(r^2+s^3)$; by (S5.66) alone, $At=O(s)$, which gives the crude bound $|\det H_y|\le Cs\Psi$, and at $M$ and $S$ the rows along $t$ are $O(r)$, which gives $|\det H_M|+|\det H_S|\le Cr\Psi$. The entries $A[t,e_h]$ and $A[\omega,e_h]$ are $O(s)$ by (S5.66)–(S5.67), and $A[e_h,e_h]=O(1)$. Expanding the $3\times 3$ determinant in the basis $(t,\omega,e_h)$ along the last row, every term is $O(r^2+s^3)$. The Gram determinant of this basis is $\rho^2$, which gives (S5.62).

*Conditional moments.* Under the conditioning on the pins and on $(f(y),\nabla f(y))=(u,0)$, equivalently on $V=v^*(u)$, Lemma S5.39(b) bounds every conditional moment of a power of $\Psi$ by $C(1+\mathcal{M}_\Pi+\mathcal{M}(u))^q$, and $\mathcal{M}_\Pi$ is bounded on compact marks (§S5.8). Multiplying the three envelopes, with $W_r\le|\det H_M||\det H_S|$, gives (S5.63). A bound of these moments by a power of $1+\rho^{-1}$, through a lower bound $\rho^{10}$ for the third-point block, would not serve here: that lower bound holds only in the contact limit.

The argument uses only Taylor's formula and Lemma S5.39(b). So it holds for every kernel satisfying Hypothesis S5.11, with constants depending on the kernel. □

The envelopes (S5.61)–(S5.62) give (3.16) of the main text. In the proof of Lemma S5.23, the factor $(1+\mathcal{M}(u))^q$ in (S5.63) is absorbed by the density (Lemma S5.40(d),(e)). The factor $\rho^{-6}$ is at most $\rho_0^{-6}$ where $\rho\ge\rho_0$, it is absorbed by the axis suppression of Lemma S5.40(d) where $s\le\rho<\rho_0$, and the crude bounds are used where $\rho<s$. The symbolic check `checks/s5_envelope_check.py` confirms the cancellations behind (S5.61) and (S5.62) for polynomial fields of degree seven: in each of the four determinants (the in-plane part and the hard-direction part, at an endpoint and at the third point), every monomial in $r$ and $s$ is within the stated envelope.

**Lemma S5.23** (singular-near count)**.** *For the side-24 field, uniformly for $t\in S^2$ and on compact positive mark sets,*

$$\mathbb{E}_r^{MS}N_{\mathrm{sing}}\le C_{\mathrm{sing}}\,r^3. \tag{S5.70}$$

*Proof.* By the Kac–Rice formula, $\mathbb{E}^0[W_rN_{\mathrm{sing}}]$ is the integral over $y=x+s\omega$ in region (ii) and over $u$ in the window of

$$J(y,u)=\mathbb{E}^0\big[W_r|\det H_y|\;\big|\;f(y)=u,\ \nabla f(y)=0\big]\,p_{(f(y),\nabla f(y))|\mathrm{pins}}(u,0).$$

By (S5.63) and (S5.56), $J\le C\,E(s,\rho)\,s^{-7}D$, where $E(s,\rho)=\min\{r^2s,\ \rho^{-6}[r(r+s^2)]^2(r^2+s^3)\}$, $D=\varepsilon^{-5}e^{-c_*/\varepsilon^2}$ if $\rho<\rho_0$ and $s<s_0$, and $D=1$ otherwise; here $\varepsilon^2=\rho^2+s^2$, $\rho_0,s_0\le\frac12$ are chosen with $\rho_0^2+s_0^2\le\varepsilon_1^2$, and $r$ is at most the $r_0$ of Lemma S5.40(e) for these $\rho_0,s_0$. The window has width $\kappa r^3/6$, the volume element is $s^2\,ds\,d\omega$, and near the axis $d\omega\le 2\rho\,d\rho\,d\phi$. We bound the parts of region (ii) in turn.

*$s\ge s_0$.* Here $J\le Cr^2s\cdot s^{-7}\le C(s_0)r^2$, so this part contributes at most $Cr^5$ to $\mathbb{E}^0[W_rN_{\mathrm{sing}}]$.

*$s<s_0$, and $\rho\ge\rho_0$ or $s\le\rho<\rho_0$.* If $\rho\ge\rho_0$, then $J\le C\rho_0^{-6}s^{-7}[r(r+s^2)]^2(r^2+s^3)$. If $s\le\rho<\rho_0$, then $\varepsilon^2\le 2\rho^2$, $J\le C\rho^{-11}e^{-c_*/(2\rho^2)}s^{-7}[r(r+s^2)]^2(r^2+s^3)$, and $\int_{\rho<\rho_0}\rho^{-11}e^{-c_*/(2\rho^2)}\,d\omega\le C$. Integrating over $\omega$ and over the window, and dividing by $Z_r\ge c_Zr^2$ from (S5.8), these parts contribute to $\mathbb{E}_r^{MS}N_{\mathrm{sing}}$ at most

$$C\,r\int_{C_1r}^{\delta}s^{-5}\,[r(r+s^2)]^2(r^2+s^3)\,ds. \tag{S5.71}$$

*$s<s_0$ and $\rho<\min(s,\rho_0)$, the boundary layer at the axis.* Here $s\le\varepsilon\le\sqrt{2}\,s$, and the crude part of (S5.63) gives $J\le Cr^2s\cdot s^{-7}\cdot s^{-5}e^{-c_*/(2s^2)}$, on a set of directions of measure at most $Cs^2$. This part contributes at most $Cr^5\int_0^{s_0}s^{-7}e^{-c_*/(2s^2)}\,ds\le Cr^5$ to $\mathbb{E}^0[W_rN_{\mathrm{sing}}]$.

Expanding the polynomial in (S5.71) produces the six terms

$$r^7s^{-5},\qquad r^6s^{-3},\qquad r^5s^{-2},\qquad r^5s^{-1},\qquad r^4,\qquad r^3s^2, \tag{S5.72}$$

up to fixed numerical factors. Integration from $C_1r$ to $\delta$ bounds them respectively by $O(r^3)$, $O(r^4)$, $O(r^4)$, $O(r^5\log(1/r))$, $O(r^4)$ and $O(r^3)$. The other two parts contribute $O(r^5)/Z_r=O(r^3)$. This

proves (S5.70). Every step holds for every kernel satisfying Hypothesis S5.11 (Lemmas S5.22 and S5.40), with constants depending on the kernel, in particular for $K_{24}$. □

S5.4.5. *The fixed-distance count.*

**Lemma S5.24** (conditional density block)**.** *Let*

$$\mathcal{A}_r = \big(P_r^{\mathrm{corr}}, f(y), \nabla f(y)\big), \qquad \operatorname{dist}(y,x) > \delta, \tag{S5.73}$$

*where $P_r^{\mathrm{corr}}$ is the corrected pair pin vector; then $\operatorname{dist}(y, \{M,S\}) \geq \delta/2$. For the side-24 field, complete positive Fourier support and derivative-distribution independence give, uniformly on the compact separated configuration set,*

$$c_\delta I \leq \Sigma_{\mathcal{A}\mathcal{A}} \leq C_\delta I. \tag{S5.74}$$

*Let $L_r(b,\kappa)$ be the corrected pair-pin target and let the remote critical value be*

$$u = b - \alpha\kappa r^3/6, \qquad \alpha \in [0,1]. \tag{S5.75}$$

*The full density target is*

$$a_r(b,\kappa,\alpha) = \big(L_r(b,\kappa), u, 0\big). \tag{S5.76}$$

*On compact positive mark sets, the conditional density of the remote value and gradient given the pins satisfies*

$$p_{(f(y),\nabla f(y))|\,pair\ pins}(u,0) = \frac{p_{\mathcal{A}}(a_r)}{p_{P_r^{\mathrm{corr}}}(L_r(b,\kappa))} \leq C. \tag{S5.77}$$

*Proof.* The contact-limit corrected pair frame consists of distinct derivative coordinates at $x$, and $(f(y), \nabla f(y))$ of derivative coordinates at $y$, so the associated distributions are linearly independent; full-spectrum Gram positivity (Section S1) and compactness of the separated configuration set give (S5.74). The pins $\Pi_{r,x,t}$ and $P_r^{\mathrm{corr}}$ are related by an invertible linear map, whose Jacobian cancels between numerator and denominator, so the conditional density is the ratio in (S5.77). By (S5.74) the numerator is at most $(2\pi)^{-6} c_\delta^{-6}$. On compact marks the target $L_r(b,\kappa)$ is bounded, and the covariance of $P_r^{\mathrm{corr}}$ has uniform eigenvalue bounds (Lemma S5.14), so the denominator is bounded below. □

**Lemma S5.25** (fixed-distance count)**.** *Let $\mathcal{Y}_\delta$ be the compact side-24 configuration region with $\operatorname{dist}(y,x) > \delta$, and let $H_y$ be the Hessian at $y$. On compact positive mark sets,*

$$\mathbb{E}\big[|\det H_M \det H_S \det H_y| \;\big|\; \mathcal{A}_r = a_r\big] \leq C r^2, \tag{S5.78}$$

*and, uniformly for $t \in S^2$,*

$$\mathbb{E}_r^{MS} N_{\mathrm{far}} \leq C_{\mathrm{far}}\, r^3. \tag{S5.79}$$

*Proof.* Condition on $\mathcal{A}_r = a_r(b,\kappa,\alpha)$. All raw cross-covariances with the scaled endpoint variables of Lemma S5.9 and with $H_y$ are bounded, and only $\Sigma_{\mathcal{A}\mathcal{A}}$ is inverted. Gaussian regression therefore gives a conditional mean bounded by $C(1+|b|+\kappa)$ and a bounded positive-semidefinite conditional covariance, even if the enlarged Hessian block is degenerate. Since the product of three $3\times 3$ determinants has total degree nine, (S5.11) and Lemma S5.10 give (S5.78) on compact marks.

By the Kac–Rice formula, $\mathbb{E}^0[W_r N_{\mathrm{far}}]$ is the integral over $y \in \mathcal{Y}_\delta$ and over $u \in [b - \kappa r^3/6, b]$ of the conditional expectation in (S5.78) times the conditional density (S5.77). Hence

$$\mathbb{E}^0[W_r N_{\mathrm{far}}] \leq C\, r^2 \cdot \frac{\kappa r^3}{6} \cdot |\mathcal{Y}_\delta| \leq C r^5. \tag{S5.80}$$

Dividing explicitly by $Z_r = \mathbb{E}^0[W_r] \geq c_Z r^2$ from (S5.8) gives (S5.79). □

The bound (S5.79) is (3.17) of the main text. The count is conditional on the pins, so the argument uses the lower bound on the pin density in (S5.77); it gives no Gaussian factor in $(b,\kappa)$, and it is stated only on compact marks, which is all that Theorem S5.36 needs.

S5.5. **Capture and escape.**
Normalized potential. On a fixed compact convex chart $D^+$ consider, for $X \in \mathbb{R}$ and $Y \in \mathbb{R}^2$,

$$P(X,Y) = p_0(X) + \tfrac{1}{2}(X^2 - \tfrac{1}{4})a \cdot Y - \tfrac{1}{2}Y^\top TY + \tfrac{1}{2}X\,Y^\top WY + \tfrac{1}{6}C[Y,Y,Y] + r\,Q(X,Y), \quad \text{(S5.81)}$$

where $p_0(X) = X^3/3 - X/4 - 1/12$, $T \succ 0$ is symmetric, $W$ is symmetric, $C$ is a symmetric cubic tensor, and $Q(M) = Q(S) = 0$, $\nabla Q(M) = \nabla Q(S) = 0$ at $M = (-\frac{1}{2}, 0)$ and $S = (\frac{1}{2}, 0)$; note $P(M) = p_0(-\frac{1}{2}) = 0$. Put

$$R = 1 + |a| + \|W\| + \|C\| + \|Q\|_{C^2(D^+)} \geq 1, \qquad \tau = \lambda_{\min}(T). \quad \text{(S5.82)}$$

For the conditioned side-24 field, $T = -Q_\perp/(\kappa r)$ (see (S5.95) below; the factor $1/r$ is essential). Fix

$$K = 4096, \qquad \varepsilon_0 = \frac{1}{1024K}, \qquad \text{and assume} \qquad rR^5 \leq \varepsilon_0, \qquad \tau \geq 2KR^5 + 2R. \quad \text{(S5.83)}$$

Write $E(Y) = Y^\top TY$ and $q(Y) = |TY|$; then $q^2 \geq \tau E$ and $|Y|^2 \leq E/\tau$. The gradient of $P$ is

$$\begin{aligned} F_X &= X^2 - \tfrac{1}{4} + Xa \cdot Y + \tfrac{1}{2}Y^\top WY + rQ_X, \\ F_Y &= \tfrac{1}{2}(X^2 - \tfrac{1}{4})a - TY + XWY + \tfrac{1}{2}C[Y,Y,\cdot\,] + r\nabla_Y Q, \end{aligned} \quad \text{(S5.84)}$$

and the half-branches below are trajectories of the gradient-ascent flow $(\dot X, \dot Y) = (F_X, F_Y)$.

**Lemma S5.26** (endpoint energy slope)**.** *The Jacobian at $S$ has block form $J_S = \begin{pmatrix} A & B^\top \\ B & -T + E_D \end{pmatrix}$ with $A \geq 3/4$, $|B| \leq R$ and $\|E_D\| \leq 3R/4$ (after one fixed small-r reduction), and exactly one positive eigenvalue. Its positive eigenvector $v = (v_X, v_Y)$ satisfies*

$$E(v_Y) \leq 4\,(R^2/\tau)\,|v_X|^2. \quad \text{(S5.85)}$$

*Proof.* From $J_S v = \lambda_+ v$, the transverse block gives $(\lambda_+ I + T - E_D)v_Y = Bv_X$. Taking the inner product with $v_Y$,

$$E(v_Y) = v_Y^\top T v_Y \leq |B|\,|v_X|\,|v_Y| + \|E_D\|\,|v_Y|^2 \leq R\,|v_X|\,|v_Y| + \tfrac{3R}{4}|v_Y|^2,$$

using $\lambda_+ > 0$ and $v_Y^\top E_D v_Y \leq \|E_D\|\,|v_Y|^2$. Young's inequality $R|v_X||v_Y| \leq \frac{R^2}{\tau}|v_X|^2 + \frac{\tau}{4}|v_Y|^2$ and the energy bound $|v_Y|^2 \leq E(v_Y)/\tau$ give

$$E(v_Y) \leq \frac{R^2}{\tau}|v_X|^2 + \Big(\frac{\tau}{4} + \frac{3R}{4}\Big)|v_Y|^2 \leq \frac{R^2}{\tau}|v_X|^2 + \Big(\frac{1}{4} + \frac{3}{8}\Big)E(v_Y),$$

where $\tau \geq 2R$, from (S5.83), gives $\frac{3R}{4} \cdot \frac{1}{\tau} \leq \frac{3}{8}$. Hence $\frac{3}{8}E(v_Y) \leq \frac{R^2}{\tau}|v_X|^2$, that is, $E(v_Y) \leq \frac{8}{3}(R^2/\tau)|v_X|^2 \leq 4(R^2/\tau)|v_X|^2$. The strict final inequality leaves a fixed normalized margin, which persists under the exact $C^1$ perturbation of Lemma S5.30. □

**Lemma S5.27** (outward energy cone)**.** *Let $\xi = X - \frac{1}{2}$, $\delta_R = 1/(512R^3)$ and*

$$\mathcal{C}_E = \{0 \leq \xi \leq \delta_R,\ E(Y) \leq 4(R^2/\tau)\xi^2\}.$$

*On $\mathcal{C}_E$: $q \geq 2R\xi$ on the nonzero boundary, $|Y| \leq \xi/K$, $F_X \geq 3\xi/4$, and $G_E = E(Y) - 4(R^2/\tau)\xi^2$ satisfies*

$$\tfrac{1}{2}\dot G_E \leq -\tfrac{2}{3}q^2 < 0 \qquad \text{on } \{G_E = 0\}. \quad \text{(S5.86)}$$

*Hence the outward half-branch stays in $\mathcal{C}_E$ until $\xi = \delta_R$.*

*Proof.* On $G_E = 0$ we have $E = 4(R^2/\tau)\xi^2$, so $q \geq \sqrt{\tau E} = 2R\xi$ and $|Y| \leq \sqrt{E/\tau} = 2R\xi/\tau \leq \xi/(KR^4)$. (The full strength $\tau \geq 2KR^5$ is used here, not merely $\tau \geq 2KR$; this matters for the next bound when $R$ is large.) Then $|Xa \cdot Y| \leq (\frac{1}{2} + \delta_R)R|Y| \leq (\frac{1}{2} + \delta_R)\xi/(KR^3) \leq \xi/16$, as $(\frac{1}{2} + \delta_R)/(KR^3) \leq 1/K \leq 1/16$; next $\frac{1}{2}|Y^\top WY| \leq \frac{R}{2}|Y|^2 \leq \frac{R}{2} \cdot \frac{\xi^2}{K^2R^8} \leq \frac{\xi}{16} \cdot \frac{8\xi}{K^2R^7} \leq \xi/16$, as

$8\xi \le 8\delta_R \le 1 \le K^2R^7$; and, since $\nabla Q(S) = 0$ and $\|Q\|_{C^2} \le R$, $|Q_X| \le R(\xi + |Y|) \le R(1 + 1/K)\xi$ on $\mathcal{C}_E$, so $|rQ_X| \le 2rR\xi \le \xi/16$, as $2rR \le 2\varepsilon_0 R^{-4} \le 2\varepsilon_0 < 1/16$. Since $X^2 - \frac{1}{4} = \xi(1+\xi) \ge \xi$, we get $F_X \ge \xi - 3\xi/16 \ge 3\xi/4$.

For the flux, $\frac{1}{2}\dot{G}_E = (TY)\cdot F_Y - 4(R^2/\tau)\xi F_X$, and the moving-boundary term is nonpositive because $F_X \ge 0$. In $(TY)\cdot F_Y$ the leading term is $-q^2$. We divide the four adverse terms by $q^2$, using $q \ge 2R\xi$, $|Y| \le 2R\xi/\tau$, $|X| \le \frac{1}{2} + \delta_R$, $X^2 - \frac{1}{4} = \xi(1+\xi)$, and, for the last term, that $\nabla Q$ vanishes at $S$ by (S5.81), so that $|\nabla_Y Q| \le R(\xi + |Y|)$ on $\mathcal{C}_E$ by the $C^2$ bound:

$$\begin{aligned}
\frac{|(TY)\cdot \frac{1}{2}(X^2 - \frac{1}{4})a|}{q^2} &\le \frac{\frac{1}{2}\xi(1+\xi)Rq}{q^2} = \frac{\xi(1+\xi)R}{2q} \le \frac{1+\xi}{4} \le \frac{1+\delta_R}{4} \le \frac{257}{1024},\\
\frac{|(TY)\cdot XWY|}{q^2} &\le \frac{|X|\,R\,|Y|}{q} \le \frac{(\frac{1}{2}+\delta_R)R\,(2R\xi/\tau)}{2R\xi} = \frac{(\frac{1}{2}+\delta_R)R}{\tau} \le \frac{257}{1024K},\\
\frac{|(TY)\cdot \frac{1}{2}C[Y,Y,\cdot]|}{q^2} &\le \frac{\frac{1}{2}R|Y|^2 q}{q^2} \le \frac{R(2R\xi/\tau)^2}{2\cdot 2R\xi} = \frac{R^2\xi}{\tau^2} \le \frac{\delta_R}{4K^2R^8} \le \frac{1}{2048K^2},\\
\frac{|(TY)\cdot r\nabla_Y Q|}{q^2} &\le \frac{rR(\xi+|Y|)}{q} \le \frac{rR\xi(1+2R/\tau)}{2R\xi} = \frac{r(1+2R/\tau)}{2} \le 2\varepsilon_0.
\end{aligned} \tag{S5.87}$$

Here the second line uses $q \ge 2R\xi$ and $|Y| \le 2R\xi/\tau$, the third uses $\tau \ge 2KR^5$ and $\xi \le \delta_R = 1/(512R^3)$, and the fourth uses $r \le \varepsilon_0/R^5 \le \varepsilon_0$ and $2R/\tau \le 1/(KR^4) \le 1$. The four brackets $\frac{257}{1024}$, $\frac{257}{1024K}$, $\frac{1}{2048K^2}$ and $2\varepsilon_0$ sum to $\frac{8625610753}{34359738368} < \frac{1}{3}$ (exact rational arithmetic). Hence $\frac{1}{2}\dot{G}_E \le -(1 - \frac{1}{3})q^2$. □

**Lemma S5.28** (outward energy strip and level bound)**.** *Let*

$$\mathcal{S}_E = \{\tfrac{1}{2} + \delta_R \le X \le \tfrac{5}{4},\ E(Y) \le E_*\}, \qquad E_* = \tfrac{9}{16}(R^2/\tau).$$

*On its lateral boundary, $q \ge 3R/4$, $|Y| \le 3R/(4\tau)$, $F_X \ge 3\delta_R/4 > 0$, and*

$$\tfrac{1}{2}\dot{E} \le -\tfrac{1}{10}q^2 < 0 \qquad \text{on } \{E = E_*\}. \tag{S5.88}$$

*The branch therefore reaches $X = \frac{5}{4}$ with*

$$P(\tfrac{5}{4}, Y) > \frac{49}{192} - \frac{1}{768} = \frac{65}{256} > \frac{1}{4} > 0 = P(M): \tag{S5.89}$$

*it reaches a point strictly above the maximum level and cannot terminate at $M$.*

*Proof.* At $\xi = \delta_R$, $E \le 4(R^2/\tau)\delta_R^2 = \frac{4\delta_R^2}{9/16}E_* < E_*$, since $\delta_R = 1/(512R^3) < 3/8$. On $\{E = E_*\}$, $q \ge \sqrt{\tau E_*} = 3R/4$, and the adverse-to-$q^2$ ratios are $\frac{7}{8}$, $\frac{5}{8K}$, $\frac{3}{32K^2}$ and $\frac{4}{3}\varepsilon_0$, which sum to $\frac{1409532425}{1610612736} < \frac{9}{10}$ (exact rational arithmetic); this gives (S5.88). Invariance together with $F_X > 0$ gives arrival at $X = \frac{5}{4}$. For the level, $p_0(\frac{5}{4}) = \frac{49}{192}$, and

$$\tfrac{1}{2}Y^\top TY \le \tfrac{1}{2}E_* = \frac{9R^2}{32\tau} \le \frac{9}{64KR^3} < \frac{1}{4K};$$

this is the point of the energy adaptation: no bound on $\lambda_{\max}(T)$ is needed, because $E_*$ itself is $O(R^2/\tau)$. The remaining adverse level terms are bounded as follows. On the strip $|Y| \le \sqrt{E_*/\tau} = 3R/(4\tau) \le 3/(8KR^4)$, by (S5.83). So at $X = \frac{5}{4}$

$$\left|\tfrac{1}{2}(X^2 - \tfrac{1}{4})a\cdot Y\right| \le \tfrac{21}{32}R|Y| \le \frac{21}{32K}, \qquad \left|\tfrac{1}{2}X\,Y^\top WY\right| + \left|\tfrac{1}{6}C[Y,Y,Y]\right| \le \tfrac{5}{8}R|Y|^2 + \tfrac{R}{6}|Y|^3 \le \frac{1}{K^2},$$

and $|rQ(\frac{5}{4}, Y)| \le \frac{1}{2}rR\cdot \mathrm{dist}((\frac{5}{4}, Y), S)^2 \le \frac{1}{2}rR \le 3\varepsilon_0$, since $Q(S) = 0$, $\nabla Q(S) = 0$ and $\|Q\|_{C^2} \le R$. Passing to the exact field changes the level by less than $\eta_R \le \frac{1}{8192}$. The two fields agree at $S$ by the pins, the $C^1$ distance between their gradient fields on the convex chart $D^+$ is at most $\eta_R$

(Lemma S5.30), and the segment from $S$ to $(\frac{5}{4}, Y)$ has length less than 1. The ledger reserves $\frac{3}{8192}$ for this term. The remaining terms total $\frac{21}{32K} + \frac{1}{K^2} + 3\varepsilon_0 + \frac{3}{8192}$, and the complete bracket is

$$\frac{21}{32K} + \frac{1}{K^2} + 3\varepsilon_0 + \frac{3}{8192} + \frac{1}{4K} = \frac{9869}{16777216} < \frac{1}{768}$$

(exact rational arithmetic), which gives (S5.89). □

**Lemma S5.29** (inward capture)**.** *Let $m = 2R/\tau$, $\delta_0 = 1/(512R^4)$, and consider the tapered tube*

$$\mathcal{T}_{\text{in}} = \{-\tfrac{1}{2} < X < \tfrac{1}{2},\ |Y| \le m(\tfrac{1}{4} - X^2)\}.$$

*(i) The inward half-branch of $S$ enters $\mathcal{T}_{\text{in}}$.*
*(ii) (*Lateral flux.*) On $\{|Y| = m(\frac{1}{4} - X^2)\}$, the function $G = |Y|^2 - m^2(\frac{1}{4} - X^2)^2$ satisfies $\dot{G} < 0$.*
*(iii) (*Capture.*) The inward half-branch converges to $M$.*

*Proof.* Two facts are used throughout. First, $Rm = 2R^2/\tau \le 1/(KR^3)$ and $m \le 1$, by (S5.83). Second, since $\nabla Q(M) = \nabla Q(S) = 0$ and $\|Q\|_{C^2} \le R$, on the tube

$$|\nabla Q(X,Y)| \le R(\tfrac{1}{2} - |X| + |Y|) \le R(2+m)(\tfrac{1}{4} - X^2) \le 3R(\tfrac{1}{4} - X^2), \tag{S5.90}$$

because $\frac{1}{2} - |X| = (\frac{1}{4} - X^2)/(\frac{1}{2} + |X|) \le 2(\frac{1}{4} - X^2)$ and $|Y| \le m(\frac{1}{4} - X^2)$. Hence, on the tube, $|F_X - (X^2 - \frac{1}{4})| \le (\frac{1}{4} - X^2)(\frac{Rm}{2} + \frac{Rm^2}{8} + 3rR)$, and

$$F_X \le -(\tfrac{1}{4} - X^2)\Big(1 - \frac{Rm}{2} - \frac{Rm^2}{8} - 3rR\Big) < 0. \tag{S5.91}$$

(i) The inward half-branch is the other half of the one-dimensional unstable manifold of $S$, so it leaves $S$ tangent to $-v$, where $v$ is the positive eigenvector of Lemma S5.26. The proof of that lemma gives $E(v_Y) \le \frac{8}{3}(R^2/\tau)|v_X|^2$, and $|v_Y|^2 \le E(v_Y)/\tau$, so $|v_Y|/|v_X| \le \sqrt{8/3}\,R/\tau < m$. The lateral boundary of the tube leaves $S$ with slope $m$, so the branch starts inside $\mathcal{T}_{\text{in}}$, and by (S5.91) it moves leftward.

(ii) On $G = 0$,

$$\dot{G} = 2Y \cdot F_Y + 4m^2 X(\tfrac{1}{4} - X^2)F_X.$$

The leading term is $-2Y^\top TY \le -2\tau m^2(\frac{1}{4} - X^2)^2$. The adverse terms, using $|Y| = m(\frac{1}{4} - X^2)$, $|X| \le \frac{1}{2}$, $\frac{1}{4} - X^2 \le \frac{1}{4}$ and (S5.90), are

$$\begin{aligned} |2Y \cdot \tfrac{1}{2}(X^2 - \tfrac{1}{4})a| &\le R|Y|(\tfrac{1}{4} - X^2) = mR(\tfrac{1}{4} - X^2)^2, \\ |2Y \cdot XWY| = 2|X|\,|Y^\top WY| &\le 2|X|\,m^2R(\tfrac{1}{4} - X^2)^2 \le m^2R(\tfrac{1}{4} - X^2)^2, \\ |Y \cdot C[Y,Y,\cdot]| &\le m^3R(\tfrac{1}{4} - X^2)^3 \le \tfrac{1}{4}m^3R(\tfrac{1}{4} - X^2)^2, \\ |2Y \cdot r\nabla_Y Q| &\le 6mrR(\tfrac{1}{4} - X^2)^2, \end{aligned} \tag{S5.92}$$

and dividing the last by the leading term $2\tau m^2(\frac{1}{4} - X^2)^2$ gives the ratio $3rR/(\tau m) = \frac{3}{2}r \le \frac{3}{2}\varepsilon_0 < 1/(4K)$, using $\tau m = 2R$. By the bound on $|F_X|$ above,

$$|4m^2X(\tfrac{1}{4} - X^2)F_X| \le 2m^2(\tfrac{1}{4} - X^2)^2\Big(1 + \frac{Rm}{2} + \frac{Rm^2}{8} + 3rR\Big),$$

where $3rR \le 3\varepsilon_0 < \frac{1}{8192}$. Collecting and dividing by $2\tau m^2(\frac{1}{4} - X^2)^2$, the adverse ratio is at most

$$\begin{aligned} &\frac{R}{2\tau m} + \frac{R}{2\tau} + \frac{mR}{8\tau} + \frac{1}{4K} + \frac{1}{\tau}\Big(1 + \frac{Rm}{2} + \frac{Rm^2}{8} + \frac{1}{8192}\Big) \\ &\quad = \frac{1}{4} + \frac{R}{2\tau} + \frac{R^2}{4\tau^2} + \frac{1}{4K} + \frac{1}{\tau}\Big(\frac{8193}{8192} + \frac{R^2}{\tau} + \frac{R^3}{2\tau^2}\Big), \end{aligned} \tag{S5.93}$$

where $R/(2\tau m) = R/(4R) = \frac{1}{4}$ and $m = 2R/\tau$. Since $\tau \geq 2KR^5 + 2R$ and $K = 4096$: $R/(2\tau) \leq 1/(4K)$, $R^2/(4\tau^2) \leq 1/(16K^2R^6)$, $1/\tau \leq 1/(2K)$, and the final bracket is at most $\frac{8193}{8192} + \frac{1}{2K} + \frac{1}{8K^2} \leq \frac{3}{2}$. So the total ratio is at most

$$\frac{1}{4} + \frac{1}{4K} + \frac{1}{16K^2} + \frac{1}{4K} + \frac{1}{2K}\Big(\frac{3}{2} + \frac{1}{2K} + \frac{1}{8K^2}\Big) < \frac{1}{4} + \frac{3}{K} < \frac{1}{3}.$$

Hence $\dot{G} \leq -\frac{2}{3}\tau m^2(\frac{1}{4} - X^2)^2 < 0$ on the whole lateral boundary: trajectories cannot exit through the lateral walls, near $S$ included.

(iii) At $M$ the Jacobian $J_M = \begin{pmatrix} A_M & B_M^\top \\ B_M & -T + E_D \end{pmatrix}$ is negative definite: for $w = (w_X, w_Y)$,

$$w^\top J_M w \leq -\tfrac{3}{4}w_X^2 + 2R|w_X||w_Y| - \tau|w_Y|^2 + \tfrac{3R}{4}|w_Y|^2 \leq -\tfrac{3}{8}w_X^2 - \tfrac{\tau}{2}|w_Y|^2,$$

by Young's inequality and $\tau \geq 2KR^5 + 2R$. Write $z = (X, Y)$ and $F(z) = J_M(z - M) + \mathcal{E}(z)$. The polynomial part of $\mathcal{E}$ is the quadratic part of (S5.84) at $M$, of norm at most $(1 + \frac{3}{2}R)(X + \frac{1}{2})^2 + 2R|Y|^2 \leq 3R|z - M|^2$. Only the $C^2$ norm of $Q$ is controlled, so the $rQ$ part of $\mathcal{E}$ is bounded linearly: it is $r[\nabla Q(z) - D^2Q(M)(z - M)]$, of norm at most $2rR|z - M|$, because $\nabla Q(M) = 0$. Hence the Lyapunov function $V = (X + \frac{1}{2})^2 + |Y|^2$ satisfies

$$\dot{V} = 2(z - M) \cdot F(z) \leq -\tfrac{3}{4}V + 6RV^{3/2} + 4rRV,$$

which on $\partial B_{\rho_B}(M)$, with $\rho_B := 1/(16R)$, is at most $(-\frac{3}{4} + \frac{3}{8} + 4\varepsilon_0)V < 0$. So the ball is forward invariant, and every trajectory entering it converges to the sink $M$. Take the section $X = -\frac{1}{2} + 2\delta_0$, where $\frac{1}{4} - X^2 = 2\delta_0(1 - 2\delta_0) \leq 2\delta_0$. There the tube has half-width at most $2m\delta_0 = 1/(128\,\tau R^3)$, and its points lie within $2\delta_0\sqrt{1 + m^2} \leq 4\delta_0 = \frac{1}{128R^4} < \frac{1}{16R} = \rho_B$ of $M$. By (S5.91), $X$ decreases monotonically along the inward branch inside the tube, so the branch reaches this section inside $B_{\rho_B}(M)$ and is captured. □

**Lemma S5.30** (exact-field robustness)**.** *Every strict inequality in Lemmas S5.26–S5.29 has a margin of at least $\frac{1}{10}q^2$, $\frac{2}{3}q^2$, $\frac{1}{3}$ of the leading term, or a fixed fraction of $\rho_B^2$, and the adverse brackets are at most $\frac{1}{3}$, $\frac{9}{10}$ and $\frac{1}{768}$ of their budgets. A pin-preserving $C^1$ perturbation of the gradient field of size $\eta_R = 1/(8192R^3)$ changes each flux by less than half the margin, so the conclusions of these lemmas persist. For the conditioned side-24 field, $\|F_{\text{exact}} - F\|_{C^1(D^+)} \leq C_+ r$ on the fifth-derivative good event, and $C_+ r \leq \eta_R$ for*

$$r \leq r_{\text{mat}} = \min\Big\{1,\ \big[1/(8192\,C_+\varepsilon_0^{3/5})\big]^{5/2}\Big\}. \tag{S5.94}$$

*Proof.* In $\mathcal{C}_E$ the perturbation error is $O(\eta_R\xi)$ against $q \geq 2R\xi$; on $\{E = E_*\}$ it is at most $q\eta_R \leq \frac{4\eta_R}{3R}q^2$; and in the tube it is $O(\eta_R|Y|)$ against the floor $\tau m^2(\frac{1}{4} - X^2)^2$. In each case this is less than half of the corresponding margin. The bound $\|F_{\text{exact}} - F\|_{C^1(D^+)} \leq C_+ r$ holds on the fifth-derivative good event of Lemma S5.33, and (S5.94) gives $C_+ r \leq \eta_R$. □

**Theorem S5.31** (energy-adapted capture and escape)**.** *Under (S5.81)–(S5.83) and $0 < r \leq r_{\text{mat}}$:*

*(i) the $M$-ward unstable half-branch of $S$ converges to $M$ through $\mathcal{T}_{\text{in}}$;*
*(ii) the outward half-branch remains in $\mathcal{C}_E \cup \mathcal{S}_E$ and reaches $X = \frac{5}{4}$ at a level strictly above $f(M)$, hence cannot terminate at $M$;*
*(iii) both conclusions hold for the exact conditioned field on the fifth-derivative good event.*

*No bound on $\lambda_{\max}(T)$ is assumed or used.*

*Proof.* Part (i) is Lemma S5.29; part (ii) is Lemmas S5.26, S5.27 and S5.28; part (iii) is Lemma S5.30. Only $\tau = \lambda_{\min}(T)$ enters these proofs. □

Theorem S5.31 is the capture–escape theorem of the main text, with the potential (S5.81) as (3.18a) there. The proofs of Lemmas S5.27 and S5.29 contain five local repairs to the proof in the 1 August supplement: the bound on $rQ_X$ in Lemma S5.27, and, in Lemma S5.29, the sign of the tangent eigenvector, containment near $S$ through (S5.90), the position of the capture section, and the $rQ$ part of the energy remainder. None of them changes a statement.

S5.6. **Transverse normalization, the shallow layer and adjacency failures.**
Transverse normalization. The matrix $T$ in (S5.81) must be read in scaled coordinates. If $Q_\perp$ is the physical transverse Hessian at the contact saddle, then

$$T_r = -\frac{Q_\perp}{\kappa r}, \qquad \tau_r = \lambda_{\min}(T_r) = \frac{\lambda_*(-Q_\perp)}{\kappa r}, \tag{S5.95}$$

where $\lambda_*(-Q_\perp)$ is the smallest eigenvalue of $-Q_\perp$. The factor $1/r$ is essential: physical transverse displacement is $rY$, while the fold potential is divided by $\kappa r^3$; hence a physical quadratic term $(r^2/2)\, Y^\top Q_\perp Y$ becomes $(1/(2\kappa r))\, Y^\top Q_\perp Y$. Equation (S5.95) is (3.18) of the main text.

In the pin-preserving degree-four scaled potential of the conditioned side-24 field, the remaining coupling tensors $a, W, C$ and the quartic remainder coefficients $Q_4$ have the polynomial envelope

$$R(U) = 1 + \|a\| + \|W\| + \|C\| + \|Q_4\|, \tag{S5.96}$$

where $U$ collects the random coefficients. By Theorem S5.31, the conditions

$$rR(U)^5 \le \varepsilon_0, \qquad \tau_r \ge 2KR(U)^5 + 2R(U), \tag{S5.97}$$

together with $r \le r_{\mathrm{mat}}$ and the fifth-derivative good event, imply simultaneously: (i) the $M$-ward unstable half-branch from $S$ converges to $M$; and (ii) the other half-branch reaches a fixed forward section at field value strictly above $b$ and therefore cannot terminate at $M$. The proof uses $-T_rY$ for transverse contraction, bounds every coupling by $R(U)$, and chooses the cone, strip and tube widths proportional to $R(U)/\tau_r$; condition (S5.97) makes the transverse forcing smaller than the axial margins of $X^2 - \frac{1}{4}$, and the fixed-chart $C^1$ robustness transfers the strict inequalities to the exact field.

**Lemma S5.32** (shallow layer)**.** *Because of* (S5.95)*, failure of the second inequality in* (S5.97) *inside the typed support is exactly the physical shallow-eigenvalue event*

$$\mathcal{S}_r = \Big\{0 < \lambda_* \le \kappa r\big[2KR(U)^5 + 2R(U)\big]\Big\} \subset \big\{0 < \lambda_* \le r\,\Psi(U)\big\} \tag{S5.98}$$

*on compact positive $\kappa$ sets, for a fixed polynomial $\Psi$. On the typed support the endpoint determinant product satisfies*

$$W_r \le Cr^2\big[\lambda_* + r\Psi(U)\big]^2(1 + \|U\|)^N, \tag{S5.99}$$

*and*

$$\mathbb{E}^0\big[W_r \mathbf{1}_{\mathcal{S}_r}\big] \le Cr^2 \int_0^{r\Psi(U)} \lambda^2\, d\lambda \le Cr^5, \tag{S5.100}$$

$$\mathbb{P}_r^{MS}(\mathcal{S}_r) \le Cr^3. \tag{S5.101}$$

*Proof.* At contact the transverse Hessian is $-bI_2 + \mathrm{GOE}_2$ (Lemma S5.13), jointly nondegenerate with the remaining finite jet; side-24 jet perturbation and compactness preserve bounded conditional densities and the shallow-eigenvalue coarea estimate. A bounded Gaussian density and integration in the smallest-eigenvalue coordinate, applied to (S5.99), give the unnormalized shallow-layer mass (S5.100). Dividing by $Z_r \ge c_Z r^2$ from (S5.8) yields (S5.101). □

**Lemma S5.33** (fifth-derivative control)**.** *Let $H_5$ be the supremum over $\mathbb{T}^3_{24}$ of the norm of the fifth-derivative field of $f$. On compact marks there are constants $C, c > 0$ such that*

$$\mathbb{P}^0(H_5 > u) \le C\exp(-cu^2), \qquad u > 0. \tag{S5.102}$$

*Taking $u = r^{-1}$ gives the failure bound* $\exp(-c/r^2)$ *for the complement of the* fifth-derivative good event $\{H_5 \le r^{-1}\}$.

*Proof.* For the fixed side-24 periodized Bargmann–Fock field every spectral moment is finite, and every derivative process is continuous on the compact torus. Conditioning on the corrected pair pins changes the fifth-derivative field by a uniformly bounded affine mean and a covariance dominated by the unconditional covariance. On compact marks, the Borell–TIS inequality [1] therefore gives (S5.102). □

**Lemma S5.34** (regular-domain and fifth-derivative failures)**.** *Let $F_r$ be the union of the regular-domain failure $\{rR(U)^5 > \varepsilon_0\}$ and the fifth-derivative failure $\{H_5 > r^{-1}\}$. Then*

$$\mathbb{P}_r^{MS}(F_r) = \frac{\mathbb{E}^0[W_r \mathbf{1}_{F_r}]}{Z_r} \le C\,\mathbb{P}^0(F_r)^{1/2}, \tag{S5.103}$$

*and $\mathbb{P}_r^{MS}(F_r) = o(r^3)$.*

*Proof.* The inequality in (S5.103) follows from the exact endpoint second moment $\mathbb{E}^0[W_r^2] \le C_W r^4$, the Cauchy–Schwarz inequality and $Z_r \ge c_Z r^2$, all from (S5.8). The corrected finite-jet regular-domain tail is super-polynomial, and the conditioned fifth-derivative Borell tail is $\exp(-c/r^2)$ by Lemma S5.33, so both contributions are $o(r^3)$. □

**Proposition S5.35** (adjacency and self-attachment)**.** *Uniformly for $t \in S^2$ and on compact positive mark sets, for $0 < r \le r_{\mathrm{mat}}$,*

$$\mathbb{P}_r^{MS}(A_r^c) \le C_A r^3, \qquad \mathbb{P}_r^{MS}(L_{\mathrm{same},r}) \le C_L r^3. \tag{S5.104}$$

*Proof.* Outside $\mathcal{S}_r \cup F_r$ the conditions (S5.97) and the fifth-derivative good event hold, with $T = T_r$. By Theorem S5.31(i) the $M$-ward half-branch converges to $M$, so $A_r$ holds; by Theorem S5.31(ii) the outward half-branch cannot terminate at $M$, which excludes same-maximum self-attachment. Hence both probabilities in (S5.104) are bounded by $\mathbb{P}_r^{MS}(\mathcal{S}_r) + \mathbb{P}_r^{MS}(F_r)$, which is $O(r^3)$ by Lemmas S5.32 and S5.34. □

## S5.7. **The selection theorem.**

**Theorem S5.36** (elder selection on compact marks)**.** *For every compact $I_b \subset \mathbb{R}$ and every compact $I_\kappa \subset (0,\infty)$ there are $r_0 > 0$ and $C_{\mathrm{sel}} < \infty$ such that, under the all-typed maximum–index-two-saddle Palm law (S5.1),*

$$\sup_{t \in S^2,\ b \in I_b,\ \kappa \in I_\kappa} \big[1 - p_r(t,b,\kappa)\big] \le C_{\mathrm{sel}}\, r^3, \qquad 0 < r \le r_0. \tag{S5.105}$$

*In particular, the elder-rule selection probability tends to one uniformly on compact positive mark sets.*

*Proof.* By Lemmas S5.19, S5.23 and S5.25, the three regional Palm expectations satisfy

$$\mathbb{E}_r^{MS} N_{\mathrm{col}} \le C_{\mathrm{col}} r^3, \qquad \mathbb{E}_r^{MS} N_{\mathrm{sing}} \le C_{\mathrm{sing}} r^3, \qquad \mathbb{E}_r^{MS} N_{\mathrm{far}} \le C_{\mathrm{far}} r^3, \tag{S5.106}$$

so (S5.7) gives $\mathbb{P}_r^{MS}(B_r) \le C_B r^3$. Proposition S5.35 gives $\mathbb{P}_r^{MS}(A_r^c) \le C_A r^3$ and $\mathbb{P}_r^{MS}(L_{\mathrm{same},r}) \le C_L r^3$. By Corollary S5.8,

$$1 - p_r = \mathbb{P}_r^{MS}(E_r^c) \le \mathbb{P}_r^{MS}(A_r^c) + \mathbb{P}_r^{MS}(E_r^c \cap A_r) \le \mathbb{P}_r^{MS}(A_r^c) + \mathbb{P}_r^{MS}(B_r) + \mathbb{P}_r^{MS}(L_{\mathrm{same},r}) \le C_{\mathrm{sel}} r^3,$$

with all constants uniform in $t \in S^2$ and on the compact mark set. Here $r_0$ is the least of the radii for which Lemma S2.9(iii) on $I_b \times I_\kappa$ and the results just cited hold, so that each of them applies for $0 < r \le r_0$. □

Theorem S5.36 is the bound (3.13) of the main text, proved here through explicit witnesses in $d = 3$.

*Remark* S5.37 (the parent theorem). The bound (S5.105) is also Theorem A of [7] (Theorem S12.3) at $d = 3$ and $L = 24$, and the main text cites that theorem for (3.13). It is proved there for every fixed $d \geq 2$ and $L > 0$, uniformly over compact birth and gap windows and over all orthonormal frames, with existential constants. Its selection probability, that the global superlevel elder partner of $M$ is $S$, is the $p_r$ of Definition S5.4; on $\Omega_{\mathrm{Mo}}$ the two events agree by Lemma S5.3. Theorem S5.36 is a second, three-dimensional proof that does not use [7].

*Remark* S5.38 (the Morse event under the Palm law). Lemma S5.7 is applied under $\mathbb{P}_r^{MS}$, so it needs $\mathbb{P}_r^{MS}(\Omega_{\mathrm{Mo}}) = 1$. Since $\mathbb{P}(\Omega_{\mathrm{Mo}}) = 1$, disintegration over the pin values gives this for almost every $(b, \kappa)$, which suffices for every integrated intensity. It holds for every $(b, \kappa)$ as well. Under the pins, $H_M$ and $H_S$ have nondegenerate Gaussian laws, so $M$ and $S$ are nondegenerate critical points almost surely. On each compact set of points at distance at least $1/n$ from $\{M, S\}$, the conditional density of $\nabla f(z)$ is bounded, by derivative-distribution independence (Section S1), and a Bulinskaya-type lemma (see [1], Chapter 11) excludes degenerate critical points there; the union over $n$ excludes them everywhere. Equal critical values are excluded in the same way. On pairs of points at distance at least $1/n$ from each other and from $\{M, S\}$, the field $(z, z') \mapsto (\nabla f(z), \nabla f(z'), f(z) - f(z'))$ has seven components in six variables and a bounded conditional density. For a critical point $z \notin \{M, S\}$ with $f(z) \in \{f(M), f(S)\}$, the map $z \mapsto (\nabla f(z), f(z))$ has four components in three variables.

S5.8. **Finite-scale witness densities near the pair axis.** The charts of §S5.4 are computed in the contact limit, $r \to 0$ at a fixed chart position. Near the pair axis this limit is not uniform. At a fixed positive scale the conditional density of $(f(y), \nabla f(y))$, or of $\nabla f(y)$, given the pins is positive at points $y \notin \{M, S\}$ of the axis itself, where the contact-limit densities vanish to infinite order. So a bound by $\rho^{-m}$ times a function of the scale alone cannot be integrated over the directions. This subsection proves the finite-scale estimates that the collar and singular-near counts use instead. They rest on elementary Gaussian conditioning (Lemma S5.39), on first-order expansions of the scaled third-point variables in a radial blow-up of the parameters at the axis (Lemmas S5.40 and S5.41), and on Hypothesis S5.11(2). They hold for every kernel satisfying Hypothesis S5.11, with constants depending on the kernel; the exact planar covariances are not used.

Let $\mathcal{H} \subset L^2(\Omega)$ be the closed linear span of the values $f(z)$. It contains every derivative $\partial^\alpha f(z)$, and $\|X\|^2 = \operatorname{Var} X$ for $X \in \mathcal{H}$. For $X, X' \in \mathcal{H}$ write $X \equiv X'$ if $X - X'$ lies in the span of the pin evaluations $\Pi_{r,x,t}$; covariances given the pins depend only on these classes. For a centred Gaussian vector $(\Pi, Z)$ in $\mathcal{H}$ with $\operatorname{Cov}(\Pi)$ and $\Sigma_Z = \operatorname{Cov}(Z \mid \Pi)$ nonsingular, and values $\pi$ and $z$, put $\mu_Z(\pi) = \mathbb{E}[Z \mid \Pi = \pi]$,

$$\mathcal{M}_\Pi(\pi)^2 = \pi^\top \operatorname{Cov}(\Pi)^{-1} \pi, \qquad \mathcal{M}_Z(z; \pi)^2 = (z - \mu_Z(\pi))^\top \Sigma_Z^{-1} (z - \mu_Z(\pi)),$$

the Mahalanobis norms of the conditioning values. For the pair pins at their target, $\mathcal{M}_\Pi(z_r(b, \kappa))$ is bounded uniformly in $r$, $t$ and the compact mark set. It is unchanged by the invertible passage to the corrected pins, whose covariance has a uniform eigenfloor (Lemma S5.14) and whose target is bounded.

**Lemma S5.39** (Gaussian conditioning). *Let $(\Pi, Z)$, with $\Pi \in \mathcal{H}^m$ and $Z \in \mathcal{H}^k$, be as above, and let $\pi \in \mathbb{R}^m$ and $z \in \mathbb{R}^k$.*

*(a) For each $i$, $\mathcal{M}_Z(z; \pi)^2 \geq (z_i - \mu_{Z,i}(\pi))^2 / (\Sigma_Z)_{ii}$.*

*(b) Given $(\Pi, Z) = (\pi, z)$, the field is $f = \bar{f} + g$. Here $\bar{f}$ is deterministic, and*

$$|\partial^\alpha \bar{f}(w)| \leq \|\partial^\alpha f(w)\| (\mathcal{M}_\Pi(\pi)^2 + \mathcal{M}_Z(z; \pi)^2)^{1/2}$$

*for every point $w$ and multi-index $\alpha$. The field $g$ is centred Gaussian, its law does not depend on $(\pi, z)$, and $\mathbb{E} \sup_B |\partial^\alpha g|^p$ is bounded by a constant that depends only on the bounded set $B$, on $p \geq 1$, on $\alpha$ and on $m + k$. Consequently, for every polynomial $\Psi$ in $\kappa$*

*and in the suprema over $B$ of the norms of finitely many derivatives of $f$, there are $C$ and $q$, depending only on $\Psi$, $B$, $m+k$ and the compact mark set, such that*

$$\mathbb{E}\big[|\Psi| \mid \Pi=\pi,\ Z=z\big] \le C(1+\mathcal{M}_\Pi(\pi)+\mathcal{M}_Z(z;\pi))^q.$$

*(c) For every $q \ge 0$,*

$$(1+\mathcal{M}_Z(z;\pi))^q\, p_{Z|\Pi=\pi}(z) \le C_q\,(\det\Sigma_Z)^{-1/2}\exp(-\tfrac14\mathcal{M}_Z(z;\pi)^2). \tag{S5.107}$$

*(d) Split $Z=(Z_1,Z')$ with $\Sigma_{Z'}=\mathrm{Cov}(Z'\mid\Pi)$ nonsingular. Put $\sigma_1^2=\mathrm{Var}(Z_1\mid\Pi)$ and $\tilde m=\mathbb{E}[Z_1\mid\Pi=\pi,\ Z'=z']-z_1$. Then $|\tilde m-(\mu_{Z,1}(\pi)-z_1)|\le\sigma_1\mathcal{M}_{Z'}(z';\pi)$, and if $\tilde m\neq0$, then for every $q\ge0$*

$$(1+\mathcal{M}_Z(z;\pi))^q\, p_{Z|\Pi=\pi}(z) \le \frac{C_q(1+\mathcal{M}_{Z'}(z';\pi))^q}{(\det\Sigma_{Z'})^{1/2}\,|\tilde m|}\exp\Big(-\frac{\tilde m^2}{8\sigma_1^2}\Big). \tag{S5.108}$$

*Proof.* (a) For positive definite $\Sigma$ and a vector $w$, $w^\top\Sigma^{-1}w=\sup_{a\neq0}(a^\top w)^2/(a^\top\Sigma a)$; take $a$ the $i$-th unit vector.

(b) Let $e_1,\dots,e_{m+k}$ be an orthonormal basis of the span of the coordinates of $(\Pi,Z)$ in $\mathcal{H}$. They are independent standard Gaussian variables, and $(\Pi,Z)=Ae$ with $A$ invertible. Conditioning on $(\Pi,Z)=(\pi,z)$ is conditioning on $e=e^*=A^{-1}(\pi,z)$, and $|e^*|^2=(\pi,z)^\top\mathrm{Cov}(\Pi,Z)^{-1}(\pi,z)=\mathcal{M}_\Pi(\pi)^2+\mathcal{M}_Z(z;\pi)^2$ by block inversion of $\mathrm{Cov}(\Pi,Z)$. Write $f(w)=\sum_j\langle f(w),e_j\rangle e_j+g(w)$. Then $g$ is orthogonal to $e$, hence independent of it, so its conditional law is its law; and $\bar f(w)=\sum_j\langle f(w),e_j\rangle e_j^*$. Since $\partial^\alpha\bar f(w)=\sum_j\langle\partial^\alpha f(w),e_j\rangle e_j^*$, the Cauchy–Schwarz inequality gives the bound on $\partial^\alpha\bar f$. Moreover $\partial^\alpha g=\partial^\alpha f-\sum_j\langle\partial^\alpha f,e_j\rangle e_j$, so $\sup_B|\partial^\alpha g|\le\sup_B|\partial^\alpha f|+\sigma_\alpha\sum_j|e_j|$, where $\sigma_\alpha=\|\partial^\alpha f(w)\|$ does not depend on $w$. The supremum of a derivative of the smooth stationary field has Gaussian moments of every order by the Borell–TIS inequality [1]. The bound for $\Psi$ follows by expanding the polynomial, with $\kappa$ bounded on the compact mark set.

(c) The conditional density is $(2\pi)^{-k/2}(\det\Sigma_Z)^{-1/2}\exp(-\frac12\mathcal{M}_Z^2)$, and $\sup_{u\ge0}(1+u)^qe^{-u^2/4}<\infty$.

(d) Gaussian regression gives $\tilde m-(\mu_{Z,1}(\pi)-z_1)=\mathrm{Cov}(Z_1,Z'\mid\Pi)\,\Sigma_{Z'}^{-1}\big(z'-\mu_{Z'}(\pi)\big)$, and the Cauchy–Schwarz inequality for the conditional covariance gives the first claim. Factor $p_{Z|\Pi=\pi}(z)=p_{Z'|\Pi=\pi}(z')\,p_{Z_1|\Pi=\pi,Z'=z'}(z_1)$. The second factor is the normal density with mean $z_1+\tilde m$ and variance $\tilde\sigma^2\le\sigma_1^2$, evaluated at $z_1$, and $\mathcal{M}_Z^2=\mathcal{M}_{Z'}^2+\tilde m^2/\tilde\sigma^2$. Hence $(1+\mathcal{M}_Z)^q\le2^q(1+\mathcal{M}_{Z'})^q(1+|\tilde m|/\tilde\sigma)^q$, and

$$\Big(1+\frac{|\tilde m|}{\tilde\sigma}\Big)^q\frac{e^{-\tilde m^2/(2\tilde\sigma^2)}}{\sqrt{2\pi}\,\tilde\sigma}\le C_q\,\frac{e^{-\tilde m^2/(8\tilde\sigma^2)}}{\tilde\sigma}\,e^{-\tilde m^2/(8\tilde\sigma^2)}\le C_q\sup_{u>0}\frac{e^{-\tilde m^2/(8u^2)}}{u}\cdot e^{-\tilde m^2/(8\sigma_1^2)},$$

and the supremum is $2e^{-1/2}/|\tilde m|$. With $p_{Z'|\Pi=\pi}(z')\le(2\pi)^{-(k-1)/2}(\det\Sigma_{Z'})^{-1/2}$ this is (S5.108). □

The point of (d) is that no lower bound on the conditional variance of the mismatched coordinate $Z_1$ is needed: a Gaussian density at distance $|\tilde m|$ from its mean is at most $C/|\tilde m|$ times $\exp(-\tilde m^2/(8\sigma_1^2))$, however small the variance.

**Lemma S5.40** (finite-scale singular-near frame)**.** *Assume Hypothesis S5.11, and use the notation of Lemma S5.20 in the frame $(t,v,w)$ adapted to $\omega$: $\omega=ct+\rho v$ with $\rho\ge0$ (any unit $v\perp t$ if $\rho=0$), so that $(Y,Z)=(\rho,0)$, $V_2=s^{-1}\partial_v f(y)$ and $V_3=s^{-1}\partial_w f(y)$. Put*

$$\varepsilon=(\rho^2+s^2)^{1/2},\qquad(\rho,s)=\varepsilon(\cos\theta,\sin\theta),\qquad\lambda=1-\eta^2/4,\qquad\varsigma=\operatorname{sign}c,$$

*and*

$$\tilde V_0=\varepsilon^{-2}\Big(V_0+\varsigma\,\frac{4+3\eta^2}{48}\,V_1\Big),\qquad\tilde V_i=\varepsilon^{-1}V_i\quad(i=1,2,3). \tag{S5.109}$$

*For $u$ in the window (S5.44), let $\mathcal{M}(u)=\mathcal{M}_V(v^*(u);z_r)$ be the Mahalanobis norm, under the conditional law given the pins, of the value $v^*(u)=\big(s^{-3}(u-b+\kappa r^3/12),0,0,0\big)$ that $V$ takes when*

*$f(y) = u$ and $\nabla f(y) = 0$. There are $\varepsilon_1 > 0$, $c_* > 0$, $c_V > 0$ and $C$, depending only on the kernel, on $\eta_0$ and on the compact mark set, such that the following hold uniformly in $t$, $v$, $\theta \in [0, \pi/2]$, $\eta \in [0, \eta_0]$ and both signs of $c$.*

*(a) For $\varepsilon \le \varepsilon_1$, $\tilde V_i \equiv \tilde V_i^{(0)} + R_i$ with $\|R_i\| \le C\varepsilon$, where, with all derivatives at $x$,*

$$\tilde V_0^{(0)} = -\varsigma\,\frac{\lambda}{8}\Big[\cos^2\theta\, f_{tvv} + \frac{\lambda}{3}\sin\theta\cos\theta\, f_{tttv} + \frac{\lambda^2}{60}\sin^2\theta\, f_{ttttt}\Big], \quad \tilde V_1^{(0)} = \varsigma\Big[\cos\theta\, f_{ttv} + \frac{\lambda}{6}\sin\theta\, f_{tttt}\Big],$$
$$\tilde V_2^{(0)} = \cos\theta\, f_{vv} + \frac{\lambda}{2}\sin\theta\, f_{ttv}, \qquad \tilde V_3^{(0)} = \cos\theta\, f_{vw} + \frac{\lambda}{2}\sin\theta\, f_{ttw}. \tag{S5.110}$$

*(b) For $\varepsilon \le \varepsilon_1$, $\det \mathrm{Cov}(V \mid pins) \ge c_V\, \varepsilon^{10}$.*

*(c) For $\varepsilon \le \varepsilon_1$, $\mathrm{Var}(V_1 \mid pins) \le C\varepsilon^2$ and $\big|\mathbb{E}[V_1 \mid pins = z_r(b,\kappa)] - \kappa\lambda\big| \le C\varepsilon$; after decreasing $\varepsilon_1$, this mean is at least $\kappa/2$, since $\lambda \ge \frac{3}{4}$.*

*(d) For every $q \ge 0$ and $\varepsilon \le \varepsilon_1$,*

$$(1 + \mathcal{M}(u))^q\, p_{(f(y),\nabla f(y))|pins}(u, 0) \le C_q\, s^{-7}\varepsilon^{-5}\exp(-c_*/\varepsilon^2).$$

*(e) For every $\rho_0 > 0$ and $s_0 > 0$ there are $c_V' > 0$ and $r_0 > 0$ with the following property. If $C_1 r < s \le \delta$ and $r \le r_0$, and if $\rho \ge \rho_0$ or $s \ge s_0$, then $\det \mathrm{Cov}(V \mid pins) \ge c_V'$, and so $(1 + \mathcal{M}(u))^q\, p_{(f(y),\nabla f(y))|pins}(u, 0) \le C_q\, s^{-7}$.*

*Proof. Expansion.* Fix $N = 10$. Taylor's formula at $x$ with integral remainder writes each of $f(y)$, $\nabla f(y)$, $f(M)$, $f(S)$, $\nabla f(M)$ and $\nabla f(S)$ as a combination of the derivatives $\partial^\alpha f(x)$ with $|\alpha| \le N$, whose coefficients are polynomials in $r$, $s$ and $\omega$, plus a remainder of norm at most $Cs^{N+1}$ for $f(y)$, $Cs^N$ for $\nabla f(y)$, $Cr^{N+1}$ for the pin values and $Cr^N$ for the pin gradients. The eight pin equations determine the contact coordinates $P$ of (S5.15): solving them expresses each coordinate of $P$ as a combination of the pins, with coefficients at most $Cr^{-3}$, plus a combination of the other derivatives $\partial^\alpha f(x)$ with $|\alpha| \le N$, with coefficients polynomial in $r$, plus a remainder of norm at most $Cr^{N-3}$. Substituting into $V$ and (S5.109) gives

$$\tilde V_i \equiv \sum_\alpha w_{i\alpha}\,\partial^\alpha f(x) + R_i', \qquad \|R_i'\| \le C\varepsilon^{-2}s^{N-6} \le C\varepsilon^{N-8},$$

where $\alpha$ runs over the multi-indices with $|\alpha| \le N$ outside $P$. The coefficients of $V$ have no negative power of $s$. Indeed

$$s^3 V_0 = f(y) - U^{(0)} - \tfrac{1}{2}(y - x)\cdot\nabla f(y) + \tfrac{sc}{6}\,\partial_t f(y),$$

where $U^{(0)} \equiv 0$. In $f(y) - \frac{1}{2}(y-x)\cdot\nabla f(y)$ the Taylor term of degree $j$ carries the factor $1 - j/2$, so the terms of degree two cancel; the terms of degree at most one, and those of $\frac{sc}{6}\partial_t f(y)$ of degree at most one in $y - x$, involve only coordinates of $P$. Modulo the pins, $f(x)$ and $f_t(x)$ are $O(r^4)$ and the other coordinates of $P$ are $O(r^2)$, with coefficients polynomial in $r$. In $V$, $f(x)$ carries the factor $s^{-3}$, and the other coordinates of $P$ carry factors at most $s^{-2}$. Since $r = \eta s$, every coefficient of $V_0, \dots, V_3$ is a polynomial in $s$, $\rho$, $c$ and $\eta$, where $c = \varsigma(1 - \rho^2)^{1/2}$. Modulo the pins, the random part of each $V_i$ vanishes at $s = \rho = 0$, and the linear terms in $(s, \rho)$ of $V_0$ and of $-\varsigma\frac{4+3\eta^2}{48}V_1$ coincide. Hence the $w_{i\alpha}$ are real-analytic functions of $(\varepsilon, \theta, \eta)$ on a neighbourhood of $[0, \varepsilon_1] \times [0, \pi/2] \times [0, \eta_0]$. Their values $w_{i\alpha}(0)$ at $\varepsilon = 0$ are the coefficients displayed in (S5.110). They vanish for $|\alpha| \ge 6$, since a derivative of order $j$ enters $V_0$ and $V_1$ with a coefficient $O(s^{j-3})$, and $V_2$ and $V_3$ with $O(s^{j-2})$. The script `checks/s5_finite_scale_check.py` computes the coefficients exactly for a polynomial field of degree six and both signs of $c$. It verifies that those of $V$ have no negative power of $s$, and it verifies the other statements of this paragraph. Hence $R_i = \sum_\alpha (w_{i\alpha} - w_{i\alpha}(0))\partial^\alpha f(x) + R_i'$ has norm at most $C\varepsilon$, which is (a).

*Determinant.* The symbols of the derivatives in (S5.110) are $\xi_t\xi_v^2$, $\xi_t^3\xi_v$ and $\xi_t^5$ in $\tilde V_0^{(0)}$; $\xi_t^2\xi_v$ and $\xi_t^4$ in $\tilde V_1^{(0)}$; $\xi_v^2$ and $\xi_t^2\xi_v$ in $\tilde V_2^{(0)}$; and $\xi_v\xi_w$ and $\xi_t^2\xi_w$ in $\tilde V_3^{(0)}$. None of them is a symbol of $P$. The monomials of $\tilde V_0^{(0)}$ occur in no other $\tilde V_i^{(0)}$. Its coefficient vector is nonzero: the coefficient of $f_{tvv}$ is $-\varsigma\lambda\cos^2\theta/8$ and that of $f_{ttttt}$ is $-\varsigma\lambda^3\sin^2\theta/480$. If $\cos\theta\neq 0$, the second-order monomials $\xi_v^2$ and $\xi_v\xi_w$ occur only in $\tilde V_2^{(0)}$ and $\tilde V_3^{(0)}$ respectively, and $\tilde V_1^{(0)}\neq 0$ involves no second-order derivative. If $\cos\theta=0$, then $\tilde V_1^{(0)}$, $\tilde V_2^{(0)}$ and $\tilde V_3^{(0)}$ are nonzero multiples of $f_{tttt}$, $f_{ttv}$ and $f_{ttw}$. In each case the symbols of $P,\tilde V_0^{(0)},\dots,\tilde V_3^{(0)}$ are linearly independent. By Hypothesis S5.11(2) (Theorem S1.3 for $K_{24}$, Remark S5.12 for $K_0$) their joint covariance is positive definite, and so is the Schur complement $\Sigma^{(0)}=\operatorname{Cov}(\tilde V^{(0)}\mid P)$. It is continuous in $(t,v,\theta,\eta)$ on a compact set, so $\det\Sigma^{(0)}\geq 2c_0>0$. The corrected pins span the same subspace of $\mathcal H$ as the pins, differ from $P$ by $O(r)$ (Lemma S5.13) and have a uniform eigenfloor (Lemma S5.14). So the orthogonal projections onto the two spans differ by at most $Cr\leq C\varepsilon$ in operator norm, and with (a), $\operatorname{Cov}(\tilde V\mid\text{pins})=\Sigma^{(0)}+O(\varepsilon)$. Hence $\det\operatorname{Cov}(\tilde V\mid\text{pins})\geq c_0$ for $\varepsilon\leq\varepsilon_1$. The map $V\mapsto\tilde V$ is triangular with determinant $\varepsilon^{-5}$, which gives (b) with $c_V=c_0$.

*The coordinate $V_1$.* By (a), $V_1=\ell_1(\Pi)+\varepsilon\hat V_1$, where $\ell_1(\Pi)$ is a combination of the pins and $\|\hat V_1\|\leq C$. Hence $\operatorname{Var}(V_1\mid\text{pins})\leq C\varepsilon^2$, and $\mathbb E[V_1\mid\text{pins}=z_r]=\ell_1(z_r)+\varepsilon\,\mathbb E[\hat V_1\mid\text{pins}=z_r]$, where the last mean is at most $\|\hat V_1\|\,\mathcal M_\Pi(z_r)\leq C$ by Lemma S5.39(b). The value $\ell_1(z_r)$ is that of $V_1$ for the polynomial field of degree $N$ that satisfies the pins and has all derivatives at $x$ outside $P$ equal to zero. It equals $\kappa\lambda+O(\varepsilon^2)$, in agreement with the deterministic terms of Lemma S5.20, and the check script verifies it. This gives (c).

*The density.* Given the pins, $V$ is an affine function of $(f(y),\nabla f(y))$ with Jacobian $s^{-7}$ (Lemma S5.20), and $f(y)=u$, $\nabla f(y)=0$ correspond to $V=v^*(u)$. Since $v_1^*=0$, (c) and Lemma S5.39(a) give $\mathcal M(u)^2\geq(\kappa/2)^2/(C\varepsilon^2)=4c_*/\varepsilon^2$. With (b), Lemma S5.39(c) gives (d).

*Away from the axis.* For $\rho\geq\rho_0$, the expansion above with the normalization of $V$ itself shows $V\equiv L+O(s)$ uniformly, where $L=(L_0,\dots,L_3)$ are the random residual limits of Lemma S5.20; their covariance given $P$ has determinant $\rho^{10}(3c^2+\rho^2)A_{\text{sing}}\geq\rho_0^{10}a_{\text{sing}}$ by Lemma S5.21, since $3c^2+\rho^2=3-2\rho^2\geq 1$. So there is $\bar s=\bar s(\rho_0)>0$ with $\det\operatorname{Cov}(V\mid\text{pins})\geq\frac12\rho_0^{10}a_{\text{sing}}$ for $s\leq\bar s$ and $\rho\geq\rho_0$. Put $s_1=\min(s_0,\bar s)$ and take $r_0<s_1/C_1$. For $s\in[s_1,\delta]$, the covariance $\operatorname{Cov}(V\mid\text{pins})$ is continuous in $(r,s,\omega)\in[0,r_0]\times[s_1,\delta]\times S^2$, with the corrected pins at $r=0$. It is positive definite at each point by Theorem S1.3: for $r>0$ the evaluations sit at the three distinct points $M$, $S$ and $y$, and for $r=0$ at $x$ and $y$. Compactness gives a positive lower bound for its determinant. A point with $\rho\geq\rho_0$ or $s\geq s_0$ has $s\geq s_1$, or $\rho\geq\rho_0$ and $s<s_1\leq\bar s$, so one of the two bounds applies. The density bound follows from Lemma S5.39(c) and the Jacobian $s^{-7}$. □

**Lemma S5.41** (finite-scale collar frames)**.** *Assume Hypothesis S5.11. Let $y\notin\{M,S\}$ be a point of the collar, $|y-x|\leq C_1r$, and use the frame $(t,v,w)$ adapted to $y-x$, so that $y=x+r(Xt+\rho v)$ with $\rho\geq 0$. Put $G=(r^{-2}\partial_tf(y),\ r^{-1}\partial_vf(y),\ r^{-1}\partial_wf(y))$. All constants below depend only on the kernel, on $C_1$ and on the compact mark set.*

*(a) (Away from the axis.) For every $\rho_1>0$ there are $r_0,c>0$ such that $\det\operatorname{Cov}(G\mid\text{pins})\geq c$ for $r\leq r_0$ and $\rho\geq\rho_1$.*

*(b) (Near the axis, away from the endpoints.) Suppose that $|y-M|\geq r/4$ and $|y-S|\geq r/4$. Put $\varepsilon=(\rho^2+r^2)^{1/2}$ and $(\rho,r)=\varepsilon(\cos\theta,\sin\theta)$. Then $G_1\equiv\varepsilon R_1$ and $(G_2,G_3)\equiv\varepsilon(G_2^{(0)}+R_2,\ G_3^{(0)}+R_3)$, with $\|R_1\|\leq C$, $\|R_2\|+\|R_3\|\leq C\varepsilon$ and*

$$G_2^{(0)}=\cos\theta\,f_{vv}+\tfrac12(X^2-\tfrac14)\sin\theta\,f_{ttv},\qquad G_3^{(0)}=\cos\theta\,f_{vw}+\tfrac12(X^2-\tfrac14)\sin\theta\,f_{ttw}.\tag{S5.111}$$

*The conditional means satisfy $|\mathbb E[G_1\mid\text{pins}=z_r]-\kappa(X^2-\frac14)|\leq C\varepsilon$ and $|\mathbb E[(G_2,G_3)\mid\text{pins}=z_r]|\leq C\varepsilon$. There are $\varepsilon_1\in(0,\frac18]$ and $c_*>0$ such that, for $\varepsilon\leq\varepsilon_1$, the smallest*

*eigenvalue of* $\mathrm{Cov}(G_2, G_3 \mid \textit{pins})$ *is at least* $c\varepsilon^2$ *and, with* $\mathcal{M}$ *the Mahalanobis norm of* $G$ *at* $0$ *given the pins,*

$$(1+\mathcal{M})^q\, p_{G|pins}(0) \le C_q\, \varepsilon^{-2} \exp(-c_*/\varepsilon^2) \qquad (q \ge 0).$$

(c) *(Near an endpoint.) Let* $p \in \{M, S\}$*, with* $\chi = -1$ *for* $p = M$ *and* $\chi = 1$ *for* $p = S$*. Write* $y = p + r\tau(\cos\alpha\, t + \sin\alpha\, v)$ *with* $0 < \tau < \frac{1}{4}$ *and* $0 \le \alpha \le \pi$*, in the frame adapted to* $y - p$*, and put*

$$\tilde G = \big((r^2\tau)^{-1}\partial_t f(y),\ (r\tau)^{-1}\partial_v f(y),\ (r\tau)^{-1}\partial_w f(y)\big).$$

*For every* $\gamma_1 > 0$ *there are* $r_0, c > 0$ *with* $\det \mathrm{Cov}(\tilde G \mid \textit{pins}) \ge c$ *for* $r \le r_0$ *and* $\sin\alpha \ge \gamma_1$*. Where* $\sin\alpha < 1$*, put* $\varsigma = \operatorname{sign}\cos\alpha$*,* $\varepsilon' = (\sin^2\alpha + r^2)^{1/2}$ *and* $(\sin\alpha, r) = \varepsilon'(\cos\theta', \sin\theta')$*. Then* $\tilde G_1 \equiv \varepsilon' R_1$ *and* $(\tilde G_2, \tilde G_3) \equiv \varepsilon'(\tilde G_2^{(0)} + R_2,\ \tilde G_3^{(0)} + R_3)$*, with* $\|R_1\| \le C$*,* $\|R_2\| + \|R_3\| \le C\varepsilon'$ *and*

$$\tilde G_2^{(0)} = \cos\theta'\, f_{vv} + \tfrac{1}{2}(\tau + \chi\varsigma)\sin\theta'\, f_{ttv}, \qquad \tilde G_3^{(0)} = \cos\theta'\, f_{vw} + \tfrac{1}{2}(\tau + \chi\varsigma)\sin\theta'\, f_{ttw}. \tag{S5.112}$$

*Moreover* $\big|\mathbb{E}[\tilde G_1 \mid \textit{pins} = z_r] - \kappa(\tau + \chi\varsigma)\big| \le C\varepsilon'$ *and* $\big|\mathbb{E}[(\tilde G_2, \tilde G_3) \mid \textit{pins} = z_r]\big| \le C\varepsilon'$*, and for* $\varepsilon' \le \varepsilon_1$ *the smallest eigenvalue of* $\mathrm{Cov}(\tilde G_2, \tilde G_3 \mid \textit{pins})$ *is at least* $c\,\varepsilon'^2$ *and*

$$(1+\mathcal{M})^q\, p_{\tilde G|pins}(0) \le C_q\, \varepsilon'^{-2} \exp(-c_*/\varepsilon'^2) \qquad (q \ge 0),$$

*with* $\mathcal{M}$ *the Mahalanobis norm of* $\tilde G$ *at* $0$ *given the pins.*

*Proof.* The expansions in (b) and (c) are obtained exactly as in the proof of Lemma S5.40(a): Taylor's formula at $x$, or at $p$, eliminate the contact coordinates through the pins, and expand in $\varepsilon$, or $\varepsilon'$. The coefficients have no negative power of $r$ or $\tau$. The random parts vanish at $\varepsilon = 0$, and the terms of order $\varepsilon$ of $G_2$ and $G_3$ are (S5.111); the terms of order $\varepsilon'$ of $\tilde G_2$ and $\tilde G_3$ are (S5.112), uniformly in $\tau \in [0, \frac{1}{4}]$. The pin parts of $G_1$ and $\tilde G_1$ are $\kappa(X^2 - \frac{1}{4}) + O(\varepsilon^2)$ and $\kappa(\tau + \chi\varsigma) + O(\varepsilon'^2)$. Those of $G_2$, $G_3$, $\tilde G_2$ and $\tilde G_3$ vanish: these components involve only derivatives with at least one transverse index, the coordinates of $P$ among them are determined by the transverse gradient pins, and those pins have target zero. The script `checks/s5_finite_scale_check.py` verifies these expansions for a polynomial field of degree six. The bounds on the means follow as in the proof of Lemma S5.40(c).

*The transverse pairs.* In (S5.111), if $\cos\theta \ne 0$ the second-order derivatives $f_{vv}$ and $f_{vw}$ occur one in each component; if $\cos\theta = 0$ the components are nonzero multiples of $f_{ttv}$ and $f_{ttw}$, because $|X^2 - \frac{1}{4}| = |X - \frac{1}{2}|\,|X + \frac{1}{2}| \ge \frac{\sqrt{3}}{16}$ when $\rho < \frac{1}{8}$ and both distances to the endpoints are at least $r/4$. The same holds in (S5.112), since $|\tau + \chi\varsigma| \ge \frac{3}{4}$. So, as in the proof of Lemma S5.40(b), the limits have a covariance given $P$ whose smallest eigenvalue is bounded below on the compact parameter sets, and the smallest eigenvalue of $\mathrm{Cov}(G_2, G_3 \mid \text{pins})$, respectively $\mathrm{Cov}(\tilde G_2, \tilde G_3 \mid \text{pins})$, is at least $c\varepsilon^2$, respectively $c\varepsilon'^2$.

*The densities.* Apply Lemma S5.39(d) with $Z_1 = G_1$, $Z' = (G_2, G_3)$ and $z = 0$. The mean of $Z'$ is $O(\varepsilon)$ and its covariance has smallest eigenvalue at least $c\varepsilon^2$, so $\mathcal{M}_{Z'} \le C$, and $\det \Sigma_{Z'} \ge c\varepsilon^4$. Also $\sigma_1 \le C\varepsilon$, so $|\tilde m - \kappa(X^2 - \frac{1}{4})| \le C\varepsilon$, and $|\tilde m| \ge \frac{\sqrt{3}}{32}\kappa$ for $\varepsilon \le \varepsilon_1$. Then (S5.108) gives (b). Part (c) is the same, with $\tilde G$ in place of $G$ and $|\tau + \chi\varsigma| \ge \frac{3}{4}$. We may take $\varepsilon_1$ and $c_*$ common to (b), (c) and Lemma S5.40.

*Away from the axis.* In (a), $G \equiv L + O(r)$, where $L = (L_1, L_\perp)$ are the contact limits of Lemma S5.16, whose covariance given $P$ has determinant $\rho^6(X^2 + \rho^2)A_{\mathrm{col}} \ge \rho_1^8 a_{\mathrm{col}}$ by Lemma S5.18. In (c), expand at $p$ instead of $x$: since $\nabla f(p) \equiv 0$, the random part of $\tilde G$ is a combination of derivatives at $p$ whose coefficients are polynomials in $r$, $\tau$, $\cos\alpha$ and $\sin\alpha$, plus a remainder $O(r)$ uniformly in $\tau \in [0, \frac{1}{4}]$; and derivatives at $p$ converge in $\mathcal{H}$ to those at $x$ as $r \to 0$. So $\mathrm{Cov}(\tilde G \mid \text{pins})$ is jointly continuous in $(r, \tau) \in [0, r_0] \times [0, \frac{1}{4}]$, for $\sin\alpha \ge \gamma_1$. At $r = 0$ the random part of $\tilde G$ is $\sin\alpha\big((\tau\cos\alpha + \frac{\chi}{2})f_{ttv} + \frac{\tau}{2}\sin\alpha\, f_{tvv},\ f_{vv},\ f_{vw}\big)$, which is $\tau^{-1}$ times that of the contact limit of $G$ at

$X = \frac{\chi}{2} + \tau\cos\alpha$, $\rho = \tau\sin\alpha$; here $f_{vv}$ and $f_{vw}$ occur once each, and the first component is a nonzero third-order combination, since $|\tau\cos\alpha + \frac{\chi}{2}| \geq \frac{1}{4}$. At $\tau = 0$ it is $(r^{-1}H_p[\hat{y}, t],\ H_p[\hat{y}, v],\ H_p[\hat{y}, w])$, with $\hat{y} = \cos\alpha\, t + \sin\alpha\, v$: three second derivatives at $p$ whose symbols $(\hat{y}\cdot\xi)\xi_t$, $(\hat{y}\cdot\xi)\xi_v$, $(\hat{y}\cdot\xi)\xi_w$ are independent of each other and of those of the pins. At $r, \tau > 0$ the evaluations sit at the three distinct points $M$, $S$ and $y$. In each case Theorem S1.3 gives positive definiteness. The family is continuous on a compact parameter set, which gives (a) and the first claim of (c). □

## S6. The near/far decomposition and the limit

This section proves the leading term of Theorem 2.1 of the main text, $\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3}(1 + o(1))$ as $\ell \downarrow 0$, with $c_{3,24}$ the coefficient (2.9) of the main text. The elder pairs are split at a fixed separation $\varrho$. The near part is pushed forward exactly to the lifetime variable (Proposition S6.1), and the far part is bounded (Proposition S6.2). Dominated convergence then gives the asymptotic under three hypotheses (Theorem S6.3). These hypotheses hold for the side-24 field (Proposition S6.4), and the limiting constant is the coefficient (2.9) of the main text (Corollary S6.5).

**Notation.** Throughout, $f = f_{24}$ is the normalized side-24 periodized Bargmann–Fock field on $\mathbb{T}^3_{24}$, and "density" is meant as in Convention 2.2 of the main text. The following objects are taken from earlier sections.

*Pairs* (Section S1). An ordered typed pair $(M, S)$ consists of a local maximum $M$ and an index-two critical point $S$ of $f$. For $0 < \varrho < 12$ the coordinates

$$M = x - \tfrac{r}{2}\,t, \qquad S = x + \tfrac{r}{2}\,t, \qquad x \in \mathbb{T}^3_{24}, \quad 0 < r < \varrho, \quad t \in S^2,$$

are a bijection onto the ordered pairs of points with $0 < \operatorname{dist}(M, S) < \varrho$, with $r = \operatorname{dist}(M, S)$, and

$$dM\, dS = dx\, r^2\, dr\, d\sigma_2(t); \tag{S6.1}$$

integration over the full directed sphere counts each ordered typed pair once. Almost surely $f$ is Morse with pairwise distinct critical values (Proposition 1.1 of the main text). Then each finite nonessential superlevel $H_0$ bar has exactly one creating maximum $M \neq M_*$, where $M_*$ is the global maximum, and one death saddle $D_f(M)$, and $M \mapsto (M, D_f(M))$ is a bijection from the non-global maxima onto the elder pairs.

*Palm law and elder mark* (Section S5). For $0 < r < 12$, $t \in S^2$, $b \in \mathbb{R}$ and $\kappa > 0$, the pins of the pair are $\Pi_{r,x,t} = (f(M), f(S), \nabla f(M), \nabla f(S)) \in \mathbb{R}^8$, and their target is $z_r(b, \kappa) = (b,\ b - \kappa r^3/6,\ 0,\ 0)$. Let $\mathbb{E}^0 = \mathbb{E}^0_{r,t,b,\kappa}$ be the regular conditional Gaussian law given $\Pi_{r,x,t} = z_r(b, \kappa)$, let

$$W_r = |\det H_M|\,|\det H_S|\,\mathbf{1}_{\{H_M \prec 0,\ \operatorname{ind}(H_S)=2\}}, \qquad Z_r = \mathbb{E}^0[W_r] > 0,$$

and let $\mathbb{E}^{MS}_r[F] = \mathbb{E}^0[W_r F]/Z_r$ be the determinant-weighted Palm law. The elder mark $e(f; M, S) \in \{0, 1\}$ is Borel measurable, and on the Morse locus it equals 1 if and only if $S = D_f(M)$. The selection probability is $p_r(t, b, \kappa) = \mathbb{E}^{MS}_r[e] \in [0, 1]$. With $p_\Pi$ the Gaussian density of $\Pi_{r,x,t}$, which by stationarity does not depend on $x$, put

$$\begin{aligned} K^{\mathrm{all}}_r &= p_\Pi(z_r(b,\kappa))\, Z_r, & K^{\mathrm{sel}}_r &= p_r\, K^{\mathrm{all}}_r, \\ G^{\mathrm{all}}_r &= r^4 K^{\mathrm{all}}_r, & G^{\mathrm{sel}}_r &= r^4 K^{\mathrm{sel}}_r, \end{aligned} \tag{S6.2}$$

which for each $0 < r < 12$ are nonnegative functions of the mark $(t, b, \kappa) \in S^2 \times \mathbb{R} \times (0, \infty)$.

*Contact variables* (Section S2). At fixed $(x, t, r, b)$ the saddle value $h = b - \kappa r^3/6$ satisfies $|dh| = \frac{r^3}{6}\, d\kappa$. The lifetime of the pinned pair is $\ell = \kappa r^3/6$. For fixed $\kappa > 0$ put $r_\ell(\kappa) = (6\ell/\kappa)^{1/3}$; then

$$r\, dr = \frac{6^{2/3}}{3}\,\kappa^{-2/3}\,\ell^{-1/3}\, d\ell. \tag{S6.3}$$

**The split.** Fix $\varrho \in (0, 12)$. Split the expected lifetime measure $\Lambda_{3,24}$ of Proposition 1.2 of the main text according as the elder pair's own separation is less than $\varrho$ or not: for Borel $A \subset (0, \infty)$,

$$\Lambda_\varrho^{\mathrm{near}}(A) = 24^{-3}\,\mathbb{E} \sum_{M \neq M_*} \mathbf{1}_{\{f(M)-f(D_f(M))\in A,\ \mathrm{dist}(M,D_f(M))<\varrho\}}, \tag{S6.4}$$

and $\Lambda_\varrho^{\mathrm{far}}(A)$ is defined in the same way with $\mathrm{dist}(M, D_f(M)) \geq \varrho$. Both measures are bounded by $\Lambda_{3,24}$, hence absolutely continuous, and their densities satisfy

$$\nu_{3,24}(\ell) = \nu_\varrho^{\mathrm{near}}(\ell) + \nu_\varrho^{\mathrm{far}}(\ell). \tag{S6.5}$$

In accordance with Convention 2.2 of the main text, pointwise statements below refer to the canonical marked Kac–Rice versions: (S6.7) for $\nu_\varrho^{\mathrm{near}}$, (S6.10) for $\nu_\varrho^{\mathrm{far}}$, and their sum for $\nu_{3,24}$, so that (S6.5) holds for every $\ell > 0$. The sum does not depend on $\varrho$. For $\varrho < \varrho' < 12$ and fixed $\ell$, the substitution $\kappa = 6\ell/r^3$ turns the part of (S6.7) for $\nu_{\varrho'}^{\mathrm{near}}$ with $\kappa \leq 6\ell/\varrho^3$ into $\int_\varrho^{\varrho'} r^2 \int_{S^2}\int_{\mathbb{R}} K_r^{\mathrm{sel}}(t, b, 6\ell/r^3)\, db\, d\sigma_2(t)\, dr$, which is the part of (S6.10) for $\nu_\varrho^{\mathrm{far}}$ with $\mathrm{dist}(x, y) < \varrho'$, written in the pair coordinates (S6.1).

**The near part.**

**Proposition S6.1** (exact near-density pushforward)**.** *Let $\varrho \in (0, 12)$. For every nonnegative Borel function $\varphi$ on $(0, \infty)$,*

$$\int_0^\infty \varphi(\ell)\, \nu_\varrho^{\mathrm{near}}(\ell)\, d\ell = \int_{S^2}\int_{\mathbb{R}}\int_0^\infty \int_0^\varrho \varphi\Big(\frac{\kappa r^3}{6}\Big) K_r^{\mathrm{sel}}(t, b, \kappa)\, \frac{r^3}{6}\, r^2\, dr\, d\kappa\, db\, d\sigma_2(t), \tag{S6.6}$$

*where the factor $r^3/6$ is the gap Jacobian $|dh| = (r^3/6)\, d\kappa$. The density version is, for $\ell > 0$,*

$$\nu_\varrho^{\mathrm{near}}(\ell) = \frac{6^{2/3}}{18}\, \ell^{-1/3} \int_{S^2}\int_{\mathbb{R}}\int_{6\ell/\varrho^3}^\infty \kappa^{-2/3}\, G^{\mathrm{sel}}_{r_\ell(\kappa)}(t, b, \kappa)\, d\kappa\, db\, d\sigma_2(t). \tag{S6.7}$$

*Proof.* By the bijection between non-global maxima and elder pairs, and since the elder mark equals 1 exactly on the elder pairs, almost surely

$$\sum_{M \neq M_*} \varphi(f(M) - f(D_f(M)))\, \mathbf{1}_{\{\mathrm{dist}(M,D_f(M))<\varrho\}} = \sum_{(M,S)} \varphi(f(M) - f(S))\, e(f; M, S),$$

where the sum on the right runs over the ordered typed pairs with $0 < \mathrm{dist}(M, S) < \varrho$; a pair with $e = 1$ has $f(S) < f(M)$. Multiply by $24^{-3}$ and take expectations. The left side becomes $\int \varphi\, d\Lambda_\varrho^{\mathrm{near}} = \int_0^\infty \varphi(\ell)\, \nu_\varrho^{\mathrm{near}}(\ell)\, d\ell$. For the right side, the marked two-point Kac–Rice formula for the ordered typed pair process with the bounded Borel mark $e$ (Proposition S12.22, on the compact sets $\{\delta \leq \mathrm{dist}(M, S) \leq \varrho - \delta\}$ and by monotone convergence as $\delta \downarrow 0$; the conditional expectation is the Gaussian-regression version, defined for every target and jointly measurable in $(r, t, b, \kappa)$) gives

$$24^{-3} \int_{\{0<\mathrm{dist}(M,S)<\varrho\}} \int_{\mathbb{R}} \int_{-\infty}^b \varphi(b - h)\, p_\Pi(b, h, 0, 0)\, \mathbb{E}\big[W_r\, e \mid \Pi = (b, h, 0, 0)\big]\, dh\, db\, dM\, dS, \tag{S6.8}$$

with $\Pi = (f(M), f(S), \nabla f(M), \nabla f(S))$ and $e = e(f; M, S)$. Pass to the pair coordinates by (S6.1). By stationarity the integrand does not depend on the midpoint $x$, and $24^{-3} \int_{\mathbb{T}^3_{24}} dx = 1$. At fixed $(t, r, b)$ substitute $h = b - \kappa r^3/6$: the range $h < b$ becomes $\kappa > 0$, $|dh| = (r^3/6)\, d\kappa$, $b - h = \kappa r^3/6$, and $(b, h, 0, 0) = z_r(b, \kappa)$. By the definition of the Palm law,

$$p_\Pi(z_r(b, \kappa))\, \mathbb{E}^0[W_r\, e] = p_\Pi(z_r(b, \kappa))\, Z_r\, \mathbb{E}_r^{MS}[e] = p_r\, K_r^{\mathrm{all}} = K_r^{\mathrm{sel}}.$$

This proves (S6.6).

For (S6.7), write $K_r^{\mathrm{sel}}\, \frac{r^3}{6}\, r^2 = (r^4 K_r^{\mathrm{sel}})\, \frac{r}{6} = G_r^{\mathrm{sel}}\, \frac{r}{6}$. For fixed $\kappa > 0$ the map $r \mapsto \ell = \kappa r^3/6$ is an increasing bijection of $(0, \varrho)$ onto $(0, \kappa\varrho^3/6)$ with inverse $r = r_\ell(\kappa)$, and (S6.3) holds. The

constraint $\ell < \kappa\varrho^3/6$ is $\kappa > 6\ell/\varrho^3$, and $\frac{1}{6}\cdot\frac{6^{2/3}}{3} = \frac{6^{2/3}}{18}$. All integrands are nonnegative, so Tonelli's theorem gives

$$\int_0^\infty \varphi(\ell)\,\nu_\varrho^{\mathrm{near}}(\ell)\,d\ell = \int_0^\infty \varphi(\ell)\Big[\frac{6^{2/3}}{18}\,\ell^{-1/3}\int_{S^2}\int_{\mathbb{R}}\int_{6\ell/\varrho^3}^\infty \kappa^{-2/3}\,G^{\mathrm{sel}}_{r_\ell(\kappa)}(t,b,\kappa)\,d\kappa\,db\,d\sigma_2(t)\Big]d\ell$$

for every nonnegative Borel $\varphi$. Hence the bracket is a density of $\Lambda_\varrho^{\mathrm{near}}$; it is the version obtained from the pair coordinates, which is (S6.7). □

*The kernel in Part II's notation.* In the notation of Part II (Convention S8.1), with $u = t$ and gap mark $k = \kappa/6$,

$$G_r^{\mathrm{all}}(t,b,\kappa) = \mathcal{A}_r(b,\kappa/6,t), \qquad G_r^{\mathrm{sel}}(t,b,\kappa) = p_r(t,b,\kappa)\,\mathcal{A}_r(b,\kappa/6,t) = \mathcal{A}_r^{\mathrm{eld}}(b,\kappa/6,t).$$

Indeed, the observation vector $U_r$ of Part II is the image of the pins $\Pi_{r,x,t}$ under a linear map of absolute determinant $12r^{-6}$ that carries $z_r(b,\kappa)$ to $v_r$, so $p_\Pi(z_r(b,\kappa)) = 12r^{-6}\pi_r(v_r)$. The two conditional laws coincide, so $Z_r = \mathbb{E}_Q[W_r]$, and $G_r^{\mathrm{all}} = r^4 p_\Pi(z_r)Z_r = 12\pi_r(v_r)\mathbb{E}_Q[W_r/r^2] = \mathcal{A}_r$. Finally $p_r = \mathbb{E}_Q[W_r e]/\mathbb{E}_Q[W_r]$ gives the selected identity. So the near-density version (S6.7) and the near-pair formulas of Sections S7–S10 are built from the same function.

**The far part.**

**Proposition S6.2** (far elder pairs)**.** *Let $\varrho \in (0,12)$. For every $\ell_0 \in (0,\infty)$,*

$$\sup_{0<\ell\le\ell_0} \nu_\varrho^{\mathrm{far}}(\ell) < \infty. \tag{S6.9}$$

*In particular $\nu_\varrho^{\mathrm{far}}(\ell) = O(1) = o(\ell^{-1/3})$ as $\ell \downarrow 0$.*

*Proof.* Put $\mathcal{T}_\varrho = \{(x,y) \in (\mathbb{T}^3_{24})^2 : \operatorname{dist}(x,y) \ge \varrho\}$, $\Xi_{x,y} = (f(x), f(y), \nabla f(x), \nabla f(y))$, $\xi_\ell(b) = (b,\, b-\ell,\, 0,\, 0)$ and $\mathbf{1}_{\mathrm{types}} = \mathbf{1}_{\{H_x \prec 0,\ \mathrm{ind}(H_y)=2\}}$. The marked two-point Kac–Rice formula for the ordered typed pairs $(M,S) = (x,y)$ with $(x,y) \in \mathcal{T}_\varrho$, with the bounded Borel mark $e$ (Proposition S12.22 with $\Omega = \mathcal{T}_\varrho$), followed by the change of variables $(b,h) \mapsto (b,\ell) = (b, b-h)$, which has unit Jacobian, gives the version

$$\nu_\varrho^{\mathrm{far}}(\ell) = 24^{-3}\int_{\mathcal{T}_\varrho}\int_{\mathbb{R}} p_{\Xi_{x,y}}(\xi_\ell(b))\,\mathbb{E}\big[|\det H_x \det H_y|\,\mathbf{1}_{\mathrm{types}}\,e \;\big|\; \Xi_{x,y} = \xi_\ell(b)\big]\,db\,dx\,dy, \tag{S6.10}$$

with $e = e(f;x,y)$. Since $e \le 1$, $\nu_\varrho^{\mathrm{far}}(\ell)$ is bounded by the all-typed ordered pair density at height gap $\ell$ on $\mathcal{T}_\varrho$:

$$\nu_\varrho^{\mathrm{far}}(\ell) \le 24^{-3}\int_{\mathcal{T}_\varrho}\int_{\mathbb{R}} p_{\Xi_{x,y}}(\xi_\ell(b))\,\mathbb{E}\big[|\det H_x \det H_y|\,\mathbf{1}_{\mathrm{types}} \;\big|\; \Xi_{x,y} = \xi_\ell(b)\big]\,db\,dx\,dy. \tag{S6.11}$$

This is the marked two-point Kac–Rice formula on a compact separated configuration set. By full-spectrum Gram positivity of the side-24 covariance (Section S1), the value and gradient evaluations at the distinct points $x \ne y$ are linearly independent, so the covariance of $\Xi_{x,y}$ is positive definite on $\mathcal{T}_\varrho$, and by compactness its smallest eigenvalue is bounded below there. Its largest eigenvalue is bounded above, since its entries are derivatives of $K_{24}$ of order at most two. The mean-zero target satisfies $|\xi_\ell(b)|^2 = b^2 + (b-\ell)^2 \ge b^2/2$. Hence

$$p_{\Xi_{x,y}}(\xi_\ell(b)) \le C_\varrho\, e^{-c_\varrho b^2}, \qquad (x,y) \in \mathcal{T}_\varrho,\ 0 < \ell \le \ell_0. \tag{S6.12}$$

Gaussian regression against this nondegenerate block gives conditional Hessian means affine in $(b, b-\ell)$ and uniformly bounded conditional covariances. Since $\det H_x \det H_y$ is a polynomial of degree six in the Hessian entries, the degree-six moment bound gives

$$\mathbb{E}\big[|\det H_x \det H_y| \;\big|\; \Xi_{x,y} = \xi_\ell(b)\big] \le C_\varrho(1 + (|b| + \ell)^6). \tag{S6.13}$$

Insert (S6.12) and (S6.13) in (S6.11). This bounds the density slice at $\ell$, uniformly in $0 < \ell \le \ell_0$:

$$\nu_\varrho^{\mathrm{far}}(\ell) \le C_\varrho \int_{\mathbb{R}} (1 + (|b| + \ell_0)^6) e^{-c_\varrho b^2}\, db \cdot 24^{-3}\, |\mathcal{T}_\varrho|. \tag{S6.14}$$

The right side is finite and does not depend on $\ell$, since the separated pair domain has finite volume $|\mathcal{T}_\varrho| \le 24^6$. This proves (S6.9), and then $\ell^{1/3}\nu_\varrho^{\mathrm{far}}(\ell) \to 0$ as $\ell \downarrow 0$. □

The proof bounds the elder mark by one and so controls all typed far pairs at height gap $\ell$, elder or not; it gives boundedness and no rate in $\ell$. For $0 < \ell \le 1$, the same bound, for elder pairs and for all typed far pairs, is (14.1) of [7] (Lemma S12.29). There it is stated for the pairs at distance at least the small radius $r_0$ of §S12; its proof is the one above, at every separation.

**The limit.**

**Theorem S6.3** (limit interchange)**.** *Let $\varrho \in (0, 12)$, and assume the following.*

*(H1) There is a function $G_0^{\mathrm{sel}}$ on $S^2 \times \mathbb{R} \times (0, \infty)$ with $G_r^{\mathrm{sel}} \to G_0^{\mathrm{sel}}$ pointwise as $r \downarrow 0$.*

*(H2) There is one integrable majorant for $\kappa^{-2/3} G_r^{\mathrm{sel}}$ on the whole mark space, including $\kappa \downarrow 0$: a function $\bar{G} \in L^1(S^2 \times \mathbb{R} \times (0, \infty),\, d\sigma_2(t)\, db\, d\kappa)$ with $\kappa^{-2/3} G_r^{\mathrm{sel}}(t, b, \kappa) \le \bar{G}(t, b, \kappa)$ for all $0 < r < \varrho$ and all $(t, b, \kappa)$.*

*(H3) The selected and all-typed limiting integrands coincide: $G_r^{\mathrm{all}} \to G_0^{\mathrm{all}}$ pointwise as $r \downarrow 0$, and $G_0^{\mathrm{sel}} = G_0^{\mathrm{all}}$.*

*Then the constant*

$$c_{3,24} = \frac{6^{2/3}}{18} \int_{S^2} \int_{\mathbb{R}} \int_0^\infty \kappa^{-2/3}\, G_0^{\mathrm{all}}(t, b, \kappa)\, d\kappa\, db\, d\sigma_2(t) \tag{S6.15}$$

*is finite, and as $\ell \downarrow 0$*

$$\ell^{1/3}\, \nu_\varrho^{\mathrm{near}}(\ell) \to c_{3,24}, \qquad \ell^{1/3}\, \nu_{3,24}(\ell) \to c_{3,24}. \tag{S6.16}$$

*In particular, if $c_{3,24} > 0$, then $\nu_\varrho^{\mathrm{near}}(\ell) = c_{3,24}\, \ell^{-1/3}(1 + o(1))$ and $\nu_{3,24}(\ell) = c_{3,24}\, \ell^{-1/3}(1 + o(1))$.*

*Proof.* By (H1)–(H3), $\kappa^{-2/3} G_0^{\mathrm{all}} = \lim_{r \downarrow 0} \kappa^{-2/3} G_r^{\mathrm{sel}} \le \bar{G}$ pointwise, so $c_{3,24} \le \frac{6^{2/3}}{18} \int \bar{G} < \infty$. By (S6.7),

$$\ell^{1/3}\, \nu_\varrho^{\mathrm{near}}(\ell) = \frac{6^{2/3}}{18} \int_{S^2} \int_{\mathbb{R}} \int_0^\infty \mathbf{1}_{\{\kappa > 6\ell/\varrho^3\}}\, \kappa^{-2/3}\, G_{r_\ell(\kappa)}^{\mathrm{sel}}(t, b, \kappa)\, d\kappa\, db\, d\sigma_2(t).$$

Fix $(t, b, \kappa)$ with $\kappa > 0$. As $\ell \downarrow 0$ the lower limit $6\ell/\varrho^3$ decreases to 0, so the indicator is eventually 1, and $r_\ell(\kappa) = (6\ell/\kappa)^{1/3} \downarrow 0$; by (H1) the integrand tends to $\kappa^{-2/3} G_0^{\mathrm{sel}}(t, b, \kappa)$. Where the indicator is 1, $r_\ell(\kappa) < \varrho$, so by (H2) the integrand is at most $\bar{G}(t, b, \kappa)$. Dominated convergence, applied along an arbitrary sequence $\ell_n \downarrow 0$, gives $\ell^{1/3} \nu_\varrho^{\mathrm{near}}(\ell) \to \frac{6^{2/3}}{18} \int \kappa^{-2/3} G_0^{\mathrm{sel}}$, and by (H3) the limit is $c_{3,24}$. By Proposition S6.2, $\ell^{1/3} \nu_\varrho^{\mathrm{far}}(\ell) \to 0$, and (S6.5) gives the second limit in (S6.16). If $c_{3,24} > 0$, the two limits are the stated asymptotics. □

**Proposition S6.4** (the hypotheses for the side-24 field)**.** *Let $\varrho \in (0, 12)$. For the side-24 field, (H1)–(H3) of Theorem S6.3 hold, with $G_0^{\mathrm{sel}} = G_0^{\mathrm{all}}$. More precisely, there are constants $C, c > 0$ and an integer $N$, depending on $\varrho$, such that for all $0 < r < \varrho$ and all $(t, b, \kappa) \in S^2 \times \mathbb{R} \times (0, \infty)$*

$$0 \le \kappa^{-2/3} G_r^{\mathrm{sel}}(t, b, \kappa) \le \kappa^{-2/3} G_r^{\mathrm{all}}(t, b, \kappa) \le C\big[1 + (|b| + \kappa)^N\big] e^{-c(b^2 + \kappa^2)}\, \kappa^{-2/3}, \tag{S6.17}$$

*and the right side is integrable on $S^2 \times \mathbb{R} \times (0, \infty)$, including $\kappa \downarrow 0$. Moreover $0 < c_{3,24} < \infty$.*

*Proof. Corrected pins.* Let $Y_r$ be the corrected divided-difference pin vector of the pair ((3.7)–(3.8) of the main text), the image of $\Pi_{r,x,t}$ under the raw-to-corrected linear map, whose determinant has absolute value $r^{-6}$ (Section S2). On the target $z_r(b, \kappa)$,

$$Y_r = y_r(b, \kappa) := \Big(b - \frac{\kappa r^3}{12},\, -\frac{\kappa}{6},\, 0, \ldots, 0\Big) \in \mathbb{R}^8. \tag{S6.18}$$

Let $p_r^{\mathrm{corr}}(t,b,\kappa)$ be the density of $Y_r$ at $y_r(b,\kappa)$, so that $p_\Pi(z_r(b,\kappa)) = r^{-6}\,p_r^{\mathrm{corr}}(t,b,\kappa)$, and put $\zeta_r = r^{-2}Z_r$. By (S6.2),

$$G_r^{\mathrm{all}} = r^4 \cdot r^{-6} p_r^{\mathrm{corr}} \cdot r^2 \zeta_r = p_r^{\mathrm{corr}}\,\zeta_r. \tag{S6.19}$$

By Section S2, as $r \downarrow 0$, locally uniformly on compact positive mark sets, $p_r^{\mathrm{corr}} \to p_0^{\mathrm{corr}}$, the density of the limiting corrected pin vector at $(b,-\kappa/6,0,0)$, and $\zeta_r \to \zeta_0 = \kappa^2\,\mathbb{E}[(\det Q)^2 \mathbf{1}_{\{Q\prec 0\}} \mid \text{limiting pins}]$, where $Q$ is the transverse Hessian at contact. Moreover $Z_r \geq c_Z r^2$ on each compact positive mark set, with $c_Z > 0$ depending on the set, so $\zeta_0 \geq c_Z > 0$ there.

*(H1) and (H3).* By (S6.19), $G_r^{\mathrm{all}} \to G_0^{\mathrm{all}} := p_0^{\mathrm{corr}}\,\zeta_0$ at every point of $S^2 \times \mathbb{R} \times (0,\infty)$, since every point with $\kappa > 0$ lies in a compact positive mark set. By Theorem S12.3 at $d = 3$ and $L = 24$, applied with gap mark $k = \kappa/6$, gap window $\{\kappa/6 : \kappa \in I_\kappa\}$ for a compact $I_\kappa \subset (0,\infty)$ and any orthonormal frame $R$ with $Re_1 = t$ (the conversion $k = \kappa/6$ of the paragraph *The kernel in Part II's notation* before Proposition S6.2), and whose selection probability is $p_r$ (Remark S5.37), and also by Theorem S5.36, $1 - p_r \leq Cr^3$ uniformly for $t \in S^2$ and $(b,\kappa)$ in a compact subset of $\mathbb{R} \times (0,\infty)$, so $p_r \to 1$ pointwise. Hence $G_r^{\mathrm{sel}} = p_r\,G_r^{\mathrm{all}} \to G_0^{\mathrm{all}}$. This is (H1) with $G_0^{\mathrm{sel}} = G_0^{\mathrm{all}}$, which is (H3).

*(H2).* The first two inequalities in (S6.17) hold because $0 \leq p_r \leq 1$. For the third we bound the two factors in (S6.19).

Let $\Sigma_r(t)$ be the covariance of $Y_r$; its eigenvalues do not depend on the choice of transverse frame. For $0 < r < 12$ the points $M$ and $S$ are distinct, so the raw pins are nondegenerate by full-spectrum Gram positivity (Section S1), and $\Sigma_r(t)$ is positive definite because the corrected map is invertible. As $r \downarrow 0$, $\Sigma_r(t)$ extends continuously to the covariance of the limiting corrected pins, which is positive definite (Sections S1 and S2). By compactness of $[0,\varrho] \times S^2$ there are $0 < \lambda_0 \leq \lambda_1 < \infty$ with $\lambda_0 I \leq \Sigma_r(t) \leq \lambda_1 I$ for $0 < r \leq \varrho$ and $t \in S^2$. The corrected pair target is uniformly injective in $(b,\kappa)$: with $\epsilon = r^3/12 \leq \varrho^3/12$, the quadratic form $|y_r(b,\kappa)|^2 = b^2 - 2\epsilon b\kappa + (\epsilon^2 + \frac{1}{36})\kappa^2$ has determinant $\frac{1}{36}$ and trace at most $1 + \frac{1}{36} + \frac{\varrho^6}{144}$, so

$$|y_r(b,\kappa)|^2 \geq c_\varrho\,(b^2+\kappa^2), \qquad c_\varrho = \frac{1/36}{1 + 1/36 + \varrho^6/144}.$$

Therefore

$$p_r^{\mathrm{corr}}(t,b,\kappa) \leq (2\pi)^{-4}\lambda_0^{-4}\exp\Big(-\frac{|y_r(b,\kappa)|^2}{2\lambda_1}\Big) \leq C\,e^{-c(b^2+\kappa^2)}.$$

For the weight, under the pins $\det H_M = r\,D_M$ and $\det H_S = r\,D_S$ exactly, where $D_M$ and $D_S$ are polynomials of degree three, with coefficients bounded for $0 < r \leq \varrho$, in the transverse Hessian entries at $M$ and $S$ and in scaled variables that are linear functionals of the third-derivative process on the segment from $M$ to $S$, with uniformly bounded variances (Section S2). Hence $\zeta_r = \mathbb{E}^0\big[|D_M D_S|\,\mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S)=2\}}\big]$. Under $\mathbb{E}^0$ these variables are Gaussian. Gaussian regression on the corrected pins, whose covariance is at least $\lambda_0 I$, gives conditional means that are linear in $y_r(b,\kappa)$ with coefficients bounded for $0 < r \leq \varrho$, and conditional covariances bounded by the unconditional ones. Since $D_M D_S$ has degree six and $|y_r(b,\kappa)| \leq C(|b|+\kappa)$, bounded conditional polynomial moments give $\zeta_r \leq C\big[1 + (|b|+\kappa)^6\big]$. With (S6.19), this is the third inequality in (S6.17), with $N = 6$.

*Integrability.* Since $1 + (|b|+\kappa)^N \leq 2^{N-1}(1+|b|^N)(1+\kappa^N)$, the integral of the right side of (S6.17) over $S^2 \times \mathbb{R} \times (0,\infty)$ is at most a constant times

$$4\pi \int_{\mathbb{R}} (1+|b|^N)\,e^{-cb^2}\,db \int_0^\infty (1+\kappa^N)\,\kappa^{-2/3} e^{-c\kappa^2}\,d\kappa < \infty :$$

the singularity $\kappa^{-2/3}$ produced by the lifetime pushforward is integrable at $\kappa = 0$, and the Gaussian factor controls $|b| \to \infty$ and $\kappa \to \infty$. So (H2) holds with $\bar{G}$ the right side of (S6.17). In particular

one majorant controls every neighbourhood of $\kappa = 0$, where the cubic contact degenerates; the degenerate (quartic) contacts themselves have $\kappa = 0$, a null set of marks.

*The constant.* By (H2), $c_{3,24} < \infty$ (Theorem S6.3). At every point of $S^2 \times \mathbb{R} \times (0,\infty)$, $p_0^{\mathrm{corr}} > 0$, a nondegenerate Gaussian density, and $\zeta_0 > 0$, so $G_0^{\mathrm{all}} > 0$ and (S6.15) gives $c_{3,24} > 0$. □

**Corollary S6.5** (the selected coefficient)**.** *For $0 < r < 12$ let $\psi_r = \frac{6^{2/3}}{18}\,\kappa^{-2/3} G_r^{\mathrm{all}}$ be the nonnegative all-typed rescaled contact integrand on $S^2 \times \mathbb{R} \times (0,\infty)$; the selected integrand is $p_r\,\psi_r = \frac{6^{2/3}}{18}\,\kappa^{-2/3} G_r^{\mathrm{sel}}$. Then, with integrals over $S^2 \times \mathbb{R} \times (0,\infty)$ with respect to $d\kappa\,db\,d\sigma_2(t)$,*

$$\lim_{r\downarrow 0} \int p_r\,\psi_r = \lim_{r\downarrow 0} \int \psi_r = c_{3,24}, \tag{S6.20}$$

*and the constant (S6.15) is the coefficient (2.9) of the main text. Therefore the exact finite-side coefficient (2.9) is the selected short-lifetime coefficient of the finite nonessential superlevel $H_0$ bars:*

$$\nu_{3,24}(\ell) = c_{3,24}\,\ell^{-1/3}(1 + o(1)), \qquad \ell \downarrow 0. \tag{S6.21}$$

*Proof.* Fix $\varrho \in (0,12)$. By Proposition S6.4, $0 \le p_r \le 1$, $p_r \to 1$ and $\psi_r \to \frac{6^{2/3}}{18}\kappa^{-2/3} G_0^{\mathrm{all}}$ pointwise, and for $0 < r < \varrho$ the function $\frac{6^{2/3}}{18}$ times the right side of (S6.17) is an integrable majorant common to $p_r\psi_r$ and $\psi_r$. Dominated convergence gives both limits in (S6.20), equal to (S6.15).

It remains to evaluate (S6.15). Fix $t \in S^2$, a transverse frame, and the contact variables $u_t, g_t, h_t, q_t$ of (2.2)–(2.4) of the main text. The limiting corrected pin vector is $(f, u_t, g_t, h_t)$ with target $(b, -\kappa/6, 0, 0)$, and $Q = Q_t(q_t)$. Because $K_{24}$ is even, $(f, h_t, q_t)$ is independent of $(u_t, g_t)$. Hence

$$p_0^{\mathrm{corr}} = p_{(f,h_t)}(b,0)\,p_{(u_t,g_t)}(-\kappa/6,0), \qquad \zeta_0 = \kappa^2\,\mathbb{E}\big[F(q_t) \mid f = b,\ h_t = 0\big],$$

with $F(q) = (\det Q_t(q))^2 \mathbf{1}_{\{Q_t(q)<0\}}$ as in (3.20) of the main text. The $b$-integral is (3.20) of the main text, and the $\kappa$-integral, with $\kappa^{-2/3}\kappa^2 = \kappa^{4/3}$, is (3.21) of the main text:

$$\int_{\mathbb{R}} \int_0^\infty \kappa^{-2/3}\,G_0^{\mathrm{all}}\,d\kappa\,db = p_{g_t}(0)\,\frac{\Gamma(7/6)\,72^{7/6}\,(\sigma_t^2)^{2/3}}{2\sqrt{2\pi}}\,p_{h_t}(0)\,D_t,$$

where $p_{g_t}(0) = (2\pi)^{-3/2}(\det \Sigma_g(t))^{-1/2}$ and $p_{h_t}(0) = (2\pi)^{-3/2}(\det \Sigma_h(t))^{-1/2}$. Since

$$\frac{6^{2/3}}{18} \cdot \frac{72^{7/6}}{2\sqrt{2\pi}} \cdot (2\pi)^{-3} = \frac{3}{2^{4/3}\pi^{7/2}},$$

integration over $t \in S^2$ turns (S6.15) into (2.9) of the main text, with $B_3 = 3\Gamma(7/6)/(2^{4/3}\pi^{7/2})$. Finally, (S6.21) follows from Theorem S6.3 and Proposition S6.4, since $0 < c_{3,24} < \infty$. □

## S7. A bounded remainder

This section proves that the remainder in the leading short-lifetime law is bounded, in every fixed dimension $d \ge 2$ and for every fixed side $L > 0$ (Theorem S7.9). At $d = 3$ and $L = 24$ this is Proposition 3.12 of the main text, and it gives the rate $O(\ell^{1/3})$ in (2.1) of the main text (Remark S7.22). The proof combines statements of the companion paper [7], which Section S12 states and proves, collected in Lemmas S7.1–S7.3, Hypotheses S7.4–S7.7 and Definition S7.8, with three estimates proved here: a second-order coupling of the pinned laws at separation $r$ and at contact (Lemmas S7.11–S7.13), a quadratic determinant cancellation for the $\sqrt{r}$ off-diagonal block of the rescaled endpoint Hessians (Lemma S7.15), and an unnormalized bound for the loss of elder pairing with explicit dependence on the gap mark (Lemma S7.18).

**Setting and notation.** Fix $d \geq 2$ and $L > 0$, and put $m = d-1$. Let $f$ be the centered stationary Gaussian field on the torus $\mathbb{T}_L^d = \mathbb{R}^d/(L\mathbb{Z}^d)$ with covariance

$$K_L(z) = \frac{\sum_{n\in\mathbb{Z}^d} \exp(-|z+Ln|^2/2)}{\sum_{n\in\mathbb{Z}^d} \exp(-|Ln|^2/2)}. \tag{S7.1}$$

All constants in this section may depend on $d$ and $L$; they need not be uniform in $d$ or in $L$, and no statement about an infinite-volume field is made. Constants $C < \infty$ and $c > 0$ may change from line to line.

*Pins.* We parametrize the gap by $k = \kappa/6$, where $\kappa$ is the gap mark of the main text, so that the pins (3.5) of the main text become

$$f(M) = b, \qquad f(S) = b - kr^3, \qquad \nabla f(M) = \nabla f(S) = 0, \tag{S7.2}$$

with $b \in \mathbb{R}$, $k > 0$ and $r > 0$, and the lifetime of the pair is $\ell = kr^3$. By stationarity the pair midpoint is placed at the origin. For an orthonormal frame $R = (t, n_1, \dots, n_m)$ of $\mathbb{R}^d$ with axial vector $t \in S^{d-1}$, put $M = -(r/2)t$ and $S = (r/2)t$, write points as $xt + \sum_j y_j n_j$, and denote derivatives in the frame by the subscripts $x$ and $y_j$; $\nabla_y$ and $D_y^2$ are the transverse gradient and the transverse $m \times m$ Hessian. The index of a symmetric matrix is its number of negative eigenvalues, and $\|\cdot\|$ is the operator norm.

*Contact observations.* For $r > 0$ let $U_r$ be the vector whose first four entries are the axial rows

$$\begin{gathered} \frac{f(M)+f(S)}{2}, \qquad \frac{f(S)-f(M)}{r}, \qquad \frac{f_x(S)-f_x(M)}{r}, \\ \frac{6}{r^2}\Big[f_x(M) + f_x(S) - \frac{2(f(S)-f(M))}{r}\Big], \end{gathered} \tag{S7.3}$$

followed, for $j = 1, \dots, m$, by the two transverse rows $\frac{1}{2}\big(f_{y_j}(M) + f_{y_j}(S)\big)$ and $\big(f_{y_j}(S) - f_{y_j}(M)\big)/r$. At $r = 0$ put

$$U_0 = \big(f,\ f_x,\ f_{xx},\ f_{xxx},\ f_{y_1},\ f_{xy_1},\ \dots,\ f_{y_m},\ f_{xy_m}\big)(0). \tag{S7.4}$$

The pins (S7.2) hold if and only if $U_r = v_r$, and at contact we impose $U_0 = v_0$, where

$$v_r = \big(b - kr^3/2,\ -kr^2,\ 0,\ 12k,\ 0, \dots, 0\big), \qquad v_0 = \big(b,\ 0,\ 0,\ 12k,\ 0, \dots, 0\big). \tag{S7.5}$$

For $r \geq 0$ write $\Sigma_r = \mathrm{Cov}(U_r)$, $\mathcal{C}_r(z) = \mathrm{Cov}(f(z), U_r)$ for $z \in \mathbb{T}_L^d$, and $\pi_r$ for the Lebesgue density of $U_r$. These depend on $r$ and on the frame, but not on $b$ or $k$.

*Pinned laws and selection.* For $r > 0$, $\mathbb{E}^0 = \mathbb{E}^0_{r,b,k,R}$ is the continuous Gaussian regression law of $f$ given $U_r = v_r$, that is, given the $2(d+1)$ observations in (S7.2); it is a regular conditional law, not a conditioning on an event of positive probability. Likewise $\mathbb{E}^0_{0,b,k,R}$ is the regression law of $f$ given $U_0 = v_0$. With $H_z = D^2 f(z)$ put

$$W_r = |\det H_M|\,|\det H_S|\,\mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}(H_S) = d-1\}}, \qquad Z_r = \mathbb{E}^0[W_r]. \tag{S7.6}$$

$Z_r$ is the full normalizer; it carries no further pin density, Jacobian or adjacency condition. When $0 < Z_r < \infty$ let $\mathbb{E}_r^{MS}[F] = \mathbb{E}^0[W_r F]/Z_r$ and $p_r = p_r(b,k,t) = \mathbb{E}_r^{MS}[e]$, where $e = e(f; M, S)$ is the elder mark of Hypothesis S7.4. Put

$$z_0(b,k,t) = (6k)^2\, \mathbb{E}^0_{0,b,k,R}\Big[\big(\det D_y^2 f(0)\big)^2\, \mathbf{1}_{\{D_y^2 f(0) \prec 0\}}\Big], \tag{S7.7}$$

$$\mathcal{A}_r(b,k,t) = 12\,\pi_r(v_r)\,\frac{Z_r}{r^2} \quad (r > 0), \qquad \mathcal{A}_0(b,k,t) = 12\,\pi_0(v_0)\,z_0(b,k,t). \tag{S7.8}$$

These quantities are invariant under rotations of the transverse frame, so they are functions of $t$: for $\mathcal{A}_r$ this is [7, §10] (Proposition S12.23), and for $\pi_0(v_0)$ and $z_0$ it holds because the contact pins $U_0 = v_0$ and the integrand in (S7.7) are invariant under such rotations.

*Partial-block norms and the cap event.* For $r > 0$ let $\mathcal{D}_r = [-2r, 2r] \times \bar{B}(0, 2r)$ in the frame coordinates $(x, y) \in \mathbb{R} \times \mathbb{R}^m$. For $j = 3, 4$ put

$$M_j = \max_{0 \le i \le j} \sup_{\mathcal{D}_r} \left\| \partial_x^{j-i} D_y^i f \right\|_{\mathrm{op}}, \tag{S7.9}$$

where the norm is the absolute value for $i = 0$, and for $i \ge 1$

$$\left\| \partial_x^{j-i} D_y^i f \right\|_{\mathrm{op}} = \sup_{w_1, \dots, w_i} \left| \partial_x^{j-i} D_y^i f\,[w_1, \dots, w_i] \right|,$$

the supremum being over transverse unit vectors $w_1, \dots, w_i$ chosen independently of each other. The cap event is

$$\mathcal{G}_r = \Big\{ \lambda_{\min}(-D_y^2 f(M)) > \frac{4}{3k}\, r\, M_3^2 \Big\} \cap \Big\{ r M_4 \le \frac{3k}{10} \Big\}. \tag{S7.10}$$

*Densities.* For $\ell > 0$, $\nu_{\mathrm{cand}}(\ell)$ is the $\ell$-density, per unit volume, of the expected number of *candidate pairs*: ordered pairs $(M, S)$ of a local maximum $M$ and a critical point $S$ of index $d - 1$ with $f(M) - f(S) = \ell$. Further, $\nu_{\mathrm{eld}}(\ell)$ is the density of the finite ordinary superlevel $H_0$ bars with lifetime $\ell$, the essential class of the global maximum being excluded. All birth heights, gap marks, orientations and separations on $\mathbb{T}_L^d$ are included. We write $\nu^{\mathrm{near}}$ and $\nu^{\mathrm{far}}$ for the contributions of pairs at torus distance at most $r_0$ and larger than $r_0$, where $r_0$ is fixed in (S7.12); their versions are fixed in Hypotheses S7.6 and S7.7.

**Statements used from the companion paper.** The proof uses the following statements of [7], exactly as they are stated and proved there. Section S12 states and proves them in this supplement, and each statement below names its counterpart there. Lemmas S7.1–S7.3 are analytic facts about the field and about functions satisfying the pins; Hypotheses S7.4–S7.7 are the genericity, elder-pairing and Kac–Rice statements; Definition S7.8 defines the coefficient.

**Lemma S7.1** ([7, §2]; Lemmas S12.7 and S12.8)**.** (a) $K_L(z) = \sum_{n \in \mathbb{Z}^d} a_n\, e^{2\pi i\, n \cdot z / L}$ *with*

$$a_n = \frac{\exp(-2\pi^2 |n|^2 / L^2)}{\sum_{j \in \mathbb{Z}^d} \exp(-2\pi^2 |j|^2 / L^2)} > 0, \qquad \sum_{n \in \mathbb{Z}^d} \sqrt{a_n}\, (1 + |n|)^q < \infty \quad \text{for every } q \ge 0. \tag{S7.11}$$

*Consequently the real Fourier series of $f$ converges in every $C^q$ norm in every finite $L^p$ norm, $f$ has a smooth version, and the global derivative suprema $\|f\|_{C^q(\mathbb{T}_L^d)}$ have finite moments of every order.*

(b) *Every finite list of distinct derivative-evaluation functionals $\varphi \mapsto \partial^{\mathbf{j}} \varphi(z)$, at one or several points $z$ and in any orthonormal frame, is linearly independent: its covariance matrix under $f$ is positive definite. Here a Hessian is represented by its independent symmetric entries. For several sites in a rotated frame this follows from the proof of Lemma S12.8, which reduces a combination of vanishing variance to the zero distribution and then argues site by site.*

**Lemma S7.2** (contact frame; [7, §3, (13.1) and (13.2)]; Lemmas S12.10, S12.11 and S12.26)**.** (a) *For $r > 0$, $U_r$ is an invertible linear image of*

$$\big(f(M), f_x(M), f(S), f_x(S), f_{y_1}(M), f_{y_1}(S), \dots, f_{y_m}(M), f_{y_m}(S)\big),$$

*and the transformation has absolute determinant $12\, r^{-(d+3)}$.*

(b) *There is $r_1 > 0$ such that, for $0 \le r \le r_1$ and every frame, the eigenvalues of $\Sigma_r$, and those of the covariance matrix of $U_r$ extended by the $m(m+1)/2$ independent entries of $D_y^2 f(M)$ (of $D_y^2 f(0)$ when $r = 0$), lie in a fixed compact subinterval of $(0, \infty)$.*

(c) *For $0 < r \le \min\{r_1, 1\}$, $b \in \mathbb{R}$, $k > 0$ and every frame, $|v_r| \le C(|b| + k)$ and $|v_r|^2 \ge c(b^2 + k^2)$; consequently $\pi_r(v_r) \le C \exp[-c(b^2 + k^2)]$.*

Fix

$$0 < r_0 \le \min\{1, r_1\}, \qquad r_0 < \frac{L}{4\sqrt{2}}. \tag{S7.12}$$

The cylinder $\mathcal{D}_r$ lies in the closed ball of radius $2\sqrt{2}\,r$ about the midpoint, and the exponential chart of $\mathbb{T}^d_L$ is injective on open balls of radius $L/2$; hence $\mathcal{D}_r$ is embedded in $\mathbb{T}^d_L$ for $0 < r \le r_0$. In particular $r_0$ is smaller than the injectivity radius $L/2$, and a pair at torus distance $r \le r_0$ has a unique displacement $S - M = rt$. We also take $r_0$ no larger than the radius $r_0$ fixed in §S12.13, so that the results of Section S12 quoted in Hypotheses S7.4–S7.7 hold for $0 < r \le r_0$.

**Lemma S7.3** (endpoint identities and typed-weight envelope; [7, (5.1)–(5.3), (6.1), (6.2) and §13]; Lemmas S12.14, S12.16 and S12.27)**.** *Let $0 < r \le r_0$ and let $\varphi$ be a $C^4$ function on $\mathbb{T}^d_L$ satisfying the pins (S7.2), with norms $M_3, M_4$ as in (S7.9). For $i \in \{M, S\}$ put*

$$\alpha_i = \frac{\varphi_{xx}(i)}{r}, \qquad \beta_i = \frac{\nabla_y \varphi_x(i)}{r}, \qquad A_i = D^2_y \varphi(i), \qquad \text{so that} \quad D^2\varphi(i) = \begin{pmatrix} r\alpha_i & r\beta_i^{\top} \\ r\beta_i & A_i \end{pmatrix}.$$

*Then*

$$|\alpha_i| \le \frac{M_3}{2}, \qquad \|\beta_i\| \le \frac{M_3}{2}, \qquad \|A_S - A_M\| \le r M_3, \tag{S7.13}$$

$$|\alpha_M + 6k| \le \frac{r M_4}{2}, \qquad |\alpha_S - 6k| \le \frac{r M_4}{2}, \tag{S7.14}$$

$$\frac{\det D^2\varphi(i)}{r} = \alpha_i \det A_i - r\,\beta_i^{\top} \operatorname{adj}(A_i)\,\beta_i, \tag{S7.15}$$

*the last identity holding also when $\alpha_i$ or $A_i$ vanishes. If $D^2\varphi(M) \prec 0$ and $0 < \lambda_1 \le \cdots \le \lambda_m = \Lambda$ are the eigenvalues of $-A_M$, the weight (S7.6) of $\varphi$ satisfies*

$$\frac{W_r}{r^2} \le \frac{M_3^2}{4}\,\lambda_1\Big(\lambda_1 + \frac{3}{2}\,r M_3\Big) \prod_{j=2}^{m} \lambda_j\,(\lambda_j + r M_3), \tag{S7.16}$$

*with the empty product equal to one when $m = 1$. Finally, $W_r/r^2 \le C(1 + \|\varphi\|_{C^3})^{2d}$.*

**Hypothesis S7.4** (pinned genericity and the elder mark; [7, §§8 and 14]; Lemmas S12.19(a) and S12.20)**.** (a) For $0 < r \le r_0$, $b \in \mathbb{R}$, $k > 0$ and every frame, $f$ is $\mathbb{E}^0$-almost surely a Morse function with pairwise distinct critical values; when $0 < Z_r < \infty$ the same holds $\mathbb{E}^{MS}_r$-almost surely.

(b) For $\varphi \in C^2(\mathbb{T}^d_L)$ and $z \in \mathbb{T}^d_L$ let

$$d_\varphi(z) = \sup_{\gamma} \min_{s \in [0,1]} \varphi(\gamma(s)),$$

the supremum over continuous paths $\gamma$ with $\gamma(0) = z$ and $\varphi(\gamma(1)) > \varphi(z)$, with $\sup \varnothing = -\infty$. The elder mark $e(\varphi; M, S) = \mathbf{1}_{\{d_\varphi(M) = \varphi(S)\}}$ is a Borel function of $(M, S, \varphi)$. If $\varphi$ is a Morse function with pairwise distinct critical values and $M$ is a local maximum of $\varphi$, then $e(\varphi; M, S) = 1$ if and only if $S$ is the ordinary superlevel elder death partner of $M$; for the global maximum $d_\varphi = -\infty$, so its essential class is never selected. For such $\varphi$ each finite superlevel $H_0$ bar has exactly one birth, at a local maximum, and one death, at a critical point of index $d - 1$, so the finite bars correspond one to one to the candidate pairs with $e = 1$.

**Hypothesis S7.5** (separating cylinder; [7, §§7 and 8]; Lemma S12.21 and Theorem S13.2)**.** Let $0 < r \le r_0$, $b \in \mathbb{R}$ and $k > 0$, and fix a frame. Let $\varphi \in C^4(\mathbb{T}^d_L)$ be a Morse function with pairwise distinct critical values that satisfies the pins (S7.2), and suppose that its norms (S7.9) satisfy the two inequalities defining $\mathcal{G}_r$ in (S7.10). Then $S$ is the ordinary superlevel elder death partner of $M$; that is, $e(\varphi; M, S) = 1$.

Hypothesis S7.5 is used only for $r \le r_0 < L/(4\sqrt{2})$, where the cylinder $\mathcal{D}_r$ is embedded.

**Hypothesis S7.6** (marked Kac–Rice identity; [7, §§9–11 and (13.5)]; Lemma S12.27(c) and Propositions S12.22, S12.23 and S12.28)**.** (a) Let $\Omega \subset \mathbb{T}^d_L \times \mathbb{T}^d_L$ be a compact set of ordered pairs at positive distance from the diagonal, and let $h$ be a bounded Borel function on $\Omega \times C^2(\mathbb{T}^d_L)$. Then

$$\mathbb{E} \sum_{(x,y)\in\Omega:\ \nabla f(x)=\nabla f(y)=0} h\big((x,y),f\big) = \int_\Omega p_{x,y}(0)\,\mathbb{E}\Big[|\det H_x\,\det H_y|\,h\big((x,y),f\big)\ \Big|\ \nabla f(x)=\nabla f(y)=0\Big]\,dx\,dy, \quad \text{(S7.17)}$$

where $p_{x,y}$ is the Gaussian density of $(\nabla f(x), \nabla f(y))$ and the conditional expectation is taken under the continuous Gaussian regression kernel on $C^2(\mathbb{T}^d_L)$.

(b) For $0 < r \le r_0$, $b \in \mathbb{R}$, $k > 0$ and every frame, $0 < Z_r < \infty$. Per unit volume, the expected number of ordered pairs $(M, S)$ of a local maximum $M$ and a critical point $S$ of index $d-1$ with displacement $S - M = rt$, $0 < r \le r_0$, $t \in S^{d-1}$, and with heights $f(M) = b$ and $f(S) = b - kr^3$, has intensity

$$r\,\mathcal{A}_r(b,k,t)\,dr\,db\,dk\,d\sigma(t), \quad \text{(S7.18)}$$

where $\sigma$ is the surface measure on $S^{d-1}$, not normalized to total mass one; there is no factor $1/2$, since the two points of a pair have different types. The pairs among them with $e = 1$ have intensity $r\,\mathcal{A}_r\,p_r\,dr\,db\,dk\,d\sigma(t)$.

(c) Consequently, with $r_\ell(k) = (\ell/k)^{1/3}$, the near densities have the versions

$$\ell^{1/3}\,\nu^{\mathrm{near}}_{\mathrm{cand}}(\ell) = \int \mathbf{1}_{\{k\ge \ell/r_0^3\}}\,\frac{\mathcal{A}_{r_\ell(k)}(b,k,t)}{3k^{2/3}}\,db\,dk\,d\sigma(t), \quad \text{(S7.19)}$$

$$\ell^{1/3}\,\nu^{\mathrm{near}}_{\mathrm{eld}}(\ell) = \int \mathbf{1}_{\{k\ge \ell/r_0^3\}}\,\frac{\mathcal{A}_{r_\ell(k)}(b,k,t)\,p_{r_\ell(k)}(b,k,t)}{3k^{2/3}}\,db\,dk\,d\sigma(t), \quad \text{(S7.20)}$$

with integrals over $\mathbb{R} \times (0,\infty) \times S^{d-1}$ and integrands equal to zero where the indicator vanishes.

In [7] (Propositions S12.22, S12.23, S12.24 and S12.28), part (b) is obtained from part (a), applied with the bounded Borel weight given by the elder mark and the type indicators, by disintegrating the two heights, by the change of variables to $U_r$ (Lemma S7.2(a), which turns the pin density into $12\,r^{-(d+3)}\pi_r(v_r)$), by polar coordinates $r^{d-1}dr\,d\sigma(t)$ and the height Jacobian $r^3\,db\,dk$ at fixed $r$, and by exhausting the pair domain towards the diagonal; the powers of $r$ add up to $(d-1)+3-(d+3)+2 = 1$. Part (c) follows from part (b) by the change of variables $\ell = kr^3$ at fixed $k$, for which $r\,dr = \frac{1}{3}k^{-2/3}\ell^{-1/3}\,d\ell$, and the restriction $r \le r_0$ is exactly $k \ge \ell/r_0^3$.

**Hypothesis S7.7** (far pairs; [7, §14 and (14.1)]; Lemma S12.29)**.** Let $\nu^{\mathrm{far}}_{\mathrm{cand}}$ and $\nu^{\mathrm{far}}_{\mathrm{eld}}$ be the densities of the candidate pairs, respectively of the finite bars, whose two critical points are at torus distance larger than $r_0$, in the versions given by (S7.17) with the original pins $(f(M), f(S)) = (b, b-\ell)$, whose height Jacobian is one. There is $C$ such that

$$0 \le \nu^{\mathrm{far}}_{\mathrm{eld}}(\ell) \le \nu^{\mathrm{far}}_{\mathrm{cand}}(\ell) \le C, \qquad 0 < \ell \le 1. \quad \text{(S7.21)}$$

The near and far populations together contain every candidate pair, and the separation $r_0$ itself is a null set [7, §14] (Lemma S12.29).

**Definition S7.8** (the coefficient; [7, (13.6), §15 and (15.2)]; Propositions S12.28 and S12.31)**.** Put

$$\begin{aligned} c_{d,L} &= \int_{\mathbb{R}\times(0,\infty)\times S^{d-1}} \frac{\mathcal{A}_0(b,k,t)}{3k^{2/3}}\,db\,dk\,d\sigma(t) \\ &= 4\int_{\mathbb{R}\times(0,\infty)\times S^{d-1}} k^{-2/3}\,\pi_0(v_0)\,z_0(b,k,t)\,db\,dk\,d\sigma(t). \end{aligned} \quad \text{(S7.22)}$$

By [7, (13.6) and §15] (Propositions S12.28 and S12.31), $0 < c_{d,L} < \infty$ and

$$c_{d,L} = \frac{\Gamma(7/6)}{24^{1/3}\sqrt{\pi}} \int_{S^{d-1}} p_G(0)\, p_{V_t}(0)\, \tau_t^{4/3}\, D_t\, d\sigma(t), \tag{S7.23}$$

where $G = \nabla f(0)$ and $V_t = D^2 f(0)\, t$, that is $(f_{xx}, f_{xy_1}, \dots, f_{xy_m})(0)$ in the frame, have Gaussian densities $p_G$ and $p_{V_t}$; $\tau_t^2 = \mathrm{Var}(\partial_t^3 f(0) \mid G = 0)$; and $D_t = \mathbb{E}[(\det D_y^2 f(0))^2\, \mathbf{1}_{\{D_y^2 f(0) \prec 0\}} \mid V_t = 0]$. The factor 24 in (S7.23) comes from the Jacobian and the gamma integral; it is the same for every $L$ and is not the side length.

**The theorem.**

**Theorem S7.9** (bounded remainder)**.** *Let $d \geq 2$ and $L > 0$, and let*

$$\nu_{\mathrm{cand}} = \nu_{\mathrm{cand}}^{\mathrm{near}} + \nu_{\mathrm{cand}}^{\mathrm{far}}, \qquad \nu_{\mathrm{eld}} = \nu_{\mathrm{eld}}^{\mathrm{near}} + \nu_{\mathrm{eld}}^{\mathrm{far}}$$

*be the versions, given by Hypotheses S7.6 and S7.7, of the densities per unit volume of all candidate pairs and of all finite ordinary superlevel $H_0$ bars on $\mathbb{T}_L^d$, with all birth heights, all gap marks $k > 0$, all orientations and all separations, and with the essential class of the global maximum excluded. There are $\ell_* > 0$ and $C < \infty$ such that, for $0 < \ell \leq \ell_*$,*

$$\begin{aligned} &\left|\nu_{\mathrm{cand}}(\ell) - c_{d,L}\, \ell^{-1/3}\right| \leq C, \\ &0 \leq \nu_{\mathrm{cand}}(\ell) - \nu_{\mathrm{eld}}(\ell) \leq C, \\ &\left|\nu_{\mathrm{eld}}(\ell) - c_{d,L}\, \ell^{-1/3}\right| \leq C. \end{aligned} \tag{S7.24}$$

*Remark* S7.10 (scope of Theorem S7.9). Since $c_{d,L} > 0$, (S7.24) gives relative errors $O(\ell^{1/3})$: $\nu_{\mathrm{eld}}(\ell) = c_{d,L}\, \ell^{-1/3}(1 + O(\ell^{1/3}))$, and likewise for $\nu_{\mathrm{cand}}$. The theorem does not assert that the bounded remainder $\nu_{\mathrm{eld}}(\ell) - c_{d,L}\ell^{-1/3}$ converges as $\ell \downarrow 0$; it identifies no second coefficient and gives no numerical value of $C$ or $\ell_*$. The difference $\nu_{\mathrm{cand}} - \nu_{\mathrm{eld}}$ is shown to be bounded, not to be $O(\ell^{2/3})$; the latter rate is proved in [7, Theorem B] (Theorem S12.4) only on compact windows of births and gaps, and a bounded far contribution need not vanish. The proof needs neither a lower bound for $Z_r/r^2$ uniform over all $b$ and $k$ nor a constant in an estimate $1 - p_r \leq Cr^3$ uniform over all $b$ and $k$. Its two main ingredients are the quadratic cancellation of Lemma S7.15 for the $\sqrt{r}$ off-diagonal block of the rescaled Hessians, and the bound (S7.42), which is unnormalized and keeps the dependence on $k$ explicit.

**Symmetric contact observations and a second-order coupling.**

**Lemma S7.11** (second-order contact estimates)**.** (a) *Let $\varphi$ be a $C^7$ function. As $r \to 0$, uniformly over frames, the four axial rows (S7.3) of $\varphi$ satisfy*

$$\begin{aligned} \frac{\varphi(M) + \varphi(S)}{2} &= \varphi + \frac{r^2}{8}\, \varphi_{xx} + O(r^4), \\ \frac{\varphi(S) - \varphi(M)}{r} &= \varphi_x + \frac{r^2}{24}\, \varphi_{xxx} + O(r^4), \\ \frac{\varphi_x(S) - \varphi_x(M)}{r} &= \varphi_{xx} + \frac{r^2}{24}\, \varphi_{xxxx} + O(r^4), \\ \frac{6}{r^2}\Big[\varphi_x(M) + \varphi_x(S) - \frac{2(\varphi(S) - \varphi(M))}{r}\Big] &= \varphi_{xxx} + \frac{r^2}{40}\, \varphi_{xxxxx} + O(r^4), \end{aligned} \tag{S7.25}$$

*with the derivatives on the right evaluated at 0, and each $O(r^4)$ bounded by $r^4$ times a constant multiple of the supremum on the segment $[M, S]$ of the derivatives of $\varphi$ of order at most seven. The two transverse rows satisfy the first two expansions with $\varphi$ replaced by $\varphi_{y_j}$. In particular each entry of $U_r - U_0$ computed from $\varphi$ is bounded by $Cr^2$ times the supremum of the derivatives of $\varphi$*

*of order at most five; the fourth axial row differs from $\varphi_{xxx}(0)$ by a term of order $r^2$, not by an error of order $r$ or by an unscaled residual.*

(b) *For every finite $p$, uniformly over frames and $0 < r \le 1$, $\|U_r - U_0\|_{L^p} \le C_p r^2$; and for every $q \ge 0$, $\sup_z \max_{|\mathbf{j}|\le q} |\mathrm{Cov}(\partial^{\mathbf{j}} f(z), U_r - U_0)| \le C_q r^2$.*

(c) *Uniformly over frames and $0 \le r \le r_0$, and independently of $b$ and $k$,*

$$\Sigma_r - \Sigma_0 = O(r^2), \qquad \Sigma_r^{-1} - \Sigma_0^{-1} = O(r^2), \qquad \|\mathcal{C}_r - \mathcal{C}_0\|_{C^q} = O(r^2) \quad (q \ge 0). \tag{S7.26}$$

*Proof.* (a) Let $g(x) = \varphi(xt)$ be the restriction to the axis. Expand $g$ and $g'$ at $\pm r/2$ by Taylor's formula with integral remainder about 0. All rows are centered (symmetric averages or antisymmetric differences), so the odd powers of $r$ cancel, and the coefficients of $r^0$ and $r^2$ are those displayed. For the fourth row,

$$g'(-\tfrac{r}{2}) + g'(\tfrac{r}{2}) = 2g' + \frac{r^2}{4}\, g''' + \frac{r^4}{192}\, g^{(5)} + O(r^6),$$
$$\frac{2}{r}\Big(g(\tfrac{r}{2}) - g(-\tfrac{r}{2})\Big) = 2g' + \frac{r^2}{12}\, g''' + \frac{r^4}{960}\, g^{(5)} + O(r^6),$$

so the row equals

$$\frac{6}{r^2}\Big[\frac{r^2}{6}\, g''' + \frac{r^4}{240}\, g^{(5)} + O(r^6)\Big] = g''' + \frac{r^2}{40}\, g^{(5)} + O(r^4).$$

The transverse rows are the first two patterns applied to $\varphi_{y_j}$. Stopping the expansions one order earlier, with integral remainder, gives the bounds by $Cr^2$ times derivatives of order at most five.

(b) Represent $f$ by its real Fourier series with independent standard Gaussian coefficients and amplitudes $\sqrt{a_n}$ (Lemma S7.1(a)). A derivative of order $j$ of a mode of frequency $n$ is bounded by $(2\pi|n|/L)^j$ in every direction, so by (a) the centered-rule error of each mode is at most $Cr^2(1+|n|)^5$, uniformly over frames. Minkowski's inequality and the summability (S7.11) give $\|U_r - U_0\|_{L^p} \le C_p r^2$; no random-grid approximation is used. The same argument with products of coefficients, $\mathrm{Cov}(\partial^{\mathbf{j}} f(z), U_r - U_0)$ being a sum over modes of $a_n$ times a derivative of the mode at $z$ times its centered-rule error, gives the bound on cross-covariances, uniformly in $z$ and in the frame.

(c) By Lemma S7.2(b), $\Sigma_r$ and $\Sigma_r^{-1}$ are bounded uniformly for $0 \le r \le r_0 \le r_1$ and over frames. By (b) and the Cauchy–Schwarz inequality, $\Sigma_r - \Sigma_0 = \mathrm{Cov}(U_r - U_0, U_r) + \mathrm{Cov}(U_0, U_r - U_0) = O(r^2)$, and then $\Sigma_r^{-1} - \Sigma_0^{-1} = \Sigma_r^{-1}(\Sigma_0 - \Sigma_r)\Sigma_0^{-1} = O(r^2)$. Finally $\|\mathcal{C}_r - \mathcal{C}_0\|_{C^q} = \sup_z \max_{|\mathbf{j}|\le q} |\mathrm{Cov}(\partial^{\mathbf{j}} f(z), U_r - U_0)| = O(r^2)$ by (b). Only covariances enter, so these estimates do not depend on $b$ or $k$. □

**Lemma S7.12** (second-order coupling)**.** *Let $F$ be a smooth copy of the unconditioned field $f$ on some probability space, write $U_r$ and $U_0$ for its contact observations, and for $0 < r \le r_0$, $b \in \mathbb{R}$, $k > 0$ and a frame put*

$$F_r = F + \mathcal{C}_r \Sigma_r^{-1}(v_r - U_r), \qquad F_0 = F + \mathcal{C}_0 \Sigma_0^{-1}(v_0 - U_0). \tag{S7.27}$$

*Then $F_r$ has the law of $f$ under $\mathbb{E}^0_{r,b,k,R}$, and $F_0$ the law of $f$ under $\mathbb{E}^0_{0,b,k,R}$. With $P = 1 + |b| + k$, for every finite $p \ge 1$ and every $q \ge 0$,*

$$\big\| \|F_r - F_0\|_{C^q} \big\|_{L^p} \le C_{p,q}\, r^2 P, \qquad \big\| 1 + \|F_r\|_{C^q} + \|F_0\|_{C^q} \big\|_{L^p} \le C_{p,q}\, P. \tag{S7.28}$$

This is a coupling of laws on one probability space. It does not assert that two conditioned fields are equal, nor that endpoint variables are independent.

*Proof.* The field $F - \mathcal{C}_r \Sigma_r^{-1} U_r$ is uncorrelated with, hence independent of, the Gaussian vector $U_r$; so $F_r$ is the Gaussian regression of $F$ on $U_r = v_r$, and similarly for $F_0$. For $r \le 1$, $|v_r - v_0| =$

$|(-kr^3/2, -kr^2, 0, \dots, 0)| \le Cr^2k$, and $|v_r| + |v_0| \le C(|b| + k)$ by Lemma S7.2(c). Write

$$F_r - F_0 = (\mathcal{C}_r\Sigma_r^{-1} - \mathcal{C}_0\Sigma_0^{-1})(v_r - U_r) + \mathcal{C}_0\Sigma_0^{-1}[(v_r - v_0) - (U_r - U_0)].$$

By (S7.26), $\|\mathcal{C}_r\Sigma_r^{-1} - \mathcal{C}_0\Sigma_0^{-1}\|_{C^q} \le Cr^2$ and $\|\mathcal{C}_0\Sigma_0^{-1}\|_{C^q} \le C$; moreover $\|U_r\|_{L^p} \le C_p$ and $\|U_r - U_0\|_{L^p} \le C_p r^2$ by Lemma S7.11(b). This gives the first bound in (S7.28). The second follows from $\|F_r\|_{C^q} \le \|F\|_{C^q} + \|\mathcal{C}_r\Sigma_r^{-1}\|_{C^q}(|v_r| + |U_r|)$, the same bound for $F_0$, and Lemma S7.1(a). □

**Lemma S7.13** (pin densities)**.** *For $0 < r \le r_0$, $b \in \mathbb{R}$, $k > 0$ and every frame, with $P = 1 + |b| + k$,*

$$\pi_r(v_r) \le C\exp[-c(b^2 + k^2)], \qquad |\pi_r(v_r) - \pi_0(v_0)| \le Cr^2P^2\exp[-c(b^2 + k^2)]. \tag{S7.29}$$

*Proof.* The first bound is Lemma S7.2(c). For the second, interpolate covariance and target linearly: for $\theta \in [0, 1]$ let $\Sigma_\theta = (1-\theta)\Sigma_0 + \theta\Sigma_r$, $v_\theta = (1-\theta)v_0 + \theta v_r = (b - \theta kr^3/2,\ -\theta kr^2,\ 0,\ 12k,\ 0, \dots, 0)$, and let $\pi_\theta$ be the centered Gaussian density with covariance $\Sigma_\theta$. The eigenvalues of $\Sigma_\theta$ stay in the compact interval of Lemma S7.2(b). The target keeps its coordinate $12k$, and its other nonzero height coordinate is $b - \theta kr^3/2$; since $b^2 \le 2(b - \theta kr^3/2)^2 + k^2/2$ for $r \le 1$, $|v_\theta|^2 \ge c(b^2 + k^2)$, and $|v_\theta| \le CP$. Differentiating the logarithm of the Gaussian density,

$$\frac{d}{d\theta}\log\pi_\theta(v_\theta) = -\tfrac{1}{2}\operatorname{tr}(\Sigma_\theta^{-1}(\Sigma_r - \Sigma_0)) + \tfrac{1}{2}\,v_\theta^\top\Sigma_\theta^{-1}(\Sigma_r - \Sigma_0)\Sigma_\theta^{-1}v_\theta - v_\theta^\top\Sigma_\theta^{-1}(v_r - v_0),$$

and by (S7.26) and $|v_r - v_0| \le Cr^2k$ the trace term is $O(r^2)$, the quadratic term $O(r^2P^2)$ and the target term $O(r^2kP)$. Hence $|d\pi_\theta(v_\theta)/d\theta| \le Cr^2P^2\pi_\theta(v_\theta) \le Cr^2P^2\exp[-c(b^2 + k^2)]$, and integrating over $\theta \in [0, 1]$ gives (S7.29), with a possibly smaller $c$. This handles arbitrarily large targets. □

**Index-filtered determinants.** For a real symmetric $n \times n$ matrix $H$ and an integer $j$ put

$$\Phi_j(H) = |\det H|\,\mathbf{1}_{\{H \text{ has exactly } j \text{ negative eigenvalues}\}}, \tag{S7.30}$$

with $\Phi_j(H) = 0$ when $H$ is singular. The inertia indicator itself is not Lipschitz; the filtered determinant is.

**Lemma S7.14.** *For real symmetric $n \times n$ matrices $X$, $Y$ and every $j$,*

$$|\Phi_j(X) - \Phi_j(Y)| \le n\,\max(\|X\|, \|Y\|)^{n-1}\,\|X - Y\|. \tag{S7.31}$$

*Proof.* Telescoping over columns, $\det X - \det Y = \sum_{i=1}^n \det(Y_1, \dots, Y_{i-1}, X_i - Y_i, X_{i+1}, \dots, X_n)$, where $X_i$, $Y_i$ are the columns. Every column of a matrix has Euclidean norm at most its operator norm, so Hadamard's inequality gives

$$|\det X - \det Y| \le n\,\max(\|X\|, \|Y\|)^{n-1}\,\|X - Y\|. \tag{S7.32}$$

If $\Phi_j(X) = \Phi_j(Y) = 0$ there is nothing to prove. If both are nonzero, $X$ and $Y$ are nonsingular with exactly $j$ negative eigenvalues, their determinants have the sign $(-1)^j$, and $|\Phi_j(X) - \Phi_j(Y)| = |\det X - \det Y|$. If exactly one is nonzero, say $\Phi_j(X)$, then either $Y$ is singular, and we put $Z = Y$, or $Y$ is nonsingular with a number of negative eigenvalues different from $j$; in the latter case the eigenvalues of $(1-s)X + sY$ depend continuously on $s \in [0, 1]$ and the number of negative ones can change only where one of them vanishes, so some $Z = (1 - s_0)X + s_0Y$ is singular. In both cases $|\Phi_j(X) - \Phi_j(Y)| = |\det X| = |\det X - \det Z|$, and (S7.32) applied to $X$ and $Z$, with $\|Z\| \le \max(\|X\|, \|Y\|)$ and $\|X - Z\| \le \|X - Y\|$, gives (S7.31). □

**Lemma S7.15** (quadratic off-diagonal cancellation)**.** *Let $A$ be a real symmetric $(n-1) \times (n-1)$ matrix, $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}^{n-1}$, and for $s \ge 0$ put*

$$K_s = \begin{pmatrix} \alpha & \sqrt{s}\,\beta^\top \\ \sqrt{s}\,\beta & A \end{pmatrix}.$$

*Then, for every $j$ and every $r \geq 0$,*

$$|\Phi_j(K_r) - \Phi_j(K_0)| \leq r\,|\beta^\top \operatorname{adj}(A)\,\beta|. \tag{S7.33}$$

*Proof.* For all $s \geq 0$, including singular $A$,

$$\det K_s = \alpha \det A - s\,\beta^\top \operatorname{adj}(A)\,\beta : \tag{S7.34}$$

for invertible $A$ the right side is $\det A\,(\alpha - s\beta^\top A^{-1}\beta)$, and both sides are polynomials in the entries. Thus $\det K_s$ is affine in $s$ with slope $-\beta^\top \operatorname{adj}(A)\beta$. If both $K_0$ and $K_r$ are nonsingular with the same number of negative eigenvalues, either both filtered determinants vanish or both determinants have the same sign, and $|\Phi_j(K_r) - \Phi_j(K_0)| \leq |\det K_r - \det K_0| = r|\beta^\top \operatorname{adj}(A)\beta|$. If exactly one endpoint contributes, say $K_{s_1}$ with $s_1 \in \{0, r\}$, then the other endpoint is singular, or it has a different number of negative eigenvalues and, by continuity of the eigenvalues of $K_s$ in $s$, some $K_{s_0}$ with $s_0$ between 0 and $r$ is singular. Then $\Phi_j(K_{s_1}) = |\det K_{s_1} - \det K_{s_0}| = |s_1 - s_0|\,|\beta^\top \operatorname{adj}(A)\beta| \leq r|\beta^\top \operatorname{adj}(A)\beta|$. If neither endpoint contributes there is nothing to prove; this includes the case in which the determinant vanishes identically. □

The proof uses no inverse of $A$ and no lower bound on its eigenvalues. Applying (S7.31) to the entries $\sqrt{r}\,\beta$ would give only a bound of order $\sqrt{r}$; (S7.33) is linear in $r$, also when $\alpha$ (below, the gap mark $k$) tends to zero.

**The all-mark intensity and its contact error.**

**Lemma S7.16** (contact error of the normalizer)**.** *There is an integer $N$, depending only on $d$, such that for $0 < r \leq r_0$, $b \in \mathbb{R}$, $k > 0$ and every frame, with $P = 1 + |b| + k$,*

$$\frac{Z_r}{r^2} \leq C(k + r)^2 P^N, \qquad \left|\frac{Z_r}{r^2} - z_0(b, k, t)\right| \leq C\,r\,(k + r)\,P^N. \tag{S7.35}$$

*Proof.* Work in the coupling of Lemma S7.12. Put $A_0 = D_y^2 F_0(0)$ and, for $i \in \{M, S\}$, $A_i = D_y^2 F_r(i)$, $\alpha_i = \partial_x^2 F_r(i)/r$, $\beta_i = \nabla_y \partial_x F_r(i)/r$ and $H_i = D^2 F_r(i)$. With $\Delta_r = \operatorname{diag}(\sqrt{r}, I_m)$,

$$K_i := \Delta_r^{-1} H_i \Delta_r^{-1} = \begin{pmatrix} \alpha_i & \sqrt{r}\,\beta_i^\top \\ \sqrt{r}\,\beta_i & A_i \end{pmatrix}, \qquad \frac{|\det H_i|}{r} = |\det K_i|. \tag{S7.36}$$

By Sylvester's law of inertia, $K_i$ and $H_i$ have the same number of negative eigenvalues, so the weight (S7.6) of $F_r$ is $W_r = r^2\,\Phi_d(K_M)\,\Phi_{d-1}(K_S)$; since $F_r$ has the law $\mathbb{E}^0$, $Z_r/r^2 = \mathbb{E}[\Phi_d(K_M)\,\Phi_{d-1}(K_S)]$.

The exact gradient and height pins give, by (S7.13)–(S7.14) applied to $F_r$,

$$|\alpha_M + 6k| + |\alpha_S - 6k| \leq rM_4, \qquad \|\beta_i\| \leq M_3/2. \tag{S7.37}$$

Let $T$ be a sufficiently large constant multiple of $1 + k + \|F_r\|_{C^4} + \|F_0\|_{C^4} + r^{-2}\|F_r - F_0\|_{C^2}$. Then $T \geq 1 + k$, and $\|T\|_{L^p} \leq C_p P$ for every finite $p$ by (S7.28). Since

$$\|A_i - A_0\| \leq \|D_y^2 F_r(i) - D_y^2 F_r(0)\| + \|D_y^2 F_r(0) - D_y^2 F_0(0)\| \leq \tfrac{r}{2}\|F_r\|_{C^3} + \|F_r - F_0\|_{C^2},$$

since $r \leq 1$, and since $M_3, M_4 \leq C\|F_r\|_{C^4}$,

$$\begin{gathered} \|A_i - A_0\| \leq rT, \qquad |\alpha_M + 6k| + |\alpha_S - 6k| \leq CrT, \\ \|A_i\| + \|A_0\| + \|\beta_i\| + M_3 + M_4 \leq CT. \end{gathered} \tag{S7.38}$$

No independence of $T$ and $A_0$ is needed.

Use $j = d$ at $M$ and $j = d - 1$ at $S$, with the reference matrices $\operatorname{diag}(-6k, A_0)$ at $M$ and $\operatorname{diag}(6k, A_0)$ at $S$. First remove the off-diagonal block: by Lemma S7.15 with $s = r$, and since $\|\operatorname{adj}(A)\| \leq \|A\|^{m-1}$ for symmetric $m \times m$ matrices,

$$|\Phi_j(K_i) - \Phi_j(\operatorname{diag}(\alpha_i, A_i))| \leq r\,\|\beta_i\|^2\,\|A_i\|^{m-1} \leq CrT^d.$$

Then compare with the reference matrix: $\|\operatorname{diag}(\alpha_i, A_i) - \operatorname{diag}(\mp 6k, A_0)\| \leq |\alpha_i \pm 6k| + \|A_i - A_0\| \leq CrT$ (upper signs at $M$), and both matrices have norm at most $CT$ because $T \geq 1 + k$; so

Lemma S7.14 bounds the difference of their filtered determinants by $d\,(CT)^{d-1}\,CrT = CrT^d$. Hence each endpoint has a filtered-determinant error $O(rT^d)$. The block determinant (S7.15) also gives

$$\Phi_j(K_i) \le C(k+r)T^d, \qquad \Phi_d(\mathrm{diag}(-6k, A_0)) \le CkT^{d-1}, \qquad \Phi_{d-1}(\mathrm{diag}(6k, A_0)) \le CkT^{d-1}.$$

The matrix $\mathrm{diag}(-6k, A_0)$ has exactly $d$ negative eigenvalues, and $\mathrm{diag}(6k, A_0)$ exactly $d-1$, if and only if $A_0 \prec 0$; no other transverse inertia gives both prescribed indices. So the reference product is

$$\Phi_d(\mathrm{diag}(-6k, A_0))\,\Phi_{d-1}(\mathrm{diag}(6k, A_0)) = (6k)^2 \det(A_0)^2\,\mathbf{1}_{\{A_0 \prec 0\}},$$

whose expectation is $z_0(b,k,t)$ by (S7.7), because $F_0$ has the law $\mathbb{E}^0_{0,b,k,R}$. Multiplying the two endpoint estimates,

$$\left|\Phi_d(K_M)\Phi_{d-1}(K_S) - (6k)^2 \det(A_0)^2 \mathbf{1}_{\{A_0 \prec 0\}}\right| \le CrT^d \cdot C(k+r)T^d + CkT^{d-1} \cdot CrT^d \le Cr(k+r)T^{2d},$$

and $\Phi_d(K_M)\Phi_{d-1}(K_S) \le C(k+r)^2 T^{2d}$. Taking expectations and using $\mathbb{E}T^{2d} \le CP^{2d}$ gives (S7.35) with $N = 2d$. □

**Corollary S7.17** (all-mark intensity)**.** *There is a function*

$$\Psi(b,k) = C_\Psi\,(1 + |b| + k)^N \exp\!\left[-c_\Psi(b^2 + k^2)\right], \tag{S7.39}$$

*with constants $C_\Psi, c_\Psi > 0$ and an integer $N$, such that for all $b \in \mathbb{R}$, $k > 0$, $t \in S^{d-1}$ and $0 < r \le r_0$*

$$0 \le \mathcal{A}_r \le (k+r)^2\,\Psi, \qquad 0 \le \mathcal{A}_0 \le k^2\,\Psi, \qquad |\mathcal{A}_r - \mathcal{A}_0| \le r(k+r)\,\Psi. \tag{S7.40}$$

*Proof.* The first bound follows from (S7.29) and (S7.35). In the coupling $\|A_0\| \le \|F_0\|_{C^2}$, so (S7.7) and (S7.28) give $z_0 \le 36k^2\,\mathbb{E}\|A_0\|^{2m} \le Ck^2P^{2m}$; and $\pi_0(v_0) \le C\exp[-c(b^2+k^2)]$ because $\Sigma_0$ is positive definite (Lemma S7.2(b)) and $|v_0|^2 = b^2 + 144k^2$. This gives the second bound. For the third,

$$|\mathcal{A}_r - \mathcal{A}_0| \le 12\,\pi_r(v_r)\left|\frac{Z_r}{r^2} - z_0\right| + 12\,|\pi_r(v_r) - \pi_0(v_0)|\,z_0.$$

By (S7.29) and (S7.35) the first term is at most $Cr(k+r)P^N\exp[-c(b^2+k^2)]$, and the second is at most $Cr^2P^2 \cdot k^2P^{2m}\exp[-c(b^2+k^2)]$. Since $r^2k^2 \le r(k+r)\cdot rk \le r(k+r)P$ for $r \le 1$, the factor $r^2k^2$ is absorbed by $r(k+r)$ times a further power of $P$. Increasing $N$ and decreasing $c$ gives (S7.40). □

The bounds (S7.35)–(S7.40) hold for all $b \in \mathbb{R}$, $k > 0$ and $0 < r \le r_0$; no division by $k$ or by $Z_r$ occurs in their proofs. Enlarging $C_\Psi$ and $N$ and decreasing $c_\Psi$ if necessary, the same function $\Psi$ also serves in Lemma S7.18.

**The unnormalized cap-loss bound.**

**Lemma S7.18** (unnormalized cap loss)**.** *For $0 < r \le r_0$, $b \in \mathbb{R}$, $k > 0$ and every frame put*

$$\mathcal{B}_r = \mathcal{B}_r(b,k,t) = 12\,\pi_r(v_r)\,\mathbb{E}^0\Big[\frac{W_r}{r^2}\,\mathbf{1}_{\mathcal{G}_r^c}\Big]. \tag{S7.41}$$

*Then $\mathcal{A}_r(1-p_r) \le \mathcal{B}_r \le \mathcal{A}_r$, and, with $\delta = r/k$,*

$$\mathcal{B}_r \le \delta^3\,\Psi(b,k) \qquad \text{whenever } \delta \le 1. \tag{S7.42}$$

*The constants in (S7.42) do not depend on $b$, $k$, $r$ or the frame; the growth in the target is contained in $\Psi$.*

The bound (S7.42) is unnormalized: it does not give a bound on $1 - p_r$ that is uniform over all $b$ and $k$.

*Proof. Comparison.* By Hypothesis S7.6(b), $0 < Z_r < \infty$, so $\mathcal{A}_r(1-p_r) = 12\,\pi_r(v_r)\,\mathbb{E}^0[(W_r/r^2)(1-e)]$. By Hypothesis S7.4(a), $\mathbb{E}^0$-almost surely $f$ is a Morse function with pairwise distinct critical values, and it satisfies the pins; by Hypothesis S7.5, $e = 1$ on $\mathcal{G}_r$. Hence $W_r(1-e) \le W_r \mathbf{1}_{\mathcal{G}_r^c}$ $\mathbb{E}^0$-almost surely, which gives $\mathcal{A}_r(1-p_r) \le \mathcal{B}_r$; and $\mathcal{B}_r \le \mathcal{A}_r$ because $\mathbf{1}_{\mathcal{G}_r^c} \le 1$.

*Joint density.* Append the $m(m+1)/2$ independent entries of $A_M = D_y^2 f(M)$ to $U_r$. By Lemma S7.2(b) the joint covariance has eigenvalues in a fixed compact subinterval of $(0,\infty)$, for $0 \le r \le r_0$ and all frames. With $|v_r|^2 \ge c(b^2+k^2)$ from Lemma S7.2(c), the joint density at $(v_r, A)$ therefore satisfies

$$\pi_r(v_r)\,\rho_r(A) \le C \exp\bigl[-c(b^2+k^2+\|A\|_F^2)\bigr], \tag{S7.43}$$

where $\rho_r$ is the density of $A_M$ under $\mathbb{E}^0$ in the Lebesgue coordinates of symmetric matrices (their independent entries) and $\|\cdot\|_F$ is the Frobenius norm. The bound is uniform in the eigenvectors of $A$.

*Residual.* Regress the whole field on the appended vector $(U_r, A_M)$, as in [7, §4] (Lemma S12.13). The centered residual $g$ is independent of that entire vector, and its $C^4$ norm has moments of every finite order, bounded uniformly in $0 < r \le r_0$ and in the frame, by Lemma S7.1(a), the uniform bounds of Lemma S7.2(b) on the inverse joint covariance, and the uniform bounds on the cross-covariances of $f$ and its derivatives with $(U_r, A_M)$ (Lemma S7.11(b) and the smoothness of $K_L$). Given $U_r = v_r$ and $A_M = A$, the regression mean has $C^4$ norm at most $C(|v_r|+\|A\|) \le C(P+\|A\|)$, where $P = 1 + |b| + k$. Put $J = 1 + \|g\|_{C^4}$ and $T = P + J$. On $\{A_M \prec 0\}$ write $B = -A_M$, with eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_m = \Lambda$. Since $\|A_M\|_F \le \sqrt{m}\,\Lambda$ there, $M_3$ and $M_4$ are at most $K_*(T+\Lambda)$ for a constant $K_*$ (absorbing $\sqrt{m}$). The variable $J$ is independent of $B$, although the eigenvalues of $B$ are not independent of each other.

On the typed support, $\mathcal{G}_r^c$ is the union of the *depth failure* $\{\lambda_1 \le \frac{4}{3k} r M_3^2\}$ and the *fourth-derivative exception* $\{rM_4 > 3k/10\}$.

*Depth failure when $m \ge 2$.* Depth failure implies

$$0 < \lambda_1 \le D_*\,\delta\,(T+\Lambda)^2, \qquad D_* = \tfrac{4}{3}K_*^2.$$

Put $\Upsilon = T + \Lambda$. In the envelope (S7.16), for $r \le 1$, $M_3^2/4 \le K_*^2\Upsilon^2/4$, $\frac{3}{2} r M_3 \le E_*\, r\Upsilon$ with $E_* = \frac{3}{2}K_*$, and $\lambda_j(\lambda_j + rM_3) \le (1+K_*)\Upsilon^2$ for $j \ge 2$. Thus all factors other than the two soft factors $\lambda_1$ and $\lambda_1 + \frac{3}{2} r M_3$ are at most $C\Upsilon^{2m}$, and $W_r/r^2 \le C\Upsilon^{2m}\lambda_1(\lambda_1 + E_* r\Upsilon)$. In the eigenvalue coordinates of symmetric matrices, Lebesgue measure is a constant multiple of $\prod_{i<j}|\lambda_j - \lambda_i|\,d\lambda_1 \cdots d\lambda_m$ times a finite measure on the eigenvectors [7, §7] (Lemma S12.17); this is a change of variables for Lebesgue measure, not a claim that $A_M$ is a GOE matrix. On the ordered positive domain the Vandermonde factor is at most $\Lambda^{m(m-1)/2}$, and (S7.43) gives a Gaussian bound in all eigenvalues, uniformly in the eigenvectors. Condition on $J$. Since $\Upsilon$ does not involve $\lambda_1$ when $m \ge 2$, we may extend the $\lambda_1$-integration to $[0, D_*\delta\Upsilon^2]$ and drop the Gaussian factor in $\lambda_1$; then

$$\int_0^{D_*\delta\Upsilon^2} \lambda\,(\lambda + E_* r\Upsilon)\,d\lambda = \frac{D_*^3}{3}\,\delta^3\,\Upsilon^6 + \frac{E_* D_*^2}{2}\,r\,\delta^2\,\Upsilon^5,$$

and $r\delta^2 = k\delta^3 \le P\delta^3$, so both terms are $\delta^3$ times a polynomial in $P$, $J$ and $\Lambda$. The Gaussian integration in $\lambda_2, \dots, \lambda_m$ and the finite moments of $J$ give

$$12\,\pi_r(v_r)\,\mathbb{E}^0\Bigl[\frac{W_r}{r^2}\,\mathbf{1}_{\{\text{depth failure}\}}\Bigr] \le \delta^3\,C P^N \exp\bigl[-c(b^2+k^2)\bigr].$$

The largest eigenvalue is integrated, not frozen at a chosen value, and no lower cutoff on $\lambda_2$ is used: every intersection of higher corank is included.

*Depth failure when $m = 1$.* Now $\lambda_1 = \Lambda = \lambda$, and the quadratic inequality is instead

$$\lambda \le D_*\delta(T+\lambda)^2 \quad\Longrightarrow\quad \lambda \le 4D_*\delta T^2 \quad\text{or}\quad \lambda > \frac{1}{4D_*\delta}, \tag{S7.44}$$

because $(T+\lambda)^2 \le 2T^2+2\lambda^2$ and $2D_*\delta\lambda^2 \le \lambda/2$ when $\lambda \le 1/(4D_*\delta)$. On the near branch, $\delta \le 1$ and $T \ge 1$ give $M_3 \le K_*(T+4D_*T^2) \le CT^2$, so $W_r/r^2 \le CT^4\lambda(\lambda+CrT^2)$ by (S7.16); direct scalar integration,

$$\int_0^{4D_*\delta T^2} \lambda\,(\lambda+CrT^2)\,d\lambda = \frac{64D_*^3}{3}\,\delta^3T^6 + 8CD_*^2\,r\,\delta^2\,T^6,$$

together with (S7.43), $r\delta^2 = k\delta^3$ and the moments of $J$, again gives at most $\delta^3 CP^N \exp[-c(b^2+k^2)]$. On the far branch $1 \le (4D_*\delta\lambda)^4$, so the fourth-moment Markov inequality with the actual weight gives

$$12\,\pi_r(v_r)\,\mathbb{E}^0\Big[\frac{W_r}{r^2}\,\mathbf{1}_{\mathrm{far}}\Big] \le (4D_*\delta)^4\,12\,\pi_r(v_r)\,\mathbb{E}^0\Big[\frac{W_r}{r^2}\,\lambda^4\Big] \le \delta^4\,CP^N\exp\big[-c(b^2+k^2)\big];$$

here the joint moment is bounded by a polynomial Gaussian integral from (S7.43), or directly by $W_r/r^2 \le C(1+\|f\|_{C^3})^{2d}$ (Lemma S7.3), $\lambda \le \|f\|_{C^2}$ and (S7.28). Since $\delta \le 1$, $\delta^4 \le \delta^3$. The far branch is kept, not discarded.

*Fourth-derivative exception.* The condition $rM_4 > 3k/10$ means $M_4 > 3/(10\delta)$, so $1 \le (10\delta M_4/3)^4$ there, and

$$12\,\pi_r(v_r)\,\mathbb{E}^0\Big[\frac{W_r}{r^2}\,\mathbf{1}_{\{rM_4>3k/10\}}\Big] \le \Big(\frac{10\delta}{3}\Big)^4\,12\,\pi_r(v_r)\,\mathbb{E}^0\Big[\frac{W_r}{r^2}\,M_4^4\Big] \le \delta^4\,CP^N\exp\big[-c(b^2+k^2)\big],$$

with the joint moment bounded as before. These are weighted pointwise Markov inequalities; no Cauchy–Schwarz step is used.

Adding the bounds for depth failure and for the exception, and enlarging the common polynomial degree and the constants of $\Psi$, proves (S7.42). □

**Integration over the marks.** Since $\Psi$ contains a Gaussian factor,

$$\bar\Psi := \sup_{k>0}\int_{\mathbb{R}} \Psi(b,k)\,db < \infty, \qquad \int_0^\infty\!\!\int_{\mathbb{R}} \Psi(b,k)\,db\,dk < \infty, \tag{S7.45}$$

and $\sigma(S^{d-1}) < \infty$. At a lifetime $\ell > 0$ we write $r = r_\ell(k) = (\ell/k)^{1/3}$ and $a = \ell/r_0^3$, so that $r \le r_0$ exactly when $k \ge a$, and

$$k^{-2/3}\,rk = \ell^{1/3}, \qquad k^{-2/3}\,r^2 = \ell^{2/3}\,k^{-4/3}, \qquad \delta = \frac{r}{k} = \ell^{1/3}k^{-4/3}. \tag{S7.46}$$

**Lemma S7.19** (near candidate density)**.** *For $0 < \ell \le 1$,*

$$\big|\ell^{1/3}\,\nu_{\mathrm{cand}}^{\mathrm{near}}(\ell) - c_{d,L}\big| \le C\ell^{1/3}. \tag{S7.47}$$

*Proof.* By (S7.19) and (S7.22),

$$\ell^{1/3}\nu_{\mathrm{cand}}^{\mathrm{near}}(\ell) - c_{d,L} = \int_{\{k\ge a\}} \frac{\mathcal{A}_r - \mathcal{A}_0}{3k^{2/3}}\,db\,dk\,d\sigma(t) - \int_{\{k<a\}} \frac{\mathcal{A}_0}{3k^{2/3}}\,db\,dk\,d\sigma(t).$$

By (S7.40) and (S7.45), the missing contact mass is at most $\sigma(S^{d-1})\,\bar\Psi\int_0^a k^{4/3}\,dk/3 \le Ca^{7/3}$. In the first integral, (S7.40) bounds the integrand by $r(k+r)\Psi/(3k^{2/3})$, whose two monomials are given by (S7.46). The monomial $\ell^{1/3}$ contributes at most $C\ell^{1/3}$ by (S7.45). For the monomial $\ell^{2/3}k^{-4/3}$, the integral over $a \le k \le 1$ is at most $3a^{-1/3}\sigma(S^{d-1})\bar\Psi$ and the integral over $k \ge 1$ is finite; it contributes at most $C\ell^{2/3}a^{-1/3} + C\ell^{2/3} = Cr_0\ell^{1/3} + C\ell^{2/3}$. Finally $a^{7/3} = r_0^{-7}\ell^{7/3} \le C\ell^{1/3}$ for $\ell \le 1$. This proves (S7.47). □

The rate in (S7.47) is proved from the quantitative bounds (S7.40); it is not a rate for the dominated-convergence argument of [7, §13] (Proposition S12.28), which gives the limit only. The lower cutoff $a = \ell/r_0^3$ must be kept: replacing it by zero in the term $\ell^{2/3}k^{-4/3}$ would produce a divergent integral.

**Lemma S7.20** (near pairing loss)**.** *Let $\ell_1 \in (0,1)$ satisfy $\ell_1^{5/6} < r_0^3$. For $0 < \ell \le \ell_1$,*

$$0 \le \ell^{1/3}(\nu_{\mathrm{cand}}^{\mathrm{near}}(\ell) - \nu_{\mathrm{eld}}^{\mathrm{near}}(\ell)) \le C\ell^{1/3}. \tag{S7.48}$$

*Proof.* Let $\eta = \ell^{1/6}$. The choice of $\ell_1$ gives $a < \eta < 1$. By (S7.19)–(S7.20),

$$\ell^{1/3}(\nu_{\mathrm{cand}}^{\mathrm{near}} - \nu_{\mathrm{eld}}^{\mathrm{near}})(\ell) = \int_{\{k \ge a\}} \frac{\mathcal{A}_r\,(1-p_r)}{3k^{2/3}}\, db\, dk\, d\sigma(t) \ge 0.$$

On $a \le k \le \eta$ use $\mathcal{A}_r(1-p_r) \le \mathcal{A}_r \le (k+r)^2\Psi \le (2k^2+2r^2)\Psi$, by Lemma S7.18 and (S7.40). Since $k^{-2/3}k^2 = k^{4/3}$ and $k^{-2/3}r^2 = \ell^{2/3}k^{-4/3}$, this region contributes at most

$$C\big[\eta^{7/3} + \ell^{2/3}a^{-1/3}\big] \le C\big[\ell^{7/18} + \ell^{1/3}\big]. \tag{S7.49}$$

On $k \ge \eta$, $\delta = \ell^{1/3}k^{-4/3} \le \ell^{1/9} \le 1$ by (S7.46), so the loss estimate (S7.42) applies, for all $b$ and also for large $k$. With $\mathcal{A}_r(1-p_r) \le \mathcal{B}_r \le \delta^3\Psi$ the integrand is at most $\delta^3\Psi/(3k^{2/3}) = \ell k^{-14/3}\Psi/3$. Its integral over $\eta \le k \le 1$ is at most $C\ell\eta^{-11/3} = C\ell^{7/18}$, and its integral over $k \ge 1$ is at most $C\ell$, by (S7.45). Since $7/18 > 1/3$ and $\ell \le 1$, (S7.48) follows. □

No lower bound for the normalized law is used in the region of small $k$; there the comparison $\mathcal{A}_r(1-p_r) \le \mathcal{A}_r$ suffices.

*Proof of Theorem S7.9.* Let $\ell_* = \ell_1$ be as in Lemma S7.20, and let $0 < \ell \le \ell_*$. By Lemma S7.19 and (S7.21),

$$\big|\nu_{\mathrm{cand}}(\ell) - c_{d,L}\ell^{-1/3}\big| \le \ell^{-1/3}\big|\ell^{1/3}\nu_{\mathrm{cand}}^{\mathrm{near}}(\ell) - c_{d,L}\big| + \nu_{\mathrm{cand}}^{\mathrm{far}}(\ell) \le C.$$

By Lemma S7.20 and (S7.21),

$$0 \le \nu_{\mathrm{cand}}(\ell) - \nu_{\mathrm{eld}}(\ell) = (\nu_{\mathrm{cand}}^{\mathrm{near}} - \nu_{\mathrm{eld}}^{\mathrm{near}})(\ell) + (\nu_{\mathrm{cand}}^{\mathrm{far}} - \nu_{\mathrm{eld}}^{\mathrm{far}})(\ell) \le C.$$

The third inequality in (S7.24) follows from the first two. By Hypotheses S7.4(b), S7.6 and S7.7, $\nu_{\mathrm{cand}}$ and $\nu_{\mathrm{eld}}$ are versions of the densities of all candidate pairs and of all finite bars. □

**Corollary S7.21** (counts and moments of short bars)**.** *Let $0 < \varepsilon \le \ell_*$. Let $N_{\mathrm{eld}}(0,\varepsilon]$ be the number of finite bars with lifetime at most $\varepsilon$, and $N_{\mathrm{rej}}(0,\varepsilon]$ the number of candidate pairs with gap at most $\varepsilon$ that are not finite bars. Per unit volume,*

$$L^{-d}\,\mathbb{E}N_{\mathrm{eld}}(0,\varepsilon] = \tfrac{3}{2}\,c_{d,L}\,\varepsilon^{2/3} + O(\varepsilon), \qquad 0 \le L^{-d}\,\mathbb{E}N_{\mathrm{rej}}(0,\varepsilon] \le C\varepsilon. \tag{S7.50}$$

*For $q > -2/3$, the sums $\mathcal{N}_q(\varepsilon) = L^{-d}\,\mathbb{E}\sum_{\ell_i \le \varepsilon} \ell_i^q$, over the lifetimes $\ell_i$ of the finite bars and over the gaps $\ell_i$ of the candidate pairs, both satisfy*

$$\mathcal{N}_q(\varepsilon) = c_{d,L}\,\frac{\varepsilon^{q+2/3}}{q+2/3} + O(\varepsilon^{q+1}). \tag{S7.51}$$

*For $q > -1$, the same sum over the candidate pairs that are not finite bars is at most $C_q\,\varepsilon^{q+1}$.*

*Proof.* Integrate (S7.24) against $\ell^q\,d\ell$ over $(0,\varepsilon]$: for $q > -2/3$, $\int_0^\varepsilon \ell^{q-1/3}\,d\ell = \varepsilon^{q+2/3}/(q+2/3)$, and for $q > -1$, $\int_0^\varepsilon C\ell^q\,d\ell = C\varepsilon^{q+1}/(q+1)$. The case $q = 0$ is (S7.50). □

The last statement of Corollary S7.21 concerns the counting measure of the candidate pairs that are not finite bars directly, not a difference of two infinite expectations, and it is meaningful also for $-1 < q \le -2/3$. It shows that inverse lifetimes $\ell^{-p}$ are integrable against these pairs for $p < 1$; this is a sufficient range, not an exact threshold. For the finite bars and for the candidate pairs separately the threshold is $p < 2/3$: by (S7.24) their densities are at least $\frac{1}{2}c_{d,L}\ell^{-1/3}$ for small $\ell$, so $\mathbb{E}\sum_{\ell_i \le \varepsilon} \ell_i^{-p}$ is finite for $p < 2/3$ and infinite for $p \ge 2/3$.

**The side-24 field.**

*Remark* S7.22 ($d=3$, $L=24$). Let $d=3$ and $L=24$. Then (S7.1) is the covariance (1.1) of the main text (its denominator is $Z_{24}=\sum_n e^{-288|n|^2}$), $f$ is the field $f_{24}$, $k=\kappa/6$ and $\ell=kr^3=\kappa r^3/6$, and $\nu_{\mathrm{eld}}$ is the lifetime density $\nu_{3,24}$ of the finite nonessential superlevel $H_0$ bars per unit volume (Convention 2.2 of the main text). The constant $c_{3,24}$ of Definition S7.8, that is, the constant of [7, (15.2)] (Proposition S12.31) at $d=3$ and $L=24$, is the constant $c_{3,24}$ of (2.9) of the main text. Theorem S7.9 is the statement of Theorem R of record [R3] of the main text at every fixed $d\geq 2$ and $L>0$. The proof of Proposition 3.12 of the main text identifies that proposition with Theorem R at $d=3$ and $L=24$, and §§5.1 and 6 of the main text state that its leading law is the instance $d=3$, $L=24$ of [7, Theorem C] (Theorem S12.5), whose coefficient is [7, (15.2)] (Proposition S12.31). The two formulas also agree term by term: in the notation of (2.2)–(2.8) of the main text, $p_G(0)=(2\pi)^{-3/2}\det\Sigma_g(t)^{-1/2}$, $p_{V_t}(0)=(2\pi)^{-3/2}\det\Sigma_h(t)^{-1/2}$, $\tau_t^2=144\,\sigma_t^2$ because $u_t=-\partial_t^3 f_{24}/12$, and $D_t$ in (S7.23) is $D_t$ of (2.8); since

$$(2\pi)^{-3}\cdot 144^{2/3}\cdot\frac{\Gamma(7/6)}{24^{1/3}\sqrt{\pi}}=\frac{3\,\Gamma(7/6)}{2^{4/3}\pi^{7/2}}=B_3,$$

(S7.23) becomes (2.9) of the main text, with $dt$ there the surface measure on $S^2$.

Theorem S7.9 therefore gives $C,\ell_*>0$ with

$$\left|\nu_{3,24}(\ell)-c_{3,24}\,\ell^{-1/3}\right|\leq C,\qquad 0<\ell\leq\ell_*,$$

which is Proposition 3.12 of the main text. Since $c_{3,24}>0$, this is $\nu_{3,24}(\ell)=c_{3,24}\,\ell^{-1/3}(1+O(\ell^{1/3}))$, the rate in (2.1) of the main text. In this bound the near-contact error (Lemma S7.19), the near pairing loss (Lemma S7.20) and the far pairs (Hypothesis S7.7) each contribute $O(1)$. Theorem S7.9 does not prove that $\nu_{3,24}(\ell)-c_{3,24}\ell^{-1/3}$ tends to zero as $\ell\downarrow 0$.

## Part II. The three-term law

### S8. Vanishing and rate of the remainder

This section and the next prove the first links of Theorem 2.3 of the main text. Theorem S8.6 identifies the constant term of the bounded remainder of Theorem R (§S7): it is the equal-height mass $B_{d,L}$ for the candidate density and 0 for the elder density. Theorem S8.9 shows that the elder remainder is $O(\ell^{1/4})$. Both theorems hold for every fixed dimension $d\geq 2$ and side $L>0$; at $d=3$, $L=24$ the elder density is $\nu_{3,24}$.

#### S8.1. **Setting and inputs.**

**Convention S8.1** (notation for Part II)**.** (a) *Field and constants.* Fix $d\geq 2$ and $L>0$, and let $f$ be the centred stationary Gaussian field of variance one on the torus $\mathbb{T}_L^d=\mathbb{R}^d/(L\mathbb{Z}^d)$ with covariance $K_L(z)=\sum_{n\in\mathbb{Z}^d}e^{-|z+Ln|^2/2}/\sum_{n\in\mathbb{Z}^d}e^{-|Ln|^2/2}$; for $d=3$, $L=24$ this is (1.1) of the main text. Put $m:=d-1$. Unless stated otherwise, $C,c,N$ are positive constants depending on $d$ and $L$ only; they may change from line to line. The index of a symmetric matrix is its number of negative eigenvalues.

(b) *Densities.* $\nu_{\mathrm{cand}}(\ell)$ is the $\ell$-density, per unit volume, of the expected number of ordered pairs $(M,S)$ of a local maximum $M$ and a critical point $S$ of index $d-1$ with $f(M)-f(S)=\ell>0$, all births, gaps and separations included; $\nu_{\mathrm{eld}}(\ell)$ is the same density for the pairs that are finite ordinary superlevel $H_0$ bars (the essential class excluded); $\rho_{\mathrm{rej}}:=\nu_{\mathrm{cand}}-\nu_{\mathrm{eld}}$. All three are taken in their canonical marked Kac–Rice density versions (Convention 2.2 of the main text and §S1; for general $d,L$ the same construction, [7, §9]; Proposition S12.22(b), (S12.51) and Lemma S12.29). At $d=3$, $L=24$, $\nu_{\mathrm{eld}}=\nu_{3,24}$. A superscript near, $\rho$, far, $\rho$ or $[\rho,\rho')$ restricts to pairs with $\mathrm{dist}(M,S)<\rho$, $\geq\rho$ or in $[\rho,\rho')$. The constant $c_{d,L}>0$ is the

leading coefficient of [7, (15.2)] (Proposition S12.31), so that $\nu_{\mathrm{cand}}(\ell) \sim \nu_{\mathrm{eld}}(\ell) \sim c_{d,L}\ell^{-1/3}$ [7, Theorem C] (Theorem S12.5); at $d = 3$, $L = 24$ it equals $c_{3,24}$ of (2.9) of the main text, both being the leading coefficient of $\nu_{3,24}$.

(c) *Pinned pairs.* By stationarity the midpoint of the pair is the origin. For a direction $u \in S^{d-1}$ (the $t$ of (3.4) of the main text; we write $u$, as in (2.13) of the main text), an orthonormal frame $\Theta$ of $u^{\perp}$, and coordinates $(x, y) \in \mathbb{R} \times \mathbb{R}^m$ along $u$ and $\Theta$, the pins are

$$M = -\tfrac{r}{2}u, \quad S = \tfrac{r}{2}u, \qquad f(M) = b, \quad f(S) = b - kr^3, \qquad \nabla f(M) = \nabla f(S) = 0,$$

with $b \in \mathbb{R}$ and gap mark $k > 0$; in the notation of (3.5) of the main text $k = \kappa/6$, and the lifetime is $\ell = kr^3$. Put $P := 1+|b|+k$. The observation vector $U_r$ has the axial rows $\frac{1}{2}(f(M)+f(S))$, $(f(S)-f(M))/r$, $(\partial_x f(S)-\partial_x f(M))/r$, $\frac{6}{r^2}[\partial_x f(M)+\partial_x f(S)-2(f(S)-f(M))/r]$ and the transverse rows $\frac{1}{2}(\partial_{y_j} f(M)+\partial_{y_j} f(S))$, $(\partial_{y_j} f(S)-\partial_{y_j} f(M))/r$; the map from the raw pins $(f(M), \nabla f(M), f(S), \nabla f(S))$ to $U_r$ has absolute determinant $12r^{-(d+3)}$, and the target of $U_r$ is $v_r = (b - kr^3/2,\ -kr^2,\ 0,\ 12k,\ 0, \dots, 0)$. At $r = 0$, $U_0 = (f, \partial_x f, \partial_x^2 f, \partial_x^3 f, \nabla_y f, \nabla_y \partial_x f)(0)$ with target $v_0 = v_0(b, k) = (b, 0, 0, 12k, 0, \dots, 0)$. We write $\pi_r$, $\pi_0$ for the Gaussian densities of $U_r$, $U_0$, $Q = Q_{r,b,k}$ for the continuous Gaussian regression law of the field given $U_r = v_r$, and call the regression law given $U_0 = v_0$ the *contact law*.

(d) *Weights.* For a symmetric matrix $H$ let $F_j(H) := |\det H|$ if $H$ has exactly $j$ negative eigenvalues and $F_j(H) := 0$ otherwise (so $F_j(H) = 0$ if $H$ is singular). With $H_i = D^2 f(i)$ $(i \in \{M, S\})$, the typed weight is $W_r := |\det H_M \det H_S|\, \mathbf{1}\{H_M \prec 0,\ \operatorname{ind} H_S = d - 1\}$ and $Z_r := \mathbb{E}_Q W_r$. The scaled endpoint Hessians are

$$K_i := D_r^{-1} H_i D_r^{-1} = \begin{pmatrix} \alpha_i & \sqrt{r}\,\beta_i^{\top} \\ \sqrt{r}\,\beta_i & A_i \end{pmatrix}, \qquad \alpha_i = \frac{\partial_x^2 f(i)}{r}, \quad \beta_i = \frac{\nabla_y \partial_x f(i)}{r}, \quad A_i = D_y^2 f(i),$$

with $D_r = \operatorname{diag}(\sqrt{r}, I_m)$; thus $|\det H_i| = r|\det K_i|$, $K_i$ is congruent to $H_i$, and $W_r/r^2 = F_d(K_M)F_{d-1}(K_S)$.

(e) *Elder mark.* For $M \in \mathbb{T}_L^d$, $d_f(M) := \sup\{\min_{s\in[0,1]} f(\gamma(s)) : \gamma \in C([0,1]; \mathbb{T}_L^d),\ \gamma(0) = M,\ f(\gamma(1)) > f(M)\}$ (with $\sup \emptyset = -\infty$) is the maximin connection level, and $e := \mathbf{1}\{d_f(M) = f(S)\}$. The mark $e$ is Borel [7, §8] (Lemma S12.20; cf. §S5); on the Morse locus with distinct critical values, $e = 1$ exactly when $S$ is the elder death saddle of $M$.

(f) *Kernels.* $\mathcal{A}_r(b, k, u) := 12\pi_r(v_r)\mathbb{E}_Q[W_r/r^2]$, $\mathcal{A}_r^{\mathrm{eld}} := 12\pi_r(v_r)\mathbb{E}_Q[(W_r/r^2)e]$ and $\mathcal{A}_r^{\mathrm{rej}} := \mathcal{A}_r - \mathcal{A}_r^{\mathrm{eld}} = 12\pi_r(v_r)\mathbb{E}_Q[(W_r/r^2)(1-e)]$. At contact, $A_0$ denotes the transverse Hessian $D_y^2 f(0)$ under the contact law, $z_0(b, k, u) := \mathbb{E}[(6k)^2(\det A_0)^2 \mathbf{1}\{A_0 \prec 0\}]$ and $\mathcal{A}_0(b, k, u) := 12\pi_0(v_0)z_0$. Finally $H(b, k) := CP^N e^{-c(b^2+k^2)}$, with $N$ enlarged when necessary.

(g) *Two-site kernels.* For $y \in \mathbb{T}_L^d \setminus \{0\}$ let $O_y = (f(0), \nabla f(0), f(y), \nabla f(y))$, $v_{b,\ell} = (b, 0, b - \ell, 0)$, $p_y$ the density of $O_y$, $Q_{y,b,\ell}$ the regression law given $O_y = v_{b,\ell}$, $W = |\det H_0 \det H_y|\, \mathbf{1}\{H_0 \prec 0,\ \operatorname{ind} H_y = d - 1\}$ and $e_y = \mathbf{1}\{d_f(0) = f(y)\}$. The unmarked candidate kernel, the equal-height kernel and the equal-height mass are

$$\Psi_\ell^{\mathrm{cand}}(b, y) = p_y(v_{b,\ell})\, \mathbb{E}_{Q_{y,b,\ell}}[W], \qquad \Psi_0(b, y) = p_y(v_{b,0})\, \mathbb{E}_{Q_{y,b,0}}[W],$$

$$B_{d,L} = \int_{\mathbb{T}_L^d \setminus \{0\}} \int_{\mathbb{R}} \Psi_0(b, y)\, db\, dy.$$

(h) *Radii.* $r_R \in (0, \frac{1}{2}]$ is the radius of Proposition S8.2, $r_K$ that of Proposition S8.3, and $r_Z := r_K$. The *admissible radius* is $r_0^* := \min\{r_R, r_K, r_Z\}$, decreased if necessary so that $r_0^* < L/(4\sqrt{2})$, the embedding radius of the cap cylinder (each of the two radius statements remains true for a smaller radius; for $L = 24$ nothing changes, as $r_R \le 1 < 24/(4\sqrt{2})$). The dominating variable is

$$T := 1 + k + \|F_r\|_{C^7} + \|F_0\|_{C^7} + r^{-2}\|F_r - F_0\|_{C^2},$$

for the coupling $(F_r, F_0)$ of Proposition S8.2.

The marked Kac–Rice disintegration (§S2; [7, §§9–10, (10.2)]; Propositions S12.22 and S12.23) gives the candidate intensity $r\mathcal{A}_r\,dr\,db\,dk\,d\sigma(u)$ per unit volume, with $\sigma$ the surface measure, and the elder and rejected intensities with $\mathcal{A}_r^{\mathrm{eld}}$, $\mathcal{A}_r^{\mathrm{rej}}$ in place of $\mathcal{A}_r$. Pushing forward at fixed $r$ with $dk = d\ell/r^3$ gives, for $0 \le \rho < \rho' \le r_R$, $\ell > 0$ and $\bullet \in \{\mathrm{cand}, \mathrm{eld}, \mathrm{rej}\}$ (with $\mathcal{A}_r^{\mathrm{cand}} := \mathcal{A}_r$),

$$\nu_\bullet^{[\rho,\rho')}(\ell) = \int_\rho^{\rho'} \int_{\mathbb{R}} \int_{S^{d-1}} r^{-2}\mathcal{A}_r^\bullet(b, \ell/r^3, u)\, d\sigma(u)\, db\, dr. \tag{S8.1}$$

The leading coefficient is $c_{d,L} = \int_{S^{d-1}} \int_{\mathbb{R}} \int_0^\infty \mathcal{A}_0(b,k,u)\,(3k^{2/3})^{-1}\,dk\,db\,d\sigma(u)$ (§S7; [7, (13.6)]; Proposition S12.28); the substitution $k = \ell/r^3$, with $|dk/dr| = 3\ell r^{-4}$ and the order of the endpoints reversed, turns it into

$$c_{d,L}\,\ell^{-1/3} = \int_0^\infty \int_{\mathbb{R}} \int_{S^{d-1}} r^{-2}\mathcal{A}_0(b, \ell/r^3, u)\, d\sigma(u)\, db\, dr. \tag{S8.2}$$

Translating the pins to 0 and $y = ru$, the polar factor $r^{d-1}$, the pin density $12r^{-(d+3)}\pi_r(v_r)$ and the determinant scale $r^2$ give, for $0 < r \le r_0^*$,

$$r^{-2}\mathcal{A}_r(b, \ell/r^3, u) = r^{d-1}\Psi_\ell^{\mathrm{cand}}(b, ru), \qquad r^{-2}\mathcal{A}_r(b, 0, u) = r^{d-1}\Psi_0(b, ru), \tag{S8.3}$$

where $\mathcal{A}_r(b,0,u)$ is defined with the pinned law at $k = 0$. On $\{\mathrm{dist}(0,y) \ge \rho\}$ the height map $(b,\ell) \mapsto (b, b-\ell)$ has absolute Jacobian one, and

$$\nu_{\mathrm{cand}}^{\mathrm{far},\rho}(\ell) = \int_{\mathrm{dist}(0,y)\ge\rho} \int_{\mathbb{R}} \Psi_\ell^{\mathrm{cand}}(b,y)\, db\, dy, \qquad \nu_{\mathrm{eld}}^{\mathrm{far},\rho}(\ell) = \int_{\mathrm{dist}(0,y)\ge\rho} \int_{\mathbb{R}} p_y(v_{b,\ell})\, \mathbb{E}_{Q_{y,b,\ell}}[We_y]\, db\, dy. \tag{S8.4}$$

Separation boundaries are Lebesgue-null in the pair variables, so the ranges in (S8.1)–(S8.4) may be combined freely.

We use three groups of results that are proved elsewhere.

**Proposition S8.2** (the contact coupling and the bounds of Theorem R)**.** *There is $r_R \in (0, \frac12]$, with $r_R \le r_0$ of (S7.12), such that, for $0 < r \le r_R$, $b \in \mathbb{R}$ and every frame:*

*(i) Let $F$ be an unconditioned copy of the field, $\Sigma_r = \mathrm{Cov}(U_r)$ and $C_r(z) = \mathrm{Cov}(F(z), U_r)$. Then $F_r := F + C_r\Sigma_r^{-1}(v_r - U_r(F))$ has law $Q$ and $F_0 := F + C_0\Sigma_0^{-1}(v_0 - U_0(F))$ has the contact law. The matrices $\Sigma_r$ are uniformly positive definite and bounded, $\Sigma_r - \Sigma_0$, $\Sigma_r^{-1} - \Sigma_0^{-1}$ and $\|C_r - C_0\|_{C^q}$ are $O(r^2)$, and these covariance facts do not depend on the target, so (i) and (ii) hold for all $k \ge 0$. For every finite $p \ge 1$ and every derivative order $q$,*

$$\|F_r - F_0\|_{L^p(C^q)} \le C_{p,q}\, r^2 P, \qquad \big\|1 + \|F_r\|_{C^q} + \|F_0\|_{C^q}\big\|_{L^p} \le C_{p,q}\, P. \tag{S8.5}$$

*In particular $\|T\|_{L^p} \le C_p P$ for every $p$.*

*(ii) $\pi_r(v_r) \le Ce^{-c(b^2+k^2)}$ and $|\pi_r(v_r) - \pi_0(v_0)| \le Cr^2P^2e^{-c(b^2+k^2)}$.*

*(iii) For a symmetric $m \times m$ matrix $A$, $\alpha \in \mathbb{R}$, $\beta \in \mathbb{R}^m$ and $t \ge 0$, the matrix $K_t = \left(\begin{smallmatrix} \alpha & \sqrt{t}\beta^\top \\ \sqrt{t}\beta & A \end{smallmatrix}\right)$ satisfies, also for singular $A$,*

$$\det K_t = \alpha \det A - t\,\beta^\top \mathrm{adj}(A)\,\beta. \tag{S8.6}$$

*(iv) For $k \ge 0$, with $f = F_r$ and $A_0 := D_y^2F_0(0)$: $\|A_i - A_0\| \le (\frac{r}{2} + r^2)T \le rT$, $\|A_i\| + \|A_0\| + \|\beta_i\| \le CT$, $|\alpha_M + 6k| + |\alpha_S - 6k| \le CrT$, $F_d(K_M), F_{d-1}(K_S) \le C(k+r)T^d$, and*

$$\frac{Z_r}{r^2} \le C(k+r)^2P^N, \qquad 0 \le \mathcal{A}_r \le (k+r)^2H, \qquad 0 \le \mathcal{A}_0 \le k^2H, \qquad |\mathcal{A}_r - \mathcal{A}_0| \le r(k+r)H. \tag{S8.7}$$

*Proof.* These are the coupling, the pin-density estimates, the block-determinant lemma and the weight bounds in the proof of Theorem R (§S7); Theorem R and its proof hold for every fixed $d \geq 2$ and $L > 0$. These bounds hold also at $k = 0$: no division by $k$ or by $Z_r$ occurs in the proofs of (S7.35)–(S7.40) (see the remark after (S7.40)); the covariance $\Sigma_r$ does not depend on the target, so the Gaussian bounds hold with $k = 0$; the endpoint identities (S12.17) of Lemma S12.14 are deterministic and hold for every value of the gap mark, including $k = 0$; and the block-determinant comparison (iii) holds also for singular blocks. The variable $T$ dominates the dominating variable used there (which is built from $C^4$ norms), so its bounds hold with $T$ after adjusting $C$. By Taylor's formula, $\|A_i - A_0\| \leq \frac{r}{2}\|F_r\|_{C^3} + \|F_r - F_0\|_{C^2} \leq (\frac{r}{2} + r^2)T$, and $\frac{r}{2} + r^2 \leq r$ as $r \leq \frac{1}{2}$; a smaller radius keeps every statement. □

**Proposition S8.3** (gap-mark-explicit weight bounds)**.** *Let $G_r := \{\lambda_{\min}(-A_M) > 4rM_3^2/(3k)\} \cap \{rM_4 \leq 3k/10\}$ be the cap region, where $M_3, M_4$ are the local $C^3$ and $C^4$ seminorms of the field on the cap cylinder* [7, §§4,7] *(Definition S13.1). There are $r_K > 0$, $C$ and $N$ such that, for every $b \in \mathbb{R}$, $k > 0$, every frame and $0 < r \leq r_K$,*

$$\mathbb{E}_Q[W_r/r^2] \leq C(k+r)^2 P^N, \qquad \mathbb{E}_Q[(W_r/r^2)\mathbf{1}_{G_r^c}] \leq C\,\frac{r^3}{k}\,P^N \quad \text{whenever } r \leq \min\{k, r_K\}. \tag{S8.8}$$

*Moreover, if also $r < L/(4\sqrt{2})$, then $1 - e \leq \mathbf{1}_{G_r^c}$ $Q$-almost surely on $\{W_r > 0\}$.*

*Proof.* The first bound is that of (S8.7), as $Z_r = \mathbb{E}_Q W_r$, and the second is Lemma S8.16; take $r_K := \min\{r_R, r_0\}$ with the radius $r_0$ of (S7.12). Together the two bounds are Theorem K of record [R7] of the main text. The last statement is the deterministic cap theorem [7, §§7–8] (Lemma S12.21 and Theorem S13.2), which needs the cap cylinder to be embedded, that is, $r < L/(4\sqrt{2})$ (Definition S12.2): on $G_r$ and on the typed support, the global elder partner of $M$ is $S$, and $Q$ is almost surely Morse with distinct critical values [7, §8] (Lemma S12.19(a)). □

**Proposition S8.4** (the equal-height kernel)**.** *The following hold.*

*(i)* $0 < B_{d,L} < \infty$, *and, for the canonical marked Kac–Rice density versions,* $\rho_{\mathrm{rej}}(\ell) \to B_{d,L}$ *as* $\ell \downarrow 0$.

*(ii) For fixed $y \neq 0$ and $b$, the covariance of $Q_{y,b,\ell}$ does not depend on $\ell$; on one probability space $f_\ell = f_0 + \ell h_y$, with $f_0$ of law $Q_{y,b,0}$ and $h_y$ a deterministic smooth function with $h_y(0) = 0$, $h_y(y) = -1$, $\nabla h_y(0) = \nabla h_y(y) = 0$. The two endpoint Hessians are almost surely nonsingular, and $\Psi_\ell^{\mathrm{cand}}(b, y) \to \Psi_0(b, y)$ as $\ell \downarrow 0$ for every $y \neq 0$ and $b \in \mathbb{R}$.*

*(iii) For every $\rho \in (0, L/2]$ there are $C_\rho, c_\rho > 0$ with $0 \leq \Psi_\ell^{\mathrm{cand}}(b, y) \leq C_\rho(1 + |b|)^{2d} e^{-c_\rho b^2}$ for $\mathrm{dist}(0, y) \geq \rho$ and $0 < \ell \leq 1$.*

*Proof.* Item (iii) is the nondegenerate far bound [7, §14, (14.1)] (Lemma S12.29). Items (i) and (ii) are Theorem Z of record [R7] of the main text; they are proved in §S8.6, item (ii) and the bounds $0 < B_{d,L} < \infty$ directly, and the limit in (i) from (S8.16) and Theorem S8.9, whose proofs use this proposition only through (ii), (iii) and the finiteness of $B_{d,L}$ (the proof of (S8.17) then uses the limit). □

## S8.2. **The even contact expansion.**

**Lemma S8.5** (even contact expansion)**.** *There are $C, N$ such that for all $b \in \mathbb{R}$, $k > 0$, $u \in S^{d-1}$ and $0 < r \leq r_0^*$,*

$$\left|\frac{Z_r}{r^2} - z_0(b, k, u)\right| \leq Cr^2 P^N, \qquad |\mathcal{A}_r - \mathcal{A}_0| \leq r^2 H(b, k). \tag{S8.9}$$

The first-order term of (S8.7) is absent: the two endpoint corrections enter the typed product with opposite signs and cancel identically (Steps 1–3 below), and on the layers where the inertia can flip the product of the two typed determinants is $O(r^2)$ (Step 4).

*Proof.* Work under the coupling of Proposition S8.2(i). Write $f = F_r$, $f_j = \partial_x^j f(0)$, and $O(x)$ for a quantity bounded by $C|x|$ with $C = C(d)$; $\|T\|_{L^p} \le C_p P$ by (S8.5) with $q = 7$ and $q = 2$.

*Step 1 (axial pins to second order).* The third and fourth rows of $U_r$ take the pinned values 0 and $12k$. Taylor's formula with integral remainder (the rules are centred, so the odd orders cancel) gives

$$\frac{\partial_x f(S) - \partial_x f(M)}{r} = f_2 + \frac{r^2}{24} f_4 + O(r^4 T) = 0,$$
$$\frac{6}{r^2}\Big[\partial_x f(M) + \partial_x f(S) - \frac{2(f(S) - f(M))}{r}\Big] = f_3 + \frac{r^2}{40} f_5 + O(r^4 T) = 12k.$$

Substituting both identities in $\partial_x^2 f(\mp r/2) = f_2 \mp \frac{r}{2} f_3 + \frac{r^2}{8} f_4 + O(r^3 T)$ gives

$$\alpha_M = -6k + \frac{r}{12} f_4 + \rho_M, \qquad \alpha_S = 6k + \frac{r}{12} f_4 + \rho_S, \qquad |\rho_M| + |\rho_S| \le C r^2 T. \tag{S8.10}$$

(For a quintic profile the expansion is exact: $\alpha_{M,S} = \mp 6k + \frac{r}{12} f_4 \mp \frac{r^2}{120} f_5$.) The first-order coefficients coincide, so

$$\begin{aligned} \alpha_M \alpha_S &= -36k^2 + 6k(\rho_M - \rho_S) + r^2\Big(\frac{f_4}{12}\Big)^2 + \frac{r f_4}{12}(\rho_M + \rho_S) + \rho_M \rho_S \\ &= -36k^2 + O(r^2(1+k)T^2). \end{aligned} \tag{S8.11}$$

*Step 2 (transverse blocks).* Put $B := \partial_x D_y^2 f(0)$. Since $\|D_y^2 f(0) - A_0\| \le r^2 T$ by the coupling,

$$A_M = A_0 - \tfrac{r}{2} B + O(r^2 T), \qquad A_S = A_0 + \tfrac{r}{2} B + O(r^2 T), \tag{S8.12}$$

and $\det(A_0 + V) = \det A_0 + \operatorname{tr}(\operatorname{adj}(A_0) V) + O(\|V\|^2 T^{m-2})$ gives $\det A_M \det A_S = (\det A_0)^2 + \det A_0\,[-\frac{r}{2} + \frac{r}{2}] \operatorname{tr}(\operatorname{adj}(A_0) B) + O(r^2 T^{2m}) = (\det A_0)^2 + O(r^2 T^{2m})$. The transverse difference row gives $(\nabla_y f(S) - \nabla_y f(M))/r = \nabla_y \partial_x f(0) + \frac{r^2}{24} \nabla_y \partial_x^3 f(0) + O(r^4 T) = 0$, so $\nabla_y \partial_x f(0) = O(r^2 T)$; with $\gamma := \nabla_y \partial_x^2 f(0)$,

$$\begin{aligned} \beta_M = -\tfrac{1}{2}\gamma + O(rT), \qquad \beta_S = \tfrac{1}{2}\gamma + O(rT), \\ q_i := \beta_i^\top \operatorname{adj}(A_i) \beta_i = q + O(r T^{m+1}), \qquad q := \tfrac{1}{4} \gamma^\top \operatorname{adj}(A_0) \gamma. \end{aligned} \tag{S8.13}$$

The sign of $\beta_i$ enters $q_i$ quadratically. That $\nabla_y \partial_x f(0)$ is $O(r^2)$ and not $O(r)$ is essential: an $O(r)$ value would shift $\beta_M$ and $\beta_S$ by a common $O(1)$ vector and make $q_M - q_S = O(1)$.

*Step 3 (the product of determinants).* By (S8.6) with $t = r$ (the off-diagonal entries of $K_i$ are $\sqrt{r}\beta_i$), $\det K_i = \alpha_i \det A_i - r q_i$ exactly. Hence

$$\det K_M \det K_S = \alpha_M \alpha_S \det A_M \det A_S - r(\alpha_M \det A_M\, q_S + \alpha_S \det A_S\, q_M) + r^2 q_M q_S.$$

By (S8.11) and Step 2 the first term is $-36k^2 (\det A_0)^2 + O(r^2(1+k)^2 T^{N_1})$. By (S8.10), (S8.12) and (S8.13), $\alpha_M \det A_M\, q_S = -6k \det A_0\, q + O(r(1+k)T^{N_1})$ and $\alpha_S \det A_S\, q_M = +6k \det A_0\, q + O(r(1+k)T^{N_1})$, so the middle term is $O(r^2(1+k)T^{N_1})$; the last term is $O(r^2 T^{N_1})$. Thus

$$\big|\det K_M \det K_S + 36k^2 (\det A_0)^2\big| \le C r^2 (1+k)^2 T^{N_1}. \tag{S8.14}$$

Structurally, $\det K_M = -6k \det A_0 + rY + O(r^2)$ and $\det K_S = +6k \det A_0 + rY + O(r^2)$ with the *same*

$$Y := \frac{f_4}{12} \det A_0 + 3k \operatorname{tr}\big(\operatorname{adj}(A_0) B\big) - q, \tag{S8.15}$$

so the $r$-term of the product is $6k \det A_0\,(Y - Y) = 0$. This is the reflection symmetry of the pinned family: reversing the axis exchanges $M$ and $S$, and the rows of $U_r$ are even in $r$.

*Step 4 (inertia).* For invertible $A_i$, Haynsworth's inertia additivity gives $\operatorname{inertia}(K_i) = \operatorname{inertia}(A_i) + \operatorname{inertia}(\sigma_i)$ with $\sigma_i := \alpha_i - r \beta_i^\top A_i^{-1} \beta_i$. So $\operatorname{ind} K_M = d$ if and only if $A_M \prec 0$ and $\sigma_M < 0$, and $\operatorname{ind} K_S = d - 1$ if and only if either $A_S \prec 0$ and $\sigma_S > 0$, or $\operatorname{ind} A_S = m - 1$ and $\sigma_S < 0$. Put $\mathsf{u} := C_1 r T (1 + T/k)$ with $C_1 = C_1(d) \ge 2$ large, and let $G := \{\lambda_{\max}(A_0) \le -\mathsf{u}\} \cap \{k \ge C_1 r T\}$.

On $G$: $\|A_i - A_0\| \le rT \le \mathsf{u}/C_1$, so $A_M, A_S \prec 0$; $\alpha_S \ge 6k - CrT > 0$ and $r\beta_S^\top(-A_S)^{-1}\beta_S \ge 0$, so $\sigma_S > 0$; $\alpha_M \le -6k + CrT \le -5k$ and $r\beta_M^\top(-A_M)^{-1}\beta_M \le rT^2/\lambda_{\min}(-A_M) \le 2rT^2/\mathsf{u} \le 2k/C_1$, so $\sigma_M < 0$. Hence on $G$ the indices are $d$ and $d-1$, $\mathbf{1}\{A_0 \prec 0\} = 1$, and

$$\left|F_d(K_M)F_{d-1}(K_S) - 36k^2(\det A_0)^2\mathbf{1}\{A_0 \prec 0\}\right| = \left|\,|\det K_M \det K_S| - 36k^2(\det A_0)^2\right|$$

is at most the left side of (S8.14). Off $G$ there are three cases, in each of which both typed products are $r^2$ times a polynomial in $T$:

- on $\{k < C_1 rT\}$: $F_d(K_M)F_{d-1}(K_S) \le C(k+r)^2T^{2d} \le Cr^2T^{2d+2}$ by Proposition S8.2(iv), and $36k^2(\det A_0)^2 \le Cr^2T^{2m+2}$;
- on $\{\lambda_{\max}(A_0) \ge \mathsf{u}\} \cap \{k \ge C_1 rT\}$: $\lambda_{\max}(A_M) \ge \mathsf{u} - rT > 0$, so $K_M$ is not negative definite, $F_d(K_M) = 0$, and $\mathbf{1}\{A_0 \prec 0\} = 0$; both products vanish;
- on $\{|\lambda_{\max}(A_0)| < \mathsf{u}\} \cap \{k \ge C_1 rT\}$: by Weyl's inequality $|\det A_i| \le (|\lambda_{\max}(A_0)| + rT)T^{m-1} \le 2\mathsf{u}T^{m-1}$, so $|\det K_i| \le (6k + CrT)\,2\mathsf{u}T^{m-1} + rT^{m+1} \le Ck\mathsf{u}T^{m-1}$ (as $rT^2 \le k\mathsf{u}/C_1$); hence $F_d(K_M)F_{d-1}(K_S) \le Ck^2\mathsf{u}^2T^{2m-2}$ and likewise $36k^2(\det A_0)^2\mathbf{1}\{A_0 \prec 0\} \le 36k^2\mathsf{u}^2T^{2m-2}$; with $k^2\mathsf{u}^2 = C_1^2r^2T^2(k+T)^2$ both are $\le Cr^2(k+T)^2T^{2m}$.

No smallness of the probability of these layers is needed: on the last layer each typed determinant is $O(r)$.

*Step 5 (assembly).* Taking expectations, $|Z_r/r^2 - z_0| \le \mathbb{E}[Cr^2(1+k)^2T^{N_1} + Cr^2(k+T)^2T^{2m+2}] \le Cr^2P^N$. For $\mathcal{A}_r - \mathcal{A}_0 = 12[\pi_r(v_r) - \pi_0(v_0)]Z_r/r^2 + 12\pi_0(v_0)[Z_r/r^2 - z_0]$ use Proposition S8.2(ii), the bound $Z_r/r^2 \le C(k+r)^2P^N$ of (S8.7) and the first bound: every term is $r^2$ times a polynomial in $P$ times $e^{-c(b^2+k^2)}$, that is, at most $r^2H$. □

## S8.3. **The constant term of the remainder.**

**Theorem S8.6** (Theorem R$^+$: the constant term of the remainder)**.** *For every fixed $d \ge 2$ and $L > 0$, and for the canonical marked Kac–Rice density versions (the versions for which Proposition S8.4(i) is stated; $\nu_{\mathrm{cand}}$ and $\nu_{\mathrm{eld}}$ are the unrestricted densities), as $\ell \downarrow 0$,*

$$\nu_{\mathrm{cand}}(\ell) = c_{d,L}\,\ell^{-1/3} + B_{d,L} + o(1), \tag{S8.16}$$

$$\nu_{\mathrm{eld}}(\ell) = c_{d,L}\,\ell^{-1/3} + o(1) = c_{d,L}\,\ell^{-1/3}(1 + o(\ell^{1/3})). \tag{S8.17}$$

*Moreover, for every $r_0 \in (0, r_0^*]$, as $\ell \downarrow 0$,*

$$\begin{aligned} \nu_{\mathrm{cand}}^{\mathrm{near},r_0}(\ell) - c_{d,L}\,\ell^{-1/3} &\to \int_{0<\mathrm{dist}(0,y)<r_0}\int_{\mathbb{R}} \Psi_0(b,y)\,db\,dy, \\ |\nu_{\mathrm{cand}}^{\mathrm{near},r_0}(\ell) - c_{d,L}\,\ell^{-1/3}| &\le C(r_0 + \ell^2 r_0^{-7}), \end{aligned} \tag{S8.18}$$

*the bound holding for every $\ell > 0$.*

So the bounded remainder of Theorem R has the constant term $B_{d,L}$ for the candidate density and the constant term 0 for the elder density: no expansion $\nu_{\mathrm{eld}} = c_{d,L}\ell^{-1/3} + C_0 + o(1)$ with $C_0 \ne 0$ is asserted, and the coefficient $c_1$ of §S9 is the coefficient of $\ell^{1/4}$, a different term. The theorem gives no rate in (S8.16), no numerical constants, no uniformity in $d$ or $L$, and no statement about other almost-everywhere-equal versions of the densities. It does not enlarge the scope of Proposition S8.4(i); it identifies the constant term of the remainder of Theorem R by means of it. A rate for (S8.17) is Theorem S8.9.

*Proof. The separation variable.* By (S8.1) and (S8.2),

$$\begin{aligned} \nu_{\mathrm{cand}}^{\mathrm{near},r_0}(\ell) - c_{d,L}\ell^{-1/3} &= \int_0^{r_0}\!\!\iint r^{-2}(\mathcal{A}_r - \mathcal{A}_0)(b, \ell/r^3, u)\,d\sigma\,db\,dr \\ &\quad - \int_{r_0}^{\infty}\!\!\iint r^{-2}\mathcal{A}_0(b, \ell/r^3, u)\,d\sigma\,db\,dr. \end{aligned} \tag{S8.19}$$

*Domination.* By Lemma S8.5, $|r^{-2}(\mathcal{A}_r - \mathcal{A}_0)(b, \ell/r^3, u)| \le H(b, \ell/r^3) \le Ce^{-cb^2}$, an integrable majorant on $(0, r_0) \times \mathbb{R} \times S^{d-1}$ that does not depend on $\ell$. (With the first-order bound of (S8.7) the integrand is only bounded by $(1 + \ell/r^4)H$.) The second term of (S8.19) is at most $\int_{r_0}^{\infty} r^{-2}(\ell/r^3)^2 H\, dr \le C\ell^2 r_0^{-7}$, because $\mathcal{A}_0 \le k^2 H$.

*Pointwise limit.* Fix $r, b, u$. As $\ell \downarrow 0$, $k = \ell/r^3 \downarrow 0$ and $\mathcal{A}_r(b, k, u) \to \mathcal{A}_r(b, 0, u)$: the pinned law is the regression law with target $v_r(b, k)$, affine in the target with a $k$-free covariance, so $F_r$ depends continuously on $k$ in $C^2$; the typed weight $W_r = F_d(H_M)F_{d-1}(H_S)$ is continuous in the endpoint Hessians, as $F_d$ and $F_{d-1}$ are locally Lipschitz (Lemma S7.14, where $F_j$ is written $\Phi_j$), and is dominated by a polynomial in them; and $\pi_r(v_r)$ is continuous in the target. Also $\mathcal{A}_0(b, k, u) \le k^2 H \to 0$. By (S8.3), dominated convergence in (S8.19) gives

$$\nu_{\mathrm{cand}}^{\mathrm{near},r_0}(\ell) - c_{d,L}\ell^{-1/3} \to \int_0^{r_0}\!\!\iint r^{d-1}\Psi_0(b, ru)\, d\sigma\, db\, dr = \int_{0<\mathrm{dist}(0,y)<r_0}\int_{\mathbb{R}} \Psi_0\, db\, dy,$$

and the majorant gives the bound in (S8.18).

*The far candidate density.* On the compact set $\{\mathrm{dist}(0, y) \ge r_0\}$, Proposition S8.4(iii) bounds $\Psi_\ell^{\mathrm{cand}}$ by an integrable function of $b$, uniformly in $y$ and in $0 < \ell \le 1$, and $\Psi_\ell^{\mathrm{cand}} \to \Psi_0$ pointwise by Proposition S8.4(ii). By (S8.4) and dominated convergence, $\nu_{\mathrm{cand}}^{\mathrm{far},r_0}(\ell) \to \int_{\mathrm{dist}(0,y)\ge r_0} \int \Psi_0$. Adding the near limit, $\nu_{\mathrm{cand}}(\ell) - c_{d,L}\ell^{-1/3} \to B_{d,L}$, which is finite by Proposition S8.4(i) (the cutoff boundary is spatially null). This is (S8.16).

*The elder density.* $\nu_{\mathrm{eld}} - c_{d,L}\ell^{-1/3} = (\nu_{\mathrm{cand}} - c_{d,L}\ell^{-1/3}) - \rho_{\mathrm{rej}} \to B_{d,L} - B_{d,L} = 0$ by (S8.16) and the limit in Proposition S8.4(i), which §S8.6 derives from (S8.16) and Theorem S8.9; equivalently, (S8.17) is the case $o(1)$ of (S8.20). This is (S8.17). □

The proof shows that the constant $C$ in (S8.18) depends on $d$ and $L$ only: it comes from the majorant $H(b, \ell/r^3) \le Ce^{-cb^2}$ on $r < r_0$ and from $\int_{r_0}^{\infty} r^{-2}(\ell/r^3)^2 H\, dr \le C\ell^2 r_0^{-7}$.

*Remark* S8.7 (a signed near bound). Always $\mathcal{A}_r \ge 0$ ($\pi_r \ge 0$ and $F_j \ge 0$) and $\mathcal{A}_0 \le k^2 H$. Put $r_1 := \min(\ell^{1/4}, r_0)$. In (S8.19), on $0 < r < r_1$ Lemma S8.5 gives $r^{-2}(\mathcal{A}_r - \mathcal{A}_0) \ge -Ce^{-cb^2}$, contributing at least $-Cr_1$; on $r_1 \le r < r_0$, $r^{-2}(\mathcal{A}_r - \mathcal{A}_0) \ge -\ell^2 r^{-8} H$, contributing at least $-C\ell^2 r_1^{-7}/7 = -C\ell^{1/4}/7$ when $r_1 = \ell^{1/4}$ (and nothing when $r_1 = r_0$); the far term is at least $-C\ell^2 r_0^{-7}$. Hence

$$\nu_{\mathrm{cand}}^{\mathrm{near},r_0}(\ell) - c_{d,L}\ell^{-1/3} \ge -C\big(\min(\ell^{1/4}, r_0) + \ell^2 r_0^{-7}\big).$$

A lower bound $-O(\ell^2 r_0^{-7})$ would need a positive lower bound on the near-diagonal equal-height kernel, which is not available here.

*Remark* S8.8 (where the bounded remainder goes). For the candidate density the bounded remainder of Theorem R is real: what the leading law $c_{d,L}\ell^{-1/3}$, the small-separation contact mass, misses is the equal-height mass $B_{d,L}$, spread over all separations, and by Proposition S8.4(i) all of it is rejected mass. For the elder density, the three bounded contributions in the proof of Theorem R — the first-order contact monomial $k^{-2/3} rk = \ell^{1/3}$, the terms proportional to the cutoff $r_0$, and the far term — are, respectively, an artefact of bounding the two endpoints separately (Lemma S8.5), the rejected mass at separations below $r_0$, and the far rejected mass; none of them is elder mass in the limit.

## S8.4. **The rate.**

**Theorem S8.9** (Theorem R$^{++}$: rate of the elder remainder)**.** *For every fixed $d \ge 2$ and $L > 0$, and for the canonical marked Kac–Rice density version of the elder density, there are $C < \infty$ and $\ell_0 \in (0, 1]$ depending on $d, L$ only such that*

$$\big|\nu_{\mathrm{eld}}(\ell) - c_{d,L}\,\ell^{-1/3}\big| \le C\ell^{1/4}, \qquad 0 < \ell \le \ell_0; \tag{S8.20}$$

*equivalently,* $\nu_{\mathrm{eld}}(\ell) = c_{d,L}\,\ell^{-1/3}(1+O(\ell^{7/12}))$ *as* $\ell \downarrow 0$.

No sharpness of the exponent is asserted here (see Remark S8.15), and the theorem gives no numerical $C$ or $\ell_0$, no rate for the candidate remainder in (S8.16), no rate for $\rho_{\mathrm{rej}} \to B_{d,L}$, and no uniformity in $d$ or $L$. The proof uses three estimates: a deterministic barrier (Lemma S8.10), which bounds the elder density at intermediate and at fixed separations (Lemmas S8.11 and S8.12), and a majorant for the near-pair weight at small gap marks (Lemma S8.14).

**Lemma S8.10** (deterministic barrier)**.** *Let* $f \in C^3(\mathbb{T}^d_L)$ *and* $M \in \mathbb{T}^d_L$ *with* $f(M) = b$, $\nabla f(M) = 0$ *and* $H_M = D^2 f(M) \prec 0$. *Put* $\lambda := \lambda_{\min}(-H_M)$, $K := 1 + \|D^2 f\|_{\infty,\mathrm{op}} + \|D^3 f\|_{\infty,\mathrm{op}}$ *(suprema over* $\mathbb{T}^d_L$*),* $a_L := \min(1, L/4)$ *and* $c_L := a_L^2/3$. *Then*

$$b - d_f(M) \geq c_L \lambda^3 / K^2. \tag{S8.21}$$

*Consequently, if* $d_f(M) = b - \ell$ *with* $\ell > 0$, *then* $\lambda_{\min}(-H_M) \leq (\ell K^2/c_L)^{1/3}$ *and* $|\det H_M| \leq c_L^{-1/3} \ell^{1/3} K^{d-1/3}$.

*Proof.* For a unit vector $v$ and $0 < t \leq R := a_L \lambda / K$, Taylor's theorem gives $f(M+tv) \leq b - \frac{\lambda}{2}t^2 + \frac{K}{6}t^3 \leq b - \frac{\lambda}{3}t^2$, since $t \leq \lambda/K$. The ball $B(M,R)$ is embedded ($R \leq L/4$), no point of it lies above $b$, and every path from $M$ to a point above $b$ crosses its boundary sphere, on which $f \leq b - \lambda R^2/3$. Hence $d_f(M) \leq b - \lambda R^2/3 = b - c_L \lambda^3/K^2$. For the consequence, $-H_M \succ 0$ has eigenvalues $\lambda = \lambda_1 \leq \cdots \leq \lambda_d \leq K$, so $|\det H_M| \leq \lambda K^{d-1} \leq c_L^{-1/3} \ell^{1/3} K^{d-1/3}$. □

**Lemma S8.11** (short elder bars at near separations are born at soft maxima)**.** *For* $0 < r \leq r_R$, $b \in \mathbb{R}$ *and* $k > 0$, *with* $\ell = k r^3$,

$$\mathbb{E}_Q\big[(W_r/r^2)\, e\big] \leq C \ell^{1/3} r^{-1} (k + r) P^N, \tag{S8.22}$$

*and consequently, for* $0 < \rho \leq r_0 \leq r_R$ *and* $\ell > 0$,

$$\nu_{\mathrm{eld}}^{[\rho, r_0)}(\ell) = \int_\rho^{r_0} \int_{\mathbb{R}} \int_{S^{d-1}} r^{-2} \mathcal{A}_r^{\mathrm{eld}}(b, \ell/r^3, u)\, d\sigma\, db\, dr \leq C \ell^{1/3} (\rho^{-1} + \ell \rho^{-5}). \tag{S8.23}$$

*Proof.* Under $Q$ the field is almost surely $C^\infty$. On $\{W_r > 0\} \cap \{e = 1\}$ we have $\nabla f(M) = 0$, $f(M) = b$, $H_M \prec 0$ and $d_f(M) = f(S) = b - \ell$, so Lemma S8.10 gives $|\det H_M| \leq c_L^{-1/3} \ell^{1/3} K^{d-1/3}$. On the support of $W_r$, $W_r/r^2 = (|\det H_M|/r)\, F_{d-1}(K_S)$, and $F_{d-1}(K_S) \leq C(k+r) T^d$ (Proposition S8.2(iv)). Hence $(W_r/r^2)e \leq C \ell^{1/3} r^{-1} (k+r) K^{d-1/3} T^d$ everywhere. Since $K \leq C_d(1 + \|F_r\|_{C^3})$ has $\|K\|_{L^p} \leq C_p P$ by (S8.5), Hölder's inequality gives $\mathbb{E}_Q[K^{d-1/3} T^d] \leq C P^N$, which is (S8.22). The identity in (S8.23) is (S8.1). By Proposition S8.2(ii), with $k = \ell/r^3$, $(1 + |b| + k)^N \leq 2^N (1 + |b|)^N (1 + k)^N$ and $(1+k)^N e^{-ck^2} \leq C$,

$$r^{-2} \mathcal{A}_r^{\mathrm{eld}}(b, \ell/r^3, u) \leq C \ell^{1/3} r^{-3} (\ell r^{-3} + r) P^N e^{-c(b^2+k^2)} \leq C \ell^{1/3} (\ell r^{-6} + r^{-2}) (1 + |b|)^N e^{-cb^2},$$

and $\int_\rho^{r_0} (\ell r^{-6} + r^{-2})\, dr \leq \ell \rho^{-5}/5 + \rho^{-1}$. □

**Lemma S8.12** (far elder density, crude rate)**.** *For every fixed* $r_0 \in (0, L/2]$ *there is* $C = C(d, L, r_0)$ *with*

$$\nu_{\mathrm{eld}}^{\mathrm{far}, r_0}(\ell) \leq C \ell^{1/3}, \qquad 0 < \ell \leq 1. \tag{S8.24}$$

*Proof.* Let $D_{r_0} = \{y : \mathrm{dist}(0, y) \geq r_0\}$ (compact) and use the far identity (S8.4). The covariance $\Gamma_y = \mathrm{Cov}(O_y)$ is positive definite for every $y \neq 0$ (finite-jet nondegeneracy at distinct sites, §S1; [7, §2]; Lemma S12.8) and continuous in $y$, so $\lambda_{\min}(\Gamma_y) \geq c_0 > 0$ on $D_{r_0}$ [7, §14] (Lemma S12.29). Consequently: (a) $p_y(v) \leq (2\pi c_0)^{-(d+1)} \exp(-|v|^2/(2 \operatorname{tr} \Gamma_y)) \leq C e^{-cb^2}$ at $v = v_{b,\ell}$ ($|v_{b,\ell}|^2 \geq b^2$, $\operatorname{tr} \Gamma_y \leq C$); (b) with $\psi^y := \Gamma_y^{-1} \mathrm{Cov}(O_y, f(\cdot))$, $2d+2$ smooth functions with $\|\psi_j^y\|_{C^3} \leq C$ uniformly on $D_{r_0}$, the field decomposes as $f = \sum_j O_y^j \psi_j^y + g_y$ with $g_y$ a centred Gaussian field independent of $O_y$, and under $Q_{y,b,\ell}$ the field has the law of $\sum_j v_{b,\ell}^j \psi_j^y + g_y$. Since $\|g_y\|_{C^3} \leq \|f\|_{C^3} + C|O_y|$

and all moments of $\|f\|_{C^3}$ and $|O_y|$ are finite, $\mathbb{E}\|g_y\|_{C^3}^p \le C_p$ uniformly on $D_{r_0}$. For $0 < \ell \le 1$ ($|v_{b,\ell}| \le 2|b| + 1$) this gives $K \le C(1 + |b| + \|g_y\|_{C^3})$ and

$$\mathbb{E}_{Q_{y,b,\ell}}[K^p] \le C_p(1 + |b|)^p \qquad (p \ge 1,\ y \in D_{r_0},\ 0 < \ell \le 1). \tag{S8.25}$$

On $\{W > 0\} \cap \{e_y = 1\}$ the point 0 is a nondegenerate maximum with $f(0) = b$ and $d_f(0) = f(y) = b - \ell$, so Lemma S8.10 gives $|\det H_0| \le c_L^{-1/3}\ell^{1/3}K^{d-1/3}$, while $|\det H_y| \le K^d$. So $We_y \le C\ell^{1/3}K^{2d-1/3}$ and, by (S8.25), $\mathbb{E}_{Q_{y,b,\ell}}[We_y] \le C\ell^{1/3}(1 + |b|)^{2d}$. With (a) in (S8.4), $\nu_{\mathrm{eld}}^{\mathrm{far},r_0}(\ell) \le C\ell^{1/3}\,\mathrm{vol}(D_{r_0})\int_{\mathbb{R}}(1 + |b|)^{2d}e^{-cb^2}db \le C\ell^{1/3}$. □

Lemma S8.12 needs no Hessian density and suffices for this section, because the far term is not the bottleneck of the estimate below. Keeping the determinant weight through a spectral integration gives the sharper rate $\ell^{2/3}$, which Section S10 uses.

**Proposition S8.13** (far elder density, rate $\ell^{2/3}$)**.** *For every fixed $\rho \in (0, L/2]$ there is $C = C(d, L, \rho)$ with*

$$\nu_{\mathrm{eld}}^{\mathrm{far},\rho}(\ell) \le C\ell^{2/3}, \qquad 0 < \ell \le 1. \tag{S8.26}$$

*Proof.* Let $D_\rho = \{y : \mathrm{dist}(0, y) \ge \rho\}$ and use the far identity (S8.4). Append to $O_y$ the independent entries of $H_0$ and $H_y$: $V_y = (O_y, H_0, H_y)$, with $2(d + 1) + d(d + 1)$ coordinates. These are distinct derivative evaluations at two distinct sites, so the covariance of $V_y$ is positive definite (§S1; [7, §2]; Lemma S12.8). It is continuous in $y$, so its eigenvalues lie in a fixed compact subinterval of $(0, \infty)$ on the compact set $D_\rho$. Two consequences follow.

(a) Since $|v_{b,\ell}|^2 \ge b^2$, uniformly for $y \in D_\rho$, $b \in \mathbb{R}$ and $\ell > 0$,

$$p_y(v_{b,\ell})\, p_{(H_0,H_y)|O_y=v_{b,\ell}}(H_0, H_y) \le Ce^{-c(b^2+|H_0|^2+|H_y|^2)}. \tag{S8.27}$$

(b) Regressing the field on $V_y$, as in (b) of the proof of Lemma S8.12, gives $f = m + g$ with $g$ independent of $V_y$, $\mathbb{E}\|g\|_{C^3}^q \le C_q$ and $\|m\|_{C^3} \le C(1 + |b| + |H_0| + |H_y|)$ for $0 < \ell \le 1$. So, with $K$ as in Lemma S8.10, for every $q \ge 1$

$$\mathbb{E}\big[K^q \mid V_y = (v_{b,\ell}, H_0, H_y)\big] \le C_q(1 + |b| + |H_0| + |H_y|)^q. \tag{S8.28}$$

Fix $y \in D_\rho$, $b \in \mathbb{R}$ and $0 < \ell \le 1$. On $\{W > 0\} \cap \{e_y = 1\}$ the point 0 is a nondegenerate maximum with $f(0) = b$ and $d_f(0) = f(y) = b - \ell$. So Lemma S8.10, with $\lambda = \lambda_{\min}(-H_0)$, gives $K \ge (c_L\lambda^3/\ell)^{1/2}$, and pathwise $We_y \le W\,\mathbf{1}\{K \ge (c_L\lambda^3/\ell)^{1/2}\}$. The weight $W$ and $\lambda$ are functions of $(H_0, H_y)$. Conditioning on $V_y$ and applying Markov's inequality with (S8.28) bounds the conditional probability of the last event by $\min\{1, C_q(1 + |b| + |H_0| + |H_y|)^q(\ell/(c_L\lambda^3))^{q/2}\}$. Multiply by the density (S8.27) and bound $(1+|b|+|H_0|+|H_y|)^q \le (1+|b|)^q(1+|H_0|)^q(1+|H_y|)^q$. Take the two Hessian factors outside the minimum, since $\min\{1, uv\} \le v\min\{1, u\}$ for $v \ge 1$, and integrate $H_y$ against $|\det H_y|(1 + |H_y|)^q e^{-c|H_y|^2}$, a constant. For $H_0$ use the ordered spectral coordinates $-H_0 = O\,\mathrm{diag}(\lambda_1 \le \cdots \le \lambda_d)\,O^\top$ on the cone $\{H_0 \prec 0\}$, whose Lebesgue Jacobian is a constant times $\prod_{i<j}(\lambda_j - \lambda_i)$, with finite angular volume. With $|\det H_0| = \prod_j \lambda_j$, bound the Vandermonde factor and $\prod_{j\ge2}\lambda_j$ by a polynomial in $|\lambda|$, extend $\lambda_2, \dots, \lambda_d$ to $(0, \infty)$ and integrate them out. This gives

$$p_y(v_{b,\ell})\,\mathbb{E}_{Q_{y,b,\ell}}[We_y] \le Ce^{-cb^2}\int_0^\infty \lambda_1 \min\Big\{1, A\Big(\frac{\ell}{\lambda_1^3}\Big)^{q/2}\Big\}(1 + \lambda_1)^N e^{-c\lambda_1^2}\,d\lambda_1, \qquad A = C(1 + |b|)^q. \tag{S8.29}$$

Take $q = 2$, and let $u = (A\ell)^{1/3}$ be the point where the two arguments of the minimum agree. If $u \ge 1$, bound the minimum by 1: the integral is at most $C \le Cu^2$. If $u < 1$, split at $u$. On $[0, u]$ the integral is at most $2^{N-1}u^2$. On $[u, \infty)$ it is at most

$$A\ell\int_u^\infty \lambda_1^{-2}(1 + \lambda_1)^N e^{-c\lambda_1^2}\,d\lambda_1 \le CA\ell\,(u^{-1} + 1) = Cu^3(u^{-1} + 1) \le 2Cu^2.$$

In every case the integral in (S8.29) is at most $Cu^2 = CA^{2/3}\ell^{2/3} = C(1+|b|)^{4/3}\ell^{2/3}$. Integrating over $b \in \mathbb{R}$ and $y \in D_\rho$ in (S8.4) proves (S8.26). □

The factor $\lambda_1$ in (S8.29) is the soft eigenvalue of $|\det H_0|$. It turns the barrier constraint $\lambda_1 \lesssim \ell^{1/3}$ into the mass $\ell^{2/3}$; without the determinant weight the same argument would give only $\ell^{1/3}$, as in Lemma S8.12. At $d=3$, $L=24$ and $\rho<12$, Proposition S8.13 sharpens the bound of Proposition S6.2.

**Lemma S8.14** (sign-window majorant for the near-pair weight)**.** *For $0<r\le r_R$, $b\in\mathbb{R}$ and $k\ge 0$ (the pinned family at $k=0$ included),*

$$\mathbb{E}_Q[W_r/r^2] \le C(k^2 + r^4(1+k)^2)P^N. \tag{S8.30}$$

*Consequently, for $0<\rho\le r_0^*$ and $0<\ell\le 1$,*

$$|\nu_{\mathrm{cand}}^{\mathrm{near},\rho}(\ell) - c_{d,L}\,\ell^{-1/3}| \le C(\min(\ell^{1/4},\rho) + \rho^3 + \ell^2\rho^{-7}), \tag{S8.31}$$

$$0 \le \rho_{\mathrm{rej}}^{\mathrm{near},\rho}(\ell) \le C(\ell^{1/4} + \rho^3), \tag{S8.32}$$

$$\int_{0<\mathrm{dist}(0,y)<\rho}\int_{\mathbb{R}} \Psi_0(b,y)\,db\,dy \le C\rho^3. \tag{S8.33}$$

Since $k^2 + r^4(1+k)^2 \le 2(1+r_R)^2(k+r)^2$, the bound (S8.30) is never worse than the first bound in (S8.8), and for $k<r$ it is better by the factor $k^2/r^2 + r^2$. Bounds (S8.30), (S8.32) and (S8.33) use only Steps 1–3 of the proof of Lemma S8.5; (S8.31) uses the full Lemma S8.5. In particular (S8.33) says that the radial density $r^{d-1}\Psi_0(b,ru)$ of the equal-height kernel is $O(r^2)$ near the diagonal.

*Proof. Proof of* (S8.30). Work under the coupling as in the proof of Lemma S8.5, with $A_0 = D_y^2F_0(0)$ and $\Delta := \det A_0$; nothing below divides by $k$. Steps 1–3 there hold pathwise for $0<r\le r_R$; multiplying out $\det K_i = \alpha_i \det A_i - rq_i$ with $\tau := \mathrm{tr}(\mathrm{adj}(A_0)B)$, $\det A_i = \Delta \mp \frac{r}{2}\tau + E_i'$, $|E_i'| \le Cr^2T^m$ (by (S8.12)), $\alpha_i = \mp 6k + \frac{r}{12}f_4 + \rho_i$ (by (S8.10)) and $q_i = q + O(rT^{m+1})$ (by (S8.13)) gives

$$\det K_M = -6k\Delta + rY + R_M, \qquad \det K_S = +6k\Delta + rY + R_S, \qquad |R_M| + |R_S| \le Cr^2(1+k)T^{N_1}, \tag{S8.34}$$

with $Y$ as in (S8.15) and

$$R_M = -6kE_M' - \tfrac{r^2}{24}f_4\tau + \tfrac{r}{12}f_4E_M' + \rho_M \det A_M - r(q_M - q),$$
$$R_S = 6kE_S' + \tfrac{r^2}{24}f_4\tau + \tfrac{r}{12}f_4E_S' + \rho_S \det A_S - r(q_S - q);$$

each term is at most $Cr^2(1+k)T^{N_1}$, and the $r$-linear terms $\frac{r}{12}f_4\Delta$, $3kr\tau$ and $-rq$ are the same at both endpoints. On the support of $W_r$, $\mathrm{ind}\,K_M = d$ and $\mathrm{ind}\,K_S = d-1$, so $\det K_M$ has the sign $(-1)^d$, $\det K_S$ the sign $(-1)^{d-1}$, and $\det K_M \det K_S < 0$. With $w := rY$, the product $(w - 6k\Delta + R_M)(w + 6k\Delta + R_S)$ is negative only when $w$ lies strictly between $6k\Delta - R_M$ and $-6k\Delta - R_S$; hence

$$|w| \le 6k|\Delta| + \max(|R_M|,|R_S|), \qquad |\det K_i| \le 12k|\Delta| + 2\max(|R_M|,|R_S|) \quad (i=M,S),$$

and on the support of $W_r$

$$W_r/r^2 = |\det K_M \det K_S| \le \left(12k|\Delta| + 2Cr^2(1+k)T^{N_1}\right)^2 \le C(k^2\Delta^2 + r^4(1+k)^2T^{2N_1}). \tag{S8.35}$$

Completing the square gives the same bound more directly: $\det K_M \det K_S = \left(w + \frac{1}{2}(R_M+R_S)\right)^2 - \left(6k\Delta + \frac{1}{2}(R_S - R_M)\right)^2$, so on the support of $W_r$ we have $W_r/r^2 \le \left(6k|\Delta| + \frac{1}{2}(|R_M| + |R_S|)\right)^2$. Off the support $W_r = 0$. Since $|\Delta| \le \|A_0\|^m \le T^m$, $\mathbb{E}\Delta^2 \le CP^{2m}$ and $\mathbb{E}T^{2N_1} \le CP^{2N_1}$; $W_r$ is a

functional of $F_r$, whose law is $Q$, so the pathwise bound on the coupling space bounds $\mathbb{E}_Q$. This is (S8.30).

*Proof of* (S8.31). Put $r_1 := \ell^{1/4}$, so that $r \le r_1$ if and only if $k = \ell/r^3 \ge r$, and use (S8.19) with $r_0 = \rho$. On $0 < r < \min(r_1, \rho)$, Lemma S8.5 gives $|r^{-2}(\mathcal{A}_r - \mathcal{A}_0)| \le H(b,k) \le Ce^{-cb^2}$, contributing at most $C\min(r_1,\rho)$. On $r_1 \le r < \rho$ (empty if $\rho \le r_1$), $|\mathcal{A}_r - \mathcal{A}_0| \le \mathcal{A}_r + \mathcal{A}_0$ with $\mathcal{A}_r \le C(k^2 + r^4(1+k)^2)P^N e^{-c(b^2+k^2)}$ by (S8.30) and Proposition S8.2(ii), and $\mathcal{A}_0 \le k^2 H$; so $r^{-2}|\mathcal{A}_r - \mathcal{A}_0| \le C(\ell^2 r^{-8} + r^2)(1+|b|)^N e^{-cb^2}$, which integrates to at most $C(\ell^2 r_1^{-7}/7 + \rho^3/3) = C(\ell^{1/4}/7 + \rho^3/3)$. The tail term is at most $C\ell^2\rho^{-7}$ as in the proof of Theorem S8.6.

*Proof of* (S8.32). By (S8.1), $\rho_{\mathrm{rej}}^{\mathrm{near},\rho} = \int_0^\rho \int\int r^{-2}\mathcal{A}_r^{\mathrm{rej}}$. On $0 < r \le \min(r_1,\rho)$, Proposition S8.3 (valid as $r \le k$ and $r \le \rho \le r_0^*$) gives $r^{-2}\mathcal{A}_r^{\mathrm{rej}} \le Cr^{-2}(r^3/k)P^N e^{-c(b^2+k^2)} = C(r^4/\ell)P^N e^{-c(b^2+k^2)}$, which integrates to at most $Cr_1^5/(5\ell) = C\ell^{1/4}/5$. On $r_1 < r \le \rho$, $1 - e \le 1$ and (S8.30), as in the proof of (S8.31), give at most $C(\ell^{1/4} + \rho^3)$.

*Proof of* (S8.33). By (S8.3), $r^{d-1}\Psi_0(b, ru) = r^{-2}\mathcal{A}_r(b,0,u)$, and (S8.30) at $k = 0$ with Proposition S8.2(ii) gives $\mathcal{A}_r(b,0,u) \le Cr^4(1+|b|)^N e^{-cb^2}$. So $\int_{0<\mathrm{dist}<\rho} \int \Psi_0 = \int_0^\rho \int\int r^{-2}\mathcal{A}_r(b,0,u) \le C\int_0^\rho r^2 dr = C\rho^3/3$. □

The window $|rY| \le 6k|\Delta| + O(r^2)$ is where the typed weight lives at small $k$: for $k = 0$ both scaled endpoint determinants are $rY + O(r^2)$ with the same $rY$, so a maximum/saddle pair can be typed only when $|Y| = O(r)$. Bound (S8.31) needs Lemma S8.5 on $r < \ell^{1/4}$: (S8.30) alone gives $\int_0 r^{-2}k^2\,dr = \infty$, and (S8.7) alone gives only $O(1)$.

*Proof of Theorem S8.9.* Fix $0 < \rho \le r_0^*$ and $0 < \ell \le 1$. Splitting the elder density by the separation of the pair into $(0,\rho)$, $[\rho, r_0^*)$ and $[r_0^*, \operatorname{diam}\mathbb{T}_L^d]$ (the boundaries are null), and using $\nu_{\mathrm{eld}}^{\mathrm{near},\rho} = \nu_{\mathrm{cand}}^{\mathrm{near},\rho} - \rho_{\mathrm{rej}}^{\mathrm{near},\rho}$,

$$\nu_{\mathrm{eld}}(\ell) - c_{d,L}\ell^{-1/3} = [\nu_{\mathrm{cand}}^{\mathrm{near},\rho}(\ell) - c_{d,L}\ell^{-1/3}] - \rho_{\mathrm{rej}}^{\mathrm{near},\rho}(\ell) + \nu_{\mathrm{eld}}^{[\rho,r_0^*)}(\ell) + \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell). \tag{S8.36}$$

The four terms are bounded as follows: (i) $|\nu_{\mathrm{cand}}^{\mathrm{near},\rho} - c_{d,L}\ell^{-1/3}| \le C(\min(\ell^{1/4},\rho) + \rho^3 + \ell^2\rho^{-7})$ by (S8.31); (ii) $0 \le \rho_{\mathrm{rej}}^{\mathrm{near},\rho} \le C(\ell^{1/4} + \rho^3)$ by (S8.32); (iii) $0 \le \nu_{\mathrm{eld}}^{[\rho,r_0^*)} \le C\ell^{1/3}(\rho^{-1} + \ell\rho^{-5})$ by (S8.23) with $r_0 = r_0^* \le r_R$; (iv) $0 \le \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*} \le C\ell^{1/3}$ by (S8.24) with $r_0 = r_0^*$. Choose $\rho = \ell^{1/12}$ and $\ell_0 := \min\{1, (r_0^*)^{12}\}$, so that $\rho \le r_0^*$ for $\ell \le \ell_0$. Then $\min(\ell^{1/4},\rho) = \ell^{1/4}$, $\rho^3 = \ell^{1/4}$, $\ell^2\rho^{-7} = \ell^{17/12}$, $\ell^{1/3}\rho^{-1} = \ell^{1/4}$ and $\ell^{4/3}\rho^{-5} = \ell^{11/12}$, and (S8.36) with (i)–(iv) (eight terms, five of them equal to $\ell^{1/4}$) gives

$$|\nu_{\mathrm{eld}}(\ell) - c_{d,L}\ell^{-1/3}| \le C(5\ell^{1/4} + \ell^{17/12} + \ell^{11/12} + \ell^{1/3}) \le 8C\ell^{1/4}, \qquad 0 < \ell \le \ell_0,$$

which is (S8.20). Dividing by $c_{d,L}\ell^{-1/3}$ ($c_{d,L} > 0$) gives the relative form. □

*Remark* S8.15 (what fixes the exponent). In (S8.36) the leading $\rho$-dependence is $\rho^3$ (the candidate excess and the rejected mass at separations in $(\ell^{1/4},\rho)$, both $O(r^2)$ per unit separation by Lemma S8.14) against $\ell^{1/3}\rho^{-1}$ (Lemma S8.11 above $\rho$); they balance at $\rho = \ell^{1/12}$, where both equal $\ell^{1/4}$. But $\ell^{1/4}$ also appears three times independently of $\rho$, all from the transition $r = k$ at separation $\ell^{1/4}$, below which the pinned pair is a cap pair: the excess $C\min(\ell^{1/4},\rho)$ from Lemma S8.5, the loss from Proposition S8.3 at $r \le k$, and the tail $\int_{r>\ell^{1/4}} r^{-2}k^2\,dr$ of the leading law, $\mathcal{A}_0 \asymp k^2$. The order $\ell^{1/4}$ is attained, with a negative coefficient, by Theorem S9.2: at the cusp scale $r \asymp k \asymp \ell^{1/4}$ the contact law overcounts. Note also that (S8.20) does not use Proposition S8.4(i); together with (S8.16) it re-derives $\rho_{\mathrm{rej}}(\ell) \to B_{d,L}$.

S8.5. **The cap-region bound.** This subsection proves the second bound of Proposition S8.3, which is also part of Theorem K of record [R7] of the main text. The proof is that of Lemma S7.18, with two changes: the axial entries of the endpoint Hessians are bounded by (S7.14) instead of

(S7.13), which keeps a factor $k^2$ in the weight, and the pin density is left out, so that the density of the transverse Hessian is a Gaussian with a shifted centre.

**Lemma S8.16** (the cap-region bound)**.** *Let $r_0$ be as in (S7.12). There are $C$ and $N$ such that, for $0 < r \le r_0$, $b \in \mathbb{R}$, $k \ge r$ and every frame,*

$$\mathbb{E}_Q\Big[\frac{W_r}{r^2}\,\mathbf{1}_{G_r^{\mathrm{c}}}\Big] \le C\,\frac{r^3}{k}\,P^N. \tag{S8.37}$$

*Proof.* Put $\delta = r/k \le 1$. The law $Q$ is the law $\mathbb{E}^0$ of Section S7, $G_r$ is the cap event (S7.10), and $\alpha_i, \beta_i, A_i$ are as in Lemma S7.3.

*The weight with the axial entries at their true size.* Put $a := 6k + rM_4/2$, so that $|\alpha_M|, |\alpha_S| \le a$ by (S7.14). By (S7.15), $|\det H_i|/r \le |\alpha_i|\,|\det A_i| + r\|\beta_i\|^2 \|\operatorname{adj}(A_i)\|$, and (S7.13) gives

$$\frac{W_r}{r^2} \le \Big(a\,\|f\|_{C^2}^m + \frac{r}{4}\,M_3^2\,\|f\|_{C^2}^{m-1}\Big)^2 \le C(k+r)^2\big(1+\|f\|_{C^4}\big)^{2d+2}. \tag{S8.38}$$

On the typed support $H_M \prec 0$, so $\alpha_M < 0$ and $B := -A_M \succ 0$; let $0 < \lambda_1 \le \cdots \le \lambda_m = \Lambda$ be the eigenvalues of $B$. By (S7.15), $|\det H_M| = r\big(|\alpha_M|\det B - r\beta_M^{\top}\operatorname{adj}(B)\beta_M\big) \le ra\prod_j \lambda_j$, as $\operatorname{adj}(B) \succeq 0$. By (S7.13) and Weyl's inequality the eigenvalues of $-A_S$ lie within $rM_3$ of those of $B$, so $|\det A_S| \le \prod_j(\lambda_j + rM_3)$ and $\|\operatorname{adj}(A_S)\| \le \prod_{j\ge 2}(\lambda_j + rM_3)$; with $\|\beta_S\| \le M_3/2$, (S7.15) gives $|\det H_S| \le r\big[a(\lambda_1 + rM_3) + \frac{r}{4}M_3^2\big]\prod_{j\ge 2}(\lambda_j + rM_3)$. Hence

$$\frac{W_r}{r^2} \le a\,\lambda_1\Big[a\lambda_1 + a\,rM_3 + \frac{r}{4}\,M_3^2\Big]\prod_{j=2}^{m}\lambda_j\,(\lambda_j + rM_3); \tag{S8.39}$$

for $a = M_3/2$ this is (S7.16).

*Residual and the density of $A_M$.* As in the proof of Lemma S7.18, Lemma S12.13 writes $f$ under $Q$ as a regression mean, linear in $(v_r, A_M)$, plus a centred residual $g$ that is independent of $A_M$ and whose $C^4$ norm has moments of every order, bounded uniformly in $r$ and in the frame. With $J = 1 + \|g\|_{C^4}$ and $T = P + J$ (in this proof $T$ denotes $P + J$, as in the proof of Lemma S7.18, not the variable of Convention S8.1(h)), on the typed support $M_3, M_4 \le K_*(T + \Lambda)$ for a constant $K_*$. Under $Q$ the independent entries of $A_M$ are Gaussian. Their covariance is the Schur complement of $\Sigma_r$ in the extended covariance of Lemma S7.2(b), so its eigenvalues lie in a fixed compact subinterval of $(0, \infty)$. Their mean $m_A$ is linear in $v_r$, so $\|m_A\|_F \le CP$ by Lemma S7.2(c). Hence their density is at most $C\exp(-c\|A - m_A\|_F^2)$. In the eigenvalue coordinates of Lemma S12.17, with $-A = O\,\mathrm{diag}(\lambda)O^{\top}$ and $O$ fixed, $\|A - m_A\|_F^2 \ge \sum_i(\lambda_i + q_i)^2$, where $q_i = (O^{\top}m_A O)_{ii}$ and $|q_i| \le CP$. Since $\int_0^{\infty}\lambda^p e^{-c(\lambda+q)^2}d\lambda \le C_p(1+|q|)^p$, a polynomial in $\lambda_2, \dots, \lambda_m$ integrated against the Gaussian bound in these variables is at most a polynomial in $P$, uniformly in $O$; the integral over $O$ is taken last. This replaces (S7.43), which carries the pin density.

*Depth failure, $m \ge 2$.* On the typed support, $G_r^c$ is the union of the depth failure $\{\lambda_1 \le \frac{4}{3k}rM_3^2\}$ and the exception $\{rM_4 > 3k/10\}$. Put $\Upsilon = T + \Lambda \ge 1$, which does not involve $\lambda_1$, and $D_* = \frac{4}{3}K_*^2$. Depth failure gives $\lambda_1 \le D_*\delta\Upsilon^2$, and as $r \le k$, $a \le \bar a := 6k + \frac{1}{2}rK_*\Upsilon \le C_*k\Upsilon$; since $\bar a$ and $\Upsilon$ do not involve $\lambda_1$, we replace $a$ by $\bar a$ and $M_3$ by $K_*\Upsilon$ in (S8.39) before integrating in $\lambda_1$. In (S8.39) the product over $j \ge 2$ is at most $C\Upsilon^{2m-2}$, the Vandermonde factor is at most $\Lambda^{m(m-1)/2}$, and the Gaussian factor in $\lambda_1$ is dropped. Then

$$\bar a\int_0^{D_*\delta\Upsilon^2}\lambda\Big[\bar a\lambda + \bar a rK_*\Upsilon + \frac{r}{4}K_*^2\Upsilon^2\Big]d\lambda = \frac{\bar a^2 D_*^3}{3}\,\delta^3\Upsilon^6 + \frac{\bar a^2 rK_*D_*^2}{2}\,\delta^2\Upsilon^5 + \frac{\bar a rK_*^2D_*^2}{8}\,\delta^2\Upsilon^6.$$

By $\bar a \le C_*k\Upsilon$, $\delta = r/k$ and $r^3 \le P\,r^3/k$, each term is at most $C(r^3/k)P\,\Upsilon^8$. The integration in $\lambda_2, \dots, \lambda_m$ and over $O$, and the moments of $J$, give

$$\mathbb{E}_Q\Big[\frac{W_r}{r^2}\,\mathbf{1}_{\{\lambda_1 \le \frac{4}{3k}rM_3^2\}}\Big] \le C\,\frac{r^3}{k}\,P^N.$$

*Depth failure, $m = 1$.* Now $\lambda_1 = \Lambda = \lambda$, and by (S7.44) depth failure gives $\lambda \le 4D_*\delta T^2$ or $\lambda > 1/(4D_*\delta)$. On the first branch $M_3, M_4 \le K_*(T + 4D_*T^2) \le CT^2$ and $a \le CkT^2$, and integrating (S8.39) over $[0, 4D_*\delta T^2]$ as above gives at most $C(r^3/k)PT^{10}$, hence $C(r^3/k)P^N$ after the moments of $J$. On the second branch $1 \le (4D_*\delta\lambda)^4$, so by (S8.38), $\lambda \le \|f\|_{C^2}$ and (S7.28) (under $Q$ the field is $F_r$ of Lemma S7.12, for every $0 < r \le r_0$),

$$\mathbb{E}_Q\Big[\frac{W_r}{r^2}\,\mathbf{1}_{\{\lambda > 1/(4D_*\delta)\}}\Big] \le (4D_*\delta)^4\,\mathbb{E}_Q\Big[\frac{W_r}{r^2}\,\lambda^4\Big] \le C\delta^4(k+r)^2P^N \le C\,\frac{r^3}{k}\,P^N,$$

since $\delta^4(k+r)^2 \le 4\delta^4k^2 = 4\delta\, r^3/k$.

*Exception.* On $\{rM_4 > 3k/10\}$, $1 \le (10\delta M_4/3)^4$, so by (S8.38) and (S7.28),

$$\mathbb{E}_Q\Big[\frac{W_r}{r^2}\,\mathbf{1}_{\{rM_4 > 3k/10\}}\Big] \le \Big(\frac{10\delta}{3}\Big)^4\,\mathbb{E}_Q\Big[\frac{W_r}{r^2}\,M_4^4\Big] \le C\delta^4(k+r)^2P^N \le C\,\frac{r^3}{k}\,P^N.$$

Adding the bounds for depth failure and for the exception proves (S8.37). □

With the axial bound (S7.13) in place of (S7.14), the same argument gives only $C(r/k)^3P^N$, the analogue without the pin density of the bound (S7.42) of Lemma S7.18; the factor $k^2$ kept here is the factor $(6k)^2$ of $z_0$ in (S7.7), and it is what the uses of Proposition S8.3 in Lemma S8.14 (hence Theorem S8.9), Section S9 and (S10.12) of Section S10 need.

S8.6. **The equal-height kernel.** This subsection proves items (i) and (ii) of Proposition S8.4, which are also Theorem Z of record [R7] of the main text. Item (ii) and the bounds $0 < B_{d,L} < \infty$ are direct. The limit $\rho_{\mathrm{rej}}(\ell) \to B_{d,L}$ follows from (S8.16) and (S8.20); the proof of (S8.16) uses Proposition S8.4 only through (ii), (iii) and the finiteness of $B_{d,L}$, and the proof of (S8.20) does not use it at all (cf. Remark S8.15), so there is no circularity.

*Proof of Proposition S8.4(ii).* Fix $y \neq 0$ and $b$, and let $\Gamma_y = \mathrm{Cov}(O_y)$ and $\mathcal{C}_y(z) = \mathrm{Cov}(f(z), O_y)$. By Lemma S12.8, applied to the values and gradients at the two distinct sites 0 and $y$, $\Gamma_y$ is positive definite. Let $F$ be a smooth copy of the unconditioned field and put $f_\ell := F + \mathcal{C}_y\Gamma_y^{-1}(v_{b,\ell} - O_y(F))$. The field $F - \mathcal{C}_y\Gamma_y^{-1}O_y(F)$ is uncorrelated with, hence independent of, the Gaussian vector $O_y(F)$, so $f_\ell$ has the regression law $Q_{y,b,\ell}$, and its covariance, that of $F - \mathcal{C}_y\Gamma_y^{-1}O_y(F)$, does not depend on $\ell$. Since $v_{b,\ell} - v_{b,0} = -\ell\,\mathsf{e}$, where $\mathsf{e}$ is the unit vector of the coordinate $f(y)$ of $O_y$,

$$f_\ell = f_0 + \ell\, h_y, \qquad h_y := -\mathcal{C}_y\Gamma_y^{-1}\mathsf{e},$$

a deterministic function, smooth because each component of $\mathcal{C}_y$ is a derivative of $K_L$; and $O_y(h_y) = -\Gamma_y\Gamma_y^{-1}\mathsf{e} = -\mathsf{e}$, that is, $h_y(0) = 0$, $\nabla h_y(0) = 0$, $h_y(y) = -1$ and $\nabla h_y(y) = 0$. By Lemma S12.8 again, applied to the list consisting of $O_y$ and the independent entries of $H_0 = D^2f(0)$ and $H_y = D^2f(y)$, the covariance of that list is positive definite, so the conditional covariance of the Hessian entries given $O_y$, its Schur complement, is positive definite too; hence under $Q_{y,b,\ell}$ the pair $(H_0, H_y)$ has a Lebesgue density on $\mathrm{Sym}(d)^2$, and the algebraic set $\{\det H_0 \det H_y = 0\}$, which is proper, is null. Finally, $W(f_\ell) = F_d(H_0(f_\ell))\,F_{d-1}(H_y(f_\ell))$ with $H_i(f_\ell) = H_i(f_0) + \ell D^2h_y(i)$, and $F_d$ and $F_{d-1}$ are locally Lipschitz (Lemma S7.14, where $F_j$ is written $\Phi_j$); so $W(f_\ell) \to W(f_0)$ pathwise as $\ell \downarrow 0$, and $W(f_\ell) \le C\big(1 + \|f_0\|_{C^2} + \|h_y\|_{C^2}\big)^{2d}$ for $\ell \le 1$, which has finite expectation by Lemma S7.1(a), as $\|f_0\|_{C^2} \le \|F\|_{C^2} + \|\mathcal{C}_y\Gamma_y^{-1}\|_{C^2}(|v_{b,0}| + |O_y(F)|)$ and $|O_y(F)| \le C\|F\|_{C^1}$. Dominated convergence gives $\mathbb{E}_{Q_{y,b,\ell}}[W] \to \mathbb{E}_{Q_{y,b,0}}[W]$, and $p_y(v_{b,\ell}) \to p_y(v_{b,0})$ as $p_y$ is continuous. Hence $\Psi_\ell^{\mathrm{cand}}(b,y) \to \Psi_0(b,y)$. □

*Proof of Proposition S8.4(i). Finiteness.* Split $B_{d,L}$ at the torus distance $r_0^*$. By (S8.33) at $\rho = r_0^*$, the part over $0 < \mathrm{dist}(0,y) < r_0^*$ is at most $C(r_0^*)^3$; the proof of (S8.33) uses only (S8.30) at $k = 0$, Proposition S8.2(ii) and (S8.3). On the compact set $\{\mathrm{dist}(0,y) \ge r_0^*\}$, item (ii) and item (iii) give

$\Psi_0(b,y) = \lim_{\ell\downarrow 0} \Psi_\ell^{\mathrm{cand}}(b,y) \le C(1+|b|)^{2d} e^{-cb^2}$, which is integrable in $b$; so that part is finite as well.

*Positivity.* Fix $y \neq 0$ and $b$. The density $p_y$ is positive everywhere, as $\Gamma_y$ is positive definite. Under $Q_{y,b,0}$ the pair $(H_0, H_y)$ has a positive Lebesgue density on $\mathrm{Sym}(d)^2$ (the proof of (ii)), and the set $\{H_0 \prec 0,\ \mathrm{ind}\, H_y = d-1,\ \det H_y \neq 0\}$, the typed set less the null set $\{\det H_0 \det H_y = 0\}$, is open (an index changes only where an eigenvalue vanishes) and nonempty; on it $W > 0$. Hence $\mathbb{E}_{Q_{y,b,0}}[W] > 0$ and $\Psi_0(b,y) > 0$ for every $y \neq 0$ and $b \in \mathbb{R}$, so $B_{d,L} > 0$.

*The limit.* By (S8.16), whose proof uses only (ii), (iii) and the finiteness of $B_{d,L}$ just proved, $\nu_{\mathrm{cand}}(\ell) - c_{d,L}\ell^{-1/3} \to B_{d,L}$; by (S8.20), $\nu_{\mathrm{eld}}(\ell) - c_{d,L}\ell^{-1/3} = O(\ell^{1/4}) \to 0$. Subtracting, $\rho_{\mathrm{rej}}(\ell) = \nu_{\mathrm{cand}}(\ell) - \nu_{\mathrm{eld}}(\ell) \to B_{d,L}$. □

With Lemma S8.16 and this subsection, every statement of Part II rests on Sections S1–S7, Part III and the computer-assisted enclosures described in §S9 and §S11; no analytic statement is imported from a repository record.

## S9. The cusp term

This section proves the second term of Theorem 2.3 of the main text: $\nu_{\mathrm{eld}}(\ell) = c_{d,L}\ell^{-1/3} + c_1\ell^{1/4} + o(\ell^{1/4})$ with the explicit constant $c_1 < 0$ of (2.13) of the main text (Theorem S9.2), the corresponding expansions of the candidate and rejected densities (Theorem S9.3), and enclosures of $c_1$ (§S9.7). The mechanism is a cusp crossover. At separations $r \asymp \ell^{1/4}$ the gap $\ell = kr^3$ is comparable to $r^4$; rescaled, the pinned field converges to a universal random quartic (Theorem S9.4) whose superlevel topology reduces exactly to a one-dimensional ridge (Theorem S9.5). On that ridge the pair is typed if and only if $|\varphi| < 1$ and the saddle is the elder partner of the maximum if and only if $|\varphi| < 1/3$. The notation is that of Convention S8.1; in particular $d \ge 2$ and $L > 0$ are fixed.

### S9.1. Statements.

**Convention S9.1** (cusp variables, jets and kernels)**.** (a) *Cusp variables.* $\kappa := k/r = \ell/r^4$ and $s := r\ell^{-1/4} = \kappa^{-1/4}$, so that $\ell = kr^3 = \kappa r^4$. (In this section $\kappa$ is the cusp variable, as in §5.2 of the main text, and not the gap mark of (3.5) of the main text.)

(b) *Jets.* For a $C^5$ function $f$, in the frame $(u,\Theta)$ of the pair: $f_4 := \partial_u^4 f(0)$, $\gamma := \nabla_\Theta \partial_u^2 f(0) \in \mathbb{R}^m$, $A := D_\Theta^2 f(0) \in \mathrm{Sym}(m)$, $\Delta := \det A$, and

$$Y := \frac{f_4}{12}\,\Delta - \frac{1}{4}\,\gamma^\top \mathrm{adj}(A)\,\gamma, \qquad \varphi := \frac{Y}{6\kappa\Delta} = \frac{f_4 - 3\gamma^\top A^{-1}\gamma}{72\kappa} \quad (A \text{ invertible}). \tag{S9.1}$$

($Y$ is the scalar $Y$ of (S8.15) at $k = 0$.)

(c) *The zero-gap contact law.* $\mathbb{E}_0[\,\cdot \mid b]$ is the expectation over the jets of the field under the contact law at the zero-gap target $v_0(b,0)$, that is, given $f(0) = b$, $\partial_u f(0) = \partial_u^2 f(0) = \partial_u^3 f(0) = 0$ and $\nabla_\Theta f(0) = \nabla_\Theta \partial_u f(0) = 0$; and $\pi_0(u; v_0(b,0))$ is the density of $U_0$ at $v_0(b,0)$ in the frame with axis $u$.

(d) *Cusp kernels.* For $\kappa > 0$ put $w^{\mathrm{cand}} := (36\kappa^2\Delta^2 - Y^2)\mathbf{1}\{A \prec 0,\ |Y| < 6\kappa|\Delta|\}$, $w^{\mathrm{eld}} := (36\kappa^2\Delta^2 - Y^2)\mathbf{1}\{A \prec 0,\ |Y| < 2\kappa|\Delta|\}$, $w^{\mathrm{con}} := 36\kappa^2\Delta^2\mathbf{1}\{A \prec 0\}$ and $\mathcal{C}^\bullet(b,\kappa,u) := 12\,\pi_0(u; v_0(b,0))\,\mathbb{E}_0[w^\bullet \mid b]$.

**Theorem S9.2** (Theorem CU: the cusp term of the elder law)**.** *For every $d \ge 2$ and $L > 0$, and for the canonical marked Kac–Rice density version of the elder density, as $\ell \downarrow 0$,*

$$\nu_{\mathrm{eld}}(\ell) = c_{d,L}\,\ell^{-1/3} + c_1\,\ell^{1/4} + o(\ell^{1/4}), \tag{S9.2}$$

$$c_1 = -\frac{192}{7}\,2^{1/4} \int_{S^{d-1}} \int_{\mathbb{R}} \pi_0(u; v_0(b,0))\,\mathbb{E}_0\Big[|Y|^{7/4}|\Delta|^{1/4}\mathbf{1}\{A \prec 0\} \;\Big|\; b\Big]\,db\,d\sigma(u), \tag{S9.3}$$

*and* $c_1 \in (-\infty, 0)$. *Equivalently, with* $G := \nabla f(0)$, $V_u := (\partial_u^2 f, \nabla_\Theta \partial_u f)(0)$, $t_u := \partial_u^3 f(0)$ *and* $\tau_u^2 := \mathrm{Var}(t_u \mid G = 0)$ *(odd and even jets are independent* [7, §15]*; Lemma S12.30),*

$$c_1 = -\frac{192}{7}\, 2^{1/4} \int_{S^{d-1}} p_G(0)\, p_{V_u}(0)\, (2\pi)^{-1/2} \tau_u^{-1} \times \mathbb{E}\Big[ |Y|^{7/4} |\Delta|^{1/4} \mathbf{1}\{A \prec 0\} \;\Big|\; V_u = 0,\ G = 0,\ t_u = 0 \Big]\, d\sigma(u). \quad \text{(S9.4)}$$

*So* $\nu_{\mathrm{eld}}(\ell) = c_{d,L} \ell^{-1/3}(1 + (c_1/c_{d,L})\ell^{7/12} + o(\ell^{7/12}))$.

For $d = 3$, $L = 24$, (S9.2) gives the first two terms of (2.12) of the main text, and (S9.3) is (2.13) of the main text.

**Theorem S9.3** (Theorem CU′: the candidate and rejected densities)**.** *Under the hypotheses of Theorem S9.2, as* $\ell \downarrow 0$,

$$\nu_{\mathrm{cand}}(\ell) = c_{d,L}\, \ell^{-1/3} + B_{d,L} + I^{\mathrm{cand}} \ell^{1/4} + o(\ell^{1/4}), \quad \text{(S9.5)}$$

$$I^{\mathrm{cand}} = -\frac{96}{7}\, 6^{1/4} \int_{S^{d-1}} \int_{\mathbb{R}} \pi_0(u; v_0(b,0))\, \mathbb{E}_0\Big[ |Y|^{7/4} |\Delta|^{1/4} \mathbf{1}\{A \prec 0\} \;\Big|\; b \Big]\, db\, d\sigma(u) = \frac{3^{1/4}}{2}\, c_1 < 0, \quad \text{(S9.6)}$$

$$\rho_{\mathrm{rej}}(\ell) = B_{d,L} + (I^{\mathrm{cand}} - c_1)\, \ell^{1/4} + o(\ell^{1/4}), \qquad I^{\mathrm{cand}} - c_1 = \Big(1 - \frac{3^{1/4}}{2}\Big)|c_1| > 0. \quad \text{(S9.7)}$$

*Moreover, for every* $r_0 \in (0, r_0^*]$, $\nu_{\mathrm{cand}}^{\mathrm{far}, r_0}(\ell) = \int_{\mathrm{dist}(0,y) \ge r_0} \int_{\mathbb{R}} \Psi_0(b, y)\, db\, dy + O(\ell)$.

Beyond their share of $B_{d,L}$ the far pairs contribute only $O(\ell)$; the $\ell^{1/4}$ terms are carried by the cusp scale, and the rejected density approaches $B_{d,L}$ from above. The candidate density needs no elder decision: (S9.5) uses neither Proposition S9.6, Lemma S9.8 and Proposition S8.3, nor the conclusions of Lemmas S8.11 and S8.12 (it uses the uniform far facts established in the proof of the latter); (S9.7) is (S9.5) minus (S9.2). Neither theorem gives a rate for its $o(\ell^{1/4})$ terms or any uniformity in $d$ or $L$; $B_{d,L}$ is not evaluated here, and enclosures of $c_1$ and $I^{\mathrm{cand}}$ for the Gaussian kernel and its periodizations are in §S9.7.

## S9.2. **The cusp limit field and the model.**

**Theorem S9.4** (the cusp limit field)**.** *Let* $f \in C^5(\mathbb{T}_L^d)$, $0 < r \le 1$, $\kappa > 0$, *and let* $M = -ru/2$ *and* $S = ru/2$ *be critical points of* $f$ *with* $f(M) = b$ *and* $f(S) = b - \kappa r^4$. *For* $R \ge 1$ *with* $2Rr \le L/4$ *put* $\mathcal{W}_R := [-R, R] \times \{\Xi \in \mathbb{R}^m : |\Xi| \le R\}$ *and* $\mathfrak{F}(X, \Xi) := r^{-4}[f(rXu + r^2 \Theta \Xi) - b]$, *and let*

$$\mathfrak{P}(X, \Xi) := 2\kappa (X + \tfrac{1}{2})^2 (X - 1) + \frac{f_4}{24} (X^2 - \tfrac{1}{4})^2 + \tfrac{1}{2}(X^2 - \tfrac{1}{4}) \gamma \cdot \Xi + \tfrac{1}{2} \Xi^\top A\, \Xi \quad \text{(S9.8)}$$

*with the jets* $(f_4, \gamma, A)$ *of* $f$ *at* $0$. *Then*

$$\|\mathfrak{F} - \mathfrak{P}\|_{C^2(\mathcal{W}_R)} \le C(d, R)\, r\, \|f\|_{C^5}. \quad \text{(S9.9)}$$

*Proof.* Write $\vartheta_j := \partial_u^j f(0)$, $\psi_j := \partial_u^j \nabla_\Theta f(0)$ (so $\vartheta_4 = f_4$, $\psi_2 = \gamma$), $h := r/2$ and $N_f := \|f\|_{C^5}$. Along the axis, $\vartheta(x) := f(xu) = \sum_{j \le 4} \vartheta_j x^j / j! + \rho(x)$ with $|\rho| \le N_f |x|^5/120$ and $|\rho'| \le N_f |x|^4/24$. The pins $\vartheta'(\pm h) = 0$, $\vartheta(-h) = b$, $\vartheta(h) = b - 16\kappa h^4$ give successively $\vartheta'(h) - \vartheta'(-h) = 2\vartheta_2 h + \vartheta_4 h^3/3 + O(N_f h^4) = 0$, $\vartheta'(h) + \vartheta'(-h) = 2\vartheta_1 + \vartheta_3 h^2 + O(N_f h^4) = 0$, $\vartheta(h) - \vartheta(-h) = 2\vartheta_1 h + \vartheta_3 h^3/3 + O(N_f h^5) = -16\kappa h^4$ and $\vartheta(h) + \vartheta(-h) = 2\vartheta_0 + \vartheta_2 h^2 + \vartheta_4 h^4/12 + O(N_f h^5) = 2b - 16\kappa h^4$, hence

$$\vartheta_2 = -\frac{r^2}{24} f_4 + O(N_f r^3), \qquad \vartheta_3 = 12\kappa r + O(N_f r^2),$$
$$\vartheta_1 = -\frac{3}{2}\kappa r^3 + O(N_f r^4), \qquad \vartheta_0 - b = \Big(-\frac{\kappa}{2} + \frac{f_4}{384}\Big) r^4 + O(N_f r^5). \quad \text{(S9.10)}$$

The transverse pins $\psi(\pm h)=0$ for $\psi(x):=\nabla_\Theta f(xu)=\psi_0+\psi_1 x+\gamma x^2/2+\psi_3 x^3/6+O(N_f|x|^4)$ give $\psi_0=-\frac{r^2}{8}\gamma+O(N_f r^4)$ and $\psi_1=O(N_f r^2)$. Expand $f$ at 0 to total order four in $(x,y)$: the remainder $\rho_5$ satisfies $|\partial^\alpha\rho_5(x,y)|\le CN_f(|x|+|y|)^{5-|\alpha|}$ for $|\alpha|\le 2$, and at $(x,y)=(rX,r^2\Xi)$ with $(X,\Xi)\in\mathcal{W}_R$ we have $|x|+|y|\le 2Rr$; since $\partial_X=r\partial_x$ and $\partial_\Xi=r^2\partial_y$, every derivative of order at most two of $r^{-4}\rho_5(rX,r^2\Xi)$ is $O(C_R N_f r)$. The Taylor monomials $x^iy^j$ of weighted degree $i+2|j|\ge 5$ contribute $O(C_R N_f r)$ in $C^2(\mathcal{W}_R)$ likewise. The monomials of weighted degree at most four have $(i,|j|)\in\{(0,0),(1,0),(2,0),(3,0),(4,0),(0,1),(1,1),(2,1),(0,2)\}$; inserting the pinned values,

$$\begin{aligned}
r^{-4}\Big[&\vartheta_0-b+\vartheta_1 rX+\tfrac12\vartheta_2 r^2X^2+\tfrac16\vartheta_3 r^3X^3+\tfrac{1}{24}f_4 r^4X^4\\
&\quad+\psi_0\cdot r^2\Xi+\psi_1\cdot r^3X\Xi+\tfrac12\gamma\cdot r^4X^2\Xi+\tfrac12 r^4\Xi^\top A\Xi\Big]\\
&=\Big[-\frac{\kappa}{2}-\frac{3}{2}\kappa X+2\kappa X^3\Big]+\frac{f_4}{24}\Big[\frac{1}{16}-\frac{X^2}{2}+X^4\Big]+\Big[-\frac{\gamma\cdot\Xi}{8}+\frac{\gamma\cdot X^2\Xi}{2}\Big]\\
&\quad+\frac12\Xi^\top A\Xi+O(C_R N_f r),
\end{aligned}$$

and $-\frac{\kappa}{2}-\frac32\kappa X+2\kappa X^3=2\kappa(X+\frac12)^2(X-1)$, $\frac{1}{16}-\frac{X^2}{2}+X^4=(X^2-\frac14)^2$, $-\frac18\gamma\cdot\Xi+\frac12\gamma\cdot X^2\Xi=\frac12(X^2-\frac14)\gamma\cdot\Xi$. □

**Theorem S9.5** (the model: fibre reduction and the elder window)**.** *Let $\kappa>0$, $f_4\in\mathbb{R}$, $\gamma\in\mathbb{R}^m$ and $A\in\mathrm{Sym}(m)$ negative definite; let $\mathfrak{P}$ be (S9.8), $Y,\varphi$ as in (S9.1), and $X_3:=-1/(2\varphi)$ for $\varphi\neq 0$.*

*(a) (Fibre reduction.) $\mathfrak{P}(X,\Xi)=g(X)+\frac12(\Xi-\Xi^*(X))^\top A(\Xi-\Xi^*(X))$ with $\Xi^*(X):=-\frac12(X^2-\frac14)A^{-1}\gamma$ and*

$$g(X)=\kappa\Big[2(X+\tfrac12)^2(X-1)+3\varphi(X^2-\tfrac14)^2\Big],\qquad g'(X)=6\kappa(X^2-\tfrac14)(1+2\varphi X).\tag{S9.11}$$

*(b) (Critical points and types.) The critical points of $\mathfrak{P}$ are the points $(X,\Xi^*(X))$ with $g'(X)=0$: $M=(-\frac12,0)$, $S=(\frac12,0)$ and, for $\varphi\notin\{0,\pm1\}$, $(X_3,\Xi^*(X_3))$. Further $\mathfrak{P}(M)=0$, $\mathfrak{P}(S)=-\kappa$, $\det\mathrm{Hess}\,\mathfrak{P}(M)=6\kappa(\varphi-1)\Delta=Y-6\kappa\Delta$ and $\det\mathrm{Hess}\,\mathfrak{P}(S)=6\kappa(\varphi+1)\Delta=Y+6\kappa\Delta$; $M$ is a nondegenerate maximum and $S$ a nondegenerate critical point of index $m$ if and only if $|\varphi|<1$.*

*(c) (The elder window.) Let $|\varphi|<1$, and let $d_{\mathfrak{P}}(M)$ be the maximin connection level of $M$ for $\mathfrak{P}$ on $\mathbb{R}^{1+m}$ (paths to points with $\mathfrak{P}>0$; $-\infty$ if there are none). Then*

$$d_{\mathfrak{P}}(M)=\begin{cases}-\kappa=\mathfrak{P}(S) & \text{if } |\varphi|<\frac13,\\ g(X_3)>-\kappa & \text{if } \frac13<\varphi<1,\\ -\infty & \text{if } -1<\varphi<-\frac13.\end{cases}$$

*(d) (Critical values.) $g(X_3)+\kappa=\kappa(\varphi+1)^3(3\varphi-1)/(16\varphi^3)$ and $g(X_3)=\kappa(\varphi-1)^3(3\varphi+1)/(16\varphi^3)$. At $\varphi=\frac13$, $g-g(\frac12)=\frac{\kappa}{16}(2X-1)^2(2X+3)^2$ (the saddles $\frac12$ and $-\frac32$ at equal height); at $\varphi=-\frac13$, $g=-\frac{\kappa}{16}(2X-3)^2(2X+1)^2$ (the maxima $-\frac12$ and $\frac32$ at equal height).*

So the pair is typed if and only if $|Y|<6\kappa|\Delta|$, the sign window of Lemma S8.14 at leading order, and $S$ is the elder partner of $M$ if and only if $|Y|<2\kappa|\Delta|$: the elder window is one third of the typed window.

*Proof.* (a) $\frac12(\Xi-\Xi^*)^\top A(\Xi-\Xi^*)=\frac12\Xi^\top A\Xi+\frac12(X^2-\frac14)\gamma\cdot\Xi+\frac18(X^2-\frac14)^2\gamma^\top A^{-1}\gamma$; subtracting this from (S9.8) leaves $2\kappa(X+\frac12)^2(X-1)+\frac{1}{24}(f_4-3\gamma^\top A^{-1}\gamma)(X^2-\frac14)^2$, and $\frac{1}{24}(f_4-3\gamma^\top A^{-1}\gamma)=3\kappa\varphi$. Direct differentiation gives $g'$.

(b) In the coordinates $(X, Z)$, $Z := \Xi - \Xi^*(X)$ (a smooth diffeomorphism with unit Jacobian), $\mathfrak{P} = g(X) + \frac{1}{2}Z^\top AZ$. The critical points are $\{Z = 0,\ g' = 0\}$, and at a critical point the Hessian changes by a congruence of determinant one, so $\det \mathrm{Hess}_{(X,\Xi)} \mathfrak{P} = g''\Delta$ and the inertia is that of $\mathrm{diag}(g'', A)$. By (S9.11), $g''(-\frac{1}{2}) = 6\kappa(\varphi - 1)$ and $g''(\frac{1}{2}) = 6\kappa(\varphi + 1)$, and $6\kappa\varphi\Delta = Y$.

(c) For every $t$, $\{\mathfrak{P} > t\} = \{(X, \Xi^*(X) + Z) : g(X) > t,\ Z^\top(-A)Z < 2(g(X) - t)\}$: over each $X$ with $g(X) > t$ a nonempty open ellipsoid, and empty otherwise. Hence there is a point above 0 over $X$ if and only if $g(X) > 0$; every path has $\min \mathfrak{P} \le \min g(X(\cdot))$, and the lifted path $X \mapsto (X, \Xi^*(X))$ has $\mathfrak{P} = g$. So $d_{\mathfrak{P}}(M) = \max(d_-, d_+)$, where $d_-$ (resp. $d_+$) is the supremum, over $x < -\frac{1}{2}$ (resp. $x > -\frac{1}{2}$) with $g(x) > 0$, of the minimum of $g$ on the segment between $-\frac{1}{2}$ and $x$ ($\sup \emptyset = -\infty$). On $(-\frac{1}{2}, \frac{1}{2})$, $g' < 0$ (as $1 + 2\varphi X \ge 1 - |\varphi| > 0$), so $g$ falls from 0 to $-\kappa$. *Case $0 \le \varphi < \frac{1}{3}$:* on $(\frac{1}{2}, \infty)$, $g' > 0$ and $g \to +\infty$, so $d_+ = -\kappa$; on $(-\infty, -\frac{1}{2})$ either $\varphi = 0$ and $g' > 0$ (so $g < 0$ there and $d_- = -\infty$), or $X_3 < -\frac{3}{2}$ is a local minimum with $g(X_3) < -\kappa$ by (d) and $g \to +\infty$ as $X \to -\infty$, so $d_- = g(X_3) < -\kappa$. *Case $\frac{1}{3} < \varphi < 1$: $d_+ = -\kappa$* as before; $X_3 \in (-\frac{3}{2}, -\frac{1}{2})$ is a local minimum, $g \to +\infty$ to the left and $g(X_3) > -\kappa$ by (d), so $d_- = g(X_3)$. *Case $-\frac{1}{3} < \varphi < 0$:* on $(-\infty, -\frac{1}{2})$, $g' > 0$, so $d_- = -\infty$; on $(\frac{1}{2}, \infty)$, $g$ rises to the local maximum $X_3 > \frac{3}{2}$ with $g(X_3) > 0$ by (d), so $d_+ = -\kappa$. *Case $-1 < \varphi < -\frac{1}{3}$: $d_- = -\infty$*; now $X_3 \in (\frac{1}{2}, \frac{3}{2})$ and $g(X_3) < 0$, so $g < 0$ on $(-\frac{1}{2}, \infty)$, $d_+ = -\infty$, and $M$ is the unique global maximum.

(d) Direct substitution of $X_3$ in (S9.11) and expansion. □

S9.3. **Stability of the elder decision.**

**Proposition S9.6** (stability)**.** *Fix $\delta \in (0, 1/20)$, put $R_\delta := 30\delta^{-2}$ and $\mathcal{W} := [-3, 2] \times \{|\Xi| \le R_\delta\}$, and let $\mathcal{K}_\delta$ be the set of $(\kappa, f_4, \gamma, A)$ with $\kappa \in [\delta, \delta^{-1}]$, $A \preceq -\delta I$, $\|A\| + |\gamma| + |f_4| \le \delta^{-1}$ and $|\varphi| \in [0, \frac{1}{3} - \delta] \cup [\frac{1}{3} + \delta, 1 - \delta]$. There is $\varepsilon = \varepsilon(\delta, m) > 0$ with the following property. Let $f \in C^2(\mathbb{T}^d_L)$, let $M = -ru/2$ and $S = ru/2$ be critical points of $f$ with $f(M) = b$ and $f(S) = b - \kappa r^4$, let the map $\Phi : (X, \Xi) \mapsto rXu + r^2\Theta\Xi$ be injective on $\mathcal{W}$ (so that, being a local diffeomorphism, it maps the interior and boundary of every subregion of $\mathcal{W}$ onto the interior and boundary of its image), let $(\kappa, f_4, \gamma, A) \in \mathcal{K}_\delta$, and let $\|\mathfrak{F} - \mathfrak{P}\|_{C^2(\mathcal{W})} \le \varepsilon$, where $\mathfrak{F} = r^{-4}(f \circ \Phi - b)$ and $\mathfrak{P} = \mathfrak{P}_{\kappa, f_4, \gamma, A}$ is (S9.8) (operator norms on the derivatives). Then $M$ is a nondegenerate local maximum, $S$ is a nondegenerate critical point of index $m$, and*

$$e(f) = \mathbf{1}\{d_f(M) = f(S)\} = \mathbf{1}\{|\varphi| < 1/3\}. \tag{S9.12}$$

*Proof.* Constants $C_\delta, \mu_\delta, c_\delta, \eta_\delta, \dots$ depend only on $\delta$ and $m$ (by continuity on the compact set $\mathcal{K}_\delta$).

*Step 1 (the ridge of $\mathfrak{F}$).* On $\mathcal{W}$, $\partial^2_\Xi \mathfrak{F} \preceq A + \varepsilon I \preceq -\frac{\delta}{2} I$ for $\varepsilon \le \delta/2$, and $|\Xi^*(X)| \le \frac{35}{8}\delta^{-2}$ for $X \in [-3, 2]$. Since $\mathfrak{F}(X, \cdot) \le g(X) - \frac{\delta}{2}|\cdot - \Xi^*(X)|^2 + \varepsilon$ and $\mathfrak{F}(X, \Xi^*(X)) \ge g(X) - \varepsilon$, every maximizer of $\mathfrak{F}(X, \cdot)$ over $|\Xi| \le R_\delta$ lies within $2(\varepsilon/\delta)^{1/2} < 1$ of $\Xi^*(X)$, hence in the interior; by strict concavity it is unique, $\Xi_{\mathfrak{F}}(X)$, and the first-order identity $A(\Xi_{\mathfrak{F}} - \Xi^*) = \partial_\Xi \mathfrak{P}(X, \Xi_{\mathfrak{F}}) = \partial_\Xi(\mathfrak{P} - \mathfrak{F})(X, \Xi_{\mathfrak{F}})$ improves this to $|\Xi_{\mathfrak{F}} - \Xi^*| \le \varepsilon/\delta$. By the implicit function theorem $\Xi_{\mathfrak{F}} \in C^1$, and $g_{\mathfrak{F}}(X) := \mathfrak{F}(X, \Xi_{\mathfrak{F}}(X))$ is $C^2$ with $g'_{\mathfrak{F}} = \partial_X \mathfrak{F}(\cdot, \Xi_{\mathfrak{F}})$ and $g''_{\mathfrak{F}} = [\partial^2_X \mathfrak{F} - \partial_{X\Xi}\mathfrak{F}(\partial^2_\Xi \mathfrak{F})^{-1}\partial_{\Xi X}\mathfrak{F}](\cdot, \Xi_{\mathfrak{F}})$; the same formulas hold for $g$ with $(\mathfrak{P}, \Xi^*)$. Hence $\|g_{\mathfrak{F}} - g\|_{C^2([-3,2])} \le C_\delta \varepsilon$.

*Step 2 (connectivity reduces to one dimension).* For every $t$, each nonempty $X$-fibre of $\mathcal{W} \cap \{\mathfrak{F} > t\}$ is convex (a ball cut by a superlevel set of a concave function) and contains $\Xi_{\mathfrak{F}}(X)$, and it is nonempty if and only if $g_{\mathfrak{F}}(X) > t$. So the path components of $\mathcal{W} \cap \{\mathfrak{F} > t\}$ are the parts over the components of $\{X \in [-3, 2] : g_{\mathfrak{F}}(X) > t\}$, and the supremum of $\mathfrak{F}$ over such a part is the supremum of $g_{\mathfrak{F}}$ over the interval.

*Step 3 (transverse boundary).* On $|\Xi| = R_\delta$, $|\Xi - \Xi^*| \ge 25\delta^{-2}$, so $\mathfrak{F} \le \max_{[-3,2]} g - 312\delta^{-3} + \varepsilon$; since $|g| \le 280\kappa \le 280\delta^{-1}$ on $[-3, 2]$ and $\delta^{-2} > 400$, $\mathfrak{F} < -2\kappa - 1$ there.

*Step 4 (types).* The points $M \leftrightarrow (-\frac{1}{2}, 0)$ and $S \leftrightarrow (\frac{1}{2}, 0)$ are critical points of $\mathfrak{F}$, so $\Xi_{\mathfrak{F}}(\pm\frac{1}{2}) = 0$, $g'_{\mathfrak{F}}(\pm\frac{1}{2}) = 0$, $g_{\mathfrak{F}}(-\frac{1}{2}) = 0$ and $g_{\mathfrak{F}}(\frac{1}{2}) = -\kappa$ exactly. As $g''(-\frac{1}{2}) = 6\kappa(\varphi - 1) \le -6\delta^2$ and $g''(\frac{1}{2}) \ge 6\delta^2$,

for $C_\delta\varepsilon < 3\delta^2$ the Schur complement $g''_{\mathfrak{F}}$ of $\partial^2_\Xi\mathfrak{F} \prec 0$ in $\operatorname{Hess}\mathfrak{F}$ is negative at $M$ and positive at $S$; by Haynsworth's inertia additivity $M$ is a nondegenerate maximum and $S$ has index $m$, for $\mathfrak{F}$ and hence for $f$.

*Step 5 (the decision).* Let $\tau_\delta := \delta/10$ and let $Z_g$ be the set of *all* real zeros of $g'$: $\pm\frac12$ and, for $\varphi \neq 0$, $X_3$, wherever it lies. All are nondegenerate ($g''(X_3) = 3\kappa(1-\varphi^2)/\varphi$), $|X_3 \mp \frac12| \geq \delta/2$, $|g''| \geq c_\delta > 0$ on the $\tau_\delta$-neighbourhoods of the points of $Z_g$, and $|g'| \geq \mu_\delta > 0$ at the points of $[-3,2]$ at distance at least $\tau_\delta$ from $Z_g$. For $C_\delta\varepsilon < \frac12\min(\mu_\delta, c_\delta\tau_\delta)$, $g'_{\mathfrak{F}}$ has exactly the zeros $\pm\frac12$ and, when $X_3 \in [-3+\tau_\delta, 2-\tau_\delta]$, one more nondegenerate zero $X_3^{\mathfrak{F}}$ within $\tau_\delta$ of $X_3$, with the sign pattern of $g'$ elsewhere on $[-3,2]$ away from the $\tau_\delta$-neighbourhoods of $Z_g$.

*Case E, $|\varphi| \leq \frac13 - \delta$.* By Theorem S9.5, $g$ increases on $[-\frac32, -\frac12]$ (for $\varphi > 0$ because $X_3 < -\frac32 - 4\delta$), decreases on $[-\frac12, \frac12]$ and increases on $[\frac12, X_R]$, where $X_R := 2$ if $\varphi \geq 0$ and $X_R := \min(2, X_3 - \tau_\delta)$ if $\varphi < 0$; and $g(-\frac32) = \kappa(-5+12\varphi) \leq -\kappa(1+12\delta)$, $g(X_R) \geq \eta_\delta > 0$ ($g(2) = \kappa(12.5+42.1875\varphi) \geq 1.9\kappa$ for $\varphi \geq -\frac14$, and for $\varphi < -\frac14$, $X_3 \leq 2$ and $g(X_3 - \tau_\delta) \geq g(X_3) - C\kappa\tau_\delta^2 \geq 11\kappa\delta$ since $g(X_3) \geq 12\kappa\delta$ by Theorem S9.5(d)). The same holds for $g_{\mathfrak{F}}$, with $-\kappa(1+11\delta)$ and $\eta_\delta/2$. *Path:* the ridge curve $X \mapsto \Phi(X, \Xi_{\mathfrak{F}}(X))$, $X \in [-\frac12, X_R]$, runs from $M$ through $S$ to a point where $f > b$, and along it $f - b = r^4 g_{\mathfrak{F}} \geq -\kappa r^4 = f(S) - b$; so $d_f(M) \geq f(S)$. *Trap:* let $D := (-\frac32, \frac12) \times \{|\Xi| < R_\delta\}$ and $D_{\mathrm{phys}} := \Phi(D)$, so $\partial D_{\mathrm{phys}} = \Phi(\partial D)$. On $\partial D$: $\mathfrak{F} \leq g_{\mathfrak{F}}(-\frac32) < -\kappa$ at $X = -\frac32$; $\mathfrak{F} < -2\kappa - 1$ at $|\Xi| = R_\delta$; and $\mathfrak{F}(\frac12, \Xi) \leq g_{\mathfrak{F}}(\frac12) = -\kappa$ at $X = \frac12$, with equality only at $\Xi = 0$, that is, at $S$. So $f \leq f(S)$ on $\partial D_{\mathrm{phys}}$. The path component $\mathcal{C}$ of $\{f > f(S)\}$ containing $M$ does not meet $\partial D_{\mathrm{phys}}$ (the first exit of a path in $\mathcal{C}$ from $D_{\mathrm{phys}}$ would land on it), hence $\mathcal{C} \subset D_{\mathrm{phys}}$, and by Step 2 its supremum is $b + r^4 \sup g_{\mathfrak{F}}$ over the component of $\{g_{\mathfrak{F}} > -\kappa\} \cap (-\frac32, \frac12)$ containing $-\frac12$, which is $b + r^4 g_{\mathfrak{F}}(-\frac12) = b$. So $\mathcal{C}$ contains no point above $b$, and every path from $M$ to a point above $b$ leaves $\mathcal{C}$ and meets $\{f \leq f(S)\}$: $d_f(M) \leq f(S)$. Hence $d_f(M) = f(S)$ and $e = 1$.

*Case $R^+$, $\varphi \in [\frac13 + \delta, 1 - \delta]$.* $X_3 \in (-\frac32, -\frac12 - \frac{\delta}{2})$ is a nondegenerate local minimum of $g$; $g$ decreases on $[-3, X_3]$ and increases on $[X_3, -\frac12]$; $g(X_3) \geq -\kappa + 0.44\delta\kappa$ by Theorem S9.5(d); $g(-3) = \kappa(-50 + 229.6875\varphi) \geq 26\kappa$. For $g_{\mathfrak{F}}$ the same holds with $g_{\mathfrak{F}}(X_3^{\mathfrak{F}}) \geq -\kappa + 0.4\delta\kappa$. The ridge curve from $M$ to $X = -3$ ends where $f > b$ and has minimum $b + r^4 g_{\mathfrak{F}}(X_3^{\mathfrak{F}}) > f(S)$: $d_f(M) > f(S)$ and $e = 0$.

*Case $R^-$, $\varphi \in [-1 + \delta, -\frac13 - \delta]$.* $X_3 \in (\frac12 + \frac{\delta}{2}, \frac32)$; $g$ increases on $[-\frac32, -\frac12]$, decreases on $[-\frac12, \frac12]$, increases on $[\frac12, X_3]$ and decreases on $[X_3, 2]$; $g(-\frac32) \leq -9\kappa$, $g(2) \leq -1.5\kappa$ and $g(X_3) \leq -1.5\delta\kappa < 0$. Put $t_* := -\frac54\kappa$ and $D' := (-\frac32, 2) \times \{|\Xi| < R_\delta\}$. Then $\mathfrak{F} < t_*$ on $\partial D'$ (at the ends by the values of $g_{\mathfrak{F}}$, on the transverse boundary by Step 3), the component of $\{f > b + r^4 t_*\}$ containing $M$ lies in $\Phi(D')$, and its supremum is at most $b + r^4 \max(g_{\mathfrak{F}}(-\frac12), g_{\mathfrak{F}}(X_3^{\mathfrak{F}})) = b$. So $d_f(M) \leq b + r^4 t_* < f(S)$ (or $M$ is the global maximum and $d_f(M) = -\infty$), and $e = 0$. □

Only window information is used: every conclusion is either an explicit path inside $\Phi(\mathcal{W})$ or a trap whose boundary lies inside it, so the landscape outside the window cannot change the decision. No Morse–Smale, separatrix or genericity assumption is made, and $e$ is computed exactly. Theorems S9.4, S9.5 and Proposition S9.6 are deterministic statements about $C^5$ functions; only the law of $(f_4, \gamma, A)$, and hence $c_1$, depends on the field.

## S9.4. **The kernels at the cusp scale.**

**Theorem S9.7** (kernel limits at fixed $\kappa$)**.** *For fixed $b \in \mathbb{R}$, $u \in S^{d-1}$ and $\kappa > 0$, as $r \downarrow 0$,*

$$r^{-2}\mathcal{A}_r(b, \kappa r, u) \to \mathcal{C}^{\mathrm{cand}}, \qquad r^{-2}\mathcal{A}_r^{\mathrm{eld}}(b, \kappa r, u) \to \mathcal{C}^{\mathrm{eld}}, \qquad r^{-2}\mathcal{A}_0(b, \kappa r, u) \to \mathcal{C}^{\mathrm{con}}. \tag{S9.13}$$

*Proof.* Work under the coupling of Proposition S8.2 with $k = \kappa r$ and the dominating variable $T$ of Convention S8.1(h). The contact law of $F_0$ has target $v_0(b, \kappa r)$, which differs from $v_0(b, 0)$ only in the coordinate $12k$; the regression mean is affine in the target and the covariance does not

depend on it, so $F_0 = F_0^0 + 12k\,\chi_0$ with $F_0^0$ of target $v_0(b,0)$ and $\chi_0$ a fixed smooth function. The variables of $\mathbb{E}_0$ are the jets of $F_0^0$, and those of $F_0$ differ from them by deterministic $O(\kappa r)$ vectors.

(i) *Weights.* By (S8.34), $\det K_M = r(Y_r - 6\kappa\Delta_0) + R_M$ and $\det K_S = r(Y_r + 6\kappa\Delta_0) + R_S$ with $|R_i| \le Cr^2(1+k)T^{N_1}$, $\Delta_0 = \det A_0$ and $Y_r$ the $Y$ of (S8.15). As $r \to 0$, $Y_r \to Y$ and $\Delta_0 \to \Delta$ (the variables of $\mathbb{E}_0$) in every $L^p$, so $\det K_i/r \to Y \mp 6\kappa\Delta$ in every $L^p$.

(ii) *Types.* $\{W_r > 0\} = \{\operatorname{ind} K_M = d,\ \operatorname{ind} K_S = d-1\}$. On $\{A \prec 0,\ |Y| \ne 6\kappa|\Delta|\}$ and for small $r$, $A_M, A_S \prec 0$ and, by Haynsworth's inertia additivity, the indices are $m + \mathbf{1}\{\sigma_i < 0\}$ with $\sigma_i = \det K_i / \det A_i$; with $\operatorname{sign}\Delta = (-1)^m$ this gives, in the limit, typed if and only if $|Y| < 6\kappa|\Delta|$. Off $\{A \prec 0\}$ a typed configuration needs $A_M \prec 0$, hence $\lambda_{\max}(A_0) \le CrT$, an event whose probability tends to 0 (the deterministic $O(\kappa r)$ shift between $A_0$ and $A$ included). So $\mathbf{1}\{W_r > 0\} \to \mathbf{1}\{A \prec 0,\ |Y| < 6\kappa|\Delta|\}$ in probability.

(iii) *Marks.* By Theorem S9.4, $\|\mathfrak{F}_r - \mathfrak{P}_r\|_{C^2(\mathcal{W})} \le C_\delta rT$ for the rescaled $F_r$ and the model $\mathfrak{P}_r$ built from the jets of $F_r$. On $\{(\kappa, \text{jets of } F_r) \in \mathcal{K}_\delta,\ C_\delta rT \le \varepsilon(\delta)\}$ Proposition S9.6 gives $e = \mathbf{1}\{|\varphi_r| < 1/3\}$, with $\varphi_r$ built from the jets of $F_r$, and $\varphi_r \to \varphi = Y/(6\kappa\Delta)$ in probability. The typed event minus this good event has probability at most

$$\mathbb{P}(C_\delta rT > \varepsilon(\delta)) + \mathbb{P}(|\text{jets}(F_r) - \text{jets}(F_0^0)| > \eta_\delta) + \mathbb{P}\big((\kappa, \text{jets}(F_0^0)) \notin \mathcal{K}_{2\delta},\ A \prec 0,\ |Y| < 6\kappa|\Delta|\big) + o(1),$$

with $\eta_\delta$ so small that $\eta_\delta$-perturbations of points of $\mathcal{K}_{2\delta}$ stay in $\mathcal{K}_\delta$. The first two terms tend to 0 as $r \to 0$, and the third tends to 0 as $\delta \to 0$: the jets of $F_0^0$ form a nondegenerate Gaussian vector (finite-jet nondegeneracy, §S1), and given $(A, \gamma)$ the variable $f_4$ has a nondegenerate Gaussian law, so $\varphi$ has a continuous distribution and $\mathbb{P}(|\varphi| \in (\frac13 - 2\delta, \frac13 + 2\delta) \cup (1 - 2\delta, 1))$, $\mathbb{P}(\lambda_{\max}(A) \in (-2\delta, 0))$ and the probability of large jets all vanish as $\delta \to 0$.

(iv) *Expectations.* On the typed event $W_r/r^2 = -\det K_M \det K_S$, and by (S8.35) $W_r/r^2 \le C(\kappa^2 r^2 \Delta_0^2 + r^4(1+k)^2 T^{2N_1})$, so $(W_r/r^2)/r^2$ is bounded in $L^2$ and uniformly integrable. With (i)–(ii) for the candidate limit and (i)–(iii) for the elder limit (note $|\varphi| < \frac13 \iff |Y| < 2\kappa|\Delta|$), $\mathbb{E}_Q[W_r/r^2]/r^2 \to \mathbb{E}_0[w^{\text{cand}} \mid b]$ and $\mathbb{E}_Q[(W_r/r^2)e]/r^2 \to \mathbb{E}_0[w^{\text{eld}} \mid b]$.

(v) *Densities.* $\pi_r(v_r) \to \pi_0(u; v_0(b,0))$ by Proposition S8.2(ii) and the continuity of $\pi_0$ in the target, and $\mathcal{A}_0(b, \kappa r, u)/r^2 = 12\pi_0(v_0(b,\kappa r))\, 36\kappa^2\, \mathbb{E}[\Delta^2 \mathbf{1}\{A \prec 0\} \mid v_0(b,\kappa r)] \to \mathcal{C}^{\text{con}}$. □

**Lemma S9.8** (the typed window has small probability)**.** *Let $\beta := \min(1/8, 1/(4m))$. There are $r_1 > 0$, $C$ and $N$ such that, for $0 < r \le r_1$, $b \in \mathbb{R}$ and $0 \le k \le 1$,*

$$\mathbb{E}_Q[W_r/r^2] \le C\big(k^2 + r^4\big)\big(\min(1, k/r)^{1/2} + r^\beta\big)P^N. \tag{S9.14}$$

Without the second factor, (S9.14) is (S8.30).

*Proof.* Let $J$ be the vector of all partial derivatives of $f = F_r$ at 0 of orders at most 8 in the frame $(u, \Theta)$, $J = (J_{\text{pin}}, f_4, J')$, where $J_{\text{pin}} := (f, \partial_u f, \partial_u^2 f, \partial_u^3 f, \nabla_\Theta f, \nabla_\Theta \partial_u f)(0)$ and $J'$ is the rest. ($J'$ must exclude the pinned jets: with $\partial_u^2 f(0)$ in $J'$, the pin row $(\partial_x f(S) - \partial_x f(M))/r$ would determine $\frac{r^2}{24} f_4$ up to $O(r^7)$.)

*Conditional density of $f_4$.* Under $Q$, $(f_4, J')$ is Gaussian with the law of $(f_4, J')$ given $U_r = v_r$, so $\operatorname{Var}_Q(f_4 \mid J') = \operatorname{Var}(f_4 \mid J', U_r)$. As $r \to 0$, $U_r \to U_0 = J_{\text{pin}}$ in $L^2$ of the field, and $(J', J_{\text{pin}})$ and $(f_4, J', J_{\text{pin}})$ are families of distinct partial derivatives at one point, hence nondegenerate. The Gram matrix of $(J', U_r)$ therefore stays uniformly invertible for $r \le r_1$ (uniformly over frames), and $\operatorname{Var}(f_4 \mid J', U_r) \to \operatorname{Var}(f_4 \mid J', J_{\text{pin}}) > 0$. So the conditional density of $f_4$ given $J'$ under $Q$ is bounded by a constant $C_0$.

*Affine structure.* Expanding $\alpha_i$, $A_i$ and $\beta_i$ ($x_i = \mp r/2$) to order 8 in $x_i$ and inserting the pin identities to the same order ($\partial_u^2 f(0) = -\frac{r^2}{24} f_4 + (\text{terms in } J') + \dots$, $\partial_u^3 f(0) = 12k - \frac{r^2}{40}\partial_u^5 f(0) + \dots$, $\nabla_\Theta \partial_u f(0) = (\text{terms in } J')$; the transverse pins involve no $f_4$), $\det K_i = \alpha_i \det A_i - r\beta_i^\top \operatorname{adj}(A_i)\beta_i = d_i(J') f_4 + c_i(J') + \rho_i$, with $d_i(J') = \frac{r}{12}\det A_i(J')$ and $|\rho_i| \le Cr^6(1 + \|f\|_{C^9})^{N_2}$. Let $E_1 :=$

$\{\|F_r\|_{C^9} + T \le r^{-\eta}\}$ with $\eta := 1/(8\max(N_1, N_2))$ ($N_1, N_2 \ge m+1$, enlarged if necessary, so that $m\eta < 1/8$). On $E_1$, $|\rho_i| \le Cr^5$, and $\mathbb{P}(E_1^c) \le C_q r^q P^{N_q}$ for every $q$, by (S8.5).

*The window.* Let $\Delta := \det D^2_\Theta f(0)$ (a function of $J'$) and $B_1 := \{|\Delta| \ge r^{1/2}\}$. On $B_1 \cap E_1$ and for small $r$, $|\det A_i(J') - \Delta| \le Cr^{1-m\eta} \le |\Delta|/2$, so $\det A_M(J')$ and $\det A_S(J')$ have the sign of $\Delta$ and $|d_i| \ge \frac{r}{24}|\Delta| \ge r^{3/2}/24$. Write $\Lambda_i(x) := d_i x + c_i$, so that $\det K_i = \Lambda_i(f_4) + \rho_i$, and $z_i := -c_i/d_i$. On the typed event the two determinants have opposite signs (proof of Lemma S8.14) while $d_M, d_S$ have the same sign, so $f_4$ lies within $\max_i |\rho_i|/|d_i| \le Cr^5/\min_i |d_i|$ of the segment between $z_M$ and $z_S$; hence $\{W_r > 0\} \cap B_1 \cap E_1 \subset \{f_4 \in I(J')\}$, where $I(J')$ is the $J'$-measurable interval spanned by $z_M, z_S$, enlarged by $Cr^5/\min_i |d_i|$. To bound its length, note $z_S - z_M = \Lambda_M(f_4)/d_M - \Lambda_S(f_4)/d_S$; on the typed event the sign window gives $|\det K_i| \le 12k|\Delta_0| + 2\max_j |R_j|$ (proof of Lemma S8.14), so $|\Lambda_i(f_4)| \le 12k|\Delta| + Cr^{2-N_1\eta}$ on $E_1$ (the swap $\Delta_0 \to \Delta$ costs $12k|\Delta_0 - \Delta| \le Cr^2 T^m \le Cr^{2-m\eta}$, absorbed as $N_1 \ge m+1$, and $|\rho_i| \le Cr^5$). With $|d_i| \ge \frac{r}{24}|\Delta|$ and $|\Delta| \ge r^{1/2}$,

$$|I(J')| \le \frac{48(12k|\Delta| + Cr^{2-N_1\eta})}{r|\Delta|} + Cr^{7/2} \le C(k/r + r^{1/2-N_1\eta}) \le \lambda := C(k/r + r^{1/4}).$$

Put $I(J') := \mathbb{R}$ on the $J'$-measurable set $\{d_M d_S \le 0\}$. The event $\{|I(J')| \le \lambda\}$ is $J'$-measurable, and $\{W_r > 0\} \cap B_1 \cap E_1 \subset \{f_4 \in I(J')\} \cap \{|I(J')| \le \lambda\}$; $E_1$ and the sign window involve the coupling variables $T$ and $\Delta_0$, so this inclusion holds on the coupling space, while the dominating event depends on $F_r$ alone. Therefore $\mathbb{P}_Q(\{W_r > 0\} \cap B_1 \cap E_1) \le \mathbb{E}[\mathbf{1}\{|I(J')| \le \lambda\}\mathbb{P}_Q(f_4 \in I(J') \mid J')] \le C_0\lambda$.

*Small determinant.* Under $Q$, $D^2_\Theta f(0)$ is a nondegenerate Gaussian symmetric matrix and $\Delta$ a polynomial of degree $m$ in it, with $\mathbb{E}_Q[\Delta^2] \ge c > 0$ uniformly in $(r, b, k,$ frame$)$: the top ($m$-th) Wiener-chaos component of $\det(\mu + G)$ is the Wick product of $\det G$, which does not depend on the mean $\mu$. By the Carbery–Wright inequality [3], $\mathbb{P}_Q(|\Delta| < r^{1/2}) \le Cr^{1/(2m)}$.

*Weights.* On the typed event $W_r/r^2 \le C(k^2\Delta_0^2 + r^4(1+k)^2T^{2N_1})$ by (S8.35), with $L^2$ norm at most $C(k^2 + r^4)P^N$. By the Cauchy–Schwarz inequality,

$$\mathbb{E}_Q[(W_r/r^2)\mathbf{1}_{B_1\cap E_1}] \le C(k^2+r^4)P^N \min(1, C_0\lambda)^{1/2} \le C(k^2+r^4)P^N(\min(1, k/r)^{1/2} + r^{1/8}),$$
$$\mathbb{E}_Q[(W_r/r^2)\mathbf{1}_{B_1^c}] \le C(k^2+r^4)P^N r^{1/(4m)}, \qquad \mathbb{E}_Q[(W_r/r^2)\mathbf{1}_{E_1^c}] \le C(k^2+r^4)P^N r^q.$$

Summing gives (S9.14). □

S9.5. **Proof of Theorem S9.2.** Fix $s_* \ge 1$ and $\varepsilon_0 \in (0, \beta/(12(3+\beta)))$, with $\beta$ as in Lemma S9.8; put $\rho_1 := \ell^{1/12-\varepsilon_0}$, and let $\ell$ be so small that $s_*\ell^{1/4} < \rho_1 \le \min(r_1, r_0^*)$ and $s_*\ell^{1/4} \le r_K$. By (S8.1) and (S8.2),

$$\nu_{\mathrm{eld}}(\ell) - c_{d,L}\ell^{-1/3} = I_1 + I_2 + I_3 + I_4 + I_5,$$

where, with all integrands evaluated at $(b, \ell/r^3, u)$ and integrated against $d\sigma(u)\,db\,dr$,

$$I_1 := \int_{r<s_*\ell^{1/4}} r^{-2}(\mathcal{A}_r^{\mathrm{eld}} - \mathcal{A}_0), \qquad I_2 := \int_{s_*\ell^{1/4}\le r<\rho_1} r^{-2}(\mathcal{A}_r^{\mathrm{eld}} - \mathcal{A}_0), \qquad I_3 := \int_{\rho_1\le r<r_0^*} r^{-2}\mathcal{A}_r^{\mathrm{eld}},$$
$$I_4 := -\int_{r\ge\rho_1} r^{-2}\mathcal{A}_0, \qquad I_5 := \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell).$$

*The term $I_1$.* With $r = s\ell^{1/4}$, $I_1 = \ell^{1/4}\int\!\!\int\!\!\int_0^{s_*}[r^{-2}(\mathcal{A}_r^{\mathrm{eld}} - \mathcal{A}_0)](b, \ell/r^3, u)\,ds\,db\,d\sigma$, and at fixed $s$ the integrand is evaluated at $\kappa = s^{-4}$, so by Theorem S9.7 it converges to $(\mathcal{C}^{\mathrm{eld}} - \mathcal{C}^{\mathrm{con}})(b, s^{-4}, u)$. Domination: for $s \le 1$ ($\kappa \ge 1$, i.e. $r \le k$), $|r^{-2}(\mathcal{A}_r - \mathcal{A}_0)| \le H(b,k) \le Ce^{-cb^2}$ by Lemma S8.5, and $r^{-2}(\mathcal{A}_r - \mathcal{A}_r^{\mathrm{eld}}) = 12\pi_r\mathbb{E}_Q[(W_r/r^2)(1-e)]/r^2 \le Ce^{-c(b^2+k^2)}P^N(r^3/k)/r^2 = Ce^{-c(b^2+k^2)}P^N s^4 \le Ce^{-c'b^2}$ by Proposition S8.3 (valid as $r \le \min(k, r_K)$); for $1 \le s \le s_*$,

$r^{-2}\mathcal{A}_r \le C(\kappa^2+r^2)P^N e^{-c(b^2+k^2)} \le Ce^{-c'b^2}$ by Lemma S8.14 and Proposition S8.2(ii), and $r^{-2}\mathcal{A}_0 \le C\kappa^2 e^{-cb^2}$. By dominated convergence,

$$I_1 = \ell^{1/4}\Big[\int_{S^{d-1}}\int_{\mathbb{R}}\int_0^{s_*}(\mathcal{C}^{\mathrm{eld}}-\mathcal{C}^{\mathrm{con}})(b,s^{-4},u)\,ds\,db\,d\sigma(u)+o(1)\Big].$$

*The term $I_2$.* Here $\kappa=\ell r^{-4}\le 1$. By Lemma S9.8 with Proposition S8.2(ii) and $\mathcal{A}_0\le k^2H$, $r^{-2}(\mathcal{A}_r^{\mathrm{eld}}+\mathcal{A}_0)\le C[(\kappa^2+r^2)(\min(1,\kappa)^{1/2}+r^\beta)+\kappa^2]e^{-c'b^2}$. The integrals are $\int_{s_*\ell^{1/4}}^\infty \kappa^2 dr = \ell^{1/4}s_*^{-7}/7$, $\int_{s_*\ell^{1/4}}^\infty \kappa^{5/2}dr=\ell^{1/4}s_*^{-9}/9$, $\int_{s_*\ell^{1/4}}^\infty \kappa^2 r^\beta dr \le C\ell^{1/4+\beta/4}s_*^{-7+\beta}$, $\int_0^{\rho_1} r^2\kappa^{1/2}dr=\ell^{1/2}\rho_1$ and $\int_0^{\rho_1} r^{2+\beta}dr\le \rho_1^{3+\beta}$; the last two are $o(\ell^{1/4})$ because $(\frac{1}{12}-\varepsilon_0)(3+\beta)>\frac14$ by the choice of $\varepsilon_0$. So $|I_2|\le C\ell^{1/4}s_*^{-7}+o(\ell^{1/4})$.

*The terms $I_3, I_4, I_5$.* Lemma S8.11 gives $0\le I_3\le C\ell^{1/3}(\rho_1^{-1}+\ell\rho_1^{-5})=O(\ell^{1/4+\varepsilon_0})$; $|I_4|\le C\ell^2\rho_1^{-7}=o(\ell^{1/4})$; Lemma S8.12 gives $0\le I_5\le C\ell^{1/3}$.

*Conclusion.* The lim sup and lim inf of $\ell^{-1/4}(\nu_{\mathrm{eld}}(\ell)-c_{d,L}\ell^{-1/3})$ lie within $Cs_*^{-7}$ of $\int\int\int_0^{s_*}(\mathcal{C}^{\mathrm{eld}}-\mathcal{C}^{\mathrm{con}})\,ds\,db\,d\sigma$. The integrand equals $-12\pi_0\mathbb{E}_0[\mathrm{loss}(s)\mid b]$ with $\mathrm{loss}(s):=Y^2\mathbf{1}\{|Y|<2s^{-4}|\Delta|\}+36s^{-8}\Delta^2\mathbf{1}\{|Y|\ge 2s^{-4}|\Delta|\}$ on $\{A\prec 0\}$ (and 0 off it); loss $\ge 0$, and with $s_1:=(2|\Delta|/|Y|)^{1/4}$, since $36\Delta^2 s_1^{-8}=9Y^2$,

$$\int_0^\infty \mathrm{loss}(s)\,ds = Y^2 s_1+\frac{36}{7}\Delta^2 s_1^{-7}=\frac{16}{7}Y^2 s_1=\frac{16}{7}\,2^{1/4}|Y|^{7/4}|\Delta|^{1/4}. \tag{S9.15}$$

By Tonelli's theorem the $s$-integral over $(0,\infty)$ converges absolutely; letting $s_*\to\infty$ gives (S9.2) with $c_1=-12\int\int\pi_0\mathbb{E}_0[\frac{16}{7}2^{1/4}|Y|^{7/4}|\Delta|^{1/4}\mathbf{1}\{A\prec 0\}\mid b]\,db\,d\sigma$, which is (S9.3). The expectation is finite (Gaussian moments; $\pi_0$ has Gaussian decay in $b$) and strictly positive, because $Y\neq 0$ almost surely and $\{A\prec 0\}$ has positive probability under the nondegenerate contact law; so $c_1<0$. The form (S9.4) follows from the factorization $\pi_0(u;v_0(b,0))=p_{(f,V_u)}(b,0)\,p_G(0)\,(2\pi)^{-1/2}\tau_u^{-1}$ (the last factor is the density of $t_u$ given $G=0$, at 0) and integration over $b$ [7, §15] (Proposition S12.31). □

S9.6. **The candidate and rejected densities.** For the candidate density no elder decision is needed, and the part of the remainder that is not carried by the cusp scale, the equal-height mass $B_{d,L}$, is reached at a Lipschitz rate in the gap, because the gap enters the pinned law affinely. By (S8.3), (S8.1) and the definition of $B_{d,L}$, for every $r_0\in(0,r_0^*]$,

$$B_{d,L}=\int_0^{r_0}\int_{\mathbb{R}}\int_{S^{d-1}} r^{-2}\mathcal{A}_r(b,0,u)\,d\sigma\,db\,dr+\int_{\mathrm{dist}(0,y)\ge r_0}\int_{\mathbb{R}}\Psi_0\,db\,dy. \tag{S9.16}$$

**Lemma S9.9** (the gap enters Lipschitz)**.** *There are $C,c,N$ such that for $0<r\le r_0^*$, $b\in\mathbb{R}$, $u\in S^{d-1}$ and $0\le k\le 1$,*

$$|\mathcal{A}_r(b,k,u)-\mathcal{A}_r(b,0,u)|\le C(k^2+kr^2)(1+|b|)^N e^{-cb^2}, \tag{S9.17}$$

*and for every $r_0\in(0,r_0^*]$ there is $C_{r_0}$ such that, for $\mathrm{dist}(0,y)\ge r_0$ and $0\le\ell\le 1$,*

$$|\Psi_\ell^{\mathrm{cand}}(b,y)-\Psi_0(b,y)|\le C_{r_0}\,\ell\,(1+|b|)^N e^{-cb^2}. \tag{S9.18}$$

*Proof. Affine structure.* The coupling of Proposition S8.2(i) realizes $Q_{r,b,k}$ as $F_r=F+C_r\Sigma_r^{-1}(v_r-U_r)$ with one unconditioned copy $F$ for all $(b,k)$, and the target $v_r=(b-kr^3/2,-kr^2,0,12k,0,\dots,0)$ is affine in $k$. Hence, writing $F_r^{(k)}$ for the field with gap mark $k$, $F_r^{(k)}=F_r^{(0)}+k\,\omega_r$, where $\omega_r:=C_r\Sigma_r^{-1}\delta v_r$ and $\delta v_r:=(-r^3/2,-r^2,0,12,0,\dots,0)$. The function $\omega_r$ is deterministic and pinned with $b=0$ and $k=1$ (that is, $U_r(\omega_r)=\delta v_r$), and $\|\omega_r\|_{C^7}\le\|C_r\|_{C^7}\|\Sigma_r^{-1}\|\,|\delta v_r|\le C$ uniformly in $r\le r_0^*$ and frames. Put $T_0:=1+\|F_r^{(0)}\|_{C^7}+\|\omega_r\|_{C^7}$; then $\|F_r^{(0)}\|_{C^7}\le\|F\|_{C^7}+C(|b|+|U_r|)$, so $\|T_0\|_{L^p}\le C_p(1+|b|)$, and $\|F_r^{(k')}\|_{C^7}\le T_0$ for $0\le k'\le 1$.

*The typed product along the gap.* The scaled endpoint Hessians are linear in the field, so $K_i(k')=K_i(0)+k'K_i[\omega_r]$ with $\|K_i[\omega_r]\|\le C$ (by (S8.10) for the pinned $\omega_r$, $\alpha_i[\omega_r]=\mp 6+O(r)$) and

$\|K_i(k')\| \le CT_0$. Hence $\det K_i(k')$ is a polynomial in $k'$ with $|\partial_{k'} \det K_i| = |\operatorname{tr}(\operatorname{adj} K_i(k')\, K_i[\omega_r])| \le CT_0^{d-1}$. The typed product $w(k') := F_d(K_M(k'))F_{d-1}(K_S(k'))$ is continuous on $[0,k]$: an index changes only where an eigenvalue crosses zero, where the corresponding determinant, hence $w$, vanishes. Between the finitely many zeros of the polynomial $p := \det K_M \det K_S$ (if $p \equiv 0$ then $w \equiv 0$), $w = \pm p$ or $w = 0$, so for every realization

$$|w(k) - w(0)| \le \int_0^k |\partial_{k'}(\det K_M \det K_S)|\, \mathbf{1}\{\text{typed}(k')\}\, dk'.$$

On typed$(k')$ the two determinants have opposite signs, and Steps 1–3 of the proof of Lemma S8.5 with the remainder bound of (S8.34), applied to the pinned $C^7$ function $F_r^{(k')}$ with $D_y^2 F_r^{(k')}(0)$ in place of $A_0$, give $\det K_{M,S}(k') = \mp 6k'\Delta' + rY' + R_{M,S}$ with $|R_i| \le Cr^2 T_0^{N_1}$; so, as in the proof of Lemma S8.14, $|\det K_i(k')| \le 12k'T_0^m + 2Cr^2T_0^{N_1}$. Hence $|\partial_{k'}(\det K_M \det K_S)| \le C(k' + r^2)T_0^N$ on typed$(k')$, so $|w(k) - w(0)| \le C(k^2 + kr^2)T_0^N$, and taking expectations, $|Z_r/r^2(b,k,u) - Z_r/r^2(b,0,u)| \le C(k^2 + kr^2)(1+|b|)^N$.

*The density factor.* $\pi_r(v_r(b,\cdot))$ is a Gaussian density with covariance uniformly invertible for $r \le r_0^*$, evaluated at a target affine in $k$: $|\pi_r(v_r(b,k)) - \pi_r(v_r(b,0))| \le Ck(1+|b|)e^{-cb^2}$. With $Z_r/r^2 \le C(k^2 + r^4)(1+|b|)^N$ (Lemma S8.14) and $\mathcal{A}_r = 12\pi_r(v_r)Z_r/r^2$, (S9.17) follows ($k^3 \le k^2$, $kr^4 \le kr^2$).

*Proof of* (S9.18). The same argument with the two-site regression in place of the near one: by Proposition S8.4(ii), $f_\ell = f_0 + \ell h_y$ with $f_0$ of law $Q_{y,b,0}$ and $h_y$ deterministic. On the compact set $D_{r_0} = \{\operatorname{dist}(0,y) \ge r_0\}$ the uniform facts of the proof of Lemma S8.12 apply: $\lambda_{\min}(\operatorname{Cov} O_y) \ge c_0 > 0$, the regression functions $\psi_j^y$ are bounded in $C^3$ uniformly (so $h_y = -\psi^y_{f(y)}$, the regression function of the coordinate $f(y)$, has $\sup_y \|h_y\|_{C^2} \le C_{r_0}$), the Hessian moments under $Q_{y,b,\ell}$ are at most $C_p(1+|b|)^p$ uniformly in $y \in D_{r_0}$ and $\ell \in [0,1]$ ((S8.25) at $\ell = 0$ by the same proof), and $|\partial_\ell p_y(v_{b,\ell})| = p_y(v_{b,\ell})|(\Gamma_y^{-1}v_{b,\ell})_{f(y)}| \le C_{r_0}(1+|b|)e^{-cb^2}$. The typed product $W(f_\ell)$ is continuous and piecewise polynomial in $\ell$ for the same reason, with $|\partial_\ell(\det H_0 \det H_y)| \le C(\|f_0\|_{C^2} + \|h_y\|_{C^2})^{2d-1}\|h_y\|_{C^2}$. □

*Proof of Theorem S9.3.* Fix $s_* \ge 1$ and $r_0 \in (0, r_0^*]$, and let $\ell$ be so small that $s_*\ell^{1/4} < r_0$. By (S8.1), (S8.2), (S8.4) and (S9.16),

$$\nu_{\text{cand}}(\ell) - c_{d,L}\ell^{-1/3} - B_{d,L} = J_1 - J_2 + J_3 - J_4 + J_5,$$

where, with the inner integrals over $d\sigma(u)\, db$,

$$J_1 := \int_{r<s_*\ell^{1/4}} \iint r^{-2}(\mathcal{A}_r - \mathcal{A}_0)(b, \ell/r^3, u), \qquad J_2 := \int_{r<s_*\ell^{1/4}} \iint r^{-2}\mathcal{A}_r(b,0,u),$$

$$J_3 := \int_{s_*\ell^{1/4}\le r<r_0} \iint r^{-2}[\mathcal{A}_r(b,\ell/r^3,u) - \mathcal{A}_r(b,0,u)], \qquad J_4 := \int_{r\ge s_*\ell^{1/4}} \iint r^{-2}\mathcal{A}_0(b,\ell/r^3,u),$$

$$J_5 := \int_{\operatorname{dist}(0,y)\ge r_0} \int_{\mathbb{R}} (\Psi_\ell^{\text{cand}} - \Psi_0)\, db\, dy.$$

The term $J_1$ is $I_1$ of §S9.5 with $\mathcal{A}_r$ in place of $\mathcal{A}_r^{\text{eld}}$: Lemma S8.5 gives the majorant $|r^{-2}(\mathcal{A}_r - \mathcal{A}_0)| \le H(b,k) \le Ce^{-cb^2}$ on the whole range, and the pointwise limit is the candidate part of Theorem S9.7 (steps (i), (ii), (iv), (v); Proposition S9.6 is not used), so $J_1 = \ell^{1/4}[\iiint_0^{s_*}(\mathcal{C}^{\text{cand}} - \mathcal{C}^{\text{con}})(b, s^{-4}, u)\, ds\, db\, d\sigma + o(1)]$. By (S8.30) at $k=0$, $0 \le r^{-2}\mathcal{A}_r(b,0,u) \le Cr^2(1+|b|)^N e^{-cb^2}$, so $J_2 = O(s_*^3\ell^{3/4})$. On $r \ge s_*\ell^{1/4}$ we have $k = \ell/r^3 \le 1$, and (S9.17) gives $|J_3| \le C\int_{s_*\ell^{1/4}}^\infty (\ell^2 r^{-8} + \ell r^{-3})\, dr \le C(\ell^{1/4}s_*^{-7} + \ell^{1/2}s_*^{-2})$. Since $\mathcal{A}_0 \le k^2 H$, $0 \le J_4 \le C\ell^{1/4}s_*^{-7}$, and $|J_5| \le C_{r_0}\ell$ by (S9.18), which is also the far statement of the theorem. Hence the lim sup and lim inf of $\ell^{-1/4}(\nu_{\text{cand}}(\ell) - c_{d,L}\ell^{-1/3} - B_{d,L})$ lie within $Cs_*^{-7}$ of $\iiint_0^{s_*}(\mathcal{C}^{\text{cand}} - \mathcal{C}^{\text{con}})\, ds\, db\, d\sigma$. The integrand is

$-12\pi_0\mathbb{E}_0[\min(Y^2, 36s^{-8}\Delta^2)\mathbf{1}\{A \prec 0\} \mid b]$, because $w^{\mathrm{cand}} - w^{\mathrm{con}} = -\min(Y^2, 36\kappa^2\Delta^2)$ on $\{A \prec 0\}$; with $s_2 := (6|\Delta|/|Y|)^{1/4}$,

$$\int_0^\infty \min(Y^2, 36s^{-8}\Delta^2)\, ds = Y^2 s_2 + \frac{36}{7}\Delta^2 s_2^{-7} = \frac{8}{7}\, Y^2 s_2 = \frac{8}{7}\, 6^{1/4}|Y|^{7/4}|\Delta|^{1/4}. \tag{S9.19}$$

Letting $s_* \to \infty$ gives (S9.5) with (S9.6); $I^{\mathrm{cand}}/c_1 = \frac{8}{7}6^{1/4}/(\frac{16}{7}2^{1/4}) = 3^{1/4}/2$. Subtracting (S9.2) from (S9.5) gives (S9.7). □

*Remark* S9.10 (the rejected coefficient). The difference of the two losses, $(36\kappa^2\Delta^2 - Y^2)\mathbf{1}\{2\kappa|\Delta| \le |Y| < 6\kappa|\Delta|\} \ge 0$, is the weight of the typed cusp-scale pairs outside the elder window ($\frac{1}{3} \le |\varphi| < 1$), and $\int_0^\infty(\mathrm{loss} - \min(Y^2, 36s^{-8}\Delta^2))\, ds = \frac{16}{7}Y^2 s_1 - \frac{8}{7}Y^2 s_2 = \frac{8}{7}(2^{5/4} - 6^{1/4})|Y|^{7/4}|\Delta|^{1/4}$. So the excess of the rejected density over $B_{d,L}$ in (S9.7) is the rejected part of the cusp-scale typed pairs.

*Remark* S9.11 (the crossover). At fixed $\kappa$ the elder share of the typed weight is the weight of $\{|\varphi| < \frac{1}{3}\}$ inside $\{|\varphi| < 1\}$, with weight $36\kappa^2\Delta^2(1 - \varphi^2)$. As $\kappa \to \infty$ (the cap regime $r \ll k$) the rejected weighted mass concentrates on $|\Delta| \lesssim |Y|/\kappa$ and is of order $1/\kappa = r/k$, the scaling $r^3/k$ of the mass in Proposition S8.3. As $\kappa \to 0$ the typed weight is $O(\kappa)$ of the contact weight, and within it the elder share tends to 13/27, since $\int_{|y|<c\kappa|\Delta|}(36\kappa^2\Delta^2 - y^2)\, dy = \kappa^3|\Delta|^3(72c - \frac{2}{3}c^3)$ and $\frac{416/3}{288} = \frac{13}{27}$ for $c = 2$ against $c = 6$. The equal-height limit of Proposition S8.4(i), in which all candidates are rejected, is the different limit $k \to 0$ at fixed $r$, which does not commute with this one. The cusp scale $r \asymp k$ is the crossover between the two regimes, and the elder rule on it is the single number 1/3.

S9.7. **The value of $c_1$.** The integral (S9.3) is defined for every $d \ge 1$; for $d = 1$, $A$ is empty, $\Delta = 1$ and $Y = f_4/12$, and (S9.3) is the coefficient $C_1$ of the one-dimensional law of §5.4 of the main text. We enclose it for the Gaussian kernel $e^{-|z|^2/2}$ on $\mathbb{R}^d$ and transfer the enclosure to the periodized fields of side $L \ge 24$. Write $c_{d,\infty}$ for the leading coefficient $c_{d,L}$ computed for the Gaussian kernel on $\mathbb{R}^d$ (for $d = 3$ it is (2.11) of the main text), and $c_1^{(\infty)}$, $c_1^{(L)}$ for (S9.3) computed for the Gaussian kernel and for the field of Convention S8.1(a).

**Lemma S9.12** (the zero-gap contact law of the Gaussian kernel)**.** *For the Gaussian kernel on $\mathbb{R}^d$ and every direction $u$, under $\mathbb{E}_0[\,\cdot \mid b]$: $f_4 \sim N(-3b, 24)$; $\gamma \sim N(0, 2I_m)$; $A = -bI_m + G$ with $G$ symmetric, $G_{jj} \sim N(0, 2)$ and $G_{jk} \sim N(0, 1)$ for $j < k$; all of these are independent; and $\pi_0(u; v_0(b, 0)) = (2\pi)^{-(d+1)}12^{-1/2}e^{-3b^2/4}$.*

*Proof.* For the Gaussian kernel $K$ and multi-indices $\mathsf{a}, \mathsf{b}$, $\mathrm{Cov}(\partial^{\mathsf{a}} f, \partial^{\mathsf{b}} f) = (-1)^{|\mathsf{b}|}\partial^{\mathsf{a}+\mathsf{b}}K(0)$, with $\partial^{\mathsf{g}}K(0) = \prod_i(-1)^{\mathsf{g}_i/2}(\mathsf{g}_i - 1)!!$ for even $\mathsf{g}$ and 0 otherwise. The odd jets $(\partial_u f, \nabla_\Theta f, \partial_u^3 f, \gamma)$ are uncorrelated with the even jets. Each $\gamma_j$ has variance 3, covariance $-1$ with $\partial_{\theta_j} f$ and no other correlation with the pinned odd jets or with the other $\gamma$'s, so given the pins its variance is 2; $\tau_u^2 = \mathrm{Var}(\partial_u^3 f \mid \nabla f = 0) = 15 - 9 = 6$, so the density of $\partial_u^3 f$ given $\nabla f = 0$ is $(12\pi)^{-1/2}$ at 0, and $p_G(0) = (2\pi)^{-d/2}$. The even vector $(f, V_u)$ has covariance $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 3 \end{smallmatrix}\right) \oplus I_m$, with determinant 2 and inverse $(1,1)$ entry 3/2, so $p_{(f,V_u)}(b, 0) = (2\pi)^{-(d+1)/2}2^{-1/2}e^{-3b^2/4}$. Regression on $(f, V_u) = (b, 0)$ gives $\mathbb{E}f_4 = -3b$, $\mathrm{Var}\, f_4 = 105 - 81 = 24$, $\mathbb{E}A_{jj} = -b$, $\mathrm{Var}\, A_{jj} = 3 - 1 = 2$, $\mathrm{Cov}(A_{jj}, A_{kk}) = 1 - 1 = 0$, $\mathrm{Var}\, A_{jk} = 1$ and $\mathrm{Cov}(f_4, A_{jj}) = -3 - (-3) = 0$; all other covariances vanish by the parity of the individual indices. Multiplying the three density factors gives $\pi_0$. □

**Theorem S9.13** (enclosures for the Gaussian kernel)**.** *For the Gaussian kernel on $\mathbb{R}^d$, the integral $c_1 = c_1^{(\infty)}$ of (S9.3) and $I^{\mathrm{cand}} = \frac{3^{1/4}}{2}c_1$ satisfy*

| $d$ | *quantity* | *enclosure* |
|---|---|---|
| 1 | $c_1$ | $[-0.22760635877558, -0.22760635877504]$ |
| | $I^{\mathrm{cand}}$ | $[-0.14977340698364, -0.14977340698328]$ |
| | $I^{\mathrm{cand}} - c_1$ | $[0.077832951791707, 0.077832951791985]$ |
| 2 | $c_1$ | $[-0.269398825674, -0.269398825672]$ |
| | $I^{\mathrm{cand}}$ | $[-0.177274396795, -0.177274396793]$ |
| | $I^{\mathrm{cand}} - c_1$ | $[0.0921244288786, 0.0921244288791]$ |
| 3 | $c_1$ | $[-0.211848347, -0.211848346]$ |
| | $I^{\mathrm{cand}}$ | $[-0.139404052, -0.139404051]$ |
| | $I^{\mathrm{cand}} - c_1$ | $[0.0724442943, 0.0724442950]$ |

*Moreover* $c_{2,\infty} \in [0.0734069193059, 0.0734069193062]$, $c_{3,\infty} \in [0.0417759318405, 0.0417759318407]$, *and*

$$c_1/c_{2,\infty} \in [-3.66993776909, -3.66993776907], \qquad c_1/c_{3,\infty} \in [-5.07106215, -5.07106213].$$

*The ratios are enclosed from the unrounded enclosures of the computation,*

$$c_1 \in [-0.2693988256731888, -0.2693988256725268] \quad (d = 2),$$
$$c_1 \in [-0.2118483463910089, -0.2118483460317414] \quad (d = 3),$$

*and from* $c_{2,\infty} = 0.07340691930603427103\ldots$ *and* $c_{3,\infty} = 0.04177593184059834334\ldots$, *the values of the closed form of* $c[K_{\mathrm{ref}}]$ *in Theorem S11.5; dividing the rounded entries of the table gives slightly wider intervals. For* $d = 1$, $c_1 = -\frac{384}{7}\, 2^{1/4}\, 12^{-9/4} (2\pi)^{-2} \sqrt{4\pi/3}\, 30^{7/8}\, 2^{7/8}\, \Gamma(11/8)/\sqrt{\pi}$.

**Lemma S9.14** (Gauss–Legendre error on an ellipse)**.** *Let* $F$ *be analytic in the open Bernstein ellipse* $\frac{a+b}{2} + hE_\rho$ *of* $[a, b]$, *where* $h = \frac{b-a}{2}$ *and* $E_\rho$ *is the ellipse with foci* $\pm 1$ *and sum of semi-axes* $\rho > 1$, *and let* $|F| \le M_F$ *there. Then the* $n$*-point Gauss–Legendre rule* $Q_n$ *on* $[a, b]$ *satisfies*

$$\Big| \int_a^b F - Q_n F \Big| \le h \cdot 4M_F \Big(1 + \frac{1}{4n^2 - 1}\Big) \frac{\rho^{-2n}}{1 - \rho^{-2}}.$$

*Proof.* On $[-1, 1]$ the Chebyshev coefficients of $F$ satisfy $|a_k| \le 2M_F \rho^{-k}$. The rule $Q_n$ is exact up to degree $2n - 1$ and, by symmetry, on odd $T_k$. For even $k$, $|\int_{-1}^{1} T_k| \le 2/(k^2 - 1)$ and $|Q_n T_k| \le \sum w = 2$. Summing over even $k \ge 2n$ and scaling to $[a, b]$ gives the bound. □

*Proof of Theorem S9.13 (method).* The enclosures are computer-assisted; we describe the method, which is that of the certificate in record [R12] of the main text.

*Reduction.* The kernel is isotropic, so the $u$-integral is $|S^{d-1}|$ times the value at $u = e_1$ ($|S^0| = 2$, $|S^1| = 2\pi$, $|S^2| = 4\pi$), and Lemma S9.12 makes the inner expectation an explicit Gaussian integral. The power is written as $|y|^p = K_p^{-1} \int_0^\infty (1 - \cos \omega y)\, \omega^{-1-p}\, d\omega$ for $0 < p < 2$, $K_p = \pi/(2\Gamma(1 + p) \sin(\pi p/2))$, at $p = 7/4$; by Tonelli's theorem, applied to the nonnegative function $(1 - \cos \omega Y)\omega^{-11/4}$ times the nonnegative determinant and cone weight, $\mathbb{E}|Y|^{7/4} = K_{7/4}^{-1} \int_0^\infty (1 - \operatorname{Re} \mathbb{E} e^{i\omega Y})\, \omega^{-11/4}\, d\omega$, the bounded characteristic function being integrated against the finite Gaussian weight at each fixed $\omega$, and the two terms are never split into separately divergent integrals. Given $(A, b)$, $Y$ is a Gaussian linear term in $f_4$ plus a quadratic form in $\gamma$, so its characteristic function is explicit, and the $b$-integral is Gaussian. For $d = 1$, averaging over $b$ makes $f_4$ a centred normal of variance 30, and $\mathbb{E}|Z|^{7/4} = 2^{7/8}\Gamma(11/8)/\sqrt{\pi}$ gives the closed form. For $d = 2$ the integrals over $b$ and over $a = -A > 0$ are done in closed form, which leaves an explicit constant times a one-dimensional integral with an analytic integrand. For $d = 3$, write $A \prec 0$ through its eigenvalues $-a_1 \le -a_2 < 0$ (Lebesgue measure on $2 \times 2$ symmetric matrices is

$|\lambda_1-\lambda_2|\,d\lambda_1 d\lambda_2 d\theta$, $\theta\in[0,\pi)$); the substitution $a_1=s^4$, $a_2=a_1v^4$, $\omega=t^4/a_1$ gives

$$c_1=-\frac{192}{7}\,2^{1/4}\cdot 4\pi\,(2\pi)^{-11/2}\,12^{-1/2}\cdot\tfrac{1}{2}\sqrt{4\pi/5}\cdot\pi\cdot K_{7/4}^{-1}\,T_3,$$
$$T_3=64\int_0^\infty ds\int_0^1 dv\int_0^\infty dt\; s^{20}v^4(1-v^4)\,e^{-s^8E(v)}\,\frac{1-\operatorname{Re}\Phi}{t^8},$$

with $E(v)=(2v^8-v^4+2)/10\ge 3/16$ and $\Phi=(1-i\omega a_1)^{-1/2}(1-i\omega a_2)^{-1/2}\exp(-i\omega a_1a_2(a_1+a_2)/20-23\omega^2(a_1a_2)^2/240)$. The integrands are written in a cancellation-free form $(1-\operatorname{Re}\Phi=-(e^{\mathcal R}-1)+e^{\mathcal R}\,2\sin^2(\mathcal I/2)$ with $\mathcal R+i\mathcal I=\log\Phi$, expressed through the analytic functions $(e^x-1)/x$, $\log(1+x)/x$, $\arctan(x)/x$ and $\sin(x)/x$, so that no step divides by a small quantity.

*Quadrature with certified error.* Each one-dimensional integral is computed by Gauss–Legendre rules with the error bound of Lemma S9.14. The bound $M_F$ is obtained by interval evaluation over complex boxes covering the closed Bernstein ellipse, with one representation of the integrand on all the boxes of an ellipse. If that representation evaluates without exception on every box, each principal branch being evaluated off its cut, it is analytic on a neighbourhood of the ellipse; since it agrees with the integrand on the real interval, it is the analytic continuation of the integrand there, by the identity theorem. On the real axis $\operatorname{Re}\Phi=\frac{1}{2}(\Phi+\Phi^*)$ with $\Phi^*(z)=\overline{\Phi(\bar z)}$, and this form is the one continued. The nodes are certified in exact rational arithmetic and the weights are enclosed. For $d=3$ the region $s\le 3$, $t\le 15$ is cut into 247 boxes adapted to the branch points at $|v|=1/t$: 76 boxes, on which the nonnegative integrand is below $10^{-12}$ in total, are enclosed by $[0,\text{volume}\times\sup]$, and the other 171 by tensor Gauss–Legendre rules, each to $10^{-11}$; the tails $t>15$ (through $|\Phi|\le t^{-2}$ on the real axis and $\int_0^\infty s^{20}e^{-s^8E}ds=\Gamma(21/8)/(8E^{21/8})$) and $s>3$ (below $10^{-280}$) are bounded explicitly. For $d=2$ fifteen panels are each computed to $5\cdot10^{-16}$, with an explicit tail. On a box of the $d=3$ cover, with $I_j$ the integral and $Q_j$ the rule in the $j$-th variable, $I_1I_2I_3-Q_1Q_2Q_3=(I_1-Q_1)I_2I_3+Q_1(I_2-Q_2)I_3+Q_1Q_2(I_3-Q_3)$, and each term is bounded by Lemma S9.14 in one variable, uniformly in the others, as the weights are positive.

*Arithmetic.* All operations are in binary64 interval arithmetic with one-ulp outward rounding; $\pi$, $\ln 2$, exp, log, arctan, sin, cos and $\log\Gamma$ are computed from series with explicit remainder bounds, and no library transcendental function is trusted. The results are widened by $10^{-12}$ relative and rounded outward; the coefficients $c_{d,\infty}$ are enclosed from their closed forms. Floating-point recomputations by routes that avoid the Mellin identity give $-0.269398825672854$ ($d=2$) and $-0.211848346210999$ ($d=3$), inside the enclosures. □

**Lemma S9.15** (transfer to the torus)**.** *Let $d\in\{2,3\}$ and $L\ge 24$, and let $C_L$ be the covariance matrix of the $N$ jets entering (S9.3) for the field with covariance $\sum_{n\in\mathbb Z^d}e^{-|z+Ln|^2/2}$, normalized to variance one or not, in an orthonormal frame $(u,\Theta)$, and $C_0$ that of the Gaussian kernel; here $N=9$ for $d=2$ and $N=14$ for $d=3$, counting the $2d+2$ pinned coordinates and the free jets $(f_4,\gamma,A)$. Put $E:=(764\cdot24^8\cdot4374+105\cdot54)\cdot10^{-125}\le 3.7\cdot10^{-108}$, $\mu:=1/4$, $\varepsilon:=NE/\mu$ and $\bar N:=N/2+1$. Then $(1-\varepsilon)C_0\preceq C_L\preceq(1+\varepsilon)C_0$ in every frame, and*

$$|c_1^{(L)}/c_1^{(\infty)}-1|\le\eta:=8\bar N\varepsilon,$$

*with $\varepsilon\le 1.4\cdot10^{-106}$, $\eta\le 5.9\cdot10^{-105}$ for $d=2$ and $\varepsilon\le 2.1\cdot10^{-106}$, $\eta\le 1.4\cdot10^{-104}$ for $d=3$.*

*Proof. Image bound.* Every covariance entry is $(-1)^{|\mathsf b|}D^{|\mathsf a|+|\mathsf b|}K(0)$ contracted with frame vectors, of total order $q\le 8$ ($\operatorname{Cov}(f_4,f_4)=\partial_u^8K(0)$). Let $K_\infty(x):=e^{-|x|^2/2}$. For unit $v_i$, $D^qK_\infty(x)[v_1,\dots,v_q]$ is $K_\infty(x)$ times a sum over the partial matchings of $\{1,\dots,q\}$ of products of the factors $-\langle v_i,v_j\rangle$ and $-\langle x,v_i\rangle$; so $|D^qK_\infty(x)[v]|\le T_8|x|^8K_\infty(x)$ for $|x|\ge 1$, with $T_8=764$ partial matchings of eight points, and $|D^qK_\infty(0)[v]|\le 7!!=105$. For $d\le 3$ the shell $\max_i|n_i|=j$ has at most $27j^3$ points, all with $|n|^2\le 3j^2$, and consecutive shell terms shrink by at least $\frac12$; so $\sum_{n\ne0}|n|^8e^{-288|n|^2}\le 4374e^{-288}$ and $S_{24}-1\le 54e^{-288}$, where $S_L=\sum_n e^{-L^2|n|^2/2}$. Also $e^{288/125}>10$, so $e^{-288}<10^{-125}$, and

$L^8 e^{-L^2|n|^2/2}$ decreases in $L$ once $L|n| \geq 3$, so the bound at $L = 24$ holds for every $L \geq 24$. Writing $D^q K_L(0) - D^q K_\infty(0) = S_L^{-1} \sum_{n\neq 0} D^q K_\infty(Ln) + D^q K_\infty(0)(S_L^{-1} - 1)$, every entry of $C_L - C_0$ is at most $E$ in absolute value (the unnormalized periodization has only the first term).

*Sandwich.* $C_0 - \frac{1}{4}I$ is positive definite (an exact $LDL^\top$ factorization has all pivots positive; $\lambda_{\min}(C_0) \approx 0.258$ for $d = 3$), so $|x^\top (C_L - C_0)x| \leq NE|x|^2 \leq \varepsilon\, x^\top C_0 x$.

*Transfer.* Write the integral of (S9.3) at direction $u$ as $I_u(C) = \int db \int dW\, \mathsf{F}(W)\, p_C(b, 0, W)$, where $p_C$ is the centred Gaussian density with covariance $C$, $2d + 1$ pinned coordinates are fixed at 0, $W$ are the free jets and $\mathsf{F} = |Y|^{7/4}|\Delta|^{1/4}\mathbf{1}\{A \prec 0\} \geq 0$; $\pi_0 \mathbb{E}_0[\mathsf{F} \mid b]$ is exactly this slice integral of the joint density. $\mathsf{F}$ is homogeneous of degree $2d - \frac{1}{4}$ in the free jets ($Y$ has degree $d$ and $\Delta$ degree $d - 1$), so the substitution $(b, W) \mapsto \sqrt{a}\,(b, W)$ gives $I_u(aC_0) = a^{-5/8} I_u(C_0)$. From $(1 - \varepsilon)C_0 \preceq C \preceq (1 + \varepsilon)C_0$ follow $\det C \geq (1 - \varepsilon)^N \det C_0$, $C^{-1} \succeq C_0^{-1}/(1 + \varepsilon)$ and symmetrically, hence the pointwise sandwich

$$\Big(\frac{1 - \varepsilon}{1 + \varepsilon}\Big)^{N/2} p_{(1-\varepsilon)C_0} \leq p_C \leq \Big(\frac{1 + \varepsilon}{1 - \varepsilon}\Big)^{N/2} p_{(1+\varepsilon)C_0}.$$

Integrating against $\mathsf{F} \geq 0$ gives $I_u(C_L)/I_u(C_0) \in [\varrho^{-\bar{N}}, \varrho^{\bar{N}}]$ with $\varrho = (1 + \varepsilon)/(1 - \varepsilon)$; as $\log \varrho \leq 4\varepsilon$ and $e^x \leq 1 + 2x$ on $[0, 1]$, this lies within $8\bar{N}\varepsilon$ of 1. Since $I_u(C_0)$ does not depend on $u$, integrating over $S^{d-1}$ preserves the bound. □

**Corollary S9.16** (the periodized fields)**.** *Let $L \geq 24$. For $d = 3$,*

$$c_1^{(L)} \in [-0.211848347,\ -0.211848346], \qquad I^{\mathrm{cand},(L)} \in [-0.139404052,\ -0.139404051],$$

*and, with the enclosure $c_{3,24} \in [0.04177593184059834334,\ 0.04177593184059834335]$ quoted in Theorem 2.3(iii) of the main text,*

$$c_1^{(24)}/c_{3,24} \in [-5.07106215,\ -5.07106213].$$

*For $d = 2$,*

$$c_1^{(L)} \in [-0.269398825674,\ -0.269398825672], \quad I^{\mathrm{cand},(L)} \in [-0.177274396795,\ -0.177274396793],$$

*and, with $c_{2,24} \in [0.07340691930603427103,\ 0.07340691930603427104]$,*

$$c_1^{(24)}/c_{2,24} \in [-3.66993776909,\ -3.66993776907].$$

*Proof.* Theorem S9.13 and Lemma S9.15; in binary64 the factor $[1 - \eta, 1 + \eta]$ is enclosed by the one-ulp interval around 1, and the intervals are rounded outward. The ratios are formed, as in Theorem S9.13, from the unrounded enclosures of $c_1$ stated there, the factor $[1 - \eta, 1 + \eta]$ and the quoted enclosures of $c_{3,24}$ and $c_{2,24}$. □

*Remark* S9.17 (how the enclosed ratio is read). What Corollary S9.16 encloses for $d = 3$, $L = 24$ is the ratio of the integral (S9.3), evaluated for the side-24 field, to $c_{3,24}$. Its reading as the relative correction $(c_1/c_{3,24})\ell^{7/12}$ in (S9.2), that is, in (2.14) of the main text, uses two inputs that the enclosure itself does not supply: Theorem S9.2, which identifies the integral as the coefficient of $\ell^{1/4}$ in $\nu_{3,24}$, and the enclosure of $c_{3,24}$ quoted in Corollary S9.16. Likewise, the intervals for $I^{\mathrm{cand}}$ enclose $\frac{3^{1/4}}{2}$ times the integral; their reading as the coefficient of $\ell^{1/4}$ in $\nu_{\mathrm{cand}}$ uses Theorem S9.3.

## S10. The third-order terms

This section expands the candidate and the elder lifetime densities to third order, in every dimension $d \geq 2$ and on every torus side $L > 0$: with remainder $O(\ell^{4/11})$ (Theorems S10.1 and S10.2) and then $O(\ell^{3/7})$ (Theorems S10.4 and S10.5). The three-term law (2.12) of the main text is the case $d = 3$, $L = 24$ of (S10.31) (Remark S10.8). All constants are existential.

S10.1. **Setting and notation.** *Field and densities.* Fix $d \geq 2$, $L > 0$, $m = d-1$; $f$ is the centred stationary Gaussian field on $\mathbb{T}_L^d = \mathbb{R}^d/(L\mathbb{Z}^d)$ with covariance $K_L(z) = \sum_n e^{-|z+Ln|^2/2}/\sum_n e^{-|Ln|^2/2}$ (sums over $n \in \mathbb{Z}^d$; for $d = 3$, $L = 24$ this is $K_{24}$ of (1.1) of the main text and $f = f_{24}$). $\nu_{\mathrm{cand}}(\ell)$ counts ordered pairs (local maximum, critical point of index $m$) with height gap $\ell$, $\nu_{\mathrm{eld}}(\ell)$ the finite nonessential superlevel $H_0$ bars of lifetime $\ell$, and $\rho_{\mathrm{rej}} := \nu_{\mathrm{cand}} - \nu_{\mathrm{eld}}$; all are per unit volume, in their canonical marked Kac–Rice density versions (Convention 2.2 of the main text; [7]; Proposition S12.22(b), (S12.51) and Lemma S12.29), and $\nu_{\mathrm{eld}} = \nu_{3,24}$, $\nu_{\mathrm{cand}} = \nu_{3,24}^{\mathrm{cand}}$ for $d = 3$, $L = 24$. $c_{d,L} > 0$ is the leading coefficient of [7, (15.2)] (Proposition S12.31; $c_{3,24}$ of (2.9) of the main text). Constants $C, c, N$ ($C_p$ for an exponent $p$) depend only on $d, L$ unless stated otherwise and change from line to line. $\|g\|_{C^q}$ is the supremum over $\mathbb{T}_L^d$ of the operator norms of the derivatives of order $\leq q$.

*Pinned pairs.* For $u \in S^{d-1}$ fix an orthonormal frame $\Theta$ of $u^\perp$ and write points as $xu + \Theta y$, $(x, y) \in \mathbb{R} \times \mathbb{R}^m$. For $r > 0$, $b \in \mathbb{R}$, $k \geq 0$ the pins are

$$M = -\tfrac{r}{2}u, \quad S = \tfrac{r}{2}u, \qquad f(M) = b, \quad f(S) = b - kr^3, \qquad \nabla f(M) = \nabla f(S) = 0, \tag{S10.1}$$

so the gap is $\ell = kr^3$. By stationarity the midpoint is 0; in (3.4)–(3.5) of the main text, $t = u$, $\Theta = (n_1, n_2)$ for $d = 3$, and the parameter $\kappa$ of (3.5) there equals $6k$. In this section $\kappa$ is instead the *cusp variable*

$$\kappa := k/r = \ell/r^4, \qquad s := r\ell^{-1/4} = \kappa^{-1/4}, \qquad \text{and} \qquad P := 1 + |b| + k. \tag{S10.2}$$

$U_r$ is the vector of pin rows of [7, (3.1)] (Definition S12.9 and Lemma S12.10) (centred difference rules of the values and gradients at $M, S$; an invertible image of the pins with absolute Jacobian $12r^{-(d+3)}$), and (S10.1) reads $U_r = v_r := (b - kr^3/2, -kr^2, 0, 12k, 0, \dots, 0)$. $\pi_r(v_r)$ is the density of $U_r$ at $v_r$ and $Q = Q_{r,b,k}$ the Gaussian regression law given $U_r = v_r$. At $r = 0$ the rows become the *pinned jets* $U_0 = (f, \partial_u f, \partial_u^2 f, \partial_u^3 f, \nabla_\Theta f, \nabla_\Theta \partial_u f)(0)$, with target $v_0(b, k) = (b, 0, 0, 12k, 0, \dots, 0)$ and density $\pi_0(u; v_0(b, k))$; the *contact law at* $v_0(b, k)$ is the law of the jets at 0 given $U_0 = v_0(b, k)$, $\mathbb{E}_0[\,\cdot \mid v_0(b, k)]$ its expectation, and $\mathbb{E}_0[\,\cdot \mid b] := \mathbb{E}_0[\,\cdot \mid v_0(b, 0)]$.

*Weights and kernels.* $F_j(H)$ is $|\det H|$ if the symmetric matrix $H$ has exactly $j$ negative eigenvalues and 0 otherwise. With $H_i = D^2 f(i)$, $i \in \{M, S\}$, and $D_r = \mathrm{diag}(r^{1/2}, I_m)$ put $K_i = D_r^{-1} H_i D_r^{-1}$, so $|\det H_i|/r = |\det K_i|$, and $W_r = |\det H_M \det H_S| \mathbf{1}\{H_M \prec 0, \ \mathrm{ind}\, H_S = m\}$, so $W_r/r^2 = F_d(K_M) F_{d-1}(K_S)$. The elder mark is $e = \mathbf{1}\{d_f(M) = f(S)\}$ with the maximin level $d_f(M) = \sup\{\min_t f(\gamma(t)) : \gamma(0) = M, \ f(\gamma(1)) > f(M)\}$ ($\sup \varnothing = -\infty$); it is Borel and on the Morse locus it is the elder pairing [7] (Lemma S12.20). The kernels, functions of $(b, k, u)$, are

$$\mathcal{A}_r = 12\pi_r(v_r)\mathbb{E}_Q[W_r/r^2], \quad \mathcal{A}_r^{\mathrm{eld}} = 12\pi_r(v_r)\mathbb{E}_Q[(W_r/r^2)e], \quad \mathcal{A}_r^{\mathrm{rej}} = \mathcal{A}_r - \mathcal{A}_r^{\mathrm{eld}}, \quad \mathcal{A}_0 = 12\pi_0(v_0)z_0, \tag{S10.3}$$

with $z_0 = 36k^2 \mathbb{E}_0[(\det A)^2 \mathbf{1}\{A \prec 0\} \mid v_0(b, k)]$. For a $C^5$ function $g$ the jets at 0 in the frame $(u, \Theta)$ are $f_4 = \partial_u^4 g(0)$, $\gamma = \nabla_\Theta \partial_u^2 g(0) \in \mathbb{R}^m$, $A = D_\Theta^2 g(0)$, $\Delta = \det A$, $B = \partial_u D_\Theta^2 g(0)$, and

$$Y = \tfrac{f_4}{12}\Delta - \tfrac{1}{4}\gamma^\top \mathrm{adj}(A)\gamma, \qquad \varphi = \frac{Y}{6\kappa\Delta} = \frac{f_4 - 3\gamma^\top A^{-1}\gamma}{72\kappa} \quad (A \text{ invertible}). \tag{S10.4}$$

Under $\mathbb{E}_0$ these are the jets of the field under the contact law.

*Cusp kernels.* For $\kappa > 0$ put $w_\kappa(A, Y) = (36\kappa^2(\det A)^2 - Y^2)_+ \mathbf{1}\{A \prec 0\} = (36\kappa^2\Delta^2 - Y^2)\mathbf{1}\{A \prec 0, |Y| < 6\kappa|\Delta|\}$,

$$\begin{aligned} w_\kappa^{\mathrm{eld}}(A, Y) &= (36\kappa^2\Delta^2 - Y^2)\mathbf{1}\{A \prec 0, \ |Y| < 2\kappa|\Delta|\}, \\ \mathrm{loss}_\kappa(A, Y) &= 36\kappa^2\Delta^2 \mathbf{1}\{A \prec 0\} - w_\kappa^{\mathrm{eld}}(A, Y) \\ &= \big[Y^2 \mathbf{1}\{|Y| < 2\kappa|\Delta|\} + 36\kappa^2\Delta^2 \mathbf{1}\{|Y| \geq 2\kappa|\Delta|\}\big]\mathbf{1}\{A \prec 0\}. \end{aligned} \tag{S10.5}$$

The cusp kernels of §S9 are $\mathcal{C}^\bullet(b,\kappa,u) = 12\pi_0(u;v_0(b,0))\mathbb{E}_0[w^\bullet \mid b]$ with $w^{\mathrm{cand}} = w_\kappa$, $w^{\mathrm{eld}} = w_\kappa^{\mathrm{eld}}$, $w^{\mathrm{con}} = 36\kappa^2\Delta^2\mathbf{1}\{A \prec 0\}$, so

$$\mathcal{C}^{\mathrm{cand}} - \mathcal{C}^{\mathrm{con}} = -12\pi_0\mathbb{E}_0[\min(Y^2, 36\kappa^2\Delta^2)\mathbf{1}\{A \prec 0\} \mid b], \qquad \mathcal{C}^{\mathrm{eld}} - \mathcal{C}^{\mathrm{con}} = -12\pi_0\mathbb{E}_0[\mathrm{loss}_\kappa \mid b] \le 0, \tag{S10.6}$$

and the cusp coefficients of §S9 are

$$\begin{aligned} c_1 &= \int_{S^{d-1}}\int_{\mathbb{R}}\int_0^\infty (\mathcal{C}^{\mathrm{eld}} - \mathcal{C}^{\mathrm{con}})(b, s^{-4}, u)\, ds\, db\, d\sigma(u) < 0, \\ I^{\mathrm{cand}} &= \int_{S^{d-1}}\int_{\mathbb{R}}\int_0^\infty (\mathcal{C}^{\mathrm{cand}} - \mathcal{C}^{\mathrm{con}})(b, s^{-4}, u)\, ds\, db\, d\sigma(u) = \tfrac{3^{1/4}}{2} c_1; \end{aligned} \tag{S10.7}$$

for $d = 3$, $L = 24$, evaluating the $s$-integral (§S9) turns $c_1$ into (2.13) of the main text.

*Two-site kernels.* For $y \neq 0$ let $O_y = (f(0), \nabla f(0), f(y), \nabla f(y))$, $v_{b,\ell} = (b, 0, b-\ell, 0)$, $p_y$ the density of $O_y$, $Q_{y,b,\ell}$ the regression law given $O_y = v_{b,\ell}$, $W = |\det H_0 \det H_y|\mathbf{1}\{H_0 \prec 0,\ \mathrm{ind}\, H_y = m\}$, and $e_y$ the elder mark of $(0,y)$. Put $\Psi_\ell^{\mathrm{cand}}(b,y) = p_y(v_{b,\ell})\mathbb{E}_{Q_{y,b,\ell}}[W]$, $\Psi_0 = \Psi_0^{\mathrm{cand}}$, the equal-height mass $B_{d,L} = \int_{\mathbb{T}_L^d\setminus\{0\}}\int_{\mathbb{R}} \Psi_0\, db\, dy \in (0,\infty)$, and

$$\nu_{\mathrm{cand}}^{\mathrm{far},\rho}(\ell) = \int_{\mathrm{dist}(0,y)\ge\rho}\int_{\mathbb{R}} \Psi_\ell^{\mathrm{cand}}\, db\, dy, \qquad \nu_{\mathrm{eld}}^{\mathrm{far},\rho}(\ell) = \int_{\mathrm{dist}(0,y)\ge\rho}\int_{\mathbb{R}} p_y(v_{b,\ell})\, \mathbb{E}_{Q_{y,b,\ell}}[W e_y]\, db\, dy. \tag{S10.8}$$

S10.2. **Inputs.** *From §S7.* With $r_R \in (0, \frac12]$ of Proposition S8.2, for $0 < r \le r_R$: the pinned and contact laws are realized on one probability space by $F_r \sim Q_{r,b,k}$ and $F_0$ (whose jets at 0 have the contact law at $v_0(b,k)$), affine images of one copy of the field with coefficients bounded uniformly in $r$ and the frame; $\Sigma_r = \mathrm{Cov}(U_r)$ is uniformly positive definite; and for $p \ge 1$, $q \ge 0$,

$$\|F_r - F_0\|_{L^p(C^q)} \le C_{p,q} r^2 P, \qquad \big\|1 + \|F_r\|_{C^q} + \|F_0\|_{C^q}\big\|_{L^p} \le C_{p,q} P, \tag{S10.9}$$

$$\pi_r(v_r) \le Ce^{-c(b^2+k^2)}, \qquad |\pi_r(v_r) - \pi_0(v_0(b,k))| \le Cr^2P^2e^{-c(b^2+k^2)}. \tag{S10.10}$$

In $K_i = \begin{pmatrix} \alpha_i & r^{1/2}\beta_i^\top \\ r^{1/2}\beta_i & A_i \end{pmatrix}$, $\alpha_i = \partial_x^2 F_r(i)/r$, $\beta_i = \nabla_y\partial_x F_r(i)/r$, $A_i = D_y^2 F_r(i)$, one has $\det K_i = \alpha_i \det A_i - r\beta_i^\top \mathrm{adj}(A_i)\beta_i$. We use the dominating variable (it dominates the one of §S8)

$$T := 1 + k + \|F_r\|_{C^9} + \|F_0\|_{C^9} + r^{-2}\|F_r - F_0\|_{C^4}, \qquad \|T\|_{L^p} \le C_p P; \tag{S10.11}$$

with $A_0 := D_y^2 F_0(0)$, $\|A_i - A_0\| \le (\frac{r}{2} + r^2)T \le rT$ for $r \le \frac12$, $F_j(K_i) \le C(k+r)T^d$, and $0 \le \mathcal{A}_0 \le k^2 H(b,k)$, $H(b,k) := CP^N e^{-c(b^2+k^2)}$.

*From §S8.* The admissible radius satisfies $0 < r_0^* \le r_R \le \frac12$, $r_0^* < L/(4\sqrt{2})$ and $r_0^* \le r_K$, where $r_K > 0$ is the radius of the gap-mark-explicit weight bound for the cap region $G_r$; as $1 - e \le \mathbf{1}_{G_r^c}$ $Q$-almost surely on $\{W_r > 0\}$ for $r < L/(4\sqrt{2})$, that bound gives, for $0 < r \le \min\{k, r_0^*\}$,

$$\mathbb{E}_Q[(W_r/r^2)(1-e)] \le C(r^3/k)P^N. \tag{S10.12}$$

For every $r_0 \in (0, r_0^*]$, with inner integrals $d\sigma(u)\, db$ over $S^{d-1}\times\mathbb{R}$,

$$\begin{aligned} \nu_{\mathrm{cand}}(\ell) &= \int_0^{r_0}\!\!\iint r^{-2}\mathcal{A}_r(b, \tfrac{\ell}{r^3}, u)\, dr + \nu_{\mathrm{cand}}^{\mathrm{far},r_0}(\ell), \qquad c_{d,L}\ell^{-1/3} = \int_0^\infty\!\!\iint r^{-2}\mathcal{A}_0(b, \tfrac{\ell}{r^3}, u)\, dr, \\ \nu_{\mathrm{eld}}(\ell) &= \int_0^{r_0}\!\!\iint r^{-2}\mathcal{A}_r^{\mathrm{eld}}(b, \tfrac{\ell}{r^3}, u)\, dr + \nu_{\mathrm{eld}}^{\mathrm{far},r_0}(\ell), \end{aligned} \tag{S10.13}$$

and $r^{-2}\mathcal{A}_r(b,0,u) = r^{d-1}\Psi_0(b, ru)$, so that

$$B_{d,L} = \int_0^{r_0}\!\!\iint r^{-2}\mathcal{A}_r(b,0,u)\, dr + \int_{\mathrm{dist}(0,y)\ge r_0}\int_{\mathbb{R}} \Psi_0\, db\, dy. \tag{S10.14}$$

With $f_4, \gamma, B$ the jets of $F_r$, $\Delta_0 = \det A_0$ and $\widehat{Y} = \frac{f_4}{12}\Delta_0 + 3k\,\mathrm{tr}(\mathrm{adj}(A_0)B) - \frac{1}{4}\gamma^\top \mathrm{adj}(A_0)\gamma$, pathwise for $0 < r \le r_R$,

$$\det K_M = -6k\Delta_0 + r\widehat{Y} + R_M, \quad \det K_S = 6k\Delta_0 + r\widehat{Y} + R_S, \quad |R_M| + |R_S| \le Cr^2(1+k)T^{N_1}. \tag{S10.15}$$

Its proof is a pathwise Taylor expansion of $F_r$ about its own jets at 0, with remainders polynomial in $\|F_r\|_{C^7}$ and $k$ for $r \le 1$; the coupling enters only through $\|D_y^2 F_r(0) - A_0\| \le r^2 T$. So, with $A = D_y^2 F_r(0)$, $\Delta = \det A$, $\mathcal{N}_r := 1 + k + \|F_r\|_{C^9}$ and $Y_r := \frac{f_4}{12}\Delta + 3k\,\mathrm{tr}(\mathrm{adj}(A)B) - \frac{1}{4}\gamma^\top \mathrm{adj}(A)\gamma$, pathwise for all $0 < r \le r_0^*$, $k \ge 0$,

$$\det K_M = -6k\Delta + rY_r + R'_M, \quad \det K_S = 6k\Delta + rY_r + R'_S, \quad |R'_M| + |R'_S| \le Cr^2(1+k)\mathcal{N}_r^{N_1}. \tag{S10.16}$$

*Sign window:* on $\{W_r > 0\}$, $\det K_M \det K_S < 0$; so if $\det K_M = w - a + R_M$, $\det K_S = w + a + R_S$, then $|w| \le |a| + \max_i |R_i|$ and $|\det K_i| \le 2|a| + 2\max_j |R_j|$. For $0 < r \le r_R$, $k \ge 0$, and $0 < \rho \le r_0^*$,

$$\mathbb{E}_Q[W_r/r^2] \le C(k^2 + r^4(1+k)^2)P^N, \qquad \int_{0<\mathrm{dist}(0,y)<\rho} \int_{\mathbb{R}} \Psi_0(b,y)\,db\,dy \le C\rho^3. \tag{S10.17}$$

*Barrier:* if $g \in C^3(\mathbb{T}_L^d)$, $\nabla g(M) = 0$, $D^2 g(M) \prec 0$, then $g(M) - d_g(M) \ge c_L \lambda^3 / K_g^2$, with $\lambda = \lambda_{\min}(-D^2 g(M))$, $K_g = 1 + \sup\|D^2 g\| + \sup\|D^3 g\|$, $c_L = a_L^2/3$, $a_L = \min(1, L/4)$. So on $\{W_r > 0\} \cap \{e = 1\}$,

$$\lambda_{\min}(-H_M) \le (\ell K_f^2 / c_L)^{1/3}. \tag{S10.18}$$

For every fixed $r_0 \in (0, L/2]$ there is $C = C(d, L, r_0)$ with

$$\nu_{\mathrm{eld}}^{\mathrm{far}, r_0}(\ell) \le C\ell^{1/3}, \qquad 0 < \ell \le 1. \tag{S10.19}$$

*From §S9.* For $0 < r \le r_0^*$, $0 \le k \le 1$, and for $r_0 \in (0, r_0^*]$, $\mathrm{dist}(0,y) \ge r_0$, $0 \le \ell \le 1$,

$$\begin{aligned} |\mathcal{A}_r(b,k,u) - \mathcal{A}_r(b,0,u)| &\le C(k^2 + kr^2)(1+|b|)^N e^{-cb^2}, \\ |\Psi_\ell^{\mathrm{cand}}(b,y) - \Psi_0(b,y)| &\le C_{r_0}\ell(1+|b|)^N e^{-cb^2}. \end{aligned} \tag{S10.20}$$

The cusp polynomial, its fibre reduction and the values of its ridge are recalled in §S10.7.

*The far elder rate.* By Proposition S8.13, for every $\rho \in (0, L/2]$ there is $C = C(d, L, \rho)$ with

$$\nu_{\mathrm{eld}}^{\mathrm{far}, \rho}(\ell) \le C\ell^{2/3}, \qquad 0 < \ell \le 1. \tag{S10.21}$$

*From [7] (Lemmas S12.8 and S12.30).* Any finite list of distinct derivative evaluation functionals at distinct sites is linearly independent, and so is any finite list of distinct partial derivatives at one site in a rotated frame; odd and even derivatives at a point are independent.

S10.3. **Results.** Let $\mathcal{A}_2$ be the fold coefficient of Lemma S10.12 ($\mathcal{A}_r = \mathcal{A}_0 + r^2\mathcal{A}_2 + O(r^3(1+k^{-1}))$ at fixed $(b,k,u)$, $\mathcal{A}_2(b,\cdot,u) \in C^1[0,\infty)$). The fold-scale coefficient, the constant $c_2$ evaluated in §S11, is the Hadamard finite part at $k = 0$ of $\frac{1}{3}\iiint \mathcal{A}_2 k^{-4/3}$:

$$c_2 := \frac{1}{3}\int_{S^{d-1}} \int_{\mathbb{R}} \int_0^\infty [\mathcal{A}_2(b,k,u) - \mathcal{A}_2(b,0,u)]\,k^{-4/3}\,dk\,db\,d\sigma(u). \tag{S10.22}$$

**Theorem S10.1** (the candidate density to third order)**.** *For every $d \ge 2$ and $L > 0$, and for the canonical marked Kac–Rice density version of $\nu_{\mathrm{cand}}$, as $\ell \downarrow 0$,*

$$\nu_{\mathrm{cand}}(\ell) = c_{d,L}\ell^{-1/3} + B_{d,L} + I^{\mathrm{cand}}\ell^{1/4} + c_2\ell^{1/3} + O(\ell^{4/11}), \tag{S10.23}$$

*and (S10.22) converges absolutely. In particular $\nu_{\mathrm{cand}}(\ell) - c_{d,L}\ell^{-1/3} - B_{d,L} - I^{\mathrm{cand}}\ell^{1/4} = O(\ell^{1/3})$.*

**Theorem S10.2** (the elder density to third order)**.** *For every $d \geq 2$ and $L > 0$, and for the canonical marked Kac–Rice density version of $\nu_{\mathrm{eld}}$, as $\ell \downarrow 0$,*

$$\nu_{\mathrm{eld}}(\ell) = c_{d,L}\ell^{-1/3} + c_1\ell^{1/4} + c_2\ell^{1/3} + \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) + O(\ell^{4/11}), \qquad 0 \leq \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) \leq C\ell^{1/3}. \tag{S10.24}$$

*Consequently*

$$\nu_{\mathrm{eld}}(\ell) = c_{d,L}\ell^{-1/3} + c_1\ell^{1/4} + O(\ell^{1/3}), \tag{S10.25}$$

$$\nu_{\mathrm{eld}}(\ell) \geq c_{d,L}\ell^{-1/3} + c_1\ell^{1/4} + c_2\ell^{1/3} - C\ell^{4/11}, \tag{S10.26}$$

*and, by* (S10.21) *at $\rho = r_0^*$,*

$$\nu_{\mathrm{eld}}(\ell) = c_{d,L}\ell^{-1/3} + c_1\ell^{1/4} + c_2\ell^{1/3} + O(\ell^{4/11}). \tag{S10.27}$$

*More precisely, for $0 < \theta \leq 4/11$, $\nu_{\mathrm{eld}}(\ell) - c_{d,L}\ell^{-1/3} - c_1\ell^{1/4} - c_2\ell^{1/3} = O(\ell^{\theta})$ if and only if $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) = O(\ell^{\theta})$ (both hold for $\theta \leq 1/3$), and the three-term law with remainder $o(\ell^{1/3})$ holds if and only if $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) = o(\ell^{1/3})$. Only* (S10.27) *uses* (S10.21).

**Corollary S10.3** (the rejected density)**.** *For every $d \geq 2$ and $L > 0$, and for the canonical marked Kac–Rice density versions, as $\ell \downarrow 0$,*

$$\rho_{\mathrm{rej}}(\ell) = B_{d,L} + (I^{\mathrm{cand}} - c_1)\ell^{1/4} - \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) + O(\ell^{4/11}). \tag{S10.28}$$

*In particular: (1) $\rho_{\mathrm{rej}}(\ell) = B_{d,L} + (I^{\mathrm{cand}} - c_1)\ell^{1/4} + O(\ell^{1/3})$; (2) $\rho_{\mathrm{rej}}(\ell) \leq B_{d,L} + (I^{\mathrm{cand}} - c_1)\ell^{1/4} + C\ell^{4/11}$; (3) by* (S10.21)*, $\rho_{\mathrm{rej}}(\ell) = B_{d,L} + (I^{\mathrm{cand}} - c_1)\ell^{1/4} + O(\ell^{4/11})$.*

**Theorem S10.4** (the candidate density, remainder $\ell^{3/7}$)**.** *For every $d \geq 2$ and $L > 0$, and for the canonical marked Kac–Rice density version of $\nu_{\mathrm{cand}}$, as $\ell \downarrow 0$,*

$$\nu_{\mathrm{cand}}(\ell) = c_{d,L}\ell^{-1/3} + B_{d,L} + I^{\mathrm{cand}}\ell^{1/4} + c_2\ell^{1/3} + O(\ell^{3/7}). \tag{S10.29}$$

**Theorem S10.5** (the elder density, remainder $\ell^{3/7}$)**.** *For every $d \geq 2$ and $L > 0$, and for the canonical marked Kac–Rice density version of $\nu_{\mathrm{eld}}$, as $\ell \downarrow 0$,*

$$\nu_{\mathrm{eld}}(\ell) = c_{d,L}\ell^{-1/3} + c_1\ell^{1/4} + c_2\ell^{1/3} + \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) + O(\ell^{3/7}), \tag{S10.30}$$

*and, by* (S10.21) *at $\rho = r_0^*$,*

$$\nu_{\mathrm{eld}}(\ell) = c_{d,L}\ell^{-1/3} + c_1\ell^{1/4} + c_2\ell^{1/3} + O(\ell^{3/7}). \tag{S10.31}$$

*The elder density from the intermediate separations (the part of the integral in* (S10.13) *with $\ell^{2/9} \leq r \leq r_0^*$, at $r_0 = r_0^*$) is $O(\ell^{4/9}\log(1/\ell))$. For $0 < \theta \leq 3/7$, $\nu_{\mathrm{eld}}(\ell) - c_{d,L}\ell^{-1/3} - c_1\ell^{1/4} - c_2\ell^{1/3} = O(\ell^{\theta})$ if and only if $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) = O(\ell^{\theta})$. Only* (S10.31) *uses* (S10.21).

**Corollary S10.6** (the rejected density, remainder $\ell^{3/7}$)**.** *For every $d \geq 2$ and $L > 0$, and for the canonical marked Kac–Rice density versions, as $\ell \downarrow 0$, with the second form by* (S10.21)*,*

$$\rho_{\mathrm{rej}}(\ell) = B_{d,L} + (I^{\mathrm{cand}} - c_1)\ell^{1/4} - \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) + O(\ell^{3/7}) = B_{d,L} + (I^{\mathrm{cand}} - c_1)\ell^{1/4} + O(\ell^{3/7}). \tag{S10.32}$$

**Proposition S10.7** (the typed window)**.** *For $0 < r \leq r_0^*$, $b \in \mathbb{R}$, $u \in S^{d-1}$ and $0 \leq k \leq r$ (so $\kappa \leq 1$),*

$$\mathbb{E}_Q[W_r/r^2] \leq Cr^2(\kappa + r)^2(\kappa + r\log(2/r))P^N, \tag{S10.33}$$

$$r^{-2}\mathcal{A}_r(b,k,u) \leq C(\kappa + r)^2(\kappa + r\log(2/r))(1 + |b|)^N e^{-cb^2}. \tag{S10.34}$$

*At $k = 0$, $r^{d-1}\Psi_0(b, ru) \leq Cr^3\log(2/r)(1 + |b|)^N e^{-cb^2}$, and*

$$\int_{0<\mathrm{dist}(0,y)<\rho}\int_{\mathbb{R}} \Psi_0(b,y)\,db\,dy \leq C\rho^4\log(2/\rho), \qquad 0 < \rho \leq r_0^*. \tag{S10.35}$$

*Remark* S10.8 (the case $d=3$, $L=24$). For $d=3$, $L=24$, (S10.31) is (2.12) of the main text, with $c_1$ of (2.13) there and $c_2$ of (S10.22); dividing by $c_{3,24}\ell^{-1/3}$ gives (2.14) there, with relative remainder $O(\ell^{3/7+1/3})=O(\ell^{16/21})$. As $I^{\mathrm{cand}}-c_1=(1-3^{1/4}/2)|c_1|>0$, (S10.32) is (2.18) of the main text. Theorems S10.4 and S10.5 are proved by modifying the proofs of Theorems S10.1 and S10.2.

S10.4. **Gaussian and elementary facts.**

**Lemma S10.9** (eigenvalue and determinant layers)**.** *Let $G$ be a random element of $\mathrm{Sym}(m)\cong\mathbb{R}^{m(m+1)/2}$ whose law has a density bounded by $C_D\exp(-c_D|G-\mu|^2)$. For every $n\ge 0$ there is $C=C(m,n,C_D,c_D)$, independent of $\mu$, such that for $\varepsilon>0$, with $n':=n+m(m-1)/2$,*

$$\mathbb{E}[(1+\|G\|)^n\mathbf{1}\{|\lambda_{\max}(G)|<\varepsilon\}]\le C(1+|\mu|)^{n'}\varepsilon,\qquad \mathbb{E}[(1+\|G\|)^n\mathbf{1}\{|\det G|<\varepsilon\}]\le C(1+|\mu|)^{n'}\varepsilon. \tag{S10.36}$$

*Proof.* In eigenvalue coordinates $G=O\,\mathrm{diag}(\lambda)O^\top$, Lebesgue measure is $c_m|V(\lambda)|\,d\lambda\,dO$, $V(\lambda)=\prod_{i<j}(\lambda_i-\lambda_j)$, $dO$ Haar measure (Weyl's integration formula). First bound: integrate $\lambda_{\max}$ over $(-\varepsilon,\varepsilon)$; the remaining integrand $(1+\max|\lambda_i|)^n|V|C_De^{-c_D|\mathrm{diag}(\lambda)-O^\top\mu O|^2}$ has integral over the other eigenvalues and $O$ at most $C(1+|\mu|)^{n'}$ uniformly in $\lambda_{\max}$, as the Vandermonde factor costs at most $(1+\max|\lambda_i|)^{m(m-1)/2}$, absorbed by the Gaussian at the price of the power of $1+|\mu|$. Second bound: on $\{|\det G|<\varepsilon\}$ let $i^*$ index an eigenvalue of least modulus and $D:=\prod_{j\ne i^*}|\lambda_j|>0$. Then $|\lambda_{i^*}|<\varepsilon/D$ and $|\lambda_{i^*}-\lambda_j|\le 2|\lambda_j|$, so $|V|\le 2^{m-1}D|V'|$, $V'$ the Vandermonde product of the other eigenvalues. Integrating $\lambda_{i^*}$ over an interval of length $2\varepsilon/D$ against a density $\le C_D$ gives $\le 2^mC_D\varepsilon|V'|$; $D$ cancels, and the remaining integral of $(1+\max|\lambda_j|)^n|V'|$ against the Gaussian bound in the other eigenvalues is $\le C(1+|\mu|)^{n'}$. Sum over $i^*$. For $m=1$ both bounds are the bounded density of a Gaussian variable. □

**Lemma S10.10** (conditional moments)**.** *If $J_0$ is a finite vector of derivatives of $F_r$ at $0$ whose covariance under $Q$ is invertible uniformly for $r\le r_0^*$, then $\mathbb{E}[T^p\mid J_0]\le C_p(P+|J_0|)^p$ for $p\ge 1$, with $T$ of (S10.11).*

*Proof.* $F_r$, $F_0$, $(F_r-F_0)/r^2$ are affine images of one Gaussian field $F$ with coefficients bounded uniformly in $r$. Given $J_0$, $F$ is Gaussian; the regression coefficients are bounded uniformly in $r$, so the conditional mean has $C^q$ norms $\le C(P+|J_0|)$; the conditional covariance is at most $\mathrm{Cov}\,F$, so by Anderson's inequality the conditional $C^q$-norm moments of the centred part are dominated by those of $F$. (The hypothesis is needed: for a pinned jet such as $\partial_u^2F_r(0)$, of $Q$-variance $O(r^4)$, conditioning on an $O(1)$ value forces $f_4\asymp r^{-2}$.) □

**Lemma S10.11** (a $\kappa$-free local Lipschitz bound)**.** *For $a\ge 0$ and $y,y'\in\mathbb{R}$,*

$$|(a^2-y^2)_+-(a^2-y'^2)_+|\le|y-y'|(|y|+|y'|). \tag{S10.37}$$

*Consequently, for every symmetric $A$ and $\kappa>0$, $|w_\kappa(A,Y)-w_\kappa(A,Y')|\le|Y-Y'|(|Y|+|Y'|)$.*

*Proof.* Let $|y|\le|y'|$. If $|y'|\le a$, the left side is $|y^2-y'^2|\le|y-y'|(|y|+|y'|)$; if $|y|\le a<|y'|$, it is $a^2-y^2\le y'^2-y^2\le|y'-y|(|y|+|y'|)$; if $a<|y|$, both terms vanish. For $w_\kappa$ take $a=6\kappa|\det A|$ on $\{A\prec 0\}$; off $\{A\prec 0\}$ both weights vanish. □

S10.5. **The fold expansion.**

**Lemma S10.12** (fold expansion)**.** *There are $C,c,N$ and a function $\mathcal{A}_2(b,k,u)$ such that for $0<r\le\min(r_0^*,\frac12)$, $k>0$, $b\in\mathbb{R}$, $u\in S^{d-1}$,*

$$|\mathcal{A}_r(b,k,u)-\mathcal{A}_0(b,k,u)-r^2\mathcal{A}_2(b,k,u)|\le Cr^3(1+k^{-1})P^Ne^{-c(b^2+k^2)}. \tag{S10.38}$$

*Moreover $\mathcal{A}_2(b,\cdot,u)$ extends to a $C^1$ function on $[0,\infty)$ with*

$$|\mathcal{A}_2|+|\partial_k\mathcal{A}_2|\le CP^Ne^{-c(b^2+k^2)},\qquad \mathcal{A}_2(b,0,u)=-12\pi_0(u;v_0(b,0))\,\mathbb{E}_0[Y^2\mathbf{1}\{A\prec 0\}\mid b]. \tag{S10.39}$$

*Proof.* Let $f = F_r$ and $A = D_y^2 f(0)$. *Step 1 (free jets and the product polynomial).* The *free* jets $J'$ are the partial derivatives of $f$ at 0 of order $\le N_0 := 9$ in the frame $(u, \Theta)$ other than the pinned jets ($A$, $f_4$ are free). For $r \in (0, r_0^*]$, $(J', U_r)$ is an invertible image of distinct derivative functionals at the distinct sites $0, \pm\frac{r}{2}u$, and at $r = 0$, $(J', U_0)$ is a list of distinct jets at one point; both are linearly independent [7] (Lemma S12.8). The Gram matrix of $(J', U_r)$ is continuous on the compact $[0, r_0^*] \times \{\text{frames}\}$, hence uniformly invertible, and so is its Schur complement $\mathrm{Cov}_Q(J')$. Each pin row is a centred difference rule; Taylor's formula with integral remainder writes it as the pinned jet plus a polynomial in $r$ and higher jets, with error $\le Cr^{N_0-3}T$ (e.g. $(\partial_x f(S) - \partial_x f(M))/r = \partial_u^2 f(0) + \frac{r^2}{24}\partial_u^4 f(0) + \cdots$). Under $Q$ the rows equal their targets, so each pinned jet is a polynomial in $(J', r, b, k)$ up to $O(r^{N_0-3}T)$ (the $1/r$, $1/r^2$ of the rows are absorbed as the rules are centred). Expanding the entries of $K_M, K_S$ at $x = \mp r/2$ to the same order gives, for $r \le 1$, a polynomial $\Omega(J'; r, k)$ with

$$-\det K_M \det K_S = \Omega(J'; r, k) + \varrho, \qquad |\varrho| \le Cr^4 T^{n_0} \tag{S10.40}$$

(the explicit $1/r$ in $\alpha_i$ disappears once $\partial_u^2 f(0)$ is replaced by its pin expansion $-\frac{r^2}{24}f_4 - \cdots$). Write $\Omega = \Omega_0 + r\Omega_1 + r^2\Omega_2 + r^3\Omega_3(J'; r, k)$ with $|\Omega_3| \le C(1+k)^2(1+|J'|)^{n_0}$. By (S10.16) (the same $Y_r$ at both ends),

$$\Omega_0 = 36k^2\Delta^2, \qquad \Omega_1 = -[(-6k\Delta)Y_r + Y_r(6k\Delta)] = 0, \qquad \Omega_2|_{k=0} = -Y_r^2|_{k=0}, \tag{S10.41}$$

the last because at $k = 0$ both determinants are $rY_r + O(r^2)$. These are identities of polynomials in $J'$: for pinned polynomial fields of degree $\le N_0 - 1$ with arbitrary free jets, $\varrho = 0$ and $T < \infty$, so the coefficients of $r^0, r^1$ and (at $k = 0$) $r^2$ in $\Omega - 36k^2\Delta^2$ and $\Omega + r^2 Y_r^2$ vanish identically.

*Step 2 (the typed indicator and its layers).* Let $\upsilon := C_\star rT(1 + T/k)$ and $\mathcal{G} := \{\lambda_{\max}(A) \le -\upsilon\} \cap \{k \ge C_\star rT\}$, $C_\star = C_\star(d) \ge 2$ large: the good event of §S8 with $A$ for $A_0$ (its proof uses only $\|A_i - A\| \le \frac{r}{2}\|F_r\|_{C^3} \le rT$ and $\|\beta_i\| \le CT$). On $\mathcal{G}$, by Haynsworth's inertia additivity, $K_M, K_S$ have indices $d, d-1$ and $A \prec 0$, so $F_d(K_M)F_{d-1}(K_S) = -\det K_M \det K_S = \Omega \mathbf{1}\{A \prec 0\} + \varrho$. On $\{\lambda_{\max}(A) \ge rT\}$, $\lambda_{\max}(A_M) \ge rT - \frac{r}{2}T > 0$, so $F_d(K_M) = 0 = \mathbf{1}\{A \prec 0\}$. The rest of $\mathcal{G}^c$ lies in $\{|\lambda_{\max}(A)| < \upsilon\} \cup \{k < C_\star rT\}$, where the three cases of the good-event argument bound $F_d(K_M)F_{d-1}(K_S)$ and $|\Omega|\mathbf{1}\{A \prec 0\}$ (via (S10.40)) by $\omega(T) := Cr^2(k+T)^2T^{N_2}$. Hence $|\mathbb{E}_Q[F_d(K_M)F_{d-1}(K_S)] - \mathbb{E}_Q[\Omega \mathbf{1}\{A \prec 0\}]| \le Cr^4P^N + \mathbb{E}[\omega(T)(\mathbf{1}\{k < C_\star rT\} + \mathbf{1}\{|\lambda_{\max}(A)| < \upsilon\})]$. First, $\mathbb{E}[\omega(T)\mathbf{1}\{T > k/(C_\star r)\}] \le (C_\star r/k)\mathbb{E}[\omega(T)T] \le Cr^3k^{-1}P^N$. Second, split by $T \in [2^j, 2^{j+1})$, with $\upsilon_j := C_\star r 2^{j+1}(1 + 2^{j+1}/k)$, and use Lemma S10.10 with $J_0 = A$:

$$\mathbb{E}[\omega(T)\mathbf{1}\{|\lambda_{\max}(A)| < \upsilon\}] \le \sum_{j \ge 0} \omega(2^{j+1})\, C_p 2^{-jp}\, \mathbb{E}[(P + \|A\|)^p \mathbf{1}\{|\lambda_{\max}(A)| < \upsilon_j\}].$$

Under $Q$, $A$ is Gaussian with mean $O(P)$ and uniformly nondegenerate covariance (Step 1), so by Lemma S10.9 the expectation is $\le CP^{2p+m(m-1)/2}\upsilon_j$, and for $p > N_2 + 4$ the sum is $\le Cr^3(1 + k^{-1})P^N$. Thus

$$\mathbb{E}_Q[F_d(K_M)F_{d-1}(K_S)] = \mathbb{E}_Q[\Omega(J'; r, k)\mathbf{1}\{A \prec 0\}] + O(r^3(1 + k^{-1})P^N). \tag{S10.42}$$

*Step 3 (smoothness of the pinned law in $r$).* Under $Q$, $J' \sim N(\mu_r, V_r)$, built from $\mathrm{Cov}(J', U_r)$, $\mathrm{Cov}(U_r)$ and $v_r$. The rows of $U_r$ are centred, hence even in $r$, so the covariances are smooth even functions of $r$ and $V_r$ is uniformly nondegenerate. As $v_r = v_0 + r^2(0, -k, 0, \dots) + r^3(-k/2, 0, \dots)$, $\mu_r = \mu_0 + r^2\mu_2 + r^3\mu_3 + O(r^4 P)$ with $|\mu_3| \le Ck$, and $V_r = V_0 + r^2 V_2 + O(r^4)$. For polynomial $\Psi$, $\mathbb{E}_{N(\mu,V)}[\Psi(J')\mathbf{1}\{A \prec 0\}]$ is smooth in $(\mu, V)$ (differentiate the Gaussian density), with derivatives bounded by Gaussian moments. So $\mathbb{E}_Q[\Omega_0 \mathbf{1}\{A \prec 0\}] = 36k^2(e_0 + r^2 e_2 + O(r^3(1+k)P^N))$ with $e_0 = \mathbb{E}_0[\Delta^2 \mathbf{1}\{A \prec 0\} \mid v_0(b,k)]$, $\mathbb{E}_Q[\Omega_2 \mathbf{1}\{A \prec 0\}] = \mathbb{E}_0[\Omega_2 \mathbf{1}\{A \prec 0\} \mid v_0(b,k)] + O(r^2 P^N)$, and the $r^3$-term contributes $O(r^3(1+k)^2P^N)$ (at $r = 0$ the free jets have the contact law at $v_0(b,k)$).

Likewise $\pi_r(v_r) = \pi_0(v_0) + r^2\pi_2 + O(r^3(1+k)P^N e^{-c(b^2+k^2)})$, the density of $U_r$ being smooth in $(r^2, v)$. Multiplying, with (S10.42) and $\mathcal{A}_0 = 12\pi_0(v_0)36k^2e_0$, gives (S10.38) with

$$\mathcal{A}_2 := 12\big[\pi_2 \cdot 36k^2e_0 + \pi_0(v_0(b,k))\big(36k^2e_2 + \mathbb{E}_0[\Omega_2\mathbf{1}\{A \prec 0\} \mid v_0(b,k)]\big)\big]. \tag{S10.43}$$

Each ingredient is a polynomial in $k$, or a Gaussian expectation whose target is affine in $k$, times $\pi_0(v_0(b,k)) \le Ce^{-c(b^2+k^2)}$; this gives the bounds in (S10.39) and the $C^1$ extension. At $k=0$ the first two terms vanish, and by (S10.41) the third is $-12\pi_0(v_0(b,0))\mathbb{E}_0[Y^2\mathbf{1}\{A \prec 0\} \mid b]$, since under the contact law $Y_r|_{k=0} = Y$. □

S10.6. **The candidate cusp kernel and the overlap.**

**Lemma S10.13** (the candidate cusp kernel with a rate)**.** *There are $C, c, N$ such that for $0 < r \le \min(r_0^*, \frac{1}{2})$, $b \in \mathbb{R}$, $u \in S^{d-1}$ and $\kappa > 0$ with $k := \kappa r \le 1$,*

$$|r^{-2}(\mathcal{A}_r - \mathcal{A}_0)(b, \kappa r, u) - (\mathcal{C}^{\mathrm{cand}} - \mathcal{C}^{\mathrm{con}})(b, \kappa, u)| \le Cr(1+\kappa)(1+|b|)^N e^{-cb^2}; \tag{S10.44}$$

*in particular*

$$|r^{-2}(\mathcal{A}_r - \mathcal{A}_0)(b, \kappa r, u) - (\mathcal{C}^{\mathrm{cand}} - \mathcal{C}^{\mathrm{con}})(b, \kappa, u)| \le Cr(1+\kappa)^2(1+|b|)^N e^{-cb^2}. \tag{S10.45}$$

*Proof.* Work on the coupling space with $T$, $\Delta_0$ and $\widehat{Y}$ as in §S10.2.

*Step 1 (pathwise comparison).* By (S10.15), $\det K_M = r(\widehat{Y} - 6\kappa\Delta_0) + R_M$ and $\det K_S = r(\widehat{Y} + 6\kappa\Delta_0) + R_S$. Put $\delta_i := \det K_i / r$ and $\eta := Cr(1+k)T^{N_1} \ge |R_i|/r$; then $r^{-2}F_d(K_M)F_{d-1}(K_S) = |\delta_M\delta_S|\mathbf{1}\{\text{typed}\}$. *Case 1, $\lambda_{\max}(A_0) < -rT$:* $A_M, A_S \prec 0$, and by Haynsworth $\operatorname{ind} K_i = m + \mathbf{1}\{\varsigma \det K_i < 0\}$, $\varsigma := (-1)^m = \operatorname{sign}\Delta_0$. So typed $\iff$ $\varsigma\delta_M < 0 < \varsigma\delta_S$, while $(\varsigma\Delta_0 = |\Delta_0|)$ $|\widehat{Y}| < 6\kappa|\Delta_0| \iff \varsigma(\widehat{Y} - 6\kappa\Delta_0) < 0 < \varsigma(\widehat{Y} + 6\kappa\Delta_0)$. If both hold, $|\delta_M\delta_S - (\widehat{Y} - 6\kappa\Delta_0)(\widehat{Y} + 6\kappa\Delta_0)| \le 2\eta(|\widehat{Y}| + 6\kappa|\Delta_0|) + \eta^2 \le 24\eta\kappa|\Delta_0| + \eta^2$. If exactly one holds, for some $i$ the factor $\widehat{Y} \mp 6\kappa\Delta_0$ and $\delta_i$ differ in sign or one vanishes, so $|\widehat{Y} \mp 6\kappa\Delta_0| \le \eta$ and the other factor is $\le 12\kappa|\Delta_0| + \eta$; both weights are then $\le 2\eta(12\kappa|\Delta_0| + 3\eta)$. *Case 2, $\lambda_{\max}(A_0) > rT$:* $\lambda_{\max}(A_M) > 0$, so $F_d(K_M) = 0$, and $\mathbf{1}\{A_0 \prec 0\} = 0$. *Case 3, $|\lambda_{\max}(A_0)| \le rT$:* $|\Delta_0| \le rT \cdot T^{m-1}$, so $w_\kappa(A_0, \widehat{Y}) \le 36\kappa^2 r^2 T^{2m}$; if typed, the sign window gives $|\delta_i| \le 12\kappa|\Delta_0| + 2\eta$, so $|\delta_M\delta_S| \le C(\kappa r + \eta)^2 T^{2m}$. In all cases, as $|\Delta_0| \le T^m$ and $\kappa r = k \le 1$,

$$|r^{-2}F_d(K_M)F_{d-1}(K_S) - w_\kappa(A_0, \widehat{Y})| \le Cr(1+\kappa)T^N; \tag{S10.46}$$

indeed Case 1's bounds are linear in $\kappa$, Case 3's only $\kappa^2$ is $36\kappa^2r^2T^{2m} = 36k^2T^{2m}$, and the $k^2$ and $\eta^2$ parts are absorbed by $k^2 \le k = \kappa r$ and $r^2 \le r$.

*Step 2 (to the contact jets).* Let $Y_0 := \frac{f_4^{(0)}}{12}\Delta_0 - \frac{1}{4}(\gamma^{(0)})^\top \operatorname{adj}(A_0)\gamma^{(0)}$, built from the jets of $F_0$. As the $C^4$-jets of $F_r, F_0$ differ by $\le r^2T$, $|\widehat{Y} - Y_0| \le 3k|\operatorname{tr}(\operatorname{adj}(A_0)B)| + Cr^2T^N$, and $|\widehat{Y}| + |Y_0| \le CT^N$ $(|f_4|, |\gamma|, \|B\|, \|A_0\| \le T,\ k \le 1)$. By Lemma S10.11, $|w_\kappa(A_0, \widehat{Y}) - w_\kappa(A_0, Y_0)| \le C(k + r^2)T^{N'}$, and $k + r^2 = r(\kappa + r) \le r(1+\kappa)$. (The global Lipschitz constant $12\kappa|\det A|$ of $Y \mapsto w_\kappa(A, Y)$, from $|Y^2 - Y'^2| \le (|Y| + |Y'|)|Y - Y'|$ on the window, gives instead $C\kappa(\kappa r + r^2)T^N$, enough for (S10.45).) As $(A_0, Y_0)$ has the contact law at $v_0(b,k)$, (S10.46) and $\mathbb{E}[T^N] \le CP^N$ give $|r^{-2}\mathbb{E}_Q[W_r/r^2] - \mathbb{E}_0[w_\kappa(A, Y) \mid v_0(b,k)]| \le Cr(1+\kappa)P^N$.

*Step 3 (densities and the target).* By (S10.10) and $\mathbb{E}_0[w_\kappa] \le C(1+\kappa)^2P^N$ the density step costs $Cr^2P^2(1+\kappa)^2P^N$, and $r^2(1+\kappa)^2 = (r+k)^2 \le 2(r+k) = 2r(1+\kappa)$ as $r + k \le 2$. With the exact identity $r^{-2}\mathcal{A}_0(b, \kappa r, u) = 12\pi_0(v_0(b,k))36\kappa^2\mathbb{E}_0[\Delta^2\mathbf{1}\{A \prec 0\} \mid v_0(b,k)]$,

$$\begin{aligned} r^{-2}(\mathcal{A}_r - \mathcal{A}_0)(b, \kappa r, u) = &-12\pi_0(v_0(b,k))\mathbb{E}_0[\min(Y^2, 36\kappa^2\Delta^2)\mathbf{1}\{A \prec 0\} \mid v_0(b,k)] \\ &+ O(r(1+\kappa)P^N e^{-c(b^2+k^2)}). \end{aligned}$$

Move the target to $v_0(b,0)$ by parity [7] (Lemma S12.30): even- and odd-order jets at 0 are independent; the rows $f, \partial_u^2 f, \nabla_\Theta \partial_u f$ are even and $\partial_u f, \partial_u^3 f, \nabla_\Theta f$ odd, with target parts $(b, 0, 0)$

and $(0, 12k, 0)$. So the conditional law of the even jets $(A, f_4)$ does not depend on $k$; the odd jets have $k$-independent covariance and mean shift $12kw$, $w$ fixed, so $\gamma = \bar\gamma + 12kw_\gamma$; and $\pi_0(v_0(b,k)) = p_{\rm even}(b,0,0)p_{\rm odd}(0,12k,0)$, $p_{\rm odd}(0,12k,0) = p_{\rm odd}(0,0,0)(1+O(k^2))$. Since $|\min(Y^2,a) - \min(\bar Y^2,a)| \le (|Y|+|\bar Y|)|Y-\bar Y|$ for $a \ge 0$ (this is (S10.37)) and the shift moves $Y$ by $\le Ck(|\bar\gamma|+k)\|A\|^{m-1}$, the expectation moves by $O(kP^N)$, with no growth in $\kappa$; so the two targets differ by $O(k(1+|b|)^N e^{-cb^2}) = O(\kappa r(1+|b|)^N e^{-cb^2})$. Collecting, with $P^N e^{-c(b^2+k^2)} \le C(1+|b|)^N e^{-cb^2}$, gives (S10.44), and (S10.45) follows as $1+\kappa \le (1+\kappa)^2$. □

**Lemma S10.14** (overlap)**.** *For $\kappa \ge 1$,*

$$0 \le (\mathcal{C}^{\rm cand} - \mathcal{C}^{\rm con})(b,\kappa,u) - \mathcal{A}_2(b,0,u) \le C\kappa^{-1}(1+|b|)^N e^{-cb^2}. \tag{S10.47}$$

*Proof.* By (S10.39) and (S10.6) the difference is $12\pi_0\mathbb{E}_0[(Y^2 - 36\kappa^2\Delta^2)_+ \mathbf{1}\{A \prec 0\} \mid b] \ge 0$, at most $12\pi_0\mathbb{E}_0[Y^2\mathbf{1}\{|\Delta| < |Y|/(6\kappa)\} \mid b]$. Split by $|Y| \in [2^{j-1}, 2^j)$, $j \ge 1$, and $|Y| < 1$ for $j = 0$. Given $A$, $(f_4, \gamma)$ is Gaussian with mean affine in $(A, b)$ and bounded covariance, and $Y$ is polynomial in them with coefficients polynomial in $A$, so $\mathbb{P}(|Y| \ge t \mid A) \le C_p t^{-p}(1 + |b| + \|A\|)^{n_p}$. Under $\mathbb{E}_0[\,\cdot \mid b]$, $A$ is a nondegenerate Gaussian symmetric matrix with mean $O(1+|b|)$, so by Lemma S10.9

$$\begin{aligned}\mathbb{E}_0[Y^2\mathbf{1}\{|\Delta| < \tfrac{|Y|}{6\kappa}\}] &\le \sum_{j\ge 0} 4^j \mathbb{E}_0\big[\mathbf{1}\{|\det A| < \tfrac{2^j}{6\kappa}\}\min(1, C_p 2^{-(j-1)p}(1+|b|+\|A\|)^{n_p})\big] \\ &\le \sum_j 4^j C_p 2^{-(j-1)p}(1+|b|)^N \tfrac{2^j}{\kappa}.\end{aligned}$$

Take $p > 3$; $\pi_0 \le Ce^{-cb^2}$ completes the proof. □

S10.7. **The elder decision.** This subsection is deterministic. Fix $f \in C^5(\mathbb{T}^d_L)$ and $\mathcal{N} \ge \max(1, \|f\|_{C^5})$; let $0 < r \le r_Q := \min\{\frac{1}{10}, r_L\}$, $r_L$ the positive root of $6r + 10r^2 = L/8$; let $\kappa > 0$ and let $M = -\frac{r}{2}u$, $S = \frac{r}{2}u$ be critical points of $f$ with $f(M) = b$, $f(S) = b - \kappa r^4$. Put $\Phi(X,\Xi) := rXu + r^2\Theta\Xi$ ($X \in \mathbb{R}$, $\Xi \in \mathbb{R}^m$), $\mathfrak{F} := r^{-4}(f\circ\Phi - b)$, let $(f_4, \gamma, A)$ be the jets of $f$ at 0 and

$$\mathfrak{P}(X,\Xi) = 2\kappa(X+\tfrac12)^2(X-1) + \tfrac{f_4}{24}(X^2-\tfrac14)^2 + \tfrac12(X^2-\tfrac14)\gamma\cdot\Xi + \tfrac12\Xi^\top A\Xi \tag{S10.48}$$

the cusp polynomial of §S9. When $A \prec 0$, $\Xi^*(X) := -\frac12(X^2-\frac14)A^{-1}\gamma$ and the ridge is $g(X) := \mathfrak{P}(X,\Xi^*(X)) = \kappa[2(X+\frac12)^2(X-1) + 3\varphi(X^2-\frac14)^2]$, $\varphi$ as in (S10.4) (§S9).

**Lemma S10.15** (weighted Taylor bounds)**.** *There is $C_1 = C_1(d)$ such that on $\{|X| \le 3,\ r|\Xi| \le 1\}$*

$$\begin{aligned}&|\mathfrak{F}-\mathfrak{P}|,\ |\partial_X(\mathfrak{F}-\mathfrak{P})| \le C_1\mathcal{N}r(1+|\Xi|)^2, \qquad \|\partial_\Xi^2(\mathfrak{F}-\mathfrak{P})\| \le C_1\mathcal{N}r, \\ &|\partial_\Xi(\mathfrak{F}-\mathfrak{P})|,\ |\partial_X^2(\mathfrak{F}-\mathfrak{P})|,\ |\partial_X\partial_\Xi(\mathfrak{F}-\mathfrak{P})| \le C_1\mathcal{N}r(1+|\Xi|).\end{aligned} \tag{S10.49}$$

*Proof.* Follow the proof of the cusp limit in §S9, keeping track of $\Xi$. (i) *Pin corrections:* with $\vartheta_j = \partial_u^j f(0)$, $\psi_j = \partial_u^j \nabla_\Theta f(0)$, the pinned jets differ from the values that produce $\mathfrak{P}$ by $O(\mathcal{N}r^5)$ $(\vartheta_0 - b)$, $O(\mathcal{N}r^4)$ $(\vartheta_1)$, $O(\mathcal{N}r^3)$ $(\vartheta_2)$, $O(\mathcal{N}r^2)$ $(\vartheta_3)$, $O(\mathcal{N}r^4)$ $(\psi_0)$, $O(\mathcal{N}r^2)$ $(\psi_1)$, with constants independent of $\kappa$ (which enters only through the exact pins); rescaled they contribute $O(\mathcal{N}r)(1+|X|^3) + O(\mathcal{N}r)|X||\Xi| + O(\mathcal{N}r^2)|\Xi|$, and their derivatives obey (S10.49). (ii) *Dropped monomials* $x^iy^j$ with $i + 2|j| \ge 5$, $i + |j| \le 4$, i.e. $(i,|j|) \in \{(3,1),(1,2),(2,2),(0,3),(1,3),(0,4)\}$, coefficients $\le C\mathcal{N}$: each becomes $r^aX^i\Xi^j$, $a = i+2|j|-4 \ge 1$; on $r|\Xi| \le 1$, $r^a|\Xi|^q \le r|\Xi|^{q-a+1}$ for $q \ge a-1$ and $\le r$ otherwise, and $|j| - a + 1 = 5 - i - |j| \le 2$. A $\Xi$-derivative lowers $q$ by one; two $X$-derivatives need $i \ge 2$, and then $5 - i - |j| \le 1$; one $X$-derivative keeps $(1+|\Xi|)^2$ $(\partial_X(rX\Xi^2) = r\Xi^2)$. (iii) *Remainder:* $|\partial^\alpha\rho_5(x,y)| \le C\mathcal{N}(|x|+|y|)^{5-|\alpha|}$ $(|\alpha| \le 2)$; at $(rX, r^2\Xi)$, $|x|+|y| \le 4r$, so $|\partial_X^p\partial_\Xi^q[r^{-4}\rho_5(rX, r^2\Xi)]| = r^{p+2q-4}|\partial_x^p\partial_y^q\rho_5| \le C\mathcal{N}4^5 r^{1+q}$. □

**Lemma S10.16** (Schur complements)**.** *Let $D, D' \in \mathrm{Sym}(m)$ with $D \preceq -\lambda I$, $D' \preceq -\frac{\lambda}{2} I$, $\lambda > 0$, let $a, a' \in \mathbb{R}$, $w, w' \in \mathbb{R}^m$, $\varsigma := a - w^\top D^{-1} w$, $\varsigma' := a' - w'^\top D'^{-1} w'$. Then*

$$|\varsigma' - \varsigma| \le |a' - a| + \tfrac{2}{\lambda}|w' - w|(|w| + |w'|) + \tfrac{2}{\lambda^2}|w|^2 \|D' - D\|. \tag{S10.50}$$

*Proof.* $w'^\top D'^{-1} w' - w^\top D^{-1} w = (w' - w)^\top D'^{-1}(w' + w) + w^\top D'^{-1}(D - D')D^{-1} w$, with $\|D'^{-1}\| \le 2/\lambda$, $\|D^{-1}\| \le 1/\lambda$. □

**Lemma S10.17** (the elder decision with explicit margins)**.** *Suppose $A \preceq -\lambda I$, $\lambda > 0$, and put $\Gamma := |\gamma|/\lambda$, $\varepsilon^* := 80 C_1 \mathcal{N} r (1+\Gamma)^2$, $R := 5(1+\Gamma) + 3((\kappa+1)/\lambda)^{1/2}$. Assume* (Q1) *$\mathcal{N}(3r + r^2 R) \le \lambda/2$ and $6r + 2r^2 R \le L/4$;* (Q2) *$5r(1+\Gamma) \le 1$ and $8 C_1 \mathcal{N} r (1 + 5\Gamma) \le \lambda$;* (Q3) *$\varepsilon^* \le \kappa/4$;* (Q4) *$|\varphi| \le 1$ and $||\varphi| - \frac{1}{3}| > 2\varepsilon^*/(3\kappa)$. Then*

$$e(f) = \mathbf{1}\{d_f(M) = f(S)\} = \mathbf{1}\{|\varphi| < 1/3\}. \tag{S10.51}$$

*Proof.* Put $M_1 := \frac{35}{8}\Gamma$, so $|\Xi^*(X)| \le \frac{1}{2} \cdot \frac{35}{4} |A^{-1}\gamma| \le M_1$ on $[-3, 2]$.

*Step 1 (window).* Let $\mathcal{W} := [-3, 2] \times \{|\Xi| \le R\}$. $\Phi$ is linear and injective and $\operatorname{diam} \Phi(\mathcal{W}) \le 5r + 2r^2 R < L/4$ by (Q1), so $\Phi$ embeds a neighbourhood of $\mathcal{W}$ in $\mathbb{T}_L^d$ and maps interiors and boundaries of subregions of $\mathcal{W}$ onto those of their images. On $\mathcal{W}$, $\partial_\Xi^2 \mathfrak{F} = D_\Theta^2 f(\Phi(X, \Xi)) \preceq A + \|D^3 f\|(3r + r^2 R) I \preceq -\frac{\lambda}{2} I$ by (Q1).

*Step 2 (the ridge).* Fix $X \in [-3, 2]$. The ball $\{|\Xi - \Xi^*(X)| \le 1\}$ lies in $\{|\Xi| \le 5(1+\Gamma)\} \subset \mathcal{W}$, where $r|\Xi| \le 1$ by (Q2). As $\partial_\Xi \mathfrak{P}(X, \Xi^*) = 0$, (S10.49) and (Q2) give $|\partial_\Xi \mathfrak{F}(X, \Xi^*)| \le C_1 \mathcal{N} r (1 + M_1) \le \lambda/8$, so $\mathfrak{F}(X, \Xi) \le \mathfrak{F}(X, \Xi^*) + \frac{\lambda}{8} - \frac{\lambda}{4} < \mathfrak{F}(X, \Xi^*)$ on $|\Xi - \Xi^*| = 1$. Hence the unique maximizer $\Xi_{\mathfrak{F}}(X)$ of the strictly concave $\mathfrak{F}(X, \cdot)$ over $\{|\Xi| \le R\}$ lies in the open unit ball around $\Xi^*$, and strong concavity gives $|\Xi_{\mathfrak{F}} - \Xi^*| \le \frac{2}{\lambda} |\partial_\Xi (\mathfrak{F} - \mathfrak{P})(X, \Xi^*)| \le \frac{1}{4}$. The ridge $g_{\mathfrak{F}}(X) := \mathfrak{F}(X, \Xi_{\mathfrak{F}}(X))$ is $C^2$, and $g_{\mathfrak{F}}''$ is the Schur complement of $\partial_\Xi^2 \mathfrak{F}$ in $\operatorname{Hess} \mathfrak{F}$ at $(X, \Xi_{\mathfrak{F}})$; likewise $g''$ with $(\mathfrak{P}, \Xi^*)$. From $g - |\mathfrak{F} - \mathfrak{P}|(X, \Xi^*) \le \mathfrak{F}(X, \Xi^*) \le g_{\mathfrak{F}}(X) \le \mathfrak{P}(X, \Xi_{\mathfrak{F}}) + |\mathfrak{F} - \mathfrak{P}|(X, \Xi_{\mathfrak{F}}) \le g + |\mathfrak{F} - \mathfrak{P}|(X, \Xi_{\mathfrak{F}})$, $\varepsilon_0 := \sup_{[-3,2]} |g_{\mathfrak{F}} - g| \le C_1 \mathcal{N} r (\frac{5}{4} + M_1)^2 \le \varepsilon^*$. Apply Lemma S10.16 with $(a, w, D) = (\partial_X^2 \mathfrak{P}, X\gamma, A)$ at $(X, \Xi^*)$ and $(a', w', D')$ the blocks of $\operatorname{Hess} \mathfrak{F}$ at $(X, \Xi_{\mathfrak{F}})$: as $\partial_X^2 \mathfrak{P}$ depends on $\Xi$ only through $\gamma \cdot \Xi$, $|a' - a| \le C_1 \mathcal{N} r [(\frac{5}{4} + M_1) + 2\Gamma(1 + M_1)]$; $|w' - w| \le C_1 \mathcal{N} r (\frac{5}{4} + M_1) \le \frac{5\lambda}{32}$ by (Q2); $\|D' - D\| \le C_1 \mathcal{N} r$; $|w| \le 3|\gamma|$. So

$$\begin{aligned} \varepsilon_2 := \sup_{[-3,2]} |g_{\mathfrak{F}}'' - g''| &\le C_1 \mathcal{N} r \big[ (\tfrac{5}{4} + M_1)(\tfrac{21}{16} + 12\Gamma) + 2\Gamma(1 + M_1) + 18\Gamma^2 \big] \\ &= C_1 \mathcal{N} r \big[ \tfrac{105}{64} + \tfrac{2911}{128}\Gamma + \tfrac{317}{4}\Gamma^2 \big] \le \varepsilon^*, \end{aligned}$$

this inequality and $(\frac{5}{4} + M_1)^2 \le 80(1 + \Gamma)^2$ holding coefficientwise.

*Step 3 (reduction to the ridge).* For every $t$, each nonempty $X$-fibre of $\mathcal{W} \cap \{\mathfrak{F} > t\}$ is convex, contains $\Xi_{\mathfrak{F}}(X)$, and is nonempty iff $g_{\mathfrak{F}}(X) > t$. So the path components of $\mathcal{W} \cap \{\mathfrak{F} > t\}$ are the parts over the components of $\{X \in [-3, 2] : g_{\mathfrak{F}}(X) > t\}$, and the supremum of $\mathfrak{F}$ over such a part is that of $g_{\mathfrak{F}}$ over the interval. On $|\Xi| = R$, $|\Xi - \Xi_{\mathfrak{F}}(X)| \ge R - M_1 - \frac{1}{4} \ge 3((\kappa + 1)/\lambda)^{1/2}$, so by strong concavity

$$\mathfrak{F}(X, \Xi) \le g_{\mathfrak{F}}(X) - \tfrac{9}{4}(\kappa + 1) \qquad (|\Xi| = R). \tag{S10.52}$$

*Step 4 (pins).* $M = \Phi(-\frac{1}{2}, 0)$, $S = \Phi(\frac{1}{2}, 0)$ are critical points, so $\Xi_{\mathfrak{F}}(\pm\frac{1}{2}) = 0$, $g_{\mathfrak{F}}(-\frac{1}{2}) = 0$, $g_{\mathfrak{F}}(\frac{1}{2}) = -\kappa$, $g_{\mathfrak{F}}'(\pm\frac{1}{2}) = 0$, and the same for $g$. So $h := g_{\mathfrak{F}} - g$ has $|h| \le \varepsilon_0$ and $|h(X)| \le \frac{\varepsilon_2}{2}(X \mp \frac{1}{2})^2$ on $[-3, 2]$.

*Step 5 (margins of the model ridge).* Let $X_3 := -1/(2\varphi)$. By the ridge identities of §S9 ($g' = 6\kappa(X^2 - \frac{1}{4})(1 + 2\varphi X)$, $g(X_3) + \kappa = \kappa(\varphi + 1)^3(3\varphi - 1)/(16\varphi^3)$, $g(X_3) = \kappa(\varphi - 1)^3(3\varphi + 1)/(16\varphi^3)$): (M1) $g + \kappa = \kappa(X - \frac{1}{2})^2[2(X + 1) + 3\varphi(X + \frac{1}{2})^2]$, $g = \kappa(X + \frac{1}{2})^2[2(X - 1) + 3\varphi(X - \frac{1}{2})^2]$. (M2) If $|\varphi| < \frac{1}{3}$, $g + \kappa \ge \kappa(X - \frac{1}{2})^2$ on $[-\frac{1}{2}, X_R]$, where $X_R := 2$ for $\varphi \ge -\frac{1}{4}$ and $X_R := X_3 = \frac{1}{2|\varphi|} \in (\frac{3}{2}, 2)$ for $-\frac{1}{3} < \varphi < -\frac{1}{4}$ (for $\varphi \ge 0$ the bracket is $\ge 2(X + 1) \ge 1$; for $\varphi < 0$ it is $1 + (X + \frac{1}{2})(2 - 3|\varphi|(X + \frac{1}{2})) \ge 1$

while $X+\frac12 \le \frac{2}{3|\varphi|}$, which holds up to $X_3$ when $|\varphi| \le \frac13$). (M3) If $\varphi \le \frac13$, $g \le -\kappa(X+\frac12)^2$ on $[-\frac32,\frac12]$ (the bracket is $\le 2(X-1)+(X-\frac12)^2$, convex, $=-1$ at both ends). (M4) For $\frac13 \le \varphi \le 1$: $g(X_3)+\kappa-\frac32\kappa(\varphi-\frac13) = \frac{\kappa(3\varphi-1)[(\varphi+1)^3-8\varphi^3]}{16\varphi^3} = \frac{\kappa(3\varphi-1)(1-\varphi)(7\varphi^2+4\varphi+1)}{16\varphi^3} \ge 0$; for $-1 \le \varphi \le -\frac13$ the same identity in $\psi=-\varphi$ gives $-g(X_3) \ge \frac32\kappa(|\varphi|-\frac13)$; for $-\frac13 \le \varphi < 0$, $\psi=-\varphi$: $g(X_3) = \frac{\kappa(1+\psi)^3(1-3\psi)}{16\psi^3}$ and $g(X_3)-12\kappa(\frac13-\psi) = \frac{\kappa(1-3\psi)[(1+\psi)^3-64\psi^3]}{16\psi^3} = \frac{\kappa(1-3\psi)^2(21\psi^2+6\psi+1)}{16\psi^3} \ge 0$. (M5) $g(2)=\kappa(\frac{25}{2}+\frac{675}{16}\varphi)$, $g(-\frac32)=\kappa(-5+12\varphi)$, $g(-3)=\kappa(-50+\frac{3675}{16}\varphi)$: so $g(2) \ge \frac{125}{64}\kappa$ for $\varphi \ge -\frac14$; $g(2) \le -\frac{25}{16}\kappa$ and $g(-\frac32) \le -9\kappa$ for $\varphi \le -\frac13$; $g(-\frac32)+\kappa = -12\kappa(\frac13-\varphi)$; $g(-3) \ge \frac{425}{16}\kappa$ for $\varphi \ge \frac13$. (M6) For $\frac13 < \varphi \le 1$, $g$ decreases on $[-3,X_3]$ and increases on $[X_3,-\frac12]$; for $-1 \le \varphi < -\frac13$, $g$ increases on $[\frac12,X_3]$ and decreases on $[X_3,2]$.

*Step 6 (the decision).* By (Q3)–(Q4), $\varepsilon_2 \le \varepsilon^* \le \kappa/4$ and $\varepsilon_0 \le \varepsilon^* < \frac32\kappa||\varphi|-\frac13|$. *Case E,* $|\varphi| < \frac13$. Path: on $[-\frac12,X_R]$, (M2) and Step 4 give $g_{\mathfrak F}+\kappa \ge (\kappa-\frac{\varepsilon_2}{2})(X-\frac12)^2 \ge 0$, and $g_{\mathfrak F}(X_R) > 0$ (for $\varphi \ge -\frac14$, $g(2) \ge \frac{125}{64}\kappa > \frac{\kappa}{4} \ge \varepsilon_0$; otherwise $g(X_3) \ge 12\kappa(\frac13-|\varphi|) > \varepsilon_0$ by (M4)). So the ridge curve $X \mapsto \Phi(X,\Xi_{\mathfrak F}(X))$, $X \in [-\frac12,X_R]$, runs from $M$ through $S$ to a point above $b$ with $f \ge f(S)$ along it: $d_f(M) \ge f(S)$. Trap: let $\mathcal D := (-\frac32,\frac12)\times\{|\Xi|<R\}$, $\mathcal D_{\rm phys} := \Phi(\mathcal D)$. At $X=-\frac32$, $\mathfrak F \le g_{\mathfrak F}(-\frac32) \le -\kappa-12\kappa(\frac13-\varphi)+\varepsilon_0 < -\kappa$, as $\varepsilon_0 < 12\kappa(\frac13-\varphi)$ (for $\varphi \le 0$ the margin is $\ge 4\kappa$). At $X=\frac12$, $\mathfrak F \le g_{\mathfrak F}(\frac12) = -\kappa$, with equality only at $\Xi=0$, i.e. at $S$. On $[-\frac32,\frac12]$, (M3) and Step 4 give $g_{\mathfrak F} \le -(\kappa-\frac{\varepsilon_2}{2})(X+\frac12)^2 \le 0$, so at $|\Xi|=R$, (S10.52) gives $\mathfrak F \le -\frac94(\kappa+1) < -\kappa$. So $f \le f(S)$ on $\partial\mathcal D_{\rm phys}$, and the path component $\mathcal C$ of $\{f > f(S)\}$ containing $M$ lies in $\mathcal D_{\rm phys}$; by Step 3 its supremum is $b+r^4\sup g_{\mathfrak F}$ over an interval in $(-\frac32,\frac12)$, which is $\le b$. So every path from $M$ to a point above $b$ leaves $\mathcal C$: $d_f(M) \le f(S)$, and $e=1$. *Case $R^+$,* $\frac13 < \varphi \le 1$. By (M6), (M4), $\min_{[-3,-1/2]} g = g(X_3) \ge -\kappa+\frac32\kappa(\varphi-\frac13)$, so $\min_{[-3,-1/2]} g_{\mathfrak F} > -\kappa$; and $g_{\mathfrak F}(-3) \ge \frac{425}{16}\kappa-\frac{\kappa}{4} > 0$ by (M5). The ridge curve from $M$ to $X=-3$ ends above $b$ and stays above $f(S)$: $d_f(M) > f(S)$, $e=0$. *Case $R^-$,* $-1 \le \varphi < -\frac13$. Put $t^* := -\frac54\kappa$, $\mathcal D' := (-\frac32,2)\times\{|\Xi|<R\}$. $g_{\mathfrak F} \le 0$ on $[-\frac32,2]$: on $[-\frac32,\frac12]$ by (M3) and Step 4, on $[\frac12,2]$ as (M6), (M4) give $g \le g(X_3) \le -\frac32\kappa(|\varphi|-\frac13) < -\varepsilon_0$. On $\partial\mathcal D'$: $g_{\mathfrak F}(-\frac32) \le -9\kappa+\frac{\kappa}{4} < t^*$ and $g_{\mathfrak F}(2) \le -\frac{25}{16}\kappa+\frac{\kappa}{4} < t^*$ by (M5); at $|\Xi|=R$, $\mathfrak F \le -\frac94(\kappa+1) < t^*$ by (S10.52). So the component $\mathcal C'$ of $\{f > b+r^4t^*\}$ containing $M$ lies in $\Phi(\mathcal D')$ and has supremum $\le b$; every path from $M$ to a point above $b$ leaves $\mathcal C'$, so $d_f(M) \le b+r^4t^* < f(S)$ (or $d_f(M)=-\infty$), and $e=0$. □

S10.8. **The elder cusp kernel.** Here $0 < r \le r_0^*$, $f=F_r$ under $Q, A, \Delta, f_4, \gamma, B, Y_r, \mathcal N_r$ are as in (S10.16), $J'$ is the vector of free jets of order $\le 9$ (Lemma S10.12), $J'' := J' \setminus \{f_4\}$, $J''' := J'' \setminus \{A\}$, $\lambda := \lambda_{\min}(-A)$ on $\{A \prec 0\}$, $Y' := Y_r - 3k\,\mathrm{tr}(\mathrm{adj}(A)B)$ and $\varphi_r := (f_4 - 3\gamma^\top A^{-1}\gamma)/(72\kappa)$, so $Y' = 6\kappa\Delta\varphi_r$.

**Lemma S10.18** (free jets under the pinned law)**.** *(i) Given $J''$, $f_4$ is Gaussian under $Q$ with variance in $[c,C]$ and mean of modulus $\le C(P+|J''|)$; its conditional density is at most a constant $C_0$, and for every $J''$-measurable interval $I$ and $p \ge 0$*

$$\mathbb E_Q[(1+|f_4|)^p \mathbf 1\{f_4 \in I\} \mid J''] \le C_p(P+|J''|)^p|I|. \tag{S10.53}$$

*(ii)* $\mathbb E_Q[\mathcal N_r^p \mid J'] \le C_p(P+|J'|)^p$.

*(iii) Given $J'''$, $A$ is Gaussian under $Q$ with covariance between $cI$ and $CI$ and mean of modulus $\le C(P+|J'''|)$; hence, with $n' = n+m(m-1)/2$,*

$$\mathbb E_Q[(1+\|A\|)^n \mathbf 1\{|\lambda_{\max}(A)| < \varepsilon\} \mid J'''] \le C(P+|J'''|)^{n'}\varepsilon, \tag{S10.54}$$

*and likewise with $\{|\Delta|<\varepsilon\}$ in place of $\{|\lambda_{\max}(A)|<\varepsilon\}$.*

*(iv) Under $Q$, $J' \sim N(\mu_r,V_r)$; under the contact law at $v_0(b,k)$, $J' \sim N(\mu_0,V_0)$; $|\mu_r-\mu_0| \le Cr^2P$, $\|V_r-V_0\| \le Cr^2$, $V_r, V_0 \succeq cI$.*

(v) *Under the contact law at $v_0(b,k)$ the even free jets (among them $A, f_4$) are independent of the odd ones (among them $\gamma$); the law of the even jets does not depend on $k$; the odd jets are $N(12kw, \Sigma_o)$ with $w, \Sigma_o$ independent of $k$; and $\pi_0(v_0(b,k)) = p_e(b,0,0)p_o(0,12k,0)$, $p_o$ a centred Gaussian density.*

(vi) *If $\kappa r \le 1$, then $|r^{-2}F_d(K_M)F_{d-1}(K_S) - w_\kappa(A, Y_r)| \le Cr(1+\kappa)\mathcal{N}_r^N$.*

*Proof.* (i) By Step 1 of Lemma S10.12, $\mathrm{Cov}_Q(J')$ is uniformly invertible on $(0, r_0^*]$; the mean is $\mathbb{E}_Q J' = \mathrm{Cov}(J', U_r)\Sigma_r^{-1} v_r$, so $|\mathbb{E}_Q J'| \le CP$. $\mathrm{Var}_Q(f_4 \mid J'')$ is a Schur complement in $\mathrm{Cov}_Q(J')$, between $\lambda_{\min}(\mathrm{Cov}_Q J') \ge c$ and $\mathrm{Var}_Q(f_4) \le C$; the conditional mean, $\mathbb{E}_Q f_4 + \mathrm{Cov}_Q(f_4, J'')\,\mathrm{Cov}_Q(J'')^{-1}(J'' - \mathbb{E}_Q J'')$, has bounded coefficients; and the Gaussian factor absorbs polynomial weights. (ii) is Lemma S10.10 with $J_0 = J'$. (iii) As (i), with Lemma S10.9 conditionally. (iv) is Step 3 of Lemma S10.12. (v) is the parity factorization (Step 3 of Lemma S10.13). (vi) is (S10.46) with $(A, \mathcal{N}_r)$ for $(A_0, T)$: Step 1 there uses only the expansion, here (S10.16), and $\|A_i - A\| \le \frac{r}{2}\|F_r\|_{C^3} \le r\mathcal{N}_r$. □

**Lemma S10.19** (two Gaussian comparisons)**.** *Fix $n$, $p$ and $0 < c_G \le C_G$.* (a) *Let $J \sim N(\mu, V)$ and $\tilde J \sim N(\tilde\mu, \tilde V)$ on $\mathbb{R}^n$ with $c_G I \preceq V, \tilde V \preceq C_G I$, and let $G$ be Borel with $|G(j)| \le H(1+|j|)^p$. Then $|\mathbb{E}G(\tilde J) - \mathbb{E}G(J)| \le CH(1+|\mu|+|\tilde\mu|)^{p+2}(|\tilde\mu - \mu| + \|\tilde V - V\|)$.* (b) *Let $E$ and $O$ be independent, $O \sim N(\xi, \Sigma)$ on $\mathbb{R}^{n_o}$, $c_G I \preceq \Sigma \preceq C_G I$, $|\xi| \le \xi_0$. Let $G(e,-o) = G(e,o)$ and $|G(e,o)| \le h(e)(1+|o|)^p$ with $\mathbb{E}[h(E)] < \infty$. Then $|\mathbb{E}G(E,O) - \mathbb{E}G(E,O_0)| \le C|\xi|^2\mathbb{E}[h(E)]$, where $O_0 \sim N(0,\Sigma)$ is independent of $E$ and $C$ depends also on $\xi_0$.*

*Proof.* (a) Interpolate $(\mu_t, V_t)$ linearly; $c_G I \preceq V_t \preceq C_G I$, and $\partial_t \log p_t(j) = (j-\mu_t)^\top V_t^{-1}\delta\mu + \frac12 (j-\mu_t)^\top V_t^{-1}\delta V V_t^{-1}(j-\mu_t) - \frac12 \operatorname{tr}(V_t^{-1}\delta V)$, so $|\partial_t p_t| \le C(|\delta\mu| + \|\delta V\|)(1+|j-\mu_t|^2)p_t$; integrate $|G||\partial_t p_t|$ over $j$ and $t \in [0,1]$. (b) $\chi(\xi) := \mathbb{E}G(E, O_\xi)$ is smooth and even (substitute $o \mapsto -o$), so $\nabla\chi(0) = 0$ and $|\chi(\xi) - \chi(0)| \le \frac12 |\xi|^2 \sup_{|\zeta|\le\xi_0} \|\nabla^2\chi(\zeta)\|$; differentiating the density twice gives $\|\nabla^2\chi(\zeta)\| \le C\mathbb{E}[h(E)]$, and the integrability of $h(E)$ justifies the differentiation. No regularity of $G$ is needed. □

**Lemma S10.20** (the elder cusp kernel with a rate)**.** *There are $r_2 > 0$ and $C, c, N$ such that for $0 < r \le r_2$, $b \in \mathbb{R}$, $u \in S^{d-1}$ and $\kappa > 0$ with $\kappa r \le 1$,*

$$|r^{-2}(\mathcal{A}_r^{\mathrm{eld}} - \mathcal{A}_0)(b, \kappa r, u) - (\mathcal{C}^{\mathrm{eld}} - \mathcal{C}^{\mathrm{con}})(b, \kappa, u)| \le C(r(1+\kappa) + \kappa^2 r^{3/2} + \kappa^3 r^2)(1+|b|)^N e^{-cb^2}, \tag{S10.55}$$

*and in particular, as $\kappa^2 r^{3/2} + \kappa^3 r^2 \le 2\kappa^2 r$ for $r \le 1$, $\kappa r \le 1$,*

$$|r^{-2}(\mathcal{A}_r^{\mathrm{eld}} - \mathcal{A}_0)(b, \kappa r, u) - (\mathcal{C}^{\mathrm{eld}} - \mathcal{C}^{\mathrm{con}})(b, \kappa, u)| \le Cr(1+\kappa)^2(1+|b|)^N e^{-cb^2}. \tag{S10.56}$$

*One may take $r_2 = \min(r_0^*, \frac12, r_Q)$.*

*Proof. Step E1 (pathwise comparison).* Write $\mathbf{F}$ for $F_d(K_M)F_{d-1}(K_S)$ and put

$$\mathsf{w}(J') := w_\kappa(A, Y_r)\mathbf{1}\{|\varphi_r| < \tfrac13\}, \qquad r^{-2}\mathbf{F}e - \mathsf{w} = (r^{-2}\mathbf{F} - w_\kappa)e + w_\kappa(e - \mathbf{1}\{|\varphi_r| < \tfrac13\}).$$

Lemma S10.18(vi) bounds the first term by $Cr(1+\kappa)\mathcal{N}_r^N$. On $\{w_\kappa > 0\}$: if the pair is not typed, $\mathbf{F} = 0$ and Lemma S10.18(vi) gives $w_\kappa \le Cr(1+\kappa)\mathcal{N}_r^N$; if it is typed and (Q1)–(Q4) hold for $f = F_r$, $\mathcal{N} = \mathcal{N}_r$ ($\mathcal{N}_r \ge \max(1, \|F_r\|_{C^5})$, $r \le r_Q$, $f(S) = b - \kappa r^4$), Lemma S10.17 gives $e = \mathbf{1}\{|\varphi_r| < \frac13\}$. With $\mathfrak{B}$ the event that (Q1)–(Q4) fail,

$$|r^{-2}\mathbf{F}e - \mathsf{w}(J')| \le 2Cr(1+\kappa)\mathcal{N}_r^N + w_\kappa(A, Y_r)\mathbf{1}_{\mathfrak{B}}. \tag{S10.57}$$

*Step E2 (wrong decisions).* The conditions of Lemma S10.17 on $r$ alone hold as $r \le r_Q$; the others are monotone in $\mathcal{N}$ and $\lambda$. With $\mathcal{N} = \mathcal{N}_r$: (Q1) holds if $3\mathcal{N}r$, $5\mathcal{N}r^2(1+\Gamma)$, $3\mathcal{N}r^2((\kappa+1)/\lambda)^{1/2}$ are each $\le \lambda/6$ and (given $6r + 10r^2 \le L/8$) $10r^2|\gamma|/\lambda$, $6r^2((\kappa+1)/\lambda)^{1/2}$ are each $\le L/16$; this follows from $\lambda \ge 18\mathcal{N}r$, $\lambda \ge 60\mathcal{N}r^2$, $\lambda^2 \ge 60\mathcal{N}r^2|\gamma|$, $\lambda \ge 18^{2/3}2^{1/3}\mathcal{N}r$ (as $r(\kappa+1) \le 2$),

$\lambda \ge 160r^2|\gamma|/L$, $\lambda \ge (96/L)^2 r^4(\kappa+1)$. (Q2) holds if $\lambda \ge 10r|\gamma|$, $\lambda \ge 16C_1\mathcal{N}r$, $\lambda^2 \ge 80C_1\mathcal{N}r|\gamma|$. (Q3) holds if $\kappa \ge 1280C_1\mathcal{N}r$ and $\lambda^2 \ge 1280C_1\mathcal{N}r|\gamma|^2/\kappa$, by $(1+\Gamma)^2 \le 2(1+\Gamma^2)$. As $\kappa r \le 1 \le \mathcal{N}$ gives $r^2|\gamma| \le r|\gamma| \le |\gamma|(\mathcal{N}r/\kappa)^{1/2}$, and $r^4(\kappa+1) \le 2r$, all hold if $\kappa \ge C_2\mathcal{N}_r r$ and $\lambda \ge \lambda_2(\mathcal{N}_r) := C_2[\mathcal{N}_r r + (\mathcal{N}_r r|\gamma|)^{1/2} + |\gamma|(\mathcal{N}_r r/\kappa)^{1/2}]$, $C_2 = C_2(d,L)$. So $\mathfrak{B} \subset \mathfrak{B}_\lambda \cup \mathfrak{B}_\kappa \cup \mathfrak{B}_4 \cup \mathfrak{B}_5$, with $\mathfrak{B}_\lambda = \{\lambda < \lambda_2(\mathcal{N}_r)\}$, $\mathfrak{B}_\kappa = \{\kappa < C_2\mathcal{N}_r r\}$, $\mathfrak{B}_4 = \{||\varphi_r| - \frac{1}{3}| \le \frac{2\varepsilon^*}{3\kappa}\}$, $\mathfrak{B}_5 = \{|\varphi_r| > 1\}$. Now $w_\kappa \le 36\kappa^2\Delta^2 \mathbf{1}\{A \prec 0, |Y_r| < 6\kappa|\Delta|\}$, and $Y_r$ is affine in $f_4$ with slope $\Delta/12$ and $J''$-measurable intercept, so $\{|Y_r| < 6\kappa|\Delta|\} = \{f_4 \in I_{\mathrm{typ}}\}$, a $J''$-measurable interval of length $144\kappa$. The thresholds of $\mathfrak{B}_\lambda, \mathfrak{B}_\kappa, \mathfrak{B}_4$ depend on $\mathcal{N}_r$: on $\{\mathcal{N}_r \in [2^j, 2^{j+1})\}$ replace $\mathcal{N}_r$ by $2^{j+1}$, bound $\mathbf{1}\{\mathcal{N}_r \ge 2^j\}$ by Lemma S10.18(ii), integrate $f_4$ by (S10.53); each $j$ carries $C_p 2^{-jp}(P + |J''|)^p$, summable for $p$ large. $\mathfrak{B}_\lambda$: integrating $f_4$ over $I_{\mathrm{typ}}$ gives $\min(1, 144C_0\kappa)$; on $\{\lambda < \lambda_2\}$, $\Delta^2 \le \lambda_2^2\|A\|^{2m-2}$, and (S10.54) given $J'''$ ($\lambda_2$ depends on $|\gamma|$, $\gamma \in J'''$) gives a factor $\lambda_2$; so the contribution is $\le C\kappa^2 \min(1,\kappa) \sum_j 2^{-jp}\mathbb{E}[\lambda_2(2^{j+1})^3(P + |J'''|)^N]$, and the three terms of $\lambda_2^3$ give $\kappa^2\min(1,\kappa)(r^3 + r^{3/2} + (r/\kappa)^{3/2})P^N \le Cr(1+\kappa)^2P^N$, using $\kappa r \le 1$, $r \le 1$; more precisely it is $\le C(r + \kappa^2 r^{3/2})P^N$, since $\kappa^2 r^3 = k^2 r \le r$ and $\kappa^{1/2}r^{3/2} = k^{1/2}r \le r$. $\mathfrak{B}_\kappa$: $\mathbb{P}(\mathcal{N}_r > \kappa/(C_2 r) \mid J') \le \min(1, C_3(r/\kappa)^3(P + |J'|)^3)$, so the contribution is $\le C\kappa^2 \min(1,\kappa)\min(1,(r/\kappa)^3)P^N \le Cr^3P^N$. $\mathfrak{B}_5$ (on $\{w_\kappa > 0\}$): $|Y_r| < 6\kappa|\Delta| < |Y'|$, so $6\kappa|\Delta| - |Y_r| < \tau := 3k|\operatorname{tr}(\operatorname{adj}(A)B)|$ and $w_\kappa \le 12\kappa|\Delta|\tau$; and $|Y'| \in (6\kappa|\Delta|, 6\kappa|\Delta| + \tau)$ puts $f_4$ in two intervals of total length $24\tau/|\Delta|$. The contribution is $\le 288C_0\kappa\mathbb{E}[\tau^2(P + |J''|)^p] \le C\kappa^3 r^2 P^N \le C\kappa^2 r P^N$.

$\mathfrak{B}_4$: $f_4$ lies in two $J''$-measurable intervals $I_4$ of total length $192\varepsilon^*$, $\varepsilon^* = 80C_1\mathcal{N}_r r(1+\Gamma)^2$, $\Gamma = |\gamma|/\lambda$ (with the layer value of $\mathcal{N}_r$, absorbed by $2^{-jp}$). With $\Delta^2(1+\Gamma)^2 \le 2\Delta^2 + 2|\gamma|^2\|A\|^{2m-2}$ (as $\Delta^2 \le \lambda^2\|A\|^{2m-2}$) the contribution is $\le Cr\kappa^2 P^N$. Sharper: (Q1)–(Q3) hold off $\mathfrak{B}_\lambda \cup \mathfrak{B}_\kappa$, so only $\mathfrak{B}_4' := \mathfrak{B}_4 \setminus (\mathfrak{B}_\lambda \cup \mathfrak{B}_\kappa)$ matters, and

$$\mathbb{E}_Q[w_\kappa(A, Y_r)\mathbf{1}_{\mathfrak{B}_4'}] \le C\kappa r P^N \qquad \text{for every } \kappa > 0. \tag{S10.58}$$

Indeed, on $\mathfrak{B}_4'$ (Q3) holds, so $||\varphi_r| - \frac{1}{3}| \le \frac{2\varepsilon^*}{3\kappa} \le \frac{1}{6}$, i.e. $\frac{1}{6} \le |\varphi_r| \le \frac{1}{2}$ and $|f_4 - 3\gamma^\top A^{-1}\gamma| = 72\kappa|\varphi_r| \ge 12\kappa$; the weight vanishes unless $A \prec 0$, and then $|\gamma^\top A^{-1}\gamma| \le |\gamma|^2/\lambda$; hence

$$\mathfrak{B}_4' \cap \{w_\kappa > 0\} \subset \{|f_4| \ge 6\kappa\} \cup \{\lambda \le |\gamma|^2/(2\kappa)\}. \tag{S10.59}$$

On the first piece, $w_\kappa \le 36\kappa^2\Delta^2 \le 6\kappa\Delta^2|f_4|$, and integrating $f_4$ over $I_4$ by (S10.53) gives $\le C\kappa\Delta^2\varepsilon^*(P + |J''|)^p$; as $\Delta^2(1+\Gamma)^2 \le (\lambda + |\gamma|)^2\|A\|^{2m-2}$, polynomial in $J''$, this is $\le C\kappa r P^N$. On the second, integrating $f_4$ over $I_4$ gives $\le C\kappa^2 r(\lambda + |\gamma|)^2\|A\|^{2m-2}(P + |J''|)^p$, with $(\lambda + |\gamma|)^2 \le 2\|A\|^2 + 2|\gamma|^2$ free of $\kappa$; conditioning on $J''' \ni \gamma$ and using $\{\lambda \le \frac{|\gamma|^2}{2\kappa}\} \cap \{A \prec 0\} \subset \{|\lambda_{\max}(A)| \le \frac{|\gamma|^2}{2\kappa}\}$, (S10.54) gives the factor $C(P + |J'''|)^{n'}|\gamma|^2/(2\kappa)$, so this piece is $\le C\kappa r P^N$. Collecting, $\mathbb{E}_Q[w_\kappa(A, Y_r)\mathbf{1}_{\mathfrak{B}}] \le C(r(1+\kappa) + \kappa^2 r^{3/2} + \kappa^3 r^2)P^N$, and the cruder bounds give $\le Cr(1+\kappa)^2P^N$.

*Step E3 (removing the k-term).* $\mathsf{w}_0(J') := w_\kappa^{\mathrm{eld}}(A, Y') = w_\kappa(A, Y')\mathbf{1}\{|\varphi_r| < \frac{1}{3}\}$, as $|\varphi_r| < \frac{1}{3} \iff |Y'| < 2\kappa|\Delta|$. By Lemma S10.11, $|\mathsf{w} - \mathsf{w}_0| \le 3k|\operatorname{tr}(\operatorname{adj}(A)B)|(|Y_r| + |Y'|) \le Ck(1 + |J'|)^N$, so $\mathbb{E}_Q|\mathsf{w} - \mathsf{w}_0| \le CkP^N = C\kappa r P^N$ (the Lipschitz constant $12\kappa|\Delta|$ gives instead $|\mathsf{w} - \mathsf{w}_0| \le 12\kappa|\Delta| \cdot 3k|\operatorname{tr}(\operatorname{adj}(A)B)|$, i.e. $C\kappa^2 r P^N$).

*Step E4.* $|\mathsf{w}_0| \le 36\kappa^2|J'|^{2m}$, so Lemma S10.18(iv) and Lemma S10.19(a) give $|\mathbb{E}_Q[\mathsf{w}_0] - \mathbb{E}_0[\mathsf{w}_0 \mid v_0(b,k)]| \le C\kappa^2 r^2 P^N$.

*Step E5 (densities).* By (S10.10) and $\mathbb{E}_Q[r^{-2}\mathbf{F}e] \le C(1+\kappa)^2P^N$ (sign window), Steps E1–E4 give $r^{-2}\mathcal{A}_r^{\mathrm{eld}}(b, \kappa r, u) = 12\pi_0(v_0(b,k))\mathbb{E}_0[w_\kappa^{\mathrm{eld}}(A, Y) \mid v_0(b,k)] + O(r(1+\kappa)^2P^N e^{-c(b^2+k^2)})$; subtracting the exact $r^{-2}\mathcal{A}_0(b, \kappa r, u) = 12\pi_0(v_0(b,k))\mathbb{E}_0[36\kappa^2\Delta^2\mathbf{1}\{A \prec 0\} \mid v_0(b,k)]$,

$$r^{-2}(\mathcal{A}_r^{\mathrm{eld}} - \mathcal{A}_0)(b, \kappa r, u) = -12\pi_0(v_0(b,k))\mathbb{E}_0[\mathrm{loss}_\kappa(A, Y) \mid v_0(b,k)] + O(r(1+\kappa)^2P^N e^{-c(b^2+k^2)}). \tag{S10.60}$$

*Step E6 (the target $v_0(b,k) \to v_0(b,0)$).* $\mathrm{loss}_\kappa$ depends on the even jets $(A, f_4)$, and on the odd jet $\gamma$ only through $\gamma^\top \operatorname{adj}(A)\gamma$, so it is even in the odd jets; and $0 \le \mathrm{loss}_\kappa \le 9Y^2$ (on

$\{|Y| \ge 2\kappa|\Delta|\}$, $36\kappa^2\Delta^2 \le 9Y^2$). Lemma S10.19(b) applies with $E = (A, f_4)$, $O$ the odd jets, $h(A, f_4) = C(1 + |f_4|)^2(1 + \|A\|)^{2m}$, $p = 4$, $\mathbb{E}[h] \le C(1 + |b|)^N$; by Lemma S10.18(v), with $\xi = 12kw$ ($|\xi| \le 12|w|$ as $k \le 1$), $|\mathbb{E}_0[\mathrm{loss}_\kappa \mid v_0(b, k)] - \mathbb{E}_0[\mathrm{loss}_\kappa \mid v_0(b, 0)]| \le Ck^2(1 + |b|)^N$, with no growth in $\kappa$, and $p_o(0, 12k, 0) = p_o(0, 0, 0)(1 + O(k^2))$. So the right side of (S10.60) is $(\mathcal{C}^{\mathrm{eld}} - \mathcal{C}^{\mathrm{con}})(b, \kappa, u) + O(k^2(1 + |b|)^N e^{-cb^2}) + O(r(1 + \kappa)^2 P^N e^{-c(b^2+k^2)})$; as $k^2 = \kappa^2 r^2 \le \kappa r$, $r^2(1 + \kappa)^2 \le 2r(1 + \kappa)$ and $P^N e^{-ck^2} \le C(1 + |b|)^N$, collecting all steps gives (S10.55); the cruder bounds give (S10.56) directly. □

**Lemma S10.21** (elder overlap)**.** *For $\kappa \ge 1$,*

$$0 \le (\mathcal{C}^{\mathrm{cand}} - \mathcal{C}^{\mathrm{eld}})(b, \kappa, u) \le C\kappa^{-1}(1 + |b|)^N e^{-cb^2}. \tag{S10.61}$$

*Consequently, with Lemma S10.14, $|(\mathcal{C}^{\mathrm{eld}} - \mathcal{C}^{\mathrm{con}})(b, \kappa, u) - \mathcal{A}_2(b, 0, u)| \le C\kappa^{-1}(1 + |b|)^N e^{-cb^2}$.*

*Proof.* $\mathcal{C}^{\mathrm{cand}} - \mathcal{C}^{\mathrm{eld}} = 12\pi_0\mathbb{E}_0[(36\kappa^2\Delta^2 - Y^2)\mathbf{1}\{A \prec 0,\ 2\kappa|\Delta| \le |Y| < 6\kappa|\Delta|\} \mid b]$. There $36\kappa^2\Delta^2 \le 9Y^2$ and $|\Delta| \le |Y|/(2\kappa)$, so the difference is at most $108\pi_0\mathbb{E}_0[Y^2\mathbf{1}\{|\Delta| \le |Y|/(2\kappa)\} \mid b]$, which the dyadic argument of Lemma S10.14 bounds by $C\kappa^{-1}(1 + |b|)^N e^{-cb^2}$. □

S10.9. **Intermediate separations.** The next two lemmas hold for every $0 < r \le r_0^*$; of §S10.8 they use only $\mathcal{N}_r$, $\Delta$ and (S10.16).

**Lemma S10.22** (the rescaled Hessian at $M$)**.** *Let $0 < r \le r_0^*$, $b \in \mathbb{R}$, $0 < k \le r$ (so $\kappa \le 1$), $f = F_r$ under $Q$. In the frame $(u, \Theta)$ let $\Lambda_r := \mathrm{diag}(r, 1, \dots, 1)$, $H_M = D^2 f(M)$ and $\widetilde{H} := \Lambda_r^{-1} H_M \Lambda_r^{-1}$, so $\widetilde{H}_{uu} = r^{-2}\partial_u^2 f(M)$, $\widetilde{H}_{u\Theta} = r^{-1}\partial_u \nabla_\Theta f(M)$, $\widetilde{H}_{\Theta\Theta} = D_\Theta^2 f(M)$, $\det H_M = r^2 \det \widetilde{H}$. There are $c > 0$ and $C, C_p < \infty$ depending only on $d, L$ (and $p$) such that:* (a) *under $Q$, $\widetilde{H}$ is a Gaussian element of* $\mathrm{Sym}(d)$ *with $cI \preceq \mathrm{Cov}_Q(\widetilde{H}) \preceq CI$ and $|\mathbb{E}_Q\widetilde{H}| \le CP$, so Lemma S10.9 applies to $G = \widetilde{H}$ with $|\mu| \le CP$;* (b) *$\mathbb{E}_Q[\mathcal{N}_r^p \mid \widetilde{H}] \le C_p(P + |\widetilde{H}|)^p$ for every $p \ge 1$.*

*Proof. The exact form.* For $0 \le t \le r$ let $g(t) = f(M + tu)$, $h(t) = \nabla_\Theta f(M + tu)$. For every $C^4$ function, by Taylor's formula with integral remainder,

$$\tfrac{r^2}{6}g''(0) = [g(r) - g(0) - rg'(0)] - \tfrac{r}{3}[g'(r) - g'(0)] + \tfrac{1}{6}\int_0^r t(r - t)^2 g''''(t)\,dt,$$
$$rh'(0) = [h(r) - h(0)] - \int_0^r (r - t)h''(t)\,dt.$$

Under $Q$, $g(0) = b$, $g(r) = b - kr^3$, $g'(0) = g'(r) = 0$, $h(0) = h(r) = 0$, so the bracketed pin combinations are $-kr^3$ and 0; with $t = rs$, listing the $uu$, $u\Theta$ and $\Theta\Theta$ blocks,

$$\widetilde{H} = \mathsf{m}_\kappa + \mathcal{G}_r(f), \qquad \mathsf{m}_\kappa := -6\kappa e_u e_u^\top,$$
$$\mathcal{G}_r(f) := \Big(\int_0^1 s(1 - s)^2\partial_u^4 f(M + rsu)\,ds;\ -\int_0^1 (1 - s)\partial_u^2\nabla_\Theta f(M + rsu)\,ds;\ D_\Theta^2 f(M)\Big). \tag{S10.62}$$

$\mathcal{G}_r$ is linear on $C^4$ functions with $|\mathcal{G}_r(g)| \le C\|g\|_{C^4}$; at $r = 0$ its blocks are $\partial_u^4 g(0)/12$, $-\partial_u^2\nabla_\Theta g(0)/2$ and $D_\Theta^2 g(0)$, i.e. $(f_4/12, -\gamma/2, A)$ for the cusp jets.

(a) Under $Q$, $f$ is $F$ conditioned on $U_r = v_r$, so $\mathrm{Cov}_Q(\widetilde{H}) = \mathrm{Cov}(\mathcal{G}_r(F) \mid U_r)$, the Schur complement of $U_r$ in the Gram matrix $\mathfrak{G}_r$ of $(\mathcal{G}_r(F), U_r)$. Each entry of $(\mathcal{G}_r(F), U_r)$ is an integral of some $\partial^\alpha F(M + rsu)$, $|\alpha| \le 4$, against a fixed finite measure in $s \in [0, 1]$ (for the difference rows by Taylor's formula), so each entry of $\mathfrak{G}_r$ is $\iint \pm\partial^{\alpha+\beta}K_L(r(s - s')u)\,\mu_i(ds)\mu_j(ds')$, continuous on the compact $[0, r_0^*] \times \{\text{frames}\}$, including $r = 0$. For $r > 0$, by the two identities read for an arbitrary $C^4$ function, $\mathcal{G}_r(F)$ is $\Lambda_r^{-1}D^2F(M)\Lambda_r^{-1}$ minus a linear function of the pins, and $U_r$ is an invertible image of the pins ($|\det| = 12r^{-(d+3)}$); so $(\mathcal{G}_r(F), U_r)$ is an invertible image of $(D^2F(M), F(M), F(S), \nabla F(M), \nabla F(S))$, distinct derivative functionals at the distinct sites $M \ne S$, linearly independent [7] (Lemma S12.8). At $r = 0$, $(\mathcal{G}_0(F), U_0)$ is an invertible image

of $\partial_u^4 F(0)$, $\partial_u^2 \nabla_\Theta F(0)$, $D_\Theta^2 F(0)$ and the six pinned jets, distinct jets at one point in the frame $(u, \Theta)$, independent [7] (Lemma S12.8). So $\mathfrak{G}_r \succeq cI$ uniformly; hence $\mathrm{Cov}_Q(\widetilde{H}) \succeq \lambda_{\min}(\mathfrak{G}_r) I \succeq cI$ (the inverse of the Schur complement is a block of $\mathfrak{G}_r^{-1}$) and $\mathrm{Cov}_Q(\widetilde{H}) \preceq \mathrm{Cov}(\mathcal{G}_r(F)) \preceq CI$. And $\mathbb{E}_Q \widetilde{H} = \mathsf{m}_\kappa + \mathrm{Cov}(\mathcal{G}_r(F), U_r)\,\mathrm{Cov}(U_r)^{-1} v_r$, with $|v_r| \le |b| + 14k$, a bounded regression coefficient and $|\mathsf{m}_\kappa| = 6\kappa \le 6$; so $|\mathbb{E}_Q \widetilde{H}| \le CP$.

(b) Conditioning $Q$ on $\widetilde{H}$ is conditioning $F$ on $(\mathcal{G}_r(F), U_r) = (\widetilde{H} - \mathsf{m}_\kappa, v_r)$, with uniformly invertible Gram matrix $\mathfrak{G}_r$. The conditional mean of $F$ has $C^9$ norm $\le C(P + |\widetilde{H}|)$, and the conditional covariance is at most $\mathrm{Cov}\, F$, so by Anderson's inequality the conditional moments of the centred part's $C^9$ norm are at most those of $F$. As $\mathcal{N}_r = 1 + k + \|F_r\|_{C^9}$ and $k \le P$, (b) follows; no coupling is needed. □

**Lemma S10.23** (the elder weight through the rescaled Hessian)**.** *For $0 < r \le r_0^*$, $b \in \mathbb{R}$ and $0 < k \le r$, with $\ell = kr^3$ and $\kappa = k/r \le 1$,*

$$\mathbb{E}_Q[(W_r/r^2)\, e] \le C\ell^{2/3} r^{-2} (\kappa + r) P^N. \tag{S10.63}$$

*Proof.* On $\{W_r > 0\} \cap \{e = 1\}$, $H_M \prec 0$, so $\widetilde{H} \prec 0$, and by (S10.18) $\lambda_{\min}(-H_M) \le (\ell K_f^2/c_L)^{1/3}$ with $K_f = 1 + \|D^2 f\|_\infty + \|D^3 f\|_\infty \le C_K \mathcal{N}_r$. As $-H_M = \Lambda_r(-\widetilde{H})\Lambda_r$, $|\Lambda_r v| \ge r|v|$ $(r \le r_0^* \le 1)$ and $-\widetilde{H} \succ 0$, $v^\top(-H_M)v \ge r^2 \lambda_{\min}(-\widetilde{H})|v|^2$, so $\lambda_{\min}(-H_M) \ge r^2 \lambda_{\min}(-\widetilde{H})$ (this can fail for $r > 1$ or indefinite $\widetilde{H}$). Hence

$$0 < \lambda_{\min}(-\widetilde{H}) \le \varepsilon(\mathcal{N}_r), \qquad \varepsilon(t) := C_\varepsilon \ell^{1/3} r^{-2} t^{2/3}, \quad C_\varepsilon := (C_K^2/c_L)^{1/3}. \tag{S10.64}$$

Now $|\det K_M| = r|\det \widetilde{H}| \le r\lambda_{\min}(-\widetilde{H})\|\widetilde{H}\|^{d-1}$, and, the pair being typed, the sign window applied to (S10.16) gives $|\det K_S| \le 12k|\Delta| + 2\max_i |R_i'| \le Cr(\kappa + r)\mathcal{N}_r^{N_2}$ $(|\Delta| \le \mathcal{N}_r^m,\ 1 + k \le \mathcal{N}_r)$. As $\{\widetilde{H} \prec 0, \lambda_{\min}(-\widetilde{H}) \le \varepsilon\} \subset \{|\lambda_{\max}(\widetilde{H})| \le \varepsilon\}$, everywhere

$$(W_r/r^2)e \le Cr^2(\kappa + r)\, \varepsilon(\mathcal{N}_r) \|\widetilde{H}\|^{d-1} \mathcal{N}_r^{N_2} \mathbf{1}\{|\lambda_{\max}(\widetilde{H})| \le \varepsilon(\mathcal{N}_r)\}.$$

On $\{\mathcal{N}_r \in [2^j, 2^{j+1})\}$, $\varepsilon(\mathcal{N}_r) \le \varepsilon_j := \varepsilon(2^{j+1})$; conditioning on $\widetilde{H}$ and using $\mathbb{P}_Q(\mathcal{N}_r \ge 2^j \mid \widetilde{H}) \le C_p 2^{-jp}(P + |\widetilde{H}|)^p$ (Lemma S10.22(b)),

$$\mathbb{E}_Q[(W_r/r^2)e] \le Cr^2(\kappa + r) \sum_{j \ge 0} 2^{(j+1)N_2 - jp} \varepsilon_j\, \mathbb{E}_Q\big[\|\widetilde{H}\|^{d-1}(P + |\widetilde{H}|)^p \mathbf{1}\{|\lambda_{\max}(\widetilde{H})| < 2\varepsilon_j\}\big].$$

By Lemma S10.22(a) and Lemma S10.9 (first bound, $d \times d$, $n = d - 1 + p$, $n' = n + d(d-1)/2$) the expectation is $\le CP^{p+n'}\varepsilon_j$, so the $j$-th summand is $\le C 2^{j(N_2 + 4/3 - p)} P^N \ell^{2/3} r^{-4}$; with $p = N_2 + 2$ the series converges and gives (S10.63). □

*Proof of Proposition S10.7.* Fix $0 < r \le r_0^*$, $b$, $u$, $0 \le k \le r$; (S10.16) (including $k = 0$), (S10.53) and Lemma S10.18(ii)–(iii) hold on $(0, r_0^*]$. *Step W1.* Let $\bar{R} := \max(|R_M'|, |R_S'|) \le C_R r^2 \mathcal{N}_r^{N_1}$ $(k \le 1)$ and $\eta := C_R r \mathcal{N}_r^{N_1} \ge \bar{R}/r$. On $\{W_r > 0\}$ the sign window applied to (S10.16) gives $|rY_r| \le 6k|\Delta| + \bar{R}$ and $|\det K_i| \le 12k|\Delta| + 2\bar{R}$; so $|Y_r| \le 6\kappa|\Delta| + \eta$ and $W_r/r^2 \le 4r^2(6\kappa|\Delta| + \eta)^2$. Hence, everywhere,

$$W_r/r^2 \le 4r^2(6\kappa|\Delta| + \eta)^2 \mathbf{1}\{|Y_r| \le 6\kappa|\Delta| + \eta\}. \tag{S10.65}$$

*Step W2.* $Y_r$ is affine in $f_4$ with slope $\Delta/12$ and $J''$-measurable intercept, so for $J''$-measurable $\eta'$, $\{|Y_r| \le 6\kappa|\Delta| + \eta'\} = \{f_4 \in I\}$ with $I$ a $J''$-measurable interval of length $144\kappa + 24\eta'/|\Delta|$ $(\Delta \ne 0$ a.s.). On $\{\mathcal{N}_r \in [2^j, 2^{j+1})\}$, $\eta \le \eta_j := C_R r 2^{(j+1)N_1}$; bounding $\mathbf{1}\{\mathcal{N}_r \ge 2^j\}$ by Lemma S10.18(ii), integrating $f_4$ by (S10.53) and using also the trivial bound 1,

$$\mathbb{E}_Q[W_r/r^2] \le Cr^2 \sum_{j \ge 0} 2^{-jp} \mathbb{E}_Q\big[(6\kappa|\Delta| + \eta_j)^2 \min(1, 144\kappa + 24\eta_j/|\Delta|)(P + |J''|)^p\big].$$

By $(x+y)^2\min(1,a+b)\le 2(x^2+y^2)(a+\min(1,b))$ $(x,y,a,b\ge 0)$ the integrand is $\le C(\kappa^2\Delta^2+\eta_j^2)(\kappa+\min(1,\eta_j/|\Delta|))(P+|J''|)^p$. *Step W3.* $\kappa^3\Delta^2$ contributes $C\kappa^3P^N$; $\kappa^2\Delta^2\min(1,\eta_j/|\Delta|)\le\kappa^2|\Delta|\eta_j$ contributes $C\kappa^2r2^{jN'}P^N$; $\kappa\eta_j^2$ contributes $C\kappa r^2 2^{2jN'}P^N$. For $\eta_j^2\min(1,\eta_j/|\Delta|)$ condition on $J'''$ and split into $\{|\Delta|<\varepsilon\}$, $\{2^i\varepsilon\le|\Delta|<2^{i+1}\varepsilon\}$ for $0\le i<\lceil\log_2(1/\varepsilon)\rceil$, and $\{|\Delta|\ge 1\}$; Lemma S10.18(iii) gives, for $0<\varepsilon\le\frac12$,

$$\mathbb{E}[(1+\|A\|)^n\min(1,\varepsilon/|\Delta|)\mid J'''] \le C(P+|J'''|)^{n'}\varepsilon\,(4+2\log_2(1/\varepsilon)). \tag{S10.66}$$

With $\varepsilon=\eta_j\ge C_Rr$, $\log_2(1/\eta_j)\le\log_2(1/(C_Rr))$, so this term contributes $Cr^3\log(2/r)2^{3(j+1)N_1}P^N$ (for $\varepsilon>\frac12$ bound the minimum by $1\le 2\varepsilon$). For $p$ large the sum over $j$ converges, and

$$\mathbb{E}_Q[W_r/r^2]\le Cr^2(\kappa^3+\kappa^2r+\kappa r^2+r^3\log(2/r))P^N\le Cr^2(\kappa+r)^2(\kappa+r\log(2/r))P^N,$$

which is (S10.33). Then (S10.34) follows from $r^{-2}\mathcal{A}_r=12\pi_r(v_r)r^{-2}\mathbb{E}_Q[W_r/r^2]$, (S10.10) and $P^Ne^{-ck^2}\le C(1+|b|)^N$. At $k=0$, $r^{d-1}\Psi_0(b,ru)=r^{-2}\mathcal{A}_r(b,0,u)$, so (S10.34) at $\kappa=0$ bounds $\Psi_0$; integrating over $b$, $u$ and $r<\rho$, with $\int_0^\rho r^3\log(2/r)\,dr=\frac{\rho^4}{4}(\log(2/\rho)+\frac14)$, gives (S10.35). □

S10.10. **Proofs of the theorems.** *Decomposition.* Fix $r_0\in(0,r_0^*]$ and split points $\rho_f<\rho_c<a$ with $\ell^{1/3}\le\rho_f<\ell^{1/4}$, $\rho_c\le r_2$, $\rho_c<r_0$, $a<r_0^*$; put $s_f:=\rho_f\ell^{-1/4}$, $s_c:=\rho_c\ell^{-1/4}$. Integrands are at $(b,\ell/r^3,u)$ unless stated otherwise; inner integrals are $db\,d\sigma(u)$. By (S10.13) and (S10.14) at $r_0$, and by (S10.13) at $r_0^*$,

$$\nu_{\mathrm{cand}}(\ell)-c_{d,L}\ell^{-1/3}-B_{d,L}=I_f+I_c+J_2+J_3-J_4+J_5, \tag{S10.67}$$

$$\nu_{\mathrm{eld}}(\ell)-c_{d,L}\ell^{-1/3}=I_f^{\mathrm{eld}}+I_c^{\mathrm{eld}}+I_m^{\mathrm{eld}}-J_4+\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell), \tag{S10.68}$$

where $I_f^{\mathrm{eld}},I_c^{\mathrm{eld}}$ are $I_f,I_c$ with $\mathcal{A}_r^{\mathrm{eld}}$ in place of $\mathcal{A}_r$ and

$$I_f=\int_0^{\rho_f}\!\!\iint r^{-2}(\mathcal{A}_r-\mathcal{A}_0),\qquad I_c=\int_{\rho_f}^{\rho_c}\!\!\iint r^{-2}(\mathcal{A}_r-\mathcal{A}_0),\qquad I_m^{\mathrm{eld}}=\int_{\rho_c}^{r_0^*}\!\!\iint r^{-2}\mathcal{A}_r^{\mathrm{eld}}\ge 0,$$

$$J_2=-\int_0^{\rho_c}\!\!\iint r^{-2}\mathcal{A}_r(b,0,u),\qquad J_3=\int_{\rho_c}^{r_0}\!\!\iint r^{-2}\big[\mathcal{A}_r(b,\tfrac{\ell}{r^3},u)-\mathcal{A}_r(b,0,u)\big],$$

$$J_4=\int_{\rho_c}^{\infty}\!\!\iint r^{-2}\mathcal{A}_0\ge 0,\qquad J_5=\int_{\mathrm{dist}(0,y)\ge r_0}\int_{\mathbb{R}}(\Psi_\ell^{\mathrm{cand}}-\Psi_0)\,db\,dy.$$

*Fold region.* Here $k=\ell/r^3\ge r$. By (S10.38), $r^{-2}(\mathcal{A}_r-\mathcal{A}_0)=\mathcal{A}_2(b,\ell/r^3,u)+E_f$ with $|E_f|\le C(r+r/k)(1+|b|)^Ne^{-cb^2}$ (as $P^Ne^{-ck^2}\le C(1+|b|)^N$), and $r/k=r^4/\ell$, so $|\int E_f|\le C(\rho_f^2+\rho_f^5/\ell)$. With $k=\ell/r^3$, $dr=-\frac13\ell^{1/3}k^{-4/3}dk$ and $k_f:=\ell/\rho_f^3$ (so $\ell^{1/3}k_f^{-1/3}=\rho_f$),

$$\int_0^{\rho_f}\mathcal{A}_2\,dr=\frac{\ell^{1/3}}{3}\int_0^{\infty}(\mathcal{A}_2(k)-\mathcal{A}_2(0))k^{-4/3}dk+\rho_f\mathcal{A}_2(b,0,u)-\frac{\ell^{1/3}}{3}\int_0^{k_f}(\mathcal{A}_2(k)-\mathcal{A}_2(0))k^{-4/3}dk.$$

The integrals converge absolutely by (S10.39) (which proves the absolute convergence of (S10.22)), and the last term is $O(\ell^{1/3}k_f^{2/3})=O(\ell\rho_f^{-2})$, uniformly with the factor $(1+|b|)^Ne^{-cb^2}$. So $I_f=c_2\ell^{1/3}+\rho_f\iint\mathcal{A}_2(b,0,u)+O(\rho_f^2+\rho_f^5/\ell+\ell\rho_f^{-2})$. For the elder density write $\mathcal{A}_r^{\mathrm{eld}}=\mathcal{A}_r-\mathcal{A}_r^{\mathrm{rej}}$: for $r\le\rho_f<\ell^{1/4}$, $k>r$ and $r\le r_2\le r_0^*\le r_K$, so (S10.12) and (S10.10) give $0\le r^{-2}\mathcal{A}_r^{\mathrm{rej}}\le Cr^{-2}(r^3/k)P^Ne^{-c(b^2+k^2)}\le C(r^4/\ell)(1+|b|)^Ne^{-cb^2}$, which integrates to $O(\rho_f^5/\ell)$; so $I_f^{\mathrm{eld}}=I_f+O(\rho_f^5/\ell)$.

*Cusp region.* Here $k\le\ell\rho_f^{-3}\le 1$ and $r\le\rho_c\le r_2$. By Lemmas S10.13 and S10.20 with $\kappa=\ell/r^4$, $r^{-2}(\mathcal{A}_r-\mathcal{A}_0)$ and $r^{-2}(\mathcal{A}_r^{\mathrm{eld}}-\mathcal{A}_0)$ are $(\mathcal{C}^{\mathrm{cand}}-\mathcal{C}^{\mathrm{con}})(b,\ell/r^4,u)$ and $(\mathcal{C}^{\mathrm{eld}}-\mathcal{C}^{\mathrm{con}})(b,\ell/r^4,u)$ up to errors whose integrals over $[\rho_f,\rho_c]$ we call $\mathcal{E}$ and $\mathcal{E}^{\mathrm{eld}}$. With $r=s\ell^{1/4}$ the main terms are $\ell^{1/4}\iint\int_{s_f}^{s_c}(\mathcal{C}^\bullet-\mathcal{C}^{\mathrm{con}})(b,s^{-4},u)\,ds$, $\bullet\in\{\mathrm{cand},\mathrm{eld}\}$. As $0\le\mathcal{C}^{\mathrm{con}}-\mathcal{C}^\bullet\le\mathcal{C}^{\mathrm{con}}\le 12\pi_0\cdot 36\kappa^2\mathbb{E}_0[\Delta^2\mid$

$b] \le Cs^{-8}(1+|b|)^N e^{-cb^2}$, the parts over $s > s_c$ are $O(s_c^{-7})$; by Lemmas S10.14 and S10.21 ($s \le s_f < 1$ means $\kappa \ge 1$), $\int_0^{s_f} (\mathcal{C}^\bullet - \mathcal{C}^{\mathrm{con}})\, ds = s_f \mathcal{A}_2(b,0,u) + O(s_f^5)$. With (S10.7), $\ell^{1/4} s_f^5 = \rho_f^5/\ell$, $\ell^{1/4} s_c^{-7} = \ell^2 \rho_c^{-7}$, and the cancellation of the terms $\pm \rho_f \iint \mathcal{A}_2(b,0,u)$,

$$\begin{aligned} I_f + I_c &= I^{\mathrm{cand}} \ell^{1/4} + c_2 \ell^{1/3} + O(\rho_f^2 + \rho_f^5/\ell + \ell \rho_f^{-2} + \ell^2 \rho_c^{-7} + \mathcal{E}), \\ I_f^{\mathrm{eld}} + I_c^{\mathrm{eld}} &= c_1 \ell^{1/4} + c_2 \ell^{1/3} + O(\rho_f^2 + \rho_f^5/\ell + \ell \rho_f^{-2} + \ell^2 \rho_c^{-7} + \mathcal{E}^{\mathrm{eld}}). \end{aligned} \tag{S10.69}$$

By (S10.45) and (S10.56), $\mathcal{E}, \mathcal{E}^{\mathrm{eld}} \le C \int_{\rho_f}^{\rho_c} r(1+\ell r^{-4})^2 dr \le C(\rho_c^2 + \ell \rho_f^{-2} + \ell^2 \rho_f^{-6})$; by (S10.44), $\mathcal{E} \le C \int_{\rho_f}^{\rho_c} r(1+\ell r^{-4})\, dr \le C(\rho_c^2 + \ell \rho_f^{-2})$; and by (S10.55),

$$\mathcal{E}^{\mathrm{eld}} \le C \int_{\rho_f}^{\rho_c} (r + \ell r^{-3} + \ell^2 r^{-13/2} + \ell^3 r^{-10})\, dr \le C(\rho_c^2 + \ell \rho_f^{-2} + \ell^2 \rho_f^{-11/2} + \ell^3 \rho_f^{-9}).$$

*Remaining candidate terms.* Since $r^{-2}\mathcal{A}_r(b,0,u) = r^{d-1}\Psi_0(b, ru)$, (S10.17) gives $|J_2| \le C\rho_c^3$; on $r \ge \rho_c$, $k \le \ell \rho_c^{-3} \le 1$, so (S10.20) gives $|J_3| \le C \int_{\rho_c}^{\infty} (\ell^2 r^{-8} + \ell r^{-3})\, dr \le C(\ell^2 \rho_c^{-7} + \ell \rho_c^{-2})$; $0 \le J_4 \le C\ell^2 \rho_c^{-7}$ as $\mathcal{A}_0 \le k^2 H$; and $|J_5| \le C_{r_0} \ell$ by (S10.20).

*Intermediate separations.* On $[\rho_c, r_0^*]$, $k \le \ell \rho_c^{-3} \le 1$; split $I_m^{\mathrm{eld}}$ at $a$. On $[\rho_c, a]$, $e \le 1$; by (S10.17) and (S10.10), $0 \le r^{-2}\mathcal{A}_r^{\mathrm{eld}} \le r^{-2}\mathcal{A}_r \le Cr^{-2}(k^2+r^4)P^N e^{-c(b^2+k^2)} \le C(\kappa^2+r^2)(1+|b|)^N e^{-cb^2}$, $\kappa = \ell r^{-4}$, which integrates to $\le C(\ell^2 \rho_c^{-7} + a^3)$; alternatively, if $\ell \rho_c^{-4} \le 1$ (so $k \le r$ there), (S10.34) gives $r^{-2}\mathcal{A}_r^{\mathrm{eld}} \le C(1+\log(2/\rho_c))(\kappa+r)^3(1+|b|)^N e^{-cb^2}$, using $(\kappa+r)^2(\kappa+r\vartheta) \le (1+\vartheta)(\kappa+r)^3$ for $\vartheta = \log(2/r) \ge 0$ and $\log(2/r) \le \log(2/\rho_c)$, and as $(\kappa+r)^3 \le 4(\kappa^3+r^3)$ this integrates to $\le C(1+\log(2/\rho_c))(\ell^3 \rho_c^{-11} + a^4)$. On $[a, r_0^*]$, if $\ell a^{-4} \le 1$ (so $k \le r$), Lemma S10.23 with (S10.10) gives $0 \le r^{-2}\mathcal{A}_r^{\mathrm{eld}} \le C\ell^{2/3} r^{-4}(\kappa+r)(1+|b|)^N e^{-cb^2} = C\ell^{2/3}(\ell r^{-8} + r^{-3})(1+|b|)^N e^{-cb^2}$, which integrates to $\le C(\ell^{5/3} a^{-7} + \ell^{2/3} a^{-2})$.

*Far part.* $0 \le \nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) \le C\ell^{1/3}$ by (S10.19) at $r_0 = r_0^* < L/(4\sqrt{2}) < L/2$; and $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*}(\ell) \le C(r_0^*)\ell^{2/3}$ for $0 < \ell \le 1$ by (S10.21) at $\rho = r_0^* \in (0, L/2]$, whose far density is the integral (S10.8).

*Proof of Theorems S10.1, S10.2 and Corollary S10.3.* Take $\rho_f = \ell^{1/4+1/44}$, $\rho_c = \ell^{1/4-1/36} = \ell^{2/9}$, $a = \ell^{2/15}$ and $\ell$ small; then $\rho_f < \rho_c < a$ as $\frac{3}{11} > \frac{2}{9} > \frac{2}{15}$, $\ell \rho_f^{-3} = \ell^{2/11} \le 1$ and $\ell a^{-4} = \ell^{7/15} \le 1$. Use (S10.45), (S10.56), and (S10.17) on $[\rho_c, a]$. The error terms $\rho_f^2$, $\rho_f^5/\ell$, $\ell\rho_f^{-2}$, $\ell^2\rho_f^{-6}$, $\rho_c^2$, $\rho_c^3$, $\ell^2\rho_c^{-7}$, $\ell\rho_c^{-2}$, $\ell$ are $\ell$ to the powers $\frac{6}{11}$, $\frac{4}{11}$, $\frac{5}{11}$, $\frac{4}{11}$, $\frac{4}{9}$, $\frac{2}{3}$, $\frac{4}{9}$, $\frac{5}{9}$, 1; the least is $\frac{4}{11} > \frac{1}{3}$, which proves (S10.23) and the last assertion of Theorem S10.1. For the elder density, $J_4 \le C\ell^{4/9}$ and $0 \le I_m^{\mathrm{eld}} \le C(\ell^{4/9} + \ell^{2/5} + \ell^{11/15} + \ell^{2/5}) \le C\ell^{2/5}$; as $\frac{4}{11} < \frac{2}{5} < \frac{4}{9}$, (S10.68) and (S10.69) give (S10.24). Then (S10.25) holds as $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*} = O(\ell^{1/3})$ absorbs $c_2\ell^{1/3}$ and $O(\ell^{4/11})$; (S10.26) as $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*} \ge 0$; (S10.27) as $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*} \le C\ell^{2/3} \le C\ell^{4/11}$. By (S10.24) the two sides of each equivalence differ by $O(\ell^{4/11})$, which is $O(\ell^\theta)$ for $\theta \le \frac{4}{11}$ and $o(\ell^{1/3})$. For Corollary S10.3, subtract (S10.24) from (S10.23): the terms $c_{d,L}\ell^{-1/3}$ and $c_2\ell^{1/3}$ cancel, giving (S10.28); (1) follows from $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*} \le C\ell^{1/3}$, (2) from $\nu_{\mathrm{eld}}^{\mathrm{far},r_0^*} \ge 0$, (3) from (S10.21). □

*Proof of Theorems S10.4, S10.5 and Corollary S10.6.* Take $\rho_f = \ell^{2/7}$, $\rho_c = \ell^{2/9}$, $a = \ell^{1/9}$ (so $s_f = \ell^{1/28}$, $s_c = \ell^{-1/36}$) and $\ell$ small; then $\ell^{1/3} < \rho_f < \ell^{1/4} < \rho_c < a$, $k \le \ell^{1/7} \le 1$ on $[\rho_f, \infty)$ and $\kappa \le \ell^{1/9} \le 1$ on $[\rho_c, \infty)$. Use (S10.44), (S10.55), and (S10.34) on $[\rho_c, a]$. The ledger (exponents of $\ell$): exactly $\frac{3}{7}$ for $\rho_f^5/\ell$ (fold layer, rejected fold mass, overlap tails), $\ell\rho_f^{-2}$ (finite part, linear cusp error $r\kappa$) and, for the elder density only, $\ell^2\rho_f^{-11/2}$ and $\ell^3\rho_f^{-9}$ (the terms $\kappa^2 r^{3/2}$, $\kappa^3 r^2$ of (S10.55)); larger for the others: $\rho_f^2$ ($\frac{4}{7}$); $\rho_c^2$, $\ell^2\rho_c^{-7}$ (cusp errors, upper tail, $J_3$, $J_4$) and $a^4\log(1/\ell)$, $\ell^{2/3}a^{-2}$ ($I_m^{\mathrm{eld}}$), all $\frac{4}{9}$; $\ell^3\rho_c^{-11}\log(1/\ell)$ ($I_m^{\mathrm{eld}}$) and $\ell\rho_c^{-2}$ ($J_3$), $\frac{5}{9}$; $\rho_c^3$ ($J_2$) and the far part under (S10.21), $\frac{2}{3}$; $\ell^{5/3}a^{-7}$ ($I_m^{\mathrm{eld}}$), $\frac{8}{9}$; $\ell$ ($J_5$), 1. In particular $0 \le I_m^{\mathrm{eld}} \le C(\ell^{5/9} + \ell^{4/9})\log(1/\ell) + C(\ell^{8/9} + \ell^{4/9}) = O(\ell^{4/9}\log(1/\ell))$. The

least exponent is $\frac{3}{7}$, which gives (S10.29), (S10.30) and the bound on the intermediate separations; the far bound gives (S10.31), and the equivalence for $\theta \le \frac{3}{7}$ follows from (S10.30). Subtracting (S10.30) from (S10.29) gives (S10.32); its second form uses (S10.21). The splits are optimal for these terms: $\rho_f = \ell^{2/7}$ balances $\rho_f^5/\ell$ against $\ell\rho_f^{-2}$, $\rho_c = \ell^{2/9}$ balances $\rho_c^2$ against $\ell^2\rho_c^{-7}$, and $a = \ell^{1/9}$ balances $a^4$ against $\ell^{2/3}a^{-2}$. □

## S11. The constant $c_2$

This section concerns the constant $c_2$ of the three-term law (2.12) of the main text. In §S11.1, $c_2$ is defined as a finite part of the fold-scale coefficient of the two-point kernel, and the mechanism that makes this finite part the coefficient of $\ell^{1/3}$ is described; the expansions that prove it are those of Section S10. In §S11.2 the corresponding coefficient expression is evaluated in closed form, with a certified enclosure, for the Gaussian kernel on $\mathbb{R}^d$, $d \le 3$. In §S11.3 the expression is written through the derivatives of order at most eight of the covariance, and an explicit bound is proved for its change under a change of covariance; for the normalized periodization of side $L \ge 24$ the change is below $10^{-60}$. In §S11.4 the two identifications on which the application to the side-24 field rests are derived from Section S10 (Corollary S11.11), and the resulting enclosures of $c_2$ and of $c_2/c_{3,24}$ follow.

S11.1. **Definition and the fold-scale mechanism.** *Notation.* We use the notation of Section S8: $m = d-1$; a direction $u \in S^{d-1}$ (the $t$ of the main text), an orthonormal frame $\Theta$ of $u^\perp$ and coordinates $(x, y) \in \mathbb{R} \times \mathbb{R}^m$; pins at $M = -\frac{r}{2}u$, $S = \frac{r}{2}u$ with

$$f(M) = b, \qquad f(S) = b - kr^3, \qquad \nabla f(M) = \nabla f(S) = 0, \qquad b \in \mathbb{R},\ k > 0, \tag{S11.1}$$

so that $k = \kappa/6$ and $\ell = kr^3$ in the notation of (3.5) and (3.11) of the main text; the observation vector $U_r$, with axial rows $\frac{1}{2}(f(M) + f(S))$, $\frac{1}{r}(f(S) - f(M))$, $\frac{1}{r}(\partial_x f(S) - \partial_x f(M))$,

$$\tau_r := \frac{6}{r^2}\Big[\partial_x f(M) + \partial_x f(S) - \frac{2}{r}(f(S) - f(M))\Big] \tag{S11.2}$$

and transverse rows $\frac{1}{2}(\nabla_y f(M) + \nabla_y f(S))$, $\frac{1}{r}(\nabla_y f(S) - \nabla_y f(M))$, whose value under (S11.1) is $v_r = (b - \frac{1}{2}kr^3, -kr^2, 0, 12k, 0, \dots, 0)$ (in the notation of (3.7) of the main text, $\tau_r = -12V_{\mathrm{corr}}^-$); the map from $(f(M), \nabla f(M), f(S), \nabla f(S))$ to $U_r$ has absolute determinant $12r^{-(d+3)}$; at $r = 0$, $U_0 = (f, \partial_x f, \partial_x^2 f, \partial_x^3 f, \nabla_y f, \nabla_y \partial_x f)(0)$ with target $v_0(b, k) = (b, 0, 0, 12k, 0, \dots, 0)$; $\pi_r$ and $\pi_0 = \pi_0(u; \cdot)$ are the densities of $U_r$ and $U_0$, and $\mathbb{E}_0[\,\cdot \mid b]$ is the expectation under the contact law at the zero-gap target $v_0(b, 0)$; $W_r = |\det H_M \det H_S| \mathbf{1}_{\{H_M \prec 0,\ \mathrm{ind}\, H_S = d-1\}}$ is the typed weight, $e$ the elder mark, and $P = 1 + |b| + k$. These objects are defined for every real, smooth, stationary covariance $K$ on $\mathbb{R}^d$ or on a torus for which the covariance of $U_r$ is nondegenerate, with

$$\mathrm{Cov}(D^\alpha f(0), D^\beta f(0)) = (-1)^{|\beta|} D^{\alpha+\beta} K(0). \tag{S11.3}$$

Two covariances occur: the reference kernel $K_{\mathrm{ref}}(z) = e^{-|z|^2/2}$ on $\mathbb{R}^d$, and its normalized periodization on $(\mathbb{R}/L\mathbb{Z})^d$,

$$K_L(z) = \Big(\sum_{n \in \mathbb{Z}^d} e^{-|Ln|^2/2}\Big)^{-1} \sum_{n \in \mathbb{Z}^d} e^{-|z+Ln|^2/2}, \tag{S11.4}$$

which for $d = 3$, $L = 24$ is (1.1) of the main text. At the midpoint put, as in (2.13) of the main text,

$$A = D_y^2 f(0), \qquad \Delta = \det A, \qquad \gamma = \nabla_y \partial_x^2 f(0), \qquad f_4 = \partial_x^4 f(0), \qquad Y = \frac{f_4}{12}\,\Delta - \frac{1}{4}\,\gamma^\top \mathrm{adj}(A)\,\gamma. \tag{S11.5}$$

For $d = 1$, $A$ is the empty matrix, $\Delta = 1$ and $Y = f_4/12$, and $S^0 = \{\pm 1\}$ carries counting measure, $|S^0| = 2$.

**Definition S11.1** (two-point kernel and signed fixed-cone surrogate)**.** For $r > 0$, $b \in \mathbb{R}$, $k > 0$ and $u \in S^{d-1}$:

(i) the *two-point kernel* and the *contact kernel* are

$$\mathcal{A}_r(b,k,u) = \frac{12\,\pi_r(v_r)\,\mathbb{E}[W_r \mid U_r = v_r]}{r^2}, \tag{S11.6}$$

$$\mathcal{A}_0(b,k,u) = 12\,\pi_0(u; v_0(b,k))\,36k^2\,\mathbb{E}[\Delta^2 \mathbf{1}_{\{A<0\}} \mid U_0 = v_0(b,k)]; \tag{S11.7}$$

$\mathcal{A}_r^{\mathrm{eld}}$ is $\mathcal{A}_r$ with $W_r$ replaced by $W_r e$;

(ii) the *signed fixed-cone surrogate* of a covariance $K$ and its birth integral are

$$\widetilde{\mathcal{A}}_r[K](b,k,u) = \frac{12\,\pi_r(v_r)\,\mathbb{E}[-\det H_M \det H_S\, \mathbf{1}_{\{A<0\}} \mid U_r = v_r]}{r^2}, \tag{S11.8}$$

$$\bar{\mathcal{A}}_r[K](k,u) = \int_{\mathbb{R}} \widetilde{\mathcal{A}}_r[K](b,k,u)\,db, \tag{S11.9}$$

with the indicator omitted for $d = 1$; $\widetilde{\mathcal{A}}_2[K]$ and $\bar{\mathcal{A}}_2[K]$ denote their coefficients of $r^2$ at $r = 0$. The same formulas define them at $k = 0$, where the pins are still nondegenerate. (S11.12) uses $\bar{\mathcal{A}}_2[K](0,u)$; under the hypothesis of Theorem S11.7, $\bar{\mathcal{A}}_2[K](\cdot,u)$ is continuous on $[0,\infty)$ by (S11.16).

On the typed event $|\det H_M|\,|\det H_S| = -\det H_M \det H_S$; the surrogate replaces the typed indicator by the fixed cone $\{A < 0\}$ of the midpoint transverse Hessian, so its integrand is a signed polynomial in Gaussian jets and it is not a count.

*The fold expansion* (Section S10, Lemma S10.12). Let $d \geq 2$, $L > 0$ and $K = K_L$. There are $r_* > 0$ and $C, c, N$ such that, for $0 < r \leq r_*$, $b \in \mathbb{R}$, $k > 0$ and $u \in S^{d-1}$,

$$\begin{aligned} |\mathcal{A}_r(b,k,u) - \mathcal{A}_0(b,k,u) - r^2\mathcal{A}_2(b,k,u)| &\leq Cr^3(1+k^{-1})P^N e^{-c(b^2+k^2)}, \\ |\mathcal{A}_2(b,k,u)| + |\partial_k \mathcal{A}_2(b,k,u)| &\leq CP^N e^{-c(b^2+k^2)}, \\ \mathcal{A}_2(b,0,u) &= -12\,\pi_0(u; v_0(b,0))\,\mathbb{E}_0[Y^2 \mathbf{1}_{\{A<0\}} \mid b], \end{aligned} \tag{S11.10}$$

where $\mathcal{A}_2(b,\cdot,u)$ is extended to a $C^1$ function on $[0,\infty)$. There is no term of order $r$.

**Definition S11.2** (the constant $c_2$ and the coefficient expression)**.** For $d \geq 2$ and the side-$L$ field,

$$c_2 := \frac{1}{3}\int_{S^{d-1}} \int_{\mathbb{R}} \int_0^\infty [\mathcal{A}_2(b,k,u) - \mathcal{A}_2(b,0,u)]\,k^{-4/3}\,dk\,db\,d\sigma(u). \tag{S11.11}$$

For $d \geq 1$ and a covariance $K$ as above, the *fixed-cone coefficient expression* is

$$c_2[K] := \frac{1}{3}\int_{S^{d-1}} \int_0^\infty [\bar{\mathcal{A}}_2[K](k,u) - \bar{\mathcal{A}}_2[K](0,u)]\,k^{-4/3}\,dk\,d\sigma(u). \tag{S11.12}$$

*Remark* S11.3. (i) By (S11.10), $|\mathcal{A}_2(b,k,u) - \mathcal{A}_2(b,0,u)| \leq C\min(k,1)\,(1+|b|)^N e^{-cb^2}$ after adjusting $C$ (the mean value theorem for $k \leq 1$, the two bounds separately for $k > 1$), so (S11.11) converges absolutely. Since $\mathcal{A}_2(b,0,u) \neq 0$ in general, $\int_0^\infty \mathcal{A}_2(b,k,u)k^{-4/3}dk$ diverges at $k = 0$, and (S11.11) is its Hadamard finite part at $k = 0$. (ii) (S11.12) is the same finite-part expression with the same spatial normalization, written for the surrogate and with the birth integral taken first; its convergence is part of Theorem S11.7. For $d = 1$ it defines an expression only; no persistence statement in $d = 1$ is derived from it here.

*Remark* S11.4 (why a finite part: fold–cusp matching). The following formal computation explains (S11.11); the expansions of Section S10 carry it out, with remainder $O(\ell^{3/7})$. Let $d \geq 2$. The near part of the elder density is $\int_{S^{d-1}} \int_{\mathbb{R}} \int_0^{r_*} r^{-2}\mathcal{A}_r^{\mathrm{eld}}(b, \ell/r^3, u)\,dr\,db\,d\sigma(u)$. Put $s = r\ell^{-1/4}$, so that $k = \ell/r^3 = \ell^{1/4}s^{-3}$ and $k/r = s^{-4}$. The integrand has two regimes.

(a) *Fold regime,* $s \ll 1$. Every typed pair is an elder pair up to a mass $O(r^3/k)$, and at fixed $k$, $r^{-2}\mathcal{A}_r = r^{-2}\mathcal{A}_0(b,k,u) + \mathcal{A}_2(b,k,u) + O(r)$ by (S11.10).

(b) *Cusp regime,* $s \asymp 1$. At fixed $s$, $r^{-2}(\mathcal{A}_r^{\mathrm{eld}} - \mathcal{A}_0)(b, rs^{-4}, u)$ converges as $r \downarrow 0$ (Section S9) to the elder cusp correction

$$\mathcal{C}(b,s,u) = -12\,\pi_0(u; v_0(b,0))\,\mathbb{E}_0\Big[(Y^2\mathbf{1}_{\{|Y|<2s^{-4}|\Delta|\}} + 36s^{-8}\Delta^2\mathbf{1}_{\{|Y|\geq 2s^{-4}|\Delta|\}})\mathbf{1}_{\{A<0\}} \;\Big|\; b\Big],$$

and $\mathcal{C}(b,s,u) \to \mathcal{A}_2(b,0,u)$ as $s \downarrow 0$, by (S11.10).

The composite approximation $r^{-2}\mathcal{A}_0(b,k,u) + \mathcal{A}_2(b,k,u) + [\mathcal{C}(b,s,u) - \mathcal{A}_2(b,0,u)]$ reproduces both regimes: for $s \ll 1$ the bracket vanishes, and for $s \asymp 1$, $k \to 0$ and $\mathcal{A}_2(b,k,u) \to \mathcal{A}_2(b,0,u)$. Its three pieces integrate separately; the substitutions $k = \ell/r^3$ and $s = r\ell^{-1/4}$ give

$$\begin{aligned}
\int_0^\infty r^{-2}\mathcal{A}_0(b,\ell/r^3,u)\,dr &= \frac{\ell^{-1/3}}{3}\int_0^\infty \mathcal{A}_0(b,k,u)\,k^{-2/3}\,dk,\\
\int_0^\infty [\mathcal{A}_2(b,\ell/r^3,u) - \mathcal{A}_2(b,0,u)]\,dr &= \frac{\ell^{1/3}}{3}\int_0^\infty [\mathcal{A}_2(b,k,u) - \mathcal{A}_2(b,0,u)]\,k^{-4/3}\,dk,\\
\int_0^\infty \mathcal{C}(b,r\ell^{-1/4},u)\,dr &= \ell^{1/4}\int_0^\infty \mathcal{C}(b,s,u)\,ds.
\end{aligned} \tag{S11.13}$$

After integration over $b$ and $u$, the first line is the leading term $c_{d,L}\ell^{-1/3}$, the second is $c_2\ell^{1/3}$, and the third is the cusp term $c_1\ell^{1/4}$ of Section S9. The subtracted constant $\mathcal{A}_2(b,0,u)$ is the limit of the cusp correction, that is, the part of the fold-scale coefficient already counted at the cusp scale. The neglected pieces are formally of order $\ell^{1/2}$: the fold and cusp remainders, the mixed term $\mathcal{A}_2(b,k,u) - \mathcal{A}_2(b,0,u) = O(k)$ at $s \asymp 1$, the intermediate separations and the far part; at a fold cutoff $r = \rho$ the interface terms $\pm\rho\iint \mathcal{A}_2(b,0,u)$ cancel because $\mathcal{C}(b,s,u) = \mathcal{A}_2(b,0,u) + O(s^4)$ as $s \downarrow 0$.

The elder rule enters only through the cusp window (the window $|\varphi| < 1/3$ of §5 of the main text): at the fold scale the rejected mass is $O(r^3/k)$, and the elder and candidate cusp losses have the same limit $\mathcal{A}_2(b,0,u)$ as $s \downarrow 0$. Hence $c_2$ is also the coefficient of $\ell^{1/3}$ of the candidate density, whose $\ell^{1/4}$ coefficient is $I^{\mathrm{cand}} = \frac{3^{1/4}}{2}c_1$ and which has the additional constant term $B_{d,L}$ ((2.17) of the main text): $c_2$ is a fold-scale quantity.

S11.2. **The Gaussian kernel: closed form and certified enclosure.** For $K_{\mathrm{ref}}$ put $c[K_{\mathrm{ref}}] := \frac{1}{3}\int_{S^{d-1}}\int_{\mathbb{R}}\int_0^\infty \mathcal{A}_0(b,k,u)\,k^{-2/3}\,dk\,db\,d\sigma(u)$, the leading constant computed by the same procedure.

**Theorem S11.5** (closed forms for the Gaussian kernel)**.** *Let* $d \in \{1,2,3\}$ *and* $K = K_{\mathrm{ref}}$. *Then*

| $d$ | $c[K_{\mathrm{ref}}]$ | $c_2[K_{\mathrm{ref}}]$ |
|---|---|---|
| 1 | $\Gamma(\frac{1}{6})\,12^{-1/6}/(6\pi^{3/2})$ | $\frac{3}{4}\,12^{1/6}\,\Gamma(\frac{5}{6})/\pi^{3/2}$ |
| 2 | $\Gamma(\frac{1}{6})\,12^{-1/6}/(9\pi^{3/2})$ | $\frac{13}{18}\,12^{1/6}\,\Gamma(\frac{5}{6})/\pi^{3/2}$ |
| 3 | $(\frac{29}{72} - \frac{\sqrt{6}}{12})\,\Gamma(\frac{1}{6})\,12^{-1/6}/\pi^{5/2}$ | $\frac{5}{48}(33 - 7\sqrt{6})\,12^{1/6}\,\Gamma(\frac{5}{6})/\pi^{5/2}$ |

*and*

$$\frac{c_2[K_{\mathrm{ref}}]}{c[K_{\mathrm{ref}}]} = \omega_d\,\frac{\Gamma(5/6)}{12^{2/3}\,\Gamma(1/6)}, \qquad \omega_1 = 54, \quad \omega_2 = 78, \quad \omega_3 = \frac{18(141 - \sqrt{6})}{25}. \tag{S11.14}$$

*For* $d = 2,3$, $c[K_{\mathrm{ref}}] = \Gamma(\frac{7}{6})(\frac{3}{2})^{1/3}D_{d-1}/(2\sqrt{3}\,\pi^{d-1/2})$ *with* $D_1 = \frac{4}{3}$, $D_2 = \frac{29}{6} - \sqrt{6}$; *for* $d = 3$ *this is* $c_{3,\infty}$ *of (2.11) of the main text. The constants lie in the intervals*

| $d$ | $c[K_{\mathrm{ref}}]$ | $c_2[K_{\mathrm{ref}}]$ |
|---|---|---|
| 1 | [0.1101103789590410, 0.1101103789590625] | [0.2300445802661503, 0.2300445802661998] |
| 2 | [0.07340691930602738, 0.07340691930604167] | [0.2215244106266632, 0.2215244106267110] |
| 3 | [0.04177593184059426, 0.04177593184060272] | [0.1612340491269447, 0.1612340491269810] |

*and*

$$\frac{c_2[K_{\mathrm{ref}}]}{c[K_{\mathrm{ref}}]} \in \begin{cases} [2.089217950577392,\ 2.089217950578207], & d=1,\\ [3.017759261945122,\ 3.017759261946299], & d=2,\\ [3.859496174547095,\ 3.859496174548665], & d=3. \end{cases}$$

*Proof (method of the certificate).* The proof is an exact computation in rational arithmetic, followed by an interval evaluation of the resulting closed forms.

*Step 1 (what is computed).* $K_{\mathrm{ref}}$ is isotropic, so the surrogate does not depend on $u$; take $u = e_1$, so that $\int_{S^{d-1}} d\sigma = |S^{d-1}|$. By the uniform Gaussian domination of Step 2 the coefficient of $r^2$ of the birth integral is the birth integral of $\widetilde{\mathcal{A}}_2$, and by Step 3 each coefficient is a polynomial in $(b,k)$ and the cone variables times a Gaussian factor, so the integrals over $b$ and $k$ in (S11.12) may be interchanged:

$$c_2[K_{\mathrm{ref}}] = \tfrac{1}{3}|S^{d-1}| \int_{\mathbb{R}} \int_0^\infty \left[\widetilde{\mathcal{A}}_2(b,k) - \widetilde{\mathcal{A}}_2(b,0)\right] k^{-4/3}\, dk\, db.$$

The fixed cone is $\{f_{yy}(0) < 0\}$ for $d = 2$ and $\{D_y^2 f(0) < 0\}$ for $d = 3$; for $d = 1$ there is no indicator.

*Step 2 (exact series in $r$).* By Taylor's formula along $e_1$ at $x = \mp r/2$, every quantity entering (S11.8) is a linear form in the jets $\partial^\alpha f(0)$ with coefficients in $\mathbb{Q}[[r]]$, and by (S11.3) the jet covariances are integers. In the rows of $U_r$ the divisions by $r$ and by $r^3$ are exact, because the low coefficients vanish identically. The pin covariance at $r = 0$ is invertible, so the conditional means and covariances given $U_r = v_r$ are computed exactly in $\mathbb{Q}[[r]]$. For $d \geq 2$ the Hessian entries are conditioned further on the transverse block $A$; for $d = 3$, transverse rotations preserve the law and $\det H_M \det H_S$, so $A$ may be taken diagonal, $A = \mathrm{diag}(\lambda_1, \lambda_2)$, with

$$\int_{\{A<0\}} g(A)\, p(A)\, dA = \pi \int_{\{\lambda_2<\lambda_1<0\}} g\, p\, (\lambda_1 - \lambda_2)\, d\lambda_1\, d\lambda_2$$

for functions $g$, $p$ invariant under conjugation by transverse rotations. By Isserlis' theorem $F := \mathbb{E}[\det H_M \det H_S \mid A,\ U_r = v_r]$ is an exact polynomial in $(r, b, k, \lambda_1, \lambda_2)$, and $\widetilde{\mathcal{A}}_r(b,k)$ is $r^{-2}$ times a constant times the integral over the cone of $F$ against the Gaussian densities of the pins at $v_r$ and of $A$ given the pins. Writing the two quadratic forms in the exponent as their values at $r = 0$ plus increments, the determinant ratio, the exponential of the increments and $F$ are expanded exactly in $r$. Since the covariances at $r = 0$ are nondegenerate, $\widetilde{\mathcal{A}}_r$ is real analytic in $r$ near 0, with a Gaussian domination uniform in small $r$; hence its Taylor coefficients are the cone integrals of the Taylor coefficients of the integrand. (Series are carried through $r^{10}$ and polynomials through $r^6$; repeating the whole derivation at orders 12 and 8 gives identical results.)

*Step 3 (exact structural facts).* The integrand vanishes to order $r^2$, and its coefficient of $r^3$ vanishes, so $\widetilde{\mathcal{A}}_r$ has no term of order $r$. The pin quadratic form at $r = 0$ is $q_0 = \frac{3}{2}b^2 + 24k^2$, with no $bk$ term, and the pin density is a constant times $e^{-q_0/2}$; the transverse exponent at $r = 0$ does not contain $k$ (parity); and the coefficients of $r^0$ and $r^2$ contain only even powers of $k$. So these two coefficients of the integrand have the form $e^{-12k^2} \sum_{l\ \mathrm{even}} a_l\, k^l$, where $a_l$ is a polynomial in $b$ and the cone variables times a Gaussian factor in them.

*Step 4 (integration in $k$).* For even $l \geq 0$,

$$\int_0^\infty k^{l-2/3} e^{-12k^2}\, dk = \tfrac{1}{2}\Gamma\big(\tfrac{l+1/3}{2}\big) 12^{-(l+1/3)/2},$$
$$\int_0^\infty \big(k^l e^{-12k^2} - [l=0]\big) k^{-4/3}\, dk = \tfrac{1}{2}\Gamma\big(\tfrac{l-1/3}{2}\big) 12^{-(l-1/3)/2}, \tag{S11.15}$$

the second for $l = 0$ by integration by parts, with $\Gamma(-\frac{1}{6}) = -6\Gamma(\frac{5}{6})$. Hence every term of $c[K_{\mathrm{ref}}]$ is a rational multiple of $\Gamma(\frac{1}{6})12^{-1/6}$ and every term of $c_2[K_{\mathrm{ref}}]$ a rational multiple of $\Gamma(\frac{5}{6})12^{-5/6}$.

*Step 5 (integration in $b$).* The $k$-free exponent is a quadratic $a_b b^2 + b\Lambda_1(\lambda) + \Lambda_0(\lambda)$; the shift $b \mapsto b - \Lambda_1/(2a_b)$ leaves the Gaussian moments $\int b^n e^{-a_b b^2} db$ and the cone exponent $\Lambda_0 - \Lambda_1^2/(4a_b)$.

*Step 6 (the cone).* For $d = 2$ the cone integrals are the half-line moments $\int_{-\infty}^0 \lambda^n e^{-g\lambda^2} d\lambda$ with $g = \frac{3}{16}$. For $d = 3$, with $\lambda_1 = -\xi_1$, $\lambda_2 = -\xi_1 - \xi_2$ ($\xi_1, \xi_2 > 0$), the exponent is $\alpha\xi_1^2 + \beta\xi_1\xi_2 + \gamma'\xi_2^2$ with $(\alpha, \beta, \gamma') = (\frac{3}{10}, \frac{3}{10}, \frac{1}{5})$ and $D_\theta := 4\alpha\gamma' - \beta^2 = \frac{3}{20}$. The quadrant moments $I_{i,j} = \int_0^\infty\int_0^\infty \xi_1^i \xi_2^j e^{-(\alpha\xi_1^2 + \beta\xi_1\xi_2 + \gamma'\xi_2^2)} d\xi_1 d\xi_2$ satisfy two integration-by-parts identities, in $\xi_1$ and in $\xi_2$; these form a $2\times 2$ linear system with determinant $D_\theta$, started from $I_{0,0} = \theta := (\frac{\pi}{2} - \arctan(\beta/\sqrt{D_\theta}))/\sqrt{D_\theta}$. Every $I_{i+1,j}$ with $j \geq 1$ is reached in two ways, and the two values agree exactly. The angle $\theta$ cancels exactly in $c[K_{\mathrm{ref}}]$ and in $c_2[K_{\mathrm{ref}}]$, which therefore lie in $\mathbb{Q}(\sqrt{6})$ times powers of $\pi$, times the Gamma factors of Step 4. This gives the table, and (S11.14) by division.

*Step 7 (exact checks).* The certificate verifies exactly: that $\theta$ occurs in no result; that for $d = 2, 3$, $c[K_{\mathrm{ref}}]$ equals $\Gamma(\frac{7}{6})(\frac{3}{2})^{1/3} D_{d-1}/(2\sqrt{3}\pi^{d-1/2}) = \Gamma(\frac{1}{6})12^{-1/6} D_{d-1}/(12\pi^{d-1/2})$ (using $\Gamma(\frac{7}{6}) = \Gamma(\frac{1}{6})/6$ and $(\frac{3}{2})^{1/3} = \sqrt{3}\, 12^{-1/6}$; for $d = 3$ this equals $2^{-23/6} 3^{-8/3}(29\sqrt{6} - 36)\Gamma(\frac{1}{6})\pi^{-5/2}$); that for $d = 1$ the two constants equal $C_0$ and $2B_2$ of the one-dimensional law of §5 of the main text; that the structural facts of Step 3 hold; and that a repetition at higher truncation orders gives identical results.

*Step 8 (evaluation).* The closed forms are evaluated in interval arithmetic with outward rounding by one unit in the last place at every operation: $\Gamma(\frac{1}{6})$ and $\Gamma(\frac{5}{6})$ from Stirling's series for $\log\Gamma$ at $q + 10$, with the remainder bounded by the first omitted term (valid on the positive real axis; [5, §5.11(ii)]), divided by the exact product $q(q+1)\cdots(q+9)$; $12^{p/q}$ as the exponential of a logarithm; $\pi$ and square roots by interval routines. The published intervals are the computed ones widened by $10^{-14}$ relative and rounded outward to sixteen significant digits; the ratio intervals are evaluated from (S11.14) and are required to meet the interval quotients of the enclosures of $c_2[K_{\mathrm{ref}}]$ and $c[K_{\mathrm{ref}}]$. The certificate repeats the exact derivation and the evaluation and compares the result with the published table, and each of five seeded defects (pin target $11k$ in place of $12k$; finite-part subtraction omitted; sign of the cone cross term; polynomial truncation order 3; $|S^1| = \pi$) makes it fail. □

*Remark* S11.6 (floating-point controls). Two computations outside the proof agree with Theorem S11.5. (i) With the surrogate computed at finite $r$ from the exact derivatives $D^g K_{\mathrm{ref}}(z) = \prod_i (-1)^{g_i} \mathrm{He}_{g_i}(z_i) e^{-z_i^2/2}$ (60-digit conditioning, no Taylor jets), the Richardson estimates $2D(r) - D(2r)$, $D(r) = (\widetilde{\mathcal{A}}_r - \mathcal{A}_0)/r^2$, converge to the exact $\widetilde{\mathcal{A}}_2(b,k)$ at the rate $O(r^2)$, at five points in each of $d = 1, 2, 3$. (ii) A floating-point evaluation of $c_2[K_{\mathrm{ref}}]$ with truncated jets, a least-squares fit in $r$ over five separations and Gauss–Legendre quadrature differs from the closed forms by $5.7 \cdot 10^{-10}$ ($d = 1$), $1.9 \cdot 10^{-9}$ and $2.6 \cdot 10^{-9}$ ($d = 2$) and $1.0$–$1.4 \cdot 10^{-9}$ ($d = 3$).

S11.3. **Finite-jet representation and transfer to the torus.** Since $c_2[K]$ is a signed finite part, its dependence on $K$ is not controlled by a positivity (sandwich) argument; it is controlled through a representation by finitely many derivatives of $K$ at 0 and a Lipschitz bound along a path of covariances.

**Theorem S11.7** (finite-jet representation)**.** *Let $d \geq 1$, let $K$ be a real smooth stationary covariance such that the covariance of the contact vector $(\nabla f(0), D^2 f(0), \partial_x^3 f(0))$, of dimension $(d+1)(d+2)/2$, is positive definite, and fix $u$ and the frame. There are $a = a(u) > 0$ and real $p_0, p_2, p_4$ (depending on $u$) with*

$$\bar{\mathcal{A}}_2[K](k,u) = e^{-ak^2}(p_0 + p_2 k^2 + p_4 k^4), \qquad k \geq 0. \tag{S11.16}$$

*The numbers $a, p_0, p_2, p_4$ are Gaussian cone moments determined by the derivatives $D^\alpha K(0)$, $|\alpha| \leq 8$. The integral (S11.12) converges absolutely, and*

$$c_2[K] = \frac{\Gamma(5/6)}{3} \int_{S^{d-1}} \Big[ -3a^{1/6} p_0 + \frac{1}{2}\, a^{-5/6} p_2 + \frac{5}{12}\, a^{-11/6} p_4 \Big]\, d\sigma(u). \tag{S11.17}$$

*No inverse of the transverse Hessian and no derivative of the cone indicator is used.*

*Proof. Step 1 (birth marginalization and parity).* In $U_r$ replace the row $\frac{1}{r}(f(S) - f(M))$ by $\frac{1}{r}(f(S) - f(M)) + \frac{r^2}{12}\tau_r = \frac{1}{2}(\partial_x f(M) + \partial_x f(S))$. This row operation has determinant 1, and the new row has target $-kr^2 + \frac{r^2}{12}\, 12k = 0$. Integrating the first row $\frac{1}{2}(f(M) + f(S))$ over $b \in \mathbb{R}$ has Jacobian 1. What remains is the vector of $2d+1$ pins

$$V_r = (G_r^+,\ G_r^-,\ \tau_r), \qquad G_r^+ = \tfrac{1}{2}(\nabla f(M) + \nabla f(S)), \quad G_r^- = \tfrac{1}{r}(\nabla f(S) - \nabla f(M)), \tag{S11.18}$$

with target $v(k) = (0_d, 0_d, 12k)$ and $V_0 = (G, Hu, f_3)$, where $G = \nabla f(0)$, $H = D^2 f(0)$, $f_3 = \partial_x^3 f(0)$. No birth coordinate remains, and disintegration gives exactly

$$\bar{\mathcal{A}}_r[K](k,u) = 12\, p_{V_r}(v(k))\, \mathbb{E}\Big[ -\frac{\det H_M \det H_S}{r^2}\, \mathbf{1}_{\{A<0\}} \,\Big|\, V_r = v(k) \Big], \tag{S11.19}$$

$p_{V_r}$ the density of $V_r$. The integrand is a polynomial in jointly Gaussian jets, so it is absolutely integrable, and this justifies the disintegration of the signed integrand; the birth integral is not the evaluation of a birth-conditioned expression at $b = 0$. Every component of $V_r$ is even in $r$. Under $z \mapsto -z$, $G_r^+$ and $\tau_r$ are odd and $G_r^-$ and $A$ are even; since $K(-z) = K(z)$, Gaussian derivative families of opposite parity are uncorrelated, hence independent. No rotational or reflection symmetry of $K$ is used. As $r \to 0$, with $f_5 = \partial_x^5 f(0)$,

$$G_r^+ = G + \tfrac{r^2}{8}\partial_x^2 G + O(r^4), \qquad G_r^- = Hu + \tfrac{r^2}{24}\partial_x^2(Hu) + O(r^4), \qquad \tau_r = f_3 + \tfrac{r^2}{40} f_5 + O(r^4), \tag{S11.20}$$

in every Gaussian $L^p$, on compact mark sets.

*Step 2 (endpoint determinants without inverses).* At the midpoint let $B = \partial_x A$, $C = \partial_x^2 A$ (derivatives of $D_y^2 f$ at 0) and $\eta = \nabla_y \partial_x^3 f(0)$. Under the pinned law at $r$, with $\varepsilon = -1$ at $M$ and $\varepsilon = +1$ at $S$, Taylor's formula gives

$$\begin{aligned} \alpha_\varepsilon &:= \tfrac{1}{r}\, \partial_x^2 f(\varepsilon \tfrac{r}{2} u) = 6\varepsilon k + \tfrac{r}{12} f_4 + \varepsilon \tfrac{r^2}{120} f_5 + O(r^3), \\ \beta_\varepsilon &:= \tfrac{1}{r}\, \nabla_y \partial_x f(\varepsilon \tfrac{r}{2} u) = \tfrac{\varepsilon}{2}\gamma + \tfrac{r}{12}\eta + O(r^2), \\ A_\varepsilon &:= D_y^2 f(\varepsilon \tfrac{r}{2} u) = A + \varepsilon \tfrac{r}{2} B + \tfrac{r^2}{8} C + O(r^3), \end{aligned} \tag{S11.21}$$

because the axial pins force $\partial_x^2 f(0) = -\frac{r^2}{24} f_4 + O(r^4)$ and $\partial_x^3 f(0) = 12k - \frac{r^2}{40} f_5 + O(r^4)$ (the latter gives the factor $\frac{1}{120}$), and the transverse gradient-difference pin forces $\nabla_y \partial_x f(0) = -\frac{r^2}{24}\eta + O(r^4)$ (which gives the factor $\frac{1}{12}$). Let $\Delta_B, \Delta_C$ be the derivatives of det at $A$ in the directions $B$ and $C$, $\Delta_{BB}$ its second derivative in the direction $B$, $J = \mathrm{adj}(A)$, and $J_B$ the derivative of adj at $A$ in the direction $B$; these are polynomials, defined also for singular $A$. Put

$$\Upsilon_1 = Y + 3k\Delta_B, \qquad \Upsilon_2 = \tfrac{3k}{4}(\Delta_C + \Delta_{BB}) + \tfrac{f_4}{24}\Delta_B + \tfrac{f_5}{120}\Delta - \tfrac{1}{12}\gamma^\top J \eta - \tfrac{1}{8}\gamma^\top J_B \gamma. \tag{S11.22}$$

The bordered-determinant identity $\det \left(\begin{smallmatrix} \alpha & \beta^\top \\ \beta & A' \end{smallmatrix}\right) = \alpha \det A' - \beta^\top \mathrm{adj}(A')\beta$, valid without invertibility, gives

$$D_\varepsilon := \frac{\det H_\varepsilon}{r} = \alpha_\varepsilon \det A_\varepsilon - r\, \beta_\varepsilon^\top \mathrm{adj}(A_\varepsilon)\, \beta_\varepsilon = 6\varepsilon k \Delta + r\Upsilon_1 + \varepsilon r^2 \Upsilon_2 + O(r^3),$$

where $H_{-1} = H_M$, $H_{+1} = H_S$, and therefore

$$-D_{-1}D_{+1} = 36k^2\Delta^2 + r^2 X_2 + O(r^3), \qquad X_2 = 12k\Delta\Upsilon_2 - \Upsilon_1^2. \tag{S11.23}$$

The term of order $r$ cancels. The term $3k\Delta_B$ of $\Upsilon_1$ is present at fixed $k$ (replacing $\Upsilon_1$ by the cusp variable $Y$ would give a wrong fold coefficient), and $f_5$ enters $X_2$ only linearly, through $\frac{1}{10}k\Delta^2 f_5$.

*Step 3 (density scores; the gap polynomial has degree four).* Order the conditioning coordinates as the odd vector $\mathcal{O}_r = (G_r^+, \tau_r)$, of dimension $d+1$, and the even vector $\mathcal{E}_r = (G_r^-, A^{\mathrm{up}})$, of dimension $d(d+1)/2$, where $A^{\mathrm{up}}$ lists the raw upper-triangular entries of $A$ (no factor $\sqrt{2}$); at $r = 0$, $\mathcal{E}_0$ consists of all raw entries of $H$. Their targets are $12k\,e$ ($e$ the last coordinate) and $x(A) = (0_d, A^{\mathrm{up}})$. By parity they are independent. Write $\operatorname{Cov}\mathcal{O}_r = \Sigma_o + r^2\Sigma_o' + O(r^4)$, $\operatorname{Cov}\mathcal{E}_r = \Sigma_e + r^2\Sigma_e' + O(r^4)$, let $g_\Sigma$ be the centered Gaussian density with covariance $\Sigma$, and put

$$\begin{aligned} p_o = (2\pi)^{-(d+1)/2}(\det\Sigma_o)^{-1/2}, \qquad a = 72\,e^\top\Sigma_o^{-1}e, \qquad q(A) = g_{\Sigma_e}(x(A)), \\ \psi_o(k) = -\tfrac{1}{2}\operatorname{tr}(\Sigma_o^{-1}\Sigma_o') + 72k^2\,e^\top\Sigma_o^{-1}\Sigma_o'\Sigma_o^{-1}e, \\ \psi_e(A) = -\tfrac{1}{2}\operatorname{tr}(\Sigma_e^{-1}\Sigma_e') + \tfrac{1}{2}\,x(A)^\top\Sigma_e^{-1}\Sigma_e'\Sigma_e^{-1}x(A). \end{aligned} \tag{S11.24}$$

Then $p_o e^{-ak^2} q(A)$ is the density of $(\mathcal{O}_0, \mathcal{E}_0)$ at $(12ke, x(A))$, and $\psi_o$, $\psi_e$ are the derivatives in $r^2$ at $r = 0$ of the logarithms of the densities of $\mathcal{O}_r$ at $12ke$ and of $\mathcal{E}_r$ at $x(A)$. Since $\Delta$ is a function of the conditioning variable $A$, differentiating (S11.19) at $r^2 = 0$ and using (S11.23) gives

$$\begin{aligned} \bar{\mathcal{A}}_2[K](k,u) &= 12\,p_o e^{-ak^2}\int_{\{A<0\}} q(A)\,h(k,A)\,dA, \\ h(k,A) &= \mathbb{E}\big[X_2 \mid \mathcal{O}_0 = 12ke,\ \mathcal{E}_0 = x(A)\big] + 36k^2\Delta^2\big[\psi_o(k) + \psi_e(A)\big]. \end{aligned} \tag{S11.25}$$

Only the nondegenerate conditioning covariances are inverted; the conditional covariances of the higher jets may be singular, and their Gaussian polynomial moments are given by Wick's formula, which needs no inverse. Conditionally on $(\mathcal{O}_0, \mathcal{E}_0) = (12ke, x(A))$, the even jets $f_4, C, \eta$ have means linear in $A$ and independent of $k$; the odd jets $\gamma, B, f_5$ have means linear in $k$ and independent of $A$ (for a non-isotropic $K$ these means need not vanish, and they are kept); the residual covariances depend on neither target; and the odd and even residual blocks are independent. By (S11.22)–(S11.23), $X_2$ has at most four free-jet factors, and after taking moments its degree in $k$ is at most 4. The substitution $k \mapsto -k$ together with the sign change of every odd jet leaves $X_2$ and the scores invariant and maps the conditional law at $k$ to that at $-k$, so $h(-k, A) = h(k, A)$. Hence

$$h(k,A) = h_0(A) + h_2(A)k^2 + h_4(A)k^4, \qquad p_j = 12\,p_o\int_{\{A<0\}} q(A)\,h_j(A)\,dA \ \ (j = 0, 2, 4), \tag{S11.26}$$

with polynomials $h_j$; this is (S11.16). Explicitly, by polynomial evaluation (no multiplication by $e^{4a}$ is needed),

$$\begin{aligned} h_0 = h(0,A), \qquad h_2 = \tfrac{1}{12}\big(16h(1,A) - 15h(0,A) - h(2,A)\big), \\ h_4 = \tfrac{1}{12}\big(h(2,A) - 4h(1,A) + 3h(0,A)\big). \end{aligned} \tag{S11.27}$$

*Step 4 (the finite part).* For $a > 0$, integration by parts with the antiderivative $-3k^{-1/3}$ of $k^{-4/3}$ (both boundary terms vanish, since $e^{-ak^2} - 1 = O(k^2)$ at 0), Euler's integral and $\Gamma(\frac{11}{6}) = \frac{5}{6}\Gamma(\frac{5}{6})$ [5, §§5.2, 5.5] give

$$\begin{aligned} \int_0^\infty (e^{-ak^2} - 1)k^{-4/3}dk = -3a^{1/6}\Gamma(\tfrac{5}{6}), \\ \int_0^\infty e^{-ak^2}k^{2/3}dk = \tfrac{1}{2}a^{-5/6}\Gamma(\tfrac{5}{6}), \qquad \int_0^\infty e^{-ak^2}k^{8/3}dk = \tfrac{5}{12}a^{-11/6}\Gamma(\tfrac{5}{6}). \end{aligned} \tag{S11.28}$$

By (S11.16), $\bar{\mathcal{A}}_2[K](k,u) - \bar{\mathcal{A}}_2[K](0,u) = p_0(e^{-ak^2} - 1) + e^{-ak^2}(p_2k^2 + p_4k^4)$ is $O(k^2)$ at 0 and bounded at infinity, so (S11.12) converges absolutely, and (S11.28) gives (S11.17).

*Derivative orders.* By (S11.3), $\Sigma_o$ and $\Sigma_e$ use derivatives of $K$ at 0 of order at most 6, and by (S11.20) $\Sigma_o'$ and $\Sigma_e'$ use order at most 8. In $X_2$ every jet other than $f_5$ has order at most 4, so its conditional means and covariances and the Wick moments use order at most 8. The jet $f_5$ enters only through its conditional mean, which uses its covariances with $G$ (order 6) and with $f_3$ (order 8); its covariance with the even block vanishes by parity, and no variance of $f_5$ (order 10) occurs. Omitting the $f_5$ term, or using only derivatives through order 6, gives a wrong coefficient. □

*Remark* S11.8 (two exact recoveries). For $K = K_{\mathrm{ref}}$, (S11.17) reproduces Theorem S11.5 in $d = 1, 2$ by hand. For $d = 1$: $\Sigma_o = \left(\begin{smallmatrix} 1 & -3 \\ -3 & 15 \end{smallmatrix}\right)$, $\Sigma_o' = \left(\begin{smallmatrix} -3/4 & 9/4 \\ 9/4 & -21/4 \end{smallmatrix}\right)$, $\Sigma_e = (3)$, $\Sigma_e' = (-\frac{5}{4})$, so $a = 12$; the conditional mean of $f_5$ is $-120k$ and $\mathrm{Var}(f_4 \mid \partial_x^2 f(0) = 0) = 30$ (not 24, which would also pin the birth height). Hence $\mathbb{E}X_2 = -\frac{5}{24} - 12k^2$, $\psi_o + \psi_e = \frac{11}{24} + 3k^2$ and $\bar{\mathcal{A}}_2 = \pi^{-3/2} e^{-12k^2}(-\frac{5}{24} + \frac{9}{2}k^2 + 108k^4)$; the bracket in (S11.17) is $\frac{9}{8}\,12^{1/6}\pi^{-3/2}$, and the factor $\frac{1}{3}|S^0| = \frac{2}{3}$ gives $\frac{3}{4}\,12^{1/6}\Gamma(\frac{5}{6})\pi^{-3/2}$. For $d = 2$: given $Hu = 0$, $A \sim N(0, \frac{8}{3})$; given $A$ and the odd pins, $\mathbb{E}f_4 = \frac{3}{4}A$, $\mathrm{Var}\, f_4 = \frac{57}{2}$, $\mathbb{E}C = -\frac{3}{4}A$, $\mathbb{E}f_5 = -120k$, $\mathbb{E}B = \mathbb{E}\gamma = \mathbb{E}\eta = 0$, $\mathrm{Var}\, B = \mathrm{Var}\,\gamma = 2$, $\mathrm{Cov}(B, \gamma) = 0$, and $\psi_o + \psi_e = \frac{23}{32} + 3k^2 - \frac{A^2}{256}$. Then $h_0 = -\frac{A^4}{256} - \frac{13A^2}{96} - \frac{3}{4}$, $h_2 = -\frac{9A^4}{64} + \frac{57A^2}{8} - 18$, $h_4 = 108A^2$; the negative-half Gaussian moments of degrees $0, 2, 4$ are $\frac{1}{2}, \frac{4}{3}, \frac{32}{3}$, so the cone integrals are $(-\frac{43}{72}, -1, 144)$; the bracket of (S11.17) applied to them, divided by $12^{1/6}$, is $\frac{13}{6}$, and multiplication by the density factor $12 p_o p_{Hu}(0) = \frac{1}{2}\pi^{-5/2}$ and by $\frac{1}{3}|S^1|$ gives $\frac{13}{18}12^{1/6}\Gamma(\frac{5}{6})\pi^{-3/2}$. Neither computation uses the exact pipeline of Theorem S11.5. Under $f \mapsto tf$, $c_2[\,\cdot\,]$ scales as $t^{-4/3}$.

**Theorem S11.9** (explicit local stability)**.** *Let $d \in \{1, 2, 3\}$ and let $K$ be a real smooth stationary covariance with*

$$\delta := \sup_{R \in O(d),\ |\alpha| \le 8} \left| D_R^\alpha K(0) - D_R^\alpha K_{\mathrm{ref}}(0) \right| \le 10^{-4}, \tag{S11.29}$$

*where $D_R^\alpha$ is the derivative of multi-index $\alpha$ in the orthonormal frame $R$. Then*

$$\left| c_2[K] - c_2[K_{\mathrm{ref}}] \right| \le 10^{50}\,\delta. \tag{S11.30}$$

*Proof.* Let $K_t = (1 - t)K_{\mathrm{ref}} + tK$, $t \in [0, 1]$, a covariance for each $t$; dots denote $d/dt$. Fix a frame. Every $t$-derivative of a covariance entry of order at most 8 is bounded by $\delta$, and all such entries are bounded in modulus by 106 (those of $K_{\mathrm{ref}}$ by $105 = \mathbb{E}Z^8$). The coefficients in (S11.20) have sums at most 1, so the same bounds hold for the entries of $\Sigma_o', \Sigma_e'$ and their derivatives. By Theorem S11.7, $a$ and $p_j$ are smooth in $t$: one differentiates the Gaussian density and the polynomial coefficients on the fixed set $\{A < 0\}$, never the cone itself, and the bounds below give integrable dominations that justify the differentiations and the use of Fubini's theorem. Write $R_A = |A^{\mathrm{up}}|$, $\langle A \rangle = 1 + R_A$, and take $|k| \le 2$, which suffices for (S11.27).

*(i) Conditioning.* The reference odd block is $\left(\begin{smallmatrix} 1 & -3 \\ -3 & 15 \end{smallmatrix}\right) \oplus I_m$, with least eigenvalue $8 - \sqrt{58} > \frac{3}{8}$; the reference raw Hessian covariance is the diagonal block $2I + \mathbf{1}\mathbf{1}^\top$ together with variance-one off-diagonal entries, with least eigenvalue at least 1. The full conditioning dimension is at most 10, so for every conditioning covariance $\Sigma$ along the path

$$\lambda_{\min}(\Sigma) \ge \tfrac{3}{8} - 10\delta > \tfrac{1}{3}, \qquad \|\Sigma^{-1}\|_2 \le 3, \qquad \|\dot{\Sigma}\| \le 10\delta, \qquad \left\|\tfrac{d}{dt}\Sigma^{-1}\right\| \le 90\delta, \tag{S11.31}$$

and $\|\Sigma_o'\|, \|\Sigma_e'\| < 1000$, $\|\dot{\Sigma}_o'\|, \|\dot{\Sigma}_e'\| \le 10\delta$.

*(ii) Regressions.* A covariance row $c$ between a higher jet and the conditioning vector has $\|c\| < 400$ and $\|\dot{c}\| < 4\delta$, so $\|\frac{d}{dt}(c\Sigma^{-1})\| \le 12\delta + 36000\delta < 40000\delta$. With conditioning sizes at most 6 (even) and 4 (odd), the conditional means are, more precisely, at most $1000 R_A$ (even jets) and $10000|k|$ (odd jets), so for every higher-jet coordinate used in $X_2$, including $f_5$,

$$|\mu| \le 20000\langle A \rangle, \qquad |\dot{\mu}| \le 2 \cdot 10^6\,\delta \langle A \rangle. \tag{S11.32}$$

The conditional covariance entries of the higher jets are at most 106 in modulus (covariance contraction and Cauchy–Schwarz), and differentiating the Schur complement,

$$|\dot{C}_{\mathrm{cond},ij}| \le (1 + 2\cdot 4\cdot 3\cdot 400 + 400^2\cdot 9\cdot 10)\delta < 10^8\delta. \tag{S11.33}$$

The variance of $f_5$ is neither estimated nor used.

*(iii) Scores and densities.* Differentiating the inverses and traces in (S11.24) (each quadratic-form derivative has norm at most $(2\cdot 90\cdot 1000\cdot 3 + 9\cdot 10)\delta = 540090\,\delta$, the largest multiplier being 72; the two constant scores have derivatives summing to at most $450150\,\delta$),

$$|\psi_o(k) + \psi_e(A)| \le 10^6(1 + k^2 + R_A^2), \qquad |\dot{\psi}_o(k) + \dot{\psi}_e(A)| \le 10^8\delta\,(1 + k^2 + R_A^2). \tag{S11.34}$$

Moreover

$$p_o \le 9, \quad |\dot{p}_o| < 1000\delta, \quad q \ge 0, \quad |\dot{q}(A)| \le 100\delta(1 + R_A^2)\,q(A), \quad \tfrac{1}{2} < a \le 216, \quad |\dot{a}| < 10000\delta, \tag{S11.35}$$

where $a > \frac{1}{2}$ because $(\Sigma_o^{-1})_{ee} \ge 1/(\Sigma_o)_{ee} \ge \frac{1}{106}$, and the bound on $\dot{q}$ uses $\dim \mathcal{E} \le 6$, $\|\Sigma_e^{-1}\| \le 3$ and $\|\dot{\Sigma}_e\| \le 6\delta$.

*(iv) Polynomials.* For $m \le 2$ the coefficient $\ell^1$ norms of $\Delta, \Delta_B, \Delta_C, \Delta_{BB}$ are at most $2, 4, 4, 4$, each entry of $\mathrm{adj}(A)$ has norm at most 1, and each quadratic or bilinear contraction in (S11.22) has norm at most 4 (these bounds include $m = 0, 1$). Hence, for $|k| \le 2$, in the coefficient norm of polynomials in the raw entries of $A$ and the free jets,

$$\|\Upsilon_1\| \le \tfrac{151}{6}, \qquad \|\Upsilon_2\| \le \tfrac{781}{60}, \qquad \|X_2\| \le \left(\tfrac{151}{6}\right)^2 + 48\cdot\tfrac{781}{60} = \tfrac{226469}{180} < 2000. \tag{S11.36}$$

Every monomial of $X_2$ has total degree at most 6 in these entries and at most four free-jet factors, and the $f_5$ term is linear in $f_5$.

*(v) Wick moments.* A Gaussian moment of $n \le 4$ variables has at most 10 partial-matching terms. With $\mu_* = 20000\langle A\rangle$, (S11.32)–(S11.33) bound each term by $\mu_*^n$ and its derivative by $400\delta\mu_*^n$ (mean derivatives at most $100\delta\mu_*$, covariance derivatives at most $\delta\mu_*^2$, at most four factors), so

$$|\mathrm{moment}| \le 10\,\mu_*^n, \qquad |\tfrac{d}{dt}\mathrm{moment}| \le 4000\,\delta\mu_*^n. \tag{S11.37}$$

A remaining monomial in $A$ costs $\langle A\rangle^j$ with $j+n \le 6$. The score term of $h$ is at most $36\cdot 4\cdot 4\cdot 10^6\cdot 5\langle A\rangle^6$ in modulus, and its derivative at most the same with $10^8\delta$ in place of $10^6$. With (S11.36) and (S11.27), whose largest coefficient $\ell^1$ norm is $\frac{8}{3} < 3$,

$$|h(k,A)| < 10^{22}\langle A\rangle^6, \quad |\dot{h}(k,A)| < 10^{25}\delta\langle A\rangle^6, \quad |h_j(A)| < 10^{23}\langle A\rangle^6, \quad |\dot{h}_j(A)| < 10^{26}\delta\langle A\rangle^6. \tag{S11.38}$$

*(vi) The cone integral, without a boundary derivative.* The density of $Hu$ at 0 is at most $3^{d/2} < 6$. Given $Hu = 0$, $A^{\mathrm{up}}$ is a centered Gaussian vector with at most 3 coordinates and covariance trace at most 318; Jensen's inequality on the eigenvalues and $\mathbb{E}Z^8 = 105$ give $\mathbb{E}|A^{\mathrm{up}}|^8 \le 105\,(318)^4$. With $(1 + R_A)^8 \le 2^7(1 + R_A^8)$,

$$\int q(A)\,(1 + R_A)^8\,dA \le 6\cdot 128\cdot(1 + 105\cdot 318^4) = 824629750641408 < 10^{15}, \tag{S11.39}$$

and the integral over the cone $\{A < 0\}$ is smaller; there is no hidden birth or $k$ mean, because the even target is $Hu = 0$. By (S11.26) and (S11.35)–(S11.39), uniformly in the frame and in $t$,

$$|p_j| < 12\cdot 9\cdot 10^{23}\cdot 10^{15} < 10^{41}, \qquad |\dot{p}_j| < 12(10^3\cdot 10^{23} + 9\cdot 10^{26} + 9\cdot 100\cdot 10^{23})10^{15}\delta < 10^{44}\delta, \tag{S11.40}$$

where the last term comes from the derivative of the Gaussian density, not of the cone.

*(vii) Conclusion.* For $\frac{1}{2} < a \le 216$, $a^{1/6} < 3$, $a^{-5/6} < 2$, $a^{-11/6} < 4$ and $a^{-17/6} < 8$, so the three weights in (S11.17) have total absolute value at most 12 and their $a$-derivatives total at most 11.

Also $\Gamma(\frac{5}{6}) < \frac{6}{5} + 1 < 3$ (split Euler's integral at 1) and $|S^{d-1}| < 16$. Hence, by (S11.17), (S11.35) and (S11.40),

$$\Big|\frac{d}{dt} c_2[K_t]\Big| \le 16(12 \cdot 10^{44} + 11 \cdot 10^4 \cdot 10^{41})\delta = 1.952 \cdot 10^{47}\,\delta < 10^{50}\delta, \tag{S11.41}$$

and integrating over $t \in [0,1]$ gives (S11.30). The constant is not optimized. □

**Corollary S11.10** (normalized periodization)**.** *Let $d \in \{1,2,3\}$. For every real $L \ge 10$,*

$$\begin{aligned} |c_2[K_L] - c_2[K_{\mathrm{ref}}]| &\le 10^{50} E_d(L), \\ E_d(L) &= 2(3^d - 1)\big[d^4L^8 + 28d^3L^6 + 210d^2L^4 + 420dL^2 + 210\big]e^{-L^2/2}. \end{aligned} \tag{S11.42}$$

*In particular, uniformly for every real $L \ge 24$,*

$$|c_2[K_L] - c_2[K_{\mathrm{ref}}]| < 10^{-60}, \tag{S11.43}$$

*and, for every real $L \ge 24$,*

$$c_2[K_L] \in \begin{cases} [0.2300445802661502,\ 0.2300445802661999], & d = 1, \\ [0.2215244106266631,\ 0.2215244106267111], & d = 2, \\ [0.1612340491269446,\ 0.1612340491269811], & d = 3. \end{cases}$$

*Proof.* For $q \le 8$ unit directional derivatives, the derivative of $K_{\mathrm{ref}}$ at $z$ with $|z| \ge 1$ is bounded in modulus by $P_8(|z|)e^{-|z|^2/2}$, $P_8(w) = w^8 + 28w^6 + 210w^4 + 420w^2 + 105$: differentiating the Gaussian produces all contractions into singletons and pairs, each unit-vector contraction has modulus at most 1, and the numbers of contractions are $1, 28, 210, 420, 105$; lower orders are bounded by the same polynomial for $w \ge 1$. At the origin every required derivative has modulus at most 105. The normalization denominator in (S11.4) is at least 1, so the change of a normalized derivative is at most the image sum $\sum_{n\neq 0}(P_8(L|n|) + 105)e^{-L^2|n|^2/2}$. In the shell $\|n\|_\infty = j \ge 1$ there are at most $(3^d - 1)j^{d-1}$ points, all with $j \le |n| \le \sqrt{d}\,j$, and every power of $j$ that occurs is at most $j^{10}$. For $L \ge 10$ the ratio of successive shell terms $j^{10}e^{-L^2j^2/2}$ is at most $2^{10}e^{-150} < \frac{1}{2}$ (as $e^{150} > 1 + 150 + 150^2/2 > 2048$), so the shell sum is less than twice its first term, and $E_d(L)$ bounds $\delta$ of (S11.29) in every frame. Each $L^j e^{-L^2/2}$, $j \le 8$, decreases for $L \ge 10$. Exact positive Taylor sums give $e^{-50} < 2 \cdot 10^{-22}$ and $e^{-288} < 10^{-125}$; hence $E_3(10) < 9.231 \cdot 10^{-11} < 10^{-4}$, and since the polynomial factor increases with $d$, every $d \le 3$ and $L \ge 10$ lies within (S11.29). Theorem S11.9 gives (S11.42). At $L = 24$ the exact polynomial factors $2(3^d - 1)[\dots]$ are 461984450376, 28868660309280 and 471182508197544 for $d = 1, 2, 3$; multiplied by $10^{50} \cdot 10^{-125}$ they give additive errors below $4.62 \cdot 10^{-64}$, $2.887 \cdot 10^{-62}$ and $4.712 \cdot 10^{-61}$, all below $10^{-60}$, and monotonicity extends this to every real $L \ge 24$. No floating-point exponential enters. Finally, widening each endpoint of the enclosures of $c_2[K_{\mathrm{ref}}]$ in Theorem S11.5 by $10^{-16}$, which exceeds the transfer error, gives the three intervals. □

S11.4. **The side-24 constant.** The results of §§S11.2–S11.3 concern the coefficient expression $c_2[K]$. Their application to the constant (S11.11) of the side-24 field, and to the law (2.12) of the main text, uses the following two identifications. Both follow from Section S10.

**Corollary S11.11** (identification of $c_2$)**.** *Let $d \ge 2$ and $L > 0$.*

*(a)* (Typed kernel and surrogate.) *There are $r_* > 0$ and $C, c, N$ such that, for $0 < r \le r_*$, $b \in \mathbb{R}$, $k > 0$ and $u \in S^{d-1}$,*

$$|\widetilde{\mathcal{A}}_r[K_L](b,k,u) - \mathcal{A}_r(b,k,u)| \le Cr^3(1 + k^{-1})P^N e^{-c(b^2+k^2)}.$$

*Hence, at every $(b,k,u)$ with $k>0$, both kernels are $\mathcal{A}_0+r^2\mathcal{A}_2+O(r^3)$, so that $\widetilde{\mathcal{A}}_2[K_L]=\mathcal{A}_2$; moreover $\bar{\mathcal{A}}_2[K_L](k,u)=\int_{\mathbb{R}}\mathcal{A}_2(b,k,u)\,db$ for every $k\ge 0$, and the constant $c_2$ of (S11.11) equals the expression $c_2[K_L]$ of (S11.12).*

*(b)* (Persistence identification.) *The constant $c_2$ of (S11.11) is the coefficient of $\ell^{1/3}$ in the expansions (S10.29) and (S10.31) of the candidate and the elder densities of the side-L field.*

*Proof.* (a) Let $r_*=\min(r_0^*,\frac12)$, as in Lemma S10.12. Let $Q=Q_{r,b,k}$ and $K_M,K_S$ be as in §S10.1, $F_r\sim Q$ and $A=D_y^2F_r(0)$, so that $\{A<0\}$ is the event $\{A\prec 0\}$ of Section S10. Since $\det H_i=r\det K_i$ $(i=M,S)$, (S11.8) reads $\widetilde{\mathcal{A}}_r[K_L]=12\pi_r(v_r)\,\mathbb{E}_Q[-\det K_M\det K_S\,\mathbf{1}\{A\prec 0\}]$, while $\mathcal{A}_r=12\pi_r(v_r)\,\mathbb{E}_Q[F_d(K_M)F_{d-1}(K_S)]$ by (S10.3). By (S10.40) and the moment bound in (S10.11), the first expectation is $\mathbb{E}_Q[\Omega(J';r,k)\mathbf{1}\{A\prec 0\}]+O(r^4P^N)$, and by (S10.42) it therefore differs from the second by $O(r^3(1+k^{-1})P^N)$. Multiplying by $12\pi_r(v_r)$, which is at most $Ce^{-c(b^2+k^2)}$ by (S10.10), gives the bound, and (S10.38) then gives the common expansion. Both error bounds are integrable in $b$, so for $k>0$ the coefficient of $r^2$ of the birth integral (S11.9) is $\int_{\mathbb{R}}\mathcal{A}_2(b,k,u)\,db$. Both sides are continuous in $k\ge 0$: the left side by (S11.16), as Theorem S11.7 applies to $K_L$ by Lemma S12.8, and the right side by (S10.39) and dominated convergence. So the identity also holds at $k=0$. Finally, by Remark S11.3(i), $|\mathcal{A}_2(b,k,u)-\mathcal{A}_2(b,0,u)|\,k^{-4/3}\le C\min(k,1)\,k^{-4/3}(1+|b|)^Ne^{-cb^2}$, which is integrable on $S^{d-1}\times\mathbb{R}\times(0,\infty)$; so, by Fubini's theorem, (S11.12) for $K=K_L$ equals (S11.11).

(b) The definition (S11.11) is (S10.22), with the same $\mathcal{A}_2$. Theorems S10.4 and S10.5 state (S10.29) and (S10.31) with this constant; the elder expansion (S10.31) uses the far elder rate (S10.21), which is Proposition S8.13. □

At $k=0$ the values in (S11.11) and (S11.12) are the continuous extensions from $k>0$. For the typed kernel this value is not its coefficient of $r^2$ at $k=0$: by Proposition S10.7, $\mathcal{A}_r(b,0,u)=O(r^5\log(1/r))$, while $\mathcal{A}_2(b,0,u)$, given in (S11.10), is in general not zero (Remark S11.3(i)). For the surrogate the two agree: Steps 1 and 3 of the proof of Lemma S10.12 do not use $k>0$, and they give $\widetilde{\mathcal{A}}_r[K_L](b,0,u)=r^2\mathcal{A}_2(b,0,u)+O(r^3P^Ne^{-cb^2})$.

**Corollary S11.12** (the constant $c_2$ at side 24)**.** *Let $d=3$ and $L\ge 24$, and let*

$$c_2^{\mathrm{ref}}:=c_2[K_{\mathrm{ref}}]=\frac{5}{48}(33-7\sqrt{6})\,12^{1/6}\,\Gamma(\tfrac56)\,\pi^{-5/2}\in[0.1612340491269447,\ 0.1612340491269810].$$

*(i)* $|c_2[K_L]-c_2^{\mathrm{ref}}|<10^{-60}$ *and* $c_2[K_L]\in[0.1612340491269446,\ 0.1612340491269811]$.

*(ii) The same holds for the constant $c_2$ of (S11.11).*

*(iii) For $L=24$,*

$$\frac{c_2}{c_{3,24}}\in[3.859496174547456,\ 3.859496174548331];\tag{S11.44}$$

*in particular $c_2/c_{3,24}=3.8595$ to four decimals, the value printed in (2.14) of the main text.*

*(iv) $c_2$ is the constant of (2.12) and (2.14) of the main text; thus (i)–(iii) give, for that constant, the enclosure of Theorem 2.3(iii) of the main text transferred to $L=24$ with additive error below $10^{-60}$, and the value $c_2/c_{3,24}\approx 3.8595$ of (2.14).*

*Proof.* (i) is Theorem S11.5 for $d=3$ together with (S11.43) and the last display of Corollary S11.10. (ii) follows from (i) and Corollary S11.11(a). For (iii), by (2.10) and (2.11) of the main text, $|c_{3,24}/c_{3,\infty}-1|<1.5\cdot10^{-118}$ and $c_{3,\infty}=0.0417759318405983433429\ldots$, so

$$0.04177593184059834334<c_{3,24}<0.04177593184059834335,$$

the enclosure quoted in Theorem 2.3(iii) of the main text. Dividing the positive intervals of (ii) and of $c_{3,24}$,

$$\frac{c_2}{c_{3,24}} \geq \frac{0.1612340491269446}{0.04177593184059834335} = 3.85949617454745675\ldots,$$
$$\frac{c_2}{c_{3,24}} \leq \frac{0.1612340491269811}{0.04177593184059834334} = 3.85949617454833046\ldots,$$

which gives (S11.44). (iv) is (ii), (iii) and Corollary S11.11(b). □

*Remark* S11.13 (scope). The bound (S11.43) is an additive error for the coefficient expression only. It is not an estimate of $\nu_{3,24}(\ell)$ at a positive lifetime, and it gives no constant for the remainder $O(\ell^{3/7})$ of (2.12) of the main text, so the size of the omitted terms at a given $\ell$ is not quantified. Near $L = 10$ the bound (S11.42) is very loose and is not an accurate approximation of the coefficient. The interval for $c_2[K_{\mathrm{ref}}]$ in Theorem S11.5 encloses the closed form, which concerns the Gaussian kernel on $\mathbb{R}^3$; the side-24 constant is reached only through Corollaries S11.10 and S11.11.

## Part III. The parent theorem

### S12. The parent theorem

This section transcribes the companion record [7], so that the statements which Sections S5–S10 quote from it are stated and proved in this supplement. The record is read as reconciled in [6]: with its congruence erratum and the reconciled wording of the Hessian limit in the proof of Proposition S12.15, with its Section 9 replaced by the repaired Kac–Rice section in Proposition S12.22, and with the explicit embedding radius of Definition S12.2. Lemma S12.19(b) is added here: [7] uses it without proof. It holds in every fixed dimension $d \geq 2$ and for every fixed side $L > 0$. Subsection S12.1 fixes the setting and states the three theorems of [7]: uniform elder selection of contact pairs on compact birth and gap windows (Theorem S12.3), the lifetime densities of these pairs (Theorem S12.4), and the leading law of the unrestricted densities (Theorem S12.5). Subsections S12.2–S12.15 give the lemmas, identities and estimates of their proofs in the order of [7], and Subsection S12.16 records what is not proved. The deterministic separating-cylinder theorem used for selection is proved in Section S13. Where [7] only sketches a step, the step is given here as sketched there; the proofs of Lemmas S12.10 and S12.14, the averaging formulas in the proof of Lemma S12.11(a), the measurability in Lemma S12.20(a) and the Hermite identity of Lemma S13.3 are written out in full. All constants are existential.

#### S12.1. **Setting and the three theorems.**

**Definition S12.1** (the field and the windows)**.** Fix $d \geq 2$ and $L > 0$, and put $m = d - 1$. Let $f$ be the centred stationary Gaussian field of variance one on the flat torus $\mathbb{T}^d_L = \mathbb{R}^d/(L\mathbb{Z}^d)$ with covariance

$$K_L(z) = \frac{\sum_{n\in\mathbb{Z}^d} \exp(-|z + Ln|^2/2)}{\sum_{n\in\mathbb{Z}^d} \exp(-|Ln|^2/2)}. \tag{S12.1}$$

No replacement by a nonperiodic covariance is made; for $d = 3$ and $L = 24$ this is the side-24 field of the main text. Fix compact intervals $\mathcal{B} = [b_-, b_+]$ of births and $\mathcal{K} = [k_-, k_+]$ of gap marks, with $0 < k_- \leq k_+ < \infty$. Unless stated otherwise, constants $C < \infty$ and $c > 0$ may depend on $d$, $L$, $\mathcal{B}$ and $\mathcal{K}$, and may change from line to line; they do not depend on the radius, on the birth $b \in \mathcal{B}$, on the gap mark $k \in \mathcal{K}$ or on the orientation.

**Definition S12.2** (pinned pairs, typed weight and selection probability)**.** Let $R$ be an orthonormal frame of $\mathbb{R}^d$ with axial unit vector $t = Re_1$. Write $x$ for the axial coordinate and $y = (y_1, \ldots, y_m)$

for the transverse coordinates in the frame $R$, centred at the origin, and denote derivatives in these coordinates by the subscripts $x$ and $y_j$; $\nabla_y$ and $D_y^2$ are the transverse gradient and the transverse $m \times m$ Hessian. Let $0 < r < L/(4\sqrt{2})$. Then the local cylinder $\mathcal{D}_r$ of Definition S13.1, centred at the origin in these coordinates, is embedded in $\mathbb{T}_L^d$: it lies in the closed ball of radius $2\sqrt{2}\,r$ about the origin, the exponential chart of $\mathbb{T}_L^d$ at any point is injective on the open ball of radius $L/2$, and $2\sqrt{2}\,r < L/2$. For $b \in \mathbb{R}$ and $k > 0$ the pins are

$$M = -\tfrac{r}{2}\,t, \quad S = \tfrac{r}{2}\,t, \qquad f(M) = b, \quad f(S) = s := b - kr^3, \qquad \nabla f(M) = \nabla f(S) = 0. \tag{S12.2}$$

The law $\mathbb{P}^0 = \mathbb{P}^0_{r,b,k,R}$, with expectation $\mathbb{E}^0$, is the continuous Gaussian regression law of $f$ at these $2(d+1)$ observations; it is not a conditioning on an event of positive probability. The index $\operatorname{ind}(H)$ of a symmetric matrix $H$ is its number of negative eigenvalues: a maximum has $d$ negative Hessian eigenvalues, and the relevant saddle has $d-1$ negative eigenvalues and one positive eigenvalue. With $H_z = D^2 f(z)$ put

$$W_r = |\det H_M \, \det H_S| \, \mathbf{1}_{\{H_M \prec 0,\ \operatorname{ind}(H_S) = d-1\}}, \qquad Z_r = \mathbb{E}^0[W_r], \qquad d\mathbb{P}_r^{MS} = \frac{W_r}{Z_r}\, d\mathbb{P}^0, \tag{S12.3}$$

and write $\mathbb{E}_r^{MS}$ for the expectation under $\mathbb{P}_r^{MS}$. The normalizer $Z_r$ is the full one: it carries no further pin density, coordinate Jacobian or adjacency condition. Finally, $p_r(b,k,R)$ is the $\mathbb{P}_r^{MS}$-probability that the global ordinary superlevel elder death partner of $M$ is $S$; the event is made precise in Lemma S12.20.

**Theorem S12.3** (Theorem A: uniform selection)**.** *There are $r_* \in (0, L/(4\sqrt{2}))$ and $C < \infty$ such that, for all $0 < r \le r_*$, $b \in \mathcal{B}$, $k \in \mathcal{K}$ and all orthonormal frames $R$,*

$$0 \le 1 - p_r(b,k,R) \le Cr^3. \tag{S12.4}$$

**Theorem S12.4** (Theorem B: compact-mark densities)**.** *Suppose that $\mathcal{B}$ and $\mathcal{K}$ have positive lengths. Count the ordered pairs $(M,S)$ of a local maximum $M$ and a critical point $S$ of index $d-1$ with $0 < \operatorname{dist}(M,S) \le r_*$, birth $f(M) \in \mathcal{B}$ and scaled gap $(f(M) - f(S))/\operatorname{dist}(M,S)^3 \in \mathcal{K}$. Let $\nu_{\mathrm{cand}}(\ell)$ be their expected density per unit volume in the lifetime $\ell = f(M) - f(S)$, and let $\nu_{\mathrm{eld}}(\ell)$ count only the pairs in which $S$ is the ordinary superlevel elder death partner of $M$. These densities admit versions with*

$$\begin{gathered} \nu_{\mathrm{cand}}(\ell) \sim c_{\mathcal{B},\mathcal{K}}\, \ell^{-1/3}, \qquad \nu_{\mathrm{eld}}(\ell) \sim c_{\mathcal{B},\mathcal{K}}\, \ell^{-1/3}, \qquad c_{\mathcal{B},\mathcal{K}} > 0, \\ 0 \le \nu_{\mathrm{cand}}(\ell) - \nu_{\mathrm{eld}}(\ell) \le C_{\mathcal{B},\mathcal{K}}\, \ell^{2/3}, \end{gathered} \tag{S12.5}$$

*as $\ell \downarrow 0$. An exact finite-dimensional Gaussian integral for $c_{\mathcal{B},\mathcal{K}}$ is (S12.41).*

**Theorem S12.5** (Theorem C: unrestricted leading densities)**.** *Let $\nu_{\mathrm{cand}}^{\mathrm{all}}(\ell)$ count all ordered pairs of a local maximum and a critical point of index $d-1$ with height difference $\ell > 0$, and let $\nu_{\mathrm{eld}}^{\mathrm{all}}(\ell)$ count all finite ordinary superlevel $H_0$ persistence bars of lifetime $\ell$, both in expectation per unit volume. These densities admit versions, the sums of the near versions (S12.51) and of the far Kac–Rice versions of Lemma S12.29, for which there is a constant $c_{d,L}$ with $0 < c_{d,L} < \infty$ such that, as $\ell \downarrow 0$,*

$$\nu_{\mathrm{cand}}^{\mathrm{all}}(\ell) \sim c_{d,L}\, \ell^{-1/3}, \qquad \nu_{\mathrm{eld}}^{\mathrm{all}}(\ell) \sim c_{d,L}\, \ell^{-1/3}. \tag{S12.6}$$

*All birth heights, all positive scaled-gap marks and all spatial separations on the torus are included in (S12.6); the essential class of the global maximum is excluded from the finite-lifetime count.*

*Remark* S12.6 (scope of Theorems S12.3–S12.5). The proof of Theorem S12.3 includes a positive lower bound $Z_r \ge z_* r^2$ for the full normalizer (Proposition S12.15) and the full transverse-matrix boundary layer, including simultaneous small transverse eigenvalues (Proposition S12.18); it does not presume isotropy under arbitrary rotations of the torus. Theorem S12.4 is a compact-window result, in particular its quantitative difference estimate. Theorem S12.5 removes the mark and

distance restrictions for the leading asymptotics only; the difference estimate $O(\ell^{2/3})$ of (S12.5) is not asserted for the unrestricted densities. Theorem S12.5 has its own domination and off-diagonal proof (Subsections S12.13 and S12.14); it is not Theorem S12.4 under a new label. Its constant $c_{d,L}$ is an explicitly identified finite Gaussian and angular integral, in which the birth and gap integrals are eliminated analytically (Proposition S12.31); no elementary closed form or numerical enclosure of it is claimed. No numerical value of $C$, $r_*$, $z_*$ or $c_{\mathcal{B},\mathcal{K}}$ is stated.

S12.2. **Positive spectrum and finite-jet rank.**

**Lemma S12.7** (positive spectrum)**.** (a) *The covariance has the Fourier expansion*

$$K_L(z) = \sum_{n\in\mathbb{Z}^d} a_n\, e^{2\pi i\, n\cdot z/L}, \qquad a_n = \frac{\exp(-2\pi^2|n|^2/L^2)}{\sum_{j\in\mathbb{Z}^d}\exp(-2\pi^2|j|^2/L^2)} > 0. \tag{S12.7}$$

(b) *Every Fourier mode has positive variance, and* $\sum_{n\in\mathbb{Z}^d}\sqrt{a_n}\,(1+|n|)^q < \infty$ *for every* $q$.

(c) *A real sine and cosine expansion of* $f$ *converges in every* $C^q$ *norm in every finite* $L^p$ *norm. In particular* $f$ *has a smooth version, and all finite moments of its global derivative suprema exist.*

*Proof.* Poisson summation gives (a); (b) is read off (S12.7); and (c) follows from (b). □

**Lemma S12.8** (finite-jet rank)**.** *Any finite list of distinct derivative-evaluation functionals, at finitely many distinct sites and with several functionals allowed at each site, is linearly independent under* $f$*: no nontrivial linear combination of them has variance zero, so the covariance matrix of the list is positive definite. At one site the same holds for derivatives taken in a rotated orthonormal frame. Here a list is called distinct if it repeats no derivative functional, and a Hessian is parametrized by its independent symmetric entries, not by* $d^2$ *entries.*

*Proof.* A linear combination of vanishing variance must annihilate every Fourier mode, because every $a_n$ is positive (Lemma S12.7). As a distribution it is a finite sum of derivatives of point masses with every Fourier coefficient zero, so it is zero. Smooth test functions supported at one site, with arbitrarily prescribed finite jets there, show that every coefficient is zero. At one site in a rotated frame $R$ the assertion also follows from a polynomial $P(R^\top n)$ that vanishes for all $n\in\mathbb{Z}^d$: the invertible rotation reduces this to a polynomial vanishing on the full integer lattice, hence the polynomial is zero. □

Lemma S12.8 gives positive covariance for the original pins at each pair of distinct points, and for every residual jet list modulo the pins used below. Finitely many observed modes, or a numerically positive sampled covariance, would not suffice for these arguments.

S12.3. **The contact frame.** Throughout, $x$ and $y_1,\dots,y_m$ are the axial and transverse coordinates of Definition S12.2, so that $M$ and $S$ are the points $x=-r/2$ and $x=r/2$ of the line $y=0$.

**Definition S12.9** (observations and contact rows)**.** For $r>0$ order the original observations as

$$O_r = \big(f(M),\, f_x(M),\, f(S),\, f_x(S),\, f_{y_1}(M),\, f_{y_1}(S),\, \dots,\, f_{y_m}(M),\, f_{y_m}(S)\big);$$

this is a permutation and orthogonal re-expression of the pinned values and gradients. Define $U_r = T_r O_r$ by the four axial rows

$$\begin{aligned} U^{(0)} &= \frac{f(M)+f(S)}{2}, \qquad U^{(1)} = \frac{f(S)-f(M)}{r}, \qquad U^{(2)} = \frac{f_x(S)-f_x(M)}{r}, \\ U^{(3)} &= \frac{6}{r^2}\Big[f_x(M)+f_x(S)-\frac{2\,(f(S)-f(M))}{r}\Big], \end{aligned} \tag{S12.8}$$

followed, for each transverse direction $j=1,\dots,m$, by the two rows

$$V_j^{(0)} = \frac{f_{y_j}(M)+f_{y_j}(S)}{2}, \qquad V_j^{(1)} = \frac{f_{y_j}(S)-f_{y_j}(M)}{r};$$

thus $U_r = (U^{(0)}, U^{(1)}, U^{(2)}, U^{(3)}, V_1^{(0)}, V_1^{(1)}, \dots, V_m^{(0)}, V_m^{(1)})$.

**Lemma S12.10** (the exact contact transformation)**.** *For $r > 0$ the linear map $T_r$ is invertible, with absolute determinant exactly*

$$|\det T_r| = \frac{12}{r^4}\Big(\frac{1}{r}\Big)^m = 12\, r^{-(d+3)}. \tag{S12.9}$$

*The prescribed pins* (S12.2) *become exactly $U_r = v_r$, where*

$$v_r = (b - kr^3/2,\ -kr^2,\ 0,\ 12k,\ 0, \dots, 0). \tag{S12.10}$$

*Proof.* In the order $(f(M), f(S), f_x(M), f_x(S))$ the four axial rows of Definition S12.9 have the matrix

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ -\frac{1}{r} & \frac{1}{r} & 0 & 0 \\ 0 & 0 & -\frac{1}{r} & \frac{1}{r} \\ \frac{12}{r^3} & -\frac{12}{r^3} & \frac{6}{r^2} & \frac{6}{r^2} \end{pmatrix}.$$

It is block lower triangular, with diagonal blocks of determinants $1/r$ and $-12/r^3$, so its determinant is $-12r^{-4}$. Each transverse pair $(V_j^{(0)}, V_j^{(1)})$ has determinant $1/r$ in $(f_{y_j}(M), f_{y_j}(S))$. Reordering the observations changes only the sign of the determinant, and the frame is orthonormal, so $|\det T_r| = 12r^{-4}r^{-m}$. At the pins (S12.2), $U^{(0)} = b - kr^3/2$, $U^{(1)} = -kr^2$, $U^{(2)} = 0$ and $U^{(3)} = \frac{6}{r^2} \cdot \frac{2kr^3}{r} = 12k$, and the transverse rows vanish. □

The transformation is exact, not approximate, and the factor 12 in (S12.9) is retained in the lifetime ledger (Proposition S12.23).

**Lemma S12.11** (contact limit and uniform nondegeneracy)**.** (a) *As $r \to 0$, $U_r$ converges in mean square, uniformly over frames, to the contact jet*

$$U_0 = (f,\ f_x,\ f_{xx},\ f_{xxx},\ f_{y_1},\ f_{xy_1},\ \dots,\ f_{y_m},\ f_{xy_m})(0). \tag{S12.11}$$

*The covariances of $U_r$, and its cross-covariances with any fixed finite derivative order, converge uniformly. The target $v_r$ converges uniformly on $\mathcal{B} \times \mathcal{K}$ to $v_0 = (b, 0, 0, 12k, 0, \dots, 0)$.*

(b) *The covariance $\Sigma_0(R)$ of $U_0$ is positive definite, its entries are continuous in $R$, and its least eigenvalue has a positive minimum over all frames. Consequently the covariance $\Sigma_r(R)$ of $U_r$ is uniformly positive definite for all sufficiently small $r$, and $\Sigma_r$ and $\Sigma_r^{-1}$ converge uniformly.*

(c) *Let $A_M = D_y^2 f(M)$, represented by its $m(m+1)/2$ independent entries, and append these entries to $U_r$. The enlarged covariance is uniformly positive definite at contact, where the appended entries are those of $D_y^2 f(0)$, and for small $r$. Under $\mathbb{P}^0$ the matrix $A_M$ has covariance uniformly bounded above and uniformly bounded away from zero, and a uniformly bounded mean. In the Lebesgue coordinates of symmetric matrices its density $\rho_r$ satisfies*

$$\rho_r(A) \le C_0 \exp(-c_0 \|A\|_F^2) \tag{S12.12}$$

*for fixed $C_0, c_0 > 0$. This bound remains valid after any change of eigenvector orientation; orthogonal invariance of the density is not required.*

*Proof.* (a) Taylor formulas with integral remainders give the mean-square convergence, uniformly over frames. Explicitly, with $\varphi(s) = f(st)$ along the axis, $U^{(1)}$ and $U^{(2)}$ are the averages of $\varphi'$ and $\varphi''$ over the pin interval $[-r/2, r/2]$, $V_j^{(1)}$ is the average of $f_{xy_j}$ along the axis, $U^{(0)}$ and $V_j^{(0)}$ are means of the two endpoint values, and

$$U^{(3)} = \int_{-r/2}^{r/2} \frac{6}{r^3}\Big(\frac{r^2}{4} - s^2\Big)\varphi'''(s)\, ds,$$

an average of $\varphi'''$ against a probability density on the pin interval. So the potentially dangerous cancellation in $U^{(3)}$ is an averaged third derivative, not a subtraction of uncontrolled random values. Uniform convergence of its covariances, and of its cross-covariances with any fixed finite derivative order, follows either from those integral formulas or from the rapidly summable Fourier series of Lemma S12.7.

(b) The functionals in (S12.11) are distinct. Hence $\Sigma_0(R)$ is positive definite by Lemma S12.8. Its entries are continuous in $R$, and the orthogonal group is compact, so its least eigenvalue has a positive uniform minimum. Consequently $\Sigma_r(R)$ remains uniformly positive definite for all sufficiently small $r$, and $\Sigma_r$, $\Sigma_r^{-1}$ converge uniformly. In detail, each coordinate of $U_r$, and of $U_r$ extended by the entries of $A_M$ as in (c), is an integral $\int \partial^\alpha f(s't)\,\eta_r(ds')$ of a derivative of $f$ along the axis against a measure $\eta_r$ with $\eta_r \to \delta_0$ as $r \to 0$; so the covariance is jointly continuous on $[0,\bar r] \times O(d)$ for a fixed $\bar r > 0$, it is positive definite at $r = 0$ by Lemma S12.8, and compactness gives a positive lower bound for its least eigenvalue. This compactness argument establishes existence, not an evaluated interval enclosure. It never equates $\Sigma_r$ at different orientations; full rotational invariance of the torus is neither assumed nor true in general.

(c) At $r = 0$ the appended functionals are $f_{y_iy_j}(0)$, $i \le j$, and none of them duplicates a functional in (S12.11). The enlarged covariance is therefore uniformly positive definite at contact and for small $r$. Its Schur complement shows that the $\mathbb{P}^0$-law of $A_M$ has covariance uniformly bounded above and uniformly bounded away from zero. Its mean is uniformly bounded, and (S12.12) follows. □

S12.4. **Conditional moments and the independent residual.** Fix $r_1 \in (0, L/(4\sqrt{2}))$ such that the conclusions of Lemma S12.11(b),(c) hold for $0 < r \le r_1$. In this subsection and the next three, "uniformly" means uniformly in $0 < r \le r_1$, $b \in \mathcal{B}$, $k \in \mathcal{K}$ and the frame.

**Lemma S12.12** (uniform conditional moments)**.** *For every finite $p$ and every derivative order $q$,*

$$\sup_{r,b,k,R} \Big\| \|f\|_{C^q(\mathbb{T}^d_L)} \Big\|_{L^p(\mathbb{P}^0)} < \infty. \tag{S12.13}$$

*Proof.* Because $\Sigma_r^{-1}$ and $v_r$ are uniformly bounded, the regression energy $v_r^\top \Sigma_r^{-1} v_r$ is uniformly bounded. For each unconditioned standardized Fourier coefficient $\xi$ of $f$, the Cauchy–Schwarz inequality for covariances bounds its conditional mean by the square root of this energy, and its conditional centred variance by one. Minkowski's inequality and the summable Fourier coefficients of Lemma S12.7(b) then give (S12.13). □

**Lemma S12.13** (the independent residual)**.** *In one full-field Gaussian coupling under $\mathbb{P}^0$,*

$$f(z) = \mu_r(z) + B_r(z) \cdot (A_M - \mathbb{E}^0 A_M) + g_r(z), \tag{S12.14}$$

*where $\mu_r = \mathbb{E}^0 f$, $B_r(z)$ is the regression coefficient of $f(z)$ on the entries of $A_M$, and the centred residual $g_r$ is jointly independent of the entire symmetric matrix $A_M$. The coefficient $B_r$ and its derivatives through order four are uniformly bounded, and $g_r$ has uniformly bounded $C^4$ moments of every finite order. Let $J_r = 1 + \|g_r\|_{C^4(\mathbb{T}^d_L)}$, with a fixed coordinate norm, and let $M_3$, $M_4$ be the partial-block norms (S13.2) of $f$ on the local cylinder $\mathcal{D}_r$. Then, with a constant $K_0$,*

$$\begin{gathered} M_3 \le K_0(J_r + \|A_M\|_F), \qquad M_4 \le K_0(J_r + \|A_M\|_F), \\ J_r \ge 1, \qquad J_r \text{ is independent of } A_M, \qquad \sup \|J_r\|_{L^p} < \infty \text{ for every finite } p. \end{gathered} \tag{S12.15}$$

*Proof.* The proof of Lemma S12.12 applies to a centred residual after further conditioning on $A_M$. The regression coefficient $B_r$ and its derivatives through order four are uniformly bounded, by the uniform inverse covariance of Lemma S12.11(c) and bounded derivative cross-covariances. Independence in (S12.14) is not asserted separately for Hessian entries, endpoint variables or sampled derivative maxima: all residual evaluations are orthogonal to $A_M$, and Gaussian independence on

a countable dense collection together with continuity gives independence of the random smooth field. The internal correlations of the residual are retained. The partial-block operator norms of Section S13 differ from the coordinate norm by finite constants depending only on $d$; hence (S12.15). □

No inverse Hessian moment is used. Gaussian pin covariances and random field Hessians are different objects.

S12.5. **Endpoint identities and the full normalizer.**

**Lemma S12.14** (endpoint identities)**.** *Let $f$ be a $C^4$ function on a neighbourhood of $\mathcal{D}_r$ that satisfies the pins (S12.2), with partial-block norms $M_3$, $M_4$ on $\mathcal{D}_r$. For $i \in \{M, S\}$ put*

$$\alpha_i = \frac{f_{xx}(i)}{r}, \qquad \beta_i = \frac{\nabla_y f_x(i)}{r}, \qquad A_i = D_y^2 f(i), \qquad \text{so that} \quad H_i = \begin{pmatrix} r\alpha_i & r\beta_i^\top \\ r\beta_i & A_i \end{pmatrix}.$$

*Then*

$$|\alpha_i| \le \frac{M_3}{2}, \qquad \|\beta_i\| \le \frac{M_3}{2}, \qquad \|A_S - A_M\|_{\mathrm{op}} \le r M_3, \tag{S12.16}$$

$$|\alpha_M + 6k| \le \frac{r M_4}{2}, \qquad |\alpha_S - 6k| \le \frac{r M_4}{2}, \tag{S12.17}$$

$$\frac{\det H_i}{r} = \alpha_i \det A_i - r\, \beta_i^\top \operatorname{adj}(A_i)\, \beta_i, \tag{S12.18}$$

*and the block identity (S12.18) is valid also when $\alpha_i$ or $A_i$ is zero.*

*Proof.* Write $a = -r/2$ and $c = r/2$ for the axial coordinates of $M$ and $S$. The gradient pins give $f_x(a,0) = f_x(c,0) = 0$, so integration by parts yields

$$f_{xx}(a,0) = -\frac{1}{r}\int_a^c (c-x) f_{xxx}(x,0)\, dx, \qquad f_{xx}(c,0) = \frac{1}{r}\int_a^c (x-a) f_{xxx}(x,0)\, dx,$$

and $\int_a^c f_{xx}(x,0)\, dx = 0$. By the last identity $f_{xx}(\cdot,0)$ has mean zero on the pin interval, so at an endpoint $e \in \{a, c\}$

$$|f_{xx}(e,0)| \le \frac{1}{r}\int_a^c |f_{xx}(e,0) - f_{xx}(x,0)|\, dx \le \frac{M_3}{r}\int_a^c |x-e|\, dx = \frac{r M_3}{2},$$

which is the bound on $\alpha_i$. The vector $\nabla_y f_x(\cdot,0)$ also has mean zero on the pin interval, since its integral is $\nabla_y f(S) - \nabla_y f(M) = 0$; the same argument, with $\|\partial_x \nabla_y f_x\| \le M_3$, bounds $\|\beta_i\|$. It uses no simultaneous vector Rolle zero. Finally $A_S - A_M = \int_a^c \partial_x D_y^2 f(x,0)\, dx$, so $\|A_S - A_M\|_{\mathrm{op}} \le r M_3$. This proves (S12.16).

For (S12.17), the Hermite identity (S13.5) and $f(S) - f(M) = -kr^3$ say that the average of $f_{xxx}(\cdot,0)$ against the weight $(x-a)(c-x)$, whose mass is $r^3/6$, equals $12k$. Hence $f_{xxx}(x_0,0) = 12k$ for some $x_0 \in [a,c]$, and $|f_{xxx}(x,0) - 12k| \le r M_4$ on $[a,c]$. Inserting this in the two identities displayed first, with $\int_a^c (c-x)\, dx = \int_a^c (x-a)\, dx = r^2/2$, gives $|\alpha_M + 6k| \le r M_4/2$ and $|\alpha_S - 6k| \le r M_4/2$. The block identity (S12.18) is the expansion of $\det H_i$ along its first row and column, written with the adjugate so that it holds also for singular $A_i$. □

**Proposition S12.15** (the full normalizer)**.** (a) *Under $\mathbb{P}^0$ the quantities $\det H_i / r$, $i \in \{M, S\}$, and $W_r/r^2$ have uniformly bounded moments of every finite order.*

(b) *Let $A_0$ be $D_y^2 f(0)$ under the regression on $U_0 = v_0$; its law is a full-support nondegenerate Gaussian law on symmetric matrices. Then, uniformly on the compact parameter set $\mathcal{B} \times \mathcal{K} \times \{\text{frames}\}$,*

$$\frac{Z_r}{r^2} \longrightarrow z_0(b,k,R) := (6k)^2\, \mathbb{E}\big[\det(A_0)^2\, \mathbf{1}_{\{A_0 \prec 0\}}\big] \qquad (r \to 0). \tag{S12.19}$$

(c) *The right side of* (S12.19) *is continuous and strictly positive, and after reducing* $r_*$ *if needed,*

$$0 < z_* \le \frac{Z_r}{r^2} \le z^* < \infty \qquad (0 < r \le r_*,\ b \in \mathcal{B},\ k \in \mathcal{K},\ \textit{all frames}). \tag{S12.20}$$

*Proof.* (a) holds by (S12.13), (S12.16) and (S12.18). In particular $\sup \mathbb{E}^0(W_r/r^2)^2 < \infty$, by (S12.18), (S12.16), $\|A_S\| \le \|A_M\| + rM_3$ and Subsection S12.4; this is the uniform integrability of $W_r/r^2$ used in (b).

(b) Here the congruence is $H_i \mapsto \Delta_r^{-1} H_i \Delta_r^{-1}$ with $\Delta_r = \operatorname{diag}(\sqrt{r}, I_m)$, because

$$\Delta_r^{-1} H_i \Delta_r^{-1} = \begin{pmatrix} \alpha_i & \sqrt{r}\,\beta_i^\top \\ \sqrt{r}\,\beta_i & A_i \end{pmatrix}, \qquad \det(\Delta_r^{-1} H_i \Delta_r^{-1}) = \frac{\det H_i}{r}, \tag{S12.21}$$

which is exactly the determinant normalization of (S12.18). Finite-dimensional Gaussian regression gives uniform convergence of the means and covariances of $A_M$ to those of $A_0$, so $A_M \to A_0$ in law. By (S12.16), (S12.17) and $\|A_S - A_M\| \le rM_3$, the differences

$$\Delta_r^{-1} H_M \Delta_r^{-1} - \operatorname{diag}(-6k, A_M), \qquad \Delta_r^{-1} H_S \Delta_r^{-1} - \operatorname{diag}(6k, A_M),$$

whose entries are $\alpha_M + 6k$, $\alpha_S - 6k$, $\sqrt{r}\,\beta_i$ and $A_S - A_M$, tend to zero in probability, uniformly over the parameters. Slutsky's theorem then gives joint convergence in law of the congruence-scaled endpoint Hessians to

$$\operatorname{diag}(-6k, A_0) \text{ at } M, \qquad \operatorname{diag}(6k, A_0) \text{ at } S,$$

and the continuous mapping theorem applies off the null set $\{\det A_0 = 0\}$. The limiting matrices are almost surely nonsingular; this follows from the full matrix density and $k \ge k_- > 0$. Thus the type indicator in (S12.3) converges in law to $\mathbf{1}_{\{A_0 \prec 0\}}$. Uniform integrability from (S12.18), as in (a), gives (S12.19), uniformly on the compact parameter set. Uniformity can be checked by a subsequence argument in law: any allegedly bad sequence of parameters has a convergent subsequence, and continuous Gaussian means and covariances, (S12.17) and uniform integrability give the stated limit along that subsequence. The discontinuity set of the type indicator has Gaussian probability zero, and (S12.12) supplies a common integrable majorant.

(c) The right side of (S12.19) is continuous, and it is strictly positive because the Gaussian law of $A_0$ has positive density on an open ball of negative definite matrices. Compactness therefore yields (S12.20) after reducing $r_*$ if needed. The bound involves neither an added pin density nor a restricted good-event normalization. □

## S12.6. **The double soft-eigenvalue factor.**

**Lemma S12.16** (the two soft factors)**.** *In the setting of Lemma S12.14, suppose that* $H_M \prec 0$, *put* $B = -A_M \succ 0$ *and let* $0 < \lambda_1 \le \cdots \le \lambda_m = \Lambda$ *be the eigenvalues of* $B$. *Then* $\alpha_M < 0$, *the negative transverse Schur complement* $B - (r/(-\alpha_M))\,\beta_M \beta_M^\top$ *is positive definite, and*

$$|\det H_M| \le r\,\frac{M_3}{2}\,\det B. \tag{S12.22}$$

*Moreover, with the empty product equal to one when* $m = 1$,

$$|\det H_S| \le r\,\frac{M_3}{2}\Big[\lambda_1 + \frac{3}{2}\,rM_3\Big] \prod_{j=2}^{m} (\lambda_j + rM_3), \tag{S12.23}$$

*and the typed weight* (S12.3) *satisfies*

$$W_r \le \frac{r^2 M_3^2}{4}\,\lambda_1\Big[\lambda_1 + \frac{3}{2}\,rM_3\Big] \prod_{j=2}^{m} \lambda_j(\lambda_j + rM_3). \tag{S12.24}$$

*Proof.* Since $H_M \prec 0$, we have $\alpha_M < 0$, and the negative transverse Schur complement satisfies

$$B - \frac{r}{-\alpha_M}\,\beta_M\beta_M^\top \succ 0.$$

Indeed $\det H_M = r\alpha_M \det B'$ with $B' = A_M - (r/\alpha_M)\,\beta_M\beta_M^\top$, and $0 \prec -B' \preceq B$ on the typed support. Determinant monotonicity on positive definite matrices therefore gives (S12.22), with $|\alpha_M| \le M_3/2$ from (S12.16). This is a cancelled-pivot bound: no estimate of $1/\alpha_M$ is taken. From $\|A_S + B\| \le rM_3$, the ordered singular values of $A_S$ are at most $\lambda_j + rM_3$, and the norm of $\mathrm{adj}(A_S)$ is at most the product of the largest $m-1$ of these bounds. With the empty product equal to one when $m = 1$, (S12.18) and (S12.16) give (S12.23). Hence the actual typed weight satisfies (S12.24). □

The second soft factor $\lambda_1 + \frac{3}{2}rM_3$ includes the mixed-square term of the saddle. Omitting it, or replacing it by an unrelated factor of order one, changes the power of the radius. No independence of Gaussian entries is needed for (S12.24).

S12.7. **Boundary-layer integration, including the corank intersections.** The cap event is

$$\mathcal{G}_r = \Big\{\lambda_{\min}(-A_M) > \frac{4}{3k}\,rM_3^2\Big\} \cap \Big\{rM_4 \le \frac{3k}{10}\Big\}. \tag{S12.25}$$

By the deterministic separating-cylinder theorem (Theorem S13.2), elder pairing and the one-branch conclusion hold on $\mathcal{G}_r$ (Lemma S12.21). On the typed support $A_M \prec 0$ automatically. It suffices to control $\mathcal{G}_r^c$ directly; no converse is inferred from a bound on a local path event.

**Lemma S12.17** (eigenvalue coordinates)**.** *In the eigenvalue coordinates $B = O\,\mathrm{diag}(\lambda)O^\top$ of symmetric $m \times m$ matrices, Lebesgue measure has volume factor equal to a fixed constant times $\prod_{i<j}|\lambda_j - \lambda_i|$ times the angular volume. Consequently, by (S12.12), on the typed support the $\mathbb{P}^0$-law of the eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_m = \Lambda$ of $B = -A_M$ is bounded by*

$$C \exp\Big(-c\sum_{i=1}^m \lambda_i^2\Big)\,\Lambda^{m(m-1)/2}\,d\lambda_1 \cdots d\lambda_m. \tag{S12.26}$$

*Proof.* One way to see the volume factor is to differentiate $B = O\,\mathrm{diag}(\lambda)O^\top$: an infinitesimal rotation in the $ij$-plane changes the off-diagonal coordinate by $(\lambda_j - \lambda_i)$ times that angular increment. The angular space is compact, repeated-eigenvalue strata are Lebesgue null, and the finite multiplicity of the parametrization is absorbed into the constant. This is a change of variables for Lebesgue measure, not a claim that the conditional matrix is a GOE matrix. Using (S12.12), angular integration and $0 \le \lambda_i \le \Lambda$ gives (S12.26). □

**Proposition S12.18** (the pairing-defect bound)**.** *For $0 < r \le r_*$, with $r_*$ reduced if needed, $b \in \mathcal{B}$, $k \in \mathcal{K}$ and all frames,*

$$\mathbb{P}_r^{MS}(\mathcal{G}_r^c) \le C_3 r^3 + C_4 r^4 \le C r^3. \tag{S12.27}$$

*Proof. Depth failure, $m \ge 2$.* In (S12.15) absorb $\sqrt{m}$ into a constant $K_*$ and set $\Upsilon = J_r + \Lambda \ge 1$; since $\|A_M\|_F \le \sqrt{m}\,\Lambda$ on the typed support, $M_3 \le K_*\Upsilon$. Failure of the depth condition is contained in

$$0 < \lambda_1 \le D_*\, r\, \Upsilon^2, \qquad D_* = \frac{4K_*^2}{3k_-}. \tag{S12.28}$$

Unlike a scalar strip argument, this width may depend on the largest eigenvalue, and that dependence is retained in the integral. The variable $J_r$, but not the eigenvalues within $B$, is independent of $B$. In (S12.24), with $r \le 1$, all factors other than $\lambda_1[\lambda_1 + \frac{3}{2}rM_3]$ are bounded by

$Cr^2\Upsilon^{2m}$, and $\frac{3}{2}rM_3 \le E_* r\Upsilon$ with $E_* = \frac{3}{2}K_*$. Condition on $J_r$ and $\lambda_2, \dots, \lambda_m$, and extend the $\lambda_1$-integration of the positive majorant from the actual interval to $[0, D_* r\Upsilon^2]$. Then

$$\int_0^{D_* r\Upsilon^2} \lambda_1 \, (\lambda_1 + E_* r\Upsilon)\, d\lambda_1 = r^3 \Big[ \frac{D_*^3}{3}\, \Upsilon^6 + \frac{E_* D_*^2}{2}\, \Upsilon^5 \Big] \le Cr^3\Upsilon^6. \tag{S12.29}$$

The Gaussian factor involving $\lambda_1$ can be dropped for this upper bound. Equations (S12.24), (S12.26) and (S12.29) give

$$\begin{aligned} \mathbb{E}^0\big[W_r\, \mathbf{1}_{\{\text{depth failure}\}}\big] &\le Cr^5\, \mathbb{E} \int (J_r + \Lambda)^{2m+6}\, \Lambda^{m(m-1)/2} \exp\Big(-c \sum_{j=2}^m \lambda_j^2\Big)\, d\lambda_2 \cdots d\lambda_m \\ &\le C' r^5. \end{aligned} \tag{S12.30}$$

The last integral is finite by Gaussian polynomial integrability and the uniform moment of $J_r$ of order $2m+6$. All of $\lambda_2, \dots, \lambda_m$ are integrated over their ordered positive domain; in particular $\lambda_2$ is not bounded away from zero. The possible intersections of corank two and higher are included, with only nonnegative polynomial powers, and no angular singularity or inverse-eigenvalue moment remains.

*Depth failure, $m = 1$.* Now the largest and the smallest eigenvalue coincide, and using (S12.29) with $\Lambda$ held fixed would be invalid. Write $\lambda = -A_M > 0$ and use $M_3 \le K_*(J_r + \lambda)$. Depth failure implies $\lambda \le D_* r(J_r + \lambda)^2 \le 2D_* r J_r^2 + 2D_* r\lambda^2$, so it is contained in the union

$$\{0 < \lambda \le 4D_* r J_r^2\} \cup \{\lambda > 1/(4D_* r)\}. \tag{S12.31}$$

The branch of large $\lambda$ is retained. On the near branch, $M_3 \le K_*(1 + 4D_*)J_r^2$ for $r \le 1$ and $J_r \ge 1$; the bounded scalar density, (S12.24) and direct integration in $\lambda$ give a numerator at most $Cr^5\, \mathbb{E} J_r^{10}$. On the far branch the fourth-moment Markov inequality with the actual weight gives

$$\mathbb{E}^0\big[(W_r/r^2)\, \mathbf{1}_{\text{far}}\big] \le (4D_* r)^4\, \mathbb{E}^0\big[(W_r/r^2)\, \lambda^4\big] \le Cr^4. \tag{S12.32}$$

Thus the scalar numerator is at most $Cr^5 + C'r^6$. This separate argument is why the statement for all dimensions does not drop the scalar far branch.

*Fourth-derivative exception.* For every dimension the exception $\{rM_4 > 3k/10\}$ is contained in $\{rM_4 > 3k_-/10\}$, and it is bounded directly by

$$\mathbb{E}^0\big[(W_r/r^2)\, \mathbf{1}_{\{rM_4 > 3k_-/10\}}\big] \le \Big[\frac{10r}{3k_-}\Big]^4\, \mathbb{E}^0\big[(W_r/r^2)\, M_4^4\big] \le Cr^4. \tag{S12.33}$$

This is a pointwise Markov bound followed by a joint moment estimate, not a Cauchy–Schwarz bound with a missing square root. All joint moments in (S12.32) and (S12.33) are finite by Subsections S12.4 and S12.5.

*Normalization.* Divide the numerators by the full lower bound (S12.20). The result is (S12.27). □

## S12.8. **Pinned genericity, the elder mark and the proof of Theorem A.**

**Lemma S12.19** (genericity)**.** (a) *Fix $r > 0$ as in Definition S12.2, $b \in \mathbb{R}$, $k > 0$ and a frame. Then $f$ is $\mathbb{P}^0$-almost surely a Morse function on $\mathbb{T}_L^d$ with pairwise distinct critical values. If $0 < Z_r < \infty$, the same holds $\mathbb{P}_r^{MS}$-almost surely.*

(b) *Without pins, $f$ is almost surely a Morse function on $\mathbb{T}_L^d$ with pairwise distinct critical values.*

*Proof.* (a) The argument applies at every fixed positive $r$ and parameter; no common probability-one set for uncountably many laws is asserted. Away from the pins consider $(\nabla f(x), D^2 f(x)v)$ with $\|v\| = 1$: its range has dimension $2d$, and its parameter manifold has dimension $2d - 1$. Modulo the pin observations these functionals have full covariance rank: gradient and Hessian

orders are distinct, and the map $H \mapsto Hv$ from symmetric matrices onto $\mathbb{R}^d$ is surjective. For two distinct unpinned points the map $(\nabla f(x), \nabla f(y), f(x) - f(y))$ has range dimension $2d+1$ and parameter dimension $2d$. For a tie with a pinned height use $(\nabla f(x), f(x) - b)$, or the analogous map with $s$, of dimensions $d+1$ and $d$.

Exhaust the relevant domains by compact sets separated from the pins and from coincident points, using finitely many sphere charts where $v$ occurs. Covariance positivity (Lemma S12.8) and compactness give bounded Gaussian densities on each such set. An elementary overdetermined-zero lemma applies: on a mesh of size $\varepsilon$, a random $C^1$ map from dimension $q$ to dimension $p > q$ whose derivative has norm at most $N$ could have a zero only if one of $O(\varepsilon^{-q})$ mesh values has norm $O(N\varepsilon)$. The density bound makes the probability $O(N^p \varepsilon^{p-q})$, which tends to zero. First fix $N$, then let $N$ increase; the random derivative norm is almost surely finite. This proves the absence of all these zeros.

The pinned Hessians have a full density on their symmetric entries for fixed distinct pins, so their determinants are almost surely nonzero. The pinned heights differ because $k, r > 0$. Countable exhaustion now proves that $f$ is $\mathbb{P}^0$-almost surely Morse with distinct critical values. The transfer to $\mathbb{P}_r^{MS}$ follows by absolute continuity and $0 < Z_r < \infty$.

(b) This part is not in [7], whose proof of Theorem C uses it (Theorem S12.5). The argument of (a) applies with no pin observations. The map $(x, v) \mapsto (\nabla f(x), D^2 f(x)v)$ on $\mathbb{T}_L^d \times S^{d-1}$ has range dimension $2d$ and parameter dimension $2d-1$. Its covariance is positive definite at every $(x, v)$: by Lemma S12.8 the gradient and the symmetric Hessian entries at one site have a positive definite covariance, and $H \mapsto Hv$ maps the symmetric matrices onto $\mathbb{R}^d$. The covariance does not depend on $x$ and is continuous in $v$, so compactness of $S^{d-1}$ gives a uniform eigenvalue floor. For $\delta > 0$, the map $(x, y) \mapsto (\nabla f(x), \nabla f(y), f(x) - f(y))$ on the compact set $\{(x, y) : \operatorname{dist}(x, y) \geq \delta\}$ has range dimension $2d+1$ and parameter dimension $2d$. Its covariance is positive definite at every point by Lemma S12.8 at the two sites $x \neq y$, and it is continuous, so compactness again gives a uniform floor. The overdetermined-zero lemma, applied in finitely many charts, shows that almost surely neither map vanishes. Thus $f$ is almost surely Morse, and taking $\delta = 1/n$, $n \in \mathbb{N}$, no two distinct critical points have the same value. For $d = 3$ and $L = 24$ this is Proposition 1.1 of the main text. □

**Lemma S12.20** (the elder mark)**.** *For $f \in C^2(\mathbb{T}_L^d)$ and $M \in \mathbb{T}_L^d$ define the maximin connection level*

$$d_f(M) = \sup\Big\{ \min_{s' \in [0,1]} f(\gamma(s')) \ : \ \gamma \in C([0,1]; \mathbb{T}_L^d),\ \gamma(0) = M,\ f(\gamma(1)) > f(M) \Big\}, \tag{S12.34}$$

*with $\sup \varnothing = -\infty$, and the elder mark $e(f; M, S) = \mathbf{1}_{\{d_f(M) = f(S)\}}$.*

*(a) For every level $h$, the event $\{d_f(M) > h\}$ is a Borel set of $(M, f)$. Thus the equality $d_f(M) = f(S)$ is a Borel event.*

*(b) On the locus of Morse functions with pairwise distinct critical values, $d_f(M) = f(S)$ expresses ordinary superlevel elder death of $M$ at $S$. An essential maximum has no older endpoint.*

*(c) On that locus each finite superlevel $H_0$ bar has exactly one birth, at a local maximum, and one death, at a merging saddle of index $d-1$. Thus the pairs with $e = 1$ are precisely the finite bars, not just a local proxy, and the essential class of the global maximum is not counted.*

*Proof.* (a) For fixed pins, the event $\{d_f(M) > h\}$ can be represented by a countable union of polygonal-path tests, with rational interior vertices and endpoints, all staying strictly above $h$ and with endpoint value above $f(M)$; strict clearances allow approximation by such paths. For variable $M$, use countably many coordinate charts and an initial segment depending continuously on $M$. These are Borel tests in $(M, f)$. A complete argument is the measurability part of the proof of Lemma S5.3, which uses neither $d = 3$ nor $L = 24$: for a fixed path $\gamma$ from 0, the functional equal to $\min_s g(\gamma(s))$ on the open set $\{g(\gamma(1)) > g(0)\}$ and to $-\infty$ elsewhere is lower

semicontinuous on $C(\mathbb{T}^d_L)$; $d_g(0)$ is the supremum of these functionals; and $d_g(x) = d_{g(\cdot+x)}(0)$. Hence $(g,x) \mapsto d_g(x)$ is lower semicontinuous on $C(\mathbb{T}^d_L) \times \mathbb{T}^d_L$. The inclusion $C^2(\mathbb{T}^d_L) \subset C(\mathbb{T}^d_L)$ is continuous, so $\{d_f(M) > h\}$ is open in $(f, M)$, and $\{d_f(M) = f(S)\}$ is Borel in $(f, M, S)$. (b) and (c) are the elder rule of ordinary superlevel $H_0$ persistence for Morse functions with distinct critical values; see [4, §3] for background. □

**Lemma S12.21** (the cap implication)**.** *Let $r$, $b$, $k$ and the frame be as in Definition S12.2 (so $0 < r < L/(4\sqrt{2})$, $b \in \mathbb{R}$ and $k > 0$), and let $f$ be a $C^4$ Morse function on $\mathbb{T}^d_L$ with pairwise distinct critical values that satisfies the pins (S12.2) and whose norms satisfy the two inequalities defining $\mathcal{G}_r$ in (S12.25). Then the global ordinary superlevel elder partner of $M$ is $S$, that is, $e(f; M, S) = 1$, and exactly one actual ascending branch of $S$ ends at $M$.*

*Proof.* On $\mathcal{G}_r$ the deterministic cap theorem (Theorem S13.2, with the gap mark $k$) controls every boundary exit and supplies a connection through $S$ to a point higher than $b$. Therefore the global ordinary elder partner of $M$ is $S$, not merely a local adjacency candidate. The cap theorem also gives exactly one actual ascending branch to $M$; it does not identify that branch with the transverse ridge. □

*Proof of Theorem S12.3.* By Lemma S12.19(a), $f$ is $\mathbb{P}^{MS}_r$-almost surely Morse with distinct critical values, and by Lemma S12.21 the elder partner of $M$ is then $S$ on $\mathcal{G}_r$. Hence $0 \le 1 - p_r(b,k,R) \le \mathbb{P}^{MS}_r(\mathcal{G}_r^c)$, and (S12.27) gives (S12.4). □

S12.9. **The marked Kac–Rice identity.** This subsection gives the weighted pair Kac–Rice identity for the global elder mark. The Borel elder mark is reached by a finite-measure argument, not by a claim that it is lower semicontinuous. No local cap event is substituted for the global mark; the cap theorem is used only to estimate the probability that the global mark fails (Proposition S12.18).

**Proposition S12.22** (marked Kac–Rice identity)**.** *Let $\Omega \subset \mathbb{T}^d_L \times \mathbb{T}^d_L$ be a compact set of ordered pairs at positive distance from the diagonal, and write $\mathsf{G}(x,y) = (\nabla f(x), \nabla f(y)) \in \mathbb{R}^{2d}$.*

(a) *For $(x,y) \in \Omega$ and $z \in \mathbb{R}^{2d}$, Gaussian regression gives a regular conditional law $\mathcal{P}_{x,y,z}$ of the whole field $f \in C^2(\mathbb{T}^d_L)$ given $\mathsf{G}(x,y) = z$, and $(x,y,z) \mapsto \int F \, d\mathcal{P}_{x,y,z}$ is continuous for every bounded continuous cylinder functional $F$. For every bounded Borel function $h$ on $\Omega \times C^2(\mathbb{T}^d_L)$,*

$$\mathbb{E} \sum_{(x,y)\in\Omega:\ \nabla f(x)=\nabla f(y)=0} h\big((x,y), f\big) = \int_\Omega p_{x,y}(0)\, \mathbb{E}\Big[ |\det H_x \, \det H_y| \, h\big((x,y), f\big) \;\Big|\; \nabla f(x) = \nabla f(y) = 0 \Big] \, dx\, dy, \quad \text{(S12.35)}$$

*where $p_{x,y}$ is the Gaussian density of $\mathsf{G}(x,y)$ and the conditional expectation is taken under $\mathcal{P}_{x,y,0}$. For $h = \mathbf{1}_A$, $A$ a Borel set of $\Omega \times C^2(\mathbb{T}^d_L)$, both sides of (S12.35) are finite Borel measures in $A$.*

(b) *The elder mark $e$ of Lemma S12.20, the Hessian index indicators and their product are bounded Borel weights on $\Omega \times C^2(\mathbb{T}^d_L)$, so (S12.35) applies to them. Disintegrating the two endpoint heights at the prescribed values $f(M) = b$, $f(S) = b - kr^3$, the Jacobian $|\det H_M \det H_S|$ and the type indicators give exactly the endpoint weight $W_r$ of (S12.3). Thus the intensity of the typed pairs contains the full-pin density times $\mathbb{E}^0[W_r]$, and the selected intensity contains the same factor times the global elder probability $p_r$. There is no second determinant factor.*

(c) *Exhausting the off-diagonal domain towards the diagonal gives the near-diagonal formula.*

*Proof. Setting.* By the positive Fourier spectrum and the distinct-site argument of Lemmas S12.7 and S12.8, the covariance of $\mathsf{G}(x,y)$ is positive definite at every pair, uniformly on $\Omega$. The sample field is smooth. The derivative of $\mathsf{G}$ with respect to $(x,y)$ is block diagonal, so its normal Jacobian is $|\det H_x \det H_y|$. Theorem 2.2 of [2] therefore gives the unweighted zero-count formula for $\mathsf{G} = 0$

on $\Omega$, and finiteness on the compact set $\Omega$. Theorem 7.1 of [2] is used only for bounded nonnegative continuous cylinder weights, for which its lower-semicontinuity hypotheses hold automatically.

*A regular conditional law on the whole $C^2$ field.* Give $C^2(\mathbb{T}^d_L)$ its usual separable Banach topology. For $(x,y) \in \Omega$ and $z \in \mathbb{R}^{2d}$, finite-dimensional Gaussian regression gives a smooth conditional mean $\bar{f}_{x,y,z}$ and a centred residual Gaussian field $\mathsf{R}_{x,y}$ such that, as random elements of $C^2(\mathbb{T}^d_L)$,

$$\operatorname{law}(f \mid \mathsf{G}(x,y) = z) = \operatorname{law}(\bar{f}_{x,y,z} + \mathsf{R}_{x,y}).$$

The residual can be constructed from the full Gaussian field by subtracting

$$\operatorname{Cov}(f, \mathsf{G}(x,y))\ \operatorname{Cov}(\mathsf{G}(x,y))^{-1}\ \mathsf{G}(x,y).$$

Uniform nonsingularity on $\Omega$ and the smooth covariance imply that this subtraction is continuous in $(x,y,z)$ in every finite $C^q$ mean norm needed here. In particular

$$\mathcal{P}_{x,y,z}(A) = \mathbb{P}\{\bar{f}_{x,y,z} + \mathsf{R}_{x,y} \in A\}, \qquad A \text{ a Borel set of } C^2(\mathbb{T}^d_L),$$

is a regular conditional probability kernel, and $(x,y,z) \mapsto \int F\, d\mathcal{P}_{x,y,z}$ is continuous for every bounded continuous cylinder functional $F$. This is a law of the whole field, not merely a family of conditional laws of finitely many cylinder coordinates.

*Two finite measures.* For Borel sets $A$ of $\Omega \times C^2(\mathbb{T}^d_L)$ define, on the almost-sure zero set of $\mathsf{G}$,

$$\mu_1(A) = \mathbb{E} \sum_{(x,y)\in\Omega:\ \mathsf{G}(x,y)=0} \mathbf{1}_A((x,y),f),$$

$$\mu_2(A) = \int_\Omega p_{x,y}(0) \int |\det D^2\varphi(x)\ \det D^2\varphi(y)|\, \mathbf{1}_A((x,y),\varphi)\, \mathcal{P}_{x,y,0}(d\varphi)\, dx\, dy.$$

Theorem 2.2 of [2] makes $\mu_1$ finite. The total mass of $\mu_2$ equals the finite unweighted Kac–Rice integral, so $\mu_2$ is finite. For every bounded nonnegative continuous cylinder function $h$, Theorem 7.1 of [2], or equivalently the ordinary Gaussian regression proof of its continuous case, gives $\int h\, d\mu_1 = \int h\, d\mu_2$; this is (S12.35) for such $h$.

*Cylinder functions generate the Borel $\sigma$-algebra.* Choose a countable dense set $\mathcal{Q} \subset \mathbb{T}^d_L$. The maps $\varphi \mapsto \partial^\alpha\varphi(q)$, $q \in \mathcal{Q}$, $|\alpha| \le 2$, together with a countable coordinate basis on $\Omega$, separate the points of $\Omega \times C^2(\mathbb{T}^d_L)$. Because the $C^2$ norm on the compact torus is recovered from suprema over the dense set for derivatives through order two, these evaluations generate the Borel $\sigma$-algebra of $\Omega \times C^2(\mathbb{T}^d_L)$. Let $\mathcal{H}$ be the set of bounded real Borel functions $h$ with $\int h\, d\mu_1 = \int h\, d\mu_2$. Since $\mu_1$ and $\mu_2$ are finite, $\mathcal{H}$ is a vector space, contains the constants, and is closed under bounded pointwise convergence by dominated convergence. It contains the algebra of bounded continuous cylinder functions. The functional monotone-class theorem therefore implies that $\mathcal{H}$ contains every bounded Borel function on $\Omega \times C^2(\mathbb{T}^d_L)$, which proves (a). This does not extend the lower-semicontinuity hypothesis of [2, Theorem 7.1] by assertion: equality of the two finite measures is first obtained on a generating algebra and then identified on the full Borel $\sigma$-algebra.

*The elder and type marks.* Lemma S12.20(a) represents the maximin connection event by countably many strict rational polygonal-path tests in finitely many torus charts. Hence $((x,y),\varphi) \mapsto \mathbf{1}_{\{d_\varphi(x)=\varphi(y)\}}$ is Borel on the locus of Morse functions with pairwise distinct critical values, with the convention for the essential maximum preserved. The Hessian index indicators are Borel functions of the 2-jet, and their product with the elder mark is therefore a bounded Borel weight on $\Omega \times C^2(\mathbb{T}^d_L)$, so (a) applies. Disintegrating the two endpoint heights with their nondegenerate joint Gaussian density conditional on $\mathsf{G}(x,y) = 0$ gives (b).

*Exhaustion.* Finally, exhaust $\Omega$ towards the diagonal. The bound of Proposition S12.23, whose proof uses only parts (a) and (b) above, is locally $r\, dr$, hence integrable at zero, and nonnegative monotone convergence gives the near-diagonal formula (c). This step uses the estimates of Subsections S12.3, S12.5 and S12.10. □

S12.10. **The exact radial ledger.**

**Proposition S12.23** (radial ledger: the dimension cancels)**.** *Let $\pi_r(R; v)$ denote the Lebesgue density of $U_r$ at $v$.*

(a) *The density of $O_r$ at the specified pins is*

$$12\, r^{-(d+3)}\, \pi_r(R; v_r). \tag{S12.36}$$

(b) *In the midpoint $z$, the directed separation $S - M = rt$, the birth $b$ and the gap mark $k$, the candidate intensity per unit midpoint volume is*

$$r\, \mathcal{A}_r(b, k, t)\, dr\, db\, dk\, d\sigma(t), \qquad \mathcal{A}_r = 12\, \pi_r(R; v_r)\, \frac{Z_r}{r^2}, \tag{S12.37}$$

*where $\sigma$ is the ordinary surface measure on $S^{d-1}$, not normalized to total mass one. The selected intensity is $r\, \mathcal{A}_r\, p_r$ with the same measures.*

(c) *$\mathcal{A}_r$ is invariant under rotations of the transverse frame, so it is a function of $t$ alone.*

(d) *Uniformly on $\mathcal{B} \times \mathcal{K} \times S^{d-1}$, and with a uniform finite upper bound $\mathcal{A}^*$,*

$$\mathcal{A}_r \longrightarrow \mathcal{A}_0 = 12\, \pi_0(R; v_0)\, z_0(b, k, R) > 0 \qquad (r \to 0). \tag{S12.38}$$

*Proof.* (a) holds because $U_r = T_r O_r$, by (S12.9). (b) The change from the ordered pair $(M, S)$ to $(z, rt)$ has absolute determinant one. Polar coordinates for the separation contribute $r^{d-1}\, dr\, d\sigma(t)$, and the height change $(f(M), f(S)) = (b, b - kr^3)$ at fixed $r$ contributes $r^3\, db\, dk$. There is no factor $1/2$: maximum–saddle pairs are ordered by their different roles. With (a) and Proposition S12.22(a),(b) (this proof does not use its part (c), whose proof uses the present proposition), the powers of $r$ are

$$\text{spatial } (d-1) + \text{height mark } 3 + \text{pin } \big(-(d+3)\big) + \text{determinant normalizer } 2 = 1.$$

(c) Although frames are used in the proof, the original zero-gradient law and the determinant weight are unchanged by an orthogonal change of the transverse coordinates. Uniformity was proved on the compact orthogonal group and descends to the sphere; no global continuous transverse frame on the sphere, which does not exist in general, is assumed. (d) holds by Subsections S12.3 and S12.5. □

The selected intensity $r\, \mathcal{A}_r\, p_r$ supplies the domination needed below; it is not assumed that a fixed-axis statement can be integrated.

S12.11. **Lifetime pushforward and the proof of Theorem B.**

**Proposition S12.24** (lifetime pushforward on compact windows)**.** *For $\ell < k_- r_*^3$, every $k \in \mathcal{K}$ gives $r = (\ell/k)^{1/3} < r_*$, and at fixed $k$*

$$r\, \frac{dr}{d\ell} = \frac{1}{3}\, k^{-2/3}\, \ell^{-1/3}. \tag{S12.39}$$

*Hence a version of the candidate density of Theorem S12.4 is*

$$\nu_{\mathrm{cand}}(\ell) = \ell^{-1/3} \int_{\mathcal{B} \times \mathcal{K} \times S^{d-1}} \frac{\mathcal{A}_{(\ell/k)^{1/3}}(b, k, t)}{3k^{2/3}}\, db\, dk\, d\sigma(t), \tag{S12.40}$$

*and $\nu_{\mathrm{eld}}$ has the additional factor $p_{(\ell/k)^{1/3}}$ in the integrand.*

*Proof.* For $k \in \mathcal{K}$, $k \geq k_-$, so $\ell < k_- r_*^3$ gives $r^3 = \ell/k < r_*^3$. At fixed $k$, $dr/d\ell = \frac{1}{3} k^{-1/3} \ell^{-2/3}$, and multiplying by $r = k^{-1/3} \ell^{1/3}$ gives (S12.39). The candidate intensity (S12.37) is $r\, \mathcal{A}_r\, dr\, db\, dk\, d\sigma(t)$ per unit midpoint volume; its pushforward under $\ell = kr^3$ at fixed $(b, k, t)$ replaces $r\, dr$ by $\frac{1}{3} k^{-2/3} \ell^{-1/3}\, d\ell$, by (S12.39), and integrating over $\mathcal{B} \times \mathcal{K} \times S^{d-1}$ gives (S12.40). The selected intensity is $r\, \mathcal{A}_r\, p_r$ with the same measures (Proposition S12.23(b)), whence the factor $p_{(\ell/k)^{1/3}}$. □

*Proof of Theorem S12.4.* Uniform convergence (S12.38), the selection bound of Theorem S12.3 and compact domination give the common positive leading coefficient

$$c_{\mathcal{B},\mathcal{K}} = 4 \int_{\mathcal{B}\times\mathcal{K}\times S^{d-1}} k^{-2/3}\, \pi_0(R; v_0)\, z_0(b,k,R)\, db\, dk\, d\sigma(t), \tag{S12.41}$$

where $z_0 = (6k)^2\, \mathbb{E}[\det(A_0)^2 \mathbf{1}_{\{A_0 \prec 0\}}]$ with the actual conditional covariance of $A_0$. The pointwise selection estimate (S12.4), with $r^3 = \ell/k$, supplies the stronger difference statement

$$0 \le \nu_{\mathrm{cand}}(\ell) - \nu_{\mathrm{eld}}(\ell) \le C\, \ell^{2/3} \int_{\mathcal{B}\times\mathcal{K}\times S^{d-1}} \frac{\mathcal{A}^*}{3k^{5/3}}\, db\, dk\, d\sigma(t), \tag{S12.42}$$

which is finite because $k \ge k_- > 0$. This proves (S12.5). □

Formula (S12.41) is an identified Gaussian integral, not an evaluated closed form; there is no isotropic GOE substitution and no unexamined truncation of image tails. The bound (S12.42) is a difference of order $\ell^{2/3}$, not a claim that either density separately has a second-order expansion with that remainder; the convergence of $\mathcal{A}_r$ to $\mathcal{A}_0$ is not assigned a numerical rate.

## S12.12. **Short bars and moments.**

**Corollary S12.25** (short bars and moments)**.** *For the spatially and mark-restricted populations of Theorem S12.4, let $N_{\mathrm{eld}}(0,\varepsilon]$ and $N_{\mathrm{cand}}(0,\varepsilon]$ be the numbers of selected and of candidate pairs with lifetime at most $\varepsilon$. Per unit volume, as $\varepsilon \downarrow 0$,*

$$\mathbb{E}N_{\mathrm{eld}}(0,\varepsilon] \sim \tfrac{3}{2}\, c_{\mathcal{B},\mathcal{K}}\, \varepsilon^{2/3}, \qquad 0 \le \mathbb{E}[N_{\mathrm{cand}}(0,\varepsilon] - N_{\mathrm{eld}}(0,\varepsilon]] \le C\varepsilon^{5/3}. \tag{S12.43}$$

*For every real $q > -2/3$,*

$$\mathbb{E} \sum_{\text{selected},\ \ell\le\varepsilon} \ell^q \sim c_{\mathcal{B},\mathcal{K}}\, \frac{\varepsilon^{q+2/3}}{q+2/3}, \tag{S12.44}$$

*and the same leading expression holds for the candidates. For $q > -5/3$ the nonselected pairs alone satisfy*

$$\mathbb{E} \sum_{\text{candidate, not selected},\ \ell\le\varepsilon} \ell^q \le C_q\, \varepsilon^{q+5/3}. \tag{S12.45}$$

*For $q \le -2/3$ the two individual expectations in (S12.44) are infinite for every sufficiently small $\varepsilon > 0$. For the populations of Theorem S12.5, with $c_{d,L}$ in place of $c_{\mathcal{B},\mathcal{K}}$, the first relation of (S12.43), the asymptotics (S12.44) and the divergence for $q \le -2/3$ hold, for the selected and for the candidate pairs separately.*

*Proof.* Integrate the density versions of Theorem S12.4. For $q \le -2/3$ the individual expectations are infinite because the densities have a positive leading term $\ell^{-1/3}$. The unrestricted statements follow from Theorem S12.5 by the same integrable-power argument. □

Equation (S12.45) concerns the counting measure of the nonselected pairs directly, not a subtraction of two infinities. On compact windows the inverse lifetime power $\ell^{-p}$ is integrable over the selected, or the candidate, pairs exactly when $p < 2/3$, by (S12.44), and over the nonselected pairs alone at least when $p < 5/3$, by (S12.45). The compact-window bounds on the nonselected pairs, the second relation of (S12.43) and (S12.45), fail for the unrestricted populations: there the nonselected density does not vanish as $\ell \downarrow 0$, because far candidate pairs of small height difference do not vanish (Remark 2.5 of the main text).

S12.13. **Removing the birth and gap cutoffs by a direct intensity majorant.** The covariance $\Sigma_r$ of $U_r$ depends on the radius and the frame, not on the imposed target. Lemma S12.11 therefore supplies a fixed $r_0 > 0$, independent of $b$ and $k$, such that for $0 < r \le r_0$ the eigenvalues of $\Sigma_r$ are bounded above and away from zero. Reduce $r_0$ below the injectivity scale of the torus, below the embedding radius $L/(4\sqrt{2})$ of Definition S12.2, and below one. Neither a lower bound for $Z_r$ uniform over all marks nor a constant in (S12.4) uniform over all marks is required.

**Lemma S12.26** (target growth)**.** *For every $b \in \mathbb{R}$, $k > 0$, $0 < r \le r_0$ and every frame, the exact target* (S12.10) *satisfies*

$$|v_r| \le C(|b| + k), \qquad |v_r|^2 \ge c\,(b^2 + k^2), \tag{S12.46}$$

*with fixed $c, C > 0$; consequently*

$$\pi_r(R; v_r) \le C \exp\bigl[-c\,(b^2 + k^2)\bigr]. \tag{S12.47}$$

*For every fixed finite $p$, uniformly over $0 < r \le r_0$ and the frames,*

$$\mathbb{E}^0(1 + \|f\|_{C^3})^p \le C_p\,(1 + |b| + k)^p. \tag{S12.48}$$

*Proof.* For the second inequality of (S12.46) it suffices to use the two coordinates $b - kr^3/2$ and $12k$ of $v_r$: if $r \le 1$, then $b^2 \le 2(b - kr^3/2)^2 + k^2/2$. The invertible covariance bound then yields (S12.47). Conditional means of the Fourier coefficients are bounded by $C|v_r|$ and their centred variances by one, and the summability and Minkowski argument of Lemma S12.12 gives the target-growth bound (S12.48). This is not the compact-target bound (S12.13) applied outside its scope; it is its explicit polynomial-growth version. □

**Lemma S12.27** (unnormalized intensity majorant)**.** (a) *For every $C^4$ function satisfying the pins* (S12.2) *with $r \le 1$, and $i \in \{M, S\}$,*

$$\frac{|\det H_i|}{r} \le C(1 + \|f\|_{C^3})^d, \qquad 0 \le \frac{W_r}{r^2} \le C(1 + \|f\|_{C^3})^{2d}. \tag{S12.49}$$

(b) *For $b \in \mathbb{R}$, $k > 0$, $0 < r \le r_0$ and every frame,*

$$0 \le \mathcal{A}_r(b, k, t) = 12\,\pi_r(R; v_r)\,\frac{Z_r}{r^2} \le C\,(1 + |b| + k)^{2d} \exp\bigl[-c\,(b^2 + k^2)\bigr] =: \Psi(b, k). \tag{S12.50}$$

(c) *For every fixed $r \in (0, r_0]$, every $b \in \mathbb{R}$ and every $k > 0$, $Z_r$ is positive and finite, so $p_r$ is defined. For every fixed $b \in \mathbb{R}$ and $k > 0$, $p_r \to 1$ as $r \to 0$.*

*Proof.* (a) follows from (S12.16) and (S12.18). These inequalities remain true when $k$ is arbitrarily small: no division by $k$ or by $Z_r$ occurs. Combining (a) with (S12.47) and (S12.48) gives the full unnormalized intensity majorant (S12.50). (c) Append both endpoint Hessians to the distinct-site pins: the full residual Gaussian density is positive on a product of an open cone of maxima and an open cone of index $d - 1$. Thus $p_r$ is defined, although no lower floor for $Z_r$ uniform over all marks has been used. For fixed $b$ and $k > 0$, Theorem S12.3 on a compact neighbourhood implies $p_r \to 1$. □

**Proposition S12.28** (near densities and the unrestricted coefficient)**.** *At a fixed lifetime $\ell > 0$ the near-pair restriction $r \le r_0$ is exactly $k \ge \ell/r_0^3$, and after the exact change of variables* (S12.39)*,*

$$\ell^{1/3}\,\nu_{\mathrm{cand}}^{\mathrm{near}}(\ell) = \int_{\mathbb{R}\times(0,\infty)\times S^{d-1}} \mathbf{1}_{\{k \ge \ell/r_0^3\}}\,\frac{\mathcal{A}_{(\ell/k)^{1/3}}(b, k, t)}{3k^{2/3}}\,db\,dk\,d\sigma(t), \tag{S12.51}$$

*with the integrand defined to be zero where the indicator vanishes. The selected near density $\nu_{\mathrm{eld}}^{\mathrm{near}}$ has the additional factor $p_{(\ell/k)^{1/3}}$ in the integrand. Both integrands are dominated by the integrable function $\Psi(b, k)/(3k^{2/3})$. As $\ell \downarrow 0$, both $\ell^{1/3}\nu_{\mathrm{cand}}^{\mathrm{near}}(\ell)$ and $\ell^{1/3}\nu_{\mathrm{eld}}^{\mathrm{near}}(\ell)$ converge to*

$$c_{d,L} = 4\int_{\mathbb{R}\times(0,\infty)\times S^{d-1}} k^{-2/3}\,\pi_0(R; v_0)\,z_0(b, k, R)\,db\,dk\,d\sigma(t), \qquad 0 < c_{d,L} < \infty. \tag{S12.52}$$

*In particular $c_{\mathcal{B},\mathcal{K}}$ increases to $c_{d,L}$ along an exhausting sequence of mark rectangles.*

*Proof.* The majorant $\Psi(b,k)/(3k^{2/3})$ is integrable: the exponent $-2/3$ of $k$ is strictly greater than $-1$ at zero, and Gaussian tails absorb the polynomial at infinity and in $b$. This handles the shrinking-$k$ boundary, where a compact-window selection constant can blow up. For fixed $b \in \mathbb{R}$, $k > 0$ and $t$, the indicator tends to one, $\mathcal{A}_r \to \mathcal{A}_0$ and $p_r \to 1$. Dominated convergence therefore gives the same limit (S12.52) for the two near densities. The step uses the unnormalized product $\pi_r Z_r$, not a bound for the normalized law $\mathbb{P}_r^{MS}$ uniform over all marks. It proves an unrestricted leading limit; it does not produce a quantitative unrestricted remainder from dominated convergence. □

S12.14. **The off-diagonal contribution and the proof of Theorem C.**

**Lemma S12.29** (far pairs)**.** *On the compact set of spatial pairs at torus distance at least $r_0$, the original full-pin covariance, of the values and gradients at the two points, is uniformly positive definite, and its inverse is uniformly bounded. For $0 < \ell \le 1$ and $b \in \mathbb{R}$, the original target $(b, 0, \dots, 0, b-\ell, 0, \dots, 0)$ has squared norm at least $b^2$, so its Gaussian density is at most $C\exp(-cb^2)$; the conditional endpoint determinant-product moments on this set are at most $C(1+|b|)^{2d}$; and the height transformation $(f(M), f(S)) = (b, b-\ell)$ has absolute determinant one. Consequently the densities $\nu_{\mathrm{cand}}^{\mathrm{far}}$ and $\nu_{\mathrm{eld}}^{\mathrm{far}}$ of the pairs at torus distance at least $r_0$ satisfy*

$$0 \le \nu_{\mathrm{eld}}^{\mathrm{far}}(\ell) \le \nu_{\mathrm{cand}}^{\mathrm{far}}(\ell) \le C, \qquad 0 < \ell \le 1. \tag{S12.53}$$

*The near and far populations cover all distinct pairs; a distance boundary may be assigned to either, as it is spatially null.*

*Proof.* Every distinct-site jet list has full rank by Lemma S12.8, and compactness applies without a diagonal singularity; this gives the covariance statements. Integrating over $b$ and over the compact domain of relative positions gives (S12.53). No bound on remote critical counts conditional on a rare cap failure is inferred: (S12.53) is an unconditional off-diagonal height-density bound, proved directly from a nonsingular covariance on a fixed compact set. □

The proof uses only the compactness of the set of pairs and the full rank of distinct-site jet lists, so (S12.53) holds, with a constant depending on $\rho$, for the pairs at torus distance at least any fixed $\rho > 0$. For $d = 3$ and $L = 24$ this is Proposition S6.2.

*Proof of Theorem S12.5.* The near and far populations cover all distinct pairs (Lemma S12.29), and stationarity converts the spatial integral into a density per unit volume. Combining (S12.51)–(S12.52) with $\ell^{1/3}$ times (S12.53) proves (S12.6). On the almost-sure locus of Morse functions with distinct critical values (Lemma S12.19(b)), the selected full-pair count is precisely the count of finite bars, by Lemma S12.20(c), and the essential class of the global maximum is not counted. □

S12.15. **Elimination of birth and gap from the leading coefficient.** This simplification uses the inversion symmetry, or parity, of a stationary real covariance, not rotational isotropy.

**Lemma S12.30** (parity factorization)**.** *At the contact point write $G = \nabla f(0)$; let $V_t = D^2 f(0)\, t$, expressed in the frame $R$, with the $d$ coordinates $(f_{xx}, f_{xy_1}, \dots, f_{xy_m})(0)$; and let $A_t = D_y^2 f(0)$ be the transverse Hessian block on $t^\perp$. All odd derivatives $(G, \partial_t^3 f(0))$ are jointly independent of the even derivatives $(f(0), V_t, A_t)$. Put*

$$\tau_t^2 = \mathrm{Var}(\partial_t^3 f(0) \mid G = 0) > 0, \qquad D_t = \mathbb{E}\big[(\det A_t)^2\, \mathbf{1}_{\{A_t \prec 0\}} \mid V_t = 0\big]. \tag{S12.54}$$

*Given $V_t = 0$, $A_t$ is an actual centred nondegenerate Gaussian symmetric matrix, not automatically a GOE matrix. The variance $\tau_t^2$, the densities $p_G(0)$ and $p_{V_t}(0)$, and $D_t$ are positive; all are continuous in the frame and invariant under changes of the transverse frame. With $\phi_\tau$ the centred*

*scalar Gaussian density of standard deviation $\tau$, the contact density and the cone moment factor as*

$$\pi_0(R; v_0) = p_{(f,V_t)}(b,0)\, p_G(0)\, \phi_{\tau_t}(12k), \qquad z_0 = (6k)^2\, \mathbb{E}\big[(\det A_t)^2\, \mathbf{1}_{\{A_t \prec 0\}} \mid f = b,\ V_t = 0\big]. \tag{S12.55}$$

*Proof.* Every odd derivative of the even covariance $K_L$ vanishes at zero, so the odd and the even derivatives at 0 are uncorrelated, hence independent. Finite-jet rank (Lemma S12.8) proves $\tau_t > 0$, the positivity of $p_G(0)$ and $p_{V_t}(0)$, and $D_t > 0$. The factorization (S12.55) follows from this independence and the target $v_0 = (b, 0, 0, 12k, 0, \dots, 0)$. □

**Proposition S12.31** (the leading coefficient)**.** *Integrating $b$ by ordinary Gaussian disintegration turns* (S12.52) *into*

$$c_{d,L} = 144 \int_{S^{d-1}} p_G(0)\, p_{V_t}(0)\, D_t \Big[ \int_0^\infty k^{4/3}\, \phi_{\tau_t}(12k)\, dk \Big]\, d\sigma(t). \tag{S12.56}$$

*The scalar integral can be evaluated exactly: with $w = 12k$ and the elementary gamma integral,*

$$144 \int_0^\infty k^{4/3}\, \phi_\tau(12k)\, dk = \frac{\Gamma(7/6)\, \tau^{4/3}}{24^{1/3}\sqrt{\pi}}. \tag{S12.57}$$

*Consequently the full leading coefficient is*

$$c_{d,L} = \frac{\Gamma(7/6)}{24^{1/3}\sqrt{\pi}} \int_{S^{d-1}} p_G(0)\, p_{V_t}(0)\, \tau_t^{4/3}\, D_t\, d\sigma(t). \tag{S12.58}$$

*Proof.* By (S12.55), the integral over $b$ of $p_{(f,V_t)}(b,0)\, \mathbb{E}[(\det A_t)^2 \mathbf{1}_{\{A_t \prec 0\}} \mid f = b, V_t = 0]$ is $p_{V_t}(0)\, D_t$, and $4 \cdot (6k)^2 k^{-2/3} = 144\, k^{4/3}$; this gives (S12.56). The substitution $w = 12k$ and the gamma integral give (S12.57), and (S12.58) follows. □

This eliminates both unbounded mark integrals. The remaining angular integral and the expectation over the negative definite cone are finite, strictly positive and specified by derivatives of the exact periodized covariance. Formula (S12.58) is an explicit finite-dimensional Gaussian representation, not an elementary closed form or a numerical enclosure. The factor 24 in it is a numerical Jacobian and gamma factor, not the side length $L$; it is the same for every $L$. Changing the field amplitude by $a > 0$ scales the coefficient by $a^{-2/3}$, consistently with $\nu_{af}(\ell) = a^{-1}\nu_f(\ell/a)$; this is an additional normalization check.

## S12.16. **Scope.**

*Remark* S12.32 (what is not proved). This section gives no numerical value of $r_*$ or $C$ on a prescribed band; no bound $1 - p_r \le Cr^3$ with a constant uniform over unbounded $b$ and $k$; no unrestricted remainder of order $\ell^{2/3}$ for the difference of the densities, which is proved only on compact windows; no elementary closed form or numerical enclosure of (S12.58); and no evaluated lower coefficient for the failure of pairing.

*Remark* S12.33 (background). [2, §§2 and 7] supplies the Gaussian Kac–Rice and expected-integral background; the extension to the full field and to Borel marks, and its application to pairs, are those of Proposition S12.22. [4, §3] supplies elder-rule background. The separating cap in every dimension is Section S13.

## S13. The marked-cylinder cap

This section proves the deterministic separating-cylinder theorem that Section S12 uses for elder selection (Lemma S12.21); it is the cap theorem on which [7] relies. Nothing in it is probabilistic. The theorem covers every dimension $d \geq 2$, every birth, every positive gap mark and every coordinate frame, with the partial-block norms of Definition S13.1. As in Section S12, $m = d - 1$, and the gap mark is $k$, so that $f(S) = f(M) - kr^3$.

**Definition S13.1** (cylinder and partial-block norms)**.** Let $d \geq 2$ and $r > 0$, and use coordinates $(x, y) \in \mathbb{R} \times \mathbb{R}^m$. The cylinder is

$$\mathcal{D}_r = [-2r, 2r] \times \bar{B}(0, 2r). \tag{S13.1}$$

For a $C^4$ function $f$ on a neighbourhood of $\mathcal{D}_r$ and $j = 3, 4$ put

$$M_j = \max_{0 \leq i \leq j} \sup_{\mathcal{D}_r} \|\partial_x^{j-i} D_y^i f\|_{\mathrm{op}}, \tag{S13.2}$$

where the norm is the absolute value for $i = 0$, and for $i \geq 1$

$$\|\partial_x^{j-i} D_y^i f\|_{\mathrm{op}} = \sup_{w_1, \ldots, w_i} |\partial_x^{j-i} D_y^i f\,[w_1, \ldots, w_i]|,$$

the supremum being over transverse unit vectors $w_1, \ldots, w_i$ chosen independently of each other. In dimension two, $M_j$ is the maximum of the absolute values of the coordinate partial derivatives of order $j$, not the larger full-tensor operator norm.

**Theorem S13.2** (separating cylinder)**.** *Let $f$ be $C^4$ on a neighbourhood of $\mathcal{D}_r$, with critical points $M = (-r/2, 0)$ and $S = (r/2, 0)$, $f(M) = b$ and $f(S) = s := b - kr^3$, where $b \in \mathbb{R}$ and $k > 0$. Put $\lambda = \lambda_{\min}(-D_y^2 f(M))$ and suppose that*

$$\lambda > \frac{4}{3k}\, r\, M_3^2, \qquad rM_4 \leq \frac{3k}{10}. \tag{S13.3}$$

*(a)* Ridge. *For every $x \in [-2r, 2r]$ the function $f(x, \cdot)$ has a unique maximizer $h(x)$ on $\bar{B}(0, 2r)$; $h$ is $C^3$, $\nabla_y f(x, h(x)) = 0$ and $h(\pm r/2) = 0$. The only critical points of $f$ in $\mathcal{D}_r$ are $M$ and $S$; $D^2 f(M)$ is negative definite, so $M$ is a maximum, and $D^2 f(S)$ has exactly one positive eigenvalue.*

*(b)* Cap. *On $\mathcal{C} = [-2r, r/2] \times \bar{B}(0, 2r)$ one has $f \leq b$, and every boundary point of $\mathcal{C}$ has value at most $s$, with equality only at $S$. The ridge path $x \mapsto (x, h(x))$, $-r/2 \leq x \leq 2r$, starts at $M$, passes through $S$ and ends at $z = (2r, h(2r))$, where $f(z) > b$; its minimum is exactly $s$.*

*(c)* Elder pairing. *Suppose that $\mathcal{D}_r$ is embedded, through an orthonormal product chart, in a compact manifold, and that $f$ extends to a Morse function on it with pairwise distinct critical values. Then every path from $M$ to a point with value greater than $b$ has minimum at most $s$, the maximin connection level of $M$ to an older point is $s$, and the ordinary superlevel elder death partner of $M$ is $S$.*

*(d)* Branches. *In the situation of* (c)*, for the gradient of a smooth Riemannian metric, exactly one ascending unstable half-branch of $S$ ends at $M$.*

*No Morse–Smale assumption and no identification of the ridge with a flow trajectory are used.*

S13.1. **The normalized gap mark.** We first prove Theorem S13.2 for $k = 1/6$. Then (S13.3) reads

$$\lambda > 8\, r\, M_3^2, \qquad rM_4 \leq \frac{1}{20}, \tag{S13.4}$$

and $s = b - r^3/6$. In Lemmas S13.3–S13.6, $f$ satisfies the hypotheses of Theorem S13.2 with $k = 1/6$, and $a = -r/2$, $c = r/2$ are the axial coordinates of the pins.

**Lemma S13.3** (the forced third derivative)**.** *For every $C^3$ function with critical points $M$ and $S$, the critical pins give the Hermite identity*

$$f(S) - f(M) = -\frac{1}{2}\int_{-r/2}^{r/2}\Big(x + \frac{r}{2}\Big)\Big(\frac{r}{2} - x\Big) f_{xxx}(x,0)\,dx. \tag{S13.5}$$

*Under (S13.4) the weighted average of $f_{xxx}$ in (S13.5) is 2. In particular $M_3 \geq 2$, and some $x_0$ in the pin interval satisfies $f_{xxx}(x_0,0) = 2$. Throughout $\mathcal{D}_r$, in the quadratic-form order,*

$$f_{xxx} \geq 2 - 5rM_4 \geq \frac{7}{4}, \qquad D_y^2 f \preceq -(\lambda - 5rM_3)\, I = -\delta I, \qquad \delta > rM_3(8M_3 - 5), \qquad 8M_3 - 5 \geq 11, \tag{S13.6}$$

*where $\delta := \lambda - 5rM_3$.*

*Proof.* Write $a = -r/2$, $c = r/2$ and $\psi(x) = f_x(x,0)$, so that $\psi(a) = \psi(c) = 0$ by the critical pins. The kernel $K(x) = \frac{1}{2}(x-a)(c-x)$ vanishes at $a$ and $c$ and has $K'' = -1$, so two integrations by parts give $\int_a^c \psi\,dx = -\int_a^c K\psi''\,dx$; this is (S13.5), since $\int_a^c \psi\,dx = f(S) - f(M)$. The positive kernel in (S13.5) has integral $r^3/12$, and $f(S) - f(M) = -r^3/6$, so the weighted average of $f_{xxx}$ is 2. This forced lower bound on $M_3$ is useful; replacing $M_3$ by the larger $1 + M_3$ is unnecessary. The partial-block norms control increments by $|\Delta x| + \|\Delta y\|$, with no factor depending on the dimension. Every point of $\mathcal{D}_r$ is within this product distance less than $5r$ of $(x_0, 0)$, and likewise less than $5r$ from $M$. Thus (S13.6) holds, with $8M_3 - 5 \geq 11$ because $M_3 \geq 2$. The Hessian inequality holds in the quadratic-form order throughout $\mathcal{D}_r$; a scalar lower bound on a diagonal entry cannot replace this matrix inequality when $d > 2$. □

**Lemma S13.4** (the vector ridge)**.** *Put $w(x) = \nabla_y f(x,0)$ and $q = 8M_3 - 5 \geq 11$. For $|x| \leq 2r$,*

$$\|w(x)\| \leq \frac{M_3}{2}\Big|x^2 - \frac{r^2}{4}\Big| \leq \frac{15}{8}\,M_3 r^2 < 2M_3 r^2, \tag{S13.7}$$

$$\|w'(x)\| \leq \frac{M_3}{r}\int_{-r/2}^{r/2} |x - x'|\,dx' \leq 2M_3 r. \tag{S13.8}$$

*For every $x \in [-2r, 2r]$, the function $f(x,\cdot)$ is $\delta$-strongly concave on $\bar{B}(0, 2r)$, and its maximum there is attained at a unique interior point $h(x)$, which solves $\nabla_y f(x, h(x)) = 0$. Moreover $h(\pm r/2) = 0$, $h$ is $C^3$, and*

$$\|h(x)\| \leq \frac{\|w(x)\|}{\delta} < \frac{2r}{q}, \tag{S13.9}$$

$$h' = -(D_y^2 f)^{-1}\partial_x \nabla_y f, \qquad \|h'\| < \frac{2rM_3 + M_3\|h\|}{\delta} < \frac{2}{q} + \frac{2}{q^2} =: \omega, \qquad \omega \leq \frac{24}{121}, \tag{S13.10}$$

$$M_3\,\omega = \frac{1}{4}\Big(1 + \frac{6}{q} + \frac{5}{q^2}\Big) \leq \frac{48}{121}. \tag{S13.11}$$

*Proof.* The function $w$ vanishes at the two pins, and $\|w''\| \leq M_3$. Applying the scalar two-node interpolation remainder to every unit-vector projection of $w$ gives (S13.7), including outside the pin interval: there the bound follows from the interpolation remainder on the convex hull of the three points, not from an estimate valid only inside. Also $\int_{-r/2}^{r/2} w'(x')\,dx' = 0$. Subtract this average from $w'(x)$, use the Lipschitz constant $M_3$ and integrate $|x - x'|$; this gives the first inequality of (S13.8). At the end of the cylinder,

$$\int_{-r/2}^{r/2} |2r - x'|\,dx' = 2r^2,$$

so $\|w'(x)\| \leq 2M_3 r$ on $|x| \leq 2r$, and the bound is sharp. This vector average argument does not posit a common point at which every component of $w'$ vanishes.

For fixed $x$, $f(x,\cdot)$ is $\delta$-strongly concave on the transverse ball by (S13.6). On its boundary $\|y\| = 2r$,

$$\nabla_y f(x,y)\cdot y \le \|w(x)\|\,\|y\| - \delta\|y\|^2 < 0.$$

The maximum on the compact ball is therefore interior (moving inward increases the value at a boundary point), and strict concavity makes it unique. The maximizer $h(x)$ solves $\nabla_y f(x,h(x)) = 0$, has $h(\pm r/2) = 0$ and is $C^3$ by the implicit function theorem. Strong monotonicity gives (S13.9). The derivative of $h$ is given in (S13.10), and (S13.8), (S13.9) and the bound on the mixed derivative give the bound for $\|h'\|$ in (S13.10). Since $M_3 = (q+5)/8$, (S13.11) follows. All these inequalities use transverse operator norms; there is no hidden coordinate sum depending on the dimension. □

**Lemma S13.5** (the ridge derivative)**.** *Put $g(x) = f(x,h(x))$ and $F = g' = f_x(x,h(x))$ on $[-2r,2r]$, and $v = (1,h')$. Then, with the derivatives of $f$ evaluated at $(x,h(x))$,*

$$F'' = f_{xxx} + 3f_{xxy}[h'] + 3f_{xyy}[h',h'] + f_{yyy}[h',h',h'], \tag{S13.12}$$

*and*

$$F'' > \frac{7}{4} - \frac{2554128}{1771561} = \frac{2184415}{7086244} > \frac{1}{4}. \tag{S13.13}$$

*The function $F$ vanishes exactly at $a$ and $c$; it is negative between them and positive outside. The only critical points of $f$ in $\mathcal{D}_r$ are $M$ and $S$. The Schur complement of $D_y^2 f$ in $D^2 f$ is $F'(a) < 0$ at $M$ and $F'(c) > 0$ at $S$, so $M$ is a maximum and $D^2 f(S)$ has exactly one positive eigenvalue.*

*Proof.* Differentiating the transverse critical equation $\nabla_y f(x,h(x)) = 0$ gives $(D^2 f\, v)_y = 0$. The vector $v' = (0,h'')$ is purely transverse, so $D^2 f[v,v'] = (D^2 f\, v)\cdot v' = 0$, and every term of the third derivative of $g$ that involves $h''$ is zero. Hence $F'' = D^3 f[v,v,v]$, which is the exact identity (S13.12). It is valid for a vector $h$, and it does not assert that the ridge is an integral curve of the gradient. By (S13.10) and (S13.11), the adverse part of (S13.12) is at most

$$\frac{48}{121}\Big[3 + 3\cdot\frac{24}{121} + \Big(\frac{24}{121}\Big)^2\Big] = \frac{2554128}{1771561},$$

and (S13.6) gives (S13.13). The function $F$ has the prescribed zeros $a$ and $c$, and strict convexity makes them its only zeros, with $F$ negative between them and positive outside. Every critical point in $\mathcal{D}_r$ lies on the ridge, hence $\mathcal{D}_r$ contains precisely the critical points $M$ and $S$. The Schur complement of $D_y^2 f$ at these points is $F'(a) < 0$, respectively $F'(c) > 0$, which proves that $M$ is a maximum and that $S$ has exactly one positive Hessian direction. □

**Lemma S13.6** (the cap boundary)**.** *Let $\mathcal{C} = [-2r, r/2] \times \bar{B}(0,2r)$ and $s = b - r^3/6$. Then $f(x,y) \le g(x) \le b$ on $\mathcal{C}$,*

$$g(-2r) \le b - \frac{9}{32}\,r^3 = s - \frac{11}{96}\,r^3, \qquad g(2r) \ge s + \frac{9}{32}\,r^3 = b + \frac{11}{96}\,r^3, \tag{S13.14}$$

*and, for $-2r \le x \le r/2$ and $\|y\| = 2r$,*

$$f(x,y) \le g(x) - \frac{\delta}{2}\,\|y - h(x)\|^2 < b - \frac{4400}{121}\,r^3 < s. \tag{S13.15}$$

*On the face $x = c$ of $\mathcal{C}$, $f(c,y) \le s - \frac{\delta}{2}\|y\|^2$, with equality only at $S$. Thus every boundary point of $\mathcal{C}$ has value at most $s$, with equality only at $S$. The ridge segment $a \le x \le 2r$ starts at $M$, passes through $S$ and ends at $z = (2r, h(2r))$, where $f(z) > b$ by (S13.14); its minimum is exactly $s$.*

*Proof.* Strong concavity gives $f(x,y) \le g(x)$, and $g(x) \le b$ on $\mathcal{C}$ by the signs of $F = g'$ in Lemma S13.5. To control the longitudinal faces, subtract $(x-a)(x-c)/8$ from $F$. By (S13.13) the difference is convex, and it vanishes at $a$ and $c$; hence it is nonnegative outside $[a,c]$. Integrating on either exterior interval gives (S13.14). These are lower and upper inequalities at the actual ridge endpoints, not Taylor fits. For a transverse boundary point $\|y\| = 2r$ of $\mathcal{C}$, (S13.6) and (S13.9)

imply (S13.15); indeed $\delta > rM_3(8M_3 - 5) \geq 22r$ and $\|y - h(x)\| > 20r/11$. This covers the entire curved side of $\mathcal{C}$, including its intersections with the end faces; in dimension two it covers both horizontal edges and their corners. On the right face $x = c$, $h(c) = 0$, which gives the stated bound there. On the left face use $f \leq g$ and (S13.14). Thus every boundary exit from $\mathcal{C}$ has value at most $s$. Finally, along the ridge segment $g$ decreases from $b$ to $s$ on $[a, c]$ and increases from $s$ to $g(2r) > b$ on $[c, 2r]$, so its minimum is exactly $s$. A path through $S$ at level $s$ is not a preemption path staying strictly above $s$. □

## S13.2. **Proof of the theorem.**

*Proof of Theorem S13.2. The case $k = 1/6$.* Parts (a) and (b) are Lemmas S13.4, S13.5 and S13.6.

(c) Every point with value greater than $b$ lies outside $\mathcal{C}$. Every continuous path from $M$ to such a point must first meet the boundary of $\mathcal{C}$, so its minimum is at most $s$, regardless of later remote excursions or re-entry. The ridge path of (b) has minimum exactly $s$ and an endpoint with value greater than $b$. Hence the maximin connection level of $M$ to an older point is $s$. For a compact Morse function with distinct critical values, ordinary superlevel-component persistence identifies this maximin value with the elder death level of $M$; the older endpoint excludes the case of the essential, global maximum. The unique critical value $s$ identifies $S$.

(d) At $S$ the Hessian is negative definite on the transverse hyperplane. The unstable eigenvector for gradient ascent has a nonzero $x$-component: its Hessian quadratic form is positive, whereas that hyperplane is strictly negative. This argument also holds for a smooth positive definite Riemannian metric. One half-branch enters the cap $\mathcal{C}$ at height greater than $s$; the other enters the outside. The first cannot exit, because its height increases and every boundary value is at most $s$. Its $\omega$-limit is a critical point in the compact cap; gradient monotonicity and the finiteness of the critical set make this limit a single point. It cannot be $S$, so it is $M$. The other branch can neither enter $\mathcal{C}$ nor tend to its interior point $M$. Exactly one branch ends at $M$.

*General $k > 0$.* Apply the case $k = 1/6$ to $f/(6k)$. Its pins have gap $r^3/6$ and the arbitrary birth $b/(6k)$. Its derivative norms and its transverse curvature are those of $f$ divided by $6k$, so (S13.4) for $f/(6k)$ is exactly (S13.3) for $f$. The geometric ridge is unchanged. □

*Remark* S13.7 (margins for general $k$). For general $k > 0$ the scaling in the proof gives the following margins: the third derivative $F''$ of the ridge exceeds $3k/2$; the longitudinal drop $b - g(-2r)$ and rise $g(2r) - s$ are at least $\frac{27}{16}\, kr^3$, which exceeds the specified gap $kr^3$ by $\frac{11}{16}\, kr^3$; and the radial drop on the curved side of $\mathcal{C}$ exceeds $\frac{26400}{121}\, kr^3$. An arbitrary orthonormal frame can be used in this pathwise argument as long as its product chart is embedded; on the torus $\mathbb{T}^d_L$ this holds whenever $r < L/(4\sqrt{2})$ (Definition S12.2). Scaling or rotating the function here is not a proof that the conditional covariance, the pin energy or the normalizer of a random field is invariant under that operation.

## References


1. Robert J. Adler and Jonathan E. Taylor, *Random fields and geometry*, Springer Monographs in Mathematics, Springer, New York, 2007.
2. Diego Armentano, Jean-Marc Azaïs, and José R. León, *On a general Kac–Rice formula for the measure of a level set*, 2023, arXiv:2304.07424v3.
3. Anthony Carbery and James Wright, *Distributional and $L^q$ norm inequalities for polynomials over convex bodies in $\mathbb{R}^n$*, Math. Res. Lett. **8** (2001), 233–248.
4. Justin Curry, *The fiber of the persistence map for functions on the interval*, 2017, arXiv:1706.06059v2.
5. *NIST digital library of mathematical functions*, `https://dlmf.nist.gov/`, Sections 5.2, 5.5 and 5.11(ii).
6. Dylan Roy, *D1 Theorem A selection chain: source-bound reconciliation*, 2026, Repository record D1-CHAIN-RECONCILIATION-20260928-v1 (delegated AI work), 28 September 2026, `reviews/d1_chain_reconciliation_20260928/RECONCILIATION.md`, git blob `75da2597`. It reads the companion record *Uniform matrix-cap selection and unrestricted short-lifetime asymptotics* with the congruence erratum `imports/lifetime_parent_20260925/`

ERRATUM_CONGRUENCE.md (blob 213594d6), with its Section 9 replaced by reviews/d1_section9_borel_repair_20260925/REPAIR.md (blob fe9b9ce4), and with the cap proof imports/lifetime_parent_20260925/MARKED_CYLINDER_CAP_PROOF.md (blob 0633aca3). Available at https://github.com/d6g8k5htny-coder/Math-.

7. ———, *Uniform matrix-cap selection and unrestricted short-lifetime asymptotics*, 2026, Companion manuscript (delegated AI work), 24 September 2026. Available at https://github.com/d6g8k5htny-coder/Math-.